Deterministic Pushdown Normalization

From Completion-Aware DPDAs to Geodesic Compilers and a General Theory

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THEORY • CONSTRUCTION • PROOFS • GENERALIZATION

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AUTOMATA THEORY • NORMALIZATION • COMPLETION • STRUCTURE

Research Monograph

Deterministic Pushdown Normalization

From Completion-Aware DPDAs to Geodesic Compilers and a General Theory

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Research period

2023–2026

Edition note

This volume synthesizes a sequence of research manuscripts developed between 2023 and 2026 into a single integrated theory. The monograph preserves the chronological logic of discovery while consolidating repeated arguments, separating representation-specific results from representation-independent ones, and recording computational certificates beside the symbolic results they support.

Proof and validation note

Unless a result is explicitly identified as computer-assisted, the mathematical claims in this volume rest on symbolic arguments. Computational audits are used as independent consistency checks; Appendix D records their scope, numerical outcomes, and verification status without making the symbolic theorems depend on implementation details.

Version

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Preface

This book grew from a single deterministic pushdown automaton.

In 2023 I was studying the binary counting language

$$L_{2,1} = \{w \in \{0,1\}^* : N_1(w) = 2N_0(w)\}.$$

The first object of interest was not a general formalism but a concrete machine: a center state, two side states, and a stack manipulated by cancellation and replacement rules. The design was initially understood operationally. What made it the starting point of a larger research program was the realization that its state and stack were carrying more information than acceptance alone. After a prefix had been read, the configuration encoded a precise residual quantity and, after completion normalization, a shortest way of repairing that residual.

That observation supplied the first transition from a machine diagram to a mathematical theory. The stack ceased to be merely storage. It became a representation of completion demand. The side states ceased to be auxiliary control. They became finite corrections to the meaning of the stack. The cancellation transitions became local steps in a directed shortest-completion geometry.

The chapters that follow preserve this order of discovery. The original completion-aware DPDA is first analyzed on its own terms. Its reachable stores and state potentials lead to an exact residual invariant. Residuals then give a quotient coordinate for prefixes and a directed geometry of shortest continuations. That geometry supports progressively more economical realizations and, after the representation class is fixed carefully, exact resource frontiers. A related resource question then leaves the original coordinate system and becomes a representation-independent stack-alphabet hierarchy problem.

The research subsequently branches. Higher-dimensional counting systems turn shortest completions into cubical and local-action complexes. That geometry then admits a further abstraction: finite-phase periodic action systems support a finite cubical geodesic compiler for distance, optimal branching, shortest prefixes, and complete local optimal compatibility. The associated structure theory shows that higher-order compatibility is genuinely strict across the class: at every prescribed order new information can first appear, product constructions amplify the discrepancy Booleanly, and the resulting fiber viewpoint shows that this freedom is not merely a product artifact. On one fixed structural simplex, exact distance and all compatibility data through order $k-1$ can remain frozen while the entire order-k layer ranges over every k-uniform hypergraph. A different branch follows one cancellation pattern into Fuss–Catalan defect geometry and then into chronological and p-adic arithmetic. These branches are retained because they record genuine consequences of the completion viewpoint, but they are not allowed to obscure the central line of the book.

The central line returns to pushdown construction. For coprime positive p, q, a centered real-time literal-demand DPDA generalizes the original 2:1 architecture. Its finite-control rail and literal stack expose a WRITE–SEARCH–CANCEL normalization calculus. At that point a distinction that had been implicit from the beginning becomes explicit: the semantic process can be unique while its physical pushdown realization is not. This leads to the theory of exact semantic lifts, sections, representation debt, LIFO locality, normalization schedules, and causal obstruction. The six reduced 2:1 schedules are then upgraded to an arbitrary-coprime compiler and an exact causal-hypergraph classification, including the precise scope of downward closure and the category dependence of the named $A \to B$ persistence statement. Exact-lift minimality is sharpened before the General Theory by strict coordinate quotients, protected interfaces, bounded bi-recoding, and minimal protected cores. The same physical comparison layer then supports a separate confluence analysis: presentation schedules, literal completion fibers, finite-control sheets, and protected quotient classes are treated as four different rewrite objects rather than being conflated under one semantic normalizer.

The later mathematical branches are also pushed beyond their earlier endpoints. The optimal-fiber topology is no longer confined to rank two: a realization theorem identifies finite lattice-convex configurations as exactly the completion-total optimal fibers up to affine lattice equivalence, and product, apex, and prism operations give a rank-raising calculus with higher tori and unbounded higher homology at fixed active rank. The chronological-carry branch is extended from the CCR-3 jet to a typed fourth-order family in which the new connected quadruple coefficient is not forced by the third jet; its stationary p-divisible sector is one-parameter, exactly identifiable in the fourth column, and invisible to the common fourth-jet Stirling shadow modulo p. Finally, the representation-independent resource branch is sharpened from a state lower bound to the exact unrestricted state–stack antichain, and the raw one-work-symbol bounded-push deformation is determined exactly without promoting that typed result to arbitrary recognizers.

The final abstraction is deliberately broader than weighted counting languages. In the General Theory, a deterministic semantic system is given first, together with a physical policy domain and an exact projection from physical configurations to semantic states. Questions about canonical sections, local realizability, safe and fatal debt, schedule conflicts, synthesis, and verification can then be asked without assuming that the semantic state space is the integer residual line or that the physical machine is one of the centered $p{:}q$ constructions. The weighted-balance family serves as a demanding test laboratory for the abstract theory, not as its definition.

After that abstraction was completed, the residual and completion viewpoints produce several post-theory branches. The first counts the physical CANCEL/WRITE history of centered rational normalization itself: primitive cancellation becomes a Schubert statistic, repeated normalization factors through Bizley and Raney structures, and phase normality compresses arbitrary scale to an exact one-counter process. A bridge then separates these operational statistics from the intrinsic path statistics of the next branch, where rational-Dyck return events become coarea barrier contacts and the run–return enumeration problem is solved before abstraction to a contact-preserving barrier-compression framework. The shortest-completion map also generates the Farey-recursive arithmetic branch and its universal scalar spectra. Later Parts quantify fixed-semantics LIFO chronology, extend rank-two optimal-fiber topology to physical holonomy and sharp carrier cost, prove unrestricted state complexity, and finally synthesize the exact protected-contract Pareto frontiers. These post-theory branches are placed after the General Theory because they are consequences of the semantics already developed, not motivations invented after the fact.

Two editorial principles govern this monograph.

First, discovery history is preserved where it clarifies the mathematics. A later theorem is not retroactively presented as if it had motivated an earlier construction. In particular, the original 2023 machine remains the causal starting object, even when later chapters reveal cleaner coordinates or smaller realizations. The compact machines are conclusions of analysis, not substitutes for the original motivation.

Second, scope is treated as part of the theorem. Several different notions of "optimality" occur in this program: raw residual-faithful resource optimality, optimality inside a deferred-SEARCH literal-demand model, and lower bounds for broader DPDA representation classes. They are not interchangeable. Likewise, normalization to a common terminal representative is not promoted to a general Church–Rosser statement, and a fixed-policy causal obstruction is not silently upgraded to an unrestricted impossibility theorem. The book repeatedly marks these boundaries because many of the strongest results are strong precisely because their model assumptions are explicit.

The same principle applies to computation. Exhaustive programs were useful in three roles: discovering finite patterns, auditing symbolic formulas over large bounded regions, and certifying explicitly finite reductions. The first two roles do not replace proof. Where a theorem contains a genuinely finite computer-assisted component, the certificate and its scope are stated as such. Appendix D records the computational validation and distinguishes independently rerun audits, reconstructed validation runs, source-reported numerical checks, and purely symbolic results.

The book therefore has two simultaneous purposes. It is a consolidation of a sequence of papers, with repeated proofs merged and notation normalized. It is also a single mathematical narrative: a path from one completion-aware DPDA to a general language for separating semantic evolution from physical normalization.

After the post-theory consequence and synthesis branches, the final chapters return to the original $C/L/R$ machine. This return is intentional. It allows the reader to see, in the same small automaton, the residual split between state and stack, literal completion demand, SEARCH debt, kernel materialization, schedule choice, operational event statistics, physical holonomy, and representation-sensitive resource tradeoffs. The aim of the abstraction is not to make the machine disappear. It is to explain why the machine was doing more than was initially visible.

Alp Eren Bütün
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v7 integration note. This edition preserves the established discovery spine and extends the geodesic-compiler Part from prescribed-order separation and Boolean product amplification to universal k-fiber realization. The integration reuses the book's existing periodic-system, structural-cube, face-guard, and signature machinery rather than repeating the standalone article's self-contained preliminaries. The new material introduces k-fibers, proves their finite compilation, gives the universal two-phase construction and exact distinguished-layer capacity, and records the $s = 3, k = 2$ worked fiber and independent regression evidence. Unpublished companion manuscripts remain outside the bibliography; paper-specific introductions and declarations are not reproduced, and provenance is expressed through the monograph's own chapter structure and external literature boundaries.

Abstract

This monograph develops a theory of deterministic pushdown normalization from a concrete completion-aware DPDA for the binary language

$$L_{2,1} = \{w : N_1(w) = 2N_0(w)\}$$

to an abstract framework for exact semantic lifts, bounded LIFO locality, representation debt, causal scheduling, synthesis, and verification.

The first part analyzes the original 2:1 machine through the signed residual

$$\rho_{p,q}(w) = qN_1(w) - pN_0(w).$$

Its control state and logical stack satisfy an exact state-potential invariant, and end-of-input normalization converts the residual into a canonical shortest completion. The resulting completion problem has a directed-geodesic interpretation and, for general coprime p, q, a unique minimum Parikh demand characterized by the primitive kernel vector $K = (q, p)$. These coordinates lead to exact results on reachable stores, visibility, residual quotients, compact realizations, and resource frontiers. The monograph carefully separates raw residual-faithful optimality, model-relative deferred-SEARCH optimality, and representation-independent DPDA lower bounds.

The completion viewpoint is then extended in two directions. Higher-rank counting systems yield rectangular, cubical, and local-action completion complexes, including examples with nontrivial topology. The shortest-path layer is subsequently abstracted to positive-cost finite-phase periodic action systems. We develop a finite cubical geodesic compiler that recovers exact distance, all optimal first moves, arbitrary shortest-prefix membership, the complete local optimal-compatibility complex, canonical routes, and all shortest paths implicitly from finite arithmetic data. We also develop a structure theory for the information encoded by this compiler. For every prescribed order k, adjacent states can have equal shortest distance and identical compatibility through order $k-1$ while first differing at order k; independent products amplify one common lower-order signature into 2^m distinct next-order signatures. We then introduce k-fibers of the signature truncation and prove a universal realization theorem: for every $s \geq k \geq 2$, one fixed completion-total periodic system has a common order-$(k-1)$ signature whose complete order-k fiber is the full power set of the $\binom{s}{k}$ faces of a distinguished structural s-simplex. Hence every k-uniform hypergraph occurs while exact distance and all lower-order compatibility remain unchanged, and the exact distinguished-layer capacity is $2^{\binom{s}{k}}$. Thus no fixed summary $(d(v), \mathcal{H}_{\leq h}(v))$ universally reconstructs the full local optimal-compatibility complex, and lower-order optimal compatibility imposes no universal restriction on the next-order layer. A separate cancellation branch produces Fuss–Catalan defect geometry and a chronological carry refinement whose coefficient structure admits an exact Lagrange–Bürmann derivation and a p-adic interpretation.

Returning to machine construction, the book gives a centered real-time literal-demand DPDA for every coprime positive ratio p:q. The construction reveals WRITE, SEARCH,

and CANCEL as a normalization calculus. At the semantic level, the outstanding demand evolves uniquely; at the physical level, LIFO order and finite-control sheets create representation debt. Quantity debt, order debt, and control/sheet debt are separated explicitly. The semantic quotient is unique while physical sectioned lifts need not be. Starting from the six admissible reduced 2:1 schedules, an arbitrary-coprime compiler and Eight-Site theorem yield an exact causal-hypergraph classification for the canonical weighted-balance family. Downward closure is proved exactly in that family, shown not to be universal in unrestricted General-Theory schedule families, and the named $A \to B$ persistence question is resolved by separating schedule-presentation data from lift data. The exact-lift theory is then refined by strict coordinate quotients, protected interfaces, bounded bi-recoding, and finite minimal protected cores. This layer is completed by an explicit confluence hierarchy: schedule activation is classified for every coprime ratio, the entire completion-word fiber has a convergent two-debt rewrite, independent finite-control sheets admit an exact product reduction, and lexicographically anchored protected-core descent satisfies a genuine Newman criterion without collapsing protected physical information.

The final theory abstracts away from weighted counting. A deterministic semantic system $(M, \Sigma, \{U_a\})$ is related to a physical policy domain by an exact projection Θ satisfying

$$\Theta\delta_a = U_a\Theta.$$

A canonical section induces an idempotent retraction, and bounded local observation leads to capacity and obstruction theorems. Behavioral safety is characterized by a greatest forward-invariant hull; fatal schedules are controlled by reachable unsafe fibers and robust conflict supports. The earlier exact schedule families provide stress tests for the abstract theory, while the arbitrary-coprime classification has already been completed in the concrete weighted-balance category before this abstraction. The framework also gives section-synthesis criteria, forced-debt lower bounds, regular saturation methods, and a deterministic pushdown equivalence fallback for exact fixed-policy verification.

The first post-theory Part enumerates the physical events of centered rational normalization. Primitive CANCEL has an exact binomial/Schubert description, with the coprime diagonal $(1, 1)$ treated separately; rational-Dyck specialization exposes Narayana windows, repeated WRITE histories factor through Bizley and Raney structures, and a phase-normal theorem compresses the arbitrary-scale process to an exact one-counter recurrence. The following rational-path Part deliberately changes viewpoint from operational event statistics to intrinsic path statistics: the identity $R = g\rho_{p,q}$ turns returns into coarea barrier contacts and yields the determinant solution of the Dai–Fu–Qiu run/return problem, then a contact-preserving barrier-compression framework with fixed-composition, arbitrary-boundary, and parking-function applications. A separate arithmetic branch studies the shortest-completion minimizer map across Farey-related ratios and obtains an ordered parent-to-mediant recursion together with universal scalar spectra.

Four later branches sharpen the separation between semantic information and physical realization. Fixed-Parikh chronology exhibits $O(1)$, $\sqrt{n}$, and n LIFO scales. Optimal-fiber topology is first completed in rank two and then raised in rank: every finite lattice-convex configuration is realizable as an optimal completion fiber, products realize higher tori, apex and prism extensions have different homotopy effects, and fixed active rank with fixed toral fundamental group still permits unbounded higher homology. The physical layer sends $\pi_1 \to \delta \to K_0$ and distinguishes nonabelian holonomy from toral correction, with sharp LIFO carrier and repair/debt costs. The third-order chronological-carry branch is also extended exactly to order four: a connected quadruple contributes one independent stationary parameter, its four-band Hessenberg and normalized-column transport are explicit, and all parameters share the same fourth-jet Stirling shadow modulo p despite exact one-row identifiability. Returning to the natural balance family, the complete unrestricted state–stack frontier is $\{(p+q,2),(\max(p,q),3)\}$, while the one-work-symbol raw bounded-push branch has an exact state–push staircase and the three-symbol optimum survives with push at most two. Finally, protected-contract refinement combines the relevant endpoint theorems into exact typed Pareto frontiers, with bounded bi-recoding separating harmless coordinate changes from genuine resource reductions.

Throughout, symbolic proofs remain primary. Large computational audits and finite certificates are recorded separately as validation evidence. The result is a unified account of how completion semantics, pushdown representation, shortest-path compilation, normalization chronology, and their later structural, enumerative, topological, and arithmetic consequences interact—from a single 2023 DPDA to geodesic compilers, a general theory, and new mathematics generated by its residual and completion semantics.

Contents

List of Figures

Notation and Conventions

This page records the notation used across several layers of the monograph. Symbols that are intentionally local to a chapter are defined again where they are used.

Words and weighted balance

$N_a(w)$ — Number of occurrences of the symbol a in the word w.

$L_{p,q}$ — For coprime positive integers p, q,

$$L_{p,q} = \{w \in \{0,1\}^* : qN_1(w) = pN_0(w)\}.$$

$\rho_{p,q}(w)$ — Signed residual

$$\rho_{p,q}(w) = qN_1(w) - pN_0(w).$$

Thus reading 0 changes the residual by $-p$, while reading 1 changes it by $+q$.

m — The total weight $m = p + q$ when the $p{:}q$ family is under discussion.

K — Primitive nonnegative kernel vector

$$K = (q, p),$$

which preserves the weighted completion equation.

$\mu(r)$ — The canonical minimum nonnegative completion-demand vector associated with residual r; when written (x, y), x counts appended zeros and y appended ones.

$\eta(x, y)$ — The canonical literal stack word associated with demand (x, y) in the relevant orientation, typically a fixed two-block representative such as 1^y0^x.

Original and centered machines

C, L, R — Center, left, and right states of the original 2:1 completion-aware DPDA. In the original-machine chapters their state potentials are $\Phi(C) = 0$, $\Phi(L) = 3$, and $\Phi(R) = -3$.

$D(S)$ — Original-machine logical-stack weight

$$D(S) = 2N_0(S) - N_1(S).$$

Together with the state potential it represents the prefix residual.

h_{ctr} — Centered-rail half-width

$$h_{\mathrm{ctr}} = \left\lfloor \frac{p+q-1}{2} \right\rfloor.$$

The subscript distinguishes this scalar from the WSC exchange vector below.

D_d — State at centered-rail coordinate d; $D_0 = C$. In the centered construction the state potential is proportional to dm.

WRITE–SEARCH–CANCEL coordinates

P — Physical count shadow of a literal stack or physical representation.

Θ — Canonical semantic demand/count target represented by a physical configuration. In the abstract General Theory the same letter denotes the semantic projection from physical configurations to semantic states.

Q — Quantity defect, typically

$$Q = \Theta - P.$$

Ω — Order defect: the part of physical noncanonicity that is invisible to Parikh counts but visible to LIFO order.

$\mathbf{h}$ — WSC exchange direction

$$\mathbf{h} = (1, -1).$$

It is typographically distinct from h_{ctr}.

M_t — Kernel-materialization count at time t in a physical WSC factorization.

General Theory

M — Semantic state space of a deterministic semantic system. This use of M is local to the General Theory and should not be confused with the WSC scalar M_t.

Σ — Input alphabet.

$U_a : M \to M$ — Semantic update induced by input symbol a.

C — Physical policy/configuration domain in the abstract General Theory. This is a context-local reuse of the letter C; it is not the center state of the earlier machine chapters.

$\delta_a : C \to C$ — Physical update under input symbol a, on the policy domain where it is defined.

$\Theta : C \to M$ — Exact semantic projection satisfying

$$\Theta\delta_a = U_a\Theta.$$

$\sigma : M \to C$ — Chosen canonical section, with $\Theta\sigma = \mathrm{id}_M$.

N — Canonical retraction

$$N = \sigma\Theta.$$

It satisfies $N^2 = N$ on its domain.

$\mathcal{D}_H$ — Immediate-disagreement or bad set associated with a fixed physical policy or schedule H, as defined in the relevant chapter.

$\mathcal{B}_H$ — Behavioral safe hull for policy H, equivalently the greatest forward-invariant subset that avoids eventual observable disagreement under the stated hypotheses.

$\mathrm{Adm}(\mathcal{E})$ — Admissible schedule family for a schedule system or environment $\mathcal{E}$.

Rational-path enumeration

$\mathcal{D}_{m,n}$ — The set of $m \times n$ rational Dyck paths. In Parts XII–XIII, m may denote a rectangle width; this local use is distinct from the earlier shorthand $m = p + q$ when that shorthand is explicitly in force.

$\mathcal{U}(P) = (u_1, \ldots, u_n)$ — The coarea sequence of a rational Dyck path, where u_i is the starting column of the ith north step.

b_i — The rational-Dyck upper coarea barrier

$$b_i = \left\lfloor \frac{m(i-1)}{n} \right\rfloor.$$

A return is an upper-barrier contact $u_i = b_i$.

run, $\widetilde{\mathrm{run}}$, ret — Ordinary rational run, ratio-run, and return statistics. The sampled notation run_d unifies the two run specializations only where explicitly stated.

$\alpha \vDash n$ — An ordered vertical-run composition. The operators RetComp and ReturnSet denote the corresponding exact return-component and return-position sets in the fixed-composition chapters.

Farey shortest-completion arithmetic

$\mu_{p,q}(r)$ — The unique shortest nonnegative completion vector

$$\mu_{p,q}(r) = \operatorname*{arg\,min}_{x,y\geq 0}\{x + y : px - qy = r\}.$$

This is the ratio-indexed form of the earlier canonical minimum-demand map $\mu(r)$.

$L_{p,q}(r)$ — The scalar completion length $|\mu_{p,q}(r)|_1$.

$((c,u),(v,d))$ — The oriented Farey parents of a proper coprime pair (p,q), normalized in Part XIV by

$$(p,q) = (c,u) + (v,d), \qquad pu - qc = qv - pd = 1.$$

Scope conventions

"Optimal", "minimal", and "unique" are always read relative to the model specified in the theorem. In particular, the monograph distinguishes raw residual-faithful resource frontiers, deferred-SEARCH literal-demand width, and unrestricted DPDA recognition resources. A common terminal normal form is not, by itself, a claim of Church–Rosser confluence.

The symbol $\dashv$ denotes an explicit right endmarker. Completion callouts selected by $\dashv$ are not ordinary competing ε-moves.

Post-theory notation added in v3

$\mathcal{F}_{gq,gp}$	Balanced fixed-Parikh fiber used in the random-ordering experiment.
$\kappa_{p,q}$	Physical rematerialization constant of the block-boundary lift chain.
$\mathcal{O}_A(r)$	Optimal minimum-completion Parikh fiber for matrix system A and residual r.
$L_A(r)$	Active trade lattice generated by differences of optimal completions.
$I_A(r)$	Filled shortest-prefix quotient at residual r.
$Q_{\min}(L_{p,q})$	Minimum number of control states for the stated deterministic-pushdown recognition convention.

Prologue — The Machine Before the Theory

This monograph begins with a machine, not with an abstract theory.

The starting problem is the binary counting language

$$L_{2,1} = \{w \in \{0,1\}^* : N_1(w) = 2N_0(w)\}.$$

At the level of arithmetic, the condition is elementary. One can measure the imbalance between the numbers of zeros and ones by a signed linear expression. At the level of language recognition, this already suggests counter-like memory.

The research program developed in this monograph began somewhere more concrete. In 2023, a deterministic pushdown design was constructed for this language using a center state, two side states, and a stack whose contents were manipulated by cancellation and replacement rules. The machine was not originally designed as an instance of a general theory of semantic fibers, local realizability, normalization debt, or causal schedules. Those notions did not yet exist in the project.

What made the design mathematically productive was a more elementary observation: after a prefix had been processed, the machine appeared to remember not only whether the prefix was currently balanced, but also **what was still needed to make it balanced**.

That observation changed the direction of the work.

The first question was therefore not whether $L_{2,1}$ was deterministic context-free. The more useful questions were:

- Which stack contents can actually occur?
- Why do the side states matter arithmetically?
- What exactly is the cancellation operation doing?
- When the input ends away from balance, does the machine encode a continuation that repairs the deficit?
- Is that continuation shortest?
- Why does looking only a small distance into the stack reveal information about the entire store?
- Can the same structure be built for a general ratio $p{:}q$?
- If several machines realize the same counting semantics, which aspects of their design are forced and which are representational choices?

The answers do not arrive all at once. They are the subject of the book.

The development will repeatedly return to the original machine. First it will be analyzed as a concrete DPDA. Its state and stack behavior will then be reinterpreted arithmetically. The arithmetic will become a geometry of shortest completions. That geometry will lead to more efficient pushdown representations and exact resource frontiers. A resource question will then escape the original representation class and lead to a stack-alphabet hierarchy theorem for general deterministic pushdown automata. The completion geometry will branch into higher-dimensional topology and then into finite geodesic compilation with a strict hierarchy of higher-order optimal compatibility, while one local cancellation transition will generate a separate Fuss–Catalan defect geometry and, later, an arithmetic p-adic offshoot.

After those branches, the book will return to the machine-construction problem. A uniform real-time $p{:}q$ DPDA will explain the arithmetic hidden inside the original center/left/right architecture. WRITE, SEARCH, and CANCEL will then cease to be merely names for transition patterns and become components of a normalization calculus. Finally, the distinction between a unique semantic process and its nonunique physical pushdown realizations will lead to a general theory of locality, representation debt, causal schedules, synthesis, and verification.

The final chapters return once more to the 2023 design.

The aim is therefore not to replace the original machine by abstraction. It is to understand how much theory was already latent in it.

Part I

The Origin

Research provenance. Parts I–II begin from the author's 2023 completion-aware machine and synthesize the first formal stages of the research sequence. Recognition by counter-like deterministic devices and related one-counter questions are classical background rather than novelty claims here [53, 40, 18].

Chapter 1

The 2023 Completion-Aware DPDA

1.1 The motivating language

For a binary word w, let $N_0(w)$ and $N_1(w)$ denote the numbers of zeros and ones in w. The motivating language is

$$L_{2,1} = \{w \in \{0,1\}^* : N_1(w) = 2N_0(w)\}.$$

Equivalently,

$$2N_0(w) - N_1(w) = 0.$$

It is useful to define already, without yet turning it into a general theory, the imbalance

$$\rho_{2,1}(w) = N_1(w) - 2N_0(w).$$

Thus $\rho_{2,1}(w) = 0$ exactly when $w \in L_{2,1}$.

The original 2023 DPDA was built directly at the machine level. Its characteristic structure consists of three semantic core states:

$$C, \qquad L, \qquad R,$$

which we call the **center**, **left**, and **right** states.

The center is the balanced operational hub. The side states are not independent counters. They record a finite correction to the meaning of the physical stack. This interpretation will be proved rather than assumed.

Before doing so, we must distinguish two formal versions of the original design.

1.2 Two formal variants of the same core design

There are two versions of the 2023 machine that we will preserve throughout the monograph.

The first excludes the empty input and recognizes

$$L_{2,1} \setminus \{\varepsilon\}.$$

The second includes the empty input and recognizes the standard language

$$L_{2,1}.$$

Their center/left/right recognition logic is the same. Their difference lies in the initialization and acceptance wrapper.

This distinction is worth preserving for two reasons.

First, it records the actual development of the design rather than silently collapsing two formal choices into one picture. Second, it separates a semantic question—how the $C/L/R$ core represents imbalance—from an acceptance-convention question—whether ε is accepted by the wrapper.

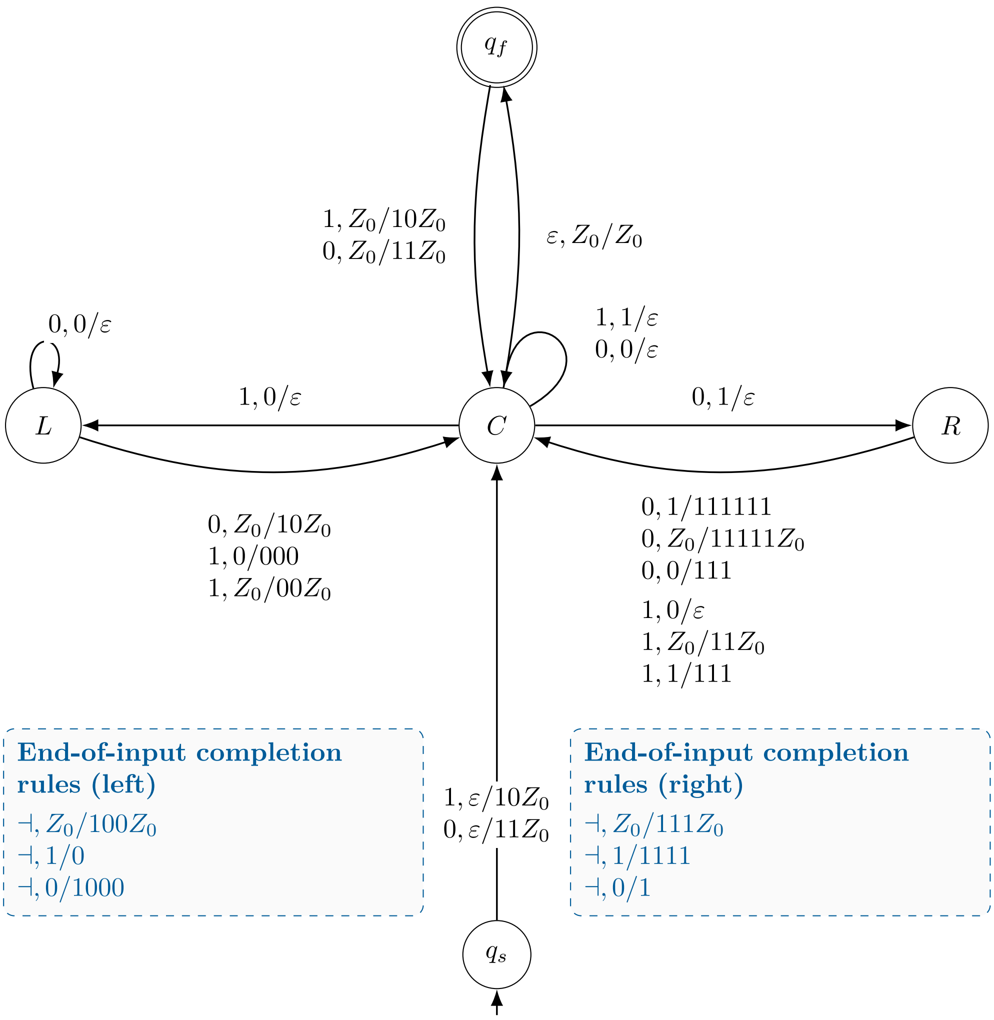


Figure 1.1: Formalized redrawing of the original 2:1 DPDA variant with the empty input excluded. Solid arrows are ordinary recognition transitions; dashed callout boxes contain only right-endmarker completion rules. The listed left-state completion rule with top 1 is retained for fidelity but has an unreachable source condition.

The logical core has states C, L, R. The wrapper contains the source state q_s and accepting state q_f.

The principal recognition transitions are the following.

From C:

$$1, 0/\varepsilon : C \to L,$$

$$0, 1/\varepsilon : C \to R,$$

and the center cancellation loops

$$1, 1/\varepsilon, \qquad 0, 0/\varepsilon.$$

The left state has the characteristic cancellation loop

$$\boxed{0, 0/\varepsilon}$$

and returns toward the center through the replacement rules

$$0, Z_0/10Z_0,$$

$$1, 0/000,$$

$$1, Z_0/00Z_0.$$

The right state returns toward the center through

$$0, 1/111111,$$

$$0, Z_0/11111Z_0,$$

$$0, 0/111,$$

$$1, 0/\varepsilon,$$

$$1, Z_0/11Z_0,$$

$$1, 1/111.$$

The wrapper moves between C and q_f at empty logical stack and initializes nonempty inputs appropriately.

This version does not accept ε.

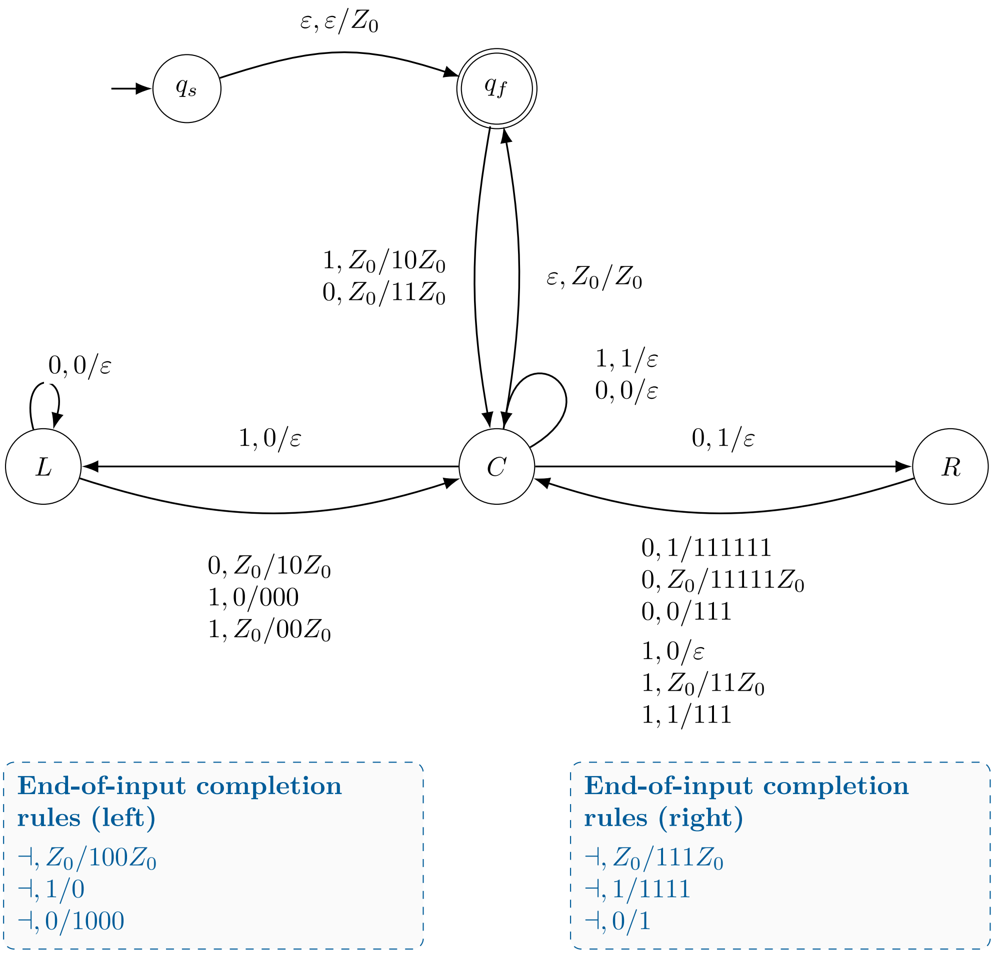


Figure 1.2: Formalized redrawing of the original 2:1 DPDA variant with the empty input included. The center/left/right recognition logic and end-of-input completion rules are identical to Figure 1.1; only the initialization/final-state wrapper changes so that ε is accepted.

The $C/L/R$ transition system is unchanged.

The difference is the initialization/final-state wrapper. In particular, an initial

$$\varepsilon, \varepsilon/Z_0$$

wrapper move allows the physical machine to represent the empty prefix at the accepting bottom-marker configuration.

Consequently this version recognizes

$$L_{2,1},$$

including ε.

The two figures will remain separate in the final monograph, but their common recognition core will be analyzed only once.

1.3 Recognition and completion are different operations

The original design also contains rules whose purpose is not to consume another binary input symbol, but to expose the continuation represented by a nonbalanced final configuration.

These are **completion rules**.

In the formal monograph model, they are enabled only when a distinguished right endmarker

$$\dashv$$

is the next unread symbol.

For the left state, the completion list is

$$\dashv, Z_0/100Z_0,$$

$$\dashv, 1/0,$$

$$\dashv, 0/1000.$$

For the right state, it is

$$\dashv, Z_0/111Z_0,$$

$$\dashv, 1/1111,$$

$$\dashv, 0/1.$$

The rule with left-state top 1 is retained as part of the original completion list, but we will prove shortly that its source condition is unreachable.

The endmarker convention is essential. If these operations were interpreted as ordinary ε-moves, some state/top pairs would also have input-consuming moves, producing a conflict with standard DPDA determinism. With the right endmarker, recognition on 0 or 1 and completion on $\dashv$ are disjoint cases.

Lemma 1.1 — Endmarker determinism.

If every post-input completion rule is interpreted as a $\dashv$-labeled transition to a completion sink with no outgoing transitions, then the completion-extended machine is deterministic in the standard DPDA sense.

Proof.

Fix a state/top pair.

During binary recognition, the next input symbol is 0 or 1. A completion rule is unavailable.

After binary input has been exhausted, the next input symbol is $\dashv$. The binary recognition rules are unavailable, and at most one completion rule applies to the current state/top pair.

Thus every instantaneous configuration has at most one successor. The completion sink introduces no further conflict. □

This formalization will be used consistently in the rest of the monograph.

1.4 Removing the wrapper from the semantic analysis

The physical states q_s and q_f serve initialization and acceptance. They should not be confused with the semantic $C/L/R$ core.

Write a physical stack as

$$SZ_0, \qquad S \in \{0,1\}^*,$$

where S is the **logical stack**.

Define the core projection

$$\pi(C, SZ_0) = (C, S),$$

$$\pi(L, SZ_0) = (L, S),$$

$$\pi(R, SZ_0) = (R, S),$$

together with

$$\pi(q_s, \varepsilon) = (C, \varepsilon),$$

$$\pi(q_f, Z_0) = (C, \varepsilon).$$

Thus the empty prefix and every balanced prefix accepted at q_f receive the same zero-residual semantic core meaning.

From now on, unless the wrapper itself matters, “the configuration after a prefix” means its core-projected configuration.

1.5 The reachable stack is much more restricted than it looks

The transition table contains replacements of several different lengths. One might therefore expect the machine to generate a complicated set of stack words.

It does not.

Let

$$\mathcal{S}_C = 0^* \cup 1^* \cup \{10\},$$

$$\mathcal{S}_L = 0^*,$$

$$\mathcal{S}_R = 1^* \cup \{0\}.$$

Theorem 1.2 — Reachable-store normal form.

For every binary prefix processed by either formal variant, if

$$(q, S)$$

is its core-projected configuration, then

$$S \in \mathcal{S}_q.$$

In particular:

- every reachable left-state stack is unary 0^k;
- every reachable right-state stack is either unary 1^k or the exceptional word 0;
- every reachable center stack is unary or the exceptional mixed word 10.

Proof.

We proceed by induction on the number of consumed input symbols.

The projected empty configuration is

$$(C, \varepsilon),$$

and

$$\varepsilon \in 0^* \cap 1^*.$$

Assume the current projected configuration belongs to the stated family.

The center loops remove matching top symbols. Hence they preserve unary forms.

A center mismatch on top 0 moves to L and removes that top. The resulting left-state stack is therefore still in 0^*.

A center mismatch on top 1 moves to R and removes that top. The resulting right-state stack lies in 1^*, with the special bottom-related construction accounting for the exceptional reachable 0 case.

The left-state loop

$$0, 0/\varepsilon$$

preserves 0^*. Its return transitions create only the allowed center forms 0^*, 1^*, or 10.

The right-state return transitions similarly produce only the allowed center families.

No recognition transition enters L with top 1. Therefore a left-state source condition with logical top 1 is unreachable.

All reachable configurations remain in the stated families. □

The theorem is stronger than a convenient regular-language description. It tells us that the apparently global question

> “Does a zero or a one occur somewhere in the stack?”

may have a bounded local answer.

Before proving that, we need to understand why the side states exist.

1.6 The state carries part of the arithmetic

Define the stack weight

$$D(S) = 2N_0(S) - N_1(S).$$

Define the state potential

$$\Phi(C) = 0,$$

$$\Phi(L) = 3,$$

$$\Phi(R) = -3.$$

Theorem 1.3 — Original-machine state-potential invariant.

After every consumed prefix w, let

$$(q, S)$$

be the core-projected configuration. Then

$$\boxed{\rho_{2,1}(w) = D(S) + \Phi(q).}$$

Proof.

For the empty prefix,

$$\rho_{2,1}(\varepsilon) = 0,$$

and the projected configuration is

$$(C, \varepsilon),$$

so

$$D(\varepsilon) + \Phi(C) = 0.$$

It remains to compare the change of the two sides under each recognition transition.

Reading 0 decreases $\rho_{2,1}$ by 2. Reading 1 increases it by 1.

Each stack replacement changes $D(S)$, and every state change changes Φ. The transition table is constructed so that

$$\Delta\big(D(S) + \Phi(q)\big) = \Delta\rho_{2,1}.$$

For the transition that will later generate the cancellation geometry,

$$(L, 0S) \xrightarrow{0} (L, S),$$

the stack weight decreases by 2, while the state potential remains 3. Thus the right-hand side changes by -2, exactly matching the change induced by consuming the relevant symbol under the selected residual sign convention.

The remaining transition types satisfy the same identity by direct finite verification. The complete transition-by-transition audit is recorded in Appendix A. □

Interpretation

The logical stack alone does **not** always equal the current arithmetic deficit.

The side state stores a finite correction:

$$+3 \quad \text{in } L,$$

$$-3 \quad \text{in } R.$$

This explains a configuration such as

$$(L, \varepsilon).$$

Its empty logical stack does not mean zero residual. The missing value is carried by the state.

This is the first place where the machine reveals a distinction that will become much more important later:

$$\boxed{\text{semantic information} = \text{stack information} + \text{finite-control information.}}$$

At this point, however, we keep the observation concrete.

1.7 End-of-input completion transfers the state correction into the stack

Suppose the binary input has been exhausted.

If the projected configuration is already in C, no side-state correction is required.

If it is in L or R, the unique endmarker completion rule converts the side-state potential into a normalized stack representation.

The result has an exceptionally simple form.

Theorem 1.4 — Original-machine shortest-completion extraction.

After end-of-input normalization, the logical stack is

$$\widehat{S}(w) = \begin{cases} 0^m, & \rho_{2,1}(w) = 2m, \quad m \geq 0, \\ 10^{m+1}, & \rho_{2,1}(w) = 2m+1, \quad m \geq 0, \\ 1^m, & \rho_{2,1}(w) = -m, \quad m \geq 1. \end{cases}$$

When interpreted as a continuation word, $\widehat{S}(w)$ is a shortest suffix that completes w to $L_{2,1}$.

Proof.

By Theorems 1.2 and 1.3, every side-state configuration belongs to a one-parameter stack family and differs from its stack weight by exactly the fixed state potential.

The endmarker completion rule removes that finite state correction and returns the representation to the zero-potential center convention.

Consequently, after normalization,

$$D(\widehat{S}(w)) = \rho_{2,1}(w).$$

The reachable normalized center family is

$$0^* \cup 1^* \cup \{10^+\},$$

so the displayed residual-to-stack map is injective.

Now let a completion contain x zeros and y ones. The completed word must satisfy

$$N_1(w) + y = 2\big(N_0(w) + x\big).$$

Since

$$\rho_{2,1}(w) = N_1(w) - 2N_0(w),$$

this is equivalent to

$$\boxed{2x - y = \rho_{2,1}(w).}$$

The three displayed normalized forms give the unique minimum nonnegative Parikh pair for this equation. Hence the continuation selected by the normalized stack has minimum possible length. □

The important discovery is independent of notation:

$$\boxed{\text{the machine's end configuration can be decoded as a shortest repair word.}}$$

This is the observation from which the completion theory begins.

1.8 A worked run: the prefix 111

Consider the empty-input-excluded presentation and ignore Z_0 in the logical-stack notation.

Starting in the center:

$$(C, 111 \dashv, \varepsilon) \vdash (C, 11 \dashv, 10) \vdash (C, 1 \dashv, 0) \vdash (L, \dashv, \varepsilon).$$

The consumed prefix has

$$\rho_{2,1}(111) = 3.$$

The physical logical stack is empty, but the machine is in L, whose potential carries the missing correction.

The endmarker completion rule

$$\dashv, Z_0/100Z_0$$

materializes the continuation

$$100.$$

Indeed,

$$111100$$

contains four ones and two zeros, so

$$111100 \in L_{2,1}.$$

No shorter continuation can repair the count imbalance.

Thus a side state that initially looks like an implementation detail has a precise role: it stores a finite amount of the completion demand outside the stack until the end-of-input normalizer materializes it.

1.9 The first structural surprise: two stack cells are enough

For a word v, let

$$\operatorname{alph}(v) = \{a \in \{0,1\} : a \text{ occurs in } v\}.$$

Let

$$\operatorname{pref}_d(v)$$

denote the prefix of v of length $\min(d, |v|)$.

Theorem 1.5 — Tight depth-two visibility.

For every completion-normalized stack S,

$$\boxed{\operatorname{alph}(S) = \operatorname{alph}(\operatorname{pref}_2(S)).}$$

Depth two is necessary.

Proof.

By Theorem 1.4, every normalized stack belongs to one of the families

$$0^*, \qquad 1^*, \qquad 10^+.$$

Every symbol that occurs anywhere in one of those words therefore occurs within the first two positions.

Depth one does not suffice. The normalized words

$$1$$

and

$$10$$

have the same top symbol but different global symbol supports.

Hence the exact visibility depth is two. □

This theorem formalizes an intuition present in the original design: a question that appears to require searching the entire stack can, on the reachable normalized family, be answered from only the first two cells.

The reason is not that arbitrary pushdown stacks are locally transparent. It is that the machine's reachable stores occupy a highly restricted language.

That distinction will matter repeatedly later.

1.10 The cancellation transition that will reappear much later

One transition deserves to be marked now even though its deeper significance will not be developed until Part V:

$$\boxed{(L, 0S) \xrightarrow{0} (L, S).}$$

At the machine level it is simply a cancellation operation.

At the arithmetic level it participates in the state-potential invariant.

Much later, when prefixes are represented as nonnegative 2-Dyck paths, the same transition will be characterized by an exact family of landing heights. That observation will generate correction signatures, an intrinsic defect, a Catalan boundary, and a binary cancellation skeleton.

We do not need any of that theory yet.

For now we record only the transition and its exact place in the original machine.

1.11 What has been learned without leaving the original machine

We can now answer several questions that were not visible from the transition diagram alone.

First, the reachable stack language is extremely small.

Second, the side states carry an exact finite arithmetic potential.

Third, the post-input completion rules can be formalized deterministically with a right endmarker.

Fourth, the final state/stack configuration can be normalized into a shortest continuation.

Fifth, global symbol support of the normalized store is exactly visible at depth two.

These facts suggest that the original DPDA is not merely storing an arbitrary history of the input. Its configuration is representing a much smaller mathematical object.

The next chapter identifies that object.

The progression will be:

$$\boxed{\begin{array}{c}\text{physical state and stack} \longrightarrow \text{residual}\\ \longrightarrow \text{minimum completion demand} \longrightarrow \text{directed completion geometry.}\end{array}}$$

Only after that geometry is understood will we return to the engineering question:

> Can the representation itself be improved?

That question will eventually produce the improved and scoped-optimal DPDA constructions, the exact resource frontiers, and the later general $p{:}q$ architectures.

Chapter 2

What the Machine Is Really Storing

Chapter 1 stayed deliberately close to the 2023 transition diagram. We saw that the reachable stores are highly restricted, that the side states carry a finite arithmetic potential, and that an end-of-input normalization converts a nonbalanced configuration into a shortest continuation.

Those observations suggest a simpler object behind the physical machine.

The purpose of this chapter is to identify it.

The central distinction will be:

$$\boxed{\begin{array}{c}\text{physical configuration} \longrightarrow \text{arithmetic residual}\\ \longrightarrow \text{minimum completion demand.}\end{array}}$$

The first object depends on a particular DPDA representation. The second and third do not.

That separation is the first decisive abstraction in the monograph.

2.1 From the 2:1 imbalance to the general p:q residual

The original language satisfies

$$N_1(w) = 2N_0(w).$$

The natural general family is

$$L_{p,q} = \{w \in \{0,1\}^* : qN_1(w) = pN_0(w)\},$$

where

$$p, q \geq 1, \qquad \gcd(p, q) = 1.$$

The coprimality assumption removes a redundant common scale. Nothing about the language changes if p and q are divided by their greatest common divisor.

Define the residual

$$\boxed{\rho_{p,q}(w) = qN_1(w) - pN_0(w).}$$

Thus

$$w \in L_{p,q} \iff \rho_{p,q}(w) = 0.$$

Reading a symbol changes the residual by a fixed amount:

$$\rho_{p,q}(w0) = \rho_{p,q}(w) - p,$$

$$\rho_{p,q}(w1) = \rho_{p,q}(w) + q.$$

For the motivating ratio $(p, q) = (2, 1)$,

$$\rho_{2,1}(w) = N_1(w) - 2N_0(w),$$

exactly the convention used in Chapter 1.

The residual is already a substantial simplification. Instead of retaining the whole Parikh pair

$$(N_0(w), N_1(w)),$$

we retain one integer.

But the stronger observation is that this one integer determines exactly what is needed on the right to reach the language.

2.2 The completion equation

Let w be a prefix with residual

$$r = \rho_{p,q}(w).$$

Suppose a continuation z contains

$$x = N_0(z)$$

zeros and

$$y = N_1(z)$$

ones.

Then

$$wz \in L_{p,q}$$

exactly when

$$q\big(N_1(w) + y\big) = p\big(N_0(w) + x\big).$$

Rearranging gives the basic completion equation.

Lemma 2.1 — Completion equation.

For every prefix w and continuation z with Parikh pair (x, y),

$$\boxed{z \in w^{-1}L_{p,q} \iff px - qy = \rho_{p,q}(w).}$$

Proof.

Expand the defining equality

$$qN_1(wz) = pN_0(wz).$$

Since

$$N_0(wz) = N_0(w) + x, \qquad N_1(wz) = N_1(w) + y,$$

we obtain

$$qN_1(w) + qy = pN_0(w) + px.$$

Therefore

$$px - qy = qN_1(w) - pN_0(w) = \rho_{p,q}(w).$$

The steps are reversible. □
This equation is the semantic heart of the original completion mechanism.
For $p = 2, q = 1$, it becomes

$$2x - y = r.$$

Thus the Chapter 1 example

$$w = 111, \qquad r = 3,$$

requires

$$2x - y = 3.$$

The minimum nonnegative solution is

$$(x, y) = (2, 1).$$

Hence every shortest completion has two zeros and one one.
There are three such words:

$$100, \qquad 010, \qquad 001.$$

The original DPDA selects 100.
This distinction will be important throughout the book:

$$\boxed{\text{minimum Parikh demand is unique;} \quad \text{a shortest word need not be unique.}}$$

2.3 The primitive balanced cycle

The completion equation

$$px - qy = r$$

has an elementary but fundamental symmetry.
If (x, y) is a solution, then

$$(x + q,\ y + p)$$

is another solution, because

$$p(x + q) - q(y + p) = px - qy.$$

The vector

$$\boxed{K = (q, p)}$$

is therefore the primitive nonnegative kernel direction.

Its meaning is transparent:

- adding q zeros contributes pq;
- adding p ones contributes $-pq$;
- the two contributions cancel.

At the word level, any word containing exactly q zeros and p ones has net residual change zero.

Its length is

$$\boxed{p + q.}$$

Later this will become the fundamental directed cycle length of the residual graph.

At this stage we need only its Diophantine role.

2.4 Every completion lies on one affine ray

Because $\gcd(p, q) = 1$, the integer solutions of

$$px - qy = r$$

form a one-dimensional affine lattice.

Proposition 2.2 — Affine completion ray.

For every

$$r \in \mathbb{Z},$$

there is a unique nonnegative solution

$$\mu(r) = (x_r, y_r)$$

minimizing

$$x + y$$

subject to

$$px - qy = r.$$

Every nonnegative solution is

$$\boxed{(x_r + qk,\ y_r + pk), \qquad k \in \mathbb{N}_0.}$$

Consequently the possible completion lengths are exactly

$$\boxed{\Delta(r) + (p + q)k, \qquad k \in \mathbb{N}_0,}$$

where

$$\Delta(r) = x_r + y_r.$$

Proof.

Since p and q are coprime, Bézout's identity guarantees an integer solution

$$(x_0, y_0)$$

to

$$px - qy = r.$$

Every integer solution has the form

$$(x_0 + qk,\ y_0 + pk), \qquad k \in \mathbb{Z}.$$

For sufficiently large k, both coordinates are nonnegative.
Let k_* be the least integer for which this happens, and define

$$(x_r, y_r) = (x_0 + qk_*,\ y_0 + pk_*).$$

Every other nonnegative solution is obtained by replacing k_* with

$$k_* + j, \qquad j \in \mathbb{N}_0.$$

Therefore every other nonnegative solution adds

$$j(q, p).$$

Its length increases by

$$j(p + q).$$

Hence (x_r, y_r) is the unique minimum-length nonnegative Parikh pair. □
This proposition gives the first exact picture of the completion space:

$$\boxed{\mu(r), \quad \mu(r) + K, \quad \mu(r) + 2K, \quad \mu(r) + 3K, \ldots}$$

The minimum demand is the first lattice point of an affine ray inside the nonnegative quadrant.

The later geometry of the monograph begins here.

2.5 Minimum completion demand

We now give the central object a name.

Definition 2.3 — Minimum completion demand.

For residual

$$r \in \mathbb{Z},$$

let

$$\boxed{\mu_{p,q}(r) = (x_r, y_r)}$$

be the unique nonnegative solution of

$$px - qy = r$$

minimizing

$$x + y.$$

Define the minimum completion length

$$\boxed{\Delta_{p,q}(r) = x_r + y_r.}$$

When p, q are fixed, we write simply

$$\mu(r), \qquad \Delta(r).$$

The word **demand** is intentional.
The pair

$$(x_r, y_r)$$

does not describe the prefix already read. It describes what the prefix still requires:

- x_r additional zeros;
- y_r additional ones.

The physical DPDA need not store those two integers literally.
The semantic claim is only that a configuration representing residual r implicitly represents the unique pair $\mu(r)$.
For $L_{2,1}$,

$$2x - y = r.$$

The minimum demand has the closed form

$$\mu(2m) = (m, 0), \qquad m \geq 0,$$

$$\mu(2m + 1) = (m + 1, 1), \qquad m \geq 0,$$

and

$$\mu(-m) = (0, m), \qquad m \geq 1.$$

These are exactly the three families already materialized by the original 2023 DPDA:

$$0^m, \qquad 10^{m+1}, \qquad 1^m.$$

So the stack forms discovered operationally in Chapter 1 are not arbitrary normal forms.
They are physical encodings of $\mu(r)$.

2.6 A canonical shortest word

The pair

$$\mu(r) = (x_r, y_r)$$

is unique, but its letters can usually be ordered in several ways.
For semantic decoding it is convenient to choose one fixed representative.
The minimum pair has a simple modular description.

Theorem 2.4 — Canonical two-block shortest completion.

Let $r > 0$.
There is a unique

$$y_r \in \{0, \dots, p-1\}$$

such that

$$qy_r \equiv -r \pmod{p}.$$

Set

$$x_r = \frac{r + qy_r}{p}.$$

Then

$$\boxed{C(r) = 1^{y_r} 0^{x_r}}$$

is a shortest completion word.
For $r < 0$, let

$$x_r \in \{0, \dots, q-1\}$$

be the unique solution of

$$px_r \equiv r \pmod{q}.$$

Set

$$y_r = \frac{px_r - r}{q}.$$

Then

$$\boxed{C(r) = 0^{x_r} 1^{y_r}}$$

is a shortest completion.
Finally,

$$C(0) = \varepsilon.$$

Proof.
Suppose first that $r > 0$.
The congruence

$$qy \equiv -r \pmod{p}$$

has a unique solution modulo p because q is invertible modulo p.
Selecting the representative

$$0 \leq y_r < p$$

makes

$$x_r = \frac{r + qy_r}{p}$$

a nonnegative integer.

Any other integer solution differs by the kernel vector

$$(q, p).$$

Subtracting one copy of K would make the bounded coordinate y_r negative. Therefore this is the first nonnegative point on the affine completion ray and hence the unique minimum Parikh pair.

The argument for $r < 0$ is symmetric, now bounding the zero coordinate modulo q.

The displayed block word is simply a fixed ordering of the unique minimum pair. □

The canonical word is a **decoder convention**, not a claim that other shortest orderings are invalid.

For example, for

$$p = 3, \qquad q = 2, \qquad r = 4,$$

we solve

$$2y \equiv -4 \pmod 3.$$

Thus

$$y = 1, \qquad x = \frac{4+2}{3} = 2.$$

Hence

$$\mu(4) = (2, 1)$$

and

$$C(4) = 100.$$

The other shortest words with the same Parikh pair are equally valid as completions.

The canonical map simply chooses one of them reproducibly.

2.7 The original DPDA implements this decoder at 2:1

We can now restate the Chapter 1 completion theorem in semantic language.

Let

$$c = (q, S)$$

be any reachable core configuration of the original 2:1 DPDA.

Its residual is decoded by

$$\boxed{\Theta_{\text{orig}}(c) = D(S) + \Phi(q).}$$

Thus

$$\Theta_{\text{orig}}(c) = \rho_{2,1}(w)$$

for the prefix w that reached c.

The endmarker normalizer removes the side-state correction and produces

$$\widehat{S}(c).$$

Chapter 1 proved that

$$\widehat{S}(c) = \begin{cases} 0^m, & r = 2m, \\ 10^{m+1}, & r = 2m+1, \\ 1^m, & r = -m, \end{cases}$$

where

$$r = \Theta_{\mathrm{orig}}(c).$$

But these are exactly the canonical two-block words

$$C(r)$$

for

$$p = 2, \qquad q = 1.$$

Therefore the original machine realizes the diagram

$$\boxed{c \xmapsto{\Theta_{\mathrm{orig}}} r \xmapsto{\mu} (x_r, y_r) \xmapsto{C} C(r).}$$

The machine is not being claimed to be an output transducer.

Rather, its configuration is a mathematical certificate from which the shortest canonical continuation can be decoded.

This distinction between **stored information** and **explicit output** will remain important later.

2.8 Physical information versus semantic information

We can now answer the question posed at the end of Chapter 1.

What is the machine really storing?

Not the entire prefix.

Not even the complete Parikh pair of the prefix.

At the semantic level, it stores only the residual

$$r.$$

From r, the minimum completion demand

$$\mu(r)$$

is uniquely determined.

From $\mu(r)$, the canonical shortest continuation

$$C(r)$$

is determined by convention.

This gives three layers.

Layer 1 — Physical representation

A concrete configuration such as

$$(L, 0^k),$$

$$(R, 1^k),$$

or

$$(C, 10)$$

belongs to the chosen 2023 machine.

Layer 2 — Semantic residual

The decoder

$$\Theta_{\text{orig}}$$

maps that configuration to one integer

$$r.$$

Layer 3 — Completion semantics

The residual determines

$$\mu(r), \qquad \Delta(r), \qquad C(r).$$

The crucial fact is that Layer 1 is not unique in principle.

Later Parts construct different physical DPDAs representing the same Layer 2 and Layer 3 objects.

That is the beginning of the distinction between semantics and representation.

The exact-lift, section, and semantic-fiber formalism is developed later in the book.

For now it is enough to observe that a complicated state/stack configuration may be merely a coordinate system for a much simpler arithmetic state.

2.9 The residual transition system

Because the residual changes by fixed amounts,

$$r \xrightarrow{0} r - p,$$

$$r \xrightarrow{1} r + q.$$

Thus every binary prefix traces a path on the integer line.

For the motivating language,

$$r \xrightarrow{0} r - 2,$$

$$r \xrightarrow{1} r + 1.$$

This graph is independent of the original DPDA's stack representation.

The 2023 machine is one physical realization of motion over this arithmetic system.
That observation immediately suggests a new question:

> If the semantic state is only one integer, why does the physical machine need the particular stack words and side states that it uses?

There are two very different possible answers.

One possibility is that the physical complexity is largely accidental and can be reduced by choosing better coordinates.

The other is that LIFO locality forces some of it.

Both possibilities will eventually occur.

Before studying resources, however, we need to understand the shortest paths in the residual graph.

2.10 Completion length as distance

The quantity

$$\Delta(r) = x_r + y_r$$

has an immediate graph interpretation.

It is the length of a shortest directed path from residual r to residual 0.

Indeed:

- appending a zero follows the edge
$$r \to r - p;$$
- appending a one follows
$$r \to r + q;$$
- a continuation reaches the language exactly when the resulting residual is zero.

Therefore

$$\boxed{\Delta(r) = \mathrm{dist}^{\to}(r, 0).}$$

We will prove and exploit this metric statement systematically in later chapters.

Here it already changes the interpretation of the minimum demand.

The pair

$$\mu(r) = (x_r, y_r)$$

does not merely count missing symbols.

It counts the two types of edges used by a shortest path to zero.

The canonical completion

$$C(r)$$

selects one distinguished shortest path among all orderings of those edges.

Thus the completion problem has become a geodesic problem.

2.11 How many shortest completions are there?

Because every shortest completion has the same Parikh pair

$$(x_r, y_r),$$

the number of shortest completion words is

$$\boxed{\binom{x_r + y_r}{x_r}.}$$

This simple count is the first indication that shortest completion has an internal geometry.

If both coordinates are positive, there are several ways to interleave the required zeros and ones.

For the prefix 111,

$$\mu(3) = (2, 1),$$

so there are

$$\binom{3}{1} = 3$$

shortest completions.

The original machine chooses the boundary ordering

$$100.$$

Later we will see that different shortest orderings are connected by elementary commuting moves, and the left branch of the original machine realizes a complete ladder of such shortest schedules at configuration level.

The multiplicity of shortest words is therefore not noise.

It is the first object that turns completion arithmetic into geometry.

2.12 Why the minimum pair is a better semantic object than the canonical word

It would be possible to define the semantics directly by the canonical word

$$C(r).$$

For the later theory, however, the pair

$$\mu(r) = (x_r, y_r)$$

is more fundamental.

There are several reasons.

First, the completion equation is linear in (x, y).

Second, the kernel direction

$$K = (q, p)$$

acts naturally on Parikh pairs.

Third, shortest-path descent becomes coordinate deletion:

- consuming a geodesic zero removes one unit of the zero demand;
- consuming a geodesic one removes one unit of the one demand.

Fourth, later physical representations will use different word orderings while preserving the same minimum pair.

Finally, in higher-dimensional counting languages the minimum demand will cease to be a single pair and become a fiber of optimal Parikh vectors. That generalization is visible only if we separate the Parikh object from its chosen word ordering now.

Thus our semantic hierarchy is

$$\boxed{r \quad\longrightarrow\quad \mu(r) \quad\longrightarrow\quad C(r),}$$

with the residual and minimum demand carrying the mathematics, and the canonical word supplying a convenient physical or decoding coordinate.

2.13 A first exact notion of redundancy

Suppose two different prefixes u and v have the same residual:

$$\rho_{p,q}(u) = \rho_{p,q}(v) = r.$$

Then both prefixes have exactly the same completion equation

$$px - qy = r.$$

Consequently they have:

- the same minimum demand;
- the same minimum completion length;
- the same set of feasible completion Parikh pairs;
- the same canonical shortest continuation.

Already, therefore, their past histories have become irrelevant to the completion problem.

This foreshadows the quotient theorem proved later:

$$u^{-1}L_{p,q} = v^{-1}L_{p,q} \iff \rho_{p,q}(u) = \rho_{p,q}(v).$$

We postpone the full quotient interpretation to Chapter 4, where it will be connected to directed geometry.

For the present chapter, the important message is simpler:

$$\boxed{\text{the residual is a sufficient statistic for all right-completion arithmetic.}}$$

2.14 The original machine viewed through the new semantics

We can now revisit several Chapter 1 observations without introducing any new transition rule.

The side states

The state potential means that a side state stores part of the residual outside the logical stack.

For example,

$$(L, \varepsilon)$$

represents

$$r = 3.$$

This is why the stack may be physically empty even though the completion demand is nonzero.

Endmarker materialization

The completion rule does not create new semantic information.

It changes coordinates.

Before the endmarker, the residual may be split between:

$$\text{stack weight} + \text{state potential}.$$

After normalization, the state potential is zero and the entire minimum demand is visible in the canonical stack word.

Tight depth-two visibility

The normalized words

$$0^*, \qquad 1^*, \qquad 10^+$$

are exactly the 2:1 canonical completion words.

Their global support is visible in the first two positions because the canonical decoder has only these three structural forms.

Thus the depth-two theorem is not an isolated stack-language coincidence.

It is a property of the chosen canonical ordering of the minimum demand.

This distinction will later become crucial when we compare alternative physical representations.

2.15 The first major conceptual compression

The entire original configuration can now be summarized by the following commuting picture:

$$\begin{array}{ccc}
\text{reachable DPDA configuration} & \xrightarrow{\Theta_{\mathrm{orig}}} & r \in \mathbb{Z} \\
 & & \downarrow \mu \\
 & & (x_r, y_r) \\
 & & \downarrow C \\
 & & \text{canonical shortest continuation}
\end{array}$$

The physical DPDA is rich.

The semantic state is one-dimensional.

The completion demand is exact.

This is the first compression of the research program.

It also exposes the next structural question.

If

$$\Delta(r)$$

is the shortest distance to balance, what does a single input symbol do to that distance?

When does reading a zero move one step closer to acceptance?

When does reading a one do so?

Can both actions be shortest simultaneously?

And do the original DPDA's cancellation and completion moves physically realize those shortest paths?

Those questions lead directly to the **minimum-demand geodesic descent** and the **completion ladder**.

That is the subject of Chapter 3.

Part II

Completion Becomes Geometry

Research provenance. This Part develops the configuration ladder, minimum-demand geometry, and rectangular shortest-completion theory from the exact structural analysis of the original machine and its quotient/completion semantics. The use of shortest-path structure in pushdown graphs has prior literature; the claim here is the exact arithmetic specialization and configuration-level geometry of this family [12, 14].

Chapter 3

Minimum Demand and the Completion Ladder

Chapter 2 replaced the original machine's physical configuration by a simpler semantic object:

$$r \longmapsto \mu(r) = (x_r, y_r).$$

For the motivating language,

$$L_{2,1} = \{w : N_1(w) = 2N_0(w)\},$$

the residual evolves by

$$r \xrightarrow{0} r - 2, \qquad r \xrightarrow{1} r + 1.$$

The minimum completion demand tells us how many zeros and ones are still required. We now ask what happens to that demand when another input symbol is read.

The answer reveals that the original DPDA is traversing shortest paths.

3.1 Geodesic input symbols

Fix

$$p = 2, \qquad q = 1.$$

For residual r, write

$$\mu(r) = (x_r, y_r),$$

and

$$\Delta(r) = x_r + y_r.$$

Recall:

$$\mu(2m) = (m, 0),$$

$$\mu(2m + 1) = (m + 1, 1),$$

and

$$\mu(-m) = (0, m).$$

Define the residual updates

$$T_0(r) = r - 2,$$

$$T_1(r) = r + 1.$$

A symbol $a \in \{0, 1\}$ will be called **geodesic at residual** r if reading it decreases the remaining shortest-completion distance by exactly one:

$$\boxed{\Delta(T_a(r)) = \Delta(r) - 1.}$$

This is the semantic version of the question:

> Does this next input symbol perform one unit of the completion work that was already required?

Lemma 3.1 — Minimum-demand geodesic descent.

Let

$$\mu(r) = (x_r, y_r).$$

If

$$x_r > 0,$$

then

$$\boxed{\mu(r-2) = (x_r - 1, y_r)}$$

and

$$\Delta(r-2) = \Delta(r) - 1.$$

If

$$y_r > 0,$$

then

$$\boxed{\mu(r+1) = (x_r, y_r - 1)}$$

and

$$\Delta(r+1) = \Delta(r) - 1.$$

Therefore the set of geodesic first symbols is

$$\boxed{\mathrm{GeoFirst}(r) = \{0 : x_r > 0\} \cup \{1 : y_r > 0\}.}$$

Proof.

Suppose

$$x_r > 0.$$

Since

$$2x_r - y_r = r,$$

we have

$$2(x_r - 1) - y_r = r - 2.$$

Thus

$$(x_r - 1, y_r)$$

is a completion demand for residual $r - 2$.
If $r - 2$ had a completion pair of length strictly less than

$$x_r + y_r - 1,$$

adding one zero to that pair would produce a completion of r shorter than

$$\mu(r),$$

contradicting minimality.
Hence

$$\mu(r - 2) = (x_r - 1, y_r).$$

The argument for $y_r > 0$ is symmetric:

$$2x_r - (y_r - 1) = r + 1.$$

Again, any shorter completion after reading 1 would yield a shorter completion before reading it.

Thus precisely those symbols whose coordinate in $\mu(r)$ is positive decrease the minimum distance by one. □

This lemma has a useful interpretation.
If the minimum demand is

$$(x, y),$$

then every geodesic input symbol simply deletes one required unit:

$$0 : (x, y) \mapsto (x - 1, y),$$

$$1 : (x, y) \mapsto (x, y - 1).$$

At the semantic level, shortest completion is therefore almost trivial.

The interesting question is whether the physical DPDA realizes the same simple coordinate deletion.

3.2 The left branch as an exact shortest-path process

Consider the reachable left configurations

$$c_k = (L, 0^k), \qquad k \geq 0.$$

From the state-potential invariant,

$$\rho(c_k) = 2k + 3.$$

Therefore

$$\boxed{\mu(c_k) = (k + 2, 1)}$$

and

$$\boxed{\Delta(c_k) = k + 3.}$$

So every such configuration requires:

- $k + 2$ zeros;
- one one.

The left cancellation transition

$$(L, 0^k) \xrightarrow{0} (L, 0^{k-1})$$

does exactly what Lemma 3.1 predicts:

$$(k + 2, 1) \mapsto (k + 1, 1).$$

It decreases the shortest-completion distance by exactly one.
This gives the original cancellation loop a new meaning.
It is not merely deleting matching physical stack symbols.
It is executing one edge of a shortest path in completion space.
The boundary sequence

$$(L, \varepsilon) \xrightarrow{0} (C, 10) \xrightarrow{0} (R, 0) \xrightarrow{1} (C, \varepsilon)$$

continues the same exact descent.
Thus the entire route from a left configuration to balance can be read geodesically.

3.3 The ladder sources are actually reachable

The configurations

$$(L, 0^k)$$

used in the next theorem are not hypothetical legal stack patterns.
They are reachable.
The prefix

$$111$$

reaches

$$(L, \varepsilon).$$

Two additional ones increase the left unary tail by one unit through the machine's center-side transition pattern.

Inductively,

$$\boxed{1^{2k+3} \text{ reaches } (L, 0^k).}$$

This reachability matters.

The geometry we are about to describe is therefore a geometry of actual machine computations, not merely of syntactically possible configurations.

3.4 Every position of the required one is shortest

From

$$(L, 0^k),$$

the minimum demand is

$$(k+2, 1).$$

Hence every shortest continuation contains exactly one 1 and $k+2$ zeros.

There are

$$k+3$$

possible positions for that unique one:

$$1\,0^{k+2},$$

$$0\,1\,0^{k+1},$$

$$0^2\,1\,0^k,$$

$$\dots,$$

$$0^{k+2}1.$$

The remarkable fact is that the original DPDA realizes **every one of these shortest schedules**, and it realizes them in a rectangular ladder of configurations.

Theorem 3.2 — Exact geodesic ladder.

For every

$$k \geq 0,$$

define the lower row

$$\Psi_k(i, 0) = \begin{cases} (L, 0^{k-i}), & 0 \leq i \leq k, \\ (C, 10), & i = k+1, \\ (R, 0), & i = k+2, \end{cases}$$

and the upper row

$$\Psi_k(i,1) = (C, 0^{k+2-i}), \qquad 0 \le i \le k+2.$$

Then:

1. every horizontal ladder edge is a solid input-0 transition;
2. every vertical ladder edge is a solid input-1 transition;
3. every monotone path from
$$(L, 0^k)$$
to
$$(C, \varepsilon)$$
consumes the minimum demand
$$(k+2, 1);$$
4. therefore every such path is shortest;
5. the shortest continuation words are exactly

$$\boxed{0^i 1\, 0^{k+2-i}, \qquad 0 \le i \le k+2.}$$

There are exactly

$$\boxed{k+3}$$

shortest continuations.

Proof.

Along the lower row, while a zero remains on the left stack, the horizontal edge is the cancellation loop

$$(L, 0^m) \xrightarrow{0} (L, 0^{m-1}).$$

At the lower boundary,

$$(L, \varepsilon) \xrightarrow{0} (C, 10),$$

followed by

$$(C, 10) \xrightarrow{0} (R, 0).$$

Along the upper row, each horizontal edge is a center cancellation

$$(C, 0^m) \xrightarrow{0} (C, 0^{m-1}).$$

The vertical edges consume the unique required 1.

For an interior left configuration,

$$(L, 0^m) \xrightarrow{1} (C, 0^{m+2}).$$

At the lower boundary,

$$(L, \varepsilon) \xrightarrow{1} (C, 00).$$

At

$$(C, 10),$$

the input 1 removes the visible 1 and reaches

$$(C, 0).$$

At

$$(R, 0),$$

the input 1 reaches

$$(C, \varepsilon).$$

Thus the entire two-row grid is physically realized by the DPDA.

Every source-to-target monotone path contains exactly:

$$k + 2$$

horizontal zero edges and one vertical one edge.

Hence every path consumes the unique minimum Parikh demand

$$(k + 2, 1).$$

By Chapter 2, no shorter continuation exists.

The position of the vertical edge determines the position of the unique 1 in the continuation, giving exactly

$$0^i 1\, 0^{k+2-i}, \qquad 0 \le i \le k + 2.$$

□

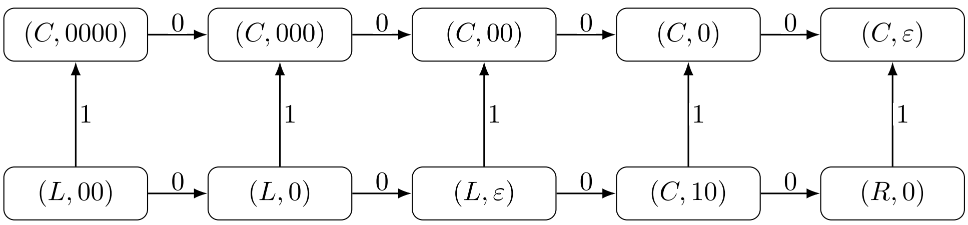


Figure 3.1: The exact configuration ladder for $k = 2$. The source $(L, 00)$ has residual 7 and minimum demand $(4, 1)$. Its five shortest continuations are $10000, 01000, 00100, 00010, 00001$. The upper boundary is canonical completion; the lower boundary postpones the unique required 1 maximally.

The source is

$$(L, 00),$$

with residual

$$7$$

and minimum demand

$$(4, 1).$$

Its five shortest completions are

$$10000,$$

$$01000,$$

$$00100,$$

$$00010,$$

$$00001.$$

The upper route materializes the required 1 as early as possible.
The lower route postpones it as long as possible.
Both are shortest.
Everything between them is shortest as well.

3.5 Local commuting diamonds

A ladder square gives two ways to consume one zero and one one:

$$01$$

or

$$10.$$

The machine reaches the same configuration either way.

Lemma 3.3 — Local commuting diamonds.
At every square of the geodesic ladder,

$$\boxed{\delta^*(c, 01) = \delta^*(c, 10).}$$

Proof.
For an interior left configuration

$$(L, 0^m), \qquad m \geq 1,$$

one route is

$$(L, 0^m) \xrightarrow{0} (L, 0^{m-1}) \xrightarrow{1} (C, 0^{m+1}).$$

The other is

$$(L, 0^m) \xrightarrow{1} (C, 0^{m+2}) \xrightarrow{0} (C, 0^{m+1}).$$

Both end at the same configuration.
The two boundary squares are checked directly from the corresponding transition rules:

- at (L, ε), both 01 and 10 reach the same center configuration;
- at $(C, 10)$, both orders reach (C, ε).

These are exactly the ladder squares. □
The point is not that 0 and 1 commute as input symbols globally.
They do not.
The point is much more specific:

Within this shortest-completion interval, adjacent geodesic actions commute physically.

3.6 Parikh synchronization

Because adjacent 01 and 10 factors can be exchanged locally, any two shortest prefixes that have consumed the same numbers of zeros and ones reach the same ladder position.

Corollary 3.4 — Geodesic Parikh synchronization.

Let u and v be prefixes of shortest continuations from

$$(L, 0^k).$$

If

$$N_0(u) = N_0(v)$$

and

$$N_1(u) = N_1(v),$$

then

$$\boxed{\delta^*((L, 0^k), u) = \delta^*((L, 0^k), v).}$$

In particular, every shortest continuation reaches

$$(C, \varepsilon).$$

Proof.

Inside the ladder, repeatedly exchange adjacent factors

$$01 \leftrightarrow 10$$

using Lemma 3.3.

The consumed Parikh pair uniquely determines the grid coordinate. □

This is the first place where the physical machine's computation depends only on the consumed Parikh pair **inside a geodesic interval**, even though the machine is still a stack automaton processing an ordered word.

That phenomenon will later generalize.

3.7 Completion and cancellation are opposite boundaries of one object

The original machine suggested two seemingly different strategies.

One strategy is to expose or materialize the canonical completion as early as possible.

The other is to continue cancelling zeros on the left before finally consuming the required one.

The ladder shows that these are not different semantic processes.

They are boundary schedules of the same shortest-path rectangle.

Corollary 3.5 — Completion–cancellation boundary duality.

For the source

$$(L, 0^k),$$

the canonical completion

$$\boxed{10^{k+2}}$$

and the maximal-cancellation continuation

$$\boxed{0^{k+2}1}$$

are the two extremal boundary paths of the same geodesic ladder.
The word **duality** here is operational.
One boundary performs the unique required 1 first.
The other performs it last.
Both consume the same minimum demand.
This is a major reinterpretation of the original design:

completion and cancellation are scheduling choices inside one geodesic object.

3.8 A commutation potential

For residual r, define

$$\Xi(r) = x_r y_r$$

when

$$\mu(r) = (x_r, y_r).$$

For the left-ladder source,

$$\mu(2k+3) = (k+2, 1),$$

so

$$\Xi(2k+3) = k+2.$$

This number has several simultaneous meanings.
It is:

1. the number of elementary squares in the two-row ladder;
2. the maximum number of geodesic zeros that can be consumed before the unique one becomes unavoidable;
3. the number of adjacent $01 \leftrightarrow 10$ swaps needed to move the unique one from the first position to the last.

Corollary 3.6 — Cancellation–diamond identity.

For every reachable left configuration,

Ξ = diamond count = cancellation horizon = extremal adjacent-swap distance.

This identity is small, but it is conceptually useful.

A quantity defined purely from minimum-demand arithmetic,

$$x_r y_r,$$

simultaneously measures:

- a piece of configuration-space geometry;
- a cancellation horizon in the DPDA;
- a word-reordering distance.

The different descriptions are beginning to converge.

3.9 Why this is more than a counting argument

At the arithmetic level, it is easy to observe that a demand

$$(k+2, 1)$$

has

$$k+3$$

possible shortest word orderings.

The DPDA theorem says more.

It says that the machine realizes these orderings in a coherent configuration-level structure:

$$\boxed{\text{shortest words} \longleftrightarrow \text{monotone paths} \longleftrightarrow \text{reachable configurations.}}$$

Adjacent word commutations become literal commuting squares of machine transitions.

This is why the completion ladder matters.

It is the first exact bridge between:

- Diophantine completion arithmetic;
- shortest-path geometry;
- pushdown-store evolution.

3.10 The canonical completion is a boundary choice

The original endmarker normalizer selects

$$10^{k+2}$$

from the $k+3$ shortest completions of a left source.

Nothing about the minimum Parikh demand itself forces the 1 to occur first.

The ordering

$$10^{k+2}$$

is a canonical representation choice.

The geodesic ladder makes this distinction visible:

$$\boxed{\text{semantic minimum demand} \neq \text{unique physical schedule.}}$$

This sentence will acquire much deeper meanings later.

In P9, the original and centered machines will differ by eager versus lazy normalization chronology.

In P10, semantic carry chronology will be separated from physical materialization chronology.

In P11, several reduced physical lifts with the same semantics will coexist.

At this early stage, however, the ladder already contains the seed of that distinction.

3.11 Limits of the ladder description

The ladder gives a complete operational picture for the motivating left branch.

It does not yet answer three broader questions.

First, when we look at an arbitrary residual, how much of its shortest-path structure can be inferred from a bounded part of the physical stack?

Second, do equal residuals really represent identical future language behavior, not merely identical completion lengths?

Third, what replaces this two-row ladder for a general ratio

$$p{:}q,$$

where a shortest demand may contain several copies of both symbols?

These questions lead in two directions:

$$\boxed{\text{quotient and visibility}}$$

and

$$\boxed{\text{general } p{:}q \text{ rectangles.}}$$

The next chapter takes the first direction.

Chapter 4

Quotients, Geodesics, and What Can Be Seen Locally

Chapter 2 showed that the residual

$$\rho_{p,q}(w) = qN_1(w) - pN_0(w)$$

determines the entire arithmetic of right completion. Chapter 3 then showed, in the original 2:1 machine, that shortest completions are realized by an actual configuration-level ladder of geodesic schedules.

We now strengthen both observations.

The residual does not merely determine the length or Parikh vector of a shortest repair. It determines the **entire future right language** of the prefix.

At the same time, the canonical shortest-completion word has a bounded visible frontier: one need not inspect its unbounded tail to know which types of geodesic action are available.

These two facts connect three levels:

$$\boxed{\text{prefix quotient} \longleftrightarrow \text{residual vertex} \longleftrightarrow \text{shortest-completion geometry.}}$$

4.1 The residual classifies all future behavior

For a language L and prefix u, the left quotient

$$u^{-1}L = \{z : uz \in L\}$$

is the set of all continuations that lead from u into L.

For $L_{p,q}$, Chapter 2 already gave the completion equation

$$z \in u^{-1}L_{p,q} \iff pN_0(z) - qN_1(z) = \rho_{p,q}(u).$$

The right-hand side depends on u only through its residual.

That observation is exact.

Lemma 4.1 — Every integer residual is reachable.

For every

$$r \in \mathbb{Z},$$

there exists a word

$$w \in \{0,1\}^*$$

such that

$$\rho_{p,q}(w) = r.$$

Proof.

Since

$$\gcd(p,q) = 1,$$

Bézout's identity gives integers x_0, y_0 satisfying

$$qy_0 - px_0 = r.$$

Every pair

$$(x_0 + qk,\ y_0 + pk)$$

has the same value of $qy - px$.

For sufficiently large k, both coordinates are nonnegative. Any binary word with those numbers of zeros and ones therefore has residual r. □

Thus the semantic residual graph will genuinely contain every integer, not merely a subset reached by special prefixes.

Theorem 4.2 — Prefix quotient classification.

For all

$$u, v \in \{0,1\}^*,$$

$$\boxed{u^{-1}L_{p,q} = v^{-1}L_{p,q} \iff \rho_{p,q}(u) = \rho_{p,q}(v).}$$

Proof.

If the residuals agree, the completion equation is identical for u and v. Hence every continuation completes one prefix exactly when it completes the other.

Conversely, suppose

$$\rho_{p,q}(u) = r, \qquad \rho_{p,q}(v) = s, \qquad r \neq s.$$

Choose any nonnegative solution

$$(x, y)$$

of

$$px - qy = r,$$

for example the minimum completion demand $\mu(r)$. Let z be any word with that Parikh pair.

Then

$$z \in u^{-1}L_{p,q}.$$

But the same pair cannot also satisfy

$$px - qy = s.$$

Therefore

$$z \notin v^{-1}L_{p,q}.$$

The quotients differ. □

This theorem gives the residual a stronger interpretation than the one used so far.

It is not merely a useful numeric invariant.

It is an exact coordinate for right-continuation behavior:

$$\boxed{\text{same residual} \iff \text{same future language.}}$$

Left quotients themselves are classical objects. The role of the theorem in this monograph is not to claim a new general quotient theory. Its importance is that the physical DPDA configurations developed from the original machine can later be shown to encode this quotient coordinate exactly.

4.2 The quotient graph

Define the directed labeled graph

$$\mathcal{G}_{p,q}$$

with vertex set

$$\mathbb{Z}$$

and edges

$$r \xrightarrow{0} r - p,$$

$$r \xrightarrow{1} r + q.$$

By Theorem 4.2 and Lemma 4.1, the vertices of this graph may equally well be regarded as the prefix left quotients of $L_{p,q}$.

Thus the arithmetic picture

$$\cdots, -2, -1, 0, 1, 2, \cdots$$

is simultaneously a quotient graph.

A word labels a path by applying its symbols from left to right.

The acceptance quotient is the vertex

$$0.$$

4.3 Completion length is exactly directed distance

Chapter 2 interpreted the minimum completion length

$$\Delta(r)$$

as a shortest path to zero.

The quotient graph allows a slightly stronger translation-invariant form.

Theorem 4.3 — Directed completion metric.

Let

$$d_{\mathcal{G}}(r, s)$$

denote directed shortest-path distance in $\mathcal{G}_{p,q}$. Then

$$\boxed{d_{\mathcal{G}}(r, s) = \Delta(r - s).}$$

Proof.
A path from r to s containing x zeros and y ones has displacement

$$s - r = qy - px.$$

Equivalently,

$$px - qy = r - s.$$

Thus the Parikh vectors of directed r-to-s paths are exactly the nonnegative solutions of the completion equation for residual $r - s$.
Minimizing path length

$$x + y$$

is therefore exactly the minimum-completion problem. □
The geometry is homogeneous:

$$d_{\mathcal{G}}(r, s)$$

depends only on the difference

$$r - s.$$

The language-specific shortest-completion problem is therefore one distinguished family of distances in a translation-invariant directed graph.

4.4 The fundamental balanced cycle

A directed closed walk has zero net residual displacement.
If it contains x zeros and y ones, then

$$qy - px = 0.$$

Coprimality forces a rigid form.

Theorem 4.4 — Fundamental directed cycle.
Every nonempty directed closed walk satisfies

$$N_0 = qk, \qquad N_1 = pk$$

for some integer

$$k \geq 1.$$

Consequently every directed cycle length is a positive multiple of

$$p + q,$$

and the shortest possible directed cycle length is

$$\boxed{g_{p,q} = p + q.}$$

Proof.
A closed walk has

$$qN_1 = pN_0.$$

Since p and q are coprime,

$$N_0 = qk, \qquad N_1 = pk$$

for some nonnegative integer k.
A nonempty walk has $k \geq 1$.
Conversely, any ordering of q zeros and p ones has net residual displacement zero and therefore gives a closed walk of length $p + q$. □
The primitive kernel vector

$$K = (q, p)$$

from Chapter 2 has now become a geometric object:

$$\boxed{K = \text{Parikh vector of a shortest directed balanced cycle.}}$$

This is the global analogue of cancellation.

4.5 Balanced-cycle reduction

Suppose a word z has Parikh pair

$$(x, y).$$

If

$$x \geq q \qquad \text{and} \qquad y \geq p,$$

then one full balanced kernel packet

$$(q, p)$$

can be removed without changing the displacement of the word.
Repeat this as often as possible.

Theorem 4.5 — Balanced-cycle reduction.
Let a word have Parikh pair

$$(x, y).$$

Define

$$k = \min\left(\left\lfloor \frac{x}{q} \right\rfloor, \left\lfloor \frac{y}{p} \right\rfloor\right).$$

Then

$$(x', y') = (x - kq,\ y - kp)$$

has the same residual displacement as (x, y) and is the unique minimum Parikh pair for that displacement.

Proof.

Subtracting

$$(q, p)$$

preserves

$$px - qy.$$

By maximality of k,

$$x' < q \qquad \text{or} \qquad y' < p.$$

Every integer solution with the same displacement differs by an integer multiple of

$$(q, p).$$

No further positive multiple can be removed while remaining nonnegative.

Therefore (x', y') is the first nonnegative point on the affine solution ray and hence the unique minimum pair. □

This theorem gives a global quotient-graph interpretation of the minimum demand:

$$\boxed{\begin{array}{c}\text{minimum demand} = \text{Parikh vector after deleting}\\ \text{every removable balanced cycle.}\end{array}}$$

The original machine's cancellation behavior was local and representation-specific.

Balanced-cycle reduction is semantic and representation-independent.

4.6 Which words are geodesic?

A word is geodesic between its endpoints precisely when no balanced kernel packet can be deleted.

Corollary 4.6 — Exact geodesic domain.

A word z is shortest among all words with the same residual displacement if and only if

$$\boxed{N_0(z) < q \quad \text{or} \quad N_1(z) < p.}$$

Equivalently,

$$\mathrm{Geo}_{p,q} = \{z : N_0(z) < q\} \cup \{z : N_1(z) < p\}.$$

The geodesic-word language is therefore regular.

The criterion is strikingly simple.

A word ceases to be shortest exactly when it contains enough of **both** symbol types to hide one complete balanced cycle in its Parikh vector.

Consequently every binary word of length less than

$$p + q$$

is geodesic.

The first possible failure occurs exactly at the fundamental cycle length.

4.7 Exact quotient spheres

Let

$$S_n = \{r \in \mathbb{Z} : d_{\mathcal{G}}(r, 0) = n\}.$$

A length-n Parikh pair has the form

$$(x, n - x).$$

Its residual displacement is

$$(p + q)x - qn,$$

which is strictly increasing in x. Hence distinct values of x give distinct residual vertices.

Combining this with the geodesic criterion gives an exact growth law.

Proposition 4.7 — Exact quotient spheres.

For every

$$n \geq 0,$$

$$\boxed{|S_n| = \min(n + 1, p + q).}$$

Therefore

$$\boxed{\sum_{n \geq 0} |S_n| z^n = \frac{1 - z^{p+q}}{(1 - z)^2}.}$$

Proof.

If

$$n < p + q,$$

every length-n Parikh pair is geodesic, so all

$$n + 1$$

possible zero counts give distinct vertices at distance n.

If

$$n \geq p + q,$$

a length-n pair is geodesic precisely when

$$x < q$$

or

$$n - x < p.$$

These two regions are disjoint and contain respectively q and p pairs.

Hence exactly

$$p + q$$

residuals occur at distance n. □

The graph therefore has linear growth only until the first balanced cycle appears. After that, every directed sphere has constant size

$$p + q.$$

This small exact formula will later help distinguish semantic growth from the sometimes much larger physical state/stack resources used by a particular lift.

4.8 What the original two-cell observation really reveals

Chapter 1 proved that the original 2:1 normalized completion word exposes its global symbol support within two stack cells.

Chapter 3 gave a more operational interpretation.

For a reachable configuration c, define

$$\mathrm{GeoFirst}(c)$$

to be the set of symbols that begin some shortest continuation from the residual represented by c.

If

$$\mu(r) = (x_r, y_r),$$

then

$$\boxed{\mathrm{GeoFirst}(r) = \{0 : x_r > 0\} \cup \{1 : y_r > 0\}.}$$

For the original machine, the reachable qualitative roles are:

$$(C, \varepsilon) \mapsto \varnothing,$$

$$(C, 0^m) \mapsto \{0\},$$

$$(C, 1^m) \mapsto \{1\},$$

$$(C, 10) \mapsto \{0, 1\},$$

every reachable left configuration maps to

$$\{0, 1\},$$

and every reachable right configuration maps to

$$\{1\}.$$

Thus the old "bounded stack search" intuition can be stated exactly.

Theorem 4.8 — Exact depth-two geodesic locality of the original machine.

For the motivating 2:1 DPDA:

1. the complete geodesic first-action set is determined by the control state and the first two logical stack symbols;
2. depth one is insufficient;

3. after end-of-input normalization,

$$\boxed{\mathrm{alph}(\widehat{S}(c)) = \mathrm{GeoFirst}(c);}$$

4. no fixed finite observation depth determines the full quantitative demand vector on all reachable configurations.

Proof.

The reachable-store normal form from Chapter 1 leaves only the center forms

$$\varepsilon, \quad 0^m, \quad 1^m, \quad 10,$$

together with the fixed qualitative left and right roles.

At the center, two visible symbols distinguish exactly the four possible geodesic-action sets:

$$\varnothing, \quad \{0\}, \quad \{1\}, \quad \{0, 1\}.$$

Depth one fails because

$$(C, 1)$$

and

$$(C, 10)$$

have the same state and top symbol but different geodesic first-action sets.

End-of-input normalization produces the canonical shortest continuation represented by the residual, and its symbol support is exactly the set of positive coordinates of the minimum demand.

Finally, arbitrarily large reachable unary center stacks

$$(C, 0^m)$$

share every fixed top-d observation once $m \geq d$, yet their demands

$$(m, 0)$$

have different magnitudes.

Thus qualitative geometry is exactly two-local, while quantitative demand is not fixed-depth local. □

This is the precise form of an important distinction:

$$\boxed{\text{bounded local observation can reveal the shape of the shortest-path choice}}$$

without revealing

$$\boxed{\text{the unbounded distance still remaining.}}$$

4.9 General canonical action visibility

For general $p{:}q$, recall the canonical shortest word

$$C(r) = \begin{cases} 1^{y_r} 0^{x_r}, & r > 0, \\ 0^{x_r} 1^{y_r}, & r < 0, \\ \varepsilon, & r = 0. \end{cases}$$

The minimum pair satisfies the bounded-coordinate property:

$$r > 0 \Longrightarrow 0 \le y_r < p,$$

$$r < 0 \Longrightarrow 0 \le x_r < q.$$

Therefore the first change of symbol, if it occurs, cannot be arbitrarily deep.

Definition 4.9 — Canonical visibility depth.
Let $d_{p,q}$ be the least integer d such that for every residual r,

$$\mathrm{alph}(C(r)) = \mathrm{alph}(\mathrm{pref}_d(C(r))).$$

Theorem 4.10 — Exact bounded visibility.

$$\boxed{d_{p,q} = \max(p, q).}$$

Proof.
Suppose first that

$$r > 0.$$

Then

$$C(r) = 1^{y_r} 0^{x_r}$$

with

$$y_r < p.$$

If a zero occurs, the first zero appears no later than position

$$p.$$

Thus depth p suffices on the positive side.
For

$$r < 0,$$

the symmetric argument shows that depth q suffices.
Hence

$$d_{p,q} \le \max(p, q).$$

For tightness, suppose

$$p > q.$$

Take residual

$$r = q.$$

The minimum solution is

$$(x_r, y_r) = (q, p - 1),$$

so

$$C(q) = 1^{p-1}0^q.$$

Depth $p - 1$ sees only ones even though zeros occur later. Therefore

$$d_{p,q} \geq p.$$

If

$$q > p,$$

the symmetric residual

$$r = -p$$

gives

$$C(-p) = 0^{q-1}1^p,$$

forcing

$$d_{p,q} \geq q.$$

For

$$p = q = 1,$$

depth one is plainly necessary and sufficient.
Therefore

$$d_{p,q} = \max(p, q).$$

□
Since

$$\mathrm{alph}(C(r)) = \mathrm{GeoFirst}(r),$$

the same theorem can be read operationally:

The complete set of locally shortest first actions is visible within $\max(p, q)$ canonical symbols.

For the original machine,

$$(p, q) = (2, 1),$$

so

$$d_{2,1} = 2,$$

exactly reproducing the original tight depth-two theorem.

4.10 A second bounded quantity: the completion frontier

Visibility asks one question:

> How far into a canonical shortest completion must we inspect before every symbol type that occurs anywhere in it has appeared?

There is another bounded quantity that will become important when we redesign the physical DPDA.

Write positive and negative residuals in Euclidean form:

$$r = pm + s, \qquad 0 \le s < p,$$

and

$$r = -(qm + t), \qquad 0 \le t < q.$$

Define the finite residue-dependent bases

$$A_s = C(s),$$

$$B_t = C(-t),$$

with

$$A_0 = B_0 = \varepsilon.$$

Then

$$\boxed{C(pm + s) = A_s 0^m,}$$

and

$$\boxed{C(-(qm + t)) = B_t 1^m.}$$

Thus every canonical completion decomposes into:

$$\boxed{\text{bounded residue-dependent frontier} + \text{unbounded unary tail.}}$$

Definition 4.11 — Completion frontier.

Define

$$K_{p,q} = \max\left(\max_{0\le s<p} |A_s|, \max_{0\le t<q} |B_t|\right).$$

Theorem 4.12 — Exact almost-cycle frontier.

For every coprime pair (p, q),

$$\boxed{K_{p,q} = \begin{cases} 0, & (p,q) = (1,1), \\ p + q - 1, & \text{otherwise.} \end{cases}}$$

Proof.

For

$$(p, q) = (1, 1),$$

the only residue bases are

$$A_0 = B_0 = \varepsilon,$$

so

$$K_{1,1} = 0.$$

Now assume the coprime pair is nontrivial. Then

$$p \neq q.$$

The bounded-coordinate formulas for $C(r)$ imply

$$K_{p,q} \leq p + q - 1.$$

If

$$p > q,$$

the base residual

$$q$$

has

$$A_q = C(q) = 1^{p-1}0^q,$$

whose length is

$$p + q - 1.$$

If

$$q > p,$$

the symmetric base is

$$B_p = C(-p) = 0^{q-1}1^p,$$

again of length

$$p + q - 1.$$

Hence the upper bound is attained. □
Since the shortest balanced cycle length is

$$g_{p,q} = p + q,$$

we obtain, for every nontrivial coprime pair,

$$\boxed{K_{p,q} = g_{p,q} - 1.}$$

The largest finite frontier is literally one symbol short of a complete balanced cycle.

4.11 One extremal word, two different bounds

The same almost-cycle word makes both visibility and frontier tight.

If

$$p > q,$$

the witness is

$$C(q) = 1^{p-1}0^q.$$

If

$$q > p,$$

the witness is

$$C(-p) = 0^{q-1}1^p.$$

But the two parameters measure different things.

Visibility depth

$$\boxed{d_{p,q} = \max(p, q)}$$

asks:

> How many leading symbols must be inspected before the **qualitative support** of the canonical shortest completion is known?

Completion frontier

$$\boxed{K_{p,q} = p + q - 1}$$

for nontrivial pairs asks:

> How large can the complete **residue-dependent non-unary base** be before only an unbounded unary magnitude remains?

These should not be conflated.

In particular, their numerical values are generally different even though one extremal word witnesses both lower bounds.

This distinction is crucial for later resource arguments.

A finite observation depth is not the same thing as a finite amount of residue information that can be moved into control.

4.12 The next representation idea is now forced

We have reached a useful decomposition:

$$C(r) = \text{finite residue-dependent base} + \text{unbounded unary tail}.$$

The finite base has exact worst-case size

$$K_{p,q}.$$

The unary tail stores the unbounded magnitude.

This immediately suggests a new physical representation:

$$\boxed{\text{store the bounded frontier in finite control}}$$

and

$$\boxed{\text{leave only the unbounded unary magnitude on the pushdown.}}$$

Such a factorization can be lossless because the pair

$$\text{control residue role} + \text{unary tail length}$$

reconstructs the exact residual.

Once the residual is known, Theorem 4.2 gives the exact prefix quotient, Theorem 4.3 gives the exact distance to acceptance, and Chapter 2 gives a canonical shortest continuation.

This will eventually yield a compact physical realization.

But before we redesign the machine, one more piece of geometry is needed.

The 2:1 ladder of Chapter 3 was only two rows high because its minimum demand contained a single 1.

For a general residual,

$$\mu(r) = (x_r, y_r)$$

may have several copies of both symbols.

What is the full shortest-path interval then?

The answer is a rectangle.

That is the subject of Chapter 5.

Chapter 5

General p:q Completion Rectangles

The 2:1 ladder of Chapter 3 had two rows because every left-ladder source had minimum demand

$$(k+2,1).$$

Only one 1 was required, so the position of that single action was the only scheduling choice.

For a general ratio p:q, the minimum demand

$$\mu(r)=(x_r,y_r)$$

may contain several copies of both symbols.

The ladder then becomes a rectangle.

This chapter identifies that rectangle exactly and shows that three objects coincide:

$$\boxed{\text{minimum-demand coordinates} \longleftrightarrow \text{directed shortest-path interval} \longleftrightarrow \text{commuting schedule geometry.}}$$

5.1 Remaining demand as a grid coordinate

Fix coprime positive integers

$$p,q$$

and residual

$$r.$$

Let

$$\mu(r)=(x_r,y_r).$$

Suppose a shortest completion has so far consumed

$$i$$

zeros and

$$j$$

ones.
Its current residual is

$$r - pi + qj.$$

If the prefix is still geodesic, the remaining demand should be

$$(x_r - i,\ y_r - j).$$

Thus the natural coordinate set is

$$[0, x_r] \times [0, y_r],$$

where the point

$$(i, j)$$

records how much of the unique minimum Parikh demand has already been consumed.
A horizontal step consumes a zero:

$$(i, j) \to (i + 1, j).$$

A vertical step consumes a one:

$$(i, j) \to (i, j + 1).$$

The source is

$$(0, 0),$$

and the balanced target is

$$(x_r, y_r).$$

The only possible problem is that two different grid coordinates might correspond to the same residual before the target is reached.
The minimum-demand property prevents exactly that.

5.2 The shortest interval is an actual rectangle

Theorem 5.1 — Rectangular shortest-completion geometry.
Let

$$\mu(r) = (x_r, y_r).$$

The map

$$\boxed{\varphi_r(i, j) = r - pi + qj}$$

is injective on

$$[0, x_r] \times [0, y_r].$$

Consequently the directed shortest-path interval from r to 0 in the residual graph is exactly the rectangular grid

$$\boxed{[0, x_r] \times [0, y_r].}$$

Every monotone source-to-target path is a shortest completion, and every shortest completion occurs in this way.
Proof.
Suppose two rectangle coordinates satisfy

$$r - pi + qj = r - pi' + qj'.$$

Then

$$p(i - i') = q(j - j').$$

Since

$$\gcd(p, q) = 1,$$

there is an integer t such that

$$i - i' = qt, \qquad j - j' = pt.$$

If

$$t \neq 0,$$

then the rectangle contains two points separated by at least one full kernel vector

$$(q, p).$$

In particular, the side lengths must satisfy

$$x_r \geq q, \qquad y_r \geq p.$$

But then

$$(x_r - q,\ y_r - p)$$

would be a shorter nonnegative solution of the same completion equation

$$px - qy = r,$$

contradicting the minimality of

$$\mu(r).$$

Hence

$$t = 0,$$

so

$$i = i', \qquad j = j'.$$

Thus φ_r is injective.
Every monotone path from

$$(0, 0)$$

to

$$(x_r, y_r)$$

consumes exactly x_r zeros and y_r ones, hence exactly the minimum demand.

Conversely, every shortest completion has that same unique minimum Parikh vector, so its sequence of consumed zero/one counts traces one monotone path through the rectangle. □

This theorem generalizes the Chapter 3 ladder without changing its meaning.

When

$$y_r = 1,$$

the rectangle has exactly two rows.

When

$$x_r = 0$$

or

$$y_r = 0,$$

it degenerates to a one-dimensional corridor.

When both coordinates are positive, the residual has a genuine scheduling fork.

5.3 Exact number of shortest completions

A monotone path in an

$$x_r \times y_r$$

rectangle consists of:

- x_r horizontal moves;
- y_r vertical moves.

Therefore the number of source-to-target paths is immediate.

Corollary 5.2 — Exact shortest-completion count.

For every residual r,

$$\boxed{|\operatorname{GeoComp}(r)| = \binom{x_r + y_r}{x_r} = \binom{x_r + y_r}{y_r}.}$$

Equivalently,

$$\operatorname{GeoComp}(r) = \{z : N_0(z) = x_r,\ N_1(z) = y_r\}.$$

The statement is stronger than "there exists a shortest completion."

It says that the unique minimum demand completely classifies **all** shortest words.

Their only freedom is order.

5.4 Exact number of commuting squares

Every elementary unit square of the rectangle corresponds to one local choice between

$$01$$

and

$$10.$$

There are

$$x_r y_r$$

such squares.
Define

$$\boxed{\Xi(r) = x_r y_r.}$$

Corollary 5.3 — Exact commutation potential.
The number of elementary commuting squares in the shortest-completion rectangle of residual r is

$$\boxed{\Xi(r) = x_r y_r.}$$

Thus the same arithmetic quantity introduced in the 2:1 ladder has a general meaning.
It measures the number of elementary places at which the two required symbol types can interchange their local chronological order while remaining inside the shortest interval.
For the Chapter 3 sources,

$$\mu(2k+3) = (k+2, 1),$$

so

$$\Xi(2k+3) = k+2,$$

exactly recovering the number of ladder diamonds.

5.5 A worked 3:2 rectangle

Take

$$p = 3, \qquad q = 2, \qquad r = 2.$$

The completion equation is

$$3x - 2y = 2.$$

The minimum nonnegative solution is

$$\boxed{\mu(2) = (2, 2).}$$

Thus

$$\Delta(2) = 4.$$

The shortest interval is

$$[0,2] \times [0,2].$$

It contains

$$(2+1)(2+1) = 9$$

residual vertices.
There are

$$\binom{4}{2} = 6$$

shortest completion words:

$$0011, \quad 0101, \quad 0110, \quad 1001, \quad 1010, \quad 1100.$$

The canonical two-block completion is

$$\boxed{C(2) = 1100.}$$

The rectangle contains

$$2 \cdot 2 = 4$$

elementary commuting squares.

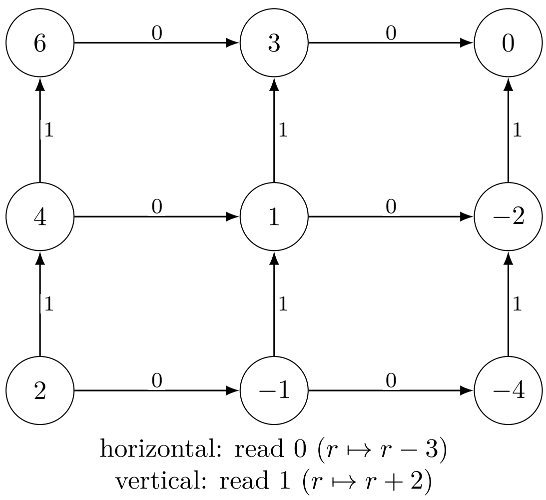


Figure 5.1: The 3:2, $r = 2$ shortest-completion rectangle. Horizontal edges read 0 and subtract 3 from the residual; vertical edges read 1 and add 2. The nine residual vertices form four commuting squares, and every monotone path from residual 2 to residual 0 is one of the six shortest completions with Parikh pair $(2,2)$.

A useful feature of this example is that neither symbol type occurs only once. The two-row ladder has become a genuinely two-dimensional scheduling space.

5.6 Canonical order as a boundary normal form

For

$$r > 0,$$

our canonical representative is

$$C(r) = 1^{y_r} 0^{x_r}.$$

Among all shortest words with Parikh pair

$$(x_r, y_r),$$

this is one boundary ordering: every required 1 is performed before every required 0.
For

$$r < 0,$$

the canonical representative

$$C(r) = 0^{x_r} 1^{y_r}$$

uses the opposite boundary orientation.
The rest of the shortest words lie between these extremal orderings.
For positive residuals, repeatedly apply the adjacent rewrite

$$01 \longrightarrow 10.$$

Each rewrite preserves the Parikh pair and therefore remains inside the same shortest-completion rectangle.

If we count inversions of the form "a 0 occurring before a 1," every rewrite decreases that count by one.

The unique normal form is

$$1^{y_r} 0^{x_r} = C(r).$$

The maximum possible inversion count is

$$x_r y_r.$$

Thus:

$$\boxed{\begin{aligned} \Xi(r) &= \text{rectangle-square count} \\ &= \text{maximum adjacent-swap normalization distance.} \end{aligned}}$$

For negative residuals the symmetric rewrite

$$10 \longrightarrow 01$$

normalizes toward

$$0^{x_r} 1^{y_r}.$$

The canonical shortest word is therefore not arbitrary.
It is a boundary normal form for a finite commuting geometry.

5.7 Exactly when does a geodesic fork occur?

A residual has a genuine first-action fork exactly when both minimum-demand coordinates are positive.

For positive residuals, the bounded coordinate satisfies

$$0 \le y_r < p.$$

The completion congruence gives

$$qy_r \equiv -r \pmod{p}.$$

Since q is invertible modulo p,

$$y_r = 0 \iff p \mid r.$$

The negative case is symmetric.

Corollary 5.4 — Exact geodesic-fork classification.

For

$$r > 0,$$

$$\boxed{\mathrm{GeoFirst}(r) = \begin{cases} \{0\}, & p \mid r, \\ \{0,1\}, & p \nmid r. \end{cases}}$$

For

$$r < 0,$$

$$\boxed{\mathrm{GeoFirst}(r) = \begin{cases} \{1\}, & q \mid r, \\ \{0,1\}, & q \nmid r. \end{cases}}$$

At

$$r = 0,$$

the set is empty.

This gives an exact arithmetic classification of local shortest-path shape.

There are only three nonterminal qualitative possibilities:

$$\boxed{\text{zero corridor}, \qquad \text{one corridor}, \qquad \text{commuting fork.}}$$

The original 2:1 machine has forks only on the positive side because

$$q = 1.$$

Every negative residual is divisible by q, so its shortest path is a one-way 1-corridor.

The one-sided appearance of the original transition diagram is therefore arithmetic, not merely graphical.

5.8 First obstruction at length $p + q$

Chapter 4 proved that a word is geodesic precisely when

$$N_0 < q \qquad \text{or} \qquad N_1 < p.$$

This immediately gives the first scale at which a word can fail to be shortest.
If

$$n < p + q,$$

a length-n word cannot simultaneously contain at least q zeros and at least p ones.
Therefore:

$$\boxed{\text{every binary word of length } n < p + q \text{ is geodesic.}}$$

At length

$$p + q,$$

the first failures appear.
They are exactly the words with Parikh vector

$$(q, p),$$

that is, the orderings of one primitive balanced cycle.
There are

$$\boxed{\binom{p+q}{q}}$$

such first non-geodesic words.
This is another appearance of the same threshold:

$$\boxed{\begin{aligned} p + q &= \text{kernel packet length} = \text{fundamental cycle length} \\ &= \text{first possible non-geodesic length.} \end{aligned}}$$

5.9 Exact geodesic-word growth

The quotient-sphere count of Chapter 4 counted **distinct residual endpoints** at a given directed distance.

We now count something different:

> How many length-n binary words themselves are geodesic between their endpoints?

Let

$$G_{p,q}(n)$$

denote that number.

Theorem 5.5 — Exact geodesic-word growth.

For

$$0 \le n < p+q,$$

$$\boxed{G_{p,q}(n) = 2^n.}$$

For

$$n \ge p+q,$$

$$\boxed{G_{p,q}(n) = \sum_{i=0}^{q-1} \binom{n}{i} + \sum_{j=0}^{p-1} \binom{n}{j}.}$$

The first non-geodesic words occur at length

$$p+q$$

and are exactly the

$$\binom{p+q}{q}$$

orderings of the primitive balanced Parikh vector

$$(q,p).$$

If

$$(p,q) \ne (1,1)$$

and

$$M = \max(p,q),$$

then

$$\boxed{G_{p,q}(n) \sim \frac{n^{M-1}}{(M-1)!}.}$$

For

$$(p,q) = (1,1),$$

$$G_{1,1}(0) = 1, \qquad G_{1,1}(n) = 2 \quad (n \ge 1).$$

Proof.

A length-n word with i zeros is geodesic exactly when

$$i < q$$

or

$$n - i < p.$$

For

$$n < p + q,$$

these inequalities cannot both fail, so every word is geodesic.
For

$$n \geq p + q,$$

the two geodesic regions are disjoint:

- words with fewer than q zeros;
- words with fewer than p ones.

Counting them gives the two fixed binomial tails.
For a nontrivial coprime pair,

$$p \neq q.$$

The larger fixed binomial tail has highest index

$$M - 1.$$

Hence the dominant term is

$$\binom{n}{M-1} \sim \frac{n^{M-1}}{(M-1)!}.$$

The balanced case $(1, 1)$ reduces to the two unary words at every positive length. □
This theorem shows a sharp change of growth.
Before the first balanced cycle, geodesic words grow exponentially:

$$2^n.$$

After the threshold, only two bounded-width binomial tails survive, and growth becomes polynomial.

Again, the primitive balanced cycle is the obstruction.

5.10 Distinct paths versus distinct quotient vertices

Two counting functions have now appeared:

$$|S_n| = \min(n + 1, p + q)$$

for quotient vertices at directed distance n, and

$$G_{p,q}(n)$$

for geodesic words of length n.
They measure different kinds of multiplicity.
The first forgets word order and remembers only the endpoint residual.
The second counts all geodesic orderings.
Inside a fixed shortest-completion rectangle, the same distinction appears as:

$$\boxed{\text{grid vertices} \quad \text{versus} \quad \text{monotone paths.}}$$

The semantic quotient collapses many word histories into one residual coordinate.

The rectangle restores exactly the scheduling information relevant to a shortest interval.

This interplay between quotienting and lifting will become a recurring theme later in the monograph.

5.11 The $2{:}1$ ladder is now completely explained

Return to the Chapter 3 source

$$(L, 0^k).$$

Its residual is

$$2k + 3,$$

and its minimum demand is

$$(k + 2, 1).$$

The general rectangle is therefore

$$[0, k + 2] \times [0, 1].$$

That is precisely a two-row ladder.

Its path count is

$$\binom{k+3}{1} = k + 3.$$

Its square count is

$$(k + 2) \cdot 1 = k + 2.$$

The positive residual is odd, hence not divisible by $p = 2$, so it has a fork:

$$\mathrm{GeoFirst}(2k + 3) = \{0, 1\}.$$

Every feature of the original operational ladder is therefore a specialization of the general minimum-demand rectangle.

This is an important methodological point.

We did not begin by defining a rectangle and then force the original machine into it.

The rectangle was discovered by first analyzing a concrete cancellation/completion ladder in the machine and then identifying the arithmetic that generalized it.

That discovery order will be preserved in the monograph.

5.12 From geometry back to representation

We now have two complementary pieces of structure.

Chapter 4 gave the canonical completion decomposition

$$\boxed{C(r) = \text{finite residue frontier} + \text{unbounded unary tail.}}$$

This suggests moving the finite frontier into control.

Chapter 5 gives the full shortest-path rectangle

$$\boxed{[0, x_r] \times [0, y_r].}$$

This gives an exact semantic graph that a physical DPDA representation ought to realize.

These facts create a new engineering objective.

Can we construct a pushdown representation in which:

1. the exact residual is decodable;
2. the exact prefix quotient is therefore known;
3. the unbounded magnitude remains stack-like;
4. the finite residue information is moved into control;
5. ordinary input transitions reproduce the residual graph as faithfully as possible?

The first attempt will work semantically but reveal a boundary defect.

That defect will turn out to be useful.

It will teach us that the problem is not the residual semantics.

The problem is the coordinate system.

The next chapter begins that redesign.

Part III

Better Coordinates and Exact Resource Frontiers

Research provenance. This Part consolidates the coordinate constructions and exact residual-faithful resource frontiers developed in the preceding structural stages of the monograph. State/stack-symbol tradeoffs and limited pushdown alphabets have a substantial descriptional-complexity literature [26, 25, 43, 49]; the frontier statements below are therefore kept explicitly scoped to their representation classes.

Chapter 6

The First General Physical Lift and Its Defects

The previous chapters separated two objects that were entangled in the original 2023 machine:

semantic residual and physical pushdown representation.

The residual graph is now completely explicit:

$$r \xrightarrow{0} r - p, \qquad r \xrightarrow{1} r + q.$$

Its shortest intervals are rectangles, its quotient coordinate is one integer, and its canonical completion has a bounded finite frontier followed by an unbounded unary tail.

This makes a new design problem unavoidable.

> Can the residual graph itself be lifted into a compact DPDA configuration graph?

The first answer is yes—but only after a deterministic normalization step at a finite boundary.

That apparent imperfection will be useful. It will identify exactly what the first coordinates got wrong.

6.1 Finite phase plus unary magnitude

Let

$$m = \max(p, q).$$

The idea is to move the bounded residue information into finite control while leaving only an unbounded magnitude on the stack.

Use stable control states

$$R_0, \ldots, R_{m-1}$$

and stack alphabet

$$\Gamma_4 = \{0, 1, X, Z_0\}.$$

The positive and negative ends are represented by different unary symbols:

- 0 for the positive end;
- 1 for the negative end.

The additional symbol X is a negative-side sentinel.
For

$$0 \le s < p,$$

define

$$a_s = \begin{cases} 0, & s = 0, \\ s - p, & 1 \le s < p. \end{cases}$$

For

$$0 \le t < q,$$

define

$$b_t = \begin{cases} 0, & t = 0, \\ q - t, & 1 \le t < q. \end{cases}$$

These are not arbitrary constants. They choose one finite base residual for each positive and negative residue class.

6.2 The stable four-symbol encoding

Define

$$\eta(0) = (R_0, Z_0).$$

If

$$r > 0,$$

let

$$s = r \bmod p$$

and write uniquely

$$r = pk + a_s, \qquad k \ge 1.$$

Set

$$\boxed{\eta(r) = (R_s, 0^k Z_0).}$$

If

$$r < 0,$$

let

$$t = (-r) \bmod q$$

and write uniquely

$$r = -qk + b_t, \qquad k \geq 1.$$

Set

$$\boxed{\eta(r) = (R_t, 1^k X).}$$

The representation is injective.

The control state records the finite phase. The unary tail records the unbounded quotient magnitude. The sign is visible physically from the top/sentinel convention.

This is the first systematic realization of the Chapter 4 factorization

$$\text{finite boundary role} + \text{unbounded unary magnitude.}$$

6.3 Stable rules away from the boundary

On the positive side, input 0 should subtract p. Deep inside the unary tail, this is simply a pop:

$$\delta(R_s, 0, 0) = (R_s, \varepsilon).$$

Input 1 adds q. Let

$$s' = (s + q) \bmod p$$

and define

$$c_s = \frac{a_s + q - a_{s'}}{p}.$$

Then

$$pk + a_s + q = p(k + c_s) + a_{s'}.$$

Thus the correct stable rule is

$$\delta(R_s, 1, 0) = (R_{s'}, 0^{c_s+1}).$$

The negative side is symmetric.

Let

$$t' = (t + p) \bmod q$$

and

$$d_t = \frac{p - b_t + b_{t'}}{q}.$$

Then

$$\delta(R_t, 1, 1) = (R_t, \varepsilon)$$

and

$$\delta(R_t, 0, 1) = (R_{t'}, 1^{d_t+1}).$$

Away from the finite boundary, the design is already raw: one input symbol produces the exact next stable residual encoding.

The only difficulty appears when a unary tail is completely consumed.

6.4 Forced boundary normalization

Suppose a positive configuration has only one unary 0 left.

Consuming that 0 exposes

$$Z_0.$$

For residue class $s = 0$, this is already the correct zero configuration.

For

$$1 \le s < p,$$

the configuration

$$(R_s, Z_0)$$

is only transient. A unique forced normalization moves it to the stable encoding of

$$a_s = s - p < 0.$$

Likewise, on the negative side, consuming the last 1 exposes

$$X,$$

and a unique forced normalization moves to the stable encoding of the corresponding nonnegative base residual b_t.

This motivates a semantic macro-step.

Definition 6.1 — Normalized input macro-step.

Write

$$c \overset{a}{\Rightarrow} c'$$

when the machine:

1. consumes the binary symbol a;
2. performs the unique forced finite normalization chain, if one is enabled;
3. stops at the next stable encoded configuration.

The normalization consumes no additional binary input and introduces no choice.

6.5 Exact normalized residual conjugacy

Theorem 6.2 — Normalized residual–configuration conjugacy.

For every

$$r \in \mathbb{Z},$$

$$\boxed{\eta(r) \overset{0}{\Rightarrow} \eta(r-p), \qquad \eta(r) \overset{1}{\Rightarrow} \eta(r+q).}$$

Consequently, for every binary word u,

$$\boxed{\widehat{\delta}^*(\eta(r), u) = \eta(r - pN_0(u) + qN_1(u))\,.}$$

Thus η is a label-preserving graph isomorphism between the residual graph and the stable configuration graph **modulo deterministic normalization**.
Proof.
Let

$$r = pk + a_s > 0.$$

On input 0, if $k > 1$, one unary 0 is popped and the result is immediately

$$\eta(r - p).$$

If $k = 1$, the pop reaches the finite boundary. The uniquely forced normalization sends the exposed boundary pair to the stable encoding of $r - p$.
On input 1,

$$pk + a_s + q = p(k + c_s) + a_{s'},$$

so the positive stable rule lands directly in

$$\eta(r + q).$$

The negative case is symmetric. Input 1 pops one unary 1, with the $k = 1$ case followed by its unique sentinel normalization; input 0 uses the identity defining d_t.
At residual zero the two binary moves are defined directly toward the encodings of $-p$ and q.
The one-step identities imply the word formula by induction. □
This theorem is already a major improvement in understanding.
The original $C/L/R$ machine encoded the residual through a more specialized stack/state interaction.
The new construction makes the quotient coordinate explicit:

finite residual phase in control + unbounded quotient magnitude on a unary tail.

6.6 The abstract rectangles lift to physical configurations

Let

$$\mu(r) = (x_r, y_r).$$

Define

$$\widehat{\Psi}_r(i, j) = \eta(r - pi + qj).$$

Corollary 6.3 — Exact normalized configuration rectangle.
The map

$$\widehat{\Psi}_r$$

is a label-preserving isomorphism from

$$[0, x_r] \times [0, y_r]$$

onto the normalized shortest-path interval from

$$\eta(r)$$

to

$$\eta(0).$$

Every elementary rectangle square is a genuine normalized commuting diamond:

$$\boxed{\widehat{\delta}^*\big(\widehat{\Psi}_r(i,j),01\big) = \widehat{\Psi}_r(i+1,j+1) = \widehat{\delta}^*\big(\widehat{\Psi}_r(i,j),10\big).}$$

The shortest-completion rectangle is therefore not merely an arithmetic diagram.
It is realized by actual DPDA configurations.
The qualification is important:

$$\boxed{\text{one rectangle edge may be} \quad \text{one input move + forced normalization.}}$$

The original 2:1 ladder was stronger: its relevant edges were ordinary binary moves.
The question is now whether the general construction can be made equally literal.

6.7 A concrete 3:2 boundary crossing

Take

$$p = 3, \qquad q = 2, \qquad r = 4.$$

Then

$$\mu(4) = (2,1).$$

The normalized rectangle contains

$$\eta(4) = (R_1, 00Z_0),$$

$$\eta(1) = (R_1, 0Z_0),$$

$$\eta(-2) = (R_0, 1X),$$

$$\eta(6) = (R_0, 00Z_0),$$

$$\eta(3) = (R_0, 0Z_0),$$

$$\eta(0) = (R_0, Z_0).$$

The lower horizontal edge

$$1 \xrightarrow{0} -2$$

is semantically correct, but in these coordinates the binary pop first exposes a transient boundary pair before the forced normalization reaches

$$\eta(-2).$$

The rectangle remains exact.
The edge is simply not yet **raw**.

6.8 Raw residual conjugacy

We now strengthen the representation requirement.

Definition 6.4 — Raw residual–configuration conjugacy.

An injective encoding

$$\zeta : \mathbb{Z} \to \text{DPDA configurations}$$

is a raw residual conjugacy if

$$\boxed{\delta(\zeta(r), 0) = \zeta(r - p), \qquad \delta(\zeta(r), 1) = \zeta(r + q)}$$

for every residual r, using exactly one ordinary binary-input transition and no normalization step.

A raw defect of an encoding is therefore an edge on which this stronger identity fails.

The four-symbol construction is normalized-exact.

We now ask exactly where it fails to be raw.

6.9 The defect set is finite and exact

Proposition 6.5 — Exact four-symbol raw-defect set.

For the four-symbol encoding η,

$$\boxed{\mathcal{D}_{p,q} = \{(s, 0) : 1 \leq s < p\} \cup \{(-j, 1) : 1 \leq j \leq q\}.}$$

Therefore

$$\boxed{|\mathcal{D}_{p,q}| = p + q - 1.}$$

Every residual edge outside this set is already one raw binary transition between stable configurations.

Proof.

On the positive side, input 1 always uses a stable rule.

Input 0 is also raw whenever at least two unary zeros are present. The only problem occurs when the source residual is one of

$$1, 2, \ldots, p - 1.$$

The last zero is popped, leaving the transient state/bottom pair

$$(R_s, Z_0)$$

instead of the required negative encoding.

These are exactly the $p - 1$ positive defects.

On the negative side, input 0 is always raw.

Input 1 is raw while at least two unary ones remain. At the final unary one, the pop exposes X, and the unique normalization is still required.

The affected residuals are

$$-q, -(q - 1), \ldots, -1,$$

giving exactly q negative defects.

The two families are disjoint. □

The count

$$p + q - 1$$

is numerically the same as the completion-frontier scale of Chapter 4 for nontrivial coprime pairs.

The meanings are different.

One counts the largest bounded canonical completion base.

The other counts the raw boundary defects of this particular four-symbol encoding.

The equality is evidence that the same finite arithmetic boundary is being seen from two physical viewpoints—not that the notions are identical.

6.10 Why the defects cannot be patched locally in the same coordinates

It is tempting to keep the encoding η and merely change the boundary transition rules.

That does not work.

Proposition 6.6 — Boundary obstruction for the fixed encoding.

Keeping the four-symbol physical representations $\eta(r)$, no standard top-rewrite PDA transition function can make all edges in

$$\mathcal{D}_{p,q}$$

raw while preserving the already correct behavior on the adjacent deeper unary tails.

Hence every defect of Proposition 6.5 is unavoidable **within these coordinates**.

Proof.

Fix

$$1 \leq s < p.$$

Compare

$$\eta(p + s) = (R_s, 00Z_0)$$

with

$$\eta(s) = (R_s, 0Z_0).$$

The two sources have the same state and visible top symbol.

Correct realization of

$$p + s \xrightarrow{0} s$$

forces the transition on the local pair

$$(R_s, 0)$$

to pop one zero and remain in state R_s.

But the same deterministic rule applied to

$$(R_s, 0Z_0)$$

must produce

$$(R_s, Z_0),$$

whereas the correct target is the negative encoding

$$\eta(s-p).$$

The transition cannot distinguish the two hidden suffixes.

Thus the positive boundary edge cannot be raw without breaking the already correct deeper edge.

The negative case is analogous. Compare

$$(R_t, 11X)$$

and

$$(R_t, 1X).$$

Correct deep behavior forces the visible-1 rule to pop. Applied at the boundary, the same pop leaves

$$(R_t, X),$$

not the required nonnegative target.

The obstruction depends only on the fixed state/top encoding and one-top locality. □

This is the real lesson of the defect theorem.

The problem is not that we failed to invent a clever enough boundary rule.

The local rule is already determined by the deeper tail.

6.11 Which shortest rectangles are already raw?

Corollary 6.7 — Raw rectangle criterion in the four-symbol orientation.

The entire shortest-completion rectangle from residual r is raw under η exactly when

$$\boxed{r = 0 \quad \text{or} \quad (r > 0 \text{ and } p \mid r).}$$

Interpretation. If

$$r > 0$$

and

$$p \mid r,$$

then

$$\mu(r) = (r/p, 0),$$

so the rectangle degenerates to a pure zero corridor and never crosses a defective boundary.

Every positive residual with a genuine two-symbol fork eventually exposes one of the positive boundary defects.

Every negative nonzero rectangle contains the final edge

$$-q \xrightarrow{1} 0,$$

which is defective in this orientation.
The asymmetry is representational.
Zero shares the positive sentinel Z_0, while the negative side uses X.
That observation points directly toward the cure.

6.12 The defect is coordinate-relative

The fixed-coordinate obstruction proves:

$$\boxed{\text{these coordinates cannot be made globally raw.}}$$

It does **not** prove:

$$\boxed{\text{four stack symbols are necessary.}}$$

Those are very different statements.
The first is a theorem.
The second would be an unjustified extrapolation.
The right design question is therefore not:

> What extra symbol should we add?

It is:

> Can we align the finite residual bases differently so that the sign-changing boundary is itself an ordinary top rewrite?

The answer is yes.
Even better, the new coordinates remove the extra sentinel X.
The next chapter performs that coordinate change.

Chapter 7

From Boundary Defects to a Raw Three-Symbol DPDA

Chapter 6 ended with a precise failure.

The four-symbol representation

$$\eta$$

was semantically exact, but at

$$p + q - 1$$

boundary edges an ordinary binary transition could not land directly in the next stable residual encoding.

The obstruction was not global.

It was local:

the same state/top pair had to serve both a deep unary tail and a sign boundary.

The first coordinates placed the finite residual bases on the wrong side of that boundary.

We now shift those bases.

The result is stronger in three ways:

1. every residual edge becomes one raw input transition;
2. the extra sentinel X disappears;
3. the construction still uses only
$$\max(p, q)$$
control states.

This is the first major example in the monograph where understanding the mathematics of a DPDA leads directly to a better DPDA.

7.1 Before using two work symbols: the unary baseline

It is useful to know what can be done with only one work-stack symbol.

Let

$$\Gamma_2 = \{A, Z_0\},$$

where Z_0 is a permanent bottom marker.

Every encoded stack then has the form

$$A^h Z_0.$$

The state must therefore distinguish every finite role that cannot be carried by stack-symbol type.

Theorem 7.1 — Exact one-work-symbol raw-conjugacy complexity.

For every coprime

$$p, q \geq 1,$$

any ε-free permanent-bottom raw residual conjugacy using only one work symbol requires

$$\boxed{|Q| \geq p + q.}$$

There is a right-endmarker construction attaining

$$\boxed{|Q| = p + q, \qquad |\Gamma| = 2.}$$

Hence $p + q$ is exact in this model.

Proof idea

Input 0 partitions the residual line into p bi-infinite orbits modulo p.

Far out on the positive end, injectivity forces the stack height to escape every bounded region. Since a single top rewrite can decrease work height by at most one, the eventual input-0 dynamics must contain enough negative-drift state phases to distinguish all p positive residue classes.

Thus at least p state roles are required.

Symmetrically, input 1 forces at least q state roles on the negative end.

With only one work symbol, those two tail families cannot safely share the same control roles: otherwise repeated negative drift would send infinitely many distinct residuals into finitely many bottom configurations.

Therefore the roles are disjoint:

$$|Q| \geq p + q.$$

A matching construction uses state families

$$P_0, \ldots, P_{p-1}$$

and

$$N_0, \ldots, N_{q-1},$$

with the single work symbol storing only unary magnitude.

The full transition table is recorded in Appendix B.

The theorem establishes an exact baseline:

$$\boxed{\text{one work symbol} \Longrightarrow p + q \text{ states.}}$$

The next question is whether a second work symbol can let the positive and negative phase roles share control states.

It can.

7.2 Boundary-compatible coordinates

Assume first

$$p \geq q.$$

Use only

$$\boxed{\Gamma_3 = \{0, 1, Z_0\}.}$$

Use control states

$$\boxed{Q_3 = \{R_0, \dots, R_{p-1}\}.}$$

There are exactly

$$p = \max(p, q)$$

states.
For nonnegative residuals, use ordinary Euclidean division:

$$r = pk + s, \qquad k \geq 0, \qquad 0 \leq s < p.$$

Define

$$\boxed{\zeta(r) = (R_s, 0^k Z_0).}$$

For negative residuals, write uniquely

$$r = u - qh, \qquad h \geq 1, \qquad 0 \leq u < q.$$

Define

$$\boxed{\zeta(r) = (R_u, 1^h Z_0).}$$

This is the decisive shift.
The negative side no longer terminates in a separate sentinel X.
Its last unary 1 sits directly above the ordinary bottom marker.
The states

$$R_0, \dots, R_{q-1}$$

are deliberately shared between:

- nonnegative boundary roles;
- negative residue phases.

The sign remains visible because:

- a negative configuration has top 1;
- a positive nonboundary configuration has top 0;
- a nonnegative boundary configuration has top Z_0.

Thus state sharing does not destroy injectivity.

7.3 Positive-tail arithmetic

For

$$0 \le s < p,$$

define

$$s^{+} = (s + q) \bmod p$$

and

$$c_s = \frac{s + q - s^{+}}{p}.$$

Since

$$0 < q \le p,$$

we have

$$c_s \in \{0, 1\}.$$

On a positive unary tail:

$$\boxed{\delta(R_s, 0, 0) = (R_s, \varepsilon),}$$

and

$$\boxed{\delta(R_s, 1, 0) = (R_{s^+}, 0^{c_s+1}).}$$

The arithmetic identity is

$$pk + s + q = p(k + c_s) + s^{+}.$$

So the visible rewrite is exactly the residual update.

7.4 The repaired sign boundary

For the nonnegative boundary, define

$$u_s = (s - p) \bmod q$$

and

$$h_s = \frac{u_s + p - s}{q}.$$

Then

$$s - p = u_s - q h_s.$$

Therefore input 0 at the bottom marker can cross directly into the negative encoding:

$$\boxed{\delta(R_s, 0, Z_0) = (R_{u_s}, 1^{h_s} Z_0).}$$

No transient pair is created.
No normalization follows.

The boundary itself is now raw.
Input 1 at a nonnegative boundary uses

$$\boxed{\delta(R_s, 1, Z_0) = (R_{s^+}, 0^{c_s} Z_0).}$$

Again the target is already the exact stable encoding.

7.5 Negative-tail arithmetic and the second repaired boundary

For

$$0 \le u < q,$$

define

$$u^- = (u - p) \bmod q$$

and

$$d_u = \frac{u^- - u + p}{q}.$$

On the negative tail:

$$\boxed{\delta(R_u, 1, 1) = (R_u, \varepsilon),}$$

and

$$\boxed{\delta(R_u, 0, 1) = (R_{u^-}, 1^{d_u+1}).}$$

Now observe what happens when the negative stack has only one 1:

$$(R_u, 1Z_0) \xrightarrow{1} (R_u, Z_0).$$

But by definition

$$(R_u, Z_0) = \zeta(u).$$

The final negative 1-pop therefore crosses the sign boundary automatically.
The very rule that was forced by the deep negative tail is now also the correct boundary rule.
This is exactly what the four-symbol coordinates failed to achieve.

7.6 Global raw conjugacy

Theorem 7.2 — Three-symbol raw residual–configuration conjugacy.
For every coprime

$$p, q \ge 1,$$

there is an ε-free right-endmarker DPDA with

$$\boxed{|Q| = \max(p, q), \qquad |\Gamma| = 3,}$$

whose stack alphabet is

$$\{0, 1, Z_0\}$$

and whose injective encoding

$$\zeta : \mathbb{Z} \to Q \times \Gamma^*$$

satisfies

$$\boxed{\delta(\zeta(r), 0) = \zeta(r - p), \qquad \delta(\zeta(r), 1) = \zeta(r + q)}$$

for every residual r.

Hence, for every word w,

$$\boxed{\delta^*(\zeta(r), w) = \zeta(r - pN_0(w) + qN_1(w))\,.}$$

The right endmarker is accepted exactly at

$$\zeta(0).$$

Proof.

Assume

$$p \geq q.$$

The nonnegative and negative encodings are injective within their own Euclidean decompositions.

They are mutually disjoint physically because the negative family has top 1, whereas the nonnegative family has top 0 or Z_0.

If

$$r = pk + s$$

with

$$k \geq 1,$$

input 0 pops one zero:

$$pk + s - p = p(k - 1) + s.$$

Input 1 uses

$$pk + s + q = p(k + c_s) + s^+.$$

If

$$k = 0,$$

the same input-1 identity works at Z_0, while the boundary identity

$$s - p = u_s - qh_s$$

makes input 0 land directly in the negative encoding.

Now let

$$r = u - qh < 0.$$

Input 1 pops one unary 1.
If

$$h > 1,$$

the result is still negative.
If

$$h = 1,$$

the pop leaves

$$(R_u, Z_0) = \zeta(u),$$

which is exactly the nonnegative target.
Finally,

$$u - p = u^- - qd_u$$

gives

$$r - p = u^- - q(h + d_u),$$

so the input-0 rewrite produces exactly the required deeper negative encoding.
No case uses an ε-move.
For

$$q > p,$$

exchange the roles of 0 and 1, exchange p and q, and reverse the residual sign. □
This theorem removes both weaknesses of Chapter 6 at once:

$$\boxed{(\max(p, q), 4, \text{normalized macro-edges})}$$

becomes

$$\boxed{(\max(p, q), 3, \text{raw input edges}).}$$

7.7 The 3:2 boundary is now literal

Return to

$$p = 3, \qquad q = 2, \qquad r = 4.$$

The new encodings include

$$\zeta(4) = (R_1, 0Z_0),$$

$$\zeta(1) = (R_1, Z_0),$$

$$\zeta(-2) = (R_0, 1Z_0),$$

$$\zeta(0) = (R_0, Z_0).$$

The old defective edge

$$1 \xrightarrow{0} -2$$

is now exactly

$$\boxed{\delta(R_1, 0, Z_0) = (R_0, 1Z_0).}$$

The old negative normalization edge

$$-2 \xrightarrow{1} 0$$

is now the ordinary pop

$$\boxed{\delta(R_0, 1, 1) = (R_0, \varepsilon),}$$

which exposes Z_0.
The same semantic rectangle now exists literally in the raw configuration graph.

7.8 Raw rectangles

Corollary 7.3 — Three-symbol raw configuration rectangles.
Let

$$\mu(r) = (x_r, y_r).$$

Define

$$\Psi_r^{\mathrm{raw}}(i, j) = \zeta(r - pi + qj).$$

Then

$$\Psi_r^{\mathrm{raw}}$$

is a label-preserving isomorphism from

$$[0, x_r] \times [0, y_r]$$

onto the ordinary DPDA shortest-path interval from

$$\zeta(r)$$

to

$$\zeta(0).$$

Every elementary square is a raw commuting diamond:

$$\boxed{\delta^*\big(\Psi_r^{\mathrm{raw}}(i, j), 01\big) = \Psi_r^{\mathrm{raw}}(i+1, j+1) = \delta^*\big(\Psi_r^{\mathrm{raw}}(i, j), 10\big).}$$

The Chapter 5 rectangle has now been fully lifted into one-symbol DPDA dynamics.
No quotient by normalization is needed.

7.9 What has actually been optimized?

The construction uses

$$\max(p, q)$$

states and two work symbols $0, 1$ above the bottom marker.

At this point it is tempting to call it simply "the optimal DPDA."

That wording is too broad.

We must specify the representation category.

The Chapter 7 machine is optimal under several increasingly strong statements, each tied to an explicit representation category. Unrestricted recognition complexity is treated separately in the representation-independent theory.

Theorem 7.4 — Exact homogeneous-tail frontier.

In the permanent-bottom, ε-free raw-conjugacy model with asymptotically homogeneous positive and negative tails:

$$\boxed{S_{\text{hom}}(k) = \begin{cases} p + q, & k = 1, \\ \max(p, q), & k \geq 2, \end{cases}}$$

where k is the number of work symbols.

Thus a second work symbol gives the entire possible state reduction in this model.

Additional homogeneous-tail work symbols cannot reduce the state count below

$$\max(p, q).$$

The proof uses preserved deep suffixes and the cyclic action of:

- addition by q on positive classes modulo p;
- subtraction of p on negative classes modulo q.

Coprimality forces one common positive tail symbol and one common negative tail symbol. With one work symbol the two ends must remain state-disjoint; with two symbols they may share control phases.

This is already an exact state/stack tradeoff.

But it still assumes a homogeneous-tail representation shape.

7.10 The motivating 2:1 ratio survives arbitrary three-symbol encodings

The original research program began at

$$(p, q) = (2, 1).$$

Here the stronger lower bound can be proved without assuming homogeneous tails or sign separation.

Theorem 7.5 — Exact three-symbol state complexity for 2:1.

Consider the full permanent-bottom stack alphabet

$$\Gamma_3 = \{A, B, Z_0\}$$

with:

- no ε-moves;
- injective raw residual conjugacy;
- no homogeneous-tail assumption;
- no prescribed sign-separated encoding.

Then

$$\boxed{|Q| \geq 2.}$$

The construction of Theorem 7.2 uses exactly two states, so

$$\boxed{|Q|_{\min} = 2}$$

for three-symbol raw conjugacy of the motivating 2:1 residual graph in this model.

Proof strategy

Assume one state.

Let

$$G : S_r \mapsto S_{r+1}$$

be input 1 and

$$F : S_r \mapsto S_{r-2}$$

be input 0.

On the encoded orbit,

$$\boxed{FG^2 = G^2F = \mathrm{id}.}$$

Choose an encoded residual of minimum work-stack height.

The inverse identity bounds the lengths of all possible top replacements, reducing the problem to finitely many local rewrite patterns.

If the minimum height is positive, a common inert suffix can be stripped. The remaining possibilities for the first few G-iterates are finite, and each is eliminated by one of:

- preserved-suffix impossibility;
- forced rule reuse;
- violation of minimum height;
- collision of two residuals.

If the minimum height is zero, the bottom write is likewise bounded and only finitely many cases remain. Every case again contradicts injectivity or the inverse relation.

The scope and finite-certificate framework for these lower bounds are recorded in Appendix C.

Thus one state is impossible.

This theorem is particularly important for the monograph because it ties the improved machine back to the original ratio:

The analysis of the 2023 2:1 DPDA leads to an exact two-state, three-symbol raw residual-conjugate realization.

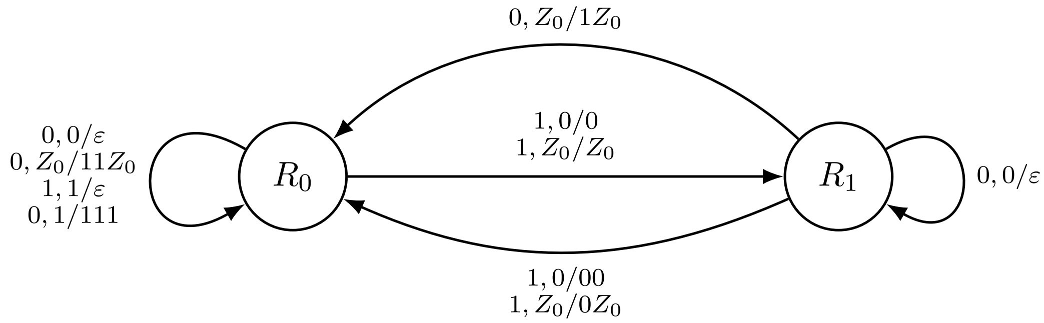


Figure 7.1: The improved two-state, three-symbol raw residual-conjugate DPDA for the motivating 2:1 residual graph. Labels have the form input, top/replacement; only transition cells occurring on the encoded residual orbit are shown. The right endmarker is accepted exactly at (R_0, Z_0).

Figure 7.1 is exact in the permanent-bottom, ε-free raw residual-conjugacy model. It is not a claim of unrestricted global optimality among all DPDAs recognizing $L_{2,1}$.

The scope words remain essential.

7.11 Further exact special ratios

The same unrestricted permanent-bottom three-symbol question can be solved at two further small ratios.

Theorem 7.6 — Exact 3:1 three-symbol state complexity.

For

$$(p, q) = (3, 1),$$

every ε-free permanent-bottom three-symbol raw conjugacy requires

$$\boxed{|Q| \geq 3.}$$

The construction of Theorem 7.2 attains three states.

Hence

$$\boxed{|Q|_{\min} = 3.}$$

The symmetric result holds for

$$(1, 3).$$

The proof again reduces an assumed smaller-state conjugacy to finitely many bounded top-rewrite possibilities, but the case structure is substantially larger than at 2:1. Its role and scope within the small-ratio finite certificates are recorded in Appendix C.

Theorem 7.7 — Computer-assisted exact 3:2 result.

For

$$(p, q) = (3, 2),$$

a finite mathematically bounded search eliminates every two-state permanent-bottom three-symbol raw conjugacy.

Therefore

$$\boxed{|Q| \geq 3,}$$

and the three-state construction is exact:

$$\boxed{|Q|_{\min} = 3.}$$

The symmetric result holds for

$$(2, 3).$$

This theorem is explicitly **computer-assisted**.

The computer is used only after the proof reduces the unrestricted-looking problem to a finite bounded search space. The survivor vectors and branch counts belong in the computational-validation appendix; the theorem should never be presented as a purely symbolic proof.

The small-ratio finite eliminations were an intermediate stage only. Chapter 8 closes the arbitrary-encoding theorem for every coprime pair by a representation-shape-free argument, so no ratio-by-ratio extension is required.

7.12 The right endmarker saves one control state in the scoped models

The constructions above use a distinguished right endmarker and accept exactly at the zero residual configuration.

If ordinary final-state acceptance on the binary alphabet is required instead, the zero configuration must use a final state that does not also occur on a nonzero residual.

In the scoped raw-conjugacy models this costs exactly one state.

Corollary 7.8 — Exact no-endmarker cost in the early raw models.

For the one-work-symbol model:

$$\boxed{|Q| = p + q + 1, \qquad |\Gamma| = 2.}$$

For the sign-separated homogeneous-tail three-symbol model:

$$\boxed{|Q| = \max(p, q) + 1, \qquad |\Gamma| = 3.}$$

Both bounds are tight.

This acceptance cost is separate from the stack-alphabet tradeoff.

It changes only how residual zero is represented.

7.13 The coordinate-shift improvement

The conclusion of Chapters 6 and 7 can now be stated cleanly.

Corollary 7.9 — Coordinate-shift improvement.

The $p + q - 1$ raw defects of the first four-symbol encoding are not a lower bound on the number of stack symbols.

They are properties of those coordinates.

A boundary-compatible shift gives

$$\boxed{(\max(p,q),\ 4,\ \text{normalized macro-steps}) \longrightarrow (\max(p,q),\ 3,\ \text{raw binary steps}).}$$

The missing ingredient was not an additional endpoint bit.
The finite bases had been aligned incorrectly with the sign-changing unary pops.
After moving the bases to adjacent residual classes:

- the same state can serve the post-pop boundary target;
- the ordinary bottom marker is sufficient;
- the extra sentinel X disappears;
- every residual edge becomes literal.

This is the first major design lesson of the monograph:

a locality defect may be a coordinate defect rather than a semantic necessity.

7.14 Three different meanings of "better"

The evolution from the original machine to the raw three-symbol construction improves the representation in several non-equivalent ways.

Semantic transparency

The residual is decoded by a simple Euclidean state/tail coordinate.

Transition literalness

Every binary residual edge is exactly one DPDA input transition.

Stack alphabet

The general raw construction needs only

$$\{0, 1, Z_0\}.$$

Control width

It uses

$$\max(p, q)$$

states.
But the exactness claims established so far belong to stated raw-conjugacy categories.
Unrestricted recognition complexity is representation-independent and is treated separately in Part XVII.
That distinction is one of the central methodological themes of the book.

7.15 Removing the representation shape

The current lower bounds still come from several different levels of assumption:

- one work symbol;
- homogeneous positive/negative tails;
- special-ratio arbitrary three-symbol encodings.

The obvious next question is:

> Are the resource points
>
> $$p + q \quad \text{and} \quad \max(p, q)$$
>
> accidents of these structured coordinates, or are they forced by raw residual conjugacy itself?

To answer that, we must stop assuming homogeneous tails.
We will keep only the semantic requirement:

> Every integer residual has one injective stable physical encoding, and every residual edge is one raw input transition.

From that weak assumption alone, the next chapter derives the complete exact frontier.

That result will turn the improved construction discovered here into a mature resource theorem.

Chapter 8

The Complete Raw Residual-Faithful Frontier

Chapter 7 produced a sharp construction:

$$\boxed{|Q| = \max(p, q), \qquad |\Gamma| = 3,}$$

with one raw binary transition for every residual edge.

It also produced exact lower bounds in several structured settings:

- one work symbol;
- homogeneous positive/negative tails;
- unrestricted three-symbol encodings for several small ratios.

The remaining question is whether the same resource points are forced **without assuming a particular stack shape**.

This chapter removes that assumption.

The result is the mature form of the early resource story:

$$\boxed{S_{\mathrm{raw}}(g) = \begin{cases} \infty, & g = 0, \\ p + q, & g = 1, \\ \max(p, q), & g \geq 2. \end{cases}}$$

The proof is deliberately representation-shape-free.

It does not assume:

- homogeneous tails;
- sign separation;
- a canonical coordinate formula;
- fixed stack depth;
- bounded replacement length.

Only raw residual conjugacy remains fixed.

8.1 Model R

Let

$$\Gamma = \Delta \cup \{Z_0\},$$

where Z_0 is a permanent bottom marker and

$$g = |\Delta|$$

is the number of work symbols.

Definition 8.1 — Model R.

A raw residual conjugacy is an injective map

$$\eta : \mathbb{Z} \to Q \times \Delta^* Z_0$$

such that, for every residual r,

$$\boxed{\delta(\eta(r), 0) = \eta(r - p), \qquad \delta(\eta(r), 1) = \eta(r + q),}$$

with no ε-normalization on encoded residual configurations.

Let

$$S_{\text{raw}}(g)$$

be the minimum number of control states among such realizations, with value ∞ if no finite-state realization exists.

The semantic restriction is exact but narrow:

$$\boxed{\text{one residual} \leftrightarrow \text{one stable physical configuration.}}$$

The physical shape of that configuration is otherwise unrestricted.

This is the right setting for deciding whether the Chapter 7 optimum was merely a feature of its chosen coordinates.

8.2 Why bounded physical height is impossible

Fix a positive residue class

$$s \bmod p.$$

Consider the encoded chain

$$\eta(s), \quad \eta(s + p), \quad \eta(s + 2p), \quad \ldots$$

or, if a finite initial segment crosses zero, begin sufficiently far into the positive tail.

Let

$$h(c)$$

denote the number of work symbols above Z_0.

Lemma 8.2 — Tail escape.

For every positive residue class $s \bmod p$,

$$\boxed{h(\eta(s + pk)) \to \infty \qquad (k \to \infty).}$$

The analogous statement holds on every negative residue class modulo q.

Proof.

Suppose infinitely many residuals on the chain were encoded at height at most H.

There are only finitely many physical configurations of bounded height because:

- Q is finite;
- Δ is finite;
- only finitely many stack words have length at most H.

Injectivity of η would then fail.

Therefore the height must escape every finite bound. □

This elementary argument replaces every earlier homogeneous-tail assumption.

No stack-shape hypothesis is available or needed here. The argument forces only unbounded growth along the residual chain.

8.3 The last occurrence of a height creates a stable suffix

Write

$$c_k = \eta(s + pk)$$

and

$$h_k = h(c_k).$$

Because

$$c_{k+1} \xrightarrow{0} c_k,$$

one ordinary PDA transition can remove at most the visible top work symbol.

Hence

$$\boxed{h_{k+1} \leq h_k + 1.}$$

This one-sided height bound, together with tail escape, creates a canonical cross-section.

Lemma 8.3 — Last-height crossing.

For every sufficiently large integer H, the sequence (h_k) has a last occurrence of height H.

If this last occurrence is at index k, then:

1. $$h_{k+1} = H + 1;$$
2. the transition
$$c_{k+1} \xrightarrow{0} c_k$$
is a pure one-symbol pop;
3. every later configuration on that residue-class tail has height $> H$;
4. all later configurations share the same bottom-H work-stack suffix.

Proof.

Tail escape implies that sufficiently large heights are eventually exceeded forever.

Because

$$h_{k+1} \leq h_k + 1,$$

the forward sequence cannot jump upward over a height H.

Thus every sufficiently large H is attained.

Choose its last occurrence.

The next height cannot be at most H, otherwise either H occurs again immediately or a later climb to values above H must pass through H once more.

Therefore

$$h_{k+1} = H + 1.$$

A one-step PDA transition that lowers height from $H + 1$ to H can only delete the visible work symbol. The lower H work symbols are untouched.

After the last height-H occurrence, every later configuration stays above H. Since subsequent transitions rewrite only the visible top region, that bottom-H suffix can never again be exposed or altered.

Hence it stabilizes forever. □

This is the key physical lemma.

No regular tail shape was assumed.

A stable suffix emerged from injectivity and one-top locality alone.

8.4 Positive residue classes synchronize

For each positive residue class

$$s \in \mathbb{Z}_p,$$

choose the last-height-H cross-section.

Let

$$\beta_s \in \Delta^H$$

be the resulting work-stack word above Z_0.

Lemma 8.4 — Positive synchronization.

For sufficiently large H,

$$\boxed{\beta_s = \beta_{s+q}}$$

for every residue class s mod p.

Consequently all p positive residue classes have the same last-height-H stack word, and their control states must be pairwise distinct.

Hence

$$\boxed{|Q| \geq p}$$

on the positive side.

Proof.

Take sufficiently deep positive configurations so that both source and target lie beyond their last-height-H crossings.

Input 1 maps residual class

$$s$$

to

$$s + q \bmod p.$$

A single top rewrite cannot modify the already stabilized bottom-H suffix.

Therefore

$$\beta_s = \beta_{s+q}.$$

Because

$$\gcd(p, q) = 1,$$

addition by q is one cycle on

$$\mathbb{Z}_p.$$

Thus all β_s are equal.

At the selected height-H cross-sections, the complete stack word is now identical across all p classes.

Since the residuals are distinct and η is injective, their control states must be distinct. □

The negative side is symmetric.

Lemma 8.5 — Negative synchronization.

At sufficiently large synchronized cross-sections on the negative tail,

$$\boxed{|Q| \geq q.}$$

Therefore every nonempty work alphabet already forces

$$\boxed{|Q| \geq \max(p, q).}$$

This is the first shape-free lower bound matching the Chapter 7 construction.

8.5 Why one work symbol is worse

Suppose now

$$g = 1.$$

Write

$$\Delta = \{A\}.$$

At a fixed height H, every work stack is literally

$$A^H Z_0.$$

The positive and negative synchronized cross-sections can therefore be chosen at the same height with exactly the same physical stack.

Lemma 8.6 — Cross-sign unary separation.

If

$$g = 1,$$

the p synchronized positive states and the q synchronized negative states must all be distinct.

Thus

$$\boxed{|Q| \geq p + q.}$$

Proof.

Choose the synchronized positive and negative cross-sections sufficiently far from zero.

All selected configurations have identical stack

$$A^H Z_0.$$

They represent

$$p+q$$

distinct residuals.

Injectivity can distinguish them only through control state.

Hence all $p+q$ states are distinct. $\square$

The reason the second work symbol is powerful is now exact.

With one work symbol, sign cannot be carried by the tail symbol, so positive and negative residue roles cannot share states.

With two work symbols, the stack itself can separate the ends, allowing state reuse.

8.6 The complete raw frontier

Theorem 8.7 — Exact raw residual-faithful frontier.

For every coprime

$$p, q \geq 1,$$

$$\boxed{S_{\mathrm{raw}}(g) = \begin{cases} \infty, & g = 0, \\ p+q, & g = 1, \\ \max(p,q), & g \geq 2. \end{cases}}$$

Proof.

If

$$g = 0,$$

the only stack is the permanent bottom marker. With finitely many control states, only finitely many physical configurations exist, so no injective encoding of all

$$r \in \mathbb{Z}$$

is possible.

For

$$g \geq 1,$$

positive and negative synchronization give

$$|Q| \geq \max(p,q).$$

For

$$g = 1,$$

cross-sign unary separation strengthens the bound to

$$p + q.$$

The matching upper constructions are exactly the Chapter 7 machines:

- one work symbol with $p + q$ states;
- two work symbols with $\max(p, q)$ states.

Extra work symbols may simply remain unused. □

This theorem changes the status of several earlier results.

The Chapter 7 homogeneous-tail lower bound was not merely an artifact of homogeneous tails.

The exact same resource points survive after all such physical-shape assumptions are removed.

Thus the improved three-symbol machine discovered from the original DPDA lies on the complete raw residual-faithful frontier.

8.7 Scope of the theorem

The theorem is strong, but its category remains explicit.

It proves exact resource complexity for **raw residual conjugacy**.

It does not prove unrestricted recognition complexity for every DPDA recognizing

$$L_{p,q}.$$

A completely different recognizer may:

- fail to give every residual one stable configuration;
- encode several residuals indirectly through longer physical histories;
- use normalization;
- use a different semantic quotient;
- leave Model R entirely.

Therefore the correct statement is:

$$\boxed{\text{the raw residual-faithful frontier is exact.}}$$

Not:

$$\boxed{\text{the unrestricted DPDA state/stack frontier is solved.}}$$

This distinction becomes even more important in Chapter 9, where the analysis moves beyond residual-faithful representation to a language-level hierarchy problem.

8.8 Acceptance without a right endmarker

In the raw right-endmarker constructions, residual zero is accepted only when the endmarker arrives.

Without a right endmarker, ordinary final-state acceptance requires a state that is final exactly at residual zero.

The synchronized lower-bound states all occur on nonzero residuals, so none may be final.

Corollary 8.8 — Exact no-endmarker raw cost.

Under standard final-state acceptance without a right endmarker,

$$\boxed{S_{\mathrm{raw}}^{\mathrm{noend}}(g) = \begin{cases} \infty, & g = 0, \\ p+q+1, & g = 1, \\ \max(p,q)+1, & g \geq 2. \end{cases}}$$

The extra state is an acceptance-wrapper cost.
It does not change the nonzero residual geometry.
This is another place where the two original Chapter 1 variants become conceptually useful: acceptance conventions and semantic core complexity should not be conflated.

8.9 Can the optimum survive bounded pushes?

The raw three-symbol construction may use fixed replacement words longer than two symbols.

Suppose we impose the standard bounded-push convention

$$\max|\alpha| \leq 2.$$

A naive compilation could add many states.
The remarkable fact is that no new state is necessary.
Assume

$$p \geq q.$$

The positive-side writes are already of length at most two because

$$c_s \in \{0, 1\}.$$

Only some negative unary writes are long.
For each negative target phase v, define a family of suffix roles

$$H_{v,j},$$

meaning:

append exactly j additional unary 1's, then enter stable state R_v.

The total number of such transient roles is exactly

$$\boxed{p-q.}$$

Now inspect the stable three-symbol construction.
Negative configurations use top 1 only in states

$$R_0, \ldots, R_{q-1}.$$

Therefore the state/top cells

$$(R_i, 1), \qquad q \leq i < p,$$

are unused by stable negative semantics.
There are exactly

$$p - q$$

such cells.

The transient suffix roles fit into them perfectly.

Theorem 8.9 — State-preserving bounded-push compiler.

For every coprime p, q, there is a deterministic right-endmarker DPDA whose stable configurations realize the residual graph and satisfy

$$\boxed{|Q| = \max(p, q), \qquad |\Gamma| = 3, \qquad \max |\alpha| \le 2.}$$

The only ε-moves are uniquely forced terminating suffix-chain moves.

Proof idea. Split every long unary write into:

1. one input-consuming push of length at most two;
2. a strictly decreasing suffix-completion chain.

The total transient role count is

$$|p - q|.$$

Exactly that many stable state/top cells are unused.

Overlay the transient roles on those cells.

Since the base machine has no consuming transition on the overlaid cells, the forced ε-moves introduce no input/ε conflict.

Every chain terminates and reconstructs exactly the macro-result of the original raw transition.

The case

$$q > p$$

is symmetric. □

This theorem is a useful warning against counting "temporary states" too early.

A transient normalization role need not require a new control state if an unused state/top cell already exists.

8.10 A stronger semantic-visibility budget

Model R asks only for exact residual faithfulness.

We now intentionally demand more.

Fix

$$B \ge \max(p, q)$$

and write every residual as

$$r = s + Bz, \qquad 0 \le s < B.$$

Define the positive and negative phase periods

$$h_+(B) = \frac{p}{\gcd(B, p)},$$

$$h_-(B) = \frac{q}{\gcd(B, q)},$$

and

$$\boxed{H(B) = h_+(B) + h_-(B).}$$

In the **phase-local architecture**, each remainder s must expose its currently active signed quotient phase through:

- the control state;
- one visible work symbol.

Let

$$C$$

be the number of available work symbols.
For one fixed remainder, one state can provide at most C distinct state/top cells.
Therefore $H(B)$ phase roles require at least

$$\left\lceil \frac{H(B)}{C} \right\rceil$$

states.

Lemma 8.10 — Phase capacity.

For each remainder class s,

$$\boxed{|Q_s| \geq \left\lceil \frac{H(B)}{C} \right\rceil.}$$

The B remainder classes are semantically distinct in this architecture.
Summing gives the exact global frontier.

Theorem 8.11 — Exact normalized phase-local frontier.

For

$$B \geq \max(p, q),$$

$$\boxed{Q_{\min}^{\text{phase}}(B, C) = B \left\lceil \frac{\frac{p}{\gcd(B,p)} + \frac{q}{\gcd(B,q)}}{C} \right\rceil.}$$

For

$$C \geq 2,$$

the matching construction can be ε-free and raw.
For

$$C = 1,$$

the same state bound is attained with one forced bottom-cell normalization at the negative-to-zero crossing.
This theorem should not be confused with Theorem 8.7.
Theorem 8.7 asks for exact residual faithfulness only.
Theorem 8.11 additionally requires a chosen signed phase to be locally visible.
A stronger representation contract can require more states.
This is not a contradiction.
It is the beginning of a general principle:

Resource complexity depends on what physical information the representation is required to expose locally.

8.11 A superlinear exact family

Take

$$p = k^2, \qquad q = k^2 - 1, \qquad B = k(k+1).$$

Then

$$H(B) = 2k - 1.$$

Hence

$$\boxed{Q_{\min}^{\text{phase}}(B, C) = k(k+1)\left\lceil \frac{2k-1}{C} \right\rceil .}$$

For fixed C,

$$Q_{\min}^{\text{phase}} = \Theta(k^3).$$

Since

$$\max(p, q) = \Theta(k^2),$$

this is

$$\boxed{\Theta\Big(\max(p, q)^{3/2}\Big) .}$$

The raw residual-faithful machine still needs only

$$\Theta(\max(p, q))$$

states.

The superlinear growth comes entirely from the extra phase-local visibility requirement.

This example makes category-relative resource claims impossible to ignore.

8.12 Direct lossless factorization: counting semantic cells

A still more structured model keeps the residual split into explicit semantic roles.

The stable roles are:

1. $p + q - 1$ boundary/bottom roles;
2. p deep positive roles with visible top 0;
3. q deep negative roles with visible top 1.

These role classes are disjoint.

Theorem 8.12 — Exact semantic-cell count.

Every exact implementation of this direct lossless unary-tail factorization requires

$$\boxed{2p + 2q - 1}$$

reachable stable state/top cells.

The canonical factorization attains this bound exactly.

The theorem counts semantic cells, not merely control states.

This is the correct unit when the representation contract explicitly requires different local roles to remain distinguishable.

8.13 Exact normalization-rule cost under push-at-most-two

Assume

$$p > q.$$

For each deep negative remainder t, the bounded-push normalization required by input 0 has a net stack-height increase

$$D_t = \left\lfloor \frac{p+t}{q} \right\rfloor - 1.$$

A terminating deterministic ε-computation that gains D persistent stack levels needs at least D distinct persistent-push rules.

Why?

If the same state/top rule created two different persistent levels, the intervening positive-drift ε-macro could repeat forever, contradicting termination.

Different source remainders also cannot share the same persistent rule, because after the common rule the lower context remains hidden and deterministic continuation would be identical, while the semantic target remainder is different.

The arithmetic sum is exact:

$$\sum_{t=0}^{q-1} D_t = p - q.$$

Theorem 8.13 — Exact ε-transition count.

In the direct lossless unary-tail factorization with:

- push-at-most-two;
- right endmarker;

the minimum number of forced normalization rules is

$$\boxed{|\delta_\varepsilon|_{\min} = |p - q|.}$$

Theorem 8.14 — Exact total transition count.

Under the same model,

$$\boxed{|\delta|_{\min} = 4(p+q) - 1 + |p - q|.}$$

Reason. The stable semantic graph forces:

$$4(p+q)-2$$

binary input rules.
The endmarker contributes one accepting rule.
The bounded-push normalization requires exactly

$$|p-q|$$

forced ε-rules.
All three lower bounds are attained simultaneously.
This is the most explicit descriptional-complexity layer so far.
We are no longer counting only states or stack symbols.
We are counting:

- semantic state/top cells;
- binary rules;
- forced normalization rules;
- total transition description size.

8.14 Four resource questions, four different answers

We can now place the resource results in a hierarchy.

Level 1 — Raw residual conjugacy

Only the residual semantics is fixed.

$$\boxed{S_{\mathrm{raw}}(g)=\infty,\ p+q,\ \max(p,q).}$$

Level 2 — Bounded-push normalized realization

The optimal three-symbol/state point survives:

$$\boxed{|Q|=\max(p,q),\quad |\Gamma|=3,\quad \max|\alpha|\le 2.}$$

The price is uniquely forced normalization.

Level 3 — Phase-local architecture

A selected signed phase must be visible from one state/top cell.

$$\boxed{Q_{\min}^{\mathrm{phase}}(B,C)=B\left\lceil\frac{H(B)}{C}\right\rceil.}$$

Level 4 — Direct lossless unary-tail factorization

The semantic roles themselves are fixed and counted.

$$\boxed{C_{\mathrm{sem,min}}=2p+2q-1,}$$

$$\boxed{|\delta_\varepsilon|_{\min}=|p-q|,}$$

$$\boxed{|\delta|_{\min} = 4(p+q) - 1 + |p-q|.}$$

These theorems do not compete.
They answer increasingly structured physical questions.

8.15 What is now genuinely settled

The early DPDA analysis has produced a complete statement:

> Within raw residual conjugacy, the improved three-symbol machine is not merely good—it is on the exact frontier.

The discovery sequence is now complete:

$$\begin{gathered}
\text{original machine} \\
\downarrow \\
\text{completion semantics} \\
\downarrow \\
\text{finite-frontier coordinates} \\
\downarrow \\
\text{four-symbol normalized lift} \\
\downarrow \\
\text{exact boundary defect} \\
\downarrow \\
\text{coordinate shift} \\
\downarrow \\
\text{three-symbol raw DPDA} \\
\downarrow \\
\boxed{\text{complete raw residual-faithful frontier.}}
\end{gathered}$$

At this point the natural resource question becomes more radical.

The entire analysis still assumes that a competing machine faithfully represents our residual semantics.

But a DPDA recognizing the same language has no obligation to use our residual coordinate at all.

It may use a completely different stack representation.

It may use deterministic ε-moves.

It may hide the arithmetic in a way our residual-faithful lower bounds do not see.

So we now ask:

> Can a stack-alphabet lower bound survive against **arbitrary equivalent deterministic pushdown representations**?

That question takes the monograph outside the original counting family and into a previously stated hierarchy problem.

That is the subject of the next Part.

Part IV

When the Resource Question Leaves Our Representation

Research provenance. This Part develops the representation-independent hierarchy question that emerges when the raw residual-faithful representation is no longer imposed; its scope is deliberately kept separate from the frontiers of the preceding Part. Related descriptional-complexity work on pushdown alphabets and determinization provides the surrounding context [43, 42, 25].

Chapter 9

From Residual Frontiers to a Representation-Independent Hierarchy Theorem

The resource story of the previous chapters has reached a natural stopping point.

Within raw residual conjugacy, the frontier is exact:

$$S_{\mathrm{raw}}(g) = \begin{cases} \infty, & g = 0, \\ p + q, & g = 1, \\ \max(p, q), & g \geq 2. \end{cases}$$

The result is strong because the lower bound no longer assumes homogeneous tails, sign-separated coordinates, or a particular stack formula.

But it still assumes something fundamental:

$$\boxed{\text{the competing machine is required to realize the residual graph itself.}}$$

A DPDA recognizing the same language need not do that.

It may encode the future language in a completely different way. It may normalize internally. It may use deterministic ε-moves between input symbols. It may never expose anything resembling our residual coordinate.

The next step in the research program was therefore not another attempt to optimize the same representation.

It was to remove the representation from the lower bound.

9.1 The methodological jump

The first two resource stages asked:

> Given the semantic object we have discovered, how efficiently can it be represented?

The new question is different:

> Can the **continuation behavior of the language itself** force a state–stack lower bound, even if an equivalent DPDA chooses an unrelated internal representation?

This distinction is the reason Chapter 8 and the present Part must not be merged.
Chapter 8 is about **representation-faithful optimality**.
This Part is about **arbitrary equivalent recognizers**.
The lower-bound invariant can therefore no longer refer to:

- our residual stack shape;
- our canonical completion word;
- our sign symbol;
- our chosen state phases.

It must be externally visible through the accepted continuation language.
Completion lengths provide exactly such an observable.

9.2 The external hierarchy question

Let

$$\mathcal{D}_{m,k}$$

denote the class of languages recognized by deterministic pushdown automata with at most:

$$m$$

control states and at most

$$k$$

pushdown symbols, in the general deterministic model where deterministic ε-moves are allowed.
The inclusion

$$\mathcal{D}_{m,k} \subseteq \mathcal{D}_{m,k+1}$$

is trivial: an extra stack symbol may simply remain unused.
The question is whether the inclusion is strict.
Earlier work had already established:

- a binary stack-alphabet hierarchy for stateless DPDAs;
- an extension to m-state **realtime** DPDAs for every $m \geq 1$.

The remaining general-model case allows deterministic ε-moves. The theorem proved in this Part establishes the strict hierarchy

$$\boxed{\mathcal{D}_{m,k} \subsetneq \mathcal{D}_{m,k+1}}$$

over a fixed binary input alphabet for

$$m \geq 2, \qquad k \geq 1,$$

under the general single-initial-symbol convention used in this Part.
The stateless case $m = 1$ is kept separate because the earlier stateless formulation uses a slightly different initial-store convention.
That model distinction must remain explicit.

9.3 Why realtime separation does not automatically survive ε-moves

A realtime DPDA processes one input symbol per transition.

Its visible state/top pair is therefore synchronized with the input position.

A general DPDA can insert a deterministic ε-computation between two input symbols.

Such an internal computation can:

- rewrite the stack;
- move information between stack and control;
- expose a deeper symbol;
- normalize a temporary encoding;
- alter the local mode before the next symbol is consumed.

Therefore a lower bound based only on what a realtime machine must expose at input boundaries may collapse when the adversary is allowed internal stack work.

This is precisely why the lower bound must reason about **raw computation positions**, including positions inside forced ε-blocks.

The proof developed in this Part will do that.

9.4 Lineage from the original DPDA

The hierarchy theorem is formally self-contained.

It does not depend on the original 2:1 transition table.

Nevertheless, its mathematical route is directly connected to the earlier program.

The progression was:

original completion-aware DPDA $\to$ shortest continuation semantics $\to$ exact state/stack tradeoffs $\to$ question of arbitrary alternative representations.

The key methodological lesson from the earlier machines was:

> Stack contents can be probed indirectly by the continuations they require.

The hierarchy witness turns that observation into an adversarial lower-bound tool.

Instead of assuming that a competitor stores a residual, we construct many prefix families whose required unary completion lengths grow at pairwise different rates.

Those rates are visible in the language itself.

The competitor may encode them however it wants.

9.5 The later 2:1 improved DPDA as the immediate precursor

For the motivating ratio $(2, 1)$, Chapter 7 obtained a later raw residual-conjugate construction with:

$$|Q| = 2, \qquad \Gamma = \{0, 1, Z_0\},$$

and no ε-normalization on binary residual edges.

This representation is more compact and more literal than the original center/left/right realization for the residual-conjugacy objective.

But that result still does not imply an unrestricted recognition lower bound.

An adversarial DPDA recognizing the same language is not required to use:

- the same two states;
- the same three stack symbols;
- the same residual coordinate;
- any stable residual encoding at all.

This is the conceptual bridge into the hierarchy problem.

9.6 Why the witness development did not end with a direct encoding trick

The route to the final binary hierarchy witness passed through two intermediate prototypes.

Four-letter prototype

The first general prototype used semantic modes

$$(i, t) \in \mathbb{Z}_m \times \mathbb{Z}_k$$

with distinct weights

$$\omega(i, t) = tm + i + 1 \in \{1, \dots, mk\}.$$

Its purpose was to isolate the product phenomenon:

$$\boxed{\begin{array}{c}\text{many externally different completion rates}\\ \Longrightarrow \text{many recurrent state/top roles.}\end{array}}$$

Three-letter prototype

A mixed-radix control symbol merged two semantic navigation coordinates.

This reduced the input alphabet but still retained a separate delimiter/cleanup phase.

Binary witness

A direct binary code of the three-letter system would have required decoder memory.

Under an exact m-state upper budget, that decoder memory is itself a resource and can ruin the comparison.

The final construction therefore changes the invariant instead of coding the alphabet.

Rather than forcing

$$mk \text{ slopes} + 1 \text{ cleanup role},$$

it forces directly

$$\boxed{mk + 1 \text{ distinct completion slopes.}}$$

The cleanup role disappears as a separate semantic mode.

The witness becomes binary without needing a phase decoder.

This is an important design lesson in its own right:

When an encoding adds resource overhead, change the observable rather than merely recoding symbols.

9.7 The general DPDA model used in the lower bound

A DPDA is

$$M = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F),$$

with partial transition function

$$\delta : Q \times \Gamma \times (\Sigma \cup \{\varepsilon\}) \to Q \times \Gamma^*.$$

We write the stack top on the right.
Thus if

$$\delta(q, X, a) = (p, \beta),$$

then

$$(q, \gamma X, av) \vdash (p, \gamma\beta, v).$$

No transition is available from an empty stack.
Determinism requires that for each state/top pair:

- there is at most one transition for each input symbol;
- if an ε-transition is enabled, then no input-consuming transition is enabled from that same state/top pair.

Acceptance is by:

$$\boxed{\text{all input consumed} + \text{final state} + \text{empty stack.}}$$

A realtime DPDA is the special case with no ε-moves.
The upper witness will be realtime.
The lower-bound opponent is allowed to be fully general.

9.8 Modes

A nonempty control-store configuration has the form

$$(q, \eta X),$$

where X is the visible top symbol.
Its **mode** is the pair

$$\boxed{[q, X].}$$

A DPDA has at most

$$\boxed{|Q||\Gamma|}$$

nonempty modes.
The product lower bound will ultimately be a mode-counting theorem.
But merely counting modes is not enough.

We must first prove that different externally visible completion slopes cannot share one recurrent mode.

That is the hard part.

9.9 The lower-bound target

For every

$$m \geq 2, \qquad k \geq 1,$$

we will construct a binary language

$$B_{m,k}$$

such that:

1. $B_{m,k}$ is recognized by a realtime DPDA with exactly

$$m$$

states and

$$k+1$$

stack symbols;

2. every general DPDA recognizing $B_{m,k}$, even with arbitrary deterministic ε-moves, satisfies

$$\boxed{|Q||\Gamma| \geq mk+1.}$$

The hierarchy corollary will then be immediate.

If a machine used at most m states and at most k stack symbols, then

$$|Q||\Gamma| \leq mk,$$

contradicting the product theorem.

The real work is therefore to build a binary witness exposing exactly

$$mk+1$$

pairwise incompatible continuation slopes.

That construction is the subject of Chapter 10.

Chapter 10

Completion Slopes and ε-Robust Pumping

Fix

$$m \geq 2, \qquad k \geq 1,$$

and write

$$\boxed{\ell = mk + 1.}$$

The witness will have ℓ distinguished live prefix families.

Each family will admit exactly one all-zero completion length after an accumulation block of 1's.

Those lengths will be affine functions with slopes

$$1, 2, \ldots, \ell.$$

The upper machine uses exactly:

$$m \text{ states}, \qquad k + 1 \text{ stack symbols}.$$

The lower bound then asks whether an arbitrary equivalent DPDA can hide those ℓ externally different rates inside fewer than ℓ state/top modes.

It cannot.

10.1 The binary realtime witness

Let

$$Q = \{q_0, \ldots, q_{m-1}\}$$

and

$$\Gamma = \{A, M_0, \ldots, M_{k-1}\}.$$

Hence

$$|Q| = m, \qquad |\Gamma| = k + 1.$$

Define the permutation

$$\pi(0) = 1, \qquad \pi(1) = 0,$$

and

$$\pi(r) = r \qquad (2 \le r < m).$$

Choose ℓ pairwise distinct slope modes

$$x_s = (r_s, X_s), \qquad 0 \le s < \ell,$$

by

$$x_0 = (q_1, A),$$

and, for

$$0 \le r < m, \qquad 0 \le t < k,$$

$$x_{1+rk+t} = (q_{\pi(r)}, M_t).$$

Thus

$$\{x_1, \ldots, x_{\ell-1}\} = Q \times \{M_0, \ldots, M_{k-1}\}.$$

The initial pair

$$(q_0, A)$$

is deliberately not a slope mode.
Define the fixed table word

$$\boxed{T = X_{\ell-1}X_{\ell-2}\cdots X_1X_0,}$$

with top on the right, so X_0 is initially visible.

10.2 Four rule families

The witness DPDA $U_{m,k}$ is realtime.

Header

$$\boxed{\delta(q_0, A, 1) = (q_1, T).}$$

The first 1 installs the entire finite mode table.

Generic zero-pop on counter symbol

For every state q_i,

$$\boxed{\delta(q_i, A, 0) = (q_i, \varepsilon).}$$

Selector transitions

For

$$0 \le s < \ell - 1,$$

$$\boxed{\delta(r_s, X_s, 0) = (r_{s+1}, \varepsilon).}$$

At the last table mode,

$$\boxed{\delta(r_{\ell-1}, X_{\ell-1}, 0) = (q_0, \varepsilon).}$$

Thus a zero can walk down the installed table.

Slope transitions

For every

$$0 \le s < \ell,$$

$$\boxed{\delta(r_s, X_s, 1) = (r_s, A^{s+1} X_s).}$$

One accumulation 1 inserts exactly $s+1$ copies of A immediately below the visible mode symbol.

The same mode remains visible.

This is the entire slope mechanism.

10.3 Upper-bound theorem

Proposition 10.1 — Realtime upper construction.

$U_{m,k}$ is a deterministic realtime DPDA with exactly

$$\boxed{m \text{ states}}$$

and

$$\boxed{k+1 \text{ stack symbols.}}$$

Proof.

There are no ε-moves.

The slope modes are pairwise distinct, so their 1-rules and selector 0-rules do not conflict.

The only apparent overlap is at

$$x_0 = (q_1, A),$$

where the selector rule agrees exactly with the generic A-pop rule.

The header uses

$$(q_0, A, 1),$$

which is not one of the slope modes.

Thus the transition function is deterministic. □

Define

$$\boxed{B_{m,k} = L(U_{m,k}).}$$

The lower bound will not require a closed combinatorial description of the entire language.

One exact slice is enough.

10.4 The exact completion-slope slice

For

$$0 \le s < \ell,$$

define

$$\boxed{N_s(n) = \ell - s + (s+1)n.}$$

Let

$$p_s = 10^s.$$

Lemma 10.2 — Exact pure-zero slice.

For every

$$0 \le s < \ell$$

and every

$$n, r \ge 0,$$

$$\boxed{10^s 1^n 0^r \in B_{m,k} \iff r = N_s(n).}$$

Proof.

After the initial 1, the machine is in mode x_0 with table T.

Each of the next s zeros removes the current table symbol and exposes the next designated slope mode.

Thus after

$$10^s$$

the visible mode is

$$x_s$$

and the remaining table contains exactly

$$\ell - s$$

symbols.

Each subsequent 1 inserts

$$s + 1$$

copies of A below the visible mode symbol.

After n such ones, the total stack height is therefore

$$\ell - s + (s+1)n.$$

Now read zeros.

The first zero removes the visible mode symbol.

Inserted A's are then popped one by one, and whenever the next table symbol is exposed the machine is already in its designated state.

Thus every remaining zero removes exactly one stack symbol.

Because acceptance requires empty stack after the input has been exhausted, the all-zero suffix is accepted exactly when its length equals the current stack height. □

Hence the ℓ prefix families

$$p_s 1^n$$

have unique pure-zero completion slopes

$$\boxed{w_s = s+1.}$$

So the language exposes:

$$\boxed{1, 2, \ldots, mk+1}$$

as pairwise distinct affine continuation rates.

10.5 The smallest new case

Take

$$m = 2, \qquad k = 1.$$

Then

$$\ell = 3.$$

The slope modes are

$$x_0 = (q_1, A),$$

$$x_1 = (q_1, M_0),$$

$$x_2 = (q_0, M_0).$$

The exact slice becomes:

$$1\,1^n 0^{3+n},$$

$$10\,1^n 0^{2+2n},$$

$$100\,1^n 0^{1+3n}.$$

The three slopes are:

$$1, \qquad 2, \qquad 3.$$

The theorem of the next chapter will imply that no DPDA with:

$$2 \text{ states}$$

and

$$1 \text{ stack symbol}$$

can recognize the same language—even if that competitor uses deterministic ε-moves.

10.6 The lower-bound opponent is arbitrary

Let

$$M = (Q, \{0, 1\}, \Gamma, \delta, q_0, Z_0, F)$$

be any general DPDA with

$$L(M) = B_{m,k}.$$

Fix one slope index s and consider the unique computation on

$$p_s 1^\omega.$$

Every finite prefix

$$p_s 1^n$$

is live because it has the accepted continuation

$$0^{N_s(n)}.$$

Therefore the computation cannot become permanently trapped in an ε-divergence before the next input symbol is consumed.

We call **raw positions** all individual computation boundaries, including positions inside forced ε-blocks.

This is essential.

The proof must not look only at input-boundary configurations.

10.7 Raw continuation principle

At any raw configuration

$$(q, \gamma),$$

determinism fixes the future computation once the unread input is fixed.

If an ε-transition is enabled, it is forced and independent of whether the next unread binary symbol is 0 or 1.

Therefore an equal raw control-store configuration can be reused with a changed future suffix.

This trivial-looking fact is the bridge from equality of physical configurations to equality of continuation behavior.

10.8 No raw configuration can repeat on a live slope ray

Lemma 10.3 — No raw configuration repetition.

Along the computation on

$$p_s 1^\omega,$$

the same full control-store configuration cannot occur twice at distinct raw positions.

Proof.

Suppose first that the two occurrences have consumed different numbers

$$n < n'$$

of accumulation ones.

The later prefix has accepted continuation

$$0^{N_s(n')}.$$

If the earlier and later raw control-store configurations are identical, determinism gives the same future computation from both.

Thus M would also accept

$$p_s 1^n 0^{N_s(n')}.$$

But the exact slice says that after

$$p_s 1^n$$

the unique all-zero completion length is

$$N_s(n),$$

and

$$N_s(n') \neq N_s(n).$$

Contradiction.

If the two repeated raw configurations occur after the same amount of input has been consumed, then the segment between them contains only ε-moves.

Determinism would repeat the same ε-cycle forever, preventing the next live input symbol from being read.

Again contradiction. □

This lemma absorbs arbitrary deterministic internal normalization into the argument.

Repeated physical configurations are impossible because the language exposes a unique continuation length.

10.9 Raw stack height must escape

Lemma 10.4 — Raw-height escape.

Along every slope ray,

$$\boxed{\text{raw stack height tends to infinity.}}$$

More explicitly, for every H, after some raw position the stack height is always greater than H.

Proof.

There are only finitely many control-store configurations of stack height at most H.

By Lemma 10.3, none can occur twice.

Therefore only finitely many raw positions have height at most H. □

The adversary may use any representation it likes.

But the infinitely many distinguishable live prefixes force its physical store to escape every bounded region.

10.10 Suffix-minimum positions

A raw position t is a **step** if its stack height is a minimum over the entire future suffix of the computation:

$$\boxed{h(t) \leq h(t') \quad \text{for every } t' \geq t.}$$

Lemma 10.5 — Infinitely many steps.

Every slope ray contains infinitely many step positions.

Proof.

The height sequence tends to infinity.

For every starting position, the future tail has a minimum.

Choose a position where that minimum occurs.

As the starting point moves arbitrarily far to the right, these minima cannot all lie in a fixed finite set. □

Step positions are the correct pumping locations because the computation never descends below their initial stack height.

Therefore the lower stack context remains hidden.

10.11 The ε-robust step pump

Lemma 10.6 — Step-pump lemma.

On every slope ray there exist:

- a mode
$$[q, X];$$
- an integer
$$c \geq 1;$$
- a nonempty word
$$\rho \in \Gamma^+;$$

such that for every lower context

$$\eta \in \Gamma^*,$$

$$\boxed{(q, \eta X) \overset{1^c}{\Longrightarrow} (q, \eta\rho X),}$$

where the macro-computation:

1. consumes exactly c input symbols 1;
2. may contain forced ε-moves;

3. never goes below the initial height

$$|\eta X|.$$

The mode $[q, X]$ is called an **up-mode**.
Proof.
There are infinitely many step positions but only

$$|Q||\Gamma|$$

possible modes.
Choose two step positions

$$t_1 < t_2$$

with the same mode

$$[q, X].$$

Write the earlier stack as

$$\eta X.$$

Because t_1 is a suffix minimum, the computation between t_1 and t_2 never goes below height

$$|\eta X|.$$

Therefore the lower prefix η is never exposed.
The later stack has form

$$\eta\rho X.$$

If

$$\rho = \varepsilon,$$

the two positions would have the same full configuration, contradicting Lemma 10.3.
Hence

$$\rho \in \Gamma^+.$$

Let c be the number of input 1's consumed between the two steps.
Because the lower context is never inspected, the same deterministic transition sequence works for every replacement lower context.
Thus the pump is context-independent.
Finally,

$$c \neq 0.$$

If it consumed no input, it would be a height-increasing ε-macro from the same mode back to itself. Determinism would repeat it forever and prevent the next live symbol from being consumed.
Therefore

$$c \geq 1.$$

□

The pump iterates:

$$\boxed{(q, \eta X) \xRightarrow{1^{jc}} (q, \eta\rho^j X) \qquad (j \geq 0).}$$

This is the technical point at which the lower bound becomes genuinely ε-robust.
The proof did not forbid internal normalization.
It found a recurrent growth mechanism inside it.

10.12 The remaining proof obligation

Every slope ray now has an up-mode.
To obtain the product lower bound, we need one final structural fact:

$$\boxed{\text{one up-mode cannot serve two different completion slopes.}}$$

If that holds, then the $\ell = mk + 1$ slopes force ℓ different modes.
The proof requires one more determinism lemma about unary completion.
That is where the next chapter begins.

Chapter 11

One Mode, One Slope

The upper witness exposes

$$\ell = mk + 1$$

different affine completion slopes.

Chapter 10 showed that every arbitrary general DPDA recognizing the witness must, along each live slope ray, develop a context-independent height-increasing up-mode.

We now prove the separation theorem:

$$\boxed{\text{one up-mode} \Longrightarrow \text{at most one completion slope.}}$$

Once this is established, the product bound and the hierarchy follow immediately.

11.1 Unary completion uniqueness

The witness slice uses all-zero continuations.

Determinism makes those continuations rigid.

Lemma 11.1 — Unary completion uniqueness.

Fix any raw control-store configuration

$$(p, \gamma)$$

of a deterministic pushdown automaton in the present model.

There is at most one integer

$$r \geq 0$$

such that input

$$0^r$$

leads from (p, γ) to acceptance.

Proof.

Suppose

$$r < r'$$

and both

$$0^r$$

and

$$0^{r'}$$

were accepted.

The two deterministic computations are identical until the shorter input has consumed all r zeros, including every forced ε-move enabled independently of the presence of further input.

The accepting run on 0^r eventually reaches:

- empty unread input;
- a final state;
- empty stack.

The run on $0^{r'}$ reaches the same empty-stack configuration while

$$r' - r > 0$$

zeros remain unread.

No transition is available from the empty stack.

Thus the longer word cannot be accepted.

Contradiction. □

This is a very small lemma, but it is exactly what converts a recurrent physical mode into a constraint on completion length.

11.2 Slope separation

Lemma 11.2 — Slope separation.

Let

$$s \neq t.$$

The slope rays

$$p_s 1^\omega \qquad \text{and} \qquad p_t 1^\omega$$

cannot have the same up-mode in any DPDA recognizing

$$B_{m,k}.$$

Proof.

Assume both rays share an up-mode

$$[q, X].$$

Take the context-independent pump

$$(q, \eta X) \overset{1^c}{\Longrightarrow} (q, \eta\rho X),$$

with

$$c \geq 1, \qquad \rho \in \Gamma^+.$$

Suppose the mode occurs on the s-ray after n_s accumulation ones with lower context γ_s, and on the t-ray after n_t accumulation ones with lower context γ_t.

After j pump iterations the two rays reach:

$$(q, \gamma_s \rho^j X)$$

and

$$(q, \gamma_t \rho^j X)$$

after respectively

$$n_s + jc$$

and

$$n_t + jc$$

accumulation ones.
The exact witness slice provides accepted zero completions of lengths

$$N_s(n_s + jc)$$

and

$$N_t(n_t + jc).$$

Now run those two zero computations in parallel.
As long as the common upper segment

$$\rho^j X$$

has not been completely removed, the two computations have:

- the same state;
- the same visible active stack suffix;
- the same deterministic local behavior.

The lower contexts γ_s and γ_t remain hidden.
Thus the common upper segment is removed after the same number

$$e_j$$

of consumed zeros.
Immediately after the final active symbol of that segment disappears, the two raw configurations have the form

$$(p_j, \gamma_s) \qquad \text{and} \qquad (p_j, \gamma_t)$$

for the same state p_j.
Because Q is finite, there is an infinite set of indices J on which

$$p_j = p$$

is constant.
For $j \in J$, the remaining accepted zero lengths are

$$R_s(j) = N_s(n_s + jc) - e_j$$

from the fixed raw configuration

$$(p, \gamma_s),$$

and

$$R_t(j) = N_t(n_t + jc) - e_j$$

from

$$(p, \gamma_t).$$

Unary completion uniqueness forces each of these two quantities to be constant on J.
Therefore

$$N_s(n_s + jc) - N_t(n_t + jc)$$

must be constant on J.
But

$$N_s(n) = \ell - s + (s+1)n,$$

so

$$\begin{aligned}&N_s(n_s + jc) - N_t(n_t + jc)\\&= (s-t)cj + O(1).\end{aligned}$$

Because

$$s \neq t$$

and

$$c \geq 1,$$

this expression is unbounded along the infinite set J.
Contradiction. □
This lemma is the structural center of the hierarchy proof.
It says that a recurrent state/top mode has an externally measurable asymptotic identity:

$$\boxed{\text{its continuation slope.}}$$

Two different slopes cannot be hidden in the same up-mode.

11.3 The product lower bound

There are

$$\ell = mk + 1$$

slope rays.
Every slope ray has an up-mode.
Distinct slopes require distinct up-modes.
A DPDA with state set Q and stack alphabet Γ has at most

$$|Q||\Gamma|$$

nonempty modes.
Everything is now ready.

Theorem 11.3 — Product lower bound.
Let

$$m \geq 2, \qquad k \geq 1.$$

If a general deterministic pushdown automaton M, with arbitrary deterministic ε-moves allowed, recognizes

$$B_{m,k},$$

then

$$\boxed{|Q_M||\Gamma_M| \geq mk + 1.}$$

Proof.
There are

$$mk + 1$$

slope rays.
By the step-pump lemma, each has an up-mode.
By slope separation, no two slope rays share an up-mode.
Therefore M has at least

$$mk + 1$$

distinct nonempty state/top-symbol modes.
But there are at most

$$|Q_M||\Gamma_M|$$

such modes.
Hence

$$|Q_M||\Gamma_M| \geq mk + 1.$$

□
This is stronger than the adjacent hierarchy statement.
It excludes an entire hyperbolic resource region:

$$\boxed{|Q||\Gamma| \leq mk.}$$

11.4 The hierarchy theorem

Corollary 11.4 — Binary adjacent stack-alphabet hierarchy.
For every

$$m \geq 2, \qquad k \geq 1,$$

$$\boxed{\mathcal{D}_{m,k} \subsetneq \mathcal{D}_{m,k+1}.}$$

The strictness is witnessed over the binary alphabet

$$\{0,1\}.$$

Proof.
The realtime witness $U_{m,k}$ uses exactly:

$$m$$

states and

$$k+1$$

stack symbols.
Therefore

$$B_{m,k} \in \mathcal{D}_{m,k+1}.$$

If

$$B_{m,k} \in \mathcal{D}_{m,k},$$

some recognizing DPDA would satisfy

$$|Q| \leq m, \qquad |\Gamma| \leq k.$$

Hence

$$|Q||\Gamma| \leq mk,$$

contradicting Theorem 11.3.
The non-strict inclusion follows by allowing an unused extra stack symbol. □
Thus deterministic ε-moves do not collapse the fixed-state stack-alphabet hierarchy in the stated general model for

$$m \geq 2.$$

11.5 What exactly has been resolved

The bibliographic claim must remain narrow.
The theorem closes the fixed-state stack-alphabet hierarchy case not covered by the earlier realtime result, under:

- the general deterministic model of this Part;
- the single-initial-symbol convention;
- fixed state budget
$$m \geq 2;$$
- binary witness alphabet.

The stateless case is not silently folded into the same theorem because its earlier formulation uses a different initial-store convention.
Thus the monograph should say:

$$\boxed{\text{the general-model fixed-state hierarchy is resolved here for } m \geq 2}$$

and separately acknowledge the known stateless result.

11.6 Why this is not a realtime lower bound in disguise

The upper witness is realtime.

The lower-bound opponent is not.

The arbitrary competitor may use deterministic ε-moves throughout.

The proof survives because it deliberately reasons at raw positions.

The chain is:

$$\boxed{\text{exact continuation slice}}$$

$$\Downarrow$$

$$\boxed{\text{no raw configuration repetition}}$$

$$\Downarrow$$

$$\boxed{\text{raw-height escape}}$$

$$\Downarrow$$

$$\boxed{\text{suffix-minimum step pump}}$$

$$\Downarrow$$

$$\boxed{\text{slope separation}}$$

$$\Downarrow$$

$$\boxed{|Q||\Gamma| \geq mk + 1.}$$

The first box is a property of the explicit binary witness.

Every later box concerns an arbitrary equivalent DPDA.

That is why the theorem is representation-independent in a way the earlier residual-frontier results were not.

11.7 Why the product, rather than states or symbols separately, appears

The proof never forces a slope into a unique state alone.

It forces it into a unique **mode**:

$$[q, X].$$

Therefore the natural capacity is:

$$\boxed{|Q||\Gamma|.}$$

This is exactly the resource unit the proof can count.

A machine may trade states against stack symbols.

The product theorem allows that trade.

What it forbids is packing more than one distinct completion slope into one recurrent up-mode.

Thus the proof yields a quantitative state–stack tradeoff rather than only a yes/no alphabet separation.

11.8 Scope and excluded claims

The theorem is deliberately scoped and does not assert the following stronger statements.

Not a full Pareto frontier

The bound

$$|Q||\Gamma| \geq mk + 1$$

does not prove that every point above the hyperbola is attainable.

It excludes one region.

It does not determine the complete state/stack Pareto frontier of the witness.

Not a bounded-push hierarchy theorem

The upper witness installs a fixed table word of length

$$\ell = mk + 1$$

in one header transition.

The general PDA model permits replacement by an arbitrary fixed stack word.

If one imposed a push-length bound, compiling the header would change the resource accounting.

No bounded-push hierarchy claim is made in this theorem; that model is outside the present scope.

Not an unrestricted lower bound for $L_{p,q}$

The witness was engineered so that its continuation slopes are maximally separable.

The theorem does not automatically transfer to the original rational counting languages.

The representation-independent state question for the natural languages $L_{p,q}$ is resolved later by protected phase pumping, which yields the exact unrestricted state bound.

Not a claim that mode arguments are themselves new

State/top modes and state–stack-symbol tradeoffs have a pre-existing literature.

The contribution here is the particular binary witness, the ε-robust slope argument, the product theorem, and the resulting resolution in the stated model.

11.9 The methodological meaning for the monograph

This Part began with a problem created by our own DPDA analysis.

The original machine and its successors showed that the same semantic quantity can be redistributed between:

- control state;
- stack symbol;
- stack height;
- normalization.

That made representation-scoped lower bounds suspicious.
The hierarchy proof answers that suspicion by abandoning the representation.
It asks only what continuations the language forces.
This produces a new lower-bound pattern:

$$\boxed{\begin{array}{c}\text{externally visible continuation rates} \to \text{recurrent physical modes}\\ \to \text{representation-independent resource count.}\end{array}}$$

This method is conceptually distinct from the raw residual frontier.
Both belong in the monograph because they answer different questions.

11.10 Transition to higher-dimensional completion geometry

We have now completed the first major resource arc.
Starting from the 2023 2:1 machine, we obtained:

$$\begin{gathered}\text{completion semantics}\\ \downarrow\\ \text{geodesic geometry}\\ \downarrow\\ \text{better coordinates}\\ \downarrow\\ \text{exact residual-faithful frontier}\\ \downarrow\\ \boxed{\text{representation-independent hierarchy theorem.}}\end{gathered}$$

The next Part changes direction.
Instead of asking how many resources are needed to realize a one-dimensional residual geometry, we ask what happens when the counting constraint itself becomes higher-dimensional.
The binary rectangle will become an optimal Parikh fiber.
The absence of equal-length trade directions will become a kernel-rank condition.
And the first nontrivial equal-length trade will create topology.
We now leave resource complexity and return to completion geometry.

Part V

Completion Geometry Beyond the Binary Line

Research provenance. This Part specializes established affine-semigroup, cubical, and vector-partition constructions to minimum-completion prefixes, optimal fibers, and the equal-length residual quotient. The surrounding theories include affine-semigroup factorization, Markov/Graver moves, squarefree-divisor complexes, cubical state/rewrite complexes, and vector-partition functions [23, 27, 17, 6, 1, 51, 28, 52]; the novelty claims concern the exact completion-theoretic results proved in the chapters below rather than those background constructions in isolation.

Chapter 12

When Minimum Demand Becomes a Fiber

The first half of the monograph was built around one scalar residual:

$$\rho_{p,q}(w) = qN_1(w) - pN_0(w).$$

That scalar residual had a unique minimum completion Parikh vector

$$\mu(r) = (x_r, y_r).$$

As a consequence, every shortest-completion interval was a rectangle.

The natural higher-dimensional question is:

> What survives if the counting constraint is no longer one scalar equation?

The answer preserves the completion viewpoint but changes the geometry fundamentally.

The minimum demand need no longer be one vector.

It can become a finite **fiber of equally short Parikh vectors**.

That is where higher-dimensional completion geometry begins.

12.1 Matrix counting languages

Let

$$\Sigma = \{a_1, \dots, a_d\}$$

and let

$$\Psi : \Sigma^* \to \mathbb{N}^d$$

be the Parikh map.

Fix an integer matrix

$$A \in \mathbb{Z}^{k \times d}.$$

Define the vector residual

$$\boxed{\rho(w) = A\Psi(w)}$$

and the matrix counting language

$$\boxed{L_A = \{w : \rho(w) = 0\}.}$$

Residuals lie in the lattice

$$A\mathbb{Z}^d.$$

The binary rational counting family is a rank-one special case.
For

$$L_{p,q},$$

take

$$A = \begin{bmatrix} -p & q \end{bmatrix}.$$

Then

$$A\Psi(w) = qN_1(w) - pN_0(w).$$

Thus the higher-dimensional theory genuinely contains the earlier one.

12.2 Completion totality

A matrix counting language need not allow every residual to be completed by appending letters.

We therefore isolate a convenient hypothesis.

Definition 12.1 — Completion totality.

We call A **completion-total** if there exists

$$h \in \mathbb{N}^d_{>0}$$

such that

$$\boxed{Ah = 0.}$$

The vector h is a strictly positive balanced Parikh vector.
It plays the role of the primitive balanced packet

$$(q, p)$$

from the binary theory, except that it need not generate the entire integer kernel.

Lemma 12.2 — Every lattice residual is completable.

Assume completion totality.
For every

$$r \in A\mathbb{Z}^d,$$

there exists

$$x \in \mathbb{N}^d$$

such that

$$\boxed{Ax = -r.}$$

Proof.

Write

$$r = Az$$

with

$$z \in \mathbb{Z}^d.$$

Because every coordinate of h is positive, for sufficiently large t,

$$th - z \in \mathbb{N}^d.$$

Since

$$Ah = 0,$$

$$A(th - z) = -Az = -r.$$

Thus $th - z$ is a nonnegative completion vector. □

This hypothesis is deliberately strong enough to make completion semantics total on the entire residual lattice.

It lets us generalize the quotient theory cleanly.

12.3 Residuals still classify right quotients

Proposition 12.3 — Matrix residual quotient classification.

Under completion totality, for all prefixes

$$u, v \in \Sigma^*,$$

$$\boxed{u^{-1}L_A = v^{-1}L_A \iff \rho(u) = \rho(v).}$$

Proof.

A continuation z completes u exactly when

$$A\Psi(z) = -\rho(u).$$

Therefore equal residuals give equal quotients.

Conversely, suppose

$$\rho(u) \neq \rho(v).$$

By Lemma 12.2, choose

$$x \in \mathbb{N}^d$$

with

$$Ax = -\rho(u).$$

Choose a word z with

$$\Psi(z) = x.$$

Then

$$uz \in L_A.$$

But if

$$vz \in L_A,$$

we would have

$$Ax = -\rho(v)$$

as well, forcing

$$\rho(u) = \rho(v).$$

Contradiction. □

The semantic lesson from Chapter 4 survives unchanged:

$$\boxed{\text{the residual is an exact coordinate for future right behavior.}}$$

What changes is the geometry of shortest return to residual zero.

12.4 Minimum completion length

For residual

$$r \in A\mathbb{Z}^d,$$

define

$$\boxed{\ell_A(r) = \min\{\mathbf{1}^\top x : x \in \mathbb{N}^d,\ Ax = -r\}.}$$

This is the shortest number of appended letters required to cancel r.

In the binary case,

$$\ell_A(r) = \Delta_{p,q}(r).$$

So far the generalization looks formal.

The decisive change appears when we ask which Parikh vectors attain this minimum.

12.5 The optimal Parikh fiber

Definition 12.4 — Optimal Parikh fiber.

Define

$$\boxed{O_A(r) = \{x \in \mathbb{N}^d : Ax = -r,\ \mathbf{1}^\top x = \ell_A(r)\}.}$$

In the binary $p{:}q$ theory,

$$O_A(r)$$

was always a singleton:

$$O_A(r) = \{\mu(r)\}.$$

In higher dimensions, several distinct nonnegative Parikh vectors can complete the same residual at the same minimum length.

Therefore the semantic object

$$\mu(r)$$

must be replaced by the entire set

$$\boxed{O_A(r).}$$

This is the first major qualitative change.

12.6 Equal-length trades

Introduce the augmented matrix

$$\boxed{C = \begin{bmatrix} A \\ \mathbf{1}^\top \end{bmatrix}.}$$

Define the integer equal-length kernel

$$\boxed{K_A = \ker_{\mathbb{Z}} C = \left\{ g \in \mathbb{Z}^d : Ag = 0,\ \mathbf{1}^\top g = 0 \right\}.}$$

A vector

$$g \in K_A$$

changes one completion Parikh vector into another while preserving:

1. residual contribution;
2. total completion length.

Thus K_A is the lattice of **equal-length trades**.

This is the higher-dimensional replacement for the binary kernel direction.

But there is a crucial difference.

For the binary minimum demand, subtracting one nonzero kernel packet always shortened the completion, so two distinct minimum Parikh vectors could not coexist.

In higher dimensions, a nonzero vector can have positive and negative coordinates while still satisfying

$$\mathbf{1}^\top g = 0.$$

Such a trade changes the distribution of letters without changing total length.

That is how an optimal fiber acquires multiple points.

12.7 Affine-semigroup viewpoint

The columns

$$C_i = \begin{bmatrix} A_i \\ 1 \end{bmatrix}$$

generate an affine semigroup.

For residual r, define the augmented optimum element

$$\boxed{b_r = (-r, \ell_A(r)).}$$

Then

$$O_A(r)$$

is exactly the factorization fiber of b_r into the generators C_i.

Every such factorization has the same last coordinate:

$$\ell_A(r).$$

This places the theory next to established affine-semigroup and factorization geometry.

The monograph uses that literature as background.

The new object of interest remains specifically the geometry of **minimum-completion schedules**.

12.8 Continuous optimum geometry is not the same as integer multiplicity

Define the continuous optimum polytope

$$P_r^A = \{x \in \mathbb{R}^d_{\geq 0} : Ax = -r,\ \mathbf{1}^\top x = \ell_A(r)\}.$$

Let

$$S(r) = \{i : \exists x \in P_r^A \text{ with } x_i > 0\}$$

be its active support.

Proposition 12.5 — Dimension of the continuous optimum polytope.

If

$$P_r^A \neq \varnothing,$$

then

$$\boxed{\dim P_r^A = |S(r)| - \operatorname{rank}_{\mathbb{R}} \begin{bmatrix} A_{S(r)} \\ \mathbf{1}^\top \end{bmatrix}.}$$

Proof.

For every active coordinate choose a feasible point positive in that coordinate.

The average of these points is positive in every active coordinate and lies in the relative interior of the nonnegative orthant restricted to $S(r)$.

Thus no additional coordinate inequality is active in the relative interior.

The dimension is therefore the number of active variables minus the rank of the affine equality system on those variables. □

The proposition is useful mainly because it prevents a common confusion.

A positive-dimensional real optimum polytope does not automatically imply many integer optimum points.

Conversely, discrete equal-length trade structure can be significant even when a simple continuous dimension count does not capture its combinatorics.

The integer fiber

$$O_A(r)$$

is the primary shortest-completion object.

12.9 The shortest-prefix down-set

In the binary rectangle, a shortest-completion prefix was described by how many required zeros and ones had already been consumed.

For a higher-dimensional optimum vector

$$x \in O_A(r),$$

every coordinatewise subvector

$$0 \le u \le x$$

is the Parikh vector of some prefix of an ordering of that optimum completion.

Define

$$B_x = \{u \in \mathbb{N}^d : 0 \le u \le x\}.$$

Now take the union over all optimum vectors.

Definition 12.6 — Shortest-prefix down-set.

$$\boxed{\mathcal{C}_A(r) = \downarrow O_A(r) = \bigcup_{x \in O_A(r)} B_x.}$$

This is the higher-dimensional replacement for one rectangle.

If the optimum is unique,

$$O_A(r) = \{x\},$$

then

$$\mathcal{C}_A(r) = B_x,$$

a literal coordinate box.

If the optimum fiber contains several vectors, the shortest-prefix region becomes a union of overlapping boxes.

The next theorem says this down-set is not merely a convenient overapproximation.

It is exact.

12.10 Exact geodesic-prefix criterion

Theorem 12.7 — Exact geodesic-prefix criterion.

For every

$$u \in \mathbb{N}^d,$$

$$\boxed{u \in \mathcal{C}_A(r) \iff \ell_A(r + Au) = \ell_A(r) - |u|_1.}$$

Equivalently,

$$\mathcal{C}_A(r)$$

is exactly the set of Parikh vectors that occur as prefixes of shortest completions of residual r.
Proof.
Suppose

$$u \in \mathcal{C}_A(r).$$

Then

$$u \le x$$

for some

$$x \in O_A(r).$$

The vector

$$x - u$$

completes residual

$$r + Au$$

and has length

$$\ell_A(r) - |u|_1.$$

If a shorter completion existed, appending it after u would produce a completion of r shorter than $\ell_A(r)$.
Therefore

$$\ell_A(r + Au) = \ell_A(r) - |u|_1.$$

Conversely, assume the displayed equality.
Let y be an optimal completion of

$$r + Au.$$

Then

$$A(u + y) = -r,$$

and

$$|u + y|_1 = |u|_1 + \ell_A(r + Au) = \ell_A(r).$$

Thus

$$u + y \in O_A(r),$$

so

$$u \le u + y.$$

Hence

$$u \in \mathcal{C}_A(r).$$

□

This theorem is the higher-dimensional form of minimum-demand descent.

The difference is that there may be several optimal endpoints x, so the geodesic-prefix set is a union of coordinate boxes rather than one box.

12.11 First actions and compatible action sets

Let e_i denote the i-th unit vector.

Corollary 12.8 — First actions.

Letter type i can begin a shortest completion exactly when

$$\boxed{\ell_A(r + Ae_i) = \ell_A(r) - 1.}$$

More generally, for

$$J \subseteq [d],$$

there exists an optimal completion using every letter type in J at least once if and only if

$$\boxed{\ell_A(r + A\mathbf{1}_J) = \ell_A(r) - |J|.}$$

Thus shortest-action compatibility can be tested intrinsically from the distance function.

This is the higher-dimensional analogue of the binary rule:

$$x_r > 0 \iff 0 \text{ is geodesic},$$

$$y_r > 0 \iff 1 \text{ is geodesic}.$$

12.12 The conceptual transition

The completion object has now changed from

$$\boxed{\text{one minimum vector}}$$

to

$$\boxed{\text{a finite optimum fiber.}}$$

Accordingly, the shortest-prefix geometry has changed from

$$\boxed{\text{one rectangle}}$$

to

$$\boxed{\text{a union of boxes.}}$$

But the residual quotient still identifies prefixes that reach the same semantic state.

If two different shortest-prefix Parikh vectors reach the same residual at the same geodesic depth, they must differ by an equal-length trade in

$$K_A.$$

That observation leads to the next object:

$$\boxed{|\mathcal{C}_A(r)|/K_A.}$$

The next chapter will show that this quotient is exactly the filled shortest-path interval.

Chapter 13

The Filled Shortest-Prefix Quotient

Chapter 12 produced two objects:

$$\mathcal{C}_A(r) = \downarrow O_A(r),$$

the exact Parikh down-set of shortest prefixes, and

$$K_A = \ker_{\mathbb{Z}} \begin{bmatrix} A \\ \mathbf{1}^\top \end{bmatrix},$$

the lattice of equal-length residual-preserving trades.

We now combine them.

The result is the higher-dimensional completion object that replaces the binary shortest rectangle.

13.1 Why vertices alone are not enough

A shortest prefix is not only a point.

If several distinct next letters can be taken in any order while remaining geodesic, the geometry contains a commuting square, cube, or higher-dimensional cell.

The binary rectangle already had this property.

A pair of actions 0 and 1 produced an elementary commuting square.

In d letters, a compatible set

$$J \subseteq [d]$$

should produce a $|J|$-dimensional cube.

Thus we should fill the shortest-prefix down-set cubically.

13.2 Cubical realization of a down-set

Let

$$D \subseteq \mathbb{N}^d$$

be a finite coordinatewise down-set.

Define

$$|D|$$

to be the cubical complex whose vertices are the elements of D.
For

$$J \subseteq [d],$$

an elementary J-oriented cube based at u is present exactly when

$$\boxed{u + \mathbf{1}_J \in D.}$$

Because D is a down-set, all intermediate vertices of that cube are automatically present.
Applied to

$$D = \mathcal{C}_A(r),$$

every cube records a set of distinct letter actions that can be executed in any order while remaining on shortest paths.
This is the filled schedule geometry before residual identifications.

13.3 Equal-length identifications

Two shortest-prefix Parikh vectors can be different while representing the same reached residual.
Let

$$u, v \in \mathcal{C}_A(r).$$

They reach the same residual exactly when

$$A(u - v) = 0.$$

But because both are shortest prefixes reaching the same residual, they also have the same geodesic depth.
Thus

$$\mathbf{1}^\top (u - v) = 0.$$

Therefore

$$u - v \in K_A.$$

This gives the correct equivalence relation.
Two equally oriented cells are identified when their base vertices differ by an element of

$$K_A.$$

Definition 13.1 — Filled shortest-geodesic quotient.
Define

$$\boxed{\mathfrak{I}_A(r) = |\mathcal{C}_A(r)|/K_A.}$$

This quotient keeps:

- every shortest prefix;
- every compatible commuting cell;

but identifies prefixes that reach the same residual at the same shortest depth.

13.4 The quotient theorem

Theorem 13.2 — Shortest-prefix quotient theorem.

The 0-skeleton of

$$\mathfrak{I}_A(r)$$

is label-preserving isomorphic to the complete directed shortest-path interval from residual r to residual 0.

Under this identification, elementary cubes are exactly compatible commuting sets of geodesic one-letter actions.

Proof.

Define

$$\pi_r : \mathcal{C}_A(r) \to A\mathbb{Z}^d$$

by

$$\boxed{\pi_r(u) = r + Au.}$$

By Theorem 12.7, every $u \in \mathcal{C}_A(r)$ is a geodesic prefix, so every image vertex lies on a shortest path from r to 0.

Conversely, every shortest path prefix has some Parikh vector u, and Theorem 12.7 places that vector in $\mathcal{C}_A(r)$.

Now suppose

$$\pi_r(u) = \pi_r(v).$$

Then

$$A(u - v) = 0.$$

Because u and v reach the same residual and both are geodesic,

$$\ell_A(r) - |u|_1 = \ell_A(\pi_r(u)) = \ell_A(\pi_r(v)) = \ell_A(r) - |v|_1.$$

Hence

$$|u|_1 = |v|_1,$$

so

$$\mathbf{1}^\top(u - v) = 0.$$

Therefore

$$u - v \in K_A.$$

Conversely, if

$$u - v \in K_A,$$

then

$$A(u - v) = 0,$$

so the two vertices have the same residual label.

Thus the vertex fibers of π_r are exactly the K_A-classes.

Finally, a J-cube based at u exists exactly when

$$u + \mathbf{1}_J \in \mathcal{C}_A(r).$$

By the geodesic-prefix criterion, every subcollection of the actions in J remains shortest, and coordinate addition commutes.

Passing to the equal-length kernel quotient therefore gives exactly the filled commuting shortest-path structure. □

This theorem is the central bridge from integer completion arithmetic to topology.

13.5 Recovery of the binary rectangle

For the binary scalar system

$$A = \begin{bmatrix} -p & q \end{bmatrix},$$

the augmented matrix is

$$\begin{bmatrix} -p & q \\ 1 & 1 \end{bmatrix}.$$

Its determinant is

$$-(p+q) \neq 0.$$

Therefore

$$\boxed{K_A = \{0\}.}$$

There are no nontrivial equal-length trades.

The optimum fiber is a singleton:

$$O_A(r) = \{\mu(r)\}.$$

Hence

$$\mathcal{C}_A(r) = [0, x_r] \times [0, y_r]$$

and no quotient identifications occur.

Thus

$$\boxed{\mathfrak{I}_A(r) = [0, x_r] \times [0, y_r].}$$

The binary rectangle is therefore the **rank-zero member** of the general theory.

This is an important reinterpretation.

The rectangle did not disappear in higher dimensions.

It became the simplest possible fiber-and-quotient case.

13.6 When is the quotient literally a box?

Theorem 13.3 — Literal-box criterion.

$$\boxed{|O_A(r)| = 1 \iff \mathfrak{I}_A(r) \text{ is a literal coordinate box.}}$$

Here "literal coordinate box" means that the quotient contains no nontrivial kernel identifications among the vertices and cells of the prequotient box.
Proof.
Suppose

$$O_A(r) = \{x\}.$$

Then

$$\mathcal{C}_A(r) = B_x.$$

Assume distinct

$$u, v \leq x$$

were identified.
Then

$$u - v \in K_A.$$

Consider

$$x - u + v.$$

It is nonnegative.
Because

$$A(u - v) = 0,$$

it has the same residual contribution as x.
Because

$$\mathbf{1}^\top (u - v) = 0,$$

it has the same length as x.
It is different from x.
Therefore it would be a second element of

$$O_A(r),$$

contradicting uniqueness.
Hence no nontrivial kernel collision occurs inside the box.
Conversely, suppose

$$x \neq y$$

both belong to

$$O_A(r).$$

Then x and y are distinct maximal vertices of $\mathcal{C}_A(r)$, both reach residual zero, and

$$x - y \in K_A.$$

Thus the quotient identifies distinct shortest schedules.

It cannot be the original literal coordinate box with all coordinate vertices distinct.

□

This theorem identifies the first source of global complexity:

$$\boxed{\text{nonunique optimum} \iff \text{nontrivial equal-length identification.}}$$

13.7 Unique optimum versus unique shortest word

The phrase "unique optimum" refers to the Parikh vector.

It does not mean there is only one shortest word.

Even when

$$O_A(r) = \{x\},$$

the letters represented by x can be ordered in many ways.

The literal box already records those schedules.

What uniqueness eliminates is a stronger ambiguity:

$$\boxed{\text{two different minimum Parikh endpoints.}}$$

Thus there are two levels of shortest-path multiplicity.

Ordering multiplicity

One Parikh vector, many permutations.

This produces a box.

Factorization multiplicity

Several distinct minimum Parikh vectors.

This creates kernel identifications among different boxes.

The second phenomenon is new beyond the binary line.

13.8 Local compatibility and global identification are different

The cubical complex

$$|\mathcal{C}_A(r)|$$

records which letter actions can coexist in one optimum and commute as schedule choices.

The quotient

$$\mathfrak{I}_A(r)$$

adds a different phenomenon:

$$\boxed{\text{globally different shortest prefixes can reach the same residual.}}$$

A later chapter will show that these two notions have different topology.
In rank one:

- every connected component of the local action-compatibility complex is contractible;
- the full global quotient can nevertheless be a circle.

Thus local coexistence and global schedule identification must not be conflated.
We are now ready to study that distinction.

13.9 The rank of equal-length trade

The integer lattice

$$K_A$$

measures the number of independent directions in which one optimal Parikh vector can trade letters for others without changing:

- residual;
- completion length.

Three regimes will become especially important.

Rank zero

$$K_A = \{0\}.$$

No equal-length trade exists.
Every optimum is unique.
The shortest interval is a literal box.
This is the binary $p{:}q$ regime.

Rank one

One primitive equal-length trade direction exists.
The optimum fiber becomes an arithmetic interval.
The global quotient may acquire its first cycle.

Rank at least two

Several independent trade directions exist.
Even the local action complex can develop holes.
The next chapters will make this hierarchy exact.

Chapter 14

Local and Global Topology

Chapter 13 produced the filled shortest-prefix quotient

$$\mathfrak{I}_A(r) = |\mathcal{C}_A(r)|/K_A.$$

Two different kinds of structure are present inside this object.

The first is **local**:

> Which distinct letter types can coexist inside one shortest completion?

The second is **global**:

> After all shortest prefixes are filled cubically and equal-length residual-preserving schedules are identified, what is the topology of the whole shortest interval?

These questions need not have the same answer.

The first genuinely new phenomenon occurs already when the equal-length kernel has rank one.

Locally, nothing topologically complicated happens.

Globally, a circle can appear.

That separation is one of the principal structural results of the higher-dimensional branch.

14.1 The local action complex

Recall the optimal Parikh fiber

$$O_A(r) = \{x \in \mathbb{N}^d : Ax = -r,\ |x|_1 = \ell_A(r)\}.$$

Define the simplicial complex

$$\boxed{\mathcal{H}_A(r) = \{J \subseteq [d] : \exists x \in O_A(r) \text{ with } x_j > 0 \text{ for every } j \in J\}.}$$

A face

$$J \in \mathcal{H}_A(r)$$

means that some shortest completion contains at least one copy of every action in J.

Equivalently, by Chapter 12,

$$J \in \mathcal{H}_A(r) \iff \ell_A(r + A\mathbf{1}_J) = \ell_A(r) - |J|.$$

Thus $\mathcal{H}_A(r)$ is the local compatibility complex of geodesic letter types.
It deliberately forgets:

- how many times an action occurs;
- the order in which actions occur;
- which shortest prefixes are globally identified.

It records only coexistence.

14.2 Squarefree-divisor interpretation

Let

$$C_i = \begin{bmatrix} A_i \\ 1 \end{bmatrix}$$

and let

$$S_C = \langle C_1, \dots, C_d \rangle$$

be the affine semigroup generated by the augmented columns.
For residual r, recall

$$b_r = (-r, \ell_A(r)).$$

Proposition 14.1 — Squarefree divisor identification.

$$\boxed{\mathcal{H}_A(r) = \left\{ J \subseteq [d] : b_r - \sum_{j \in J} C_j \in S_C \right\}.}$$

Proof.
The semigroup condition means that there exists

$$y \in \mathbb{N}^d$$

with

$$Cy = b_r - \sum_{j \in J} C_j.$$

Then

$$x = y + \mathbf{1}_J$$

satisfies

$$Cx = b_r.$$

Hence

$$x \in O_A(r)$$

and every coordinate indexed by J is positive.
The converse is obtained by subtracting $\mathbf{1}_J$ from any optimum vector supporting J.
□

Squarefree divisor complexes are established objects in affine-semigroup topology [6].
Their role here is specific:

they are exactly the local shortest-action complexes of the completion problem.

The topology of $\mathcal{H}_A(r)$ therefore measures local compatibility among geodesic actions.

14.3 Rank zero: one simplex

If

$$K_A = \{0\},$$

the optimum fiber is a singleton

$$O_A(r) = \{x\}.$$

Then

$$\mathcal{H}_A(r)$$

is simply the simplex on

$$\mathrm{supp}(x).$$

Thus rank zero has no nontrivial local topology.
This includes every binary $p{:}q$ completion problem.
The local action set may have one or two symbols, but it never develops a hole.

14.4 Rank one: the optimum fiber is an arithmetic interval

Assume now

$$K_A = \mathbb{Z}g$$

for a primitive nonzero vector

$$g \in \mathbb{Z}^d.$$

Because the optimum fiber is finite, it has the form

$$\boxed{O_A(r) = \{x_t = x_0 + tg : 0 \le t \le N\}.}$$

For each letter type i, define

$$T_i = \{t \in [0, N] \cap \mathbb{Z} : x_t(i) > 0\}.$$

Since

$$x_t(i) = x_0(i) + tg_i$$

is affine in t, the set T_i is an integer interval, possibly empty.
A set of actions

$$J$$

is locally compatible exactly when

$$\bigcap_{j \in J} T_j \neq \varnothing.$$

Thus the local action complex is the nerve of a finite family of intervals.

Theorem 14.2 — Rank-one local action topology.
If

$$\operatorname{rank}_{\mathbb{Z}} K_A \leq 1,$$

then every connected component of

$$\mathcal{H}_A(r)$$

is contractible.
In particular,

$$\boxed{\widetilde{H}_j(\mathcal{H}_A(r); \mathbb{Z}) = 0 \qquad (j \geq 1).}$$

Proof.
The rank-zero case was just described.
In rank one, $\mathcal{H}_A(r)$ is the nerve of the interval family

$$\{T_i\}.$$

Every connected component of the union of finitely many intervals is itself an interval.
By the nerve theorem [36], each connected component of the nerve is homotopy equivalent to the corresponding connected component of the union.
Hence every connected component is contractible. □
This theorem says that one equal-length trade direction is not enough to create a local homology hole.
Yet it is enough to create a global cycle.
Before proving that, we show that rank two changes the local picture immediately.

14.5 Rank two already permits a local hole

Consider the scalar weight vector

$$\boxed{\omega = (-11, -2, 1, 10)}$$

and residual

$$r = 7.$$

The minimum completion length is 5, and the complete optimum fiber is

$$\boxed{(0, 4, 1, 0), \qquad (1, 0, 4, 0), \qquad (1, 3, 0, 1).}$$

The maximal local action faces are

$$\{2, 3\}, \qquad \{1, 3\}, \qquad \{1, 2, 4\}.$$

Therefore

$$f_0 = 4, \qquad f_1 = 5, \qquad f_2 = 1.$$

There is one filled triangle, but one independent graph cycle remains unfilled.
Hence

$$\boxed{H_1(\mathcal{H}_\omega(7);\mathbb{Z}) \cong \mathbb{Z}.}$$

The augmented equal-length kernel has rank two.
This example proves that the rank-one local theorem is sharp.
The first nontrivial local H_1 can appear as soon as two independent equal-length trades are available.

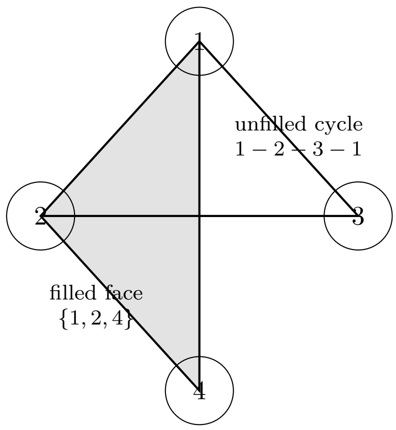


Figure 14.1: The local action complex for $\omega = (-11, -2, 1, 10)$ at residual 7. Its maximal faces are $\{2,3\}$, $\{1,3\}$, and the filled triangle $\{1,2,4\}$. The cycle $1-2-3-1$ remains unfilled, giving $H_1 \cong \mathbb{Z}$.

The figure should make the distinction visual:

- one triangular face is filled;
- another cycle is not.

This is not a global schedule-identification phenomenon.
The hole is already present in the local coexistence complex.

14.6 Rank-one global geometry

We now return to

$$K_A = \mathbb{Z}g.$$

Write the primitive generator as

$$g = g^+ - g^-,$$

where

$$g^+, g^- \in \mathbb{N}^d$$

have disjoint supports.
Let

$$O_A(r) = \{x_t = x_0 + tg : 0 \le t \le N\}.$$

Define the prequotient shortest-prefix complex

$$X = |\mathcal{C}_A(r)| = \left| \bigcup_{t=0}^{N} B_{x_t} \right|.$$

Because $\mathcal{C}_A(r)$ is a finite coordinate down-set,

$$\boxed{X \simeq *.}$$

So the topology of the final quotient cannot come from a hole already present in the prequotient region.

It must come from the kernel identification.

14.7 Overlap complexes

For

$$m \geq 1,$$

define

$$P_m = \{\sigma \subseteq X : \sigma + mg \subseteq X\},$$

where translation preserves cube orientation.

Lemma 14.3 — Exact m-step overlap.

For

$$1 \leq m \leq N,$$

translation by

$$-mg^-$$

identifies P_m with the anchored cubical down-set

$$\boxed{\bigcup_{t=0}^{N-m} B_{x_t - mg^-}.}$$

Hence P_m is nonempty and contractible.

For

$$m > N,$$

$$\boxed{P_m = \varnothing.}$$

Interpretation. The complete optimum fiber has parameter span

$$0, \ldots, N.$$

Two shortest-prefix cells can overlap after an mg translation exactly while that translation stays inside the span.

Thus the global quotient glues only finitely many neighboring translates.

The overlap itself remains topologically simple.

14.8 Periodic saturation

Define the periodic saturation

$$\boxed{\widetilde{X} = \bigcup_{k\in\mathbb{Z}} (X + kg).}$$

Let

$$X_k = X + kg.$$

Translation by g acts freely on $\widetilde{X}$.
Moreover,

$$\boxed{\widetilde{X}/\mathbb{Z} \cong \mathfrak{I}_A(r).}$$

The quotient by the integer translation action performs exactly the rank-one equal-length kernel identification.
To understand $\widetilde{X}$, we study the cover

$$\{X_k\}_{k\in\mathbb{Z}}.$$

Lemma 14.4 — Good-cover intersections.
Let

$$k_0 < \cdots < k_s.$$

Then

$$\boxed{X_{k_0} \cap \cdots \cap X_{k_s} \neq \varnothing \iff k_s - k_0 \leq N.}$$

Every nonempty finite intersection is contractible.

Proof idea. For any fixed cell, the set of translate indices k for which the cell lies in X_k is an integer interval.
Thus membership in the two extreme translates implies membership in all intermediate translates.
After translating by $-k_0 g$, the intersection reduces to the $(k_s - k_0)$-step overlap from Lemma 14.3.
That overlap is contractible exactly when the index gap is at most N. □
The nerve of this cover has vertices $\mathbb{Z}$.
A finite set of vertices spans a simplex exactly when its diameter is at most N.
For

$$N \geq 1,$$

this is the nerve of the interval cover

$$\{[k, k+N]\}_{k\in\mathbb{Z}}$$

of the real line.
Hence the nerve is contractible.

Theorem 14.5 — Contractible rank-one saturation.
If

$$N \geq 1,$$

then

$$\boxed{\widetilde{X} \simeq *.}$$

The periodic saturation is therefore a contractible universal-cover candidate.

14.9 The global circle

Theorem 14.6 — Rank-one global homotopy dichotomy.

If

$$\operatorname{rank}_{\mathbb{Z}} K_A = 1,$$

then

$$\boxed{\mathfrak{I}_A(r) \simeq \begin{cases} *, & |O_A(r)| = 1, \\ S^1, & |O_A(r)| > 1. \end{cases}}$$

Proof.

If the optimum is unique, Chapter 13 gives a literal coordinate box, hence a contractible complex.

Now suppose

$$|O_A(r)| > 1.$$

Then

$$N \geq 1.$$

By Theorem 14.5,

$$\widetilde{X}$$

is contractible.

The action of

$$\mathbb{Z}$$

by translation through g is free, properly discontinuous, and cellular.

Therefore

$$\widetilde{X} \to \widetilde{X}/\mathbb{Z} \cong \mathfrak{I}_A(r)$$

is a universal covering with deck group

$$\mathbb{Z}.$$

Hence

$$\pi_1(\mathfrak{I}_A(r)) \cong \mathbb{Z}$$

and all higher homotopy groups vanish.

Thus

$$\mathfrak{I}_A(r)$$

is a $K(\mathbb{Z}, 1)$, hence homotopy equivalent to

$$S^1.$$

□

This gives the first exact topological transition in the completion theory:

$$\boxed{\text{rank one} + \text{nonunique optimum} \Longrightarrow \text{global circle.}}$$

14.10 Local triviality, global nontriviality

Combining Theorems 14.2 and 14.6 gives the sharp distinction:

$$\boxed{\begin{array}{c}\text{rank-one local action complex:}\\ \text{every component contractible}\\[4pt] \text{rank-one global shortest-prefix quotient:}\\ \text{possibly } S^1.\end{array}}$$

The circle is not caused by a local incompatibility hole.

It is caused by globally identifying different shortest schedules through an equal-length trade.

This difference will matter later when the monograph returns to normalization schedules.

Local feasibility and global identification are fundamentally different layers of structure.

14.11 What the topology means operationally

A point of

$$\mathcal{C}_A(r)$$

records a shortest-prefix Parikh vector.

A path through the cubical complex records gradual execution of a shortest schedule.

Translation by g changes the Parikh bookkeeping without changing:

- residual;
- geodesic depth.

The quotient therefore says that two apparently different shortest schedule histories have reached the same semantic point.

When one primitive trade direction exists, those identifications wrap the contractible periodic saturation around one generator.

That is the geometric origin of the circle.

The result is not merely "some complex has homology."

It is a statement about the topology created by schedule equivalence.

The next chapter turns kernel rank into an exact alphabet-size hierarchy for scalar counting languages.

Chapter 15

Scalar Rank Hierarchy and Exact Three-Letter Geometry

The previous chapter expressed the new topology in terms of

$$\operatorname{rank}_{\mathbb{Z}} K_A.$$

For a general matrix A, this rank is an independent structural parameter.
For a scalar counting language, it has a remarkably simple form.
This produces a sharp hierarchy by alphabet size.

15.1 Scalar counting systems

Let

$$\omega = (\omega_1, \dots, \omega_d) \in \mathbb{Z}^d$$

have at least two distinct entries.
Treat

$$A = \omega$$

as a $1 \times d$ matrix.
The equal-length kernel is

$$K_\omega = \ker_{\mathbb{Z}} \begin{bmatrix} \omega \\ \mathbf{1}^\top \end{bmatrix}.$$

The two rows are linearly independent over $\mathbb{R}$.
Therefore:

$$\boxed{\operatorname{rank}_{\mathbb{Z}} K_\omega = d - 2.}$$

This single formula turns the topology of Chapter 14 into an alphabet-size threshold.

15.2 The $d = 2, 3, 4+$ hierarchy

Two letters

If

$$d = 2,$$

then

$$\operatorname{rank} K_\omega = 0.$$

Therefore:

- every optimum Parikh vector is unique;
- every shortest interval is a literal coordinate box;
- no equal-length schedule identification occurs.

This is exactly the binary theory of Chapters 2–5.

Three letters

If

$$d = 3,$$

then

$$\operatorname{rank} K_\omega = 1.$$

Therefore:

- the optimum fiber is an arithmetic interval;
- every connected component of the local action complex is contractible;
- the global quotient is either contractible or homotopy equivalent to

$$S^1.$$

Thus three letters are the first scalar alphabet size capable of creating a global shortest-schedule cycle.

Four or more letters

If

$$d \geq 4,$$

then

$$\operatorname{rank} K_\omega \geq 2.$$

Higher-rank equal-length trade geometry becomes possible.

The four-letter example of Chapter 14 already gives

$$H_1(\mathcal{H}; \mathbb{Z}) \cong \mathbb{Z}.$$

Thus four letters are the first scalar alphabet size at which a local shortest-action hole can occur.

The hierarchy is therefore:

d	shortest-completion geometry
2	rank 0: literal boxes
3	rank 1: first global S^1
≥ 4	higher rank: local holes possible

This gives a structural interpretation of alphabet size that is invisible in the original binary problem.

15.3 Three-letter normal form

Assume

$$\omega_1 < \omega_2 < \omega_3.$$

Define the adjacent gaps

$$a = \omega_3 - \omega_2, \qquad b = \omega_2 - \omega_1.$$

Let

$$h = \gcd(a, b),$$

and set

$$\alpha = \frac{a}{h}, \qquad \beta = \frac{b}{h}, \qquad m = \alpha + \beta.$$

A primitive equal-length kernel generator is

$$\boxed{g = (\alpha, -m, \beta).}$$

Indeed,

$$\alpha - m + \beta = 0,$$

and

$$\alpha\omega_1 - m\omega_2 + \beta\omega_3 = 0.$$

Thus one equal-length trade replaces m copies of the middle letter by:

- α copies of the first;
- β copies of the third.

Or vice versa.
This is the exact three-letter analogue of a primitive commuting schedule trade.

15.4 Parameterizing the optimum fiber

Write the complete optimum fiber as

$$\boxed{O_\omega(r) = \{x(t) = y + tg : 0 \leq t \leq N\},}$$

where

$$y = (y_1, y_2, y_3).$$

Thus

$$x(t) = (y_1 + \alpha t,\ y_2 - mt,\ y_3 + \beta t).$$

Let

$$z = y_2 - mN, \qquad 0 \leq z < m.$$

Define

$$P_t = (y_1 + \alpha t + 1)(y_3 + \beta t + 1).$$

Slicing the prequotient down-set by the middle coordinate gives an exact vertex count before quotienting:

$$\boxed{|\mathcal{C}_\omega(r)| = m \sum_{t=0}^{N-1} P_t + (z+1)P_N.}$$

The kernel quotient then collapses contiguous segments along the g-direction.

Because every rank-one orbit intersects the down-set in an integer interval, the quotient can be counted by:

$$\text{vertices} - \text{adjacent orbit pairs}.$$

15.5 Exact three-letter quotient vertex count

Proposition 15.1 — Exact rank-one quotient vertex count.

Let

$$\Omega(y, N)$$

be the number of vertex orbits in

$$\mathcal{C}_\omega(r)/\mathbb{Z}g.$$

If

$$N = 0,$$

then

$$\boxed{\Omega(y, 0) = (y_1 + 1)(y_2 + 1)(y_3 + 1).}$$

If

$$N \geq 1,$$

then

$$\boxed{\Omega(y, N) = mP_{N-1} + (z+1)(P_N - P_{N-1}).}$$

Proof idea. Every kernel orbit meets the down-set in a finite integer interval.

An orbit containing s vertices contributes exactly $s - 1$ adjacent pairs

$$u, \ u + g.$$

Therefore:

$$\#\text{orbits} = \#\text{vertices} - \#\text{adjacent pairs}.$$

The same middle-coordinate slicing gives the adjacent-pair count, and subtraction yields the formula. □

The result is an exact finite formula for the number of semantic shortest-prefix states after equal-length schedule identification.

15.6 Exact cubical f-vector

The same reduction works separately for every cube orientation.
For

$$J \subseteq \{1,2,3\},$$

define

$$T_J = \{t : x_i(t) \geq 1 \text{ for every } i \in J\}.$$

Because each coordinate is affine in t,

$$T_J$$

is an integer interval, possibly empty.
If nonempty, write

$$T_J = [L_J, U_J] \cap \mathbb{Z},$$

set

$$N_J = U_J - L_J,$$

and define

$$y^{(J)} = x(L_J) - \mathbf{1}_J.$$

Proposition 15.2 — Exact rank-one cubical f-vector.
The number of quotient cubes oriented by J is

$$\boxed{f_J(r) = \begin{cases} 0, & T_J = \varnothing, \\ \Omega(y^{(J)}, N_J), & T_J \neq \varnothing. \end{cases}}$$

Hence

$$\boxed{f_j(r) = \sum_{|J|=j} f_J(r), \qquad 0 \leq j \leq 3.}$$

Thus the entire finite cubical shortest-interval geometry is computable from one rank-one orbit formula.

15.7 Worked example

Take

$$\boxed{\omega = (-5,-3,1), \qquad r = 9.}$$

Then:

$$a = 4, \qquad b = 2, \qquad h = 2,$$

so

$$\alpha = 2, \qquad \beta = 1, \qquad m = 3.$$

The primitive equal-length trade is

$$\boxed{g = (2, -3, 1).}$$

The complete optimum fiber is

$$\boxed{O_\omega(9) = \{(0, 3, 0), (2, 0, 1)\}.}$$

Thus the minimum completion has two distinct Parikh realizations of the same length:

$$(0, 3, 0)$$

and

$$(2, 0, 1).$$

The quotient cubical f-vector is

$$\boxed{(f_0, f_1, f_2, f_3) = (8, 10, 2, 0).}$$

Therefore the filled quotient contains:

- 8 vertices;
- 10 labeled edges;
- 2 elementary commuting squares;
- no 3-cube.

Yet by Theorem 14.6,

$$\boxed{\mathfrak{I}_\omega(9) \simeq S^1.}$$

This is an ideal example for the monograph because the topology is nontrivial while the finite complex remains small enough to draw completely.

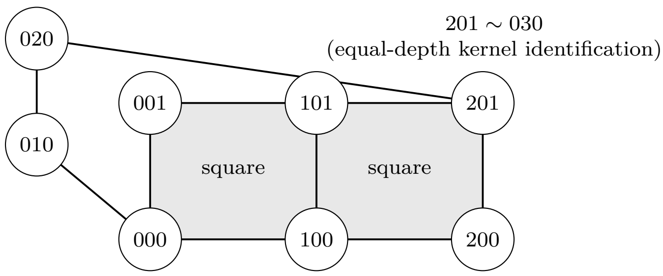


Figure 15.1: The complete rank-one quotient for $\omega = (-5, -3, 1)$ and $r = 9$. The down-set below $(2, 0, 1)$ contributes two filled squares; the path below $(0, 3, 0)$ shares the origin, and its far endpoint is identified with $(2, 0, 1)$ by the equal-length kernel translation. The quotient has $(f_0, f_1, f_2, f_3) = (8, 10, 2, 0)$ and is homotopy equivalent to S^1.

15.8 What changed relative to the binary rectangle?

In two letters, the minimum completion is one Parikh vector.

Different shortest words differ only by ordering.

In three letters, there can be two different minimum Parikh vectors:

$$x_0 \qquad \text{and} \qquad x_0 + g.$$

The prequotient geometry contains boxes beneath both.
The quotient then identifies equal-depth prefixes differing by g.
That is what creates the global circle.
Thus the transition

$$d = 2 \to d = 3$$

does not merely add another coordinate.
It adds the first equal-length trade direction.
And the transition

$$d = 3 \to d = 4$$

does not merely add one more trade.
It permits two independent trades, enough for local action topology itself to develop a hole.

15.9 Why this hierarchy belongs in the same monograph

The scalar alphabet-size hierarchy may look distant from the original 2:1 DPDA.
Its origin is nevertheless direct.
The original machine led to:

$$\text{shortest completion} \to \text{minimum Parikh demand} \to \text{rectangle}.$$

The higher-dimensional question was:

> Was the rectangle an accident of the binary alphabet?

The answer is no.
It was the rank-zero case of a general equal-length trade geometry.
The binary theory therefore sits at the first level of a topological hierarchy.
The next chapter studies a different direction: not how geometry changes with alphabet size, but how it behaves far out along a residual ray.

Chapter 16

Eventual Geodesic Arithmetic

The previous chapters studied the exact shortest-completion geometry of one fixed residual.

We now vary the residual systematically.

Fix two lattice residuals

$$r, b \in A\mathbb{Z}^d$$

and consider the ray

$$\boxed{r_n = r + nb, \qquad n \in \mathbb{N}.}$$

The minimum completion length

$$\ell_A(r_n)$$

need not be exactly linear in n.

Integer constraints can create periodic corrections.

The main result of this chapter is that those corrections eventually stabilize into a finite defect profile.

The same finite mechanism controls both:

- numerical shortest-completion distance;
- local shortest-action geometry.

16.1 The continuous directional slope

Consider the linear program

$$\boxed{\lambda_b = \min\{\mathbf{1}^\top x : Ax = -b,\ x \in \mathbb{R}^d_{\geq 0}\}.}$$

Completion totality guarantees feasibility.

Because the data are rational,

$$\lambda_b \in \mathbb{Q}_{\geq 0}.$$

The number

$$\lambda_b$$

is the continuous asymptotic completion cost per unit step in direction b.

If integer effects disappeared completely, one would expect

$$\ell_A(r + nb) \approx n\lambda_b + \text{constant}.$$

The eventual defect theory measures the exact failure of this naive linear law.

16.2 Dual certificate and reduced costs

Choose a rational dual optimum

$$y^*$$

for

$$\max\{-b^\top y : A^\top y \leq \mathbf{1}\}.$$

Thus

$$-b^\top y^* = \lambda_b.$$

Choose an integer

$$L > 0$$

clearing all denominators needed below.
Define the reduced costs

$$\boxed{\delta_i = L(1 - A_i^\top y^*).}$$

Dual feasibility gives

$$\delta_i \in \mathbb{N}.$$

If

$$x_n$$

is any optimal integer completion of r_n, then

$$Ax_n = -r_n.$$

Therefore

$$\begin{aligned} L\big(\ell_A(r_n) + (y^*)^\top r - n\lambda_b\big) &= L\left(\mathbf{1}^\top x_n - (y^*)^\top A x_n\right) \\ &= \sum_i \delta_i x_n(i). \end{aligned}$$

This motivates the exact integer defect.

Definition 16.1 — Scaled integrality defect.

$$\boxed{\Delta_r(n) = L\big(\ell_A(r_n) + (y^*)^\top r - n\lambda_b\big).}$$

Then

$$\boxed{\Delta_r(n) \in \mathbb{N}.}$$

The defect is not an asymptotic error term.
It is an exact nonnegative integer reduced-cost total.
That arithmetic discreteness is what makes eventual stabilization possible.

16.3 Zero-defect shifts

Define

$$\boxed{\mathcal{S}_b = \left\{ t \geq 1 : \exists z \in \mathbb{N}^d,\ Az = -tb,\ |z|_1 = t\lambda_b \right\}.}$$

An element

$$t \in \mathcal{S}_b$$

is a direction length for which the continuous optimum is attained integrally with zero defect.

Lemma 16.2 — Zero-defect semigroup.

$$\mathcal{S}_b$$

is nonempty and closed under addition.

Proof sketch. A rational primal optimum for direction b becomes integral after clearing denominators, giving at least one zero-defect multiple.

If

$$t_1, t_2 \in \mathcal{S}_b$$

with witnesses z_1, z_2, then

$$z_1 + z_2$$

satisfies

$$A(z_1 + z_2) = -(t_1 + t_2)b$$

and

$$|z_1 + z_2|_1 = (t_1 + t_2)\lambda_b.$$

Thus

$$t_1 + t_2 \in \mathcal{S}_b.$$

□

Define the canonical arithmetic period

$$\boxed{\tau_b = \gcd \mathcal{S}_b.}$$

A zero-defect shift by any $t \in \mathcal{S}_b$ can be appended to an existing completion without increasing the defect.

Thus $\mathcal{S}_b$ transports good integer behavior forward along the ray.

16.4 Stabilized defect profile

For each residue class

$$s \in \mathbb{Z}/\tau_b\mathbb{Z},$$

define

$$\boxed{M_s = \min\{\Delta_r(n) : n \geq 0,\ n \equiv s \pmod{\tau_b}\}.}$$

Theorem 16.3 — Stabilized defect profile.

For every residue class s, there exists

$$N_s$$

such that

$$\boxed{\Delta_r(n) = M_s}$$

whenever

$$n \geq N_s$$

and

$$n \equiv s \pmod{\tau_b}.$$

Consequently, eventually,

$$\boxed{\ell_A(r_n) = -(y^*)^\top r + n\lambda_b + \frac{M_{n \bmod \tau_b}}{L}.}$$

Proof.

Choose finitely many elements

$$t_1, \ldots, t_m \in \mathcal{S}_b$$

whose gcd is

$$\tau_b.$$

The additive numerical semigroup they generate contains every sufficiently large multiple of τ_b.

For each residue class s, choose an index n_s attaining the minimum defect M_s.

If

$$n \equiv s \pmod{\tau_b}$$

is sufficiently large, then

$$n - n_s$$

is a nonnegative combination of the t_i.

Append the corresponding zero-defect completions to an optimum at n_s.

This gives a feasible completion at n with the same defect M_s.

By definition of M_s, the defect cannot be smaller.

Hence it is exactly M_s. □

The formula has a useful interpretation:

$$\boxed{\text{eventual distance} = \text{rational linear drift} + \text{finite periodic integer defect.}}$$

The eventual arithmetic is therefore controlled by a finite cyclic table.

16.5 Why τ_b already contains the slope denominator

Write

$$\lambda_b = \frac{a}{q}$$

in lowest terms.
For every

$$t \in \mathcal{S}_b,$$

the zero-defect completion has integer length

$$t\lambda_b.$$

Hence

$$q \mid t.$$

Therefore

$$\boxed{q \mid \tau_b.}$$

The canonical zero-defect period is thus automatically compatible with the denominator of the asymptotic slope.
But the **minimal eventual affine period** can be smaller than τ_b.
The remaining reduction comes from symmetry of the stabilized defect table.

16.6 Fundamental defect period

View

$$M = (M_0, \dots, M_{\tau_b - 1})$$

as a cyclic word.
Define its fundamental circular period

$$\boxed{d_M = \min\{p \geq 1 : M_{s+p} = M_s \text{ for every } s\}.}$$

Then

$$d_M \mid \tau_b.$$

We can now determine the exact eventual distance period.

Theorem 16.4 — Exact minimal eventual distance period.
Let

$$\lambda_b = \frac{a}{q}$$

in lowest terms.
The smallest positive integer p for which there exists an integer D_p satisfying

$$\ell_A(r_{n+p}) = \ell_A(r_n) + D_p$$

for all sufficiently large n is

$$\boxed{p_{\min}(r, b) = \operatorname{lcm}(q, d_M).}$$

Moreover,

$$\boxed{p_{\min}(r, b) \mid \tau_b,}$$

and

$$\boxed{D_p = p\lambda_b.}$$

Proof.
Suppose p is an eventual affine period.
Comparing the linear terms in Theorem 16.3 forces

$$D_p = p\lambda_b.$$

Since D_p is integral,

$$q \mid p.$$

The periodic correction term must also be invariant under shift by p, so

$$d_M \mid p.$$

Therefore

$$\operatorname{lcm}(q, d_M) \mid p.$$

Conversely, let

$$p_0 = \operatorname{lcm}(q, d_M).$$

Then

$$p_0\lambda_b$$

is integral and the stabilized defect table is invariant under shift by p_0.
The formula of Theorem 16.3 gives

$$\ell_A(r_{n+p_0}) = \ell_A(r_n) + p_0\lambda_b$$

for all sufficiently large n.
Thus p_0 is the minimal eventual affine period. □
This theorem separates two arithmetic sources of periodicity:

1. the denominator of the continuous slope;
2. the intrinsic symmetry period of the integer defect profile.

The actual distance period is their least common multiple.

16.7 Eventual local action geometry

The same mechanism controls local shortest-action compatibility.
For

$$J \subseteq [d],$$

let

$$c_J = A\mathbf{1}_J.$$

Recall:

$$J \in \mathcal{H}_A(r_n) \iff \ell_A(r_n + c_J) = \ell_A(r_n) - |J|.$$

Apply Theorem 16.3 not only to the ray

$$r_n$$

but also to every finitely many shifted rays

$$r_n + c_J.$$

The same:

- dual optimum;
- zero-defect semigroup;
- canonical period τ_b;

can be used.
Only the stabilized defect table changes with J.
Therefore the truth value of

$$J \in \mathcal{H}_A(r_n)$$

is eventually determined by

$$n \bmod \tau_b.$$

Let

$$\Theta_s$$

be the stabilized local action complex on residue class s.
Define

$$\boxed{d_{\mathcal{H}} = \min\{p \geq 1 : \Theta_{s+p} = \Theta_s \text{ for every } s\}.}$$

Corollary 16.5 — Exact minimal local-geometry period.
The smallest eventual period of the local action complex is

$$\boxed{p^{\mathcal{H}}_{\min}(r, b) = d_{\mathcal{H}}.}$$

Moreover,

$$\boxed{d_{\mathcal{H}} \mid \tau_b.}$$

The first-action set is eventually periodic for the same reason.
Thus one finite defect mechanism governs both:

$$\boxed{\text{numerical shortest distance}}$$

and

$$\boxed{\text{local shortest-action geometry.}}$$

16.8 What is and is not being claimed

Eventual periodicity in semilinear and parametric integer settings is not new in itself.
The point here is more specific.
The completion program produces one exact object:

$$\Delta_r(n).$$

From it we derive:

- eventual affine distance;
- exact minimal distance period;
- eventual local action complex;
- exact minimal local-geometry period.

Thus the numerical and geometric observables are not proved periodic by unrelated arguments.
They descend from the same minimum-completion defect profile.
This is the structural contribution relevant to the monograph.

16.9 The higher-dimensional branch in one picture

The path from the original binary machine to the present theorem is now complete:

$$\boxed{\begin{array}{c}\text{original 2:1 shortest completion}\\ \downarrow \\ \text{unique minimum Parikh demand}\\ \downarrow \\ \text{binary rectangle}\\ \downarrow \\ \text{matrix residual}\\ \downarrow \\ \text{optimal Parikh fiber}\\ \downarrow \\ \text{shortest-prefix down-set}\\ \downarrow \\ \text{equal-length kernel quotient}\\ \downarrow \\ \text{rank-sensitive local/global topology}\\ \downarrow \\ \text{eventual finite defect profile.}\end{array}}$$

The binary rectangle was therefore not a small isolated trick.

It was the rank-zero member of a much larger completion geometry.

16.10 From completion geometry to finite compilation

The higher-dimensional branch has now identified optimal fibers, shortest-prefix down-sets, local action complexes, quotient topology, and eventual arithmetic. The next step is not another special counting matrix. It is to ask which part of this shortest-path structure can be compiled from finite periodic data and how much information is carried by higher-order compatibility among commuting optimal actions.

This leads to a general finite-phase periodic model. The metric backend is classical periodic shortest-path arithmetic; the new layer developed in the next Part compiles exact optimal compatibility of every structural order and then proves that no fixed lower-order truncation universally reconstructs the full local compatibility complex.

Part VI

Geodesic Compilers and Higher-Order Optimal Structure

Research provenance. This Part abstracts the shortest-completion geometry of the preceding chapters to positive-cost finite-phase periodic action systems. Periodic minimum-cost paths, semilinear and Presburger descriptions, shortest-path preprocessing, cubical state complexes, and squarefree divisor complexes are established background [109, 107, 108, 112, 115, 111, 110]. The contribution developed here is an exact finite cubical geodesic compiler for all-order local optimal compatibility, together with prescribed-order adjacent-state non-factorization, Boolean amplification, and universal k-fiber realization with exact next-order capacity.

Chapter 17

From Completion Geometry to Finite Geodesic Compilers

The preceding completion-geometry chapters started from a fixed linear counting system and studied the geometry of its shortest completions. We now isolate the mechanism that survived those examples. The resulting model is a finite-phase periodic shortest-path system: a finite phase graph carries integral translations and positive costs, while the lattice coordinate remains unbounded. This abstraction retains the arithmetic structure needed for exact completion queries but no longer assumes a particular pushdown implementation or a particular counting matrix.

The distinction between background and contribution is important. Minimum-cost paths in periodic graphs, semilinear and Presburger descriptions, and shortest-path preprocessing are established theories [109, 107, 108, 112, 115]. The contribution of this Part is their organization into an exact finite compiler for geodesic branching and all-order optimal compatibility, followed by a structure theory showing that the compiled compatibility hierarchy is universally non-collapsing across the class.

In the discovery chain of this monograph, the abstraction is reached from the earlier chapters as

$$\boxed{\begin{array}{c}\text{completion-aware 2:1 DPDA} \longrightarrow \rho_{p,q} = qN_1 - pN_0 \\ \longrightarrow \text{ minimum completion } \longrightarrow \text{ geodesic rectangles and optimal fibers} \\ \longrightarrow \text{ finite-phase periodic geodesic compilation.}\end{array}}$$

The original machine is therefore not a hypothesis of the results below; it records the concrete route by which the compiler abstraction was found.

17.1 Finite-phase periodic graphs

Let

$$\mathcal{B} = (\Phi, E)$$

be a finite directed graph. Every edge

$$e : \phi \to \psi$$

carries an integral displacement

$$g_e \in \mathbb{Z}^n$$

and a positive rational cost

$$c_e \in \mathbb{Q}_{>0}.$$

Its periodic lift has vertex set

$$\mathbb{Z}^n \times \Phi$$

and edges

$$(z, \phi) \xrightarrow{e} (z + g_e, \psi).$$

Fix a target phase $\phi_\star \in \Phi$ and target $t = (0, \phi_\star)$. Multiplying all costs by a common denominator preserves all shortest paths and scales all distances uniformly. Hence we assume

$$c_e \in \mathbb{N}_{>0}.$$

A state $v = (z, \phi)$ is *target-reachable* if there exists a path from v to t. For target-reachable states, let

$$d(v) = d_\phi(z)$$

be the minimum path cost to t. When $v = (z, \phi)$, we write $\mathrm{Reach}(v)$ as shorthand for $\mathrm{Reach}_\phi(z)$. The same convention is used for translated successor states.

Minimum-cost path problems in periodic graphs with integral gains and rational costs are classical; see Höfting and Wanke [109]. The role of the present section is to extract a finite arithmetic representation used by the compatibility compiler.

17.2 Gain-cost relations

Let

$$L_{\phi,\phi_\star} \subseteq E^*$$

be the regular language of base-graph walks from ϕ to $\phi_\star$. For $w = e_1 \cdots e_q$, define

$$G(w) = \sum_{j=1}^{q} g_{e_j}, \qquad C(w) = \sum_{j=1}^{q} c_{e_j}.$$

Set

$$R_\phi = \{(G(w), C(w)) : w \in L_{\phi,\phi_\star}\}.$$

Proposition 17.1 (Semilinear path relation)**.** *For every phase ϕ, the relation*

$$R_\phi \subseteq \mathbb{Z}^n \times \mathbb{N}$$

is effectively Presburger-definable.

Proof. The Parikh image of a regular language is effectively semilinear [107, 116]. Total gain and total cost are integral linear functions of the edge-count vector. Semilinear sets are closed under integral linear images and coincide with Presburger-definable sets [108].

We use the standard extension of semilinearity to subsets of $\mathbb{Z}^n \times \mathbb{N}$. Equivalently, signed coordinates may be represented by nonnegative coordinate pairs and projected by the integral map $x = x^+ - x^-$; this does not change Presburger definability. □

Define

$$\mathrm{Reach}_\phi(z) \iff \exists N \in \mathbb{N}\; R_\phi(-z, N).$$

Proposition 17.2 (Arithmetic distance predicate)**.** *There is an effectively constructible Presburger predicate*

$$\mathrm{Dist}_\phi(z, N)$$

such that

$$\mathrm{Dist}_\phi(z, N) \iff N = d_\phi(z)$$

for every target-reachable (z, ϕ).

Proof. Define $\mathrm{Dist}_\phi(z, N)$ by

$$R_\phi(-z, N) \wedge \neg\exists N' \left(0 \le N' < N \wedge R_\phi(-z, N')\right).$$

For a target-reachable state, the feasible costs form a nonempty subset of $\mathbb{N}$, hence possess a least element. $\square$

The finite family

$$\mathfrak{C}_{\mathrm{dist}} = \{\mathrm{Reach}_\phi, \mathrm{Dist}_\phi : \phi \in \Phi\}$$

will be called the *arithmetic distance compiler*. "Finite" refers to the description; the coordinate $z \in \mathbb{Z}^n$ remains unbounded.

17.3 Bellman guards

For a lifted edge $v \xrightarrow{e} T_e v$, define $e \in \mathrm{First}(v)$ when e is the first edge of some shortest path from v to t.

Lemma 17.3 (Optimal-edge criterion)**.** *For target-reachable v,*

$$\boxed{e \in \mathrm{First}(v) \iff \mathrm{Reach}(T_e v) \textit{ and } c_e + d(T_e v) = d(v).}$$

Proof. If a shortest path begins with e, its remaining suffix must be shortest from $T_e v$. Conversely, if the equality holds, append any shortest suffix from $T_e v$. $\square$

Lemma 17.4 (Prefix criterion)**.** *Let $w = e_1 \cdots e_q$ be executable from a target-reachable state v. Then w is a prefix of some shortest path from v if and only if*

$$\boxed{\mathrm{Reach}(T_w v) \quad \textit{and} \quad C(w) + d(T_w v) = d(v).}$$

Proof. If w occurs on a shortest path, replacing its suffix by a shorter one would shorten the full path, so the suffix is shortest. Conversely, append a shortest suffix from $T_w v$. The displayed equality makes the resulting total cost equal to $d(v)$. $\square$

17.4 Optimal unfolding

For a target-reachable state v, let $\mathrm{OptUnf}(v)$ be the directed subgraph reachable from v using only edges satisfying the Bellman guard of Theorem 17.3.

Theorem 17.5 (Finite geodesic compiler)**.** *For every fixed positive-cost finite-phase periodic graph, one can effectively construct a finite Presburger-guarded translation program such that, from every target-reachable state v,*

(i) it recovers $d(v)$;

(ii) it recovers every optimal first edge;

(iii) it decides shortest-prefix membership for every finite executable word; and

(iv) its optimal unfolding is exactly the union of all shortest v-to-t paths.

Proof. Items (i)–(iii) follow from Theorems 17.2 to 17.4. Suppose

$$v = v_0 \to v_1 \to \cdots \to v_q$$

is a path in the optimal unfolding. Each edge satisfies

$$d(v_{j-1}) = c_j + d(v_j),$$

hence

$$d(v_q) = d(v) - \sum_{j=1}^{q} c_j.$$

Every guarded successor is target-reachable, so a shortest suffix from v_q exists. Appending it yields a shortest path containing the given prefix. Conversely, every edge of a shortest path satisfies the Bellman equality by optimal substructure. □

Corollary 17.6 (Canonical and complete generation)**.** *Fix a total order on outgoing base edges. At every non-target target-reachable state, at least one optimal first edge exists. Choosing the least one gives a deterministic canonical shortest route.*

Moreover, because every cost is positive,

$$v \to v' \text{ optimal} \quad \Longrightarrow \quad d(v') < d(v).$$

Hence d is a ranking function on the optimal unfolding. Recursive branching over all optimal first edges generates all and only shortest paths and terminates on every branch.

Chapter 18

Cubical Geodesic Compilation

The local action complexes developed earlier in the higher-dimensional completion branch are recovered here as special cases. In a one-phase translation system whose structural action family is the previously fixed completion action set, the complex defined below is exactly the earlier shortest-completion local action complex. The new notation separates two layers that were intertwined in the special counting model: $\mathcal{S}_\phi$ records structural commutation, whereas $\mathcal{H}(v)$ records the faces selected by shortest-path optimality.

18.1 Periodic action systems

Let

$$I = \{1, \dots, s\}$$

be a finite action alphabet. For every $i \in I$, let

$$\tau_i : \Phi \rightharpoonup \Phi$$

be a partial phase transition. Whenever $\tau_i(\phi)$ is defined, associate

$$g_i(\phi) \in \mathbb{Z}^n, \qquad c_i(\phi) \in \mathbb{Q}_{>0}.$$

The lifted action is

$$T_i(z, \phi) = \big(z + g_i(\phi), \tau_i(\phi)\big).$$

For an executable action word $w = i_1 \cdots i_q$, T_w denotes the corresponding composition of lifted partial actions. If J is a structural cube, $T_J v$ denotes the common endpoint obtained from any permutation of the actions in J.

18.2 Structural cubes

Definition 18.1 (Structural cube)**.** Let $J \subseteq I$. We call J a *structural cube at phase ϕ* if, for every $K \subseteq J$, every permutation of the actions in K

(i) is executable from ϕ;

(ii) reaches the same final phase;

(iii) produces the same total displacement;

(iv) has the same total cost;

and, additionally,

$$K \neq L \quad \Longrightarrow \quad (\tau_K(\phi), g_K(\phi)) \neq (\tau_L(\phi), g_L(\phi)).$$

Write $\tau_K(\phi)$, $g_K(\phi)$, and $c_K(\phi)$ for the common data.

The final injectivity condition makes the $2^{|J|}$ subset vertices distinct, so the terminology "cube" is nondegenerate. Let

$$\mathcal{S}_\phi = \{J \subseteq I : J \text{ is a structural cube at } \phi\}.$$

Lemma 18.2 (Structural simpliciality)**.** *For every ϕ, $\mathcal{S}_\phi$ is a simplicial complex.*

Proof. Every defining condition is inherited by subsets, including injectivity of the subset-endpoint map. □

Because Φ and I are finite, each $\mathcal{S}_\phi$ is finite and can be computed offline from the finite action table. Cubical state complexes generated by independent local moves are classical in reconfiguration theory; see Ghrist and Peterson [111]. Our additional object is the subcomplex selected by shortest-path optimality.

18.3 Optimal compatibility

Definition 18.3 (Optimal-compatibility complex)**.** For a target-reachable state $v = (z, \phi)$ and $J \in \mathcal{S}_\phi$, define

$$J \in \mathcal{H}(v)$$

if

$$\mathrm{Reach}(T_J v)$$

and

$$\boxed{c_J(\phi) + d(T_J v) = d(v).}$$

For $h \geq 0$, define

$$\mathcal{H}_{\leq h}(v) = \{J \in \mathcal{H}(v) : |J| \leq h\}, \qquad \mathcal{H}_{=h}(v) = \{J \in \mathcal{H}(v) : |J| = h\}.$$

For later comparison across structural orders, write

$$\Gamma_h(v) = \big(d(v), \mathcal{H}_{\leq h}(v)\big).$$

The empty set belongs to both $\mathcal{S}_\phi$ and $\mathcal{H}(v)$, with $T_\varnothing v = v$ and $c_\varnothing = 0$. Hence $\mathcal{H}_{\leq 0}(v) = \{\varnothing\}$.

Proposition 18.4 (Cubical Bellman equivalence)**.** *Let $J \in \mathcal{S}_\phi$. For target-reachable $v = (z, \phi)$, the following are equivalent:*

(i) $J \in \mathcal{H}(v)$;

(ii) one permutation of the actions in J is a shortest prefix from v;

(iii) every permutation of the actions in J is a shortest prefix from v.

Proof. Every permutation of a structural cube has the same cost and endpoint. Hence the prefix criterion of Theorem 17.4 either holds for all permutations or for none. □

Proposition 18.5 (Optimal simpliciality)**.** *For every target-reachable v, $\mathcal{H}(v)$ is a simplicial subcomplex of $\mathcal{S}_\phi$.*

Proof. Let $J \in \mathcal{H}(v)$ and $K \subseteq J$. Order the actions of J so that the actions of K occur first. By Theorem 18.4 this permutation is a shortest prefix, and hence its initial K-segment is a shortest prefix. Therefore $K \in \mathcal{H}(v)$. □

18.4 Arithmetic face guards

For each defined action i at phase ϕ, let $\mathrm{Opt}_{\phi,i}(z)$ denote the Presburger Bellman guard supplied by Theorem 17.3. For $J \in \mathcal{S}_\phi$, define $\mathrm{Face}_{\phi,J}(z)$ by

$$\exists N, N' \left[\mathrm{Dist}_\phi(z, N) \wedge \mathrm{Dist}_{\tau_J(\phi)}(z + g_J(\phi), N') \wedge N = c_J(\phi) + N'\right].$$

Thus

$$J \in \mathcal{H}((z,\phi)) \iff \mathrm{Face}_{\phi,J}(z).$$

Definition 18.6 (Cubical geodesic compiler)**.** Let

$$\mathfrak{C}_{\mathrm{cube}} = \big(\{\mathrm{Reach}_\phi, \mathrm{Dist}_\phi\}, \{\mathrm{Opt}_{\phi,i}\}, \{\mathrm{Face}_{\phi,J}\}\big)$$

range over all phases, defined actions, and structural faces.

Theorem 18.7 (Finite cubical geodesic compilation)**.** *For every fixed positive-cost finite-phase periodic action system, the finite arithmetic compiler $\mathfrak{C}_{\mathrm{cube}}$ is effectively constructible and determines, at every target-reachable state,*

- *exact shortest distance;*
- *all optimal first actions;*
- *shortest-prefix membership for arbitrary finite executable words;*
- *the complete local optimal-compatibility complex;*
- *a canonical shortest route under fixed tie breaking; and*
- *an implicit exact representation of all shortest paths.*

Proof. The metric, first-action, prefix, canonical, and all-geodesic claims follow from Theorems 17.5 and 17.6. For each phase, the structural complex is finite. Membership of every structural face in $\mathcal{H}(v)$ is exactly one arithmetic face guard. Evaluating all such guards therefore recovers the complete local complex. □

18.5 Signature regions

For a subcomplex $K \subseteq \mathcal{S}_\phi$, define

$$\Omega_{\phi,K} = \{z : \mathrm{Reach}_\phi(z) \text{ and } \mathcal{H}((z,\phi)) = K\}.$$

Every such set is Presburger-definable, and

$$\boxed{\{z : \mathrm{Reach}_\phi(z)\} = \bigsqcup_K \Omega_{\phi,K}.}$$

Thus the target-reachable region admits a finite arithmetic partition by local optimal-compatibility type.

18.6 Affine-semigroup interpretation

In the one-phase unit-cost translation case, write $\mathcal{S}$ for the unique phase structural complex. The compatibility complex admits an exact divisor-complex interpretation. Let the distinct action translations be

$$a_1, \dots, a_s \in \mathbb{Z}^n$$

and let

$$D = d(r).$$

Define

$$b_i = (-a_i, 1) \in \mathbb{Z}^{n+1}, \qquad \beta_r = (r, D),$$

and the semigroup

$$\mathsf{M} = \langle b_1, \dots, b_s \rangle_{\mathbb{N}}.$$

Define the labeled squarefree divisor complex

$$\Delta_{\beta_r} = \left\{ J \subseteq [s] : \beta_r - \sum_{i \in J} b_i \in \mathsf{M} \right\}.$$

Proposition 18.8 (Homogenized semigroup correspondence)**.** *For every $J \subseteq [s]$,*

$$J \in \Delta_{\beta_r}$$

if and only if

$$|J| + d\left(r + \sum_{i \in J} a_i\right) = d(r).$$

Consequently,

$$\boxed{\mathcal{H}(r) = \mathcal{S} \cap \Delta_{\beta_r}.}$$

If every action subset is structurally nondegenerate, then $\mathcal{H}(r) = \Delta_{\beta_r}$.

Proof. Indeed,

$$\beta_r - \sum_{i \in J} b_i = \left(r + \sum_{i \in J} a_i, D - |J| \right).$$

Membership in M is therefore equivalent to the existence of a suffix consisting of exactly $D - |J|$ unit-cost actions that completes the state

$$r + \sum_{i \in J} a_i$$

to the target.

If the same state admitted a suffix of length strictly smaller than $D - |J|$, then preceding it by the J-prefix would give a completion of r of total length strictly smaller than

$$|J| + (D - |J|) = D = d(r),$$

a contradiction. Hence

$$d\left(r + \sum_{i \in J} a_i\right) = D - |J|,$$

which is equivalent to

$$|J| + d\left(r + \sum_{i \in J} a_i\right) = d(r).$$

Intersecting with the structural complex gives $\mathcal{H}(r) = \mathcal{S} \cap \Delta_{\beta_r}$. ∎

Squarefree divisor complexes are classical in affine-semigroup theory [110]; parametric Presburger methods can distinguish full divisor-complex types in affine-semigroup families [114]. We therefore do not claim novelty for divisor complexes or arithmetic definability in isolation.

18.7 Complexity of the Compiled Representation

The compiler has two distinct computational stages:

$$\boxed{\text{offline arithmetic compilation} \quad + \quad \text{online querying.}}$$

No polynomial-time claim is made for the general offline stage.

18.7.1 Query parameters

For $v = (z, \phi)$, let

$$\lambda(v) = 1 + \sum_{j=1}^{n} \lceil \log_2(1 + |z_j|) \rceil$$

be the coordinate encoding length.

Let $Q_{\text{reach}}(\lambda)$ denote the cost of evaluating target reachability and let $Q_{\text{dist}}(\lambda)$ denote the additional cost of recovering the exact distance once reachability has been established. We write

$$Q_{\text{met}}(\lambda) = Q_{\text{reach}}(\lambda) + Q_{\text{dist}}(\lambda)$$

for a complete metric query. This convention is necessary because $d(v)$ is defined only at target-reachable states. In particular, successor states appearing in optimal-action, prefix, and face queries are tested for reachability before their distance values are evaluated.

Let $Q_{\text{opt}}(\lambda)$ and $Q_{\text{face}}(\lambda)$ denote the costs of evaluating one compiled optimal-action guard and one face guard, respectively. Let $Q_{\text{first}}(\lambda)$ denote the cost of recovering the complete set of optimal first actions at one target-reachable state. Let $\deg^+_{\max}$ be the maximum outgoing base degree.

Proposition 18.9 (Distance and first-action queries)**.** *Exact distance at a target-reachable state is recovered in $Q_{\text{dist}}(\lambda(v))$. All optimal first actions are recovered in*

$$O\big(\deg^+_{\max} Q_{\text{opt}}(\lambda_*)\big) .$$

If optimal guards are evaluated through metric queries instead, then

$$O\big(\deg^+_{\max} Q_{\text{met}}(\lambda_*) + Q_{\text{dist}}(\lambda(v))\big)$$

suffices.

18.7.2 Prefix queries

Let w be an explicit executable word of length q. A single scan computes its endpoint phase, accumulated gain, and accumulated cost. For a fixed system,

$$\lambda(T_w v) = O(\lambda(v) + n \log(1 + q)) .$$

Proposition 18.10 (Prefix-query complexity)**.** *Shortest-prefix membership is decidable in*

$$O(q \ \text{poly}(\lambda_w) + Q_{\text{met}}(\lambda_w) + Q_{\text{dist}}(\lambda(v))) .$$

The linear dependence on q is unavoidable up to arithmetic costs because the explicit word must be read.

18.7.3 Face queries

A fixed structural face has precomputed phase, displacement, and cost data.

Proposition 18.11 (Face-query complexity)**.** *A specified structural face can be tested in $Q_{\text{face}}(\lambda_*)$. Using only reachability and distance recovery gives*

$$O(Q_{\text{dist}}(\lambda(v)) + Q_{\text{met}}(\lambda(T_J v)))\,.$$

Let

$$F_h(\phi) = |\{J \in \mathcal{S}_\phi : |J| \le h\}|.$$

Then

$$F_h(\phi) \le \sum_{j=0}^{h} \binom{s}{j}.$$

Thus the complete explicit truncation $\mathcal{H}_{\le h}(v)$ can be reported in

$$O(F_h(\phi) Q_{\text{face}}(\lambda_*) + |\mathcal{H}_{\le h}(v)|)\,.$$

Explicit recovery of the full complex can require exponential output, since even a full simplex has 2^s faces.

18.7.4 Canonical routes

Let

$$c_{\min} = \min_e c_e > 0.$$

Every optimal edge decreases distance by at least $c_{\min}$. Therefore any shortest route has at most

$$\left\lfloor \frac{d(v)}{c_{\min}} \right\rfloor$$

edges. If $\ell_{\text{can}}(v)$ is the canonical route length and Q_{canon} is the cost of selecting the canonical enabled optimal action, then route generation takes

$$O(\ell_{\text{can}}(v) Q_{\text{canon}}(\lambda_{\text{path}}))\,.$$

This is path-length-sensitive. With binary-encoded coordinates, an explicitly printed shortest path may itself be exponentially long in the input bit length.

18.7.5 Enumeration of all shortest paths

Let $\text{Geo}(v)$ be the finite set of shortest paths from v, and define

$$\text{OUT}(v) = 1 + \sum_{\gamma \in \text{Geo}(v)} (|\gamma| + 1).$$

The shortest-prefix tree has at most $O(\text{OUT}(v))$ nodes.

Theorem 18.12 (Output-sensitive geodesic enumeration)**.** *All shortest paths from v can be enumerated in*

$$\boxed{O\big(\text{OUT}(v)\big[Q_{\text{first}}(\lambda_{\text{path}}) + 1\big]\big)\,.}$$

The recursion depth is at most

$$\left\lfloor \frac{d(v)}{c_{\min}} \right\rfloor.$$

We describe this as output-sensitive and path-length-sensitive rather than polynomial-delay enumeration, because a single shortest path may already be exponentially long in the binary coordinate encoding.

18.7.6 Normalized arithmetic compilers

A stronger online statement is available under an explicit normalization assumption. Suppose offline preprocessing produces

(i) a finite decision list of quantifier-free Presburger regions;

(ii) an explicit affine distance output on each region; and

(iii) quantifier-free optimal-action and face guards.

Let $\|\mathfrak{C}\|$ be the encoded compiler size and $L_{\mathfrak{C}}$ the maximum coefficient bit length.

Corollary 18.13 (Online bit complexity after normalization)**.** *Under this explicit normalization assumption,* DIST*,* FIRST*, and* FACE *are computable in*

$$\mathrm{poly}\big(\|\mathfrak{C}\|, L_{\mathfrak{C}}, \lambda(v)\big).$$

For a fixed compiled system, these queries are therefore polynomial in the queried coordinate bit length.

This is a conditional normalized-compiler result; the monograph does not claim that the required normalization is polynomial-time constructible in general. General Presburger elimination can incur substantial blow-up. Recent work obtains singly exponential elimination for an existential block, but this does not imply polynomial complexity for the complete compilation pipeline [117].

Chapter 19

Prescribed-Order Separation, Boolean Amplification, and Universal k-Fibers

We now show that the higher-order compatibility compiled in Chapter 18 cannot universally be reconstructed from any fixed lower-order truncation, then sharpen that non-reconstruction into a universal realization theorem. The first construction forces one discrepancy to appear at a prescribed order; independent products amplify such discrepancies Booleanly. The final construction asks whether that freedom is intrinsic to one lower-order signature fiber and proves that, on a fixed structural simplex, the entire next-order layer can vary arbitrarily.

Fix $k \geq 1$ and define

$$M = 2^{k+1}, \qquad \delta = 2^k - 1, \qquad N = 2k.$$

Consider the one-phase unit-cost translation actions

$$a_0 = -M,$$

$$a_i = -M + 2^{i-1}, \qquad 1 \leq i \leq k,$$

and

$$a_+ = 1.$$

Let

$$A_- = \{a_0, a_1, \dots, a_k\}.$$

All action maps are translations, so they commute globally.

19.1 Completion and nondegeneracy

Lemma 19.1 (Completion-totality). *Every integer state admits a finite path to* 0.

Proof. If $x \leq 0$, repeatedly apply $a_+ = 1$. If $x > 0$, apply $a_0 = -M$ until the state becomes nonpositive and then apply a_+. □

Lemma 19.2 (Nondegeneracy of the negative family). *Distinct subsets of A_- have distinct total translations.*

Proof. For $K \subseteq A_-$, define

$$B(K) = \sum_{\substack{1 \le i \le k \\ a_i \in K}} 2^{i-1}.$$

Then

$$\sum_{a \in K} a = -|K|M + B(K).$$

If two subsets K, L have the same total translation, then

$$(|K| - |L|)M = B(K) - B(L).$$

The right side has absolute value $< M$, hence $|K| = |L|$. Then $B(K) = B(L)$, and uniqueness of binary expansion identifies all special actions. Cardinality then determines the presence of a_0. Thus $K = L$. ☐

In particular,

$$J_\star = \{a_1, \ldots, a_k\}$$

is a genuine nondegenerate structural k-cube.

19.2 Binary-deficit distance

Lemma 19.3 (Binary-deficit distance lemma)**.** *If*

$$0 \le D < 2^k$$

and

$$\ell \ge k,$$

then

$$\boxed{d(\ell M - D) = \ell.}$$

Proof. Write

$$D = \sum_{i \in S} 2^{i-1}.$$

Apply a_i once for every $i \in S$ and apply a_0 another $\ell - |S|$ times. The total translation is $-\ell M + D$, so $d(\ell M - D) \le \ell$.

On the other hand, $D < 2^k < M$, so

$$\ell M - D > (\ell - 1)M.$$

One action decreases the state by at most M, so $\ell - 1$ actions cannot suffice. ☐

Action labels used in a cubical prefix may be reused later in the completion suffix; this is ordinary path semantics.

19.3 Adjacent equal-distance states

Define

$$r = NM - \delta$$

and

$$r' = r + 1 = NM - (\delta - 1).$$

Then

$$r \xrightarrow{a_+} r'.$$

Proposition 19.4 (Equal distance)**.**

$$\boxed{d(r) = d(r') = N.}$$

Proof. Apply Theorem 19.3 with deficits δ and $\delta - 1$. □

For a negative action set J, define

$$B(J) = \sum_{\substack{1 \le i \le k \\ a_i \in J}} 2^{i-1}.$$

Proposition 19.5 (Exact common lower-order truncation)**.** *For every action set J satisfying $|J| < k$,*

$$\boxed{J \in \mathcal{H}(r) \iff J \in \mathcal{H}(r') \iff J \subseteq A_-.}$$

Consequently,

$$\boxed{\mathcal{H}_{\le k-1}(r) = \mathcal{H}_{\le k-1}(r').}$$

Proof. Suppose first that $J \subseteq A_-$ and let $q = |J| < k$. After the prefix,

$$r_J = (N - q)M - (\delta - B(J)),$$

and

$$r'_J = (N - q)M - (\delta - 1 - B(J)).$$

Since $q < k$, not all k positive binary deficits occur, so $B(J) \le \delta - 1$. Thus both remaining deficits belong to $[0, 2^k)$, while $N - q \ge k + 1$. Theorem 19.3 gives remaining distance $N - q$ at both states. Hence J is optimal at both.

Now suppose $a_+ \in J$. Let q be the number of negative actions in J. Then $|J| = q + 1 < k$. After the prefix from r, the residual exceeds $(N - q - 1)M$, because

$$M - \delta + B(J) + 1 > 0.$$

Yet an optimal suffix would have only $N - q - 1$ unit-cost steps available, each decreasing by at most M. This is impossible. The same argument at r' has an extra 1 of residual and is again impossible. If such a set is structurally degenerate, it is excluded from $\mathcal{H}$ already by definition; in either case it is not an optimal face. □

19.4 First discrepancy at order k

Proposition 19.6 (Distinguished separating face)**.** *For*

$$J_\star = \{a_1, \ldots, a_k\},$$

$$\boxed{J_\star \in \mathcal{H}(r), \qquad J_\star \notin \mathcal{H}(r').}$$

Proof. Because

$$\sum_{i=1}^{k} 2^{i-1} = \delta,$$

after applying $J_\star$ at r,

$$r + \sum_{i=1}^{k} a_i = (N - k)M = kM.$$

Theorem 19.3 with $D = 0$ gives $d(kM) = k = N - k$. Hence $J_\star$ is optimal.

At r', the same prefix leaves $kM + 1$. The required suffix would have k actions, which can decrease the residual by at most kM. Hence no such optimal suffix exists. □

Proposition 19.7 (Single-face difference through order k)**.**

$$\boxed{\mathcal{H}_{\le k}(r) = \mathcal{H}_{\le k}(r') \cup \{J_\star\}.}$$

Proof. The complexes agree below order k by Theorem 19.5. Any k-set containing a_+ fails the same maximum-descent test at both states.

A negative k-set other than $J_\star$ contains a_0 and omits exactly one special action a_i. Its deficit is $\delta - 2^{i-1}$. After the prefix, the two residuals are

$$kM - 2^{i-1}$$

and

$$kM - (2^{i-1} - 1),$$

respectively. Both deficits lie in $[0, 2^k)$, so Theorem 19.3 gives remaining distance k in both cases. □

Theorem 19.8 (Prescribed-order adjacent-state non-factorization)**.** *For every prescribed $k \ge 1$, there exists a completion-total one-phase unit-cost system of globally commuting translation actions and adjacent states $r \to r'$ such that*

$$\boxed{d(r) = d(r'),}$$

$$\boxed{\mathcal{H}_{\le k-1}(r) = \mathcal{H}_{\le k-1}(r'),}$$

and

$$\boxed{\mathcal{H}_{\le k}(r) = \mathcal{H}_{\le k}(r') \cup \{J_\star\}.}$$

Equivalently,

$$\Gamma_{k-1}(r) = \Gamma_{k-1}(r')$$

but

$$\Gamma_k(r) \ne \Gamma_k(r').$$

Thus the first compatibility discrepancy occurs exactly at the prescribed order k.

Corollary 19.9 (No fixed lower-order reconstruction)**.** *For every fixed $h \ge 0$, the data*

$$\boxed{\big(d(v), \mathcal{H}_{\le h}(v)\big)}$$

do not universally determine $\mathcal{H}(v)$.

Proof. Apply Theorem 19.8 with $k = h + 1$. □

The claim concerns this exact lower-order truncation. It does not assert insufficiency of an arbitrary auxiliary state description that might retain additional information such as the raw coordinates.

19.5 Exact full complexes

Proposition 19.10 (Full-complex description)**.**

$$\boxed{\mathcal{H}(r) = 2^{A_-}}$$

and

$$\boxed{\mathcal{H}(r') = 2^{A_-} \setminus \{J_\star, A_-\}.}$$

Proof. Every proper negative subset of size at most k is handled by Theorems 19.3, 19.5 and 19.7. For the full negative set,

$$r + \sum_{a \in A_-} a = (k-1)M.$$

If $k = 1$, this is already the target. If $k \geq 2$, its distance is $k-1$ by maximal descent and the obvious a_0-completion. Hence $A_- \in \mathcal{H}(r)$.

At r', the same prefix leaves $(k-1)M+1$, which cannot be completed in the required $k-1$ steps. No face containing a_+ is optimal. If its prefix cost already exceeds N, this is immediate; otherwise the same maximum-descent argument used above applies. □

Accordingly,

$$\boxed{\text{through order } k \text{ the complexes differ by one } k\text{-face,}}$$

whereas

$$\boxed{\text{the full complexes differ by two faces.}}$$

This distinction is retained throughout this Part.

19.6 Large optimization-induced minimal nonfaces

For $k \geq 2$, on the distinguished vertex set $J_\star$,

$$\mathcal{H}(r)|_{J_\star} = \Delta^{k-1},$$

while

$$\mathcal{H}(r')|_{J_\star} = \partial\Delta^{k-1}.$$

For $k = 1$, the distinguished vertex is present at r and absent at r'. Thus shortest-path optimality creates minimal nonfaces of arbitrarily large cardinality inside a globally commuting translation system. For $k \geq 3$, the resulting restricted complex is nonflag.

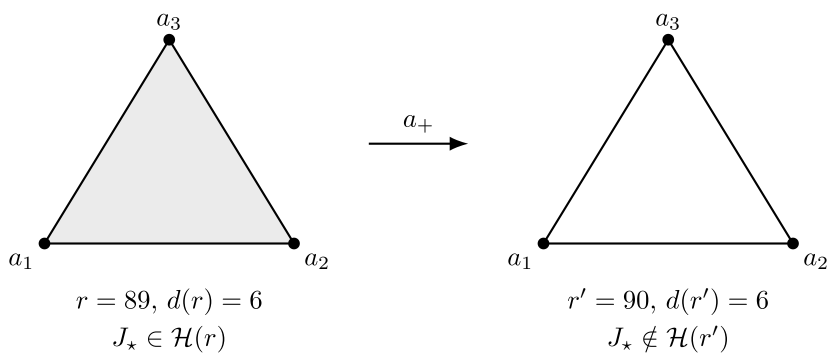


Figure 19.1: Prescribed-order separation for $k = 3$. The adjacent states $r = 89$ and $r' = 90$ have equal shortest distance and identical optimal-compatibility data through order 2. On the distinguished action set $J_\star = \{a_1, a_2, a_3\}$, the restriction of the optimal-compatibility complex changes from a filled triangle to its boundary. Thus the first discrepancy occurs exactly at order 3.

The novelty claim is not the existence of nonflag complexes or simplex boundaries in isolation; such phenomena are classical in semigroup topology [110]. The relevant constraint package is adjacency, equal shortest distance, one fixed action system, agreement of all lower compatibility orders, and first discrepancy at a prescribed order.

Remark 19.11 (Computational verification). As an independent regression check, the binary-deficit family was exhaustively verified for $1 \leq k \leq 10$. The distance formula, the exact lower-order agreement, and the full-complex identities of Theorem 19.10 matched the symbolic formulas in every case. Independent product-graph checks were also performed for representative small (k, m) pairs and recovered exactly the Boolean signatures of Theorem 19.16. These checks are not premises of any theorem.

19.7 Boolean Amplification

We now take independent products of the separating block.

19.7.1 Product formulas

Consider m periodic action systems with disjoint action alphabets.

Lemma 19.12 (Product distance)**.** *For*

$$V = (v_1, \dots, v_m),$$

$$\boxed{d(V) = \sum_{t=1}^{m} d_t(v_t).}$$

Proof. Every product path projects to a path in each factor, giving the lower bound. Interleaving shortest factor paths gives equality. □

For $v_t = (z_t, \phi_t)$, write $\mathcal{S}_t(\phi_t)$ for the structural complex in factor t, and write $\mathcal{S}(V)$ for the product structural complex at V. For an action set

$$J = J_1 \sqcup \cdots \sqcup J_m,$$

structural compatibility factors:

$$J \in \mathcal{S}(V) \iff J_t \in \mathcal{S}_t(\phi_t) \quad \forall t.$$

Distinct factor-subset tuples give distinct product endpoints, so nondegeneracy also factors.

Lemma 19.13 (Product-join lemma)**.**

$$\boxed{\mathcal{H}(V) = \mathcal{H}_1(v_1) * \cdots * \mathcal{H}_m(v_m).}$$

Proof. First, $T_J V$ is target-reachable exactly when every $T_{J_t} v_t$ is target-reachable. In that case define

$$\sigma_t = c_{J_t} + d_t(T_{J_t} v_t) - d_t(v_t) \geq 0.$$

Then

$$c_J + d(T_J V) - d(V) = \sum_{t=1}^{m} \sigma_t.$$

The sum vanishes exactly when every σ_t vanishes. □

19.7.2 Product family

Take m independent copies of the Chapter 19 block. Write

$$r^{(0)} = r, \qquad r^{(1)} = r',$$

and for $\varepsilon \in \{0,1\}^m$ define

$$R^{\varepsilon} = \big(r^{(\varepsilon_1)}, \dots, r^{(\varepsilon_m)}\big).$$

Let $J_\star^{(t)}$ be the distinguished k-face in factor t.

Proposition 19.14 (Constant lower-order signature)**.** *For every $\varepsilon, \eta \in \{0,1\}^m$,*

$$d(R^{\varepsilon}) = d(R^{\eta}) = mN$$

and

$$\boxed{\Gamma_{k-1}(R^{\varepsilon}) = \Gamma_{k-1}(R^{\eta}).}$$

Proof. Distance additivity gives the first assertion. Every factor component of an action set of total size at most $k-1$ has size at most $k-1$, where the two base states have identical compatibility. Apply Theorem 19.13. ☐

Proposition 19.15 (Exact order-k variation)**.** *There exists a fixed complex $\mathcal{K}_{k,m}$ such that*

$$\boxed{\mathcal{H}_{\le k}(R^{\varepsilon}) = \mathcal{K}_{k,m} \cup \{J_\star^{(t)} : \varepsilon_t = 0\}.}$$

Proof. If an order-$\le k$ face uses more than one factor, then every factor component has size $< k$, so its membership is determined entirely by the common lower-order base data. If all k actions lie in one factor, Theorem 19.7 says the only variable face is that factor's distinguished $J_\star$. ☐

Theorem 19.16 (Boolean fiber amplification)**.** *For every $k, m \ge 1$, there exists a completion-total one-phase unit-cost translation system containing 2^m states with one common order-$(k-1)$ signature and 2^m pairwise distinct order-k signatures.*

The variable part of the signature is precisely

$$\{J_\star^{(t)} : \varepsilon_t = 0\},$$

so the constructed order-k signatures form a Boolean lattice of rank m.

The states themselves form a directed m-cube in the underlying state graph, because changing $\varepsilon_t : 0 \to 1$ is exactly the factor action $a_+^{(t)}$. Throughout this state cube,

$$d(R^{\varepsilon}) = mN$$

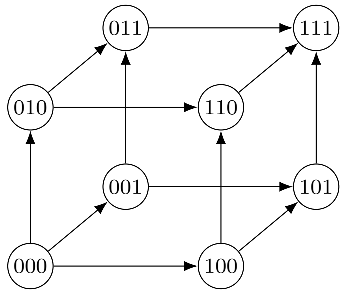
and $\Gamma_{k-1}(R^{\varepsilon})$ remain constant, whereas Γ_k distinguishes every vertex.

19.7.3 Exponential fibers

Each factor has $k+2$ action labels. Thus the product has

$$s = m(k+2)$$

actions.



$d(R^\varepsilon) = 3N$ for all ε

$\Gamma_{k-1}(R^\varepsilon)$ is constant

Γ_k distinguishes all 8 vertices

coordinate t toggles $J_\star^{(t)}$

Figure 19.2: Boolean amplification for three independent factors. The eight product states form a directed state cube. Exact distance and the complete order-$(k-1)$ signature are constant throughout the cube. Moving in coordinate t toggles exactly the distinguished order-k face $J_\star^{(t)}$, producing eight distinct order-k signatures.

Corollary 19.17 (Exponential signature fibers)**.** *For every fixed $k \geq 1$ and every $m \geq 1$, there is a system with*

$$s = m(k+2)$$

actions for which one order-$(k-1)$ signature supports at least

$$\boxed{2^m = 2^{s/(k+2)}}$$

distinct order-k signatures. Thus exponentially large fibers occur along the infinite sequence of action dimensions divisible by $k+2$.

19.7.4 Conditional discrimination

Suppose an auxiliary summary $C(v)$ is required to support an exact decoder

$$\Gamma_k(v) = F\big(\Gamma_{k-1}(v), C(v)\big)$$

on the constructed family, with no raw state coordinate supplied separately.

Corollary 19.18 (Conditional discrimination bound)**.**

$$\boxed{|\operatorname{im} C| \geq 2^m}$$

and therefore

$$\boxed{\log_2 |\operatorname{im} C| \geq m = \frac{s}{k+2}.}$$

Proof. All R^ε have the same lower-order signature but pairwise distinct order-k signatures. Hence two distinct states cannot receive the same summary value. □

This is a discrimination bound for exact summaries under the stated decoder model. It is not a general RAM-space lower bound.

19.8 From Boolean amplification to k-fibers

The product theorem proves that a common order-$(k-1)$ signature can support many different order-k extensions, but its independent bits live in disjoint factors. This leaves a sharper structural question: after exact distance and the *entire* lower-order compatibility complex have been frozen, how much freedom can remain among overlapping k-faces on one structural simplex?

Definition 19.19 (k-fiber)**.** Fix a phase ϕ, let $k \geq 1$, and let γ be a realizable value of Γ_{k-1} at phase ϕ. The *k-fiber above* γ is

$$\mathcal{F}_k(\phi, \gamma) = \{\mathcal{H}_{=k}(v) : v \text{ is in phase } \phi,\ \Gamma_{k-1}(v) = \gamma\}.$$

The word *fiber* here refers to a fiber of the truncation map on optimal-compatibility signatures. It is distinct from the optimal Parikh fibers of Chapter 12, which collect minimum completion vectors rather than higher-order compatibility layers.

For $A \subseteq \{J \in \mathcal{S}_\phi : |J| = k\}$ define the exact realization region

$$\Omega^{(k)}_{\phi,\gamma,A} = \{z : \Gamma_{k-1}((z, \phi)) = \gamma,\ \mathcal{H}_{=k}((z, \phi)) = A\}.$$

Theorem 19.20 (Finite k-fiber compilation)**.** *For every fixed positive-cost finite-phase periodic action system, every phase ϕ, every $k \geq 1$, and every realizable order-$(k-1)$ signature γ:*

(i) every region $\Omega^{(k)}_{\phi,\gamma,A}$ is effectively Presburger-definable;

(ii) nonemptiness of every such region is decidable;

(iii) the complete finite fiber $\mathcal{F}_k(\phi, \gamma)$ is effectively computable.

Proof. The structural complex $\mathcal{S}_\phi$ is finite. Exact equality $\Gamma_{k-1}((z, \phi)) = \gamma$ is a finite Boolean combination of the distance predicate and the compiled face guards for structural faces of size at most $k - 1$. Exact equality $\mathcal{H}_{=k}((z, \phi)) = A$ is a finite conjunction of positive guards for $J \in A$ and negative guards for structural k-faces not in A. Hence every exact realization region is Presburger-definable. Presburger decidability determines which of the finitely many candidate patterns occur, and those patterns are exactly the k-fiber. $\square$

This theorem is an effective classification statement rather than a polynomial-time claim. It converts the statewise compiler of Chapter 18 into a finite classifier for all possible next-order extensions of a fixed lower-order signature.

19.9 The universal two-phase construction

Fix integers

$$s \geq k \geq 2, \qquad [s] = \{1, \ldots, s\}, \qquad \mathcal{E} = \binom{[s]}{k}, \qquad m = |\mathcal{E}| = \binom{s}{k},$$

and put

$$R = \binom{s-1}{k-1}.$$

We construct one fixed two-phase periodic action system $\mathcal{U}_{s,k}$.

Let the phases be G and C, with target $(0, C)$. The coordinate set is

$$Q = \{b\} \sqcup \{p_1, \ldots, p_s\} \sqcup \{h_E : E \in \mathcal{E}\},$$

so the lattice dimension is $1+s+\binom{s}{k}$. The coordinate b balances distances across witnesses, the private coordinates p_i force subset-endpoint injectivity, and the incidence coordinates h_E carry the next-order compatibility bits.

For each $i \in [s]$, introduce a distinguished action

$$a_i : G \to G$$

with translation

$$u_i = e_{p_i} + \sum_{\substack{E \in \mathcal{E} \\ i \in E}} e_{h_E}$$

and cost

$$c(a_i) = R + 1 = \|u_i\|_1.$$

Write

$$I_\star = \{a_1, \dots, a_s\}.$$

Completion is isolated in the second phase. Introduce a switch

$$\sigma : G \to C$$

with zero translation and cost 1. For every $q \in Q$, introduce unit-cost completion actions

$$\xi_q^+, \xi_q^- : C \to C$$

with translations $+e_q$ and $-e_q$, respectively. No completion action is defined in phase G, and no distinguished action is defined in phase C.

Lemma 19.21 (Completion-totality)**.** *Every state of $\mathcal{U}_{s,k}$ is target-reachable.*

Proof. From phase C, move coordinatewise to the origin. From phase G, apply σ and then complete coordinatewise in phase C. □

19.10 Exact metric and structural isolation

Lemma 19.22 (Completion-phase metric)**.** *For every $x \in \mathbb{Z}^Q$,*

$$d(x, C) = \|x\|_1.$$

Proof. Coordinate unit moves attain $\|x\|_1$. Conversely, the total cost of any completion path is the sum of the ℓ_1-norms of its unit translations and is therefore at least the norm of the total displacement $-x$. □

Lemma 19.23 (Geodesic-phase metric)**.** *For every $x \in \mathbb{Z}^Q$,*

$$d(x, G) = 1 + \|x\|_1.$$

Proof. Switching immediately and completing coordinatewise gives the upper bound. Conversely, suppose a path first performs a distinguished prefix with total displacement U and cost C_U, then switches to phase C. Every distinguished displacement is coordinatewise nonnegative and has cost equal to its ℓ_1-norm, so $C_U = \|U\|_1$. By Lemma 19.22, the remaining cost is at least $\|x + U\|_1$. Hence the total cost is at least

$$\|U\|_1 + 1 + \|x + U\|_1 \geq 1 + \|x\|_1.$$

□

Lemma 19.24 (Distinguished structural simplex)**.** *The action set $I_\star$ spans a structural simplex in phase G.*

Proof. The distinguished actions are translations that preserve phase G, so every ordering of every subset is executable with the same displacement and cost. The p_i-coordinate of the total displacement of a subset is 1 exactly when a_i is present; therefore distinct subsets have distinct endpoints. □

Lemma 19.25 (Auxiliary structural isolation)**.** *The structural complex in phase G is*

$$\mathcal{S}_G = 2^{I_\star} \cup \{\{\sigma\}\}.$$

In particular, no structural face contains both σ and a distinguished action.

Proof. The singleton $\{\sigma\}$ is structural. If a set contains σ and some a_i, the ordering that executes σ first cannot execute a_i afterward because a_i is undefined in phase C. No completion action is defined in phase G. $\square$

19.11 Hypergraph-coded witness states

For each

$$A \subseteq \mathcal{E},$$

let $L = m + 1$ and define $v_A \in \mathbb{Z}^Q$ by

$$\begin{aligned}
(v_A)_b &= -(L - |A|), \\
(v_A)_{p_i} &= -1 \qquad (i \in [s]), \\
(v_A)_{h_E} &= -\big(k - 1 + \mathbf{1}_{E \in A}\big) \qquad (E \in \mathcal{E}).
\end{aligned}$$

Thus $(v_A)_{h_E} = -k$ when $E \in A$ and $-(k-1)$ otherwise. We regard (v_A, G) as the witness state encoding the k-uniform hypergraph $([s], A)$.

Lemma 19.26 (Equal witness distance)**.** *All witness states have the same exact shortest distance*

$$D_{s,k} = s + 2 + k\binom{s}{k}.$$

Proof. Since $L = m + 1$,

$$\|v_A\|_1 = (L - |A|) + s + m(k-1) + |A| = s + 1 + mk.$$

Apply Lemma 19.23. $\square$

For $J \subseteq [s]$, write

$$U_J = \sum_{i \in J} u_i.$$

The incidence identity driving the construction is

$$(U_J)_{h_E} = |J \cap E| \qquad (E \in \mathcal{E}). \tag{19.1}$$

Also $(U_J)_{p_i} = \mathbf{1}_{i \in J}$, $(U_J)_b = 0$, and

$$\|U_J\|_1 = |J|(R+1).$$

Lemma 19.27 (Coordinatewise geodesic criterion)**.** *For every $A \subseteq \mathcal{E}$ and every $J \subseteq [s]$,*

$$\{a_i : i \in J\} \in \mathcal{H}(v_A, G)$$

if and only if

$$U_J \le -v_A$$

coordinatewise.

Proof. By Lemma 19.23, the Bellman equality for the distinguished prefix is

$$\|U_J\|_1 + 1 + \|v_A + U_J\|_1 = 1 + \|v_A\|_1.$$

Every coordinate of v_A is nonpositive and every coordinate of U_J is nonnegative. Equality therefore holds exactly when no coordinate overshoots zero, equivalently $U_J \le -v_A$ coordinatewise. $\square$

19.12 Universal k-fiber realization

Lemma 19.28 (Complete lower-order invariance)**.** *For every $A, B \subseteq \mathcal{E}$,*

$$\mathcal{H}_{\leq k-1}(v_A, G) = \mathcal{H}_{\leq k-1}(v_B, G).$$

More precisely, every distinguished subset of cardinality at most $k-1$ is optimal at every witness state, the switch singleton $\{\sigma\}$ is optimal at every witness state, and these are all structural faces of phase G through order $k-1$.

Proof. Let $J \subseteq [s]$ with $|J| \leq k-1$. At a private coordinate,

$$(U_J)_{p_i} \leq 1 = -(v_A)_{p_i}.$$

At an incidence coordinate, (19.1) gives

$$(U_J)_{h_E} = |J \cap E| \leq |J| \leq k-1 \leq -(v_A)_{h_E}.$$

The balance coordinate receives zero distinguished displacement, so Lemma 19.27 makes every such J optimal for every A. The switch singleton is optimal because

$$1 + d(v_A, C) = 1 + \|v_A\|_1 = d(v_A, G).$$

By Lemma 19.25, there are no other structural faces in phase G. $\square$

Let

$$K_{s,k} = \mathcal{H}_{\leq k-1}(v_A, G), \qquad \gamma_{s,k} = (D_{s,k}, K_{s,k});$$

these are independent of A.

Lemma 19.29 (Exact k-face decoder)**.** *For every $A \subseteq \mathcal{E}$ and every $J \in \mathcal{E}$,*

$$\{a_i : i \in J\} \in \mathcal{H}(v_A, G) \iff J \in A.$$

Proof. At the coordinate h_J, (19.1) gives $(U_J)_{h_J} = k$. If $J \in A$, the threshold is $-(v_A)_{h_J} = k$ and is saturated exactly. If $J \notin A$, the threshold is only $k-1$, so the prefix overshoots and cannot be geodesic. For $E \neq J$ with $|E| = |J| = k$,

$$|J \cap E| \leq k-1 \leq -(v_A)_{h_E},$$

while private coordinates are saturated at most once and the balance coordinate is unchanged. Apply Lemma 19.27. $\square$

Theorem 19.30 (Universal k-Fiber Theorem)**.** *For every $s \geq k \geq 2$, there exists a fixed completion-total positive-cost two-phase periodic action system $\mathcal{U}_{s,k}$, a phase G, a distinguished structural s-simplex $I_\star$, and a realizable order-$(k-1)$ signature $\gamma_{s,k}$ whose complete k-fiber is the full power set of the distinguished k-faces:*

$$\boxed{\mathcal{F}_k(G, \gamma_{s,k}) = 2^{\binom{I_\star}{k}}.}$$

Indeed, the witness family

$$\{(v_A, G) : A \subseteq \binom{[s]}{k}\}$$

contains $2^{\binom{s}{k}}$ equal-distance states with one common complete lower-order compatibility truncation, and

$$\mathcal{H}_{=k}(v_A, G) = \big\{\{a_i : i \in J\} : J \in A\big\}.$$

Proof. Equal distance and the common order-$(k-1)$ signature follow from Lemmas 19.26 and 19.28. Since $k \geq 2$, Lemma 19.25 implies that every structural k-face in phase G lies entirely in the distinguished simplex. Hence every member of the fiber is a subset of $\binom{I_\star}{k}$. The exact decoder of Lemma 19.29 realizes every such subset as A ranges over $2^{\binom{[s]}{k}}$. $\square$

Corollary 19.31 (Uniform-hypergraph realization)**.** *For every $s \geq k \geq 2$, every k-uniform hypergraph on $[s]$ occurs as the complete order-k optimal-compatibility layer on one fixed distinguished structural simplex, while exact shortest distance and the complete optimal-compatibility structure through order $k-1$ remain unchanged.*

Corollary 19.32 (Graphs inside one 2-fiber)**.** *For every $s \geq 2$, one fixed periodic shortest-path system has a common order-1 signature whose order-2 fiber realizes every simple graph on s distinguished vertices.*

Corollary 19.33 (Overlapping-face independence)**.** *Let $\mathcal{A} \subseteq \binom{[s]}{k}$ have arbitrary intersections among its members. Inside one common order-$(k-1)$ fiber, the membership bits of all faces in $\mathcal{A}$ can be prescribed independently.*

19.13 A worked universal fiber: $s = 3$, $k = 2$

Take

$$s = 3, \qquad k = 2, \qquad \mathcal{E} = \{12, 13, 23\}.$$

Then $m = 3$, $R = 2$, and $L = 4$. In the coordinate order

$$(b, p_1, p_2, p_3, h_{12}, h_{13}, h_{23}),$$

the distinguished translations are

$$\begin{aligned} u_1 &= e_{p_1} + e_{h_{12}} + e_{h_{13}}, \\ u_2 &= e_{p_2} + e_{h_{12}} + e_{h_{23}}, \\ u_3 &= e_{p_3} + e_{h_{13}} + e_{h_{23}}, \end{aligned}$$

all of cost 3.

For

$$A = \{12, 23\},$$

the witness vector is

$$v_A = (-2, -1, -1, -1, -2, -1, -2).$$

Its ℓ_1-norm is 10, so $d(v_A, G) = 11$; the same distance holds for all eight choices of $A \subseteq \{12, 13, 23\}$. Every distinguished singleton and the switch singleton are optimal at every witness, so

$$\mathcal{H}_{\leq 1} = \{\varnothing, \{\sigma\}, \{a_1\}, \{a_2\}, \{a_3\}\}$$

is fixed. At order two, the pairs 12 and 23 saturate their threshold-2 coordinates, while 13 overshoots the threshold-1 coordinate h_{13}. Hence

$$\mathcal{H}_{=2}(v_A, G) = \big\{\{a_1, a_2\}, \{a_2, a_3\}\big\}.$$

As A ranges over all subsets of $\{12, 13, 23\}$, the common order-1 signature remains fixed while the order-2 layer runs through all eight simple graphs on the three distinguished vertices.

19.14 Exact next-order capacity

Definition 19.34 (Distinguished k-fiber capacity)**.** For $s \geq k \geq 2$, define $\mathrm{Cap}_k(s)$ to be the supremum, over all positive-cost finite-phase periodic action systems, phases ϕ, realizable order-$(k-1)$ signatures γ, and distinguished action sets $I_\star$ with $|I_\star| = s$ and $2^{I_\star} \subseteq \mathcal{S}_\phi$, of

$$\left|\left\{\mathcal{H}_{=k}(v) \cap \binom{I_\star}{k} : \Gamma_{k-1}(v) = \gamma\right\}\right|.$$

Theorem 19.35 (Exact distinguished-layer capacity)**.** *For every $s \geq k \geq 2$,*

$$\boxed{\mathrm{Cap}_k(s) = 2^{\binom{s}{k}}.}$$

Proof. An s-vertex simplex has exactly $\binom{s}{k}$ possible k-faces, so no fiber can induce more than $2^{\binom{s}{k}}$ patterns on them. Theorem 19.30 realizes every such pattern inside one common fiber. ☐

Thus the universal theorem does not merely give an exponential lower bound: relative to a fixed distinguished structural simplex, it attains the maximum combinatorially possible next-order freedom.

19.15 From separation to universality

The progression inside this Part is now exact. Prescribed-order threshold separation arranges that every prefix of size at most $k-1$ lies below two nearby thresholds while one selected k-prefix distinguishes them. Boolean amplification places independent copies in disjoint factors. Universal k-fiber realization removes the product mechanism: all variable k-faces live on one fixed set of s distinguished actions and may overlap arbitrarily.

The arithmetic replacement for factor independence is the incidence identity

$$(U_J)_{h_E} = |J \cap E|.$$

Every proper lower-order prefix sees at most $k-1$ in every incidence coordinate, while a k-set reaches the value k only in its own coordinate. The architecture separates four jobs: h_E coordinates carry the variable next-order bits, p_i coordinates enforce structural injectivity, the balance coordinate b fixes the common distance, and the completion phase C supplies completion-totality without creating mixed structural faces.

Accordingly, the progression is

$$\boxed{\begin{array}{c}\text{prescribed-order threshold separation}\\ \Downarrow\\ \text{Boolean amplification in disjoint factors}\\ \Downarrow\\ \text{universal overlapping } k\text{-fiber realization.}\end{array}}$$

19.16 Relation to realization theories and scope

The surrounding ingredients remain classical: periodic shortest paths, semilinear and Presburger descriptions, cubical state complexes, and squarefree divisor complexes were already separated from the contribution in Chapters 17–18. Shortest-path computation on a *given* metric simplicial complex is another established setting [113]; here the simplicial object is itself selected by joint shortest-prefix optimality in an infinite periodic graph. Broad realization behavior is also known for squarefree divisor complexes [110, 6], and parametric Presburger arithmetic controls complex types in affine-semigroup families [114].

No novelty is claimed for simplicial or hypergraph realization in isolation, for commuting actions forming cubes, or for Presburger definability. The structural constraint package is the point: one fixed periodic shortest-path system, one fixed phase, one fixed exact distance, one fixed complete compatibility truncation through order $k-1$, one fixed structural simplex, and every possible order-k pattern on that simplex. The capacity theorem is deliberately stated relative to the distinguished structural simplex; it makes no claim in terms of the total number of auxiliary action labels in the completion-total system.

19.17 Synthesis: finite compilation and universal next-order freedom

The compiler and structure theorems now give three levels of conclusion. First, every fixed positive-cost finite-phase periodic action system admits a finite arithmetic description of exact metric and local optimal-compatibility data. Second, the hierarchy does not collapse: for every prescribed k, a first discrepancy can occur exactly at order k, and independent products amplify such discrepancies Booleanly. Third, the ambiguity left after fixing all lower-order data can be maximal rather than sparse: one common lower-order signature can support every order-k pattern on a distinguished s-simplex.

Thus

$$\boxed{\text{shortest-path geometry is finite in description}}$$

while

$$\boxed{\begin{array}{c}\text{optimal compatibility is unbounded in informational order}\\ \text{and can be universally free at the next order.}\end{array}}$$

Equivalently, fixing exact distance and the entire compatibility hierarchy through order $k-1$ imposes no universal restriction on the next-order layer. The $\binom{s}{k}$ possible distinguished k-faces can carry exactly $\binom{s}{k}$ independent binary bits, yielding the full pattern count $2^{\binom{s}{k}}$.

The symbolic proofs above are primary. Appendix D records independent small-parameter regression checks of common distance, complete lower-order invariance, and exhaustive realization of all prescribed k-face patterns.

19.18 Returning to the original machine from the opposite direction

The compiler Part completes the outward abstraction of shortest-completion geometry. We now zoom back in.

In Chapter 1 we deliberately marked one local transition of the original 2:1 DPDA:

$$\boxed{(L, 0S) \xrightarrow{0} (L, S).}$$

So far that transition has been interpreted as:

- a stack cancellation;
- one unit of residual correction;

- one edge of a shortest-completion ladder.

But it contains another structure that the completion and compiler theories do not use. When the nonnegative 2:1 computations are drawn as lattice paths, the cancellation occurs at a precise arithmetic family of landing heights. That observation leads to an independent branch:

pushdown cancellation → correction signatures → intrinsic defect → Catalan geometry.

We now return to the 2023 machine and follow that transition.

Part VII

One Cancellation Transition Becomes a Geometry

Research provenance. This Part develops the Fuss–Catalan defect geometry and its connection back to the original cancellation transition. Generalized Dyck paths, Fuss–Catalan enumeration, and related Catalan statistics are classical or independently studied background [7, 5, 30, 48]; singularity analysis is used in its standard analytic-combinatorics role [21].

Chapter 20

Zooming Back into the Original DPDA

The previous Part generalized shortest completion outward.

We now move in the opposite direction.

Return to the original 2:1 DPDA of Chapter 1 and isolate one local transition:

$$\boxed{(L, 0S) \xrightarrow{0} (L, S).}$$

At the machine level, this is a cancellation.

At the residual level, it decreases

$$\rho = N_1 - 2N_0$$

by 2.

At the geodesic level, Chapter 3 showed that it consumes one required zero along a shortest-completion path.

There is, however, another structure hidden in the same event.

If the input word is drawn as a nonnegative 2-Dyck path, the cancellation occurs at a precise arithmetic set of landing heights.

That observation starts a second geometry.

20.1 The machine facts we reuse, not re-prove

The cancellation analysis needs only two facts already established in Chapter 1:

Reachable-store normal form

$$\mathcal{S}_C = 0^* \cup 1^* \cup \{10\},$$

$$\mathcal{S}_L = 0^*,$$

$$\mathcal{S}_R = 1^* \cup \{0\}.$$

State-potential invariant

For every consumed prefix,

$$\boxed{\rho(w) = D(S) + \Phi(q),}$$

with

$$D(S) = 2N_0(S) - N_1(S),$$

$$\Phi(C) = 0, \qquad \Phi(L) = 3, \qquad \Phi(R) = -3.$$

We do not duplicate those proofs here.

Their consequence for positive odd residuals is immediate.

Corollary 20.1 — Positive odd residuals.

Every reachable core configuration with positive odd residual at least 3 is uniquely of the form

$$\boxed{(L, 0^k)}$$

for some

$$k \geq 0,$$

and its residual is

$$\boxed{2k + 3.}$$

The left state therefore has a very precise path-height meaning.

20.2 The nonnegative path sector

Encode input symbols as steps:

$$1 \longleftrightarrow U \quad \text{of height } +1,$$

$$0 \longleftrightarrow D \quad \text{of height } -2.$$

For a prefix w, the path height is

$$\boxed{h(w) = N_1(w) - 2N_0(w) = \rho(w).}$$

A word whose prefix heights are all nonnegative and whose final height is 0 is a 2-Dyck excursion.

The cancellation statistic will first be discovered inside this nonnegative sector.

This restriction matters.

The later combinatorial theory will be defined directly on generalized Dyck paths and will no longer depend on the remaining DPDA transition table.

20.3 Exact landing-height correspondence

Immediately before a distinguished cancellation,

$$(L, 0^k) \xrightarrow{0} (L, 0^{k-1}),$$

the logical top 0 must exist, so

$$k \geq 1.$$

The pre-step residual is

$$2k + 3.$$

Reading 0 decreases the height by 2, so the down-step lands at

$$2k + 1.$$

Therefore every cancellation lands at one of:

$$3, 5, 7, \ldots$$

The converse is equally rigid.

Proposition 20.2 — Motivating $d = 2$ correspondence.

Within the nonnegative 2-Dyck sector of the original computation, the transition

$$\boxed{0, 0/\varepsilon}$$

occurs if and only if the corresponding down-step lands at height

$$\boxed{3, 5, 7, \ldots}$$

Proof.

The forward direction was just computed.
Conversely, suppose a down-step lands at

$$2k + 1 \geq 3.$$

Its pre-step height is

$$2k + 3 \geq 5.$$

This is a positive odd residual.
Corollary 17.1 therefore forces the pre-step configuration to be

$$(L, 0^k)$$

with

$$k \geq 1.$$

The enabled input-0 transition is exactly the distinguished left cancellation. □

This theorem is the only point at which the full original machine is needed to motivate the correction statistic.

Everything that follows will be stated intrinsically for paths.

20.4 From one machine event to a d-Dyck statistic

For

$$d \geq 2,$$

a d-Dyck excursion has:

- up-step U of height $+1$;
- down-step D of height $-d$;
- nonnegative prefix heights;
- final height 0.

A size-n excursion has:

$$n$$

down-steps and

$$dn$$

up-steps.
The original $d = 2$ cancellation landing heights were

$$3, 5, 7, \ldots$$

or equivalently

$$h_{\text{end}} \equiv 1 \pmod 2, \qquad h_{\text{end}} \geq 3.$$

This suggests the intrinsic generalization.

Definition 20.3 — Correction.
A down-step is a **correction** if

$$\boxed{h_{\text{end}} \equiv 1 \pmod d, \qquad h_{\text{end}} \geq d + 1.}$$

Equivalently,

$$h_{\text{end}} = d + 1 + dr$$

for a unique

$$r \in \mathbb{N}.$$

The parameter d now belongs to the combinatorial path system.
For $d > 2$, we are **not** claiming that the original DPDA itself implements this generalized rule.
The correct causal statement is:

$$\boxed{\begin{array}{c}\text{original } d = 2 \text{ cancellation} \to \text{intrinsic landing-height condition} \\ \to \text{combinatorial } d\text{-generalization.}\end{array}}$$

20.5 Chronological correction signatures

Suppose a path contains $m \geq 1$ corrections, in chronological order, landing at heights

$$c_1, \ldots, c_m.$$

Write

$$c_i = d + 1 + dr_i.$$

Definition 20.4 — Normalized correction signature.

The ordered tuple

$$\boxed{r(P) = (r_1, \ldots, r_m)}$$

is the normalized correction signature.

The order is essential.

Two paths can have the same number of corrections and even the same multiset of correction heights while having different ordered signatures.

This branch therefore remembers more than the scalar count statistic.

20.6 Path defect

If a size-n path has $m \geq 1$ corrections, define

$$\boxed{\delta(P) = n - m - 2.}$$

The form may look artificial before the realization theorem.

It is chosen so that:

- maximal-correction paths have defect 0;
- every proposed ordered signature has an exact minimum defect;
- the excess above that minimum can be increased freely by zero-to-zero padding.

The theorem in Chapter 18 will prove in particular that

$$\delta(P) \geq 0.$$

20.7 The new inverse problem

The original question was forward:

> Given a path, where do cancellations/corrections occur?

The signature theory asks the inverse question:

> Given an ordered sequence
>
> $$r = (r_1, \ldots, r_m),$$
>
> can it occur as a correction history?

And, if so:

What is the smallest path defect at which it can occur?

This is much stronger than counting corrections.
The answer is one explicit functional

$$\Delta(r).$$

To derive it, we first need the exact local price of moving from one correction height to the next.

Chapter 21

Correction Signatures and Minimum Defect

Let

$$r = (r_1, \ldots, r_m) \in \mathbb{N}^m, \qquad m \geq 1.$$

The goal is to determine exactly when r is realizable as a normalized correction signature.

The answer is built from two local costs:

1. the price of moving between consecutive corrections;
2. the price of terminating after the final correction.

The first cost contains the key nonlinearity.

21.1 The threshold function

Define

$$\boxed{\psi(a) = \begin{cases} 0, & a \leq 1, \\ a, & a \geq 2. \end{cases}}$$

Why is there a jump from zero to a?

Suppose consecutive normalized correction heights are

$$r \quad \text{and} \quad r',$$

and set

$$a = r - r'.$$

Let b be the number of non-correction down-steps strictly between the two corrections, and let u be the number of intervening up-steps.

Height balance gives

$$\boxed{u = d(b + 1 - a).}$$

This one equation explains the threshold.

21.2 Local Gap Cost

Lemma 21.1 — Local Gap Cost.

For two consecutive normalized correction heights r, r', with

$$a = r - r',$$

the number b of intervening non-correction down-steps satisfies

$$\boxed{b \geq \psi(a).}$$

The bound is sharp.

Proof.

If

$$a \leq 1,$$

choose

$$b = 0.$$

Then

$$u = d(1 - a) \geq 0.$$

The block

$$\boxed{U^{d(1-a)}D}$$

moves directly to the next correction without an intervening down-step.

Now assume

$$a \geq 2.$$

The balance equation gives

$$b \geq a - 1.$$

If

$$b = a - 1,$$

then

$$u = 0.$$

But the first intervening down-step would land again in the correction residue class, creating an unintended correction between the two prescribed ones.

Therefore

$$b \geq a.$$

Equality is attainable by the detour block

$$\boxed{UD^aU^{d-1}D.}$$

The intermediate down-steps lie one residue class away from the correction class, so they are not corrections. The final D returns exactly to the prescribed correction height. □

This theorem is the geometric origin of ψ.

A downward change of size at most 1 is free.

A drop of size at least 2 cannot be taken directly because the naive descent would create an unwanted correction.

The path must leave the correction residue class and return.

That detour costs exactly the full drop a.

21.3 Terminal Cost

Lemma 21.2 — Terminal Cost.

If the final correction has normalized height r, then at least

$$\boxed{r+2}$$

non-correction down-steps are required after it.

The bound is sharp.

Proof.

Immediately after the final correction, the height is

$$d+1+dr.$$

If b down-steps and u up-steps remain, then

$$d+1+dr+u-db=0.$$

Hence

$$u=d(b-r-1)-1.$$

Since

$$u\geq 0,$$

we must have

$$b\geq r+2.$$

Equality is attained by

$$\boxed{U^{d-1}D^{r+2}.}$$

All its down-steps land at multiples of d, so none is a correction. □

The final signature entry therefore contributes directly to the minimum defect.

21.4 Intrinsic signature defect

Define

$$\boxed{\Delta(r) = r_m + \sum_{i=1}^{m-1} \psi(r_i - r_{i+1}).}$$

This quantity contains no d.
That universality is already striking.
The physical correction heights scale as

$$d + 1 + dr_i,$$

but the normalized realizability cost is independent of the Dyck parameter.

21.5 Canonical blocks

To prove sufficiency constructively, define:

Initial block

$$\boxed{I_r = U^{2d+1+dr} D.}$$

Its unique down-step lands at correction height

$$d + 1 + dr.$$

Gap block

For $a \in \mathbb{Z}$,

$$\boxed{G_a = \begin{cases} U^{d(1-a)} D, & a \le 1, \\ U D^a U^{d-1} D, & a \ge 2. \end{cases}}$$

Terminal block

$$\boxed{T_r = U^{d-1} D^{r+2}.}$$

Padding block

$$\boxed{P = U^d D.}$$

The padding block starts and ends at height zero and creates no correction.

Proposition 21.3 — Block Purity.

Each canonical block creates exactly the advertised correction behavior and no unintended correction.

Moreover, when started at its prescribed height, every intermediate height remains nonnegative.

This proposition is what makes the global realization theorem compositional.

21.6 Defect–Signature Characterization

Theorem 21.4 — Defect–Signature Characterization.

For every

$$d \geq 2$$

and every normalized signature

$$r = (r_1, \dots, r_m),$$

the following are equivalent:

1. r is realized by a d-Dyck excursion of defect δ;
2. $$\boxed{\Delta(r) \leq \delta.}$$

Equivalently,

$$\boxed{\delta_{\min}(r) = \Delta(r).}$$

If the signature has length m, then its minimum path size is

$$\boxed{n_{\min}(r) = m + 2 + \Delta(r).}$$

Proof — lower bound. A path realizing r contains:

$$n - m$$

non-correction down-steps.
Between correction i and correction $i+1$, Lemma 18.1 forces at least

$$\psi(r_i - r_{i+1})$$

such steps.
After the last correction, Lemma 18.2 forces at least

$$r_m + 2.$$

Any non-correction down-steps before the first correction only increase the count. Therefore

$$n - m \geq 2 + r_m + \sum_{i=1}^{m-1} \psi(r_i - r_{i+1}) = 2 + \Delta(r).$$

Hence

$$\delta(P) = n - m - 2 \geq \Delta(r).$$

Proof — construction. Let

$$a_i = r_i - r_{i+1}.$$

For

$$s \geq 0,$$

define

$$\boxed{W_d(r;s) = I_{r_1} G_{a_1} \cdots G_{a_{m-1}} T_{r_m} P^s.}$$

Block Purity shows that this is a nonnegative d-Dyck excursion whose normalized correction signature is exactly r.

Its number of down-steps is

$$m + 2 + \Delta(r) + s.$$

Thus

$$\delta(W_d(r;s)) = \Delta(r) + s.$$

Taking $s = 0$ gives a minimum realization.

Taking

$$s = \delta - \Delta(r)$$

realizes every larger defect. □

This is the structural center of the cancellation branch.

It solves both directions:

$$\boxed{\text{path} \to \text{signature}}$$

and

$$\boxed{\text{signature} \to \text{minimum realizing path.}}$$

21.7 Canonical does not mean unique

Corollary 21.5 — Canonical minimum realization.

Every signature r has the explicit minimum witness

$$\boxed{W_d^{\min}(r) = W_d(r;0).}$$

The word **canonical** means:

the distinguished block construction selected by the theory.

It does **not** mean:

every minimum realizing path is identical to this word.

This scope matters.

The theorem determines the minimum defect uniquely, not the complete set of minimum path orderings.

21.8 Signature persistence

Corollary 21.6 — Signature Persistence.
A signature r is realizable at exactly the defect levels

$$\boxed{\Delta(r), \Delta(r)+1, \Delta(r)+2, \ldots}$$

because each copy of

$$P = U^d D$$

adds one defect unit without changing the correction signature.
Therefore the intrinsic defect is a **birth level**.
A signature appears for the first time at $\Delta(r)$, then persists forever.

21.9 Maximum number of corrections

Since

$$\Delta(r) \geq 0,$$

every path with $m \geq 1$ corrections satisfies

$$\boxed{m \leq n-2.}$$

Thus defect zero is exactly the maximal-correction regime.
That boundary turns out to be Catalan.

21.10 Zero defect

Corollary 21.7 — Zero-defect signatures.

$$\boxed{\Delta(r) = 0}$$

if and only if

$$\boxed{r_m = 0}$$

and

$$\boxed{r_i - r_{i+1} \leq 1 \qquad (1 \leq i < m).}$$

Reverse the sequence:

$$a_i = r_{m+1-i}.$$

Then

$$a_1 = 0$$

and

$$\boxed{a_{i+1} \leq a_i + 1.}$$

These are exactly the usual Dyck area-sequence inequalities.

Therefore the number of zero-defect signatures of length m is

$$\boxed{C_m.}$$

The Catalan family is not the whole correction theory.
It is its exact zero-cost boundary.

21.11 Defect as penalized overshoot

After reversal,

$$a_i = r_{m+1-i},$$

the defect becomes

$$\boxed{\Delta^{\leftarrow}(a) = a_1 + \sum_{\substack{1 \le i < m \\ a_{i+1} - a_i \ge 2}} (a_{i+1} - a_i).}$$

Thus ordinary Dyck area sequences are precisely the nonnegative sequences with zero cost.

Positive defect allows overshoots above the Catalan nearest-neighbor condition, but every overshoot of size at least 2 is charged by its full magnitude.

This makes the geometry intuitive:

$$\boxed{\text{Catalan behavior} = \text{zero-defect regime;}}$$

$$\boxed{\text{non-Catalan jumps} = \text{quantified correction debt.}}$$

The next chapter shows that the Catalan boundary is not only enumerative.
Its maximal-correction paths themselves contain a full binary tree.

Chapter 22

The Catalan Boundary and the Binary Cancellation Skeleton

At defect zero,

$$m = n - 2.$$

Thus a size-n path contains the maximum possible number of corrections.
The signature count is Catalan:

$$C_{n-2}.$$

But the path count contains an additional factor d:

$$dC_{n-2}.$$

This chapter explains both factors structurally.
The intrinsic correction structure is a full binary tree.
The factor d records only how the path enters that binary regime.

22.1 Two unavoidable unmarked nodes

Use the standard first-return decomposition of a nonempty d-Dyck excursion:

$$\boxed{P = UP_1UP_2 \cdots UP_dDP_{d+1}.}$$

Each internal node corresponds to one down-step.
A node is marked exactly when its down-step is a correction.
A maximal-correction size-n path has:

$$n - 2$$

marked internal nodes and therefore exactly two unmarked internal nodes.
The root is necessarily unmarked because its baseline is zero.
Before a correction baseline

$$d + 1$$

can be reached, at least one further unmarked internal node is necessary.
In the maximal-correction regime, these are the only two unmarked nodes.
That rigid entrance forces the rest of the tree into a binary skeleton.

22.2 The d entrance choices

Choose the unique nonempty pre-down-step child of the root.

If it is slot

$$j \in [d],$$

its baseline is j.

Every other root child must be empty, otherwise another unmarked internal node would be created.

At this second unmarked node, the unique child that reaches correction baseline $d+1$ without introducing an additional unmarked node is slot

$$d + 1 - j.$$

Thus there are exactly

$$\boxed{d}$$

possible entrance choices.

After those two forced unmarked nodes, the remaining structure is entirely inside the correction residue class.

22.3 Why correction nodes become binary

Suppose a correction node has baseline

$$h \equiv 1 \pmod{d}, \qquad h \geq d + 1.$$

Its pre-down-step child baselines are

$$h + 1, \ldots, h + d,$$

and its trailing child has baseline h.

Among these, the only baselines that remain in the correction class are:

$$h + d$$

and

$$h.$$

Therefore, once all unmarked nodes have been used up, a correction node has exactly two recursive correction slots:

- slot d;
- slot $d + 1$.

All other child slots must be empty.

The correction structure is therefore binary.

22.4 Binary Cancellation Skeleton

Let

$$\mathcal{B}_m$$

be the set of plane full binary trees with m internal nodes.
Let

$$\mathcal{M}_{d,n}$$

be the class of size-n maximally corrected d-Dyck excursions.

Theorem 22.1 — Binary Cancellation Skeleton.

For every

$$n \geq 3,$$

$$\boxed{\mathcal{M}_{d,n} \cong [d] \times \mathcal{B}_{n-2}.}$$

Consequently,

$$\boxed{Q_d(n, n-2) = dC_{n-2}.}$$

Proof.

From a maximal-correction path:

1. record the root entrance slot
$$j \in [d];$$
2. delete the two forced unmarked entrance nodes;
3. retain the remaining marked subtree;
4. reinterpret its recursive slots d and $d+1$ as the two ordered binary children.

This gives an element of

$$[d] \times \mathcal{B}_{n-2}.$$

Conversely, given:

- an entrance choice j;
- a plane full binary tree;

reinsert the two forced unmarked nodes and place the binary tree recursively in the only two correction-preserving child slots.

The constructions are inverse. □

The Catalan factor is therefore not a generating-function accident.

It is the count of an actual binary cancellation skeleton.

22.5 Tree coordinates

Assign child weights

$$\omega(j) = j \qquad (1 \le j \le d),$$

and

$$\omega(d+1) = 0.$$

Lemma 22.2 — Tree-coordinate lemma.

If an internal node has root-to-node child address

$$i_1 \cdots i_s,$$

then the baseline height of its associated down-step is

$$\boxed{h(v) = \sum_{\nu=1}^{s} \omega(i_\nu).}$$

For a correction node, the two recursive binary directions are:

- **high**: slot d, increment d;
- **low**: slot $d+1$, increment 0.

The two entrance weights always sum to

$$d + 1.$$

Therefore the binary root always begins at correction height

$$d + 1,$$

independently of the entrance choice.

22.6 Correction height from the binary address

Let

$$L(v)$$

be the number of high edges on the path from the binary root to a correction node v. Then:

$$\boxed{c(v) = d + 1 + dL(v),}$$

and hence its normalized correction height is

$$\boxed{r(v) = L(v).}$$

Thus the normalized signature has a purely binary-tree interpretation.
The parameter d only rescales the physical path heights.
The normalized correction values themselves count high edges.
This is another manifestation of the universality first seen in Chapter 18.

22.7 Inorder chronology

In the d-ary path traversal:

- the slot-d correction subtree is visited before the node's closing down-step;
- the slot-$(d + 1)$ correction subtree is visited after it.

Therefore correction nodes appear in ordinary binary inorder:

$$\boxed{\text{high subtree} \to \text{node} \to \text{low subtree.}}$$

Let

$$v_1, \ldots, v_m$$

be the internal binary nodes in inorder.
Then the normalized signature is

$$\boxed{r_i = L(v_i).}$$

So the chronological correction signature is exactly an inorder depth statistic on the binary skeleton.

22.8 The signature is lossless

A Catalan-counted statistic need not normally determine the full tree.
Here it does.

Theorem 22.3 — Lossless Signature Theorem.

In the zero-defect regime, the normalized correction signature uniquely determines the binary cancellation skeleton.
Proof.
Let

$$r = (r_1, \ldots, r_m)$$

satisfy the zero-defect conditions.
Let j be the first index such that

$$r_j = 0.$$

In binary inorder, the first zero is the root.
Entries before j lie in the high subtree and each carries one additional high edge.
Therefore the normalized code of the high subtree is

$$\boxed{(r_1 - 1, \ldots, r_{j-1} - 1).}$$

Entries after the root lie in the low subtree, whose code is

$$\boxed{(r_{j+1}, \ldots, r_m).}$$

Both subsequences satisfy the same zero-defect conditions.
Recursively apply the same decomposition.
This reconstructs the tree uniquely. □
Hence:

$$\boxed{\begin{array}{c}\text{binary cancellation skeleton} \longleftrightarrow \text{zero-defect signature} \\ \longleftrightarrow \text{Dyck area sequence.}\end{array}}$$

The factor d is external to this intrinsic Catalan structure.
It remembers only the entrance choice.

22.9 Area statistic

If the zero-defect signature has length m, then

$$c_i = d + 1 + dr_i.$$

Therefore

$$\boxed{\sum_{i=1}^{m} c_i = m(d+1) + d\sum_{i=1}^{m} r_i.}$$

After reversal,

$$\sum_i r_i$$

is the ordinary area statistic of the corresponding Dyck path.
If

$$\mathcal{A}_m(v) = \sum_D v^{\mathrm{area}(D)}$$

is the Dyck area polynomial, then the maximal-correction paths satisfy

$$\boxed{\sum_{P\in\mathcal{M}_{d,m+2}} v^{\sum_i c_i} = d\, v^{m(d+1)}\mathcal{A}_m(v^d).}$$

Thus the Catalan boundary is structural at three levels:

1. number of signatures;
2. binary cancellation skeleton;
3. correction-height distribution.

22.10 What the original transition has become

We began with:

$$(L, 0S) \xrightarrow{0} (L, S).$$

At defect zero, the ordered history of occurrences of that transition becomes:

$$\boxed{\text{a full binary tree encoded losslessly by its normalized correction signature.}}$$

This is a much stronger statement than “Catalan numbers appear.”
The original transition has generated a complete combinatorial coordinate system.
The next chapter moves away from individual signatures and counts:

- paths by number of corrections;
- signatures by exact intrinsic defect.

Those are different arrays and must remain separate.

Chapter 23

Exact Correction and Defect Enumeration

The structural theory has produced three different kinds of objects.

They must not be conflated.

$$\boxed{K_d \quad \text{auxiliary kernel coefficients,}}$$

$$\boxed{Q_d(n,k) \quad \text{paths with exactly } k \text{ corrections,}}$$

and

$$\boxed{N_{m,\ell} \quad \text{normalized signatures born at exact defect } \ell.}$$

The first appears naturally inside the scalar generating-function grammar.
The second is the actual path correction statistic.
The third belongs to the intrinsic signature geometry and is independent of d.

23.1 Scalar correction generating function

Let

$$G_h(z,t)$$

count d-Dyck excursions that begin and end at baseline h, never go below h, with:

- z marking down-steps;
- t marking corrections.

A down-step landing at baseline h has weight

$$\tau_h = \begin{cases} t, & h \in \{d+1, 2d+1, \ldots\}, \\ 1, & \text{otherwise.} \end{cases}$$

The standard first-return decomposition gives

$$\boxed{G_h = 1 + z\tau_h G_{h+1} \cdots G_{h+d} G_h.}$$

Above the exceptional low boundary, the marking pattern is periodic modulo d.
This collapses the infinite family to three formal series:

- F: baseline zero;
- A: unmarked tail residue;
- B: marked correction residue.

They satisfy

$$\boxed{F = 1 + zA^d F,}$$

$$\boxed{A = 1 + zA^d B,}$$

$$\boxed{B = 1 + ztA^{d-1} B^2.}$$

Define

$$\boxed{Q_d(n,k) = [z^n t^k] F(z,t).}$$

Thus $Q_d(n,k)$ is the actual correction triangle.

23.2 The auxiliary kernel and A371395

Set

$$Y = zA^{d-1}B.$$

Then

$$A = \frac{1}{1-Y}, \qquad B = \frac{1}{1-tY},$$

and

$$\boxed{Y(1-Y)^{d-1}(1-tY) = z.}$$

Define

$$\boxed{K_d = \frac{Y}{z}.}$$

Lagrange inversion gives

$$\boxed{[z^n t^k] K_d = \frac{1}{n+1} \binom{n+k}{k} \binom{dn+d-k-2}{n-k}.}$$

For

$$d = 2,$$

this becomes

$$\boxed{[z^n t^k] K_2 = \frac{1}{n+1} \binom{n+k}{k} \binom{2n-k}{n-k}.}$$

This is OEIS A371395 [47].
But K_2 is not the correction distribution.
The actual correction generating function is

$$\boxed{F_d(z,t) = \frac{1-Y}{1-2Y+tY^2}.}$$

Proposition 23.1 — Kernel–correction separation.

For $d = 2$, A371395 is the coefficient triangle of the auxiliary kernel K_2, not the path correction triangle Q_2.

This distinction is important because the two arrays diverge immediately.

A371395 is a useful algebraic waypoint.

It is not the final statistic generated by the DPDA-derived correction event.

23.3 Exact correction triangle

Eliminating B from the three-class system and writing

$$U = A - 1$$

gives

$$\boxed{U\big(1 + (1-t)U\big) = z(1+U)^{d+1}.}$$

Lagrange–Bürmann inversion yields the exact path count.

Theorem 23.2 — Exact correction triangle.

For

$$n \geq 1, \qquad k \geq 0,$$

$$\boxed{Q_d(n,k) = \mathbf{1}_{\{k=0\}} + \sum_{j=k+1}^{n-1} \frac{d(n-j)}{j} \binom{j+k-1}{k} \binom{dn-k-1}{j-k-1}.}$$

In particular,

$$\boxed{Q_d(n,k) = 0 \qquad (k > n-2).}$$

The complete Lagrange–Bürmann coefficient derivation is given in Appendix E.

The theorem statement and its structural consequences remain in the main text.

23.4 Row sums and boundary columns

Setting

$$t = 1$$

forgets the correction marking.

The row sum is the ordinary Fuss–Catalan number:

$$\boxed{\sum_{k \geq 0} Q_d(n,k) = \frac{1}{dn+1} \binom{(d+1)n}{n}.}$$

At the zero-correction boundary,

$$\boxed{Q_d(n,0) = \frac{1}{n}\sum_{j=0}^{n-1}(n-j)\binom{dn}{j}.}$$

For

$$d = 2,$$

this simplifies to

$$\boxed{Q_2(n,0) = \binom{2n-1}{n-1}.}$$

At the maximal-correction boundary,

$$\boxed{Q_d(n,n-2) = dC_{n-2}.}$$

Chapter 19 already explained the last identity bijectively through the binary cancellation skeleton.

This is a useful example of two proof styles meeting:

- algebraic coefficient extraction;
- structural combinatorial decomposition.

23.5 Intrinsic signature enumeration

Now forget the paths and enumerate normalized signatures directly.

Define

$$\boxed{N_{m,\ell} = \#\{r \in \mathbb{N}^m : \Delta(r) = \ell\}.}$$

Let

$$\boxed{S(x,q) = \sum_{m\geq 1}\sum_{\ell\geq 0} N_{m,\ell}x^m q^\ell.}$$

Here:

- x marks signature length;
- q marks exact intrinsic minimum defect.

This enumeration is fundamentally different from $Q_d(n,k)$.

A signature of defect ℓ can be realized at every larger path defect.

Thus $N_{m,\ell}$ records the level at which a signature is **born**.

23.6 Translation and first-zero decomposition

Let $Z(x,q)$ count nonempty signatures whose minimum entry is zero.

Every nonempty signature has a unique decomposition

$$r = \bar{r} + s\mathbf{1},$$

where

$$s = \min_i r_i$$

and

$$\min_i \bar{r}_i = 0.$$

Adding 1 to every entry leaves all adjacent differences unchanged and raises the terminal term by 1.

Therefore

$$\boxed{\Delta(r + \mathbf{1}) = \Delta(r) + 1.}$$

This gives

$$\boxed{Z = (1-q)S.}$$

Now cut a minimum-zero signature at its first zero.

A second auxiliary series E counts nonempty signatures ending in zero.

The first-zero decomposition yields a closed algebraic system, which simplifies to the master equation.

23.7 Defect–Signature Master Equation

Theorem 23.3 — Defect–Signature Master Equation.

The formal power series $S(x, q)$ with zero constant term is uniquely determined by

$$\boxed{(1-q)S(1+qS) = x(1+S)^2(1+q^2S).}$$

Equivalently,

$$\boxed{S = x\frac{(1+S)^2(1+q^2S)}{(1-q)(1+qS)}.}$$

The equation contains no d.

Thus the universality of the Defect–Signature theorem persists at the enumerative level.

The path parameter d changes physical height scaling and path multiplicity.

It does not change the intrinsic normalized signature birth counts.

23.8 Exact defect coefficients

Lagrange inversion gives a finite coefficient formula for every

$$N_{m,\ell}.$$

The first four defect layers simplify to:

$$\boxed{N_{m,0} = C_m,}$$

$$\boxed{N_{m,1} = \frac{3m}{m+2}C_m = C_{m+1} - C_m,}$$

$$\boxed{N_{m,2} = \frac{m(m^2+10m+1)}{(m+2)(m+3)} C_m,}$$

and

$$\boxed{N_{m,3} = \frac{5m(m^3+7m^2+2m+2)}{(m+2)(m+3)(m+4)} C_m.}$$

The zero layer is exactly Catalan.

The positive layers quantify how the signature class expands as non-Catalan overshoots are admitted.

23.9 Cumulative realizability

Define

$$\boxed{\widehat{N}_{m,\delta} = \#\{r \in \mathbb{N}^m : r \text{ is realizable at path defect } \delta\}.}$$

By signature persistence,

$$\boxed{\widehat{N}_{m,\delta} = \sum_{\ell=0}^{\delta} N_{m,\ell}.}$$

Thus:

- $N_{m,\ell}$ counts signatures born at exact intrinsic defect ℓ;
- $\widehat{N}_{m,\delta}$ counts all signatures visible by defect δ.

At defect one,

$$\boxed{\widehat{N}_{m,1} = C_{m+1}.}$$

The first cumulative layer is still exactly Catalan, one size higher.

Defect two is the first level at which a genuinely non-Catalan local overshoot such as

$$(2,0)$$

can appear.

23.10 Three arrays, three questions

The distinction can now be stated cleanly.

K_d

An auxiliary kernel series produced by the scalar grammar.

Question:

> What coefficient array appears in the algebraic series-reversion core?

$Q_d(n, k)$

Path correction distribution.
Question:

> How many size-n d-Dyck excursions contain exactly k correction events?

$N_{m,\ell}$

Intrinsic ordered signature distribution.
Question:

> How many length-m correction histories are born at exact minimum defect ℓ?

The DPDA-derived project passes through all three, but they are not interchangeable. The next chapter studies the asymptotic hierarchy of the third array.

Chapter 24

The Asymptotic Defect Hierarchy

The master equation

$$(1-q)S(1+qS) = x(1+S)^2(1+q^2S)$$

provides exact coefficients.

It also reveals a surprisingly rigid asymptotic structure.

For fixed defect ℓ, the number of length-m signatures grows on the Catalan exponential scale:

$$4^m.$$

The defect level changes only the polynomial factor.

More precisely, defect levels occur in pairs:

$$(0,1), \quad (2,3), \quad (4,5), \ldots$$

and every two defect units increase the polynomial exponent by one.

24.1 Fixed-defect generating functions

Define

$$\boxed{S_\ell(x) = [q^\ell]S(x,q) = \sum_{m\geq 1} N_{m,\ell}x^m.}$$

Set

$$\boxed{t = \sqrt{1-4x}.}$$

At defect zero, the master equation gives

$$S_0(x) = C(x) - 1,$$

hence

$$\boxed{S_0(x) = \frac{1-t}{1+t}.}$$

Thus the dominant singularity is the Catalan point

$$x = \frac{1}{4}.$$

The same singular location controls every fixed defect layer.

24.2 Algebraicity and singularity location

Proposition 24.1 — Fixed-defect algebraicity.

For every fixed

$$\ell \geq 0,$$

$$\boxed{S_\ell(x) \in \mathbb{Q}(t), \qquad t = \sqrt{1-4x}.}$$

More precisely,

$$\boxed{S_\ell(x) \in \mathbb{Q}[t, t^{-1}, (1+t)^{-1}].}$$

Therefore, on the branch $t(0) = 1$, the only possible dominant singularity is

$$\boxed{x = \frac{1}{4}.}$$

Proof idea. Let

$$\mathcal{F}(S, q, x) = (1-q)S(1+qS) - x(1+S)^2(1+q^2S).$$

At

$$q = 0,$$

the coefficient of the unknown S_ℓ in the extracted q^ℓ equation is

$$\frac{\partial \mathcal{F}}{\partial S} = 1 - 2x(1+S_0) = t.$$

Thus each new fixed-defect layer is obtained rationally from the earlier ones by dividing by powers of t.

Induction gives the claimed field. □

This proposition justifies using singularity analysis uniformly across all fixed defect levels.

24.3 Critical scaling

To extract the leading singular behavior of all layers simultaneously, set

$$\boxed{q = ty.}$$

Seek an expansion

$$\boxed{S(x, ty) = 1 + tf(y) + t^2 g(y) + O(t^3).}$$

The branch compatible with

$$S(x, 0) = \frac{1-t}{1+t}$$

must satisfy

$$f(0) = -2.$$

Substitution into the master equation gives

$$f^2 - 4yf + 8y^2 - 4 = 0.$$

The correct branch is

$$\boxed{f(y) = 2y - 2\sqrt{1 - y^2}.}$$

At the next order,

$$\boxed{g(y) = 2\frac{4y^3 - 3y + \sqrt{1 - y^2}}{\sqrt{1 - y^2}}.}$$

The even coefficients of f and odd coefficients of g determine the dominant singular terms of successive defect layers.

This is the analytic origin of the pair structure.

24.4 Pairwise leading singularities

For

$$j \geq 1,$$

one obtains

$$S_{2j}(x) = A_j t^{1-2j} + O(t^{2-2j}),$$

and

$$S_{2j+1}(x) = B_j t^{1-2j} + O(t^{2-2j}),$$

with

$$A_j = \frac{C_{j-1}}{4^{j-1}},$$

and

$$B_j = (2j + 3)A_j.$$

The key point is that defects $2j$ and $2j + 1$ have the **same singular exponent**.

Their leading constants differ, but their polynomial growth degree is identical.

24.5 Asymptotic Defect Hierarchy

Theorem 24.2 — Asymptotic Defect Hierarchy.

For every fixed

$$j \geq 0,$$

$$\boxed{N_{m,2j} \sim \frac{m^j}{j!} C_m,}$$

and

$$\boxed{N_{m,2j+1} \sim \frac{2j+3}{j!} m^j C_m.}$$

Equivalently,

$$\boxed{N_{m,2j} \sim \frac{4^m}{j!\sqrt{\pi}} m^{j-\frac{3}{2}},}$$

and

$$\boxed{N_{m,2j+1} \sim \frac{2j+3}{j!\sqrt{\pi}} 4^m m^{j-\frac{3}{2}}.}$$

Proof.

Apply the standard transfer theorem to the leading singular terms in powers of

$$t = \sqrt{1-4x}.$$

For $j \geq 1$, the singular exponent is

$$t^{1-2j} = (1-4x)^{-(j-\frac{1}{2})}.$$

The transfer coefficient, together with the explicit constants A_j, B_j, gives the displayed formulas.

The boundary case

$$j = 0$$

uses the exact identities

$$N_{m,0} = C_m$$

and

$$N_{m,1} = C_{m+1} - C_m \sim 3C_m.$$

□

Thus:

$$\boxed{\frac{N_{m,\ell}}{C_m} = \Theta\left(m^{\lfloor \ell/2 \rfloor}\right)}$$

for fixed ℓ.

24.6 The defect pairs

The first leading ratios to the Catalan scale are:

$$1, \quad 3, \quad m, \quad 5m, \quad \frac{1}{2}m^2, \quad \frac{7}{2}m^2, \quad \frac{1}{6}m^3, \quad \frac{3}{2}m^3, \ldots$$

The hierarchy is therefore:

$$\boxed{(0,1)}$$

on the Catalan polynomial scale,

then

$$\boxed{(2,3)}$$

one power of m higher,
then

$$\boxed{(4,5)}$$

two powers higher,
and so on.
Every two defect units add one asymptotically free polynomial degree.

24.7 Geometric interpretation

The theorem is analytic.

The following interpretation comes afterward.

The zero-defect Catalan condition permits adjacent reversed-signature rises of at most 1.

A true violation requires a rise of at least 2.

Such an overshoot costs at least two defect units.

Therefore the pair structure

$$(2j, 2j+1)$$

is consistent with j asymptotically free non-Catalan defect locations among m signature positions.

The factor

$$\frac{m^j}{j!}$$

has exactly the scale expected from choosing j widely separated defect locations.

This is an interpretation of the proven singularity result, not a replacement for its proof.

The monograph should keep that distinction explicit.

24.8 Cumulative asymptotics

Since

$$\widehat{N}_{m,\delta} = \sum_{\ell \le \delta} N_{m,\ell},$$

the top defect layer dominates within each fixed cumulative cutoff.

Thus:

$$\boxed{\widehat{N}_{m,2j} \sim \frac{m^j}{j!} C_m,}$$

and

$$\boxed{\widehat{N}_{m,2j+1} \sim \frac{2j+4}{j!} m^j C_m.}$$

At

$$j = 0,$$

this agrees with the exact identity

$$\widehat{N}_{m,1} = C_{m+1}.$$

24.9 What the cancellation branch has established

We can now summarize the complete branch generated by one transition of the original DPDA:

$$\boxed{0, 0/\varepsilon}$$

$$\Downarrow$$

$$\boxed{\text{odd landing-height event in the nonnegative 2-Dyck sector}}$$

$$\Downarrow$$

$$\boxed{d\text{-Dyck correction statistic}}$$

$$\Downarrow$$

$$\boxed{\text{ordered normalized signatures}}$$

$$\Downarrow$$

$$\boxed{\delta_{\min}(r) = \Delta(r)}$$

$$\Downarrow$$

$$\boxed{\text{Catalan zero-defect boundary}}$$

$$\Downarrow$$

$$\boxed{\text{lossless binary cancellation skeleton}}$$

$$\Downarrow$$

$$\boxed{\text{exact path and signature enumeration}}$$

$$\Downarrow$$

$$\boxed{\text{paired asymptotic defect hierarchy.}}$$

The DPDA was the discovery mechanism.

The final normalized signature theory is intrinsic and no longer depends on the machine's remaining states or stack alphabet.

24.10 Chronological carry as an arithmetic offshoot

The ordered correction signature remembers chronology.

Pushing that chronology one step further attaches an explicit third-order arithmetic interaction to:

- singleton correction positions;
- adjacent pairs;
- connected triples.

That construction produces the CCR-3 recurrence.

It then develops a modular Stirling shadow, a simple Hensel branch, singular extinction, and ramified p-adic recovery.

This arithmetic refinement is a genuine continuation of the correction branch, but it is not a prerequisite for the later general real-time $p{:}q$ DPDA theory.

Accordingly, the following Part is presented as a specialized offshoot.

Readers interested only in the main normalization spine may skip it and rejoin when the pushdown-machine line resumes.

Part VIII

Chronological Carry and the Arithmetic Offshoot

Research provenance. This Part develops the chronological-carry refinement and p-adic root analysis that emerge from the cancellation branch. The surrounding arithmetic literature includes carries and the p-adic behavior of Stirling numbers [16, 2, 31, 44].

Reader's guide.
Chapters 22–25 are a specialized arithmetic offshoot. They continue the ordered-correction branch of Chapters 17–21, but they are not prerequisites for the later real-time $p{:}q$ pushdown-normalization theory. Readers following only the main DPDA spine may skip directly to Chapter 26.

Chapter 25

From Ordered Corrections to CCR-3

The previous Part ended with an intrinsic correction-signature theory.

The original DPDA supplied a distinguished cancellation event.
The path interpretation ordered those events chronologically.
The Defect–Signature theorem then showed that the realization cost is local along that chronology:

$$\Delta(r) = r_m + \sum_{i=1}^{m-1} \psi(r_i - r_{i+1}).$$

Thus the correction history carries a natural path graph

$$1 - 2 - \cdots - n.$$

This is exactly the amount of pushdown-derived structure needed for the next step.

It is also the boundary of what the DPDA itself forces.

25.1 Scientific boundary

The arithmetic refinement introduced in this Part must be stated carefully.

What is inherited from the earlier DPDA/correction branch is:

1. a distinguished cancellation/correction event;
2. chronological ordering of those events;
3. nearest-neighbor locality along the ordered positions.

What is **introduced here** is a new arithmetic polarization:

1. a numerical weight for a singleton chronological slot;
2. a weight for an adjacent pair interaction;
3. a weight for the first connected triple collision.

Therefore the correct implication is

$$\boxed{\text{DPDA-derived chronological locality + specified arithmetic polarization} \implies \text{CCR-3.}}$$

The stronger statement

$$\text{original DPDA} \Longrightarrow \text{unique CCR-3 coefficients}$$

is not claimed.

This distinction is not editorial caution; it marks the exact scientific scope of the arithmetic branch.

25.2 Connected supports through order three

A connected subset of a path graph is an interval.

Therefore, up to size three, there are only three local connected support types.

Lemma 25.1 — Third-order connected support.

Every connected chronological support of cardinality at most three is one of

$$\boxed{\{j\}, \qquad \{j, j+1\}, \qquad \{j, j+1, j+2\}.}$$

These are, respectively:

- a singleton slot;
- an adjacent pair;
- a consecutive triple.

The lemma is elementary, but its role is structural: no other connected third-order motif exists on a one-dimensional chronology.

25.3 The CCR-3 recurrence

Fix a prime

$$p \geq 5$$

and define

$$\boxed{\ell_n = (p+1)n - p.}$$

Work in the truncated ring

$$\mathbb{Z}[u]/(u^4).$$

Set

$$P^{[3]}_{p,0}(u) = 1,$$

and

$$P^{[3]}_{p,-1}(u) = P^{[3]}_{p,-2}(u) = 0.$$

Definition 25.2 — CCR-3.

For

$$n \geq 1,$$

define

$$\boxed{\begin{aligned} P^{[3]}_{p,n}(u) = (1+\ell_n u)P^{[3]}_{p,n-1}(u) \\ + p(n-1)\ell_{n-1}u^2 P^{[3]}_{p,n-2}(u) \\ - p(n-2)(n-1)u^3 P^{[3]}_{p,n-3}(u) \pmod{u^4}. \end{aligned}}$$

The three nontrivial sources represent:

$$\boxed{\text{singleton} \ + \ \text{adjacent pair carry} \ + \ \text{connected triple collision.}}$$

The first rows are

$$P^{[3]}_{p,0} = 1,$$

$$P^{[3]}_{p,1} = 1 + u,$$

$$P^{[3]}_{p,2} = 1 + (p+3)u + 2(p+1)u^2,$$

and

$$P^{[3]}_{p,3} = 1 + (3p+6)u + (4p^2+15p+11)u^2 + 6(p+1)^2u^3.$$

The superscript [3] is important.

The recurrence is intentionally third-order.

It should not be extrapolated blindly into an all-column theory: at order four a genuinely new connected motif may enter.

25.4 Hessenberg realization

Let

$$H_{p,n}$$

be the $n \times n$ leading principal truncation of the infinite lower-Hessenberg matrix defined by

$$(H_p)_{i,i} = \ell_i,$$

$$(H_p)_{i,i+1} = 1,$$

$$(H_p)_{i+1,i} = -pi\ell_i,$$

$$(H_p)_{i+2,i} = -pi(i+1),$$

with all entries below the second subdiagonal equal to zero.

Define

$$D_{p,n}(u) = \det(I_n + uH_{p,n}).$$

Proposition 25.3 — Hessenberg realization.

$$\boxed{\begin{aligned} D_{p,n} = (1+\ell_n u)D_{p,n-1} \\ + p(n-1)\ell_{n-1}u^2 D_{p,n-2} \\ - p(n-2)(n-1)u^3 D_{p,n-3}. \end{aligned}}$$

Hence

$$\boxed{P^{[3]}_{p,n}(u) \equiv D_{p,n}(u) \pmod{u^4}.}$$

Proof.
Expand the determinant by the last row.
The diagonal contribution is

$$(1+\ell_n u)D_{p,n-1}.$$

The first subdiagonal closes with one superdiagonal entry and contributes

$$+p(n-1)\ell_{n-1}u^2 D_{p,n-2}.$$

The second subdiagonal closes with two consecutive superdiagonal entries and contributes

$$-p(n-2)(n-1)u^3 D_{p,n-3}.$$

No deeper band is present. □

Thus the three chronological interaction orders are encoded by a three-band Hessenberg structure.

25.5 Operator form

Use the exponential basis

$$e_i(x) = \frac{x^{i-1}}{(i-1)!}, \qquad i \geq 1.$$

Let

$$D = \partial_x, \qquad Xf = xf, \qquad N = XD,$$

and define

$$L_p = I + (p+1)N.$$

Then

$$De_i = e_{i-1},$$

$$Xe_i = ie_{i+1},$$

$$X^2 e_i = i(i+1)e_{i+2},$$

and

$$L_p e_i = \ell_i e_i.$$

Corollary 25.4 — Operator form.

The Hessenberg matrix H_p is the matrix, in the basis (e_i), of

$$\boxed{\mathcal{H}_p = D + L_p - pXL_p - pX^2.}$$

Equivalently,

$$\boxed{\mathcal{H}_p = \left[1 + (p+1)x - p(p+1)x^2\right]\partial_x + 1 - px - px^2.}$$

The determinant recurrence and differential-operator form are therefore two representations of the same CCR-3 dynamics.

25.6 What this chapter has and has not established

We have not derived CCR-3 from the original DPDA transition table.

We have established the narrower chain

$$\boxed{\begin{gathered}\text{ordered correction chronology} \to \text{path-local connected supports}\\ \to \text{specified third-order polarization} \to \text{Hessenberg/operator recurrence.}\end{gathered}}$$

The next question is arithmetic.

What do the first three fixed columns of this recurrence look like?

The answer is unexpectedly rigid: they are polynomials, and modulo p the entire third jet collapses to a universal first-kind Stirling shadow.

Chapter 26

Fixed Columns and the Universal Stirling Shadow

The CCR-3 recurrence is nonlinear in its dependence on the chronological index but finite in interaction order.

To expose its arithmetic, we now freeze the power of u and let the row index vary.

This produces a transport system for the first three fixed columns.

26.1 Column coefficients

Define

$$\boxed{C_{n,k}^{(p)} = [u^k]P_{p,n}^{[3]}(u), \qquad 0 \le k \le 3.}$$

For

$$m \ge k+1,$$

normalize by the falling factorial:

$$\boxed{\Pi_{p,k}(m) = \frac{C_{m-1,k}^{(p)}}{(m-1)_{\underline{k}}}.}$$

The normalization removes the obvious chronological placement multiplicity and reveals polynomial structure.

26.2 Transport equations

Proposition 26.1 — Transport system.

For the indicated ranges,

$$\boxed{m\Pi_{p,1}(m+1) = (m-1)\Pi_{p,1}(m) + \ell_m,}$$

$$\boxed{m\Pi_{p,2}(m+1) = (m-2)\Pi_{p,2}(m) + \ell_m\Pi_{p,1}(m) + p\ell_{m-1},}$$

and

$$\boxed{\begin{aligned} m\Pi_{p,3}(m+1) &= (m-3)\Pi_{p,3}(m) + \ell_m \Pi_{p,2}(m) \\ &\quad + p\ell_{m-1}\Pi_{p,1}(m-1) - p. \end{aligned}}$$

Proof.

Extract the coefficients of

$$u, \quad u^2, \quad u^3$$

from CCR-3.

Then substitute

$$n = m$$

and divide by the corresponding falling factorial.

The chronological factors cancel exactly, leaving the displayed first-order transport equations. □

The third source can be separated into a connected form.

Since

$$\Pi_{p,1}(m-1) = \frac{\ell_{m-1} - p}{2},$$

the genuinely new third-order source is

$$\boxed{\mathfrak{T}_{p,3}(m) = p\left[\frac{1}{2}\ell_{m-1}(\ell_{m-1} - p) - 1\right].}$$

This isolates the connected triple correction from propagated lower-order contributions.

26.3 First three fixed columns

Theorem 26.2 — First three fixed columns.

For

$$p \geq 5,$$

$$\boxed{\Pi_{p,1}(m) = \frac{(p+1)m - 2p}{2},}$$

$$\boxed{\Pi_{p,2}(m) = \frac{p+1}{24}\left[3(p+1)m^2 - (5p+1)m - 12p\right],}$$

and

$$\boxed{\Pi_{p,3}(m) = \frac{p+1}{48}F_p(m),}$$

where

$$\boxed{\begin{aligned} F_p(m) = {} & (p+1)^2 m^3 + (p^2-1)m^2 \\ & - (34p^2 + 18p)m + 56p^2 - 8p. \end{aligned}}$$

The boundary values are

$$\Pi_{p,1}(2) = 1,$$

$$\Pi_{p,2}(3) = p + 1,$$

$$\Pi_{p,3}(4) = (p + 1)^2.$$

Proof.
The first transport equation is solved directly by the displayed linear polynomial.
Substituting that result into the second equation gives a first-order recurrence with polynomial source; the displayed quadratic satisfies both the recurrence and its boundary value.
Substituting the first two solutions into the third transport equation gives a cubic source. The displayed cubic satisfies the recurrence and boundary value.
Uniqueness follows at each stage from the first-order recurrence. □
The key methodological point is that these formulas arise from transport.
They should not be presented as unexplained computer-algebra output.

26.4 Reduction modulo p

Modulo p,

$$\ell_n = (p + 1)n - p \equiv n \pmod{p}.$$

Every pair and triple interaction source carries an explicit factor of p.
Therefore those interaction terms vanish modulo p.
CCR-3 degenerates to the free recurrence

$$P^{[3]}_{p,n}(u) \equiv (1 + nu)P^{[3]}_{p,n-1}(u) \pmod{(p, u^4)}.$$

Iteration gives the universal product.

Theorem 26.3 — Stirling shadow.
For every

$$n \geq 0,$$

$$\boxed{P^{[3]}_{p,n}(u) \equiv \prod_{j=1}^{n}(1 + ju) \pmod{(p, u^4)}.}$$

Consequently, for $k \leq 3$,

$$\boxed{C^{(p)}_{m-1,k} \equiv e_k(1, \ldots, m-1) \equiv \begin{bmatrix} m \\ m-k \end{bmatrix} \pmod{p},}$$

where the bracket denotes the unsigned Stirling number of the first kind.
Thus the arithmetic interaction deformation has a universal free shadow.

26.5 Normalized shadow polynomials

The normalized fixed columns reduce to:

$$\boxed{S_1(m) = \frac{m}{2},}$$

$$\boxed{S_2(m) = \frac{m(3m-1)}{24},}$$

$$\boxed{S_3(m) = \frac{m^2(m-1)}{48}.}$$

So, as polynomial congruences,

$$\boxed{\Pi_{p,k}(m) \equiv S_k(m) \pmod{p} \qquad (k = 1, 2, 3).}$$

Important notation point

One must not derive these congruences by naively cancelling the falling factorial

$$(m-1)_{\underline{k}}$$

inside $\mathbb{F}_p$.
That denominator may vanish at exceptional residues.
The safe argument is:

1. derive the exact polynomial formulas over $\mathbb{Q}[m]$;
2. reduce those polynomial formulas modulo p.

Because

$$2, \quad 24, \quad 48$$

are units for $p \geq 5$, the reduction is legitimate.
This avoids illegal pointwise modular division.

26.6 Why the third column is the first singular one

The third normalized shadow is

$$S_3(m) = \frac{m^2(m-1)}{48}.$$

Therefore modulo p, the third-column zero set contains:

- a simple root at
$$m \equiv 1 \pmod{p};$$
- a double root at
$$m \equiv 0 \pmod{p}.$$

The exact cubic deformation $F_p(m)$ must decide what happens to these two shadow branches.
The next chapter shows that they behave completely differently.
The simple root survives integrally.
The double root becomes extinct in $\mathbb{Z}_p$.

Chapter 27

The Third-Column p-Adic Split

The third normalized column is

$$\Pi_{p,3}(m) = \frac{p+1}{48} F_p(m),$$

where

$$F_p(m) = (p+1)^2 m^3 + (p^2-1)m^2 - (34p^2+18p)m + 56p^2 - 8p.$$

Modulo p,

$$\boxed{F_p(m) \equiv m^2(m-1) \pmod p.}$$

Thus the shadow has two different types of roots:

$$m \equiv 1 \pmod p$$

is simple, while

$$m \equiv 0 \pmod p$$

is double.
The exact deformation separates them sharply.

27.1 The simple branch

At

$$m = 1,$$

$$F_p(1) \equiv 0 \pmod p,$$

and

$$\boxed{F_p'(1) \equiv 1 \pmod p.}$$

The derivative is a unit.
Hensel's lemma therefore applies in its simplest form.

Theorem 27.1 — Simple Hensel tower.
There exists a unique

$$\boxed{\alpha_p \in \mathbb{Z}_p}$$

such that

$$\alpha_p \equiv 1 \pmod p$$

and

$$F_p(\alpha_p) = 0.$$

Its expansion begins

$$\boxed{\begin{aligned}\alpha_p = 1 + 24p - 888p^2 + 58008p^3 \\ - 4715256p^4 + 428583192p^5 + O(p^6).\end{aligned}}$$

Moreover, throughout the residue class

$$m \equiv 1 \pmod p,$$

$$\boxed{\nu_p(F_p(m)) = \nu_p(m - \alpha_p).}$$

Proof.
Hensel gives existence and uniqueness.
Factor in $\mathbb{Z}_p[m]$:

$$F_p(m) = (m - \alpha_p)G_p(m).$$

Since

$$G_p(\alpha_p) = F_p'(\alpha_p)$$

is a unit, $G_p(m)$ remains a unit throughout the residue class

$$m \equiv \alpha_p \equiv 1 \pmod p.$$

Therefore the valuation is exactly the distance to the Hensel root. ☐
Thus the simple shadow branch survives exactly as expected.

27.2 The double shadow branch

Now set

$$m = pu.$$

Direct substitution gives

$$\boxed{\frac{F_p(pu)}{p} \equiv -8 \pmod p.}$$

Since $p \geq 5$,

$$-8$$

is nonzero modulo p.
Therefore:

Theorem 27.2 — Singular extinction.

For every

$$u \in \mathbb{Z}_p,$$

$$\boxed{\nu_p(F_p(pu)) = 1.}$$

Consequently, although

$$m \equiv 0 \pmod p$$

is a double root of the Stirling shadow, it has **no lift to a root modulo** p^2.
We call this phenomenon

$$\boxed{\textbf{singular extinction}.}$$

The term means exactly:

$$\boxed{\text{multiple root mod } p \quad \text{but no integral branch survives mod } p^2.}$$

It does not mean that the corresponding algebraic roots disappear from the algebraic closure.

Chapter 25 will show where they go.

27.3 Generic residue classes

If

$$m \not\equiv 0, 1 \pmod p,$$

then

$$F_p(m) \not\equiv 0 \pmod p.$$

Hence

$$\boxed{\nu_p(F_p(m)) = 0.}$$

Thus the residue classes split into three exact arithmetic regimes:

1. generic;
2. simple Hensel;
3. singularly extinct.

27.4 The scaled third column

Define

$$\boxed{Q_{m,3}^{\mathrm{CCR}} = p^{m-4}\Pi_{p,3}(m), \qquad m \geq 4.}$$

Because

$$\frac{p+1}{48}$$

is a p-adic unit for $p \geq 5$,

$$\nu_p(\Pi_{p,3}(m)) = \nu_p(F_p(m)).$$

The previous three cases therefore give the complete valuation law.

Theorem 27.3 — Complete third-column valuation law.

For

$$m \geq 4,$$

$$\boxed{\nu_p(Q^{\mathrm{CCR}}_{m,3}) = \begin{cases} m-4, & m \not\equiv 0,1 \pmod p, \\ m-3, & m \equiv 0 \pmod p, \\ m-4+\nu_p(m-\alpha_p), & m \equiv 1 \pmod p. \end{cases}}$$

This theorem is best read after the branch analysis.
The three cases are not an arbitrary valuation table.
They are exactly:

$$\boxed{\text{generic unit}}$$

$$\boxed{\text{one forced } p\text{-factor from singular extinction}}$$

$$\boxed{\text{Hensel distance to the unique integral root.}}$$

27.5 Independent consistency checks

The monograph audit independently checked the arithmetic at several levels:

- the fixed-column formulas were regenerated symbolically through several rows;
- the displayed formulas matched 15 tested symbolic column/boundary instances;
- substituting the displayed Hensel expansion into F_p cancels every coefficient through p^5;
- digitwise Hensel lifting for

$$p = 5, 7, 11, 13, 17, 19$$

  agrees through modulus p^6;
- hundreds of sample valuation checks agree with both the simple-branch and extinction laws.

These computations support the proof chain.
They do not replace it.

27.6 Ramified location of the remaining algebraic roots

The double shadow root has now disappeared from the integral p-adic locus:

$$0 \bmod p \longrightarrow \varnothing \bmod p^2.$$

But F_p is cubic.
One root has become the integral Hensel root α_p.
Two algebraic roots remain.
The discriminant will show that they have not disappeared.
They have moved into a ramified quadratic extension, with half-integral valuation.

Chapter 28

Ramified Recovery

The simple shadow root has survived inside

$$\mathbb{Z}_p.$$

The double shadow root has undergone singular extinction.

The two remaining algebraic roots are recovered in a ramified quadratic extension. Their location is encoded by the cubic discriminant.

28.1 Exact discriminant factorization

Theorem 28.1 — Discriminant and ramified field.

The discriminant of F_p factors as

$$\boxed{\mathrm{Disc}(F_p) = 4p(p+1)^2 R_p,}$$

where

$$\boxed{R_p = 9801p^5 + 22672p^4 + 32502p^3 + 14360p^2 - 927p - 8.}$$

Modulo p,

$$\boxed{R_p \equiv -8 \pmod{p}.}$$

Therefore

$$\boxed{\nu_p(\mathrm{Disc}(F_p)) = 1.}$$

The discriminant has odd valuation, so the residual quadratic factor after removing the simple Hensel root is ramified.

28.2 Removing the integral root

Factor

$$F_p(m) = (m - \alpha_p)Q_p(m)$$

over $\mathbb{Z}_p$.

The root α_p is simple and integral.

The remaining quadratic Q_p has discriminant with the same square class as the cubic discriminant, because the resultant contribution from the linear/quadratic splitting is a square.

Now

$$R_p/(-8) \equiv 1 \pmod{p}.$$

By Hensel's lemma,

$$R_p/(-8)$$

is a square in

$$\mathbb{Z}_p^\times.$$

The factors

$$4 \qquad \text{and} \qquad (p+1)^2$$

are also squares.

Hence the discriminant square class is

$$\boxed{[-2p].}$$

The quadratic splitting field is therefore

$$\boxed{\mathbb{Q}_p(\sqrt{-2p}).}$$

28.3 Half-integral roots

The quadratic factor is Eisenstein after the integral branch is removed.

Its two roots

$$\beta_{p,+}, \qquad \beta_{p,-}$$

therefore satisfy

$$\boxed{\nu_p(\beta_{p,\pm}) = \frac{1}{2}.}$$

Thus the double shadow root has not vanished algebraically.

Instead:

$$\boxed{\text{double root mod } p}$$

splits in the exact deformation into

$$\boxed{\text{two ramified roots of valuation } 1/2.}$$

This is the conceptual resolution of singular extinction.

Integral lifting fails because the two branches leave the integral valuation lattice.

28.4 Ramified square class by p mod 8

Corollary 28.2 — Ramified square class.

The ramified field is exactly

$$\boxed{\mathbb{Q}_p(\sqrt{-2p}).}$$

Its standard square-class description depends on

$$p \bmod 8.$$

For the residue classes in which -2 is a square unit, this is the standard p-ramified class.

For the remaining residue classes, it is the other ramified quadratic square class represented by a nonsquare unit times p.

This corollary is classical square-class arithmetic applied to the exact discriminant already proved.

28.5 The complete third-order arithmetic picture

The results of Chapters 22–25 can now be summarized without adding a new proof.

Corollary 28.3 — CCR-3 arithmetic geometry.

The third-order chronological carry refinement has the following structure.

1. Chronological locality allows only singleton, adjacent-pair, and consecutive-triple connected supports through order three.

2. The chosen arithmetic polarization produces the CCR-3 recurrence and its Hessenberg/operator realization.

3. The first three normalized columns are explicit polynomials.

4. Modulo p, all interaction terms disappear and

$$P^{[3]}_{p,n}(u) \equiv \prod_{j=1}^{n}(1+ju) \pmod{(p,u^4)}.$$

5. The normalized columns reduce to near-diagonal first-kind Stirling polynomials.

6. In the third column:

   - the simple shadow root
   $$1 \pmod p$$
   lifts uniquely to
   $$\alpha_p \in \mathbb{Z}_p;$$

   - the double shadow root
   $$0 \pmod p$$
   undergoes singular extinction;

- the two missing algebraic branches survive with valuation

$$1/2$$

over

$$\mathbb{Q}_p(\sqrt{-2p}).$$

This is the complete third-order arithmetic offshoot.

28.6 Novelty boundary

Nothing in this chapter requires claiming novelty for the ingredients individually.

The following are classical or established areas:

- carry processes;
- Holte-type carry matrices;
- Stirling numbers;
- p-adic valuations of Stirling numbers;
- Hensel lifting;
- Eisenstein irreducibility;
- discriminants;
- Newton polygons;
- ramified quadratic extensions;
- lower-Hessenberg production matrices.

The family-specific contribution is the chain

chronological local interactions → fixed-column polynomialization → Stirling shadow → simple lifting / singular extinction → exact ramified recovery.

The bibliographic claim is restricted to the proved family-specific chain stated here; no broader priority claim is made.

Chapter 29

Fourth-Order Chronological Carry: Connected Quadruples and the Stirling Shadow

The preceding arithmetic Part established CCR-3 and its third-jet Stirling shadow. The fourth-order problem requires a scope correction: the complete third jet and chronological path-locality determine the new connected support type, but not its numerical weight. The new datum is therefore introduced as an explicit fourth-order polarization and then identified from the normalized fourth column.

29.1 The CCR-3 baseline

Fix a prime $p \geq 5$ and define

$$\ell_n = (p+1)n - p.$$

The third-order arithmetic polarization is taken as the baseline object of this paper.

Definition 29.1 (CCR-3 baseline)**.** Work in $\mathbb{Z}[u]/(u^4)$ and set

$$P^{[3]}_{p,0}(u) = 1, \qquad P^{[3]}_{p,-1}(u) = P^{[3]}_{p,-2}(u) = 0.$$

For $n \geq 1$, define

$$\begin{aligned} P^{[3]}_{p,n}(u) = (1+\ell_n u)P^{[3]}_{p,n-1}(u) + p(n-1)\ell_{n-1}u^2 P^{[3]}_{p,n-2}(u) \\ - p(n-2)(n-1)u^3 P^{[3]}_{p,n-3}(u) \pmod{u^4}. \end{aligned} \tag{29.1}$$

The three sources are called the singleton slot, adjacent-pair interaction, and connected triple interaction.

For $0 \leq k \leq 3$ put

$$C^{(p)}_{n,k} = [u^k]P^{[3]}_{p,n}(u), \qquad \Pi_{p,k}(m) = \frac{C^{(p)}_{m-1,k}}{(m-1)_{\underline{k}}} \quad (m \geq k+1).$$

Coefficient extraction from (29.1) gives the transport equations

$$\begin{aligned} m\Pi_{p,1}(m+1) &= (m-1)\Pi_{p,1}(m) + \ell_m, \\ m\Pi_{p,2}(m+1) &= (m-2)\Pi_{p,2}(m) + \ell_m\Pi_{p,1}(m) + p\ell_{m-1}, \\ m\Pi_{p,3}(m+1) &= (m-3)\Pi_{p,3}(m) + \ell_m\Pi_{p,2}(m) \\ &\qquad + p\ell_{m-1}\Pi_{p,1}(m-1) - p. \end{aligned} \tag{29.2}$$

Proposition 29.2 (First three normalized columns)**.** *For the indicated ranges,*

$$\Pi_{p,1}(m) = \frac{(p+1)m - 2p}{2},$$

$$\Pi_{p,2}(m) = \frac{p+1}{24}\left[3(p+1)m^2 - (5p+1)m - 12p\right],$$

and

$$\Pi_{p,3}(m) = \frac{p+1}{48}F_p(m),$$

where

$$F_p(m) = (p+1)^2m^3 + (p^2-1)m^2 - (34p^2+18p)m + 56p^2 - 8p.$$

The boundary values are

$$\Pi_{p,1}(2) = 1, \qquad \Pi_{p,2}(3) = p+1, \qquad \Pi_{p,3}(4) = (p+1)^2.$$

Proof. Substitution into (29.2) verifies the three displayed polynomials. Each transport equation is first order in its current column, so the corresponding boundary value gives uniqueness. No computer-algebra interpolation is used as a proof premise. □

Modulo p, every non-singleton source in (29.1) disappears and $\ell_n \equiv n$. Hence

$$P^{[3]}_{p,n}(u) \equiv \prod_{j=1}^{n}(1+ju) \pmod{(p, u^4)}.$$

The purpose of T8 is to determine exactly what survives, and what becomes new, when the truncation is raised to u^5.

29.2 What is genuinely new at order four

Definition 29.3 (Connected chronological support)**.** A set of correction positions is *connected* if it induces a connected subgraph of the chronological path $1 - 2 - \cdots - n$.

Lemma 29.4 (Unique connected quadruple)**.** *Every connected chronological support of cardinality four is an interval of the form*

$$\{j, j+1, j+2, j+3\}.$$

Thus there is exactly one new connected support type at order four.

Proof. Every connected subset of a path graph is an interval. An interval with four vertices has the displayed form. □

The structural issue is not the shape of the new support but the arithmetic weight assigned to it. The recurrence itself gives a clean way to separate propagated lower-order terms from the genuinely new fourth-order source.

Definition 29.5 (Terminal block form through order four)**.** A path-local fourth-order extension of CCR-3 is a recurrence in $\mathbb{Z}[u]/(u^5)$ of the form

$$\begin{aligned} P_{p,n}(u) = (1+\ell_n u)P_{p,n-1}(u) + p(n-1)\ell_{n-1}u^2P_{p,n-2}(u) \\ - p(n-2)(n-1)u^3P_{p,n-3}(u) + q_{p,n}u^4P_{p,n-4}(u) \pmod{u^5}, \end{aligned} \tag{29.3}$$

with $P_{p,0} = 1$ and $P_{p,n} = 0$ for negative n. The first three terminal block weights are fixed by CCR-3; $q_{p,n}$ is the connected quadruple source.

Proposition 29.6 (Lower-order block separation)**.** *In* (29.3)*, every contribution to* $[u^4]P_{p,n}$ *obtained by appending a terminal connected block of size one, two, or three is already determined by CCR-3. The only term in the fourth coefficient not generated by those lower bands is* $q_{p,n}$ *itself.*

Proof. Extracting u^4 from (29.3) gives

$$\begin{aligned}[u^4]P_{p,n} = [u^4]P_{p,n-1} &+ \ell_n[u^3]P_{p,n-1} \\ &+ p(n-1)\ell_{n-1}[u^2]P_{p,n-2} \\ &- p(n-2)(n-1)[u]P_{p,n-3} + q_{p,n}.\end{aligned} \tag{29.4}$$

The middle three lines attach a terminal block of size one, two, or three to a previously generated lower-order history. Iterating those terms produces exactly the degree-four histories decomposable into lower connected block sizes, including the patterns 1+1+1+1, 2+1+1, 2+2, and 3+1 in chronological order. The final scalar $q_{p,n}$ is the contribution of a single terminal block of size four. By theorem 29.4, that block is the unique consecutive quadruple support. □

Remark 29.7 (Relation with cumulant language). Classical cumulants isolate connected information by Möbius inversion on the lattice of set partitions [104, 103]; mixed cumulants vanish when the variables split into independent families. That theory motivates the word “connected,” but theorem 29.6 does not identify the CCR coefficients with probabilistic cumulants. It is an internal recurrence statement about terminal interval blocks on a path.

Theorem 29.8 (Fourth-order non-identifiability from CCR-3)**.** *CCR-3 and chronological path-locality do not determine a unique fourth-order numerical weight. More precisely, for any integer sequence* $(q_{p,n})_{n\geq 4}$*, recurrence* (29.3) *has exactly the same coefficients through* u^3 *as CCR-3.*

Proof. The new term is divisible by u^4, so it cannot affect any coefficient of degree at most three. Induction on n therefore shows that every choice of $q_{p,n}$ has the same third jet as (29.1). The fourth coefficient is the first place where the choices can differ. □

This theorem is the central scope correction at order four: a numerical CCR-4 coefficient must come from an additional arithmetic polarization, not from third-order data alone.

29.2.1 Exact reconstruction of a general p-divisible fourth source

Suppose the connected source is p-divisible after the natural size-four placement factor:

$$q_{p,m} = p(m-1)_{\underline{3}}\eta_m, \qquad \eta_m \in \mathbb{Z}, \qquad m \geq 4.$$

For the corresponding solution of (29.3), write

$$C^{(q)}_{n,k} = [u^k]P_{p,n}(u), \qquad \Pi^{(q)}_{p,4}(m) = \frac{C^{(q)}_{m-1,4}}{(m-1)_{\underline{4}}}, \qquad m \geq 5.$$

Let $\Pi^{(0)}_{p,4}$ denote the zero-fourth-source solution and put

$$\Delta_m = \Pi^{(q)}_{p,4}(m) - \Pi^{(0)}_{p,4}(m).$$

Theorem 29.9 (Fourth-source reconstruction)**.** *For every $m \geq 5$,*

$$\boxed{m\Delta_{m+1} = (m-4)\Delta_m + p\eta_m.} \tag{29.5}$$

At the boundary,

$$\boxed{\Delta_5 = \frac{p\eta_4}{4}.} \tag{29.6}$$

Consequently the connected source is exactly identifiable from the fourth normalized column:

$$\boxed{\eta_m = \frac{m\Delta_{m+1} - (m-4)\Delta_m}{p} \qquad (m \geq 5),} \tag{29.7}$$

with $\eta_4 = 4\Delta_5/p$.

Proof. Extract u^4 from (29.3), put $n = m$, and divide by $(m)_{\underline{4}} = m(m-1)(m-2)(m-3)$. The lower three source bands are identical for the given source and for the zero-source recurrence because all columns of degree at most three agree by theorem 29.8. Their difference therefore cancels. The remaining fourth source contributes

$$\frac{q_{p,m}}{(m)_{\underline{4}}} = \frac{p(m-1)_{\underline{3}}\eta_m}{(m)_{\underline{4}}} = \frac{p\eta_m}{m},$$

while the preceding fourth coefficient contributes $(m-4)\Delta_m/m$. Multiplying by m gives (29.5). At $m = 4$, the only difference in $C_{4,4}$ is $q_{p,4} = 6p\eta_4$; division by $(4)_{\underline{4}} = 24$ gives (29.6). Solving (29.5) for η_m yields (29.7). □

Thus fourth-order identifiability is not restricted to the stationary model: the entire p-divisible connected chronology can be read back from the fourth column. Stationarity becomes a testable property of that column rather than merely an imposed coefficient ansatz.

29.3 The stationary p-divisible fourth-order family

We now specify one natural class in which the new fourth-order datum can be classified exactly.

Definition 29.10 (Stationary p-divisible fourth polarization)**.** A recurrence of the form (29.3) is *stationary p-divisible* if, for every $n \geq 4$,

$$p(n-1)_{\underline{3}} \;\Big|\; q_{p,n},$$

and the normalized integer quotient

$$\eta_n = \frac{q_{p,n}}{p(n-1)_{\underline{3}}}$$

is independent of n. Thus stationarity is imposed only after removing the inherited factor p and the falling-factorial normalization of a size-four terminal block.

Theorem 29.11 (Stationarity and one-parameter classification)**.** *For every prime $p \geq 5$ and every p-divisible fourth-order extension, the following are equivalent:*

(i) *the normalized connected source η_n is independent of n;*

(ii) *the fourth-column difference $\Delta_m = \Pi^{(q)}_{p,4}(m) - \Pi^{(0)}_{p,4}(m)$ is independent of $m \geq 5$;*

(iii) *for one integer* λ, *the recurrence is exactly*

$$\boxed{\begin{aligned} P^{[4,\lambda]}_{p,n}(u) &= (1+\ell_n u)P^{[4,\lambda]}_{p,n-1}(u) + p(n-1)\ell_{n-1}u^2 P^{[4,\lambda]}_{p,n-2}(u) \\ &\quad - p(n-2)(n-1)u^3 P^{[4,\lambda]}_{p,n-3}(u) \\ &\quad + \lambda p(n-3)(n-2)(n-1)u^4 P^{[4,\lambda]}_{p,n-4}(u) \end{aligned}} \pmod{u^5}, \tag{29.8}$$

with $\lambda \in \mathbb{Z}$. *In that case*

$$\Delta_m = \frac{\lambda p}{4} \qquad (m \geq 5).$$

The member $\lambda = 0$ *has no new connected size-four source, whereas every* $\lambda \neq 0$ *has a genuine connected quadruple contribution.*

Proof. The equivalence of (i) and (iii) is the definition of the normalized source: $\eta_n \equiv \lambda$ is exactly $q_{p,n} = \lambda p(n-1)_{\underline{3}}$. If (i) holds, theorem 29.9 has boundary value $\Delta_5 = \lambda p/4$, and the constant sequence $\Delta_m = \lambda p/4$ satisfies (29.5); first-order uniqueness gives (ii) and the displayed constant. Conversely, if $\Delta_m \equiv c$, then (29.7) gives $\eta_m = 4c/p$ for every $m \geq 5$, while (29.6) gives the same value for $m = 4$. Hence (ii) implies (i), completing the equivalence. □

Definition 29.12 (Primitive same-sign CCR-4)**.** The choice

$$\lambda = 1$$

is called the *primitive same-sign fourth polarization.* It is the smallest positive integral extension compatible with the sign convention for the lower Hessenberg interaction bands introduced in section 29.4.

Remark 29.13 (What $\lambda = 1$ means). Definition 29.12 is a normalization choice. It gives one concrete CCR-4 recurrence, but it is not asserted that the original DPDA uniquely forces $\lambda = 1$. The theorem-level object is the full λ-family and the exact way in which λ is visible.

29.4 Four-band Hessenberg and operator realization

The extension parameter has a direct linear-algebraic realization.

Definition 29.14 (Fourth-band Hessenberg matrix)**.** Let $H_{p,\lambda}$ be the infinite lower-Hessenberg matrix with

$$(H_{p,\lambda})_{i,i} = \ell_i, \qquad (H_{p,\lambda})_{i,i+1} = 1,$$

$$(H_{p,\lambda})_{i+1,i} = -pi\ell_i, \qquad (H_{p,\lambda})_{i+2,i} = -pi(i+1),$$
$$(H_{p,\lambda})_{i+3,i} = -\lambda pi(i+1)(i+2),$$

and all entries below the third subdiagonal equal to zero. Write $H_{p,\lambda,n}$ for its $n \times n$ leading principal truncation and

$$D^{(\lambda)}_{p,n}(u) = \det(I_n + uH_{p,\lambda,n}).$$

Theorem 29.15 (Four-band Hessenberg realization)**.** *The determinants* $D^{(\lambda)}_{p,n}$ *satisfy recurrence* (29.8) *exactly. Consequently*

$$P^{[4,\lambda]}_{p,n}(u) \equiv D^{(\lambda)}_{p,n}(u) \pmod{u^5}.$$

Proof. Expand $\det(I_n + uH_{p,\lambda,n})$ by the last row. The diagonal term gives

$$(1 + \ell_n u)D^{(\lambda)}_{p,n-1}.$$

The first subdiagonal entry closes with one superdiagonal entry and gives

$$+p(n-1)\ell_{n-1}u^2 D^{(\lambda)}_{p,n-2}.$$

The second subdiagonal closes with two consecutive superdiagonal entries and gives

$$-p(n-2)(n-1)u^3 D^{(\lambda)}_{p,n-3}.$$

The third subdiagonal closes with three consecutive superdiagonal entries. Its cofactor sign reverses the negative matrix entry, producing

$$+\lambda p(n-3)(n-2)(n-1)u^4 D^{(\lambda)}_{p,n-4}.$$

No deeper band is present. This is exactly (29.8). $\square$

Use the exponential basis

$$e_i(x) = \frac{x^{i-1}}{(i-1)!}, \qquad i \geq 1.$$

Let $D = \partial_x$, let $Xf = xf$, set $N = XD$, and define

$$L_p = I + (p+1)N.$$

Then

$$De_i = e_{i-1}, \qquad Xe_i = ie_{i+1}, \qquad X^k e_i = i(i+1)\cdots(i+k-1)e_{i+k},$$

and $L_p e_i = \ell_i e_i$.

Corollary 29.16 (Operator form)**.** *The matrix $H_{p,\lambda}$ is the matrix, in the basis (e_i), of*

$$\boxed{\mathcal{H}_{p,\lambda} = D + L_p - pXL_p - pX^2 - \lambda pX^3.} \tag{29.9}$$

Equivalently,

$$\boxed{\mathcal{H}_{p,\lambda} = \left[1 + (p+1)x - p(p+1)x^2\right]\partial_x + 1 - px - px^2 - \lambda px^3.} \tag{29.10}$$

Proof. The actions of D, L_p, XL_p, X^2, and X^3 on e_i reproduce respectively the superdiagonal, diagonal, first subdiagonal, second subdiagonal, and third subdiagonal of $H_{p,\lambda}$. Expanding $L_p = I + (p+1)XD$ gives (29.10). $\square$

Banded Hessenberg determinants and their finite-term recurrences are classical; see, for example, [105]. The point here is the exact identification of the fourth chronological parameter with the third subdiagonal and the X^3 term.

29.5 Fourth-column transport and exact identifiability

For the fourth-order family define

$$C_{n,k}^{(p,\lambda)} = [u^k]P_{p,n}^{[4,\lambda]}(u), \qquad 0 \le k \le 4,$$

and, for $m \ge k+1$,

$$\Pi_{p,k}^{(\lambda)}(m) = \frac{C_{m-1,k}^{(p,\lambda)}}{(m-1)_{\underline{k}}}.$$

Because the λ-term starts in degree four, theorem 29.8 implies

$$\Pi_{p,k}^{(\lambda)} = \Pi_{p,k} \qquad (k = 1, 2, 3).$$

Theorem 29.17 (Fourth-column transport)**.** *For $m \ge 5$,*

$$\boxed{\begin{aligned} m\Pi_{p,4}^{(\lambda)}(m+1) &= (m-4)\Pi_{p,4}^{(\lambda)}(m) + \ell_m \Pi_{p,3}(m) \\ &\quad + p\ell_{m-1}\Pi_{p,2}(m-1) - p\Pi_{p,1}(m-2) + \lambda p. \end{aligned}} \tag{29.11}$$

The last term is exactly the normalized connected quadruple source.

Proof. Extract u^4 from (29.8) and put $n = m$. Divide by

$$(m)_{\underline{4}} = m(m-1)(m-2)(m-3).$$

The four source terms simplify using

$$\frac{(m-1)_{\underline{3}}}{(m)_{\underline{4}}} = \frac{1}{m}, \qquad \frac{(m-1)(m-2)_{\underline{2}}}{(m)_{\underline{4}}} = \frac{1}{m},$$

$$\frac{(m-2)(m-1)(m-3)_{\underline{1}}}{(m)_{\underline{4}}} = \frac{1}{m}, \qquad \frac{(m-3)(m-2)(m-1)}{(m)_{\underline{4}}} = \frac{1}{m}.$$

Multiplying by m gives (29.11). □

The key payoff is that the new connected source does not distort the shape of the fourth normalized column; it translates it.

Theorem 29.18 (Connected-quadruple translation law)**.** *For every $\lambda, \mu \in \mathbb{Z}$ and every $m \ge 5$,*

$$\boxed{\Pi_{p,4}^{(\lambda)}(m) - \Pi_{p,4}^{(\mu)}(m) = \frac{p(\lambda-\mu)}{4}.} \tag{29.12}$$

Equivalently,

$$\boxed{C_{m-1,4}^{(p,\lambda)} - C_{m-1,4}^{(p,\mu)} = \frac{p(\lambda-\mu)}{4}(m-1)_{\underline{4}}.} \tag{29.13}$$

Proof. Subtract the two instances of (29.11). With

$$\Delta_m = \Pi_{p,4}^{(\lambda)}(m) - \Pi_{p,4}^{(\mu)}(m),$$

one obtains

$$m\Delta_{m+1} = (m-4)\Delta_m + p(\lambda-\mu).$$

The constant sequence

$$\Delta_m = \frac{p(\lambda-\mu)}{4}$$

satisfies this recurrence. At the first admissible row $m = 5$, direct extraction from $P_{p,4}^{[4,\lambda]}$ gives the same difference. Uniqueness of the first-order recurrence proves (29.12); multiplying by $(m-1)_{\underline{4}}$ gives (29.13). □

Corollary 29.19 (One-row identifiability)**.** *Once the lower-order polarization is fixed, any single normalized fourth-column value at an admissible row identifies λ exactly:*

$$\boxed{\lambda = \frac{4}{p}\left(\Pi^{(\lambda)}_{p,4}(m) - \Pi^{(0)}_{p,4}(m)\right).}$$

At the first raw row $n = 4$, changing λ by one changes $[u^4]P_{p,4}$ by exactly $6p$.

Thus order four is simultaneously non-identifiable from the entire third jet and exactly identifiable from one fourth-column observation.

Theorem 29.20 (Explicit normalized fourth column)**.** *For $m \geq 5$,*

$$\boxed{\Pi^{(\lambda)}_{p,4}(m) = \frac{G_{p,\lambda}(m)}{5760},} \tag{29.14}$$

where

$$\begin{aligned} G_{p,\lambda}(m) &= 15(p+1)^4m^4 + 30(p+1)^3(3p-1)m^3 \\ &\quad - 5(p+1)^2(287p^2 + 158p - 1)m^2 \\ &\quad + 2(p+1)^2(1645p^2 - 50p + 1)m \\ &\quad + 240p(6\lambda - 5p^3 - 5p^2 + 11p + 5). \end{aligned} \tag{29.15}$$

In particular,

$$\Pi^{(\lambda)}_{p,4}(5) = \frac{\lambda p + 5p^3 + 15p^2 + 13p + 4}{4}. \tag{29.16}$$

Proof. Substitute the three explicit polynomials of theorem 29.2 into the source of (29.11). The resulting source is polynomial in m of degree four. Direct expansion verifies that (29.14) satisfies (29.11); evaluation at $m = 5$ gives (29.16). The first-order transport equation then gives uniqueness. The λ-dependent part of (29.15) is $1440\lambda p$, which is exactly the translation $\lambda p/4$ of theorem 29.18. ☐

29.6 The fourth-jet Stirling shadow

The new connected parameter is exact information, but it disappears modulo p.

Theorem 29.21 (Fourth-jet Stirling shadow)**.** *For every prime $p \geq 5$, every $\lambda \in \mathbb{Z}$, and every $n \geq 0$,*

$$\boxed{P^{[4,\lambda]}_{p,n}(u) \equiv \prod_{j=1}^{n}(1 + ju) \pmod{(p, u^5)}.} \tag{29.17}$$

Consequently,

$$\boxed{C^{(p,\lambda)}_{m-1,4} \equiv e_4(1, \dots, m-1) \equiv \begin{bmatrix} m \\ m-4 \end{bmatrix} \pmod{p},} \tag{29.18}$$

where the bracket is the unsigned Stirling number of the first kind.

Proof. Modulo p, one has $\ell_n \equiv n$. Every pair, triple, and quadruple interaction source in (29.8) contains an explicit factor of p, so the recurrence reduces to

$$P^{[4,\lambda]}_{p,n}(u) \equiv (1 + nu)P^{[4,\lambda]}_{p,n-1}(u) \pmod{(p, u^5)}.$$

Iterating gives (29.17). The coefficient of u^4 in the product is the fourth elementary symmetric function of $1, \dots, m-1$, which is the near-diagonal unsigned Stirling number shown in (29.18); see, for example, [106]. ☐

Over $\mathbb{Q}[m]$ the normalized fourth shadow is

$$\boxed{S_4(m) = \frac{m(15m^3 - 30m^2 + 5m + 2)}{5760},} \tag{29.19}$$

because

$$e_4(1, \ldots, m-1) = (m-1)_{\underline{4}} S_4(m).$$

Corollary 29.22 (Normalized shadow away from $p = 5$)**.** *For every prime $p \geq 7$,*

$$\boxed{\Pi^{(\lambda)}_{p,4}(m) \equiv S_4(m) \pmod p}$$

as a polynomial congruence. The congruence is independent of λ.

Proof. Subtract (29.19) from the exact quartic (29.14). The resulting rational polynomial has an explicit factor of p. For $p \geq 7$, the denominator

$$5760 = 2^7 \cdot 3^2 \cdot 5$$

is invertible modulo p, so reduction is legitimate. $\square$

Remark 29.23 ($p = 5$ denominator resonance). The raw coefficient congruence (29.18) remains valid at $p = 5$ because it is obtained before any division. However, 5760 is divisible by 5, so one must not obtain a normalized polynomial congruence by cancelling $(m-1)_{\underline{4}}$ or reducing (29.19) coefficientwise modulo 5. This is a normalization-denominator issue, not a failure of the fourth-jet Stirling shadow.

Corollary 29.24 (Shadow blindness versus exact identifiability)**.** *All stationary p-divisible fourth polarizations have the same mod-p fourth jet, but distinct parameters have distinct exact fourth columns. Thus the Stirling shadow forgets λ, while theorem 29.19 recovers it from one exact row.*

29.7 Re-entry to the main DPDA story

The arithmetic offshoot is now complete.

We return to the main machine problem.

The monograph began with the original 2:1 center/left/right DPDA.

Chapters 2–8 extracted its shortest-demand semantics and produced compact residual-faithful machines.

But those compact machines changed the physical representation.

The operational idea encoded in the original $C/L/R$ architecture admits a uniform generalization to every $p{:}q$.

The resulting real-time centered construction is developed next.

From Chapter 26 onward, the main line will remain uninterrupted:

general real-time $p{:}q$ $\rightarrow$ Structure Theory $\rightarrow$ Uniqueness/Lift Theory $\rightarrow$ General Theory.

Part IX

Returning to the Machine: General Real-Time $p{:}q$

Research provenance. This Part develops the uniform centered real-time literal-demand construction for all coprime positive p, q.

Chapter 30

Why Static Two-Visibility Fails

We now return to pushdown machines.

This is not another residual-counter construction.

The goal is more specific:

> Reconstruct, for every coprime $p{:}q$, the operational idea hidden in the original center/left/right machine.

The original 2:1 design had three striking features:

1. a literal completion demand was physically visible in the stack;
2. a mismatch could temporarily move information into finite control;
3. a later long replacement could rematerialize that information.

The general construction will preserve this style rather than replace it by the compact Euclidean residual encoding of Chapters 7–8.

That is why its resource count is different.

30.1 We do not re-prove minimum demand

The semantic foundation is already complete.

For

$$L_{p,q} = \{w : qN_1(w) = pN_0(w)\},$$

the residual is

$$\rho = qN_1 - pN_0.$$

For every residual r, the unique minimum completion demand is

$$\mu(r) = (x_r, y_r),$$

and the primitive zero-weight kernel is

$$\boxed{K = (q, p).}$$

Every nonnegative vector of the same weight differs from the minimum demand by a nonnegative multiple of K.

Those results are imported from the earlier chapters.

The new problem is physical ordering.

30.2 A two-visible canonical word

For a demand vector

$$(x, y),$$

define

$$\boxed{\eta(x, y) = \begin{cases} (10)^y 0^{x-y}, & x \geq y, \\ (10)^x 1^{y-x}, & y > x. \end{cases}}$$

This word has an attractive property.
If both symbols occur, the stack begins with

$$10.$$

More generally, every suffix that remains mixed begins with either

$$10$$

or

$$01.$$

Thus the presence of both demand types is always visible within two stack cells.
This is the general literal-demand analogue of the original 2:1 depth-two phenomenon.
But static visibility is not enough.

30.3 The 3:2 obstruction

Consider the canonical stack

$$\boxed{1010.}$$

Both demand symbols are visible immediately.
Suppose the next input is 0, but the required zero is the **second** visible cell.
A naive LIFO strategy could:

1. pop the top 1;
2. delete the 0;
3. immediately restore the displaced 1.

This produces

$$1010 \longrightarrow 110.$$

The surviving zero has moved to depth three.
Two-cell visibility has been destroyed.
The problem is not lack of static information.
It is chronology.

30.4 Static two-visible obstruction

Lemma 30.1 — Static two-visible obstruction.

Suppose a mixed binary stack contains at least two zeros and two ones and begins with

$$10.$$

No purely static strategy can support both deletion directions while:

1. deleting either visible demand type;
2. immediately restoring a displaced first-cell blocker;
3. guaranteeing that a surviving mixed suffix still exposes both symbol types in its first two cells.

Proof.

If deleting the top 1 must leave a mixed two-visible suffix, then the next two surviving symbols must differ. Therefore the third original symbol must be 1.

If instead the second-cell 0 is deleted and the first 1 is immediately restored, preserving two-visibility requires the next surviving symbol beneath it to be 0. Therefore the third original symbol must be 0.

The same third cell cannot satisfy both conditions. □

This lemma isolates the exact reason the original architecture needs side-state memory.

A LIFO blocker sometimes must leave the stack temporarily.

30.5 SEARCH

The repair is:

$$\boxed{\text{SEARCH} = \text{move one displaced blocker into finite control.}}$$

The blocker is **not** immediately restored.

Its semantic effect is carried as finite-control debt while the remaining physical stack continues to be popped.

A later WRITE will materialize that debt.

This is the general version of the role played by L and R in the original machine.

The crucial design question becomes:

> How much signed debt must finite control be able to carry before WRITE becomes safe?

That will produce the centered state rail.

Before building the rail, we need to formalize what a debt state means semantically.

Chapter 31

Debt, Virtual Demand, and Minimal WRITE

Let

$$m = p + q.$$

A one-cell mismatch changes the relationship between the physical stack and the semantic demand by one kernel-length unit m.

This motivates a centered integer debt coordinate.

31.1 Virtual demands

For

$$d \in \mathbb{Z},$$

define the virtual demand

$$\boxed{V_d = \mu(dm).}$$

Its residual weight is

$$W(V_d) = dm.$$

A debt state D_d therefore carries finite-control potential

$$\boxed{\Phi(D_d) = dm.}$$

Let a physical logical stack S have Parikh vector

$$P(S) = (N_0(S), N_1(S)).$$

The stack and state together may contain more material than the minimum demand. The semantic projection removes any complete kernel packets.

31.2 Virtual-demand projection

Define kernel reduction

$$\mathrm{red}_{p,q}(x, y)$$

to mean repeated subtraction of

$$K = (q, p)$$

while both coordinates remain nonnegative.
The result is exactly the unique minimum demand of the same residual weight.

Definition 31.1 — Rich-to-demand projection.
For configuration

$$(D_d, S),$$

define

$$\boxed{\Theta(D_d, S) = \mathrm{red}_{p,q}\big(P(S) + V_d\big).}$$

Theorem 31.2 — Virtual-demand projection.
If

$$\boxed{\rho = W(P(S)) + dm,}$$

then

$$\boxed{\Theta(D_d, S) = \mu(\rho).}$$

Proof.
The nonnegative vector

$$P(S) + V_d$$

has weight

$$W(P(S)) + W(V_d) = W(P(S)) + dm = \rho.$$

Kernel reduction of any nonnegative vector of weight ρ is the unique minimum demand $\mu(\rho)$. □
This theorem is one of the main semantic bridges in the monograph.
It says:

$$\boxed{\text{rich state+stack configuration} \longrightarrow \text{unique shortest-demand semantics.}}$$

The rich machine may retain normalization history.
Its semantic projection does not.

31.3 The WRITE equation

Suppose one transition:

- consumes input a;
- removes visible top symbol z;
- changes debt state from D_s to D_t;
- writes replacement word u.

Let

$$e_0 = (1,0), \qquad e_1 = (0,1).$$

Ignoring the unchanged stack suffix, residual preservation forces

$$\boxed{P(u) = e_z - e_a + V_s - V_t + \kappa K}$$

for some integer κ.
At the bottom marker, the term e_z is omitted.
Thus every invariant-preserving rewrite lies on one affine kernel ray.

31.4 Minimal kernel carry

Let

$$B = e_z - e_a + V_s - V_t.$$

The first kernel shift making both coordinates nonnegative is

$$\boxed{\kappa_* = \max\left(\left\lceil -\frac{B_0}{q} \right\rceil, \left\lceil -\frac{B_1}{p} \right\rceil\right).}$$

Theorem 31.3 — Minimal kernel-carry rewrite.

Among all nonnegative invariant-preserving replacement vectors, the unique shortest one is

$$\boxed{R_* = B + \kappa_* K.}$$

Every other solution has the form

$$\boxed{R_* + \ell K, \qquad \ell \geq 1,}$$

and is longer by exactly

$$\ell(p+q)$$

symbols.
Proof.
All integer solutions differ by a multiple of the primitive kernel K.
The displayed κ_* is the least shift that places both coordinates in the nonnegative quadrant.
Every further kernel packet adds

$$q + p = m$$

symbols. □
When a physical word is needed, we write the vector in canonical order using η.

31.5 Why the historical long writes existed

The original 2:1 table contained replacements such as:

$$10, \qquad 11, \qquad 000, \qquad 111111.$$

Before the general theory, these can look ad hoc.
The WRITE equation changes their interpretation.
A long replacement is determined by:

1. the consumed input symbol;
2. the visible stack symbol removed;
3. the source debt potential;
4. the target debt potential;
5. the least kernel carry needed to make the replacement vector nonnegative.

Thus the long words are not arbitrary repair strings.
They are shortest solutions of a constrained residual-preservation equation.
This is one of the main retrospective explanations of the original machine.

31.6 From arbitrary debt to a finite rail

The virtual-demand definition allows every

$$d \in \mathbb{Z}.$$

A finite-state DPDA cannot.
The construction therefore needs a bounded debt interval.
Set

$$\boxed{h_{\mathrm{ctr}} = \left\lfloor \frac{m-1}{2} \right\rfloor.}$$

The centered core will use exactly

$$\boxed{D_{-h_{\mathrm{ctr}}}, \ldots, D_{-1}, D_0, D_1, \ldots, D_{h_{\mathrm{ctr}}}.}$$

with

$$D_0 = C.$$

Why is this width sufficient?
Because the canonical word created at every WRITE has bounded minority count.
Why is it exact in the deferred architecture?
Because one common canonical witness can exhaust the physical stack while leaving every debt value in this interval purely in finite control.
Those are the two halves of Chapters 28 and 29.

Chapter 32

The Centered Real-Time $p{:}q$ DPDA

Set

$$m = p + q, \qquad h_{\mathrm{ctr}} = \left\lfloor \frac{m-1}{2} \right\rfloor .$$

The centered state set is

$$\boxed{Q = \{D_{-h_{\mathrm{ctr}}}, \ldots, D_{-1}, D_0, D_1, \ldots, D_{h_{\mathrm{ctr}}}\},}$$

with

$$D_0 = C.$$

The stack alphabet is

$$\boxed{\{0, 1, Z_0\}.}$$

The bottom marker Z_0 remains fixed throughout binary-input processing and is removed only by the final right-endmarker acceptance move.

The construction has four binary transition roles:

$$\boxed{\text{MATCH}, \qquad \text{SEARCH}, \qquad \text{outer WRITE}, \qquad \text{bottom WRITE}.}$$

Every binary move consumes exactly one input symbol.

There are no ε-moves during binary processing.

32.1 The centered rail

MATCH

For every

$$|d| \le h_{\mathrm{ctr}}$$

and $z \in \{0, 1\}$,

$$\boxed{(D_d, zT) \xrightarrow{z} (D_d, T).}$$

A matching input simply cancels one physically present demand cell.

Interior SEARCH

For

$$d < h_{\mathrm{ctr}},$$

$$\boxed{(D_d, 0T) \xrightarrow{1} (D_{d+1}, T).}$$

For

$$d > -h_{\mathrm{ctr}},$$

$$\boxed{(D_d, 1T) \xrightarrow{0} (D_{d-1}, T).}$$

A mismatch still pops the visible cell, but instead of restoring it the state debt moves one step.

Outer-boundary WRITE

Define

$$U_+ = \eta\big(\mu((h_{\mathrm{ctr}} + 1)m)\big),$$

$$U_- = \eta\big(\mu(-(h_{\mathrm{ctr}} + 1)m)\big).$$

Then

$$\boxed{(D_{h_{\mathrm{ctr}}}, 0T) \xrightarrow{1} (C, U_+T),}$$

and

$$\boxed{(D_{-h_{\mathrm{ctr}}}, 1T) \xrightarrow{0} (C, U_-T).}$$

At the outer debt boundary, another outward mismatch triggers physical rematerialization and resets debt to zero.

Bottom WRITE

For every

$$|d| \le h_{\mathrm{ctr}},$$

$$\boxed{(D_d, Z_0) \xrightarrow{1} (C, \eta(\mu(dm + q))Z_0)\,,}$$

and

$$\boxed{(D_d, Z_0) \xrightarrow{0} (C, \eta(\mu(dm - p))Z_0)\,.}$$

When no pre-existing demand remains, the next input directly creates the exact new minimum-demand word.

Acceptance

A mandatory right endmarker is allowed only at

$$\boxed{(C, Z_0).}$$

It removes Z_0 and accepts by empty stack.
No other reachable configuration has a right-endmarker acceptance move.

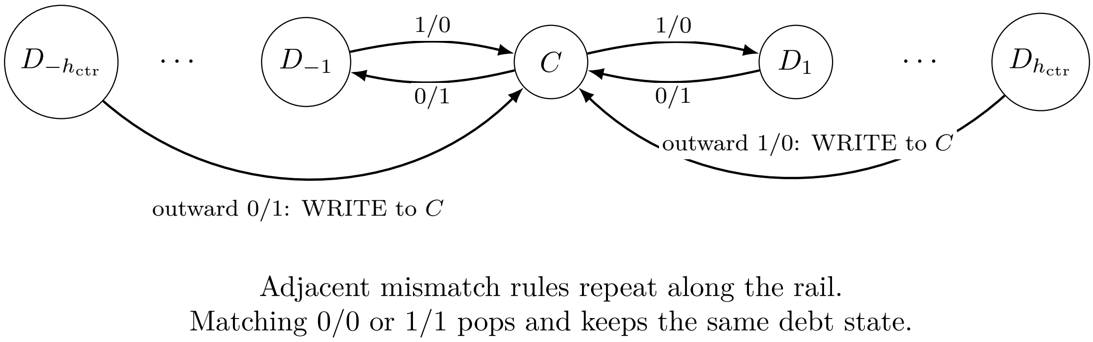


Figure 32.1: Schematic centered debt rail. Labels 1/0 and 0/1 denote input/top mismatches. Interior mismatches pop the visible cell and move debt by one; an outward mismatch at $\pm h_{\mathrm{ctr}}$ triggers WRITE back to the center.

The figure should visually emphasize:

- adjacent mismatch SEARCH moves;
- center state C;
- outer WRITE arcs back to C;
- MATCH as state-preserving pops.

32.2 Residual invariant

Let the logical stack be S.

Theorem 32.1 — Residual invariant.
After every processed binary prefix, if the centered configuration is

$$(D_d, SZ_0),$$

then

$$\boxed{\rho = W(P(S)) + dm.}$$

Proof.
Initially,

$$(C, Z_0)$$

has residual zero.
A matching 0-pop decreases stack weight by p, exactly matching the input-0 residual update.
A matching 1-pop increases stack weight by q.
For mismatch 1/0, popping one physical zero changes stack weight by $-p$, while moving

$$d \to d + 1$$

adds state potential

$$m = p + q.$$

The net change is

$$-p + m = q,$$

exactly the input-1 residual update.

The 0/1 case is symmetric:

$$q - m = -p.$$

At an outer WRITE, the replacement word is defined to carry exactly the missing boundary potential.

At bottom WRITE, the target demand is defined directly from the new residual.

Therefore every binary transition preserves the invariant. □

Combined with Chapter 27:

$$\boxed{\Theta(D_d, S) = \mu(\rho)}$$

for every reachable configuration.

32.3 Why the debt rail can stop at h_{ctr}

The key is the **minority size**

$$\mathrm{minc}(x, y) = \min(x, y).$$

Lemma 32.2 — WRITE-frontier minority bound.

Let

$$h_{\mathrm{ctr}} = \left\lfloor \frac{m-1}{2} \right\rfloor.$$

Then:

1. for every $|d| \le h_{\mathrm{ctr}}$, both bottom-WRITE demands

   $$\mu(dm + q)$$

   and

   $$\mu(dm - p)$$

   have minority size at most h_{ctr};

2. if

$$\mu((h_{\text{ctr}}+1)m) = (x_+, y_+),$$

then

$$\boxed{y_+ \leq h_{\text{ctr}} < x_+;}$$

3. if

$$\mu(-(h_{\text{ctr}}+1)m) = (x_-, y_-),$$

then

$$\boxed{x_- \leq h_{\text{ctr}} < y_-.}$$

Proof.

For the bottom-WRITE family, both residual forms can be written

$$r = jm + q$$

with

$$-h_{\text{ctr}} - 1 \leq j \leq h_{\text{ctr}}.$$

Let

$$\mu(r) = (x, y), \qquad \ell = x + y.$$

Modulo m, since

$$p \equiv -q \pmod{m},$$

the completion equation gives

$$-q\ell \equiv q \pmod{m}.$$

Because

$$\gcd(q, m) = 1,$$

$$\ell \equiv -1 \pmod{m}.$$

Hence

$$\ell = km - 1.$$

If both x and y were at least $h_{\text{ctr}} + 1$, then $\ell > m - 1$, so $k \geq 2$.
But $\mu(r)$ is kernel-free, so either

$$x < q$$

or

$$y < p.$$

Each alternative, together with $x, y \geq h_{\text{ctr}}+1$, forces the quotient index j outside the allowed interval.

Contradiction.

For the outer positive boundary, write $D = h_{\text{ctr}} + 1$. Integer solutions to weight Dm have form

$$x = D + qk, \qquad y = pk - D.$$

The minimum nonnegative solution uses the first feasible k, giving

$$y \leq h_{\text{ctr}}$$

and

$$x \geq h_{\text{ctr}} + 1.$$

The negative case is symmetric. □

This lemma explains the centered width geometrically.

Every freshly written canonical word contains at most h_{ctr} copies of its minority symbol.

The debt rail is exactly wide enough to search through that minority frontier.

32.4 Boundary guard

The centered construction also requires a boundary guarantee at $D_{h_{\text{ctr}}}$. An outward mismatch must not occur while the surviving suffix is still mixed, because writing a new canonical prefix in front of such a suffix could destroy the normal form.

The minority bound provides exactly this guarantee.

Lemma 32.3 — Boundary guard.

Assume the current stack is a suffix of a canonical word

$$\eta(x, y)$$

with

$$\min(x, y) \leq h_{\text{ctr}}.$$

Then:

1. if the state is $D_{h_{\text{ctr}}}$ and the stack begins with 0, an outward mismatch 1/0 can occur only on a unary suffix

$$\boxed{0^{t+1};}$$

2. symmetrically, at $D_{-h_{\text{ctr}}}$, an outward 0/1 mismatch can occur only on a unary suffix

$$\boxed{1^{t+1}.}$$

Proof.

Consider the positive case.

Reaching debt h_{ctr} requires at least h_{ctr} positive mismatch events.

Each such mismatch has popped one zero from the original canonical stack.

Suppose the surviving suffix begins with 0 but still contains a later 1.

Then the current zero lies inside the alternating part of $\eta(x, y)$, not in its final unary tail.

Every previously popped zero in that alternating region has a distinct preceding one that must also have been popped.

Together with:

- the current surviving zero;
- the later surviving one;

the original word would contain at least

$$h_{\mathrm{ctr}} + 1$$

zeros and at least

$$h_{\mathrm{ctr}} + 1$$

ones.

This contradicts

$$\min(x, y) \leq h_{\mathrm{ctr}}.$$

Therefore the suffix must be unary.

The negative case is symmetric. □

This is the LIFO safety theorem of the construction.

32.5 Canonical segments

Theorem 32.4 — Canonical-segment invariant.

Immediately after every WRITE, the machine is in the center state with stack

$$\boxed{\eta(\mu(r))Z_0}$$

for its current residual r.

The corresponding demand has minority size at most h_{ctr}.

Between two WRITEs, the logical stack is a suffix of that canonical word.

Proof.

Initially the logical stack is empty.

A bottom WRITE is canonical by definition, and Lemma 28.2 gives the minority bound.

Between WRITEs, MATCH and interior SEARCH only pop one pre-existing work symbol, so the stack remains a suffix of the last canonical WRITE word.

Now consider a positive outer WRITE.

By Lemma 28.3, its source suffix is

$$0^{t+1}.$$

Write

$$\mu((h_{\mathrm{ctr}} + 1)m) = (x, y).$$

Lemma 28.2 gives

$$y \le h_{\mathrm{ctr}} < x.$$

Thus the canonical word $\eta(x, y)$ ends in 0, and the WRITE produces

$$\eta(x, y)0^t = \eta(x + t, y).$$

The residual invariant shows that this vector has exactly the new residual weight.

Moreover, $y < p$, so no kernel packet can be subtracted; the vector is therefore already the minimum demand.

The negative boundary is symmetric. □

The entire reachable-store theorem has now been reduced to episodes:

$$\boxed{\text{WRITE creates a canonical word;}}$$

$$\boxed{\text{MATCH/SEARCH consume a suffix of it;}}$$

$$\boxed{\text{boundary WRITE can occur only after the mixed part is gone.}}$$

32.6 Two-cell qualitative visibility

Corollary 32.5 — Two-cell visibility.

For every reachable logical stack S,

$$\boxed{\mathrm{alph}(S) = \mathrm{alph}(\mathrm{pref}_2(S)).}$$

Equivalently, if both 0 and 1 occur anywhere in S, then both occur among the first two cells.

Proof.

Every canonical word $\eta(x, y)$, and every suffix of it, avoids

$$001 \qquad \text{and} \qquad 110.$$

Therefore a mixed suffix cannot begin with two equal symbols.

It begins with

$$10 \qquad \text{or} \qquad 01.$$

Apply Theorem 28.4. □

This is qualitative, not quantitative.

The first two cells reveal which demand types remain.

They do not reveal the unbounded magnitudes.

32.7 Exact WRITE normalization

Theorem 32.6 — Exact WRITE normalization.

After every bottom or outer-boundary WRITE,

$$\boxed{(D_d, S) = (C, \eta(\mu(\rho)))\,,}$$

where ρ is the residual after the consumed binary symbol.

WRITE therefore does more than preserve the invariant.

It returns the physical stack to the exact minimum-demand normal form.

This result is the normalization analogue of the original endmarker materialization theorem, but now it occurs online.

32.8 Zero residual

Theorem 32.7 — Zero-residual correctness.

At the end of every binary prefix,

$$\boxed{\rho = 0 \iff (D_d, S) = (C, \varepsilon).}$$

Proof.

The reverse implication follows from the invariant.

For the forward direction, consider a nonempty processed prefix.

Immediately after the most recent WRITE, the machine was at

$$(C, \eta(\mu(r_0)))$$

for some residual r_0.

Write

$$\mu(r_0) = (x, y).$$

Since then, each MATCH or SEARCH step has popped exactly one pre-existing demand cell.

Suppose the intervening input contains:

$$u$$

zeros and

$$v$$

ones, so

$$u + v \leq x + y.$$

If the current residual is zero, this segment completes r_0.

By minimality of $\mu(r_0)$, every completion requires length at least

$$x + y.$$

Hence

$$u + v = x + y.$$

The entire canonical stack has been consumed.

So

$$S = \varepsilon.$$

The residual invariant then gives

$$dm = 0,$$

hence

$$d = 0.$$

Therefore the state is C. $\square$

This proof is important because acceptance is obtained from shortest-demand minimality, not from a large case analysis of reachable stacks.

32.9 Main construction theorem

Theorem 32.8 — Real-time centered rich DPDA.

For every coprime

$$p, q \geq 1,$$

the centered schema defines a deterministic pushdown machine with:

$$\boxed{|Q| = 2\left\lfloor \frac{p+q-1}{2} \right\rfloor + 1}$$

semantic core states and stack alphabet

$$\boxed{\{0, 1, Z_0\}.}$$

Every binary transition consumes exactly one symbol.

No ε-transition is used during binary processing.

With the mandatory right endmarker, the machine recognizes

$$\boxed{L_{p,q}.}$$

Furthermore:

1. every WRITE returns to the exact minimum-demand normal form;
2. every reachable stack has two-cell qualitative visibility;
3. the residual is exactly decoded by the state-plus-stack invariant.

The state count is

$$\boxed{|Q| = \begin{cases} p+q, & p+q \text{ odd}, \\ p+q-1, & p+q \text{ even}. \end{cases}}$$

This is the exact size of the centered construction.

This is a construction-size statement rather than an unrestricted state lower bound. The representation-independent state theorem is established in Part XVII.

32.10 Shortest-demand quotient

The centered machine is physically richer than a residual counter.

Two configurations can store the same semantic demand while distributing it differently between:

- physical stack;
- finite-control debt.

Corollary 32.9 — Shortest-demand quotient.

For reachable configurations, define

$$c_1 \sim_\Theta c_2 \iff \Theta(c_1) = \Theta(c_2).$$

Then the residual/minimum-demand transition system is a semantic quotient of the rich configuration graph.

The quotient forgets normalization chronology.

The physical fiber may remember it.

That observation will later become a central theorem family in the Uniqueness/Lift Theory.

For now, it is already visible concretely.

Chapter 33

What "Optimal" Means in the Deferred-SEARCH Architecture

The centered construction uses

$$2h_{\mathrm{ctr}} + 1$$

debt states, where

$$h_{\mathrm{ctr}} = \left\lfloor \frac{p+q-1}{2} \right\rfloor .$$

Is that number necessary?

The answer is yes—but only for a precisely stated physical interface.

This theorem must not be confused with the raw residual-faithful frontier of Chapter 8.

The two results optimize different representation categories.

33.1 Canonical deferred-SEARCH model

Definition 33.1 — Canonical deferred-SEARCH realization.

A realization belongs to the canonical deferred-SEARCH model if:

1. the normalized minimum demand
$$\mu(r)$$
is physically represented by the canonical word
$$\eta(\mu(r));$$
2. during a SEARCH episode, each binary transition consuming a pre-existing demand cell removes exactly one such cell;
3. a mismatch is stored only in finite control—no auxiliary debt marker is inserted into the remaining pre-existing word;
4. in the witness episode used below, the transition consuming the final pre-existing demand cell performs no materializing push and leaves the same bottom marker Z_0;
5. debt materialization is deferred to a later transition.

The last condition is what makes the model **deferred**.

An eager implementation lies outside this model.

33.2 Residual distinguishability

Lemma 33.2 — Residual distinguishability.

If two prefixes have distinct residuals

$$r_1 \neq r_2,$$

then a common binary continuation followed by the right endmarker distinguishes them.

Proof.

Choose a word z whose Parikh vector is

$$\mu(r_1).$$

Appending z sends the first residual to zero.
The second residual becomes

$$r_2 - r_1 \neq 0.$$

The right endmarker is accepted after the first prefix and rejected after the second.
□

This is a language property.
It is not assumed by the architecture.

33.3 One witness containing enough of both symbols

Define

$$\boxed{\nu_{h_{\mathrm{ctr}}} = \begin{cases} (h_{\mathrm{ctr}}, h_{\mathrm{ctr}}), & m = p + q \text{ odd}, \\ (h_{\mathrm{ctr}}, h_{\mathrm{ctr}} + 1), & m \text{ even}. \end{cases}}$$

Its canonical word is

$$(10)^{h_{\mathrm{ctr}}}$$

in the odd case and

$$(10)^{h_{\mathrm{ctr}}}1$$

in the even case.

Lemma 33.3 — Canonical witness.

$$\boxed{\nu_{h_{\mathrm{ctr}}} = \mu(W(\nu_{h_{\mathrm{ctr}}})).}$$

Thus $\nu_{h_{\mathrm{ctr}}}$ is kernel-free and contains at least h_{ctr} zeros and h_{ctr} ones.

Proof.

If m is odd and a kernel packet

$$(q, p)$$

could be subtracted from $(h_{\mathrm{ctr}}, h_{\mathrm{ctr}})$, then

$$p \leq h_{\mathrm{ctr}}, \qquad q \leq h_{\mathrm{ctr}},$$

so

$$m = p + q \leq 2h_{\text{ctr}} = m - 1,$$

impossible.

If m is even, subtraction from $(h_{\text{ctr}}, h_{\text{ctr}} + 1)$ would require

$$q \leq h_{\text{ctr}}, \qquad p \leq h_{\text{ctr}} + 1,$$

giving again

$$m \leq 2h_{\text{ctr}} + 1 = m - 1.$$

Contradiction. □

This single canonical block contains enough of both symbol types to generate every debt in the centered interval.

33.4 Exhaust the same stack, leave different debts

Lemma 33.4 — Debt distinguishability witness.

Starting from the same canonical witness word $\eta(\nu_{h_{\text{ctr}}})$, a deferred SEARCH episode can consume the entire pre-existing work stack while leaving any debt

$$\boxed{d \in \{-h_{\text{ctr}}, -h_{\text{ctr}} + 1, \dots, h_{\text{ctr}}\}.}$$

The resulting residual is

$$\boxed{dm.}$$

Proof.

For

$$d \geq 0,$$

choose exactly d witness zero-cells and mismatch them by reading 1 instead of 0.

Match every other pre-existing cell.

Debt increases monotonically from 0 to d.

For

$$d < 0,$$

mismatch exactly $|d|$ witness one-cells by reading 0.

Because the witness contains at least h_{ctr} copies of both symbols, every debt in the interval is possible.

Let the witness contain x zeros and y ones.

If A zero-cells are mismatched by input 1 and B one-cells are mismatched by input 0, then

$$d = A - B.$$

The consumed input contains

$$x - d$$

zeros and

$$y + d$$

ones.

Starting from residual

$$px - qy,$$

the final residual is

$$px - qy + q(y + d) - p(x - d) = d(p + q) = dm.$$

□

By the deferred-model assumption, after the last witness cell is consumed, every case has the **same physical stack**:

$$\boxed{Z_0.}$$

Only finite control can distinguish the residuals.

33.5 Exact deferred width

Theorem 33.5 — Exact deferred-SEARCH control width.

Every canonical deferred-SEARCH realization requires at least

$$\boxed{2h_{\mathrm{ctr}} + 1}$$

finite-control states.

The centered construction uses exactly $2h_{\mathrm{ctr}} + 1$.

Therefore

$$\boxed{|Q|_{\min}^{\mathrm{deferred}} = 2\left\lfloor \frac{p + q - 1}{2} \right\rfloor + 1}$$

or equivalently

$$\boxed{|Q|_{\min}^{\mathrm{deferred}} = \begin{cases} p + q, & p + q \text{ odd}, \\ p + q - 1, & p + q \text{ even}. \end{cases}}$$

Proof.

Lemma 29.4 produces

$$2h_{\mathrm{ctr}} + 1$$

different residuals:

$$-hm, \ldots, -m, 0, m, \ldots, hm.$$

In every case the work stack has been exhausted and the remaining physical stack is the same bottom marker Z_0.

If two distinct residuals shared the same finite-control state, the two instantaneous configurations would be physically identical.

But Lemma 29.2 gives a common continuation that must distinguish the residuals.

Determinism from identical configurations would force identical behavior.

Contradiction.
Thus the $2h_{\mathrm{ctr}}+1$ residual debts require $2h_{\mathrm{ctr}}+1$ distinct states.
The centered rail attains the bound. □

33.6 Scope — place this beside the theorem

The theorem does **not** say:

$$\boxed{\text{every DPDA recognizing } L_{p,q} \text{ needs } 2h_{\mathrm{ctr}}+1 \text{ states.}}$$

A smaller machine may:

- abandon literal-demand storage;
- use another stack encoding;
- materialize debt eagerly;
- insert auxiliary physical markers;
- use another normalization schedule;
- leave the deferred-SEARCH interface entirely.

This scope is part of the theorem.
It is not a technical footnote.

33.7 Why Chapter 8 gives a different optimum

Chapter 8 proved the raw residual-faithful frontier:

$$S_{\mathrm{raw}}(g) = \begin{cases} \infty, & g = 0, \\ p+q, & g = 1, \\ \max(p,q), & g \geq 2. \end{cases}$$

With two work symbols, the raw residual graph can therefore be realized using only

$$\max(p,q)$$

states.
The centered deferred machine can require

$$p+q \quad \text{or} \quad p+q-1.$$

There is no contradiction.

Raw residual-faithful question

Give every residual one stable physical configuration and make every residual edge one raw input transition.

Deferred literal-demand question

Store the actual canonical minimum-demand word, search it by pure pops, and keep displaced mismatch information only in finite control until later materialization.

These are different representation contracts.

The correct conclusion is:

$$\boxed{\text{resource optimality is representation-category relative.}}$$

This observation is more important than either number individually.

It is one of the direct motivations for the later Structure and Uniqueness theories.

33.8 Why this chapter belongs before the 2:1 recovery

The width theorem tells us what the centered architecture itself costs.

Only after its category has been made explicit should we specialize back to the historical 2:1 machine.

Otherwise the reader could incorrectly interpret the three-state historical architecture as globally state-optimal.

The next chapter will instead say something more precise and more interesting:

The general centered theory reconstructs the unusual architecture of the original machine.

And the remaining transition differences will turn out to be schedule choices.

Chapter 34

Closing the First Loop: Recovering the Original 2:1 Machine

The strongest test of a proposed generalization is not that it accepts the right family.

It is that, when specialized back to the motivating case, it explains the machine from which the theory arose.

Set

$$\boxed{p = 2, \qquad q = 1.}$$

Then

$$m = p + q = 3,$$

and

$$h_{\text{ctr}} = \left\lfloor \frac{3-1}{2} \right\rfloor = 1.$$

The centered rail has exactly three states:

$$D_{-1}, \qquad D_0, \qquad D_1.$$

Identify them as

$$\boxed{D_{-1} = R, \qquad D_0 = C, \qquad D_1 = L.}$$

The state potentials become

$$\boxed{-3, \qquad 0, \qquad +3.}$$

These are exactly the state potentials discovered directly in the original DPDA.

The coincidence is not imposed.

It falls out of the general debt formula

$$\Phi(D_d) = d(p + q).$$

34.1 Center mismatch directions

From $C = D_0$:

- a mismatch 1/0 moves to $D_1 = L$;
- a mismatch 0/1 moves to $D_{-1} = R$.

Thus the general SEARCH rail reproduces the original side-state directions.
The original left state is the positive debt state.
The original right state is the negative debt state.
This gives the old L/R asymmetry a uniform arithmetic interpretation.

34.2 Left cancellation rail

At $L = D_1$, matching 0/0 is simply MATCH:

$$\boxed{(L, 0T) \xrightarrow{0} (L, T).}$$

This is exactly the original cancellation transition that generated the entire Fuss–Catalan branch.

Thus the transition that later produced correction geometry is also a literal specialization of the general MATCH rule inside positive debt.

34.3 Right locality-repair rail

At $R = D_{-1}$, the general rules reproduce the right-side locality repair behavior.

The state is not an unrelated special-purpose gadget.

It is the negative SEARCH-debt sheet required when a one-cell blocker has been displaced from the canonical demand stack.

This is the general explanation of why the original machine needed a right side at all.

34.4 Historical WRITE arithmetic

The primitive kernel is

$$\boxed{K = (1, 2).}$$

The minimal kernel-carry WRITE equation now reproduces the characteristic replacement Parikh vectors of the historical table.

Depending on:

- source debt;
- target debt;
- input;
- removed top symbol;

the first nonnegative kernel shift produces the historical families represented by words such as

$$10,$$

$$11,$$

$$00,$$

$$000,$$

$$11111,$$

$$111111,$$

together with the corresponding bottom-WRITE forms.

The precise ordering is the selected canonical literal-demand order.

Thus the long WRITE strings are recovered from the general arithmetic compiler rather than copied into the generalization by hand.

34.5 Specialization and recovery theorem

Theorem 34.1 — 2:1 **specialization and recovery.**

Specializing the centered real-time $p{:}q$ construction to

$$(p, q) = (2, 1)$$

recovers:

1. the three-state
$$C/L/R$$
skeleton;
2. the state potentials
$$0, +3, -3;$$
3. the center mismatch directions;
4. the left cancellation rail;
5. the right locality-repair rail;
6. the arithmetic of the historical WRITE words.

The only reachable transition families not reproduced literally are two right-state matching families.

Those differences are purely normalization-schedule differences.

This theorem closes the first large loop of the monograph:

$$\boxed{\text{original machine} \to \text{general theory} \to \text{original architecture reconstructed.}}$$

34.6 Lazy versus eager discharge

Consider the reachable configuration

$$(R, 0).$$

The centered specialization performs the lazy matching pop

$$\boxed{(R, 0) \xrightarrow{0}_{\text{lazy}} (R, \varepsilon).}$$

The historical machine instead immediately materializes the represented demand and moves to the center:

$$\boxed{(R, 0) \xrightarrow{0}_{\text{eager}} (C, 111).}$$

Both configurations represent the same minimum demand.

Likewise, for a unary right stack

$$(R, 1^k),$$

the centered specialization uses

$$\boxed{(R, 1^k) \xrightarrow{1}_{\text{lazy}} (R, 1^{k-1}),}$$

whereas the historical machine uses

$$\boxed{(R, 1^k) \xrightarrow{1}_{\text{eager}} (C, 1^{k+2}).}$$

Again, the semantic demand agrees.

Only the time at which virtual debt is rematerialized differs.

34.7 Canonical materialization map

Define

$$\boxed{\mathcal{N}(c) = (C, \eta(\Theta(c)))\,.}$$

This map forgets the physical distribution of demand between state and stack and returns the canonical center representation of the same shortest demand.

The historical eager successor is exactly the canonical materialization of the centered lazy successor on the two differing right-state families.

34.8 Eager–lazy semantic equivalence

Theorem 34.2 — Eager–lazy semantic equivalence.

Let

$$c_{\text{hist}}(w)$$

and

$$c_{\text{cent}}(w)$$

be the core configurations reached after the same binary prefix w by:

- the historical 2:1 machine;
- the centered 2:1 specialization.

Then

$$\boxed{\Theta(c_{\text{hist}}(w)) = \Theta(c_{\text{cent}}(w))}$$

for every prefix w.
On every transition family where the two tables differ,

$$\boxed{\delta_{\text{hist}}(c, a) = \mathcal{N}\big(\delta_{\text{cent}}(c, a)\big).}$$

Thus:

$$\boxed{\text{historical 2:1} = \text{eager discharge},}$$

while

$$\boxed{\text{centered 2:1} = \text{lazy discharge}.}$$

The semantics is the same.
The normalization chronology is different.

34.9 Historical machine as an eager refinement

Corollary 34.3 — Historical eager refinement.
Start with the centered 2:1 specialization.
Retain every specialized transition except on the two reachable right-state matching families.
On exactly those families, replace the lazy successor by its canonical materialization $\mathcal{N}$.
The resulting core is the historical 2:1 machine.
Therefore the return from the general theory is exact at the level of:

- control architecture;
- state potentials;
- WRITE arithmetic;
- shortest-demand semantics;

and becomes literal transition-table identity after choosing the historical eager normalization schedule.
Direct specialization alone is not claimed to be literal table identity.
The remaining difference is precisely the eager/lazy schedule choice.

34.10 Computational audit

The reference implementation of the centered family was independently rerun during the monograph audit for every ordered coprime pair

$$1 \leq p, q \leq 20,$$

including

$$(1,1).$$

The run covered:

$$\boxed{\text{255 ratios,}}$$

$$\boxed{\text{1,909,191 distinct reached configurations,}}$$

and

$$\boxed{\text{3,738,864 binary edges.}}$$

It checked:

1. residual invariant;
2. residual-collision consistency;
3. two-cell qualitative visibility;
4. zero-residual correctness.

Result:

$$\boxed{\text{PASS.}}$$

These computations are consistency checks.
The symbolic theorems remain the proof.

34.11 The real discovery at the end of the $p{:}q$ generalization

The general machine has done more than extend 2:1.
It has exposed a structural nonuniqueness.
The semantic evolution is rigid:

$$\boxed{\text{one residual} \to \text{one minimum demand.}}$$

But the physical normalization chronology is not rigid:

$$\boxed{\text{same semantics} \not\Rightarrow \text{unique physical schedule.}}$$

This is the hinge of the remainder of the monograph.
The Structure Theory developed in the following Part explains why WRITE, SEARCH, and CANCEL are the primitive operational roles and separates the components forced by semantic arithmetic from those forced by one-top LIFO locality.
That is the beginning of the **Structure Theory**.
From this point onward the main sequence is uninterrupted:

$$\boxed{\text{Structure Theory} \to \text{Uniqueness/Lift Theory} \to \text{General Theory.}}$$

Part X

Write–Search–Cancel as a Structural Theory

Research provenance. This Part develops the WSC decomposition, debt channels, physical materialization level, and structural factorization theory. Congruence and minimization theory for visibly pushdown systems and classical deterministic-pushdown equivalence are related but distinct background [3, 24, 50, 34].

Chapter 35

Semantic Arithmetic Before Representation

Chapter 30 ended with a fact that the original 2:1 machine could not reveal by inspection alone:

the same semantic demand can be normalized on different physical schedules.

The centered generalization is lazy on two historical right-state families.
The original machine is eager.
Both are correct.
The next task is therefore no longer to construct a machine.
It is to separate what is forced by arithmetic from what is forced by LIFO locality and what is merely a normalization choice.
This is the beginning of the Structure Theory.

35.1 Dominating kernel level

Let a prefix contain

$$a = N_0(w), \qquad b = N_1(w).$$

The primitive balanced kernel is

$$K = (q, p).$$

Define

$$\boxed{\kappa(w) = \max\left(\left\lceil \frac{a}{q} \right\rceil, \left\lceil \frac{b}{p} \right\rceil\right).}$$

Theorem 35.1 — Dominating kernel level.

The unique minimum completion demand is

$$\boxed{\mu(w) = \big(q\kappa(w) - a,\ p\kappa(w) - b\big).}$$

Equivalently,

$$\boxed{(a, b) + \mu(w) = \kappa(w)K.}$$

Proof.
If (x, y) completes w, then

$$q(b + y) = p(a + x).$$

Coprimality gives an integer k such that

$$a + x = qk, \qquad b + y = pk.$$

Thus

$$x = qk - a, \qquad y = pk - b.$$

Nonnegativity requires

$$k \geq \left\lceil \frac{a}{q} \right\rceil$$

and

$$k \geq \left\lceil \frac{b}{p} \right\rceil.$$

The least feasible k is therefore $\kappa(w)$.
The completion length is

$$(p + q)k - (a + b),$$

which is strictly increasing in k.
Hence the first dominating kernel level gives the unique minimum demand. □
Geometrically, κ is the first point

$$kK = (kq, kp)$$

on the kernel ray that dominates the current count point (a, b) coordinatewise.
The minimum demand is the slack vector from the current count point to that first dominating kernel point.

35.2 One input symbol: CANCEL plus optional carry

Let

$$e_0 = (1, 0), \qquad e_1 = (0, 1).$$

Write

$$\mu_t$$

and

$$\kappa_t$$

for the minimum demand and dominating kernel level after t input symbols.

Theorem 35.2 — Unique semantic CANCEL–carry factorization.
If the next input symbol is

$$a \in \{0, 1\},$$

then

$$\boxed{\mu_{t+1} = \mu_t - e_a + \chi_t K,}$$

where

$$\boxed{\chi_t = \kappa_{t+1} - \kappa_t \in \{0, 1\}.}$$

Moreover:

- on input 0,
$$\chi_t = 1 \iff \mu_t^{(0)} = 0;$$
- on input 1,
$$\chi_t = 1 \iff \mu_t^{(1)} = 0.$$

Proof.
Reading one symbol increments exactly one prefix-count coordinate.
Each ceiling in the formula for κ therefore changes by at most one.
Hence

$$\chi_t \in \{0, 1\}.$$

Before the input,

$$(N_0, N_1) + \mu_t = \kappa_t K.$$

After the input,

$$(N_0, N_1) + e_a + \mu_{t+1} = \kappa_{t+1} K.$$

Subtracting gives the factorization.
For input 0, the zero slack is

$$q\kappa_t - N_0.$$

The kernel level rises precisely when that coordinate is already saturated, i.e. when the zero-demand coordinate is 0.
The input-1 case is symmetric. □
Thus every semantic input step has exactly two roles:

$$\boxed{\text{CANCEL one unit of the incoming symbol}}$$

and, if the relevant demand coordinate would become negative,

$$\boxed{\text{add one primitive kernel packet.}}$$

This is realization-independent.
It is the semantic reason WRITE exists at all.

35.3 Semantic WRITE count

Every semantic carry increments κ by one.

Corollary 35.3 — Semantic WRITE count.

For every prefix,

$$\boxed{W_{\text{sem}}(w) = \kappa(w).}$$

If

$$w \in L_{p,q}$$

and

$$N_0(w) = qn, \qquad N_1(w) = pn,$$

then

$$\boxed{W_{\text{sem}}(w) = n.}$$

The number of semantic kernel arrivals is fixed by the input.
What can vary is when those arrivals are physically materialized.

35.4 Demand-length clock

Let

$$r = p + q.$$

Corollary 35.4 — Demand-length clock.

At prefix length t,

$$\boxed{|\mu_t| = r\kappa_t - t.}$$

Hence

$$\boxed{|\mu_t| \equiv -t \pmod{r}.}$$

The minimum-demand length carries a rigid modular clock.
This clock will later have a physical analogue.

35.5 Semantic chronology is unique

Corollary 35.5 — Semantic chronology uniqueness.

For a fixed input word, the entire sequences

$$(\rho_t), \qquad (\kappa_t), \qquad (\mu_t), \qquad (\chi_t)$$

are uniquely determined by the input.
They do not depend on which correct pushdown representation is used.
This is the semantic backbone of the Structure Theory.
It says:

$$\boxed{\text{the semantic chronology is unique}}$$

before we say anything about:

- stack order;
- side states;
- SEARCH debt;
- eager/lazy materialization.

The next chapter introduces the first mature abstraction separating this semantic process from its physical realization.

Chapter 36

What One-Top LIFO Locality Forbids

We now formalize a physical realization of the unique demand dynamics.

The abstraction appears here, not at the beginning of the monograph, because the reader has already seen why it is needed.

36.1 Exact literal-demand lift

Let

$$\mathcal{M}_{p,q} = \{\mu(\rho) : \rho \in \mathbb{Z}\}$$

be the semantic minimum-demand system.

Definition 36.1 — Exact real-time literal-demand lift.

An exact literal-demand lift consists of:

- a reachable physical configuration set
$$\mathcal{C};$$
- deterministic binary transitions
$$\delta_a;$$
- a semantic projection
$$\Theta : \mathcal{C} \to \mathcal{M}_{p,q};$$
- a canonical section
$$\sigma : \mathcal{M}_{p,q} \to \mathcal{C};$$

such that

$$\boxed{\Theta \circ \sigma = \mathrm{id}}$$

and

$$\boxed{\Theta(\delta_a(c)) = U_a(\Theta(c)),}$$

where U_a is the unique semantic update from Chapter 31.

The section chooses one preferred physical representative of each semantic demand.

The other physical representatives form the semantic fiber.

That vocabulary will become central later.

36.2 Fixed block orientation

For the present Structure Theory, choose the literal canonical encoding

$$\boxed{\eta(x,y) = 1^y 0^x Z_0,}$$

with stack top on the left.
A one-top real-time transition has the form

$$\boxed{XT \longrightarrow \gamma T.}$$

The suffix T is immutable during that transition.
This one fact gives an exact direct-realizability criterion.

36.3 Top-rewrite suffix criterion

Theorem 36.2 — Top-rewrite suffix criterion.
Suppose a one-top transition begins from

$$XT.$$

A desired target word S^* can be reached in that one transition if and only if

$$\boxed{T \text{ is a suffix of } S^*.}$$

Proof.
Every one-top successor has the form

$$\gamma T.$$

Therefore direct realization requires T to be a suffix of the target.
Conversely, if

$$S^* = \gamma T,$$

replace X by γ. □
This theorem is one of the simplest and most important results in the entire structural line.
It converts one-top LIFO locality into an exact string condition.

36.4 Complete obstruction classification

Apply the suffix criterion to the canonical word

$$1^y 0^x Z_0.$$

Theorem 36.3 — Canonical direct-realizability classification.
For the fixed orientation $1^y 0^x Z_0$:

1. input 1 is directly canonical-realizable for every canonical demand;

2. input 0 fails to be directly canonical-realizable exactly in two families:

Buried-CANCEL obstruction

$$\boxed{x > 0, \qquad y > 0.}$$

Carry-order obstruction

$$\boxed{x = 0, \qquad y \geq 2, \qquad q \geq 2.}$$

For the reverse block orientation, exchange 0 and 1.

Proof — input 1. If $y > 0$, pop the top 1.
If $y = 0$ and $x > 0$, the semantic target is

$$1^{p-1}0^{x+q}Z_0.$$

The untouched suffix after removing the top zero is

$$0^{x-1}Z_0,$$

which is a suffix of the target.
Thus a single top replacement suffices.
The bottom case is direct as well.

Proof — input 0. If $y = 0$ and $x > 0$, pop the top 0.
Now let $x > 0, y > 0$.
The semantic target is

$$1^y0^{x-1}Z_0.$$

But after removing the top 1, the untouched suffix is

$$1^{y-1}0^xZ_0.$$

That suffix is not a suffix of the target.
Direct canonicalization is impossible.
Finally let $x = 0$.
The carry target is

$$1^{y+p}0^{q-1}Z_0.$$

For $y = 0$, write at the bottom.
For $y = 1$, the untouched suffix is empty and direct replacement is possible.
For $y \geq 2$, the untouched suffix is

$$1^{y-1}Z_0.$$

It is a suffix of the target exactly when

$$q - 1 = 0,$$

that is,

$$q = 1.$$

So for $q \geq 2$ the second obstruction family is exact. □

36.5 Two fundamentally different failures

The classification reveals that “cannot stay canonical” has two different causes.

Buried cancellation

The symbol that semantic CANCEL wants to remove exists in the physical word, but a blocker lies above it.

This is a visibility/locality failure.

Carry-order obstruction

A semantic carry can create the correct **number** of zeros and ones in one transition, yet the immutable suffix forces those letters into the wrong order.

This is not a quantity failure.

It is an ordering failure.

Thus:

$$\boxed{\text{count-exact} \not\Rightarrow \text{canonical.}}$$

The next chapter separates the resulting representation debt into three channels.

Chapter 37

Quantity, Order, and Control Debt

A configuration may leave the canonical section in more than one way.

Parikh counts see only one of them.

The Structure Theory therefore distinguishes:

1. quantity debt;
2. order debt;
3. control/sheet debt.

These channels will remain separate through the Uniqueness and General Theory Parts.

37.1 Quantity defect

Let $S(c)$ be the literal work-stack word of a rich configuration c.

Define its physical Parikh vector

$$P(c) = \mathrm{Par}(S(c)).$$

Definition 37.1 — Quantity defect.

$$\boxed{Q(c) = \Theta(c) - P(c).}$$

A canonical configuration has

$$Q = 0.$$

But $Q = 0$ does not imply canonicality because Parikh counts forget literal order.

37.2 Universal representation conservation

Proposition 37.2 — Universal representation conservation.

For every exact transition on input a,

$$\boxed{\Delta P + \Delta Q = -e_a + \chi K.}$$

Proof.

Since

$$Q = \Theta - P,$$

$$\Delta Q = \Delta\Theta - \Delta P.$$

The semantic update is

$$\Delta\Theta = -e_a + \chi K.$$

Rearrange. □
This equation holds before we impose the later WSC lattice restrictions.
It is the universal accounting identity.

37.3 Order defect

For literal word

$$S = S_1 \cdots S_m,$$

define

$$\boxed{\Omega(S) = \#\{(i,j) : i < j,\ S_i = 0,\ S_j = 1\}.}$$

Lemma 37.3 — Block-order test.

$$\boxed{\Omega(S) = 0 \iff S \in 1^*0^*.}$$

Thus Ω measures departure from the chosen canonical block order.

37.4 Three-channel canonicality

A third channel is finite control.
A configuration can have:

- correct counts;
- correct literal order;

but still lie on a noncanonical control sheet.

Proposition 37.4 — Three-channel canonicality.
A rich configuration is canonical exactly when:

1. quantity defect is zero;
2. order defect is zero;
3. the control state is the canonical section state.

Thus:

$$\boxed{\text{canonicality} = \text{quantity} + \text{order} + \text{sheet/control}.}$$

This is the first mature answer to the question:

> Where can normalization debt live?

Proposition 37.4A — Configuration-level separation of the three defect channels.

At the level of exact semantic representations used in this theory, none of the three canonicality channels is implied by the other two.

Each channel has a witness for which the other two vanish.

- **Quantity-only defect.** In the 2:1 schedule interaction studied later, the kernel-excess configuration
$$(C, 1110)$$
is compared with canonical
$$(C, 1).$$
The control sheet and block order are canonical, but the quantity defect is one nonzero kernel packet.
- **Order-only defect.** The carry-order repair of Theorem 33.6 preserves the exact canonical Parikh vector while forcing positive inversion count
$$\Omega > 0.$$
- **Control-only defect.** Take any exact literal lift and adjoin a semantically ignored two-sheet control decoration. A balanced input can return the base machine to a canonical stack while leaving the auxiliary sheet noncanonical. Quantity and order are then canonical while control defect remains nonzero.

Therefore

$$\boxed{Q = 0, \quad \Omega = 0, \quad \text{canonical control}}$$

are genuinely independent physical requirements.

This point becomes important in the Uniqueness Theory: a count-shadow invariant alone cannot reconstruct the full physical semantic fiber.

37.5 Kernel economy

Define a literal-demand representation to be **kernel-economical** if

$$\boxed{|P(c)| \leq |\Theta(c)|}$$

for every reachable configuration.

This prohibits adding complete zero-residual kernel packets merely as redundant workspace.

Under economy, the buried-CANCEL obstruction forces an exact quantity debt.

37.6 Forced unit SEARCH

Theorem 37.5 — Forced unit SEARCH for buried cancellation.

Assume:

- one-top real-time locality;
- kernel economy;

- canonical mixed demand
  $$\Theta(c) = (x, y)$$
  with
  $$x > 0, \quad y > 0;$$
- literal encoding
  $$1^y 0^x Z_0.$$

On input 0, the one-step successor has

$$\boxed{Q' = (-1, 1) = -(1, -1).}$$

Proof.
The semantic target is

$$(x - 1, y),$$

whose length is

$$x + y - 1.$$

A one-top transition can replace only the blocking top 1 by some word γ.
The successor physical length is

$$x + y - 1 + |\gamma|.$$

Kernel economy requires this to be at most the semantic target length.
Hence

$$|\gamma| = 0.$$

The top 1 must be popped.
The physical successor counts are

$$(x, y - 1),$$

while the semantic target is

$$(x - 1, y).$$

Therefore

$$Q' = (-1, 1).$$

□
Thus SEARCH is not merely one convenient implementation trick.
Under the stated locality and economy assumptions, the buried cancellation produces exactly one forced exchange defect.

37.7 Carry obstruction with zero quantity debt

The second obstruction behaves differently.
Assume:

$$x = 0, \qquad y \geq 2, \qquad q \geq 2.$$

On input 0, the semantic target is

$$1^{y+p}0^{q-1}Z_0.$$

A one-top transition can replace the top 1 by

$$\gamma = 1^{p+1}0^{q-1}.$$

The resulting word is

$$1^{p+1}0^{q-1}1^{y-1}Z_0.$$

Theorem 37.6 — Count-exact carry-order repair.
The successor has exactly the target Parikh counts:

$$\boxed{Q' = 0.}$$

But it has nonzero order defect.
Thus the representation can be semantically count-exact and still noncanonical.

37.8 Minimum order debt

Theorem 37.7 — Minimum carry-order debt.
Under the same hypotheses, among all one-top successors with exact target counts,

$$\boxed{\Omega_{\min} = (q-1)(y-1).}$$

Proof.
The untouched suffix is

$$1^{y-1}Z_0.$$

Exact target counts force the replacement word to contain:

$$q - 1$$

zeros and

$$p + 1$$

ones.
Every one of those $q-1$ zeros lies before all $y-1$ untouched suffix ones.
Therefore at least

$$(q-1)(y-1)$$

inversions are forced.
The replacement

$$1^{p+1}0^{q-1}$$

introduces no additional inversions inside the replacement and attains the bound. $\square$

The obstruction/debt principle is now clear:

$$\boxed{\text{LIFO obstruction forces departure from the canonical section,}}$$

but the debt may be stored in:

- quantity;
- order;
- finite control;
- or several channels at once.

The next chapter isolates a count-shadow class in which the quantity arithmetic becomes rigid.

Chapter 38

The WSC Count Shadow

The preceding debt theory was general within the literal-demand setting.

We now impose an explicit representation policy to obtain a sharper arithmetic structure.

This policy is not claimed to describe every correct DPDA.

The scope is part of the theorem.

Set

$$\boxed{\mathbf{h} = (1, -1), \qquad K = (q, p), \qquad r = p + q.}$$

38.1 Phase coherence

Definition 38.1 — Phase coherence.

A reachable configuration is phase-coherent if

$$\boxed{|P(c)| \equiv |\Theta(c)| \pmod{r}.}$$

A realization is phase-coherent if every reachable configuration satisfies this congruence.

Since

$$Q = \Theta - P,$$

phase coherence is equivalent to

$$Q_0 + Q_1 \equiv 0 \pmod{r}.$$

This congruence produces the WSC lattice.

38.2 WSC defect lattice

Define

$$\boxed{\mathcal{D}_{p,q} = \mathbb{Z}\mathbf{h} + \mathbb{Z}K.}$$

Theorem 38.2 — WSC defect lattice.

$$\boxed{\mathcal{D}_{p,q} = \{(u, v) \in \mathbb{Z}^2 : u + v \equiv 0 \pmod{r}\}.}$$

Consequently every phase-coherent quantity defect has a unique decomposition

$$\boxed{Q = s\mathbf{h} + wK, \qquad s, w \in \mathbb{Z}.}$$

Proof.
Both generators have coordinate sum divisible by r:

$$(\mathbf{h})_0 + (\mathbf{h})_1 = 0,$$

$$K_0 + K_1 = r.$$

Conversely, if

$$u + v \equiv 0 \pmod{r},$$

define

$$\boxed{w = \frac{u+v}{r},}$$

and

$$\boxed{s = \frac{pu - qv}{r}.}$$

The first is integral by assumption.
Since

$$p \equiv -q \pmod{r},$$

the numerator of s is also divisible by r.
Direct substitution gives

$$(u, v) = s\mathbf{h} + wK.$$

Uniqueness follows because

$$\det(\mathbf{h}, K) = p + q = r \neq 0.$$

□

38.3 Defect coordinates

Define

$$\Phi(Q) = pQ_0 - qQ_1,$$

$$\Lambda(Q) = Q_0 + Q_1.$$

Corollary 38.3 — Defect coordinates.
For

$$Q = s\mathbf{h} + wK,$$

$$\boxed{\Phi(Q) = rs,}$$

$$\boxed{\Lambda(Q) = rw.}$$

Hence

$$\boxed{s = \frac{\Phi(Q)}{r}, \qquad w = \frac{\Lambda(Q)}{r}.}$$

The coordinate s measures the SEARCH/exchange sheet in the quantity shadow.
The coordinate w measures deferred kernel quantity.

38.4 Defect quotient

Corollary 38.4 — Defect quotient.

$$\boxed{\mathbb{Z}^2/\mathcal{D}_{p,q} \cong \mathbb{Z}_r.}$$

The homomorphism is

$$(u, v) \mapsto u + v \pmod r.$$

This quotient is a useful arithmetic shadow.
It is not the physical semantic fiber.

38.5 WSC lattice is not the full LIFO fiber

The map

$$S \mapsto \mathrm{Par}(S)$$

forgets stack order.
The quantity defect also forgets finite-control sheets.
Therefore two physical configurations can have the same coordinates

$$(s, w)$$

while having:

- different literal stack words;
- different control states;
- different future locality properties.

Thus:

$$\boxed{\text{WSC defect lattice} \neq \text{full physical representation fiber.}}$$

This warning is essential.
Chapter 38 onward will develop the missing lift/fiber theory.

38.6 Kernel economy in lattice coordinates

Since

$$|\Theta| - |P| = Q_0 + Q_1 = rw,$$

we get:

Corollary 38.5 — Kernel economy.

Within the phase-coherent class,

$$\boxed{\text{kernel economy} \iff w \geq 0.}$$

Thus the deferred-kernel coordinate acquires a direct nonnegative backlog interpretation under economy.

38.7 Abstract count-shadow fiber region

Suppose the semantic demand is

$$\Theta(c) = \mu = (x, y)$$

and

$$Q = s\mathbf{h} + wK.$$

Then

$$P = \mu - Q = (x, y) - s(1, -1) - w(q, p).$$

So

$$P_0 = x - s - qw,$$

$$P_1 = y + s - pw.$$

Proposition 38.6 — Count-shadow fiber region.

Physical nonnegativity is equivalent to

$$\boxed{pw - y \leq s \leq x - qw,}$$

with

$$\boxed{0 \leq w \leq \left\lfloor \frac{x + y}{r} \right\rfloor.}$$

For fixed w, the number of integer s-positions is

$$\boxed{x + y - rw + 1.}$$

The count-shadow fiber therefore has a finite triangular/layered shape over each semantic demand.

Again, this is an abelian shadow of the full physical fiber, not the complete fiber itself.

The next chapter gives these coordinates a temporal interpretation.

Chapter 39

Physical Materialization, Backlog, and WSC Factorization

The semantic carry schedule

$$\chi_t$$

is unique.

The physical machine may materialize those kernel packets at different times.

We now make that chronology explicit.

Let

$$Q_t = s_t h + w_t K$$

along a phase-coherent run.

39.1 Materialization level

Definition 39.1 — Physical materialization level.

Define

$$\boxed{\lambda_t = \kappa_t - w_t.}$$

The semantic system has generated κ_t kernel packets.

The quantity shadow still defers w_t packets.

Therefore λ_t is the number of kernel packets already materialized physically at the count-shadow level.

39.2 Physical length formula

Theorem 39.2 — Physical length formula.

At prefix length t,

$$\boxed{|P_t| = r\lambda_t - t.}$$

Proof.

$$|P_t| = |\Theta_t| - (Q_{t,0} + Q_{t,1}).$$

Chapter 31 gives

$$|\Theta_t| = r\kappa_t - t.$$

Chapter 34 gives

$$Q_{t,0} + Q_{t,1} = rw_t.$$

Subtract and use

$$\lambda_t = \kappa_t - w_t.$$

□

The physical stack length therefore has its own modular clock:

$$\boxed{|P_t| \equiv -t \pmod{r}.}$$

39.3 Materialization amount

Define

$$\boxed{M_t = \lambda_{t+1} - \lambda_t.}$$

Theorem 39.3 — Materialization amount from stack length.

For every phase-coherent one-top transition,

$$\boxed{M_t = \frac{\Delta|P_t| + 1}{r}.}$$

In particular,

$$\boxed{M_t \in \mathbb{N}.}$$

Proof.

Subtract the physical length formula at t and $t+1$:

$$\Delta|P_t| = rM_t - 1.$$

Thus the formula follows.

Because a one-top rewrite can delete at most the visible literal cell,

$$\Delta|P_t| \geq -1.$$

Hence

$$M_t \geq 0.$$

□

The number of kernel packets physically materialized by a transition can therefore be reconstructed from the physical stack-length change alone.

39.4 Backlog law

Theorem 39.4 — Backlog law.

$$\boxed{w_{t+1} - w_t = \chi_t - M_t.}$$

Equivalently, if

$$A_t = \kappa_t$$

is cumulative semantic arrival and

$$B_t = \lambda_t$$

is cumulative physical materialization, then

$$\boxed{w_t = A_t - B_t.}$$

This is a queue-like law:

$$\boxed{\text{backlog change} = \text{semantic arrivals} - \text{physical service.}}$$

The analogy is structural, not stochastic.

39.5 Canonical carry dichotomy

Assume phase coherence and kernel economy, and begin a transition from canonical quantity state

$$w_t = 0.$$

Theorem 39.5 — Canonical carry dichotomy.
If

$$\chi_t = 0,$$

then

$$\boxed{M_t = 0.}$$

If

$$\chi_t = 1,$$

then

$$\boxed{M_t \in \{0, 1\}.}$$

Thus a semantic carry is either:

$$\boxed{\text{eagerly materialized}}$$

or

$$\boxed{\text{deferred as exactly one kernel packet of backlog.}}$$

This is the structural generalization of the eager/lazy distinction discovered concretely in Chapter 30.

39.6 Unique physical WSC role factorization

Let

$$\alpha_t = s_{t+1} - s_t.$$

Theorem 39.6 — Unique physical WSC count-shadow factorization.

Every phase-coherent one-top transition on input a satisfies

$$\boxed{\Delta P = -e_a - \alpha_t \mathbf{h} + M_t K.}$$

The coefficients

$$\alpha_t \in \mathbb{Z}$$

and

$$M_t \in \mathbb{N}$$

are unique.

Proof.

The defect change is

$$\Delta Q = \alpha_t \mathbf{h} + (w_{t+1} - w_t)K.$$

Use the backlog law:

$$\Delta Q = \alpha_t \mathbf{h} + (\chi_t - M_t)K.$$

Insert this into the universal conservation identity

$$\Delta P + \Delta Q = -e_a + \chi_t K.$$

The $\chi_t K$ terms cancel:

$$\Delta P = -e_a - \alpha_t \mathbf{h} + M_t K.$$

Uniqueness follows from linear independence of $\mathbf{h}$ and K. □

Meaning

This is a **role decomposition**, not a syntactic transition taxonomy.

The terms mean:

$$-e_a$$

= semantic CANCEL forced by the input;

$$-\alpha\mathbf{h}$$

= SEARCH/exchange effect in the abelian quantity shadow;

$$MK$$

= physically materialized kernel packets.

One concrete PDA transition may combine several of these roles.

39.7 Reconstructing the charges

Corollary 39.7 — Reconstruction.

$$\boxed{M = \frac{\Delta|P| + 1}{r},}$$

and

$$\boxed{\alpha = -\frac{p(\Delta P_0 + e_{a,0}) - q(\Delta P_1 + e_{a,1})}{r}.}$$

Thus the WSC charges can be recovered from the physical count effect.

Corollary 39.7A — General physical phase clock.

Every phase-coherent run satisfies

$$\boxed{|P_t| \equiv -t \pmod{r}.}$$

This follows immediately from

$$|P_t| = r\lambda_t - t.$$

The semantic demand clock of Chapter 31 and the physical stack clock therefore have the same modulus $r = p + q$, even though the physical materialization schedule may be eager, lazy, or mixed.

39.8 Global materialization budget

Let an accepted word have

$$N_0 = qn, \qquad N_1 = pn.$$

Theorem 39.8 — Global materialization budget.

For a phase-coherent kernel-economical run starting canonically,

$$\boxed{\sum_t M_t = n.}$$

This equals the total semantic carry count:

$$\boxed{\sum_t \chi_t = n.}$$

Proof.

The accepted input length is

$$rn.$$

At acceptance, semantic demand is zero.
Kernel economy forces the physical literal count to zero as well.
The physical length formula gives

$$0 = r\lambda_{rn} - rn.$$

Hence

$$\lambda_{rn} = n.$$

Since $\lambda_0 = 0$,

$$\sum_t M_t = n.$$

The semantic equality follows from Chapter 31. □
The schedule may move materialization earlier or later.
The total materialized kernel budget cannot change.

39.9 Deferral capacity

Corollary 39.9 — Deferral capacity bound.

At prefix length t,

$$\boxed{\left\lceil \frac{t}{r} \right\rceil \le \lambda_t \le \kappa_t,}$$

and

$$\boxed{0 \le w_t \le \kappa_t - \left\lceil \frac{t}{r} \right\rceil.}$$

Thus the physical representation has an exact finite capacity for delaying kernel materialization at a given semantic prefix.

The next chapter moves from arithmetic chronology to semantic fibers and asks which parts of the physical presentation are unique.

Chapter 40

Unique Semantic Quotient, Nonunique Physical Presentation

The Structure Theory has now separated:

- the unique semantic demand trajectory;
- the physical normalization trajectory.

We can state that distinction as a quotient.

40.1 Semantic fibers

For semantic demand

$$m \in \mathcal{M}_{p,q},$$

define

$$\boxed{F_m = \Theta^{-1}(m).}$$

Two rich configurations are semantically equivalent when

$$\boxed{c \sim c' \iff \Theta(c) = \Theta(c').}$$

A fiber can contain several physical configurations representing the same minimum demand.

40.2 Fiber congruence

Theorem 40.1 — Fiber congruence and semantic quotient.

In every exact literal-demand lift, the relation $\sim$ is preserved by binary input transitions.

Hence

$$\boxed{\mathcal{C}/\sim \cong \mathcal{M}_{p,q}.}$$

Proof.

If

$$\Theta(c) = \Theta(c'),$$

then exactness gives

$$\Theta(\delta_a(c)) = U_a(\Theta(c)) = U_a(\Theta(c')) = \Theta(\delta_a(c')).$$

Thus successors remain in the same semantic fiber.

The quotient state is therefore exactly the common semantic demand, and the quotient transitions are the canonical demand updates. □

This is an early version of the later machine-independent semantic quotient theorem. The stronger language-level statement will come in the Uniqueness Theory.

40.3 Canonical retraction

Define

$$\boxed{N = \sigma \circ \Theta.}$$

This sends every physical configuration to the canonical section representative of its semantic state.

Proposition 40.2 — Canonical absorption.

$$\boxed{N^2 = N.}$$

Moreover, for every input word w,

$$\boxed{N \circ \delta_w = N \circ \delta_w \circ N.}$$

Proof.
Since

$$\Theta \circ \sigma = \mathrm{id},$$

$$N^2 = \sigma\Theta\sigma\Theta = \sigma\Theta = N.$$

For one symbol,

$$N\delta_a = \sigma\Theta\delta_a = \sigma U_a \Theta.$$

Also,

$$N\delta_a N = \sigma\Theta\delta_a\sigma\Theta = \sigma U_a \Theta\sigma\Theta = \sigma U_a \Theta.$$

Induct on $|w|$. □
Interpretation:

Once we normalize to the canonical section, earlier physical normalization history is erased.

This is a structural form of the eager/lazy semantic equivalence.

40.4 What is actually unique

The following are semantic and unique:

- residual;
- minimum demand;
- dominating kernel level;
- semantic carry indicator;
- canonical quotient path.

The following need not be unique:

- physical stack word;
- finite-control sheet;
- timing of materialization;
- normalization schedule;
- full pushdown presentation.

Thus the correct statement is:

$$\boxed{\text{unique semantic core;}\quad \text{nonunique physical normalization lift.}}$$

40.5 Cheap nonuniqueness already exists

Proposition 40.3 — Unrestricted physical presentation nonuniqueness.

The canonical semantic quotient does not determine a unique deterministic pushdown presentation.

Proof.

Start with any exact deterministic lift with finite control Q.

For integer

$$m \geq 2,$$

adjoin an independent cyclic coordinate

$$j \in \mathbb{Z}_m.$$

On every consumed input symbol, update

$$j \mapsto j + 1 \pmod{m}.$$

Let:

- stack dynamics;
- semantic projection;
- acceptance

ignore the extra coordinate.

The result is still an exact deterministic lift.

Its control set is

$$Q \times \mathbb{Z}_m.$$

For infinitely many suitable pairwise distinct m, these presentations have different finite-control cardinalities and are therefore non-isomorphic, while their semantic quotient is unchanged. □

This proposition proves physical presentation nonuniqueness.

But it is deliberately cheap.

It uses redundant control decoration.

It does **not** by itself distinguish genuine reduced presentations from redundant decoration. Chapters 53–54 supply the protected quotient/recoding framework that removes this ambiguity and proves existence of minimal protected cores. The conclusion used here is existence and minimality under the protected quotient/recoding framework, not a Church–Rosser assertion for arbitrary terminal protected cores.

40.6 Scope of the confluence claim

Canonical absorption says that normalizing after a fixed input word forgets earlier physical history.

It establishes endpoint coherence for the chosen normalizer. A Church–Rosser theorem for arbitrary rewrite schedules would be a different claim because the present framework fixes a deterministic normalizer rather than an arbitrary-choice rewrite relation.

Therefore the safe conclusion is:

$$\boxed{\text{canonical endpoint coherence under the chosen normalizer}}$$

not

$$\boxed{\text{global confluence.}}$$

This scope will remain important in the Uniqueness and General Theory Parts.

Chapter 41

The Original 2:1 Machine as WSC

The Structure Theory now returns to the original machine for a second reinterpretation.

Chapter 30 explained the historical transition table as an eager schedule of the centered 2:1 construction.

The present chapter goes deeper.

It interprets:

- the historical side states as SEARCH sheets;
- the historical stack-length changes as materialization bursts;
- an old transition-count identity as a global kernel budget law.

Set

$$\boxed{p = 2, \qquad q = 1, \qquad r = 3, \qquad K = (1, 2).}$$

41.1 Historical states as SEARCH sheets

The original state potentials were

$$\Phi(C) = 0,$$

$$\Phi(L) = 3,$$

$$\Phi(R) = -3.$$

Since the WSC sheet coordinate s contributes residual quantity

$$rs = 3s,$$

we obtain:

Corollary 41.1 — Historical states as SEARCH sheets.

$$\boxed{s(C) = 0, \qquad s(L) = 1, \qquad s(R) = -1.}$$

Thus the original $C/L/R$ architecture is exactly a three-sheet realization of the WSC exchange coordinate.

The side states are not merely "left" and "right."

They are the ± 1 SEARCH sheets.

41.2 Historical stack-length changes

For $r = 3$, the physical materialization formula is

$$\Delta|P| = 3M - 1.$$

Therefore:

$$M = 0 \Longrightarrow \Delta|P| = -1,$$

$$M = 1 \Longrightarrow \Delta|P| = +2,$$

$$M = 2 \Longrightarrow \Delta|P| = +5.$$

Corollary 41.2 — Historical stack actions as materialization bursts.
The characteristic historical stack-length changes

$$\boxed{-1, \qquad +2, \qquad +5}$$

are exactly the WSC materialization levels

$$\boxed{M = 0, \qquad M = 1, \qquad M = 2.}$$

Therefore a $+5$ transition is not an anomalously long push.
It is a two-kernel-packet materialization burst.
This explains why some original transition replacements were much longer than others.

41.3 The global identity $b + 2c = n$

Suppose an accepted 2:1 run has:

- b transitions with stack-length change $+2$;
- c transitions with stack-length change $+5$.

Those transitions carry materialization amounts:

$$1 \qquad \text{and} \qquad 2.$$

The global materialization budget says that an accepted word with

$$N_0 = n, \qquad N_1 = 2n$$

must satisfy

$$\sum_t M_t = n.$$

Hence:

Corollary 41.3 — Structural derivation of the historical identity.

$$\boxed{b + 2c = n.}$$

The identity is no longer a transition-table counting curiosity.
It is the 2:1 specialization of the universal materialization budget.

41.4 What Structure Theory has accomplished

The generalization and Structure Theory now give three successive interpretations of the same original machine.

Chapter 1

$$\boxed{\text{state+stack residual invariant.}}$$

Chapter 30

$$\boxed{C/L/R = \text{centered } p{:}q \text{ debt rail at } (2,1),}$$

with historical behavior as eager discharge.

Chapter 37

$$\boxed{C/L/R = \text{WSC SEARCH sheets,}}$$

and the physical stack actions are kernel-materialization bursts.
This repeated return to the original machine is intentional.
Each return explains something that was previously only observed.

41.5 The Structure Theory endpoint

The strongest conclusion is:

$$\boxed{\text{CANCEL and semantic WRITE are forced by arithmetic;}}$$

$$\boxed{\begin{array}{c}\text{SEARCH and normalization debt arise when that semantic update meets}\\ \text{one-top LIFO locality.}\end{array}}$$

The Structure Theory has shown that physical presentations are nonunique. The next structural issue is whether that nonuniqueness survives after redundant decoration has been removed.

Its explicit nonuniqueness construction can be dismissed as redundant decoration.

The next Part asks the sharper question:

> After fixing the semantic quotient, the canonical section, and an admissible LIFO morphism category, can two **reduced** physical lifts still be genuinely different?

The answer is yes.

Moreover, in the motivating 2:1 family there are not merely two schedules.

There are exactly six admissible reduced schedules in the selected three-site family, with one directed causal conflict.

That is the Uniqueness/Lift Theory.

Part XI

Semantic Uniqueness and Physical Lift Freedom

Research provenance. This Part develops the language-level semantic quotient, exact sectioned lifts, reduced-lift nonuniqueness, and causal schedule analysis. The terminology is kept separate from classical VPL congruence/minimization and DPDA equivalence results [3, 24, 50].

Chapter 42

The Semantic Core Belongs to the Language

The Structure Theory established a canonical semantic quotient inside exact literal-demand lifts.

The Uniqueness Theory strengthens that result.

The semantic core will now be derived from the **language itself**, not from a chosen WSC construction.

42.1 Right residuals

For a prefix u, define the right-continuation language

$$u^{-1}L_{p,q} = \{z : uz \in L_{p,q}\}.$$

Chapter 4 already proved that this continuation language depends only on the arithmetic residual

$$\rho(u) = qN_1(u) - pN_0(u).$$

We now promote that result to the canonical semantic theorem.

Theorem 42.1 — Right-residual classification.

For all prefixes u, v,

$$\boxed{u^{-1}L_{p,q} = v^{-1}L_{p,q} \iff \rho(u) = \rho(v).}$$

Every integer residual occurs.

Thus the reachable right-residual classes are canonically indexed by

$$\mathbb{Z}.$$

The semantic transition system is therefore

$$\boxed{\mathcal{R}_{p,q} : \qquad r \xrightarrow{0} r - p, \qquad r \xrightarrow{1} r + q.}$$

This is the language-determined behavioral core.

Corollary 42.1A — Canonical behavioral core.

The reachable right-residual classes of $L_{p,q}$ are canonically identified with

$$\boxed{\mathbb{Z}},$$

and their induced labeled transition system is exactly

$$\boxed{\mathcal{R}_{p,q}: \quad r \xrightarrow{0} r - p, \qquad r \xrightarrow{1} r + q.}$$

Thus the integer residual line is not a coordinate convention inherited from a particular DPDA. It is the canonical continuation-behavior core of the language.

42.2 Universal semantic quotient

Consider any correct deterministic real-time recognizer of $L_{p,q}$.

Look only at reachable configurations at input boundaries.

Map every such physical configuration c, reached by prefix u, to

$$\boxed{\Theta_{\mathrm{res}}(c) = \rho(u).}$$

Determinism and correctness make this map well defined on behavioral equivalence classes.

Theorem 42.2 — Universal semantic quotient.

Every correct deterministic real-time recognizer of $L_{p,q}$ admits a unique surjective transition-preserving quotient onto

$$\boxed{\mathcal{R}_{p,q}.}$$

Thus the residual system is not merely the semantic quotient of our chosen constructions.

It is the canonical behavioral quotient forced by the language.

This strengthens Chapter 36.

42.3 Minimum-demand semantics is a coordinate change

The minimum-demand system

$$\mathcal{M}_{p,q}$$

has one state

$$\mu(r)$$

for every residual r.

The map

$$\boxed{r \longmapsto \mu(r)}$$

is bijective.

It also intertwines the input transitions.

Theorem 42.3 — WSC semantic core equals the residual core.

$$\boxed{\mathcal{R}_{p,q} \cong \mathcal{M}_{p,q}.}$$

Therefore:

$$\boxed{\text{language semantics} = \text{right-residual system},}$$

while

$$\boxed{\text{completion semantics} = \text{a useful isomorphic coordinate system.}}$$

This distinction is conceptually important.

The minimum-demand representation was discovered first because it grew naturally out of the original DPDA.

But the language-level semantic object is the residual quotient.

42.4 Three levels of description

We can now separate:

Language level

$$\mathcal{R}_{p,q}.$$

This is canonical.

Semantic coordinate level

$$\mathcal{M}_{p,q}.$$

This is isomorphic to the residual system.

Physical lift level

A DPDA realization mapping onto that semantic system.

This level is not unique.

The rest of the Uniqueness Theory asks how to compare such lifts without quotienting away the very pushdown structure we want to study.

Chapter 43

Exact Sectioned Lifts and Category-Relative Minimality

If arbitrary behavioral quotienting is allowed, every correct machine collapses to the same residual core.

That is too coarse for physical comparison.

We need a category that remembers:

- the chosen semantic projection;
- the canonical section;
- the LIFO coordinates.

43.1 Reachable versus ambient configurations

Let

$$C^{\mathrm{amb}}$$

be all syntactically possible physical configurations.

Let

$$\boxed{C^{\mathrm{reach}} \subseteq C^{\mathrm{amb}}}$$

be the configurations actually reachable from the initial condition.

All minimality, fiber, and schedule claims in this Part concern reachable configurations unless explicitly stated otherwise.

This prevents unreachable syntactic artifacts from distorting the theory.

43.2 Exact sectioned lift

Definition 43.1 — Exact sectioned lift.

An exact sectioned lift is

$$\boxed{\mathcal{L} = (C^{\mathrm{reach}}, \delta, \Theta, \sigma)}$$

such that:

1. Θ is a surjective semantic projection;
2. σ is a reachable section;

3. $$\Theta \circ \sigma = \mathrm{id};$$

4. physical transitions commute exactly with semantic transitions.

The section is part of the data.

Two lifts with the same quotient but different preferred physical representatives are therefore not automatically identified.

43.3 Section-preserving morphisms

A morphism

$$F : \mathcal{L}_1 \to \mathcal{L}_2$$

is section-preserving when

$$\boxed{\Theta_2 F = \Theta_1,}$$

$$\boxed{F\delta_a^{(1)} = \delta_a^{(2)} F,}$$

and

$$\boxed{F\sigma_1 = \sigma_2.}$$

For concrete pushdown comparison, a **coordinate LIFO morphism** must also respect the finite-control and stack coordinates letterwise rather than replacing the pushdown by an arbitrary behavioral quotient.

This is the correct category for asking whether two physical lifts are genuinely the same representation.

43.4 Minimality is category-relative

Proposition 43.2 — Category-relative minimality.

There is no meaningful representation-independent notion of a "minimal physical lift" without specifying:

- the admissible physical policy;
- the allowed morphisms;
- whether the canonical section is preserved;
- which stack/control coordinates may be identified.

The same semantic system can have different minima in different categories.

This is not a weakness of the theory.

It is the same lesson already encountered in the resource frontiers:

$$\boxed{\text{physical optimality is meaningful only relative to a representation category.}}$$

The next chapter asks what one-top locality forces inside one fixed literal policy.

Chapter 44

When LIFO Locality Forces Fiber Branching

The Structure Theory proved the top-rewrite suffix criterion.

We now use it at the level of an entire observation fiber.

44.1 Uniform local rule

A deterministic one-top transition cannot inspect the hidden stack suffix.

Therefore all reachable configurations sharing the same locally visible cell must be handled by the same rule.

Theorem 44.1 — Uniform one-top rewrite criterion.

Fix one deterministic local observation class.

A single one-top rule can send every source in that class to its canonical semantic target if and only if the untouched suffix of every source is compatible with that source's canonical target under the suffix criterion.

This theorem is the fiberwise form of Chapter 32.

The obstruction is no longer one bad stack.

It is incompatibility across a family of sources that the local rule cannot distinguish.

44.2 Canonical-section closure dichotomy

Use the fixed literal section

$$\boxed{\sigma(x, y) = (C, 1^y 0^x Z_0).}$$

Theorem 44.2 — Canonical-section closure dichotomy.

The canonical literal section is transition-closed under one-top real-time updates if and only if

$$\boxed{(p, q) = (1, 1).}$$

Meaning. For the balanced one-to-one language, the literal section can remain canonical after every input step.

For every nontrivial coprime ratio, some exact semantic successor cannot be realized while staying on the canonical section.

Thus the use of noncanonical physical representatives is not merely a feature of our centered construction.

Inside this literal one-top policy, it is unavoidable.

44.3 Essential physical fiber branching

Corollary 44.3 — Essential branching.

For every nontrivial ratio

$$(p, q) \neq (1, 1),$$

every exact one-top realization of the fixed literal policy must reach some noncanonical physical configuration in the same semantic fiber as a canonical one.

Therefore:

$$\boxed{\text{nontrivial semantic fibers are forced physically}}$$

inside the policy.

This is a much stronger statement than:

"our machine happens to use side states."

It says the canonical section itself cannot be closed.

44.4 Quotient-protected branching

Could one simply identify the noncanonical representative with the canonical one?

Not if the quotient must preserve legal one-top transitions in the same policy.

Corollary 44.4 — Quotient-protected branching.

The forced branch cannot be quotiented away while preserving:

- the fixed literal section;
- the one-top LIFO policy;
- exact transition behavior.

Thus some physical branching is structurally protected inside the representation category.

But the claim remains policy-relative.

Chapter 44 will show that a broader residual re-coordination can remove this branching entirely.

Chapter 45

Two Reduced 2:1 Lifts

We now revisit the eager and lazy 2:1 machines.

Chapter 30 proved semantic equivalence.

The Structure Theory explained their chronology.

The Uniqueness Theory asks whether one is merely a redundant presentation of the other.

The answer is no.

45.1 Historical eager lift

The historical machine eagerly materializes two reachable right-state matching families.

It returns immediately to the canonical center representation.

45.2 Centered lazy lift

The centered specialization leaves those same semantic debts on the right sheet and postpones materialization.

Both lifts use:

- the same semantic residual core;
- the same three control labels C, L, R;
- the same two work symbols;
- the same canonical section.

Their difference is temporal.

45.3 Section incomparability

Theorem 45.1 — Eager–lazy section incomparability.

There is no section-preserving transition morphism

$$\mathcal{L}_{\mathrm{eager}} \to \mathcal{L}_{\mathrm{lazy}}$$

and no such morphism

$$\mathcal{L}_{\mathrm{lazy}} \to \mathcal{L}_{\mathrm{eager}}.$$

The two lifts are therefore genuinely incomparable in the chosen category.

Their semantic quotient is identical.

Their physical transition structures are not.

45.4 Coordinate reducedness

Reachable witnesses force all three control states to remain distinct.

Likewise, the two work symbols cannot be merged without collapsing physically distinguishable reachable behavior.

Proposition 45.2 — Coordinate reducedness.

Both eager and lazy lifts are reduced inside the stated:

$$\boxed{\text{three-state / two-work-symbol coordinate category.}}$$

Thus their nonisomorphism is not caused by obvious redundant states or symbols.

45.5 Morphism-relative nonuniqueness

Corollary 45.3 — Reduced lift nonuniqueness.

The same semantic system admits multiple:

- coordinate-reduced;
- section-preserving;
- non-isomorphic

physical lifts.

Therefore:

$$\boxed{\begin{array}{c}\text{same semantics} + \text{same coordinate budget} + \text{reducedness} \\ \not\Rightarrow \text{unique physical lift.}\end{array}}$$

This strengthens Chapter 36's cheap cyclic-decoration nonuniqueness into a substantive structural theorem.

The next chapter shows that there are not merely two such schedules in the selected 2:1 family.

There are exactly six.

Chapter 46

The Six 2:1 Normalization Schedules

The historical and centered machines differ at specific reachable sites.

To understand the space of schedule choices, isolate three eager-discharge sites.

46.1 Sites

Site A

Left-state 0/0 matching family.

Site B

Right-state, top 0, input 0 family.

Site C

Right-state, top 1, input 1 family.

A schedule is a bit vector

$$\boxed{(a, b, c) \in \{0, 1\}^3,}$$

where 1 means that the corresponding eager rule is enabled.

There are eight possible schedules.

Exactly six are admissible.

46.2 Exact schedule classification

Theorem 46.1 — Exact three-site schedule classification.

$$\boxed{\mathrm{Adm}_{2,1} = \{(a, b, c) \in \{0, 1\}^3 : ab = 0\}.}$$

Thus the six admissible schedules are

$$\boxed{000,\ 001,\ 010,\ 011,\ 100,\ 101.}$$

The forbidden schedules are

$$\boxed{110,\ 111.}$$

The unique minimal conflict is

$$\boxed{\{A, B\}.}$$

Site C is compatible with either side of that conflict.

46.3 Schedule geometry

Order schedules by inclusion of eager sites.

Corollary 46.2 — Schedule poset.

The admissible schedule poset is not a lattice.
Its two maximal elements are

$$\boxed{\{A, C\}}$$

and

$$\boxed{\{B, C\}.}$$

Their union

$$\{A, B, C\}$$

is inadmissible.
So no join exists inside the admissible poset.

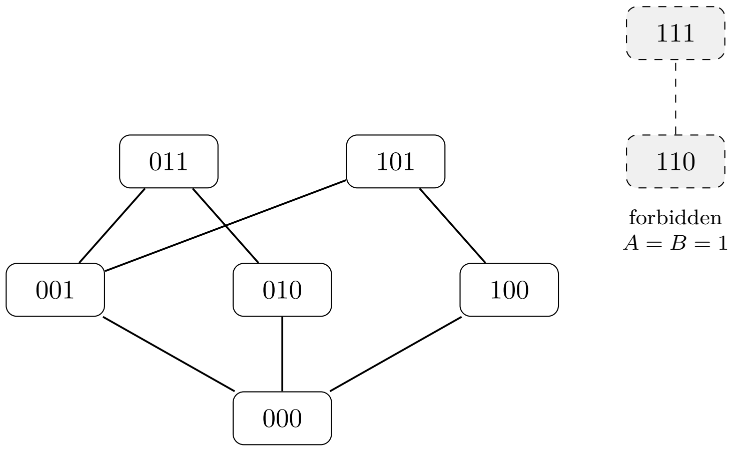


Figure 46.1: Hasse diagram of the six admissible 2:1 schedules. The schedule bits are ordered by inclusion of eager sites A, B, C. The two omitted cube vertices 110 and 111 are shown separately as forbidden because A and B cannot be simultaneously eager.

The missing 110 and 111 vertices should be shown separately as forbidden.

46.4 Six reduced lifts

Proposition 46.3 — Six coordinate-reduced schedule lifts.

All six admissible schedules define pairwise non-isomorphic coordinate-reduced lifts in the selected 2:1 category.

Thus schedule nonuniqueness is not merely eager versus lazy.

The physical realization space already contains a finite family of six reduced points.

46.5 Terminal normalization coherence

Let

$$\mathfrak{N}$$

be the stated terminal canonical normalizer.

Theorem 46.4 — Six-schedule terminal coherence.

For every admissible schedule and every common processed prefix, applying $\mathfrak{N}$ yields the same canonical semantic endpoint.

Thus all six systems have common normalized endpoint behavior.

Terminology boundary

This is **not** a Church–Rosser theorem.

The safe term is:

$$\boxed{\text{normalizer-relative terminal schedule coherence.}}$$

The six lifts are physically non-isomorphic.

They nevertheless agree after the specified terminal normalization.

Corollary 46.4A — Reduced non-isomorphism with common normalized behavior.

There exist coordinate-reduced, pairwise non-isomorphic physical lifts with identical terminal-normalization behavior on every processed prefix.

Thus even after obvious coordinate redundancy is removed,

$$\boxed{\text{common normalized semantics} \not\Rightarrow \text{unique reduced physical schedule.}}$$

This is exactly the kind of semantic/physical separation that the later General Theory will abstract.

Chapter 47

Why A Causes B

The schedule classification tells us that

$$A + B$$

is forbidden.

Its directionality is supplied by one of the clearest causal mechanisms in the corpus.

47.1 Local carry visibility

A fixed eager rule is chosen from a bounded local observation.

If two sources share that observation but require different semantic carry behavior, the same eager rule cannot be backlog-neutral on both.

Theorem 47.1 — Local carry visibility criterion.

A fixed eager local rule can remain backlog-neutral throughout an observation fiber only if the semantic carry indicator is constant on that fiber.

Thus carry aliasing is a local source of schedule failure.

47.2 Portability

For an eager rule r, define its portability set

$$\mathrm{Port}(r)$$

to be the physical source configurations on which applying that rule is safe relative to the selected canonical policy.

If F_o^α is the observation fiber exposed by schedule α, then:

Proposition 47.2 — Eager-rule portability criterion.

The eager rule is portable over that observation class exactly when

$$\boxed{F_o^\alpha \subseteq \mathrm{Port}(r).}$$

Schedule interactions therefore arise when enabling one rule changes which configurations another rule encounters.

47.3 A expands the B-fiber

Without A, site B sees only its baseline portable contexts.
Enabling A changes earlier normalization chronology.
That change exposes to B the infinite family

$$\boxed{(R, 0^m), \qquad m \geq 2.}$$

Lemma 47.3 — Schedule-induced fiber expansion.

Enabling A enlarges the first-hit observation fiber of B by the family $(R, 0^m)$, $m \geq 2$.
This is the causal mechanism.
A does not make B's local rule intrinsically different.
It changes which hidden suffixes reach the same local B-observation.

47.4 Direct canonicalization becomes impossible

For source

$$(R, 0^m)$$

on input 0, the canonical semantic target is

$$\boxed{10^{m-2}.}$$

A one-top rewrite must preserve the untouched suffix

$$\boxed{0^{m-1}.}$$

But 0^{m-1} is not a suffix of 10^{m-2}.

Theorem 47.4 — Schedule-induced direct-canonicality obstruction.

No one-top rule at site B can directly canonicalize every newly exposed source $(R, 0^m)$, $m \geq 2$.
This is the suffix-theorem diagnosis.

47.5 Forced kernel excess

Suppose the B-rule nevertheless insists on returning directly to the center while preserving residual.
Then it must create a nonminimal amount of literal demand.

Theorem 47.5 — Forced kernel-excess return.

Every such center-return repair introduces at least one extra kernel packet:

$$\boxed{(t+1)K}$$

relative to the minimum demand.
The historical replacement

$$\boxed{111}$$

realizes the minimum one-packet excess.
Thus the same conflict has three exact readings.

Corollary 47.6 — Three-way diagnosis.

Within this $A \to B$ family,

$$\boxed{\text{suffix failure}}$$

$$\iff$$

$$\boxed{\text{carry aliasing}}$$

$$\iff$$

$$\boxed{\text{forced kernel excess.}}$$

This triangle is important because the three notions are not globally identical in arbitrary representations.

They coincide here because of the specific one-top literal policy and source family.

The next chapter turns this concrete explanation into a directed causal relation.

Chapter 48

Directed Causal Scheduling

A minimal conflict such as

$$\{A, B\}$$

is set-theoretic.

It does not identify which rule caused the other to become unsafe.

The first-hit fiber expansion gives a directed notion.

48.1 First-hit exposure

For target site b and previously enabled schedule set H, let

$$\boxed{\mathcal{E}_b(H)}$$

be the first-hit exposure fiber: the configurations at which a computation first reaches site b under the schedule H.

A directed cause

$$H \rightsquigarrow b$$

occurs when H is minimal among schedule sets whose first-hit exposure contains a b-source outside the portability set of the b-rule.

Thus causality is about **newly exposed unsafe contexts**.

48.2 Exact 2:1 primary dependency

Theorem 48.1 — Exact primary dependency classification.

For the selected 2:1 three-site family,

$$\boxed{\mathcal{D}_{2,1}^{\mathrm{prim}} = \{\{A\} \rightsquigarrow B\}.}$$

So the directed causal graph is

$$\boxed{A \longrightarrow B \qquad \sqcup \qquad C.}$$

There is:

- no $B \to A$;
- no edge involving C.

Thus the forbidden pair $\{A, B\}$ has a unique primary direction.

Corollary 48.2 — Directed cause recovers the conflict.

Forgetting orientation gives the unique minimal schedule conflict:

$$\boxed{A \to B \quad \mapsto \quad \{A, B\}.}$$

48.3 Concrete causal certificate

One first-hit witness is

$$\boxed{1111100 \rightsquigarrow (R, 00).}$$

Applying the B-rule yields

$$\boxed{(C, 1110)}$$

instead of canonical

$$\boxed{(C, 1).}$$

The quantity excess is one kernel packet in the relevant sign convention:

$$\boxed{-K.}$$

Appending one more 1 reaches residual zero at the noncanonical configuration

$$\boxed{(C, 110).}$$

This gives a finite causal certificate:

1. enabling A creates a new first hit at B;
2. B's eager rule is not portable there;
3. the resulting debt is behaviorally exposed by a short continuation.

Proposition 48.2A — 2:1 causal certificate.

There exists a finite word, beginning with the witness prefix

$$\boxed{1111100,}$$

whose first use of the B-site after enabling A reaches

$$(R, 00)$$

and whose eager B-successor is the noncanonical kernel-excess configuration

$$(C, 1110).$$

A finite continuation then distinguishes this successor behaviorally from the canonical semantic successor.

Hence the directed dependency

$$\boxed{A \rightsquigarrow B}$$

admits a finite first-hit certificate.

This concrete witness is the prototype for the Fatal Frontier certificates of the General Theory.

48.4 Coordinate invariance

Suppose two lift presentations are related by a site-preserving coordinate LIFO isomorphism.

Then:

- first-hit exposure fibers correspond;
- portability sets correspond;
- directed causal hyperedges correspond.

Theorem 48.3 — Causal hypergraph invariance.

The directed causal schedule structure is invariant under site-preserving coordinate isomorphism.

Thus the $A \to B$ graph is not an artifact of renaming states or stack symbols inside the same coordinate category.

48.5 But semantics does not determine causality

A raw residual-conjugacy machine can realize the same language semantic quotient without any literal A, B, C normalization sites.

Theorem 48.4 — Semantic non-determination of causal structure.

The same semantic quotient can admit physical lifts with different causal schedule structures.

Therefore:

$$\boxed{\text{semantic quotient} \not\Rightarrow \text{causal schedule signature.}}$$

Corollary 48.5 — Coordinate/semantic separation principle.

A property can be invariant under coordinate isomorphism and still fail to be language-intrinsic.

Thus:

$$\boxed{\text{coordinate-intrinsic} \not\Rightarrow \text{semantic-intrinsic.}}$$

This distinction is central to every later obstruction theorem.

Chapter 49

Which Obstructions Are Really Forced?

The Uniqueness Theory has proved several kinds of necessity.

But necessity depends on the class of allowed representations.

This final chapter of the Part makes that dependence explicit.

49.1 Nested representation classes

Consider nested classes

$$\boxed{\mathfrak{C}_0 \subseteq \mathfrak{C}_1 \subseteq \mathfrak{C}_2.}$$

A typical interpretation is:

- $\mathfrak{C}_0$: a narrow coordinate category;
- $\mathfrak{C}_1$: a fixed literal-demand policy with more presentation freedom;
- $\mathfrak{C}_2$: a broader realization class allowing re-coordination.

For an obstruction $\mathcal{O}$, ask whether $\mathcal{O}$ is unavoidable in each class.

49.2 Obstruction monotonicity

Theorem 49.1 — Obstruction monotonicity.

If the realization class is enlarged, an essential obstruction can disappear but cannot newly become essential.

Thus:

$$\boxed{\mathrm{Ess}(\mathfrak{C}_2) \subseteq \mathrm{Ess}(\mathfrak{C}_1) \subseteq \mathrm{Ess}(\mathfrak{C}_0).}$$

For fully resolved three-level binary profiles, only

$$\boxed{111, \quad 110, \quad 100, \quad 000}$$

are possible.

49.3 Branching profile

For nontrivial semantic-fiber branching in the stated nested classes:

Theorem 49.2 — Policy-essential but broad-class removable branching.

$$\boxed{\Pi(\mathcal{O}_{\mathrm{branch}}) = 110.}$$

Thus branching is:

- essential in the narrow coordinate class;
- essential in the fixed literal policy;
- removable in a broader class containing raw residual re-coordination.

This theorem connects the early raw DPDA constructions to the later lift theory.
The compact raw representation did not contradict forced branching.
It left the literal-demand policy in which branching was forced.

49.4 Category-relative status of $A \to B$

The specific causal obstruction has an exact category-relative profile once the schedule presentation is typed separately from the underlying lift.

Proposition 49.3 — Exact category bifurcation for the classical $A \to B$ edge.

In the site-flexible fixed-literal-policy presentation class, the optional menu may vary while the baseline literal-demand lift is held fixed. The named B site can then be omitted, so

$$\boxed{\Pi_{\mathrm{flex}}(A \to B) = (1, 0, 0).}$$

If instead the designated A, B cells and the finite first-hit causal certificate are part of the preserved interface, the dependency persists:

$$\boxed{\Pi_{\mathrm{cert}}(A \to B) = (1, 1, 0).}$$

Broad exact re-coordination can remove the named phenomenon in either case. Chapter 52 gives the full presentation typing and the arbitrary-coprime extension.

49.5 Representation–obstruction separation

Theorem 49.4 — Representation-obstruction separation.

Every theorem claiming that a physical obstruction is “forced” must specify the representation class in which the necessity is proved.

This is one of the main methodological conclusions of the project. It applies to state lower bounds, stack-symbol lower bounds, fiber branching, normalization debt, schedule conflicts, and causal edges.

49.6 Computational validation note

The P11 study reports

$$\boxed{\text{458,776 assertion-level checks}},$$

including exhaustive words through length 14 for all eight schedules and the first common forbidden-schedule failure.

The exact historical executable used for that reported run is not part of the supplied research corpus. The monograph therefore records this numerical result as source-reported rather than as an independently rerun audit. The symbolic proofs do not depend on the computational validation.

49.7 Transition to General Theory

The Uniqueness Theory has now produced a family of concepts that are no longer specific to weighted binary balance:

- semantic transition system;
- exact physical lift;
- canonical section;
- semantic fiber;
- bounded local observation;
- forced branching;
- portability;
- causal schedule structure;
- representation-relative obstruction.

The next Part asks:

> Which of these notions survive for a general semantic system and a bounded-local physical normalization policy?

The answer will organize them into:

$$\boxed{\text{locality} \to \text{debt} \to \text{behavioral safety} \to \text{causality} \to \text{synthesis} \to \text{verification.}}$$

That is the General Theory.

Chapter 50

From the Six 2:1 Schedules to the Arbitrary-Coprime Compiler

The preceding chapters isolated the six reduced schedules of the original 2:1 literal-demand lift and their directed causal relation. We now keep that same shortest-completion semantics but remove the special ratio. The result is a uniform compiler and a finite arithmetic site universe for every coprime $p{:}q$.

50.1 Centered literal-demand normalization and schedule presentations

Put

$$m = p + q, \qquad h = \left\lfloor \frac{m-1}{2} \right\rfloor, \qquad K = (q, p).$$

The centered literal-demand lift has finite-control debt states

$$D_{-h}, D_{-h+1}, \ldots, D_h$$

and a center state C. The semantic residual is represented by a canonical shortest nonnegative demand vector

$$\mu(r) = \arg\min\{x + y : x, y \geq 0,\ px - qy = r\},$$

and the chosen literal stack order is denoted by $\eta(x, y)$.

For debt state D_d define the virtual demand

$$V_d = \mu(dm).$$

A local work-top observation is a triple

$$o = (d, a, z), \qquad |d| \leq h, \quad a, z \in \{0, 1\},$$

meaning that the current state is D_d, the next input letter is a, and the visible stack symbol is z.

Definition 50.1 (Schedule presentation)**.** A schedule presentation consists of a fixed mandatory centered baseline policy together with a finite optional-site menu and, for each site, a designated eager-return rule. A schedule is a subset of the menu; enabled optional rules may pre-empt the corresponding baseline transition.

Definition 50.2 (Admissibility and fatal supports)**.** A schedule H is *admissible* if the resulting deterministic machine recognizes exactly $L_{p,q}$. A support F is *fatal* if every schedule containing F is inadmissible, and is *minimal fatal* if no proper subset is fatal.

The distinction between a lift and a schedule presentation will be essential in Section 52.2: the physical transition system does not determine which optional sites are named or offered.

50.2 The uniform eager-return compiler

Let $e_0 = (1, 0)$ and $e_1 = (0, 1)$. For an observation $o = (d, a, z)$ define

$$B_o = e_z - e_a + V_d$$

and

$$\kappa(o) = \max\left\{ \left\lceil -\frac{(B_o)_0}{q} \right\rceil, \left\lceil -\frac{(B_o)_1}{p} \right\rceil \right\}.$$

Set

$$R_o = B_o + \kappa(o)K.$$

The associated local eager rule is

$$E_o : (D_d, zT) \xrightarrow{a} (C, \eta(R_o)T).$$

Theorem 50.3 (Uniform centered eager-return compiler)**.** *For every local observation o, the vector R_o is the unique shortest nonnegative replacement vector that preserves the residual invariant and returns directly to the center. Hence E_o is the canonical direct eager-return candidate at o.*

Proof. Any direct return to the center must compensate the change in the visible stack cell and the consumed input, hence its replacement vector lies in the affine lattice

$$B_o + \mathbb{Z}K.$$

The only admissible replacements are nonnegative vectors. Adding one copy of $K = (q, p)$ increases both coordinates, so the smallest integer carry making both coordinates nonnegative is exactly $\kappa(o)$. Any smaller carry has a negative coordinate, while any larger carry increases total length by $p + q$. Therefore R_o is feasible and uniquely length-minimal. $\square$

This formula recovers, for example, the six explicit $3 : 2$ eager words used in the finite stress test. The important point here is not those special values but the existence of one ratio-uniform compiler from which the whole site universe can be derived.

50.2.1 The mandatory reset envelope

Write

$$\mu((h + 1)m) = (x_+, y_+), \qquad \mu(-(h + 1)m) = (x_-, y_-).$$

Define

$$\mathcal{R}^0_{p,q} = \{\mu(jm + q) : -h - 1 \le j \le h\} \cup \{(x_+ + n, y_+) : n \ge 0\} \cup \{(x_-, y_- + n) : n \ge 0\}.$$

Lemma 50.4 (Mandatory-WRITE reset envelope)**.** *Every mandatory WRITE in the centered $p : q$ baseline lands at a canonical center pair belonging to $\mathcal{R}^0_{p,q}$.*

Proof. A bottom WRITE from D_d on input 1 produces $\mu(dm+q)$, whereas input 0 produces

$$\mu(dm-p) = \mu((d-1)m+q),$$

which gives the finite seed range $-h-1 \le j \le h$. At the positive outer boundary, the untouched suffix is a unary 0-tail, so the reset ray is $(x_+ + n, y_+)$. The negative side is symmetric. □

Definition 50.5 (Baseline-portable site)**.** A baseline episode starts at a center configuration $(C, \eta(x,y))$ with $(x,y) \in \mathcal{R}^0_{p,q}$, follows only mandatory MATCH/SEARCH transitions, and stops before the next mandatory bottom or outer WRITE. A compiler site is *baseline-portable* if its eager rule lands on the canonical center representative for every source exposed in every baseline episode.

Baseline portability is deliberately weaker than singleton admissibility: an eager reset can be safe on all baseline sources and later become unsafe on a source created by its own earlier activation. This self-exposure mechanism is responsible for the extra singleton-fatal vertices below.

50.3 The finite baseline-portable site universe

Assume first that $p \ge q \ge 2$. For $j \ge 0$, the bottom-seed formula is

$$\mu(jm+q) = (j+q, p-j-1).$$

For $j = -t < 0$, write $t = kq+s$, $0 \le s < q$; then

$$\mu(-tm+q) = \begin{cases} (0, km-1), & s = 0, \\ (q-s, p+km+s-1), & 1 \le s < q. \end{cases}$$

For $1 \le d \le h$,

$$V_d = (d+q, p-d).$$

The three nontrivial positive-debt compiler count vectors are

$$M_d = (d+q, p-d), \qquad P_d = (d+q+1, p-d-1), \qquad N_d = (d+q-1, p-d+1),$$

with imbalances

$$\Delta(M_d) = 2d+q-p,$$
$$\Delta(P_d) = 2d+q-p+2,$$
$$\Delta(N_d) = 2d+q-p-2.$$

Negative-debt compiler words are one-heavy except for the trivial forced return that is already part of the baseline.

A central reset word excludes almost every interior observation. If m is odd, the diagnostic seed is (h,h) with word $(10)^h$; if m is even, it is $(h,h+1)$ with word $(10)^h1$. Tracking untouched suffixes shows that all negative-interior candidates fail portability and leaves at most two positive-interior survivors.

Theorem 50.6 (Eight-Site Theorem)**.** *Let $p \ge q \ge 2$ be coprime and let $h = \lfloor (p+q-1)/2 \rfloor$.*

If $p+q$ is odd, the baseline-portable site universe consists exactly of the six core sites

$$(-h,0,0),\ (-h,1,0),\ (-h,1,1),\ (h-1,0,0),\ (h-1,1,0),\ (h,0,0),$$

together with

$$X = (h - q + 1, 0, 1) \iff p > 5q - 5,$$

and

$$Y = (h - q, 1, 1) \iff p > 5q - 1.$$

If $p + q$ is even, the universe consists exactly of the four core sites

$$(-h, 1, 1),\ (h, 0, 0),\ (h, 0, 1),\ (h, 1, 1),$$

together with

$$X = (h - q + 1, 0, 0) \iff p > 5q - 5,$$

and

$$Y = (h - q, 1, 0) \iff p > 5q.$$

In particular,

$$|\mathcal{S}^{\mathrm{bp}}_{p,q}| \le 8.$$

For $q > p \ge 2$, the complete list is obtained by the physical duality

$$(p, q, d, a, z) \longleftrightarrow (q, p, -d, 1 - a, 1 - z).$$

Proof. The proof is an exact arithmetic elimination. The central diagnostic removes every negative-interior observation and all but two positive-interior observations; exact boundary fibers give the core; the two survivors are decided by primitive-kernel minimality. All formulas, including the negative-seed decomposition and the parity-specific threshold simplifications, are given in Section 51.4. □

Chapter 51

Exact Causal-Hypergraph Classification for All Coprime Ratios

The compiler reduces schedule admissibility to a finite obstruction problem. The classification separates the stable high-minimum regime from the two low-minimum lines where finite fork phenomena survive.

51.1 The high-minimum regime: every core schedule is safe

Let $\mathrm{Core}_{p,q}$ denote the parity-dependent core in Theorem 50.6. For a named site A, write $\mathcal{E}_A$ for the set of visible source stack words that can occur at that site during an episode under the chosen core schedule. These source envelopes are physical reachability sets, not new semantic states.

Theorem 51.1 (Core downward closure). *If $\min(p, q) \geq 3$, then every subset $H \subseteq \mathrm{Core}_{p,q}$ is admissible. Equivalently,*

$$2^{\mathrm{Core}_{p,q}} \subseteq \mathrm{Adm}_{p,q}\,.$$

No minimal fatal support is contained entirely in the core.

Proof. By duality assume $p \geq q \geq 3$. Induct over reset episodes. Every enabled core activation returns to a canonical minimum-demand word. It therefore suffices to show that the source fibers of core sites stay within protected unary envelopes after any sequence of core activations.

If $m = 2h + 1$, label the six core sites

$$A = (-h, 0, 0),\ B = (-h, 1, 0),\ C = (-h, 1, 1),$$

$$D = (h - 1, 0, 0),\ E = (h - 1, 1, 0),\ F = (h, 0, 0).$$

Uniformly over all core schedules,

$$\mathcal{E}_A = \mathcal{E}_B = \{0\}, \qquad \mathcal{E}_C \subseteq 1^+,$$

and

$$\mathcal{E}_D, \mathcal{E}_E, \mathcal{E}_F \subseteq 0^+.$$

Negative-core resets contain fewer than $q \leq h$ zeros, so they cannot create a new negative-boundary top-0 source. Positive-core outputs have one-count at most $h-1$, so their mixed part is exhausted before the next positive-boundary source. The eager compiler has the same tail orientation as the protected source, and kernel minimality is preserved.

If $m = 2h + 2$, the four core sites satisfy

$$\mathcal{E}_A \subseteq 1^+, \qquad \mathcal{E}_B \subseteq 0^+, \qquad \mathcal{E}_C = \mathcal{E}_D = \{1\}.$$

The same minority-count argument prevents a negative reset from generating a new positive-boundary source and prevents a positive reset from generating an additional top-1 source at D_h. Hence every optional activation is canonical and kernel-reduced. The residual invariant and the baseline terminal argument then imply correctness. □

51.2 Extra-site self-exposure for $\min(p, q) \geq 3$

The extra vertices X, Y are baseline-portable precisely because their first activation stays below the primitive kernel threshold. Repeating the same activation increases a hidden alternating prefix until one full kernel packet becomes trapped.

For odd $m = 2h + 1$, define

$$S_k^X = (10)^{k(q-1)}1^{2q-2}, \qquad S_k^Y = (10)^{kq}1^{2q}.$$

For even $m = 2h + 2$, define

$$S_k^X = 0(10)^{k(q-1)}1^{2q-3}, \qquad S_k^Y = 0(10)^{kq}1^{2q-1}.$$

The first source S_0 is reached by an explicit monotone baseline prefix. Between consecutive X-activations one reads $1^{p-q+1}0$; between consecutive Y-activations one reads 1^{p-q}. Thus the alternating exponent grows by $q - 1$ or q respectively.

Lemma 51.2 (One-kernel trapping)**.** *Let*

$$k_X = \left\lceil \frac{p - (h + 2q - 2)}{q - 1} \right\rceil, \qquad k_Y = \left\lceil \frac{p - (h + 2q)}{q} \right\rceil.$$

At the first nonminimal repeated activation, the physical target differs from the canonical minimum demand by exactly one primitive kernel packet $K = (q, p)$.

Proof. At the fatal activation the one-count is $p + e$ with $0 \leq e \leq q - 1$ (and $e \leq q - 2$ in the X case), while the zero-count is at least q. Subtracting one K therefore remains nonnegative, while subtracting two cannot. Thus the target is exactly $\mu(r) + K$. □

Feeding the shortest semantic completion of r removes the minimal component and leaves exactly one balanced kernel packet physically present. The final semantic residual is zero but the machine is not in the balanced accepting configuration.

Theorem 51.3 (Complete schedule classification for $\min(p, q) \geq 3$)**.** *For every coprime p, q with $\min(p, q) \geq 3$,*

$$\boxed{\mathrm{Adm}_{p,q} = 2^{\mathrm{Core}_{p,q}}.}$$

Every present extra site is an upward-fatal singleton conflict, and these are the only minimal fatal supports.

Proof. Every core subset is admissible by Theorem 51.1. The Y self-exposure witness meets no other optional site, hence every schedule containing Y is bad. The X witness meets no core site and at most the neighboring Y observation; if Y is disabled it proves X fatal, while if Y is enabled the Y witness already proves failure. Thus every schedule containing either extra is inadmissible. □

51.3 Low-minimum regimes and the arbitrary-coprime theorem

51.3.1 The line $\min(p, q) = 2$

By duality it suffices to take $q = 2$, so $p \geq 3$ is odd. The six-site core has exactly two minimal fatal supports,

$$\{A, D\}, \qquad \{B, D\}.$$

The two possible extra sites are

$$\begin{aligned} X_1 &= (h-1, 0, 1) \quad (p \geq 7), \\ X_2 &= (h-2, 1, 1) \quad (p \geq 11). \end{aligned}$$

and both are singleton-fatal.

Theorem 51.4 (Exact $\min(p, q) = 2$ classification)**.** *For $q = 2$,*

$$\mathrm{Adm}_{p,2} = \{H \subseteq \mathcal{S}^{\mathrm{bp}}_{p,2} : X_1, X_2 \notin H,\ \{A, D\} \nsubseteq H,\ \{B, D\} \nsubseteq H\},$$

with absent extras omitted. Exactly 40 *schedules are admissible.*

Proof. Section 51.5.1 gives a self-contained reset-closure proof of sufficiency and explicit robust witnesses for the two fork pairs $\{A, D\}$ and $\{B, D\}$; the six-site core therefore has exactly 40 safe schedules. Each present extra site is an upward-fatal singleton by the one-kernel trapping certificate, so every admissible expanded schedule excludes all extras. Conversely, after the extras are excluded the machine is exactly one of the safe core schedules. Thus the admissible count remains 40 independently of how many extras are present. □

51.3.2 The exceptional line $q = 1$

For $q = 1$, negative minimum demands are unary, so the negative corridor has no order freedom. The complete baseline-portable site universe is exceptional.

For $p = 1$ it is empty. For $p = 2$ the four sites are

$$X = (0, 1, 0), \quad A = (-1, 1, 1), \quad B = (-1, 0, 0), \quad G = (1, 0, 0).$$

For odd $p = 2h + 1 \geq 3$, the always-present sites are

$$A = (-h, 1, 1), \qquad D = (h, 1, 1), \qquad E = (h, 0, 0),$$

and for $p \geq 5$ also

$$B = (h-1, 1, 0), \qquad C = (h, 0, 1).$$

For even $p = 2h \geq 4$, the sites

$$A = (-h, 1, 1),\ B = (-h, 0, 0),\ C = (-h, 1, 0),$$

$$E = (h-1, 0, 0),\ F = (h-1, 1, 0),\ G = (h, 0, 0)$$

are always present, and $D = (h-1, 1, 1)$ appears for $p \geq 6$.

Theorem 51.5 (Exact $q = 1$ schedules)**.** *Up to physical duality:*

1. $(p, q) = (1, 1)$*: the site universe is empty and there is one admissible schedule.*

2. $(2,1)$*: the minimal fatal supports are* $\{B, X\}$ *and* $\{B, G\}$*; exactly* 10 *schedules are admissible.*

3. *odd* $p \geq 3$*:*
$$\mathrm{Adm}_{p,1} = \{\varnothing, \{A\}\}.$$

4. *even* $p \geq 4$*: the maximal safe sectors are*
$$\mathcal{N} = \{A, B, C\}, \qquad \mathcal{P} = \{A, F, G\},$$
while D *and* E *are singleton-fatal when present and the cross-pairs*
$$\{B, F\}, \{B, G\}, \{C, F\}, \{C, G\}$$
are fatal. Exactly 14 *schedules are admissible.*

Proof. The site-universe elimination and the balanced fatal certificates are recorded in Section 51.5. The safe cases are proved by episode-closed reset families: for odd p, only the unary negative ray generated by A is added; for even p, every safe schedule lies in one of the two closed sectors $2^{\mathcal{N}}$ or $2^{\mathcal{P}}$. The listed singleton and cross-pair witnesses trap a primitive kernel packet at residual zero. The small case $(2, 1)$ is verified by its four-site transition geometry, with explicit witnesses 111001 and 111110001. $\square$

Theorem 51.6 (Arbitrary-coprime causal-hypergraph classification)**.** *For every coprime positive pair* (p, q)*, the canonical baseline-portable weighted-balance schedule family has an explicit finite set of minimal fatal supports given by Theorems 51.3, the* $\min = 2$ *theorem, and Theorem 51.5, together with physical duality. Consequently*
$$H \in \mathrm{Adm}_{p,q}, \quad H' \subseteq H \quad \Longrightarrow \quad H' \in \mathrm{Adm}_{p,q}\,.$$
Equivalently, $\mathrm{Adm}_{p,q}$ *is the independence complex of the exact fatal-support hypergraph, in the standard simplicial-complex sense [83].*

51.4 Arithmetic details for the Eight-Site Theorem

This appendix records the elimination argument behind Theorem 50.6; it is included so that the arbitrary-coprime vertex classification can be checked without relying on a finite search.

Put $m = p + q$ and $h = \lfloor (m-1)/2 \rfloor$, and assume $p \geq q \geq 2$. For the bottom-WRITE seed family $B_j = \mu(jm + q)$ one has, for $j \geq 0$,
$$B_j = (j + q, p - j - 1).$$
For $j = -t < 0$, write $t = kq + s$, $0 \leq s < q$. Then
$$B_{-t} = \begin{cases} (0, km - 1), & s = 0, \\ (q - s, p + km + s - 1), & 1 \leq s < q. \end{cases}$$
For $1 \leq d \leq h$,
$$V_d = \mu(dm) = (d + q, p - d).$$
Thus the positive-debt compiler count vectors are
$$M_d = (d + q, p - d), \quad P_d = (d + q + 1, p - d - 1), \quad N_d = (d + q - 1, p - d + 1),$$
with imbalances
$$2d + q - p, \qquad 2d + q - p + 2, \qquad 2d + q - p - 2.$$
Every nontrivial negative-debt compiler word is one-heavy; the only exception is the forced $d = -1$, input/top 1/0 return already contained in the baseline.

Lemma 51.7 (Central diagnostic exclusion)**.** *For $p \geq q \geq 2$, no negative-interior observation is baseline-portable. If m is odd, the only possible positive-interior survivors are*

$$(h-q+1, 0, 1), \qquad (h-q, 1, 1),$$

and if m is even the only possible positive-interior survivors are

$$(h-q+1, 0, 0), \qquad (h-q, 1, 0).$$

Proof. If $m = 2h+1$, the seed $B_{h-q} = (h, h)$ has canonical word $(10)^h$. At an interior debt d, the untouched suffix is $(10)^{h-d-1}$ under a visible top 0 and $0(10)^{h-d-1}$ under a visible top 1. A top-0 eager concatenation is canonical only when its compiler word is balanced; all relevant interior imbalances are odd. A top-1 concatenation is canonical only when the compiler has exactly one excess 1, which gives precisely the two displayed equations for d.

If $m = 2h+2$, the central seed is $(h, h+1)$ with word $(10)^h 1$. The corresponding suffixes are $(10)^{h-d-1}1$ and $0(10)^{h-d-1}1$. A top-0 candidate must have zero imbalance, while the parity of the positive-debt compiler imbalances excludes every top-1 candidate. Solving the zero-imbalance equations gives the two even-parity survivors. For negative debt, the compiler has a terminal 1-tail whereas the central diagnostic exposes an untouched 0 beyond the visible cell; the eager target is therefore not in canonical alternating-tail form. This excludes all negative-interior sites. □

Lemma 51.8 (Boundary fibers)**.** *The always-present boundary sites are exactly the parity-dependent cores in Theorem 50.6.*

Proof. For odd m, the negative boundary has a singleton top-0 source and unary top-1 sources; at $d = h-1$ every top-0 source is unary, and at $d = h$ every source is unary top-0. A source 11 at $d = h-1$ eliminates the remaining top-1 candidates. For even m, the negative boundary has unary top-1 sources but also the diagnostic source 00, while at $d = h$ the top-0 sources are unary and the only top-1 source is 1; the central sources 01 and 101 eliminate the remaining $d = h-1$ candidates. Direct substitution into the compiler gives exactly the core lists stated in Theorem 50.6. □

For the two odd-parity survivors the common compiler word is $(10)^h 1$. Their exposed source one-counts are bounded by $2q-2$ and $2q$, with equality attained by baseline sources. Kernel reduction is therefore equivalent to

$$h + 2q - 2 < p, \qquad h + 2q < p,$$

which simplify to $p > 5q-5$ and $p > 5q-1$. For even m the common compiler is $(10)^{h+1}$, the extremal untouched one-counts are $2q-3$ and $2q-1$, and the same calculation gives $p > 5q-5$ and, using coprimality of odd p, q, $p > 5q$. This completes the proof of Theorem 50.6.

51.5 Low-minimum certificates

The low-minimum cases are included explicitly because they are the only places where non-singleton fatal supports survive.

51.5.1 Self-contained odd-p:2 fork certificate

Fix $q = 2$ and write

$$p = 2h - 1, \qquad m = p + q = 2h + 1, \qquad h \geq 2.$$

For $t \geq 1$ put

$$\epsilon_t = t \bmod 2, \qquad k_t = \left\lceil \frac{t}{2} \right\rceil, \qquad \beta_t = t + pk_t, \qquad \Gamma_t = \eta(\epsilon_t, \beta_t).$$

The six core rules are, in literal stack notation,

$$\begin{array}{r|c} A : (D_{-h}, 0T) \xrightarrow{0} (C, \Gamma_h T) & B : (D_{-h}, 0T) \xrightarrow{1} (C, \Gamma_{h-1} T) \\ C : (D_{-h}, 1T) \xrightarrow{1} (C, \Gamma_h T) & D : (D_{h-1}, 0T) \xrightarrow{0} (C, (10)^h 0T) \\ E : (D_{h-1}, 0T) \xrightarrow{1} (C, (10)^{h-1} 000T) & F : (D_h, 0T) \xrightarrow{0} (C, (10)^{h-1} 000T). \end{array}$$

The arithmetic identities needed below are

$$\mu(-tm) = (\epsilon_t, \beta_t), \qquad \mu(jm + 2) = (j + 2, p - j - 1) \quad (0 \leq j \leq h),$$

$$\mu(-tm + 2) = (\epsilon_t, \beta_t - 1) \quad (1 \leq t \leq h + 1), \qquad \mu((h + 1)m) = (h + 3, h - 2).$$

They follow by direct substitution in $px - 2y = r$ and kernel reduction by $(2, p)$.

Define the canonical baseline reset family

$$\mathcal{R}_h^{\text{base}} = \{\mu(jm + 2) : -h - 1 \leq j \leq h\} \cup \{(x, h - 2) : x \geq h + 3\} \cup \{(\epsilon_{h+1}, y) : y \geq \beta_{h+1}\}.$$

Two episode-closed repertoires separate the safe schedule regimes:

$$\mathcal{R}_{\neg D,h} = \mathcal{R}_h^{\text{base}} \cup \{(\epsilon_h, y) : y \geq \beta_h\} \cup \{(\epsilon_{h-1}, \beta_{h-1})\} \cup \{(x, h - 1) : x \geq h + 2\},$$
$$\mathcal{R}_{D,h} = \mathcal{R}_h^{\text{base}} \cup \{(\epsilon_h, y) : y \geq \beta_h\} \cup \{(x, h - 1) : x \geq h + 2\} \cup \{(x, h) : x \geq h + 1\}.$$

Every pair in either repertoire has minority coordinate at most h. Consequently an episode beginning in one of these repertoires has the following exact boundary-source geometry: in $\mathcal{R}_{\neg D,h}$, a D_{-h} source with visible 0 has remaining stack exactly 0; in $\mathcal{R}_{D,h}$ it has remaining stack 0^n, $n \geq 1$, with $n \geq 2$ arising only from the D-ray. In either repertoire, D_{-h} with visible 1, D_{h-1} with visible 0, and D_h with visible 0 have unary remaining stacks. This follows because MATCH and interior SEARCH only delete the visible top cell, so every pre-reset stack is a suffix of a canonical reset word; an opposite symbol beyond the displayed unary suffix would force minority size at least $h + 1$.

For a unary source of length $n \geq 1$, the six eager targets have canonical count pairs

$$\begin{array}{r|c} A : 0 & (\epsilon_h, \beta_h) \\ B : 0 & (\epsilon_{h-1}, \beta_{h-1}) \\ C : 1^n & (\epsilon_h, \beta_h + n - 1) \\ D : 0^n & (n + h, h) \\ E : 0^n & (n + h + 1, h - 1) \\ F : 0^n & (n + h + 1, h - 1). \end{array}$$

Hence baseline resets together with any enabled subset of A, B, C, E, F preserve $\mathcal{R}_{\neg D,h}$ when D is disabled; when D is enabled but A, B are disabled, baseline resets together with C, D, E, F preserve $\mathcal{R}_{D,h}$. Therefore all $2^5 = 32$ schedules with $D = 0$ are safe, and with $D = 1$ precisely the $2^3 = 8$ schedules having $A = B = 0$ are safe.

For necessity, fix any $x \geq h+1$. Then

$$(C, \eta(x,h)) \xrightarrow{0^{2h-1}} (D_{-h}, 0^{x-h+1}),$$

so enabling D expands the common A/B source fiber from the singleton $\{0\}$ to the entire unary ray $\{0^n : n \geq 1\}$. There is a uniform prefix u_h reaching $(D_{-h}, 000)$ and meeting no optional site except D. Put

$$v_h = \begin{cases} 1^{(h-1)m/2}, & h \text{ odd}, \\ 0\,1^{2h+(h-2)m/2}, & h \text{ even}, \end{cases} \qquad u_h = v_h 1^{2h-1} 0^{2h}.$$

Then u_h has the stated property: the increment block from an empty stack advances the centered debt by two, the middle block reaches $(D_{h-1}, 00)$, the next zero invokes D, and the final zeros expose $(D_{-h}, 000)$ without meeting A, B, C, E, F earlier.

Finally, Γ_t has a suffix 1^p for every $t \geq 1$; write $\Gamma_t = z_t 1^p$. At the common source $(D_{-h}, 000)$,

$$A:\ (D_{-h}, 000) \xrightarrow{0} (C, z_h 1^p 00), \qquad B:\ (D_{-h}, 000) \xrightarrow{1} (C, z_{h-1} 1^p 00).$$

Reading z_h or z_{h-1} by MATCHes leaves $(C, 1^p 00)$. This word contains p ones and two zeros, so its residual is zero, but it is a nonempty primitive kernel packet and is therefore nonaccepting. The complete fatal witnesses

$$w_A^{(h)} = u_h 0 z_h, \qquad w_B^{(h)} = u_h 1 z_{h-1}$$

show that $\{A, D\}$ and $\{B, D\}$ are robust fatal supports. Thus the six-site core has exactly 40 admissible schedules and no other minimal fatal support.

The extra baseline-portable sites

$$X_1 = (h-1, 0, 1) \quad (p \geq 7), \qquad X_2 = (h-2, 1, 1) \quad (p \geq 11)$$

are individually upward-fatal by the one-kernel trapping mechanism proved above. Excluding them reduces the expanded problem to the proved six-site fork classification, so the admissible count remains exactly 40.

For $q = 1$ the full vertex sets are as follows. At $(2,1)$,

$$\mathcal{S}_{2,1}^{\mathrm{bp}} = \{X = (0,1,0),\ A = (-1,1,1),\ B = (-1,0,0),\ G = (1,0,0)\}.$$

For odd $p = 2h+1 \geq 3$, $A = (-h,1,1), D = (h,1,1), E = (h,0,0)$ are always present and $B = (h-1,1,0), C = (h,0,1)$ occur for $p \geq 5$. For even $p = 2h \geq 4$, the always-present sites are

$$\begin{gathered} A = (-h,1,1), \quad B = (-h,0,0), \quad C = (-h,1,0), \\ E = (h-1,0,0), \quad F = (h-1,1,0), \quad G = (h,0,0). \end{gathered}$$

and $D = (h-1,1,1)$ is present iff $p \geq 6$.

Lemma 51.9 (Explicit $q = 1$ fatal certificates)**.** *The fatal supports in Theorem 51.5 admit balanced witnesses that end at residual zero in a nonaccepting physical configuration.*

Proof. For odd $p = 2h+1$, the positive sites are witnessed by

$$w_B = 1^{p(p-2)} 0^{p-2},$$

$$w_C = 1^{2h(h+1)-1} (0\,1^{p-1})^{h-1} 0^h,$$

$$w_D = 1^{2h(h+1)} 0^p 1^{2h(h+1)+1}, \qquad w_E = 1^{2h^2+4h+1} 0^{p+1} 1^{2h(h+1)+1}.$$

The B-witness meets no other optional site; with B disabled the C- and D-witnesses are unpreempted, and with B, D disabled the E-witness is unpreempted. Site A acts only on the unary negative corridor and preserves an episode-closed unary reset family. Hence only $\varnothing$ and $\{A\}$ are admissible.

For even $p = 2h$, the two safe sectors are $\mathcal{N} = \{A, B, C\}$ and $\mathcal{P} = \{A, F, G\}$. The individually fatal sites have witnesses

$$w_D = 1^{p(p-3)}0^{p-3} \quad (h \geq 3),$$

$$w_E = 1^{(h+1)(p-1)}(0\,1^{p-1})^{h-1}0^h.$$

The four cross-pair certificates are

$$\begin{aligned} w_{BF} &= 1^{h(p+1)}0^p1^{h(p-1)}, & w_{CF} &= 1^{h(p+1)}0^{p-1}1^{h(p-3)}, \\ w_{BG} &= 1^{h(p+3)}0^{p+1}1^{h(p-1)}, & w_{CG} &= 1^{h(p+3)}0^p1^{h(p-3)}. \end{aligned}$$

They trap exactly one primitive kernel packet after the indicated mixed pair has been activated. Finally, for $(p, q) = (2, 1)$ the two balanced words 111001 and 111110001 witness $\{B, X\}$ and $\{B, G\}$ and both end at $(C, 110)$, a nonaccepting ordering of the primitive kernel packet. These certificates, together with the episode-closed safe sectors, prove the stated low-minimum classification. $\square$

Chapter 52

Downward Closure and Category-Relative Causal Persistence

The exact weighted-balance family is downward closed, but that fact is not a theorem of exact lifts in general. The correct boundary statement is obtained by separating the lift from the schedule presentation and by typing the category in which a named causal edge is preserved.

52.1 Downward closure is not universal

Theorem 51.6 is a theorem about the canonical weighted-balance family, not a theorem about arbitrary exact lifts. The distinction is real.

Consider one semantic state m with two semantic letters x, y, both acting as the identity. Let the physical states be c, d, b, all projecting to m. Assign terminal values

$$\alpha(c) = \alpha(d) = 0, \qquad \alpha(b) = 1.$$

The baseline transitions are

$$c \xrightarrow{x} c, \quad c \xrightarrow{y} c,$$
$$d \xrightarrow{x} c, \quad d \xrightarrow{y} b,$$

with b absorbing. Add two optional sites:

$$A : c \xrightarrow{x} d, \qquad B : d \xrightarrow{y} c.$$

Theorem 52.1 (Two-site failure of universal downward closure)**.** *For this exact schedule environment,*

$$\mathrm{Adm} = \{\varnothing, \{B\}, \{A, B\}\}.$$

In particular $\{A, B\}$ is admissible but $\{A\}$ is not.

Proof. The empty schedule and $\{B\}$ never leave c. Under $\{A\}$, the word xy follows $c \xrightarrow{x} d \xrightarrow{y} b$ and fails. Under $\{A, B\}$, the same first step reaches d but the enabled B rule repairs the second step and returns to c. Thus adding an optional rule can repair a failure created by another optional rule. □

This two-site example is minimal among counterexamples with an admissible empty schedule: with only one optional site, the only possible failure of downward closure would require the empty schedule to be bad and the singleton schedule good.

52.2 Causal persistence depends on the presentation category

A named edge such as $A \to B$ refers to two designated optional sites and to a first-hit or pre-emption relation between them. This data is not contained in the underlying lift alone.

Proposition 52.2 (Typing of named causal edges)**.** *A fixed exact lift does not determine a named causal edge. The optional-site menu and the local alternative rules are part of the schedule presentation, so forgetting the presentation can change the causal graph without changing the baseline lift.*

For a named edge e, let $\Pi(e) = (\pi_0, \pi_1, \pi_2)$ record persistence under (i) site-preserving coordinate isomorphism, (ii) a specified fixed-literal-policy presentation class, and (iii) broad exact physical re-coordination. Because the middle coordinate depends on what presentation data are frozen, its class is written as a subscript.

For the classical $2:1$ literal one-top policy, consider two presentation categories. In the first, the baseline policy is fixed but the optional-site menu may vary. In the second, the designated A, B cells and the finite causal certificate are preserved.

Theorem 52.3 (Exact $A \to B$ category bifurcation)**.** *For the classical* $2:1$ *baseline literal one-top policy,*

$$\Pi_{\mathrm{flex}}(A \to B) = (1, 0, 0)$$

in the site-flexible fixed-policy class, while

$$\Pi_{\mathrm{cert}}(A \to B) = (1, 1, 0)$$

in the certificate/site-preserving class. Hence the middle persistence bit has no category-independent value.

Proof. The same all-lazy baseline lift admits two presentations. With menu $\{A, B, C\}$ the classical first-hit certificate yields $A \to B$; with menu $\{A, C\}$ the named B site is absent and the directed edge cannot be formed. Hence the middle bit is 0 in the site-flexible class. In the certificate-preserving class the cells

$$1111100 \rightsquigarrow (R, 00), \qquad (R, 00) \xrightarrow[B]{0} (C, 1110)$$

and the distinguishing continuation 1 are part of the morphism data; the same non-canonical first-hit fiber therefore survives every allowed change. Broad re-coordination can remove the named phenomenon. Section 52.3 gives the typed formulation. □

52.3 Persistence signature and category typing

A schedule presentation is a triple

$$\mathfrak{P} = (\mathcal{L}_0, E, \{\delta^e\}_{e \in E}),$$

where $\mathcal{L}_0$ is the fixed baseline lift, E is the named optional-site menu, and each e carries its alternative local rule. A named directed edge therefore belongs to $\mathfrak{P}$, not to $\mathcal{L}_0$ alone.

For a named edge e define its three-level persistence signature

$$\Pi(e) = (\pi_0, \pi_1, \pi_2) \in \{0, 1\}^3,$$

where π_0 records persistence under site-preserving coordinate isomorphism, π_1 records persistence in a specified fixed-literal-policy presentation class, and π_2 records persistence under broad exact physical re-coordination. The middle class must therefore be named as part of the datum.

For the classical 2:1 edge $A \to B$, the all-lazy baseline admits both the menu $\{A, B, C\}$, where the directed dependency is present, and the menu $\{A, C\}$, where it is absent. This proves the middle bit is 0 in the site-flexible class. If instead the designated A, B cells and the finite first-hit certificate

$$1111100 \rightsquigarrow (R, 00), \qquad (R, 00) \xrightarrow[B]{0} (C, 1110)$$

with distinguishing continuation 1 are preserved, the same unsafe first-hit fiber survives every allowed presentation change; the middle bit is then 1. Broad re-coordination can eliminate the named phenomenon. Thus

$$\Pi_{\mathrm{flex}}(A \to B) = (1, 0, 0), \qquad \Pi_{\mathrm{cert}}(A \to B) = (1, 1, 0).$$

52.4 Monograph synthesis

The 2:1 schedule geometry is therefore not an isolated finite accident. It sits inside an arbitrary-coprime finite causal-hypergraph classification. What is special to the canonical weighted-balance family is the downward-closed independence-complex structure; arbitrary exact-lift schedule families need not share it. Likewise, the edge $A \to B$ is coordinate-stable but not language-intrinsic: its middle persistence bit depends on whether the optional-site presentation is held fixed. These conclusions close the schedule questions generated in the preceding lift chapters and prepare the protected comparison theory that follows.

Chapter 53

Strict Coordinate Quotients and Protected Interfaces

Exact sectioned lifts admit both genuine reductions and vacuous semantic collapses. To compare physical presentations without erasing the LIFO information under study, we first restrict reduction to coordinate quotients that preserve the exact section and the permanent bottom marker, and then declare explicitly which finite physical interface is protected.

53.1 Exact sectioned lifts and used coordinates

Let M be a semantic transition system over input alphabet Σ. An exact sectioned pushdown lift is a tuple

$$\mathcal{L} = (C^{\text{reach}}, \delta, \Theta, \sigma),$$

where C^{reach} is the reachable physical configuration set, δ is the one-top pushdown transition structure, $\Theta : C^{\text{reach}} \to M$ is the semantic projection, and $\sigma : M \to C^{\text{reach}}$ is a chosen section satisfying

$$\Theta\sigma = \mathrm{id}_M.$$

Exactness means that the physical transitions project to the semantic transitions.

For a finite-coordinate lift, define

$$Q_{\text{use}} = \{q : \exists S\ (q, S) \in C^{\text{reach}}\}$$

and let Γ_{use} be the stack symbols appearing in reachable stacks, including exactly one permanent bottom marker Z_0.

The basic resource vector is

$$\mathbf{R}(\mathcal{L}) = \big(|Q_{\text{use}}|, |\Gamma_{\text{use}}|\big).$$

Unreachable ambient states and symbols are ignored.

53.2 Strict coordinate quotients

Definition 53.1 (Strict coordinate quotient)**.** Let $\mathcal{L}_1, \mathcal{L}_2$ be exact sectioned one-top lifts of the same semantic system. A strict coordinate quotient (SCQ)

$$F : \mathcal{L}_1 \twoheadrightarrow_{\text{SCQ}} \mathcal{L}_2$$

consists of surjections

$$f_Q : Q^{(1)}_{\text{use}} \twoheadrightarrow Q^{(2)}_{\text{use}}, \qquad f_\Gamma : \Gamma^{(1)}_{\text{use}} \twoheadrightarrow \Gamma^{(2)}_{\text{use}}$$

such that

$$f_\Gamma^{-1}(Z_0) = \{Z_0\},$$

and such that the induced stack map is letterwise and non-erasing:

$$F(q, X_1 \cdots X_k Z_0) = \big(f_Q(q), f_\Gamma(X_1) \cdots f_\Gamma(X_k) Z_0\big).$$

It must satisfy

$$F(C_1^{\text{reach}}) = C_2^{\text{reach}}, \quad \Theta_2 F = \Theta_1, \quad F\sigma_1 = \sigma_2,$$

and

$$F\delta_a^{(1)} = \delta_a^{(2)} F$$

on every reachable transition.

The bottom-marker condition is essential: a work symbol may not be merged with the unique permanent bottom cell.

Theorem 53.2 (SCQ reduction theorem)**.** *SCQ morphisms compose and define a preorder. Mutual SCQ reducibility is exactly SCQ isomorphism. If*

$$\mathcal{L}_1 \twoheadrightarrow_{\text{SCQ}} \mathcal{L}_2,$$

then

$$\mathbf{R}(\mathcal{L}_2) \le \mathbf{R}(\mathcal{L}_1)$$

componentwise; if the quotient is not an isomorphism, at least one coordinate decreases strictly.

Proof. Composition is coordinatewise and preserves the transition, semantic, section, and bottom-marker conditions. If SCQ morphisms exist in both directions, finiteness and surjectivity force equality of the used state and symbol cardinalities; all coordinate maps are bijective and their inverses preserve the same structure. For monotonicity, each coordinate map is surjective. Equality in both resource coordinates forces both maps to be bijections, hence an isomorphism. □

Corollary 53.3 (Finite minimal-core existence)**.** *Every finite-coordinate exact sectioned lift has at least one SCQ-minimal quotient descendant.*

Proof. Each strict SCQ step strictly decreases a point of $\mathbb{N}^2$. No infinite strictly descending chain exists. □

Cheap finite-control decorations are automatically removed: if a redundant cyclic coordinate $j \in \mathbb{Z}_m$ is added to control but ignored by semantics and stack dynamics, projection $(q, j) \mapsto q$ is a strict SCQ quotient. On the other hand, an arbitrary behavioral quotient that erases the LIFO store is excluded by the letterwise non-erasing coordinate requirement.

53.3 Protected interfaces and maximal safe quotients

SCQ is intentionally rigid. To formulate which physical distinctions a quotient is allowed to erase, we attach a finite interface signature.

Definition 53.4 (Protected-interface lift)**.** A protected-interface lift is a pair

$$(\mathcal{L}, \lambda), \qquad \lambda : C^{\mathrm{reach}} \to \Lambda,$$

where Λ is finite and λ records physical information declared to be part of the comparison contract.

For a literal one-top policy, a useful interface is

$$\lambda_{\mathrm{lit}}(c) = \big(o_{\mathrm{lit}}(c), \chi_\sigma(c)\big),$$

where o_{lit} is the locally visible state/top observation and

$$\chi_\sigma(c) = \mathbf{1}_{\{c \in \sigma(M)\}}$$

records whether the configuration lies on the chosen canonical section.

Theorem 53.5 (Maximal interface-safe quotient criterion)**.** *Let $F : C_1^{\mathrm{reach}} \twoheadrightarrow C_2^{\mathrm{reach}}$ be surjective and let $\lambda_1 : C_1^{\mathrm{reach}} \to \Lambda$. The following are equivalent:*

1. *there is a unique $\lambda_2 : C_2^{\mathrm{reach}} \to \Lambda$ such that*

$$\lambda_1 = \lambda_2 \circ F;$$

2. *λ_1 is constant on every fiber of F.*

Hence λ-separated configurations can never be identified by an interface-safe quotient.

Proof. Necessity is immediate. Conversely define $\lambda_2(F(c)) = \lambda_1(c)$; fiber constancy makes the definition well posed, and surjectivity gives uniqueness. □

This criterion is maximal at the configuration level: it imposes exactly the identifications forbidden by the protected observable and no others.

Chapter 54

Bounded Bi-Recoding and Minimal Protected Cores

Coordinate quotients reduce resources, whereas bounded invertible recodings merely change the physical alphabet in a locally recoverable way. Treating these operations separately produces recoding-invariant minimal cores and finite protected Pareto envelopes.

54.1 Bounded bi-recoding and recoding-invariant resource spectra

A bounded invertible change of stack code should not be counted as a reduction.

Definition 54.1 (Bounded bi-recoding)**.** A bounded bi-recoding (BBR) between protected-interface lifts consists of a bijective control recoding and a non-erasing work-stack block code

$$\varphi : \Gamma_{1,\mathrm{use}} \setminus \{Z_0\} \to (\Gamma_{2,\mathrm{use}} \setminus \{Z_0\})^+$$

extended multiplicatively above the fixed bottom marker, such that:

1. $1 \leq |\varphi(X)| \leq B$ for a uniform finite B;
2. an inverse bounded non-erasing block code exists on the reachable stack languages;
3. the induced configuration bijection intertwines semantics, section, transitions, and the protected interface.

BBR is therefore an actual conjugacy of reachable dynamics. It preserves every obstruction expressible in semantic fibers, paths, section membership, and protected-interface values.

The identity code, the stipulated bounded inverse, and composition of bounded block codes show that BBR is an equivalence relation; see Section 54.2.

Define the protected recoding spectrum

$$\mathcal{R}_\lambda(\mathcal{L}) = \{(|Q_{\mathrm{use}}(\mathcal{L}')|, |\Gamma_{\mathrm{use}}(\mathcal{L}')|) : \mathcal{L}' \simeq_{\mathrm{BBR},\lambda} \mathcal{L}\}.$$

Its minimal antichain is

$$\mathcal{P}_\lambda(\mathcal{L}) = \mathrm{Min}\, \mathcal{R}_\lambda(\mathcal{L}).$$

Theorem 54.2 (Finite protected Pareto envelope)**.** *For every finite-coordinate protected lift, $\mathcal{P}_\lambda(\mathcal{L})$ is nonempty and finite and depends only on the BBR-equivalence class.*

Proof. The original presentation belongs to its spectrum, so the set is nonempty. Every nonempty subset of $\mathbb{N}^2$ has minimal elements, and Dickson's lemma [88] implies that the antichain of all componentwise-minimal elements is finite. BBR-equivalent lifts have the same spectrum by transitivity. $\square$

A normalized protected core step first chooses a representative whose resource vector lies in $\mathcal{P}_\lambda$, then takes a strict λ-preserving SCQ quotient, and finally passes to the target BBR class. By Theorem 53.2, every such step strictly decreases the normalized resource pair. Hence terminal protected BBR-classes exist. Uniqueness is not asserted.

54.2 Typed protected contracts and BBR equivalence

For complete type correctness, a protected representation contract is written

$$\mathfrak{K} = \left(\mathfrak{C}, \{\lambda_{\mathcal{L}}\}_{\mathcal{L}\in\mathfrak{C}}\right),$$

where $\mathfrak{C}$ is a BBR-closed class of finite-coordinate exact sectioned lifts of one semantic system and each $\lambda_{\mathcal{L}} : C_{\mathcal{L}}^{\mathrm{reach}} \to \Lambda$ is a protected interface. The family is required to be compatible with BBR conjugacies: if $F : \mathcal{L} \to \mathcal{L}'$ is a BBR equivalence, then $\lambda_{\mathcal{L}} = \lambda_{\mathcal{L}'} \circ F$. In the body we suppress the object subscript when no ambiguity is possible.

Proposition 54.3 (BBR is an equivalence relation)**.** *Bounded bi-recoding is reflexive, symmetric, and transitive on finite-coordinate protected lifts.*

Proof. The identity code gives reflexivity. The defining bounded non-erasing inverse on reachable stack languages gives symmetry. If block codes φ and ψ have maximum block lengths B_φ, B_ψ, then $\psi \circ \varphi$ has maximum block length at most $B_\varphi B_\psi$; the inverse is the reverse composition of the bounded inverses. The induced configuration bijections compose and continue to intertwine semantics, section, transitions, and the protected-interface family. Hence transitivity holds. $\square$

Under contract refinement $\mathfrak{K}' \succeq \mathfrak{K}$, the representation class is included and the protected interfaces refine objectwise. Therefore the resource-set inclusion used in Theorem 88.4 is literal, rather than an informal comparison between unrelated coordinate systems.

54.3 Interface with General Theory

The outcome is a comparison category strong enough to exclude semantic trivialization and weak enough to identify harmless bounded recodings. Terminal protected cores exist in every finite-coordinate class, but no uniqueness or Church–Rosser statement is asserted here. The next Part can therefore formulate General Theory over a cleaner physical comparison layer without conflating exact semantics, coordinate quotient, and recoding equivalence.

Part XII

Confluence and Normalization Choice

Research provenance. The preceding causal-schedule and protected-interface theory distinguishes semantic correctness from physical presentation. This Part makes the rewrite layer explicit: schedule activation, full completion-fiber normalization, finite-control sheets, and protected-core descent are treated as different confluence problems rather than conflated under one endpoint-coherence statement.

Chapter 55

Confluence Across Exact Pushdown Normalization Layers

The exact-lift, causal-schedule, SCQ, and BBR objects used here have already been developed in the preceding Parts. The purpose of this chapter is to add the missing rewrite-theoretic layer and to separate four questions: canonical endpoint coherence, schedule-presentation confluence, physical completion-fiber confluence, and confluence of protected quotient descent.

55.1 Exact lifts and four notions of normalization coherence

Definition 55.1 (Exact sectioned lift)**.** Let M be a deterministic semantic transition system over input alphabet Σ, with semantic transitions $U_a : M \to M$. An *exact sectioned pushdown lift* is a reachable physical transition system

$$\mathcal{L} = (C^{\mathrm{reach}}, \delta, \Theta, \sigma)$$

with semantic projection $\Theta : C^{\mathrm{reach}} \to M$, physical input transitions δ_a, and section $\sigma : M \to C^{\mathrm{reach}}$ satisfying

$$\Theta\sigma = \mathrm{id}_M, \qquad \Theta\delta_a = U_a\Theta.$$

The canonical retraction is $N = \sigma\Theta$.

Proposition 55.2 (Canonical retraction and absorption; baseline)**.** *The canonical retraction satisfies*

$$N^2 = N, \qquad \Theta N = \Theta,$$

and for every input word w,

$$N\delta_w = N\delta_w N = \sigma U_w \Theta.$$

Thus the canonical endpoint after reading w depends only on the initial semantic state. This is endpoint coherence, not a Church–Rosser theorem for an internal normalization rewrite.

Proof. Since $\Theta\sigma = \mathrm{id}_M$, one has $N^2 = \sigma\Theta\sigma\Theta = N$ and $\Theta N = \Theta$. Exactness gives $\Theta\delta_w = U_w\Theta$, hence

$$N\delta_w = \sigma U_w\Theta = \sigma U_w\Theta\sigma\Theta = N\delta_w N.$$

□

The following distinctions will be used throughout.

Definition 55.3 (Coherence notions)**.** Let $\Rightarrow$ be an internal normalization relation, distinct from the ordinary input transition δ.

1. *Endpoint coherence*: the designated normalizer sends every reachable configuration to the chosen section representative of its semantic value.
2. *Physical confluence*: every fork under $\Rightarrow$ joins to one physical configuration.
3. *Confluence modulo* $\approx$: every fork joins to configurations equivalent under a declared relation $\approx$.
4. *Section-complete normalization*: every configuration in a semantic fiber reduces to the chosen section point.

Proposition 55.4 (Bare semantic modulo-confluence is vacuous)**.** *Suppose*

$$c \Rightarrow d \quad \Longrightarrow \quad \Theta(c) = \Theta(d).$$

Then every fork $c \Rightarrow^* u$, $c \Rightarrow^* v$ already satisfies $\Theta(u) = \Theta(v)$. Thus confluence modulo the equivalence $u \equiv_\Theta v$ iff $\Theta(u) = \Theta(v)$ is automatic.

Proof. Semantic preservation is inherited by the reflexive-transitive closure. Both descendants have semantic value $\Theta(c)$. □

Consequently, a substantive modulo-confluence theorem must protect more physical information than Θ alone.

Definition 55.5 (Strictly canonicalizing physical rewrite)**.** A tuple $(C, \Rightarrow, \Theta, \sigma, D)$ is *strictly canonicalizing* if D takes values in a well-founded ordered set and:

1. $c \Rightarrow d$ implies $\Theta(c) = \Theta(d)$;
2. every rewrite strictly decreases D;
3. $D(c) = 0$ iff $c = \sigma(\Theta(c))$;
4. every noncanonical c admits a rewrite step.

Proposition 55.6 (Canonical-normal-form criterion)**.** *Every strictly canonicalizing physical rewrite is strongly normalizing and confluent. The unique normal form in the fiber $\Theta^{-1}(m)$ is $\sigma(m)$.*

Proof. Well-founded strict descent forbids infinite chains. Every maximal chain terminates, and the last two conditions imply that a terminal point is exactly the section point in its semantic fiber. Semantic preservation keeps every descendant of a fork in the same fiber, so both branches terminate at the same section representative. □

55.2 Finite schedule-activation rewriting

Let E be a finite set of optional eager-normalization sites. A schedule is a subset $H \subseteq E$, and $\mathrm{Adm}(E) \subseteq 2^E$ denotes the schedules preserving the intended recognition/normalization contract.

Definition 55.7 (Schedule activation)**.** Define

$$H \rightsquigarrow_E H \cup \{e\}$$

when $e \notin H$ and $H \cup \{e\} \in \mathrm{Adm}(E)$. A schedule is normal when no further one-site activation is admissible.

Proposition 55.8 (Termination)**.** *Schedule activation terminates for every finite E.*

Proof. The rank $|E| - |H|$ decreases by one at every step. ☐

Theorem 55.9 (Exact criterion for downward-closed schedule families)**.** *Assume* $\mathrm{Adm}(E)$ *is nonempty and downward closed. The following are equivalent:*

1. $(\mathrm{Adm}(E), \rightsquigarrow_E)$ *is confluent;*
2. *there is a unique normal form reachable from $\varnothing$;*
3. $\mathrm{Adm}(E)$ *has a unique inclusion-maximal member.*

Proof. In a downward-closed family, if $H \subseteq K$ are admissible then every intermediate set obtained by adding elements of $K \setminus H$ remains admissible. Hence rewrite normal forms are exactly the inclusion-maximal schedules and every admissible schedule reaches every maximal extension containing it. A unique maximal member therefore joins every branch. Two distinct maximal members are distinct terminal descendants of $\varnothing$ and cannot join. ☐

Suppose now that admissibility is the independence complex of a finite hypergraph $\mathcal{F}$ of minimal fatal supports:

$$\mathrm{Adm}(E) = \mathrm{Ind}(\mathcal{F}) = \{H \subseteq E : \text{ no } F \in \mathcal{F} \text{ is contained in } H\}.$$

Theorem 55.10 (Fatal-support criterion)**.** *Schedule activation is confluent iff every minimal fatal support is a singleton. If*

$$S_1 = \{e \in E : \{e\} \in \mathcal{F}\},$$

then, in the confluent case, the unique normal form is $E \setminus S_1$.

Proof. If every fatal support is singleton, admissibility is exactly $2^{E \setminus S_1}$. Conversely, let $F \in \mathcal{F}$ contain distinct x, y. Minimality makes $F \setminus \{x\}$ and $F \setminus \{y\}$ admissible. Extend each to a maximal admissible schedule. They cannot have the same maximal extension, since such an extension would contain their union F. Thus there are at least two normal forms, and theorem 55.9 gives nonconfluence. ☐

55.3 Rational schedule confluence

The canonical baseline-portable literal-demand presentation of rational $p{:}q$ normalization has a finite optional-site family whose minimal fatal supports are summarized in table 55.1. Appendix 55.7 records the schedule data and the low-width certificates needed for the present rewrite application. The important point here is not the site coordinates themselves but the size of the minimal supports.

Regime	minimal fatal-support structure	maximal-schedule count
$\min(p, q) \geq 3$	all present fatal extras are singleton; the canonical core is free	1
$\min(p, q) = 2$	two non-singleton fork supports $\{A, D\}, \{B, D\}$, plus singleton extras	2
$(p, q) = (1, 1)$	no optional site	1
$\min(p, q) = 1$, odd maximum ≥ 3	all positive optional sites singleton-fatal; one negative-boundary site is free	1
$(2, 1)$ up to duality	$\{B, X\}, \{B, G\}$	2
$\min(p, q) = 1$, even maximum ≥ 4	singleton fatal sites plus the four pairs between two negative- and two positive-sector sites	2

Table 55.1: Fatal-support geometry needed for the rational schedule-confluence theorem. Physical duality exchanges p and q.

Corollary 55.11 (The canonical 2:1 fork)**.** *For the complete four-site baseline-portable family*

$$E_{2,1} = \{X, A, B, G\},$$

one has

$$\mathrm{Adm}_{2,1} = \{H \subseteq E_{2,1} : \{B, X\} \nsubseteq H,\ \{B, G\} \nsubseteq H\}.$$

Schedule activation terminates but is not confluent. Its two normal forms are

$$\{A, B\}, \qquad \{A, X, G\}.$$

Proof. Appendix 55.7 gives the exact minimal fatal supports $\{B, X\}$ and $\{B, G\}$; the site A is free. Hence the maximal admissible schedules are exactly the two displayed sets, and theorem 55.9 gives nonconfluence. ☐

Theorem 55.12 (Exact arbitrary-coprime schedule-confluence dichotomy)**.** *For every coprime $p, q \geq 1$, activation in the canonical baseline-portable rational schedule family is confluent exactly when*

$$\boxed{\min(p, q) \geq 3 \quad \textit{or} \quad \min(p, q) = 1 \ \textit{and} \ \max(p, q) \ \textit{is odd.}}$$

It is nonconfluent exactly when

$$\boxed{\min(p, q) = 2 \quad \textit{or} \quad \min(p, q) = 1 \ \textit{and} \ \max(p, q) \ \textit{is even.}}$$

Proof. Read the minimal fatal supports from table 55.1 and apply theorem 55.10. In the high-minimum and odd boundary regimes all minimal supports are singleton. In the width-two, $(2, 1)$, and even boundary regimes a non-singleton fatal pair is present. ☐

Corollary 55.13 (One-or-two normal-form law)**.** *Every canonical rational schedule-activation system has exactly one or two normal forms. It has one precisely in the confluent regimes of theorem 55.12 and two precisely in the nonconfluent regimes.*

Proof. The maximal independent sets are visible in table 55.1. For min $= 2$, either the fork vertex D is chosen and A, B are excluded, or D is excluded and A, B are chosen, with all free sites added. The $(2, 1)$ and even boundary cases similarly split into exactly two safe sectors. The remaining regimes have one maximal admissible schedule. ☐

This classification is a presentation-level result. It says that some valid ways of adding eager normalization sites cannot be reconciled by further admissible activations. It does not yet describe rewriting inside a fixed physical semantic fiber.

55.4 The full rational completion fiber

Fix coprime $p, q \geq 1$ and use the completion equation

$$pN_0(z) - qN_1(z) = r.$$

For each $r \in \mathbb{Z}$, let

$$\mu(r) = (x_r, y_r) \in \mathbb{N}_0^2$$

be the unique minimum-length nonnegative solution. The elementary one-dimensional kernel structure gives

$$(x, y) = \mu(r) + k(q, p), \qquad k \in \mathbb{N}_0, \tag{55.1}$$

for every other nonnegative solution (x, y). Thus $K = (q, p)$ is the primitive redundant completion packet.

Define the canonical shortest completion

$$C(r) = \begin{cases} 1^{y_r} 0^{x_r}, & r \geq 0, \\ 0^{x_r} 1^{y_r}, & r < 0. \end{cases}$$

For $r = 0$ this is ε.

Definition 55.14 (Full completion-fiber rewrite)**.** Let

$$\mathcal{C}_{p,q} = \{(r, z) : r \in \mathbb{Z},\ z \in \{0,1\}^*,\ pN_0(z) - qN_1(z) = r\}.$$

Within a fixed residual fiber, generate $\Rightarrow_{p,q}$ by

$$\begin{aligned} r \geq 0 :&\quad 01 \Rightarrow 10, \quad 1^p 0^q \Rightarrow \varepsilon, \\ r < 0 :&\quad 10 \Rightarrow 01, \quad 0^q 1^p \Rightarrow \varepsilon. \end{aligned}$$

The swap discharges boundary-order debt and the block deletion removes one primitive kernel packet.

Lemma 55.15 (Semantic preservation and exact kernel debt)**.** *Every rewrite in theorem 55.14 preserves r. If $\mathrm{Par}(z) = (x, y)$, then the integer k in (55.1) is unique and*

$$\mu(r) - \mathrm{Par}(z) = -k(q, p).$$

Thus k is the exact kernel-excess quantity-debt level.

Proof. Swaps preserve Parikh counts. A deletion removes q zeros and p ones, so the value $pN_0 - qN_1$ changes by $pq - qp = 0$. The solution lattice of the coprime equation is generated by (q, p), and minimality of $\mu(r)$ forces the coefficient to be nonnegative. $\square$

Theorem 55.16 (Full completion-fiber confluence)**.** *For every coprime $p, q \geq 1$ and every residual r, the rewrite of theorem 55.14 is strongly normalizing and confluent on the entire completion fiber. Its unique normal form is $C(r)$.*

Proof. Define the sign-adapted inversion number

$$\Omega_r(z) = \begin{cases} \#\{i < j : z_i = 0, z_j = 1\}, & r \geq 0, \\ \#\{i < j : z_i = 1, z_j = 0\}, & r < 0, \end{cases}$$

and let $k(z)$ be the kernel level from theorem 55.15. Use lexicographic order on

$$D_r(z) = (k(z), \Omega_r(z)) \in \mathbb{N}_0^2.$$

A swap fixes k and decreases Ω_r by one. A kernel deletion decreases k by one. Hence every rewrite strictly decreases D_r, so the system is strongly normalizing.

Let z be irreducible. If $r \geq 0$, absence of 01 forces $z = 1^y 0^x$; if $r < 0$, absence of 10 forces $z = 0^x 1^y$. Write $(x, y) = \mu(r) + k(q, p)$. If $k \geq 1$, then the block junction contains $1^p 0^q$ in the nonnegative orientation, respectively $0^q 1^p$ in the negative orientation, contradicting irreducibility. Therefore $k = 0$ and $z = C(r)$. Conversely $C(r)$ has neither an order redex nor a removable kernel packet. Thus every fiber has one normal form, and termination implies confluence (equivalently, apply theorem 55.6). □

Corollary 55.17 (Shortest-completion stratum). *On the stratum $k = 0$, kernel deletion is unavailable and theorem 55.16 reduces to the boundary-oriented adjacent-swap normalization of shortest completions. The theorem therefore extends geodesic normalization to every nonminimal kernel level.*

Corollary 55.18 (Two-channel physical debt). *The pair*

$$(k(z), \Omega_r(z))$$

is an exact normalization debt on completion words: its first coordinate measures redundant primitive-kernel quantity and its second coordinate measures word-order debt.

The point of theorem 55.16 is physical, not merely semantic. The local relation has many possible reduction orders; the theorem proves that all of them reach the same literal word. This is strictly stronger than applying the retraction $N = \sigma\Theta$ after the fact.

55.5 Finite-control sheets and three-channel confluence

Completion words have no independent finite-control coordinate. To model the third debt channel, let W_r be the word fiber above and let $(S_r, \to_r)$ be a separately declared control-sheet rewrite system with distinguished sheet s_r^*.

Definition 55.19 (Sheet-separable normalization)**.** Set

$$\widehat{C}_r = W_r \times S_r, \qquad \widehat{\sigma}(r) = (C(r), s_r^*).$$

The combined rewrite is generated by

$$(z, s) \Longrightarrow_r (z', s) \quad \text{if } z \Rightarrow_{p,q} z',$$

and

$$(z, s) \Longrightarrow_r (z, s') \quad \text{if } s \to_r s'.$$

The two move types commute exactly because they act on disjoint coordinates.

Theorem 55.20 (Exact product reduction)**.** *For every residual r:*

1. *$\Longrightarrow_r$ terminates iff $\to_r$ terminates;*
2. *$\Longrightarrow_r$ is confluent iff $\to_r$ is confluent;*
3. $$\mathrm{NF}(\widehat{C}_r) = \{C(r)\} \times \mathrm{NF}(S_r);$$
4. *every point of $\widehat{C}_r$ reduces to $(C(r), s_r^*)$ iff every sheet reduces to s_r^*.*

Proof. The word relation is terminating and confluent with one normal form by theorem 55.16. An infinite combined chain would contain infinitely many moves in one coordinate; this is possible exactly when the sheet relation has an infinite chain. For confluence, project a combined fork to the two coordinates. Word projections join by theorem 55.16, sheet projections join exactly when $\to_r$ is confluent, and exact cross-commutation combines the joins. Conversely, any sheet fork lifts to a fork from $(C(r), s)$, so combined confluence projects back to sheet confluence. Normal forms and section reachability are coordinatewise. $\square$

Corollary 55.21 (Confluence does not imply section-complete normalization)**.** *Let S_r contain at least two sheets and let $\to_r$ be empty. Then the product rewrite is terminating and confluent but has one normal form $(C(r), s)$ for each sheet.*

Proof. The empty sheet relation is terminating and confluent. Apply theorem 55.20. $\square$

Thus

$$\boxed{\text{branching coherence} \not\Rightarrow \text{collapse of the whole semantic fiber to the section.}}$$

Theorem 55.22 (Rooted-sheet canonicalization)**.** *Suppose there is a well-founded rank $\eta_r : S_r \to W_r^{\mathrm{sh}}$ such that every sheet move strictly decreases η_r, $\eta_r(s) = 0$ iff $s = s_r^*$, and every noncanonical sheet admits a move. Then the combined rewrite is strictly canonicalizing with lexicographic debt*

$$\widehat{D}_r(z, s) = (k(z), \Omega_r(z), \eta_r(s)).$$

It is therefore strongly normalizing and confluent, with unique normal form $(C(r), s_r^*)$.

Proof. Kernel deletion decreases the first coordinate, sorting decreases the second while fixing the first, and sheet discharge decreases the third while fixing the first two. The debt is zero exactly at the section point. Apply theorem 55.6. $\square$

The separability hypothesis is deliberate. We do not assert that an arbitrary DPDA configuration fiber decomposes as word $\times$ sheet. The theorem identifies a natural class in which the independent control channel can be handled without semantically erasing it.

55.6 Protected confluence and canonical Pareto anchoring

We finally allow harmless changes of physical code while retaining declared interface information. The minimum structure needed for the argument is stated abstractly so that the protected confluence theorem is self-contained.

Definition 55.23 (Protected exact-lift class)**.** Fix a class of finite-coordinate exact sectioned lifts of one semantic system. Each lift $\mathcal{L}$ carries a finite protected interface

$$\lambda_{\mathcal{L}} : C_{\mathcal{L}}^{\mathrm{reach}} \to \Lambda.$$

A *bounded bi-recoding* (BBR) is an invertible bounded physical recoding of finite control and reachable stack blocks that intertwines semantics, section, input transitions, and λ. We write $[\mathcal{L}]$ for its BBR class.

A *strict protected coordinate quotient* (SCQ) is a surjective coordinate quotient preserving semantics, transitions, section, the unique bottom marker, and λ, and inducing a componentwise nonincreasing map on the used resource coordinates, with at least one strict decrease:

$$\mathbf{R}(\mathcal{K}) < \mathbf{R}(\mathcal{L}) \quad \text{componentwise, with at least one strict coordinate,}$$

where $\mathbf{R}(\mathcal{L}) = (|Q_{\mathrm{use}}|, |\Gamma_{\mathrm{use}}|)$.

For a BBR class $X = [\mathcal{L}]$, define its protected recoding spectrum

$$\mathcal{R}_\lambda(X) = \{\mathbf{R}(\mathcal{L}') : \mathcal{L}' \simeq_{\mathrm{BBR},\lambda} \mathcal{L}\},$$

and its Pareto envelope

$$\mathcal{P}_\lambda(X) = \operatorname{Min} \mathcal{R}_\lambda(X) \subset \mathbb{N}^2.$$

Dickson's lemma implies that this antichain is finite. Every member is realized by an actual representative because it is a spectrum point.

Definition 55.24 (Protected modulo-equivalence)**.** Let X be the state set of a normalization rewrite and let

$$\tau : X \to \Lambda^\sharp$$

be a declared protected signature. An equivalence $\approx_\tau$ is *protected* when

$$x \approx_\tau y \quad \Longrightarrow \quad \tau(x) = \tau(y).$$

At configuration level one may take $\tau = \lambda$. When the rewrite states are BBR classes, τ may be any BBR-invariant signature induced by the protected interface, such as the isomorphism type of a specified finite interface-labelled observable. Passing from representatives to BBR classes already quotients out BBR equivalence itself.

Proposition 55.25 (Terminal protected fork)**.** *In a terminating rewrite presentation equipped with a protected signature τ, if a source has terminal descendants u, v with $\tau(u) \neq \tau(v)$, then the system is not confluent modulo any protected equivalence $\approx_\tau$.*

Proof. Terminality leaves no further reduction. A modulo-join would therefore force $u \approx_\tau v$, contradicting preservation of τ. □

A subtlety arises if each reduction step is allowed to choose an arbitrary incomparable Pareto point of the current BBR class: one no longer has a single resource pair that visibly decreases along every possible path. We remove this hidden source choice canonically.

Definition 55.26 (Lex-normalized protected-core rewrite)**.** Let

$$\nu_\lambda(X) = \min_{\mathrm{lex}} \mathcal{P}_\lambda(X).$$

A step

$$X \Longrightarrow_\lambda^{\mathrm{lex}} Y$$

chooses a BBR-equivalent representative of X attaining the actual resource pair $\nu_\lambda(X)$, takes a strict protected SCQ quotient to a lift $\mathcal{K}$, and sets $Y = [\mathcal{K}]$.

Theorem 55.27 (Strong normalization of canonical protected-core descent)**.** *Every lex-normalized protected-core step strictly decreases ν_λ in lexicographic order on $\mathbb{N}^2$. Hence $\Longrightarrow_\lambda^{\mathrm{lex}}$ is strongly normalizing.*

Proof. Put $r = \nu_\lambda(X)$. A strict protected SCQ quotient from a representative attaining r has raw resource vector $r' < r$ componentwise, with at least one strict coordinate. The quotient is a representative of the target BBR class Y. Hence some Pareto point $r^\sharp \in \mathcal{P}_\lambda(Y)$ satisfies $r^\sharp \le r'$. By lexicographic minimality,

$$\nu_\lambda(Y) \le_{\mathrm{lex}} r^\sharp \le_{\mathrm{lex}} r' <_{\mathrm{lex}} r.$$

Lexicographic order on $\mathbb{N}^2$ is well-founded. □

Theorem 55.28 (Protected-core confluence criterion)**.** *On the descendant cone of any starting BBR class under $\Longrightarrow_\lambda^{\mathrm{lex}}$, the following are equivalent:*

1. *confluence;*
2. *local confluence;*
3. *exactly one terminal BBR class is reachable from the starting class.*

Proof. Termination is theorem 55.27, so Newman's lemma gives equivalence of confluence and local confluence [101, 100]. Every descendant of a terminating system reaches a terminal class. Confluence forces all terminal descendants to coincide. Conversely, if one terminal class T is reachable from the source and is the only terminal descendant of that source, then every descendant also terminates at T, because any terminal descendant of a descendant is a terminal descendant of the source. Thus any two descendants join at T. □

Corollary 55.29 (Protected signature certificate)**.** *If two terminal lex-normalized descendants have different values of a BBR-invariant protected signature τ, then the protected-core rewrite is nonconfluent from that source (and cannot be repaired by passing to any equivalence that preserves τ).*

Proof. Apply theorem 55.25. □

The canonical anchor is essential to the theorem as stated. We do not claim that a larger unanchored relation allowing arbitrary re-selection of incomparable Pareto points is strongly normalizing or confluent.

55.7 Rational fatal-support data used in Section 55.3

This appendix makes the rational schedule application self-contained at the level needed by the confluence theorem: it specifies the optional-site contract, the complete site universe, and the exact minimal-fatal-support data. No theorem in the abstract rewrite sections depends on these coordinates.

Put

$$m = p + q, \qquad h = \left\lfloor \frac{m-1}{2} \right\rfloor, \qquad K = (q, p),$$

and let

$$\mu(r) = \arg\min\{x + y : x, y \in \mathbb{N},\ px - qy = r\}$$

be the unique minimum nonnegative demand vector. Write $\eta(x, y)$ for the fixed canonical literal stack word representing the demand pair (x, y). The centered literal-demand lift has debt states $D_{-h}, \dots, D_h$ and center state C; write $V_d = \mu(dm)$. A local work-top observation is a triple

$$o = (d, a, z), \qquad |d| \le h, \quad a, z \in \{0, 1\},$$

meaning that the current state is D_d, the next input letter is a, and the visible stack symbol is z. With $e_0 = (1,0)$, $e_1 = (0,1)$, define

$$B_o = e_z - e_a + V_d, \qquad \kappa(o) = \max\left\{ \left\lceil -\frac{(B_o)_0}{q} \right\rceil, \left\lceil -\frac{(B_o)_1}{p} \right\rceil \right\}, \qquad R_o = B_o + \kappa(o)K.$$

The optional eager candidate at o consumes a, replaces the visible cell by the literal minimum-demand word representing R_o, and returns directly to C. This is the unique shortest nonnegative direct return preserving the residual invariant, because every other direct-center replacement belongs to the affine line $B_o + \mathbb{Z}K$.

A *baseline episode* begins at a canonical center demand reached by a mandatory WRITE, then follows only mandatory MATCH/SEARCH transitions, and ends before the next mandatory bottom or outer WRITE. A site is *baseline-portable* when its eager rule lands on the canonical center representative for every source exposed in every baseline episode. A schedule is a subset of the resulting finite menu; it is admissible when the deterministic machine still recognizes exactly $L_{p,q}$. A set F of sites is *minimal fatal* when every schedule containing F is bad but no proper subset has that property. Thus the terms used in table 55.1 refer to a completely specified finite presentation rather than to unnamed abstract vertices.

55.7.1 Complete site universe for $\min(p,q) \geq 2$

By physical duality it suffices to take $p \geq q \geq 2$. If m is odd, the complete baseline-portable universe consists of the six core sites

$$(-h,0,0),\ (-h,1,0),\ (-h,1,1),\ (h-1,0,0),\ (h-1,1,0),\ (h,0,0),$$

together with

$$X = (h-q+1,0,1) \iff p > 5q-5,$$
$$Y = (h-q,1,1) \iff p > 5q-1.$$

If m is even, the complete universe consists of the four core sites

$$(-h,1,1),\ (h,0,0),\ (h,0,1),\ (h,1,1),$$

together with

$$X = (h-q+1,0,0) \iff p > 5q-5,$$
$$Y = (h-q,1,0) \iff p > 5q.$$

Hence the menu has at most eight sites. For $q > p$, apply

$$(p,q,d,a,z) \longleftrightarrow (q,p,-d,1-a,1-z).$$

For completeness, the exclusion mechanism is exact rather than computational. For the bottom-WRITE seed family

$$B_j = \mu(jm+q)$$

one has, for $j \geq 0$,

$$B_j = (j+q, p-j-1),$$

while for $j = -t < 0$, writing $t = kq + s$ with $0 \leq s < q$,

$$B_{-t} = \begin{cases} (0, km-1), & s = 0, \\ (q-s, p+km+s-1), & 1 \leq s < q. \end{cases}$$

For $1 \le d \le h$, $V_d = (d+q, p-d)$ and the three nontrivial positive-debt compiler count vectors have imbalances

$$2d+q-p, \qquad 2d+q-p+2, \qquad 2d+q-p-2.$$

If $m = 2h+1$, the central seed $B_{h-q} = (h,h)$ has word $(10)^h$. At an interior debt d, the untouched suffix is $(10)^{h-d-1}$ below a visible 0 and $0(10)^{h-d-1}$ below a visible 1. A top-0 eager concatenation can be canonical only when its compiler word is balanced, while a top-1 concatenation requires exactly one excess 1; solving these conditions leaves precisely the two odd-parity candidates displayed above. If $m = 2h+2$, the central seed is $(h, h+1)$ with word $(10)^h 1$; the corresponding parity calculation eliminates every top-1 interior candidate and leaves precisely the two displayed top-0 candidates. On the negative side, every nontrivial compiler has a terminal 1-tail whereas the diagnostic exposes an untouched 0, so no negative-interior observation is portable.

The boundary source fibers give exactly the stated cores. For the two odd-parity survivors the common compiler word is $(10)^h 1$ and the extremal exposed one-counts are $2q-2$ and $2q$, so kernel reduction is equivalent to

$$h+2q-2 < p, \qquad h+2q < p,$$

which simplify to $p > 5q-5$ and $p > 5q-1$. In even parity the common compiler is $(10)^{h+1}$, the extremal counts are $2q-3$ and $2q-1$, and the same calculation gives $p > 5q-5$ and, using coprimality of odd p, q, $p > 5q$. Thus no unlisted optional site is being silently ignored.

55.7.2 High-minimum fatal supports

Assume $\min(p,q) \ge 3$. Every subset of the parity-dependent core above is admissible. Indeed, induction over reset episodes keeps every core source inside a protected unary envelope, and every enabled core activation returns to a canonical minimum-demand word. Minority counts stay below the opposite boundary threshold, so a core reset cannot create a new incompatible source on the other side.

Each present extra site X or Y is singleton-fatal. Repeated self-activation increases a hidden alternating prefix until the first nonminimal target is reached. At that first failure the target is exactly

$$\mu(r) + K,$$

not $\mu(r) + jK$ with $j \ge 2$. Feeding a shortest semantic completion for r removes the minimum component and leaves exactly one balanced primitive-kernel packet physically present at semantic residual zero. The witness for Y meets no other optional site; the witness for X meets no core site and at most the neighboring Y observation. Therefore the singleton fatality is robust under every larger schedule. Consequently the only minimal fatal supports in this regime are the present singleton extras.

55.7.3 Width two

By duality take $q = 2$ and odd $p \ge 3$, with the six odd-parity core sites labelled

$$\begin{aligned} A &= (-h,0,0), & B &= (-h,1,0), & C &= (-h,1,1), \\ D &= (h-1,0,0), & E &= (h-1,1,0), & F &= (h,0,0). \end{aligned}$$

The core has exactly the two non-singleton minimal supports

$$\{A, D\}, \qquad \{B, D\}.$$

The possible extras are

$$X_1 = (h-1, 0, 1) \quad (p \geq 7), \qquad X_2 = (h-2, 1, 1) \quad (p \geq 11),$$

and each present extra is singleton-fatal. Hence

$$\mathrm{Adm}_{p,2} = \{H : X_1, X_2 \notin H,\ \{A, D\} \not\subseteq H,\ \{B, D\} \not\subseteq H\},$$

with absent extras omitted.

Here the sufficiency and necessity can be checked directly. Write $p = 2h-1$, $m = 2h+1$, and for $t \geq 1$ put

$$\epsilon_t = t \bmod 2, \qquad k_t = \left\lceil \frac{t}{2} \right\rceil, \qquad \beta_t = t + pk_t, \qquad \Gamma_t = \eta(\epsilon_t, \beta_t).$$

The mandatory reset family is explicitly

$$\mathcal{R}_h^{\mathrm{base}} = \{\mu(jm+2) : -h-1 \leq j \leq h\} \cup \{(x, h-2) : x \geq h+3\} \cup \{(\epsilon_{h+1}, y) : y \geq \beta_{h+1}\}.$$

Two episode-closed repertoires are

$$\mathcal{R}_{\neg D,h} = \mathcal{R}_h^{\mathrm{base}} \cup \{(\epsilon_h, y) : y \geq \beta_h\} \cup \{(\epsilon_{h-1}, \beta_{h-1})\} \cup \{(x, h-1) : x \geq h+2\},$$
$$\mathcal{R}_{D,h} = \mathcal{R}_h^{\mathrm{base}} \cup \{(\epsilon_h, y) : y \geq \beta_h\} \cup \{(x, h-1) : x \geq h+2\} \cup \{(x, h) : x \geq h+1\}.$$

Every pair in either repertoire has minority coordinate at most h. Before the first reset, MATCH and interior SEARCH only delete the visible top cell, so a boundary source is a suffix of its starting canonical word. This forces the exact source geometry: with D disabled, the common A/B source at D_{-h} is the one-cell stack 0; with D enabled it is an arbitrary 0^n, $n \geq 1$; all C sources are unary 1^n and all D, E, F sources are unary 0^n.

The eager targets are therefore exactly

site	canonical target pair
A	(ϵ_h, β_h)
B	$(\epsilon_{h-1}, \beta_{h-1})$
C	$(\epsilon_h, \beta_h + n - 1)$
D	$(n+h, h)$
E, F	$(n+h+1, h-1)$.

Thus $\mathcal{R}_{\neg D,h}$ is closed under every subset of A, B, C, E, F, giving $2^5 = 32$ safe schedules with $D = 0$, while $\mathcal{R}_{D,h}$ is closed when $D = 1$ and $A = B = 0$, giving another $2^3 = 8$. This proves the 40 safe schedules without enumeration.

For necessity, the D-ray gives, for every $x \geq h+1$,

$$(C, \eta(x, h)) \xrightarrow{0^{2h-1}} (D_{-h}, 0^{x-h+1}),$$

so enabling D expands the A/B source fiber from $\{0\}$ to the whole unary ray. A uniform prefix reaches $(D_{-h}, 000)$ while meeting no optional site except D. At that common source

$$A : (D_{-h}, 000) \xrightarrow{0} (C, \Gamma_h 00), \qquad B : (D_{-h}, 000) \xrightarrow{1} (C, \Gamma_{h-1} 00).$$

Each Γ_t ends in at least p consecutive ones; writing $\Gamma_t = z_t 1^p$, subsequent MATCHes along z_t leave the nonempty stack $1^p 00$. Its residual is $2p - 2p = 0$, so it is one primitive kernel packet at semantic zero and is nonaccepting. Hence $\{A, D\}$ and $\{B, D\}$ are robust minimal fatal supports. The two maximal core schedules are $\{A, B, C, E, F\}$ and $\{C, D, E, F\}$; the extra sites are independently singleton-fatal.

55.7.4 The boundary line $q = 1$

For $(p, q) = (1, 1)$ the optional-site universe is empty. For $(2, 1)$ the complete four-site universe is

$$X = (0, 1, 0), \qquad A = (-1, 1, 1), \qquad B = (-1, 0, 0), \qquad G = (1, 0, 0),$$

and the minimal fatal supports are

$$\{B, X\}, \qquad \{B, G\}.$$

The balanced words 111001 and 111110001 are explicit fatal certificates; they end at semantic residual zero with a nonaccepting ordering of one primitive kernel packet. There are exactly ten admissible schedules and two maximal ones.

For odd $p = 2h + 1 \geq 3$, the always-present sites are

$$A = (-h, 1, 1), \qquad D = (h, 1, 1), \qquad E = (h, 0, 0),$$

and for $p \geq 5$ also

$$B = (h - 1, 1, 0), \qquad C = (h, 0, 1).$$

Every positive optional site is singleton-fatal, while A alone generates an episode-closed unary negative reset corridor. For $h \geq 2$ one may use

$$w_B = 1^{p(p-2)}0^{p-2}, \qquad w_C = 1^{2h(h+1)-1}(0\,1^{p-1})^{h-1}0^h,$$

and for every $h \geq 1$,

$$w_D = 1^{2h(h+1)}0^p1^{2h(h+1)+1}, \qquad w_E = 1^{2h^2+4h+1}0^{p+1}1^{2h(h+1)+1}.$$

The B-witness meets no other optional site; after B is excluded the C- and D-witnesses are unpreempted; after B, D are excluded the E-witness is unpreempted. The last two end at the noncanonical zero-residual kernel word $(C, 1^p0)$, and the first two likewise end at explicit nonaccepting zero-residual configurations. Hence no positive-site failure can be repaired by enlarging the schedule, whereas the unary corridor generated by A stays reset-closed. Thus

$$\mathrm{Adm}_{p,1} = \{\varnothing, \{A\}\},$$

so there is one maximal schedule.

For even $p = 2h \geq 4$, the always-present sites are

$$\begin{aligned} A &= (-h, 1, 1), & B &= (-h, 0, 0), & C &= (-h, 1, 0), \\ E &= (h - 1, 0, 0), & F &= (h - 1, 1, 0), & G &= (h, 0, 0), \end{aligned}$$

and $D = (h-1, 1, 1)$ is also present when $p \geq 6$. The vertices D and E are singleton-fatal when present, and the non-singleton minimal supports are exactly

$$\{B, F\}, \quad \{B, G\}, \quad \{C, F\}, \quad \{C, G\}.$$

After singleton-fatal sites are removed, every safe schedule lies in one of the two closed sectors

$$\mathcal{N} = \{A, B, C\}, \qquad \mathcal{P} = \{A, F, G\}.$$

For subsets of $\mathcal{N}$, the only new optional reset words are unary-one words and the source fibers satisfy $\mathcal{E}_A \subseteq 1^+$ and $\mathcal{E}_B = \mathcal{E}_C = \{0\}$; the mandatory reset family plus this unary ray is episode-closed. For subsets of $\mathcal{P}$, the only new positive reset ray is $(10)^h0^n$, $n \geq 1$, together with the unary-one negative ray from A; all positive sources remain unary zero and all targets remain canonical and kernel-reduced. Thus every subset of either sector is safe.

Necessity has uniform balanced certificates. For $h \geq 3$ one may take

$$w_D = 1^{p(p-3)}0^{p-3},$$

and for $h \geq 2$,

$$w_E = 1^{(h+1)(p-1)}(0\,1^{p-1})^{h-1}0^h.$$

The four cross-pair supports are certified by

$$w_{BF} = 1^{h(p+1)}0^p1^{h(p-1)}, \qquad w_{CF} = 1^{h(p+1)}0^{p-1}1^{h(p-3)},$$
$$w_{BG} = 1^{h(p+3)}0^{p+1}1^{h(p-1)}, \quad w_{CG} = 1^{h(p+3)}0^p1^{h(p-3)}.$$

In each cross-pair certificate the positive site first expands the negative-boundary top-0 source from 0 to 00, after which the indicated negative site traps exactly one primitive kernel packet. Consequently the only maximal schedules are $\mathcal{N}$ and $\mathcal{P}$, and the union of their subset families has $8 + 8 - 2 = 14$ schedules because the overlap is $2^{\{A\}}$. Physical duality gives the cases $p = 1$.

Combining the four regimes proves the exact fatal-support table used in table 55.1. In particular, its use in theorem 55.12 is no longer an appeal to an unnamed external classification: all vertices, the admissibility contract, the exhaustive universe, and the fatal-support mechanisms required by the confluence argument have been specified here.

Part XIII

General Theory of Exact Pushdown Normalization

Research provenance. This Part develops the abstract exact-lift framework, bounded-local obstruction theory, behavioral safety, causal scheduling, synthesis, and verification hierarchy. Weighted-balance families are stress tests rather than the definition of the theory. Classical neighboring machinery includes visibly pushdown semantics, pushdown/finite-state equivalence frameworks, pushdown saturation, and deterministic pushdown language equivalence [4, 38, 8, 50].

Chapter 56

Semantic Fibers and Canonical Retraction

The preceding Parts reached the same conclusion repeatedly from increasingly general directions:

$$\boxed{\text{semantic evolution can be unique while physical normalization is not.}}$$

The General Theory now removes the weighted-balance specifics.
Its objects are:

- an abstract semantic transition system;
- an exact physical lift;
- a chosen section;
- bounded-local physical update rules;
- explicitly stated representation categories.

The word **General** is scoped accordingly.
This Part does not claim:

- a theory of every pushdown automaton;
- existence of a global section for every correct DPDA;
- unrestricted DPDA minimization;
- unrestricted infinite-lift synthesis;
- universal schedule downward closure;
- universal confluence.

56.1 Exact semantic lift

Let M be a semantic state set with deterministic transitions

$$U_a : M \to M.$$

Let C be a reachable physical configuration set with transitions

$$\delta_a : C \to C.$$

An exact lift has a surjective semantic projection

$$\boxed{\Theta : C \to M}$$

satisfying

$$\boxed{\Theta\delta_a = U_a\Theta.}$$

Choose a section

$$\boxed{\sigma : M \to C}$$

with

$$\Theta\sigma = \mathrm{id}.$$

Define the canonical retraction

$$\boxed{N = \sigma\Theta.}$$

56.2 Canonical fibers

Theorem 56.1 — Canonical fiber and retraction.

The map N satisfies

$$\boxed{N^2 = N,}$$

$$\boxed{\Theta N = \Theta,}$$

and

$$\boxed{\mathrm{Fix}(N) = \operatorname{im}\sigma.}$$

Moreover,

$$\boxed{\Theta(c) = \Theta(c') \iff N(c) = N(c').}$$

Thus the fibers of the canonical retraction are exactly the semantic fibers.

Proof.

Idempotence follows from

$$\Theta\sigma = \mathrm{id}.$$

If $N(c) = c$, then $c = \sigma(\Theta(c))$, so c lies in the section image.

Conversely every section point is fixed.

Finally, equal semantic values clearly give equal canonical representatives, and equal canonical representatives project to equal semantic states. □

This is the final abstract version of the canonical-fiber idea first seen concretely in WSC.

56.3 Canonical absorption

Theorem 56.2 — Canonical absorption.

For every input word w,

$$\boxed{N\delta_w = N\delta_w N = \sigma U_w \Theta.}$$

Thus the canonical endpoint after reading w depends only on the initial semantic state.

All prior physical normalization history is forgotten by the final retraction.

This theorem is the general form of:

- eager/lazy semantic equivalence;
- WSC canonical absorption;
- terminal normalization coherence.

It remains weaker than an unrestricted confluence theorem.

56.4 Zero debt and raw conjugacy

A section is **transition-closed** when

$$\boxed{\delta_a \sigma = \sigma U_a}$$

for every input symbol.

Theorem 56.3 — Zero-debt/raw-conjugacy equivalence.

Under the stated reachability hypotheses, the following are equivalent:

1. the chosen section is transition-closed;
2. every reachable physical configuration lies on the section:

$$\boxed{C^{\text{reach}} = \sigma(M).}$$

Thus zero representation debt means that the reachable physical transition graph is literally the chosen semantic section.

The reachability hypotheses matter.

Mere bijectivity of Θ on reachable configurations is not enough if the chosen section uses different representatives.

56.5 Three levels now separated permanently

From this point onward we distinguish:

$$\boxed{\text{semantic state}}$$

$$\boxed{\text{canonical section representative}}$$

$$\boxed{\text{arbitrary reachable physical representative in the same fiber.}}$$

The next chapter asks whether a bounded-local physical policy can keep every semantic transition on the chosen section.

Chapter 57

Bounded-Local Realizability and the Three-Stage Obstruction

The top-rewrite suffix criterion was discovered in the literal WSC setting.

We now generalize the principle.

A bounded-local physical transition observes only finite local information and must update the hidden remainder in a restricted, tail-injective manner.

The fundamental question is:

> Can every semantic predecessor of one target be assigned a locally distinguishable canonical source whose physical update lands directly on the canonical target?

There are three distinct ways this can fail.

57.1 Uniform one-top portability

For ordinary one-top rewriting, sources with the same local observation must share one rewrite rule.

Lemma 57.1 — Uniform one-top portability.

A finite family of one-top sources can all be sent directly to their chosen canonical targets by one deterministic local rule exactly when their untouched suffixes are jointly compatible with that rule.

For a single source, this specializes to the suffix criterion of Chapter 32.

We do not re-prove the single-source theorem here.

57.2 Finite observation capacity

Let Ω_a be the finite set of local observations available on input a.

Suppose a semantic target m^* has many one-symbol predecessors:

$$U_a^{-1}(m^*).$$

If the chosen section is transition-closed, distinct predecessor sources that require different hidden-tail behavior must fit into the available observation classes.

Theorem 57.2 — Finite-observation collapse capacity.

If

$$\boxed{|U_a^{-1}(m^*)| > |\Omega_a|,}$$

then no tail-injective bounded-local policy can realize a transition-closed section over all those predecessors.

The obstruction occurs before any detailed stack alignment is considered.

It is a pure semantic-to-local capacity overflow.

Corollary 57.3 — Infinite-collapse obstruction.

If one semantic target has infinitely many relevant predecessors while the observation set is finite, total transition closure is impossible.

57.3 Bounded-d-top pushdown capacity

For a pushdown policy that can inspect at most d stack symbols, the number of local observations is bounded by the finite combinations of:

- control state;
- top window of length at most d;
- input symbol.

This yields an explicit observation-capacity lower bound in terms of

$$|Q|, \qquad |\Gamma|, \qquad d.$$

Corollary 57.4 — Bounded-d-top capacity.

A semantic collapse fiber larger than the available state/window observation capacity cannot be represented by a transition-closed bounded-d-top section.

This is the category-level generalization of the earlier finite-control debt lower bounds.

57.4 The three stages

Theorem 57.5 — Three-stage bounded-local obstruction decomposition.

Failure of direct canonical bounded-local realizability occurs at one of three ordered stages.

Stage I — Semantic capacity overflow

There are more same-target semantic predecessor roles than the observation policy can distinguish.

Stage II — Chosen-section observation collision

Capacity is sufficient in principle, but the chosen section assigns two same-target sources to the same local observation even though their canonical updates require different physical behavior.

Stage III — Tail/action alignment failure

The sources are locally separated, but the actual bounded-local rewrite cannot transform the hidden tail into the desired canonical target.

For one-top pushdown rewriting, Stage III becomes exactly the suffix/tail obstruction.

This theorem organizes what earlier papers saw separately:

- finite-control width pressure;
- fiber expansion;
- local aliasing;
- suffix failure.

57.5 One-top specialization

Corollary 57.6 — One-top specialization.

In the one-top literal setting, the Stage III test is exactly:

$$\boxed{\text{untouched suffix is a suffix of the canonical target.}}$$

Thus the simple theorem of Chapter 32 is the one-top boundary case of the general locality decomposition.

57.6 Locality–debt–recoordination trilemma

Theorem 57.7 — Trilemma.

When direct canonical realization fails, an exact implementation cannot keep all three of the following fixed:

1. the local observation regime;
2. the chosen canonical section;
3. direct canonical successor behavior.

At least one response is necessary:

$$\boxed{\text{refine observation}}$$

or

$$\boxed{\text{allow representation debt}}$$

or

$$\boxed{\text{re-coordinate the section.}}$$

This theorem summarizes a large fraction of the monograph:

- Chapters 6–8 used re-coordination;
- Chapters 26–37 used debt;
- broader observation would use more physical resources.

The next chapter asks an equally important question:

If debt is forced, is it necessarily dangerous?

The answer is no.

Chapter 58

Representation Debt and Behavioral Safety

A physical configuration can be noncanonical and still behave perfectly.

The General Theory therefore separates:

$$\boxed{\text{representation debt}}$$

from

$$\boxed{\text{behavioral incorrectness.}}$$

This distinction prevents a common mistake:

> "The machine left the canonical section, therefore it is already wrong."

That implication is false.

58.1 Terminal semantics

Let

$$f : M \to O$$

be the semantic terminal observable and

$$\alpha : C \to O$$

the physical terminal observable.

A physical configuration c is behaviorally correct for all continuations if

$$\boxed{\alpha(\delta_w(c)) = f(U_w(\Theta(c)))}$$

for every continuation w.

58.2 Immediate disagreement

Define the immediate disagreement set

$$\boxed{\mathcal{D} = \{c : \alpha(c) \neq f(\Theta(c))\}.}$$

A configuration can be outside $\mathcal{D}$ now but still have a future continuation reaching $\mathcal{D}$.

Thus immediate agreement is not enough.

Proposition 58.2A — Terminal factorization.

The identity

$$\boxed{\alpha = f\Theta}$$

holds on all physical configurations if and only if:

1. α is constant on every semantic fiber of Θ; and
2. $$\boxed{\alpha\sigma = f.}$$

Proof.

If $\alpha = f\Theta$, equal semantic projections have equal terminal values, and

$$\alpha\sigma = f\Theta\sigma = f.$$

Conversely, let c be any physical configuration. Fiber constancy gives

$$\alpha(c) = \alpha(\sigma(\Theta(c))).$$

Using $\alpha\sigma = f$,

$$\alpha(c) = f(\Theta(c)).$$

□

This proposition separates two requirements that are easy to conflate:

$$\boxed{\text{terminal correctness of the chosen section}}$$

and

$$\boxed{\text{terminal constancy across every physical representative of a semantic fiber.}}$$

58.3 Behavioral hull

Define

$$\boxed{\mathcal{B} = \{c : \alpha(\delta_w(c)) = f(U_w(\Theta(c))) \text{ for all } w\}.}$$

Theorem 58.1 — Behavioral hull characterization.

$$\boxed{\mathcal{B} = \nu X.\Phi(X) = C \setminus \mathrm{Pre}^*(\mathcal{D}).}$$

That is:

- $\mathcal{B}$ is the greatest forward-safe fixed point;
- a configuration is behaviorally safe exactly when no continuation reaches immediate disagreement.

This is the exact physical correctness object.

Corollary 58.2 — Root correctness.

A fixed policy is correct from root c_0 exactly when

$$\boxed{c_0 \in \mathcal{B}.}$$

58.4 Forward invariance

Theorem 58.3 — Forward invariance of behavioral safety.

If

$$c \in \mathcal{B},$$

then every reachable successor

$$\delta_w(c)$$

also lies in $\mathcal{B}$.

Thus safe debt remains safe under the fixed policy.

58.5 Debt birth dichotomy

Theorem 58.4 — Debt birth dichotomy.

When a transition first leaves the canonical section, the resulting debt is born in exactly one of two regimes:

1. $$\boxed{c \in \mathcal{B}}$$
   — behaviorally harmless debt;
2. $$\boxed{c \notin \mathcal{B}}$$
   — debt that is already behaviorally unsafe.

A later continuation does not transform safe debt into unsafe debt.

If a continuation eventually reveals an error, the configuration was outside $\mathcal{B}$ from the moment the unsafe debt was created.

This corrects the tempting narrative:

"debt becomes fatal later."

The correct statement is:

$$\boxed{\text{a later suffix reveals pre-existing unsafety.}}$$

58.6 Two opposite examples

Weighted-balance schedule conflict

Some noncanonical debts produced by forbidden eager schedules lie outside the behavioral hull and are exposed by short continuations.

Multi-type Dyck collapse

At a rejecting collapse boundary, bounded-locality may force noncanonical physical representatives, yet every such state can lie in an absorbing rejecting semantic fiber.

The debt is forced but harmless.

Thus:

$$\boxed{\text{forced debt} \not\Rightarrow \text{fatal debt.}}$$

The next chapter develops semantic certificates for proving harmlessness without constructing the exact physical behavioral hull.

Chapter 59

Semantic Containment Cores

The behavioral hull

$$\mathcal{B}$$

is exact.

But it lives in the full physical configuration space.

A coarser proof certificate can sometimes be obtained entirely at the semantic level.

59.1 Semantic trap containment

Let

$$S \subseteq M$$

be forward invariant:

$$U_a(S) \subseteq S.$$

Suppose every physical representative in every fiber over S is immediately terminally correct.

Theorem 59.1 — Semantic trap containment.

Under those hypotheses,

$$\boxed{\Theta^{-1}(S) \subseteq \mathcal{B}.}$$

Proof idea. A run beginning over S remains over S semantically.

Every reached physical representative lies in a fiber whose entire ambient fiber is terminally correct.

Therefore no continuation can reach disagreement. □

This is conservative because it requires **every** representative in the full fiber to be safe.

59.2 Absorbing-fiber harmlessness

Corollary 59.2 — Absorbing-fiber harmlessness.

If a semantic state is absorbing and all physical representatives in its fiber give the correct terminal observation, then the entire fiber lies in the behavioral hull.

This is the clean explanation of multi-type Dyck collapse debt:

- the debt can be forced physically;
- the semantic state is already an absorbing reject;
- the entire fiber is harmless.

59.3 Maximal full-fiber containment core

Define the semantic good set of states whose full fibers are terminally correct.

Iteratively remove any semantic state whose successor leaves the current set.

Theorem 59.3 — Maximal fiberwise semantic containment core.

There exists a greatest forward-invariant semantic set

$$\boxed{\mathcal{T}}$$

such that

$$\boxed{\Theta^{-1}(\mathcal{T}) \subseteq \mathcal{B}.}$$

This is the maximal **ambient full-fiber** semantic containment certificate.

It is deliberately conservative.

An unreachable bad representative can remove an otherwise useful semantic state from $\mathcal{T}$.

59.4 Reachable-fiber refinement

Let

$$F_m^{\text{reach}}$$

be only the representatives actually reachable under the policy.

Use those reachable fibers in the containment construction.

Theorem 59.4 — Reachable semantic containment.

There is a greatest reachable-fiber containment core

$$\boxed{\mathcal{T}^{\text{Reach}}}$$

whose reachable physical representatives are behaviorally safe.

It satisfies the conceptual hierarchy

$$\boxed{\mathcal{T} \subseteq \mathcal{T}^{\text{Reach}}.}$$

The reachable core removes ambient syntactic contamination.

59.5 Root correctness via containment

Theorem 59.5 — Root correctness via reachable containment.

If the root semantic state belongs to

$$\mathcal{T}^{\text{Reach}},$$

then the physical root lies in

$$\mathcal{B}.$$

Thus the three correctness scales are:

$$\boxed{\mathcal{T} \longrightarrow \mathcal{T}^{\text{Reach}} \longrightarrow \mathcal{B}.}$$

They answer increasingly exact questions:

- ambient semantic certificate;
- reachable semantic certificate;
- exact physical behavioral safety.

The next chapter returns to normalization schedules and uses the behavioral hull to distinguish merely structural unsafe targets from genuinely fatal ones.

Chapter 60

Safe Reset and the Fatal Frontier

The 2:1 causal theory introduced first-hit exposure and portability.

The General Theory now separates two notions:

1. a structural reset certificate;
2. exact behavioral safety.

This separation is essential because leaving a convenient reset domain need not be fatal.

60.1 Wordwise safe-reset domain

Let $\mathcal{R} \subseteq C$ be a set of physical configurations.

Informally, $\mathcal{R}$ is a safe-reset domain if every run that stays within the certified reset behavior can be normalized correctly word by word.

Proposition 60.1 — Safe-reset certificate.

If $\mathcal{R}$ is a wordwise safe-reset domain for a schedule family $\mathfrak{H}$, then every schedule whose eager activations remain within the certified reset behavior is correct.

The domain is a proof certificate.

It need not equal the exact behavioral hull.

60.2 Structural unsafe targets

For schedule H, let

$$U_{\mathcal{R}}(H)$$

be the set of actual eager activation targets that lie outside $\mathcal{R}$.

These are structurally uncertified targets.

Some may still lie in the behavioral hull.

Define that difference as behavioral slack.

The exact badness criterion therefore needs both layers.

60.3 Fatal Frontier

Theorem 60.2 — Fatal Frontier.

Assume $\mathcal{R}$ is a wordwise safe-reset domain for the full schedule family.

Then

$$\boxed{H \notin \mathrm{Adm} \iff U_{\mathcal{R}}(H) \setminus \mathcal{B}_H \neq \varnothing.}$$

Thus a schedule is bad exactly when it actually produces an eager target that is:

1. outside the reset certificate;
2. outside the exact behavioral hull.

This is the abstract form of what Chapter 44 saw concretely.

A structural deviation becomes a **fatal frontier event** only when no behavioral slack saves it.

60.4 Activation tightness

Corollary 60.3 — Activation tightness.

If every actual eager target outside $\mathcal{R}$ is behaviorally unsafe, then

$$\boxed{H \notin \mathrm{Adm} \iff U_{\mathcal{R}}(H) \neq \varnothing.}$$

In that case the structural unsafe frontier is exact.

Without activation tightness, structural detection is only sufficient, not necessary.

60.5 Site-free diagnostics

Sometimes one can certify fatality by a continuation that uses no additional optional eager site.

Theorem 60.4 — Site-free diagnostic frontier.

Under the stated safe-reset hypotheses, a finite site-free continuation can certify that an activation target is already outside the behavioral hull.

This gives a causal witness local to the activation event.

The 2:1 certificate

$$1111100 \rightsquigarrow (R, 00)$$

followed by the short exposing suffix is the motivating example.

60.6 Fatal causality

Proposition 60.5 — Fatal causality implies conflict.

If a minimal enabled site set H creates a fatal first-hit target at site b, then

$$H \cup \{b\}$$

is an inadmissible schedule set.

Thus directed fatal cause induces set-theoretic schedule conflict.

The next chapter develops the combinatorics of those conflicts and stress-tests the theory on 3:2 and an infinite odd-p:2 family.

Chapter 61

Conflict Complexes and Weighted-Balance Stress Tests

The Fatal Frontier tells us when one schedule is wrong.

We now ask how the whole family of schedules is organized.

61.1 Minimal conflicts

A conflict is a set of optional sites that cannot be enabled together.

A conflict is minimal if all of its proper subsets are admissible.

Lemma 61.1 — Minimal-conflict participation.

Every site belonging to a minimal conflict participates in some fatal mechanism for that conflict.

No member can be completely irrelevant.

Otherwise the conflict would not be minimal.

61.2 Robust fatal supports

Suppose a family of minimal site sets has fatal certificates that remain valid whenever additional sites outside the support are enabled.

These are robust fatal supports.

Theorem 61.2 — Robust fatal supports give an independence complex.

Under the robust-certificate hypotheses, the admissible schedule family is exactly the family of site subsets containing no minimal fatal support.

Therefore it is the independence complex of the hypergraph of minimal fatal supports.

The phrase **independence complex** is combinatorial packaging.

The project-specific content is the derivation of the fatal supports from physical normalization causality.

Corollary 61.3 — Finite diagnostic basis.

If the robust minimal fatal support family is finite, the entire finite schedule classification has a finite diagnostic basis.

61.3 Reset-ray fork

Theorem 61.4 — Reset-ray fork template.

Suppose one site D changes the reachable reset-ray geometry so that it exposes fatal first hits at two victim sites A and B, while schedules not containing D remain reset-safe and schedules containing D but neither victim remain safe.

Then the minimal fatal supports are

$$\boxed{\{A, D\}, \qquad \{B, D\}.}$$

The admissible schedules are the independence complex avoiding those two pairs.

This template abstracts the directed fork mechanism later verified uniformly.

61.4 Imported weighted-balance foundations

The General Theory does **not** re-prove:

- primitive-kernel minimum demand;
- centered residual exactness;
- centered canonical-episode terminal correctness.

Those results were already established in the $p{:}q$ construction.

Here they are used as a stress-test environment for the new causal machinery.

61.5 Exact 3:2 case

A finite reset analysis produces an exact certificate for the centered 3:2 schedule family.

Lemma 61.5A — Exact 3:2 finite reset certificate.

For the centered 3:2 schedule family there is an explicit finite reset repertoire whose episode closure certifies every schedule avoiding the two fatal pairs

$$\boxed{\{A, D\}, \qquad \{B, D\}.}$$

The same finite analysis produces first-hit fatal witnesses when either forbidden pair is enabled.

This lemma is the finite reset engine behind the exact schedule count; the detailed reset catalogue is retained for the technical appendix rather than repeated in the narrative.

Theorem 61.5 — Exact 3:2 schedule classification.

There are

$$\boxed{40 \text{ admissible schedules}}$$

and

$$\boxed{24 \text{ inadmissible schedules.}}$$

The minimal fatal fork is

$$\boxed{D \to A, \qquad D \to B.}$$

Equivalently, the minimal fatal supports are

$$\boxed{\{A, D\}, \qquad \{B, D\}.}$$

The technical finite reset catalogue supporting this result belongs in the appendix; the classification remains in the main narrative.

61.6 Infinite odd-p:2 family

Now let

$$q = 2$$

and

$$p \geq 3$$

be odd.
The source paper derives:

- a detailed arithmetic catalogue;
- source geometry;
- reset closure;
- uniform D-induced fiber expansion;
- reachability of a common length-three fatal frontier;
- kernel-suffix fatality.

The long arithmetic/source catalogues are appendix material.
The structural conclusion is main-text material.

Lemma 61.6A — Odd-p:2 reset closure.
Let

$$q = 2, \qquad p = 2h - 1 \geq 3.$$

There are two explicit safe reset repertoires:

- $\mathcal{R}_{\neg D}$, used when site D is absent;
- $\mathcal{R}_D$, used when D is present but A and B are absent.

They satisfy

$$\boxed{D \notin H \Longrightarrow H \in \mathrm{Adm}_{p,2},}$$

and

$$\boxed{D \in H, \quad A, B \notin H \Longrightarrow H \in \mathrm{Adm}_{p,2}\,.}$$

This accounts for

$$32 + 8 = 40$$

safe schedules before the fatal pairs are analyzed.

Lemma 61.6B — Uniform D-induced fiber expansion.
For every

$$x \geq h + 1,$$

the D-activated canonical ray satisfies

$$\boxed{(C, \eta(x, h)) \xrightarrow{0^{2h-1}} (D_{-h}, 0^{x-h+1}).}$$

Therefore enabling D expands the common A/B observation fiber from the singleton unary source

$$\{0\}$$

to

$$\boxed{\{0, 00, 000, \ldots\}.}$$

This is the uniform directed mechanism

$$\boxed{D \rightsquigarrow A, \qquad D \rightsquigarrow B.}$$

Lemma 61.6C — Uniform reachability of the length-three frontier.

For every

$$h \geq 2,$$

there is an explicit prefix u_h, using no optional eager site except D, that reaches the common fatal-frontier source

$$\boxed{(D_{-h}, 000).}$$

One convenient construction is

$$\boxed{u_h = v_h\, 1^{\,2h-1} 0^{\,2h},}$$

where v_h is the parity-dependent prefix that first reaches

$$(D_{h-1}, \varepsilon).$$

Thus the same length-three unary source is not merely syntactically possible; it is uniformly reachable across the entire odd-p:2 family.

Lemma 61.6D — Uniform kernel-suffix fatality.

For

$$p{:}2,$$

the primitive kernel is

$$\boxed{K = (2, p).}$$

The zero-weight physical kernel word

$$\boxed{1^p 00}$$

appears as a suffix in the eager A- and B-targets generated from the common source

$$(D_{-h}, 000).$$

A site-free continuation removes the non-kernel prefix and exposes this residual-zero kernel excess as behavioral disagreement.

Hence both A and B are uniformly fatal after the D-induced fiber expansion.

Appendix migration note

Two long supporting results are intentionally not expanded here:

- the **odd-p:2 arithmetic catalogue**;
- the **odd-p:2 source geometry catalogue**.

They are retained as appendix-level proof infrastructure. Their purpose is to justify the closed forms and source restrictions used in Lemmas 51.6A–51.6D, not to create a second narrative inside the main theorem.

Theorem 61.6 — Odd-p:2 fatal-fork family.

For every odd

$$p \geq 3,$$

$$\boxed{\mathrm{Adm}_{p,2} = \mathrm{Ind}\{\{A, D\}, \{B, D\}\}.}$$

Thus every member of the infinite family has:

$$\boxed{40 \text{ admissible}, \qquad 24 \text{ inadmissible}}$$

schedules and the same directed fatal fork

$$\boxed{D \to A, \qquad D \to B.}$$

This is an important stress test.
The General Theory is not merely repackaging the 2:1 example.
Its causal mechanism persists uniformly across an infinite parameter family.
The next chapter changes direction from diagnosis to construction:

Can a transition-closed section be synthesized when one exists?

Chapter 62

Rooted Section Synthesis

The theory has so far explained why canonical sections fail.

It now becomes constructive.

Given:

- a semantic system;
- a physical transition template;
- a chosen physical representative of the root;

can we synthesize a transition-closed section?

The answer is an exact path-outcome coherence criterion.

62.1 Rooted path-outcome fibers

Fix semantic root

$$m_0$$

and physical representative

$$c_0$$

with

$$\Theta(c_0) = m_0.$$

For semantic state m, define

$$\boxed{\mathcal{H}_{c_0}(m)}$$

to be the set of all physical outcomes obtained by following physical transitions along semantic words that take m_0 to m.

Different semantic paths to the same state can produce different physical endpoints.

That is exactly the synthesis obstruction.

62.2 Path-outcome coherence

Theorem 62.1 — Rooted path-outcome coherence criterion.

A rooted transition-closed section through c_0 exists if and only if

$$\boxed{|\mathcal{H}_{c_0}(m)| = 1}$$

for every reachable semantic state m.

When it exists, the section is uniquely determined by the root representative.

Proof.

If a transition-closed section exists, every semantic path to m must end at the single section point $\sigma(m)$.

Conversely, suppose every path-outcome fiber is a singleton.

Define $\sigma(m)$ to be that unique physical outcome.

If $m \xrightarrow{a} U_a(m)$, append a to any path reaching m.

Path-outcome uniqueness gives

$$\delta_a(\sigma(m)) = \sigma(U_a(m)).$$

Thus the section is transition-closed. □

Corollary 62.2 — One root representative determines the section.

For a fixed physical root candidate, there is at most one rooted transition-closed section.

62.3 Finite synthesis algorithm

For finite semantic systems, the theorem yields an exact propagation procedure.

Start from a root candidate.

Propagate its physical image along semantic edges.

Whenever a semantic state is reached again:

- if the physical endpoint agrees, continue;
- if it differs, synthesis fails.

Theorem 62.3 — Finite rooted synthesis.

For a finite semantic transition graph, this propagation decides exactly whether a rooted transition-closed section exists and constructs it when it does.

The algorithm is graph-theoretic once the physical transition template is fixed.

62.4 Short obstruction certificate

Theorem 62.4 — Short parallel-path obstruction certificate.

If finite rooted synthesis fails, there exist two semantic paths from the root to the same semantic state whose physical endpoints differ.

A shortest such pair gives a finite bounded certificate of failure.

Thus synthesis failure is witnessed by path disagreement.

62.5 Counting sections

Corollary 62.5 — Counting synthesized sections.

If the physical root fiber is finite, the number of rooted transition-closed sections equals the number of root representatives that pass the path-outcome coherence test.

The theory therefore turns section existence into a concrete search over root candidates.

62.6 Infinite semantic cores require symbolic synthesis

The coherence theorem itself does not require finiteness.

But brute-force propagation over an infinite semantic graph is not an algorithm.

The next chapter shows one important infinite case where a finite symbolic rule skeleton can be synthesized exactly.

That construction also demonstrates again that re-coordination can remove debt.

Chapter 63

Symbolic Re-Coordination and Forced-Debt Frontiers

The locality trilemma gave three responses to a failed canonical section:

$$\text{refine observation}, \quad \text{allow debt}, \quad \text{re-coordinate}.$$

This chapter gives a constructive example of the third option, then quantifies what remains unavoidable when only a partial section can be maintained.

63.1 Modular quotient/remainder synthesis

Consider integer translation semantics generated by fixed signed weights.

Choose a modulus large enough to absorb every one-step weight change.

Represent each semantic integer by:

- a finite sign/remainder state;
- a unary quotient stack.

Theorem 63.1 — Modular quotient/remainder synthesis.

There exists an explicit one-top zero-debt lift in which:

- the finite control stores the signed remainder;
- the stack stores the unbounded quotient;
- every semantic translation is realized directly;
- the chosen section is transition-closed.

This is a general constructive example of:

$$\boxed{\text{re-coordination removes normalization debt.}}$$

Scope

The construction is:

- pseudo-polynomial in the numerical weight bound;
- not polynomial in the bit length of weights;
- not a state-minimality theorem;
- not a general algorithm for arbitrary infinite lifts.

63.2 Finite symbolic skeleton

Theorem 63.2 — Finite symbolic skeleton and global path-outcome coherence.

The modular representation is governed by a finite set of symbolic local rules.

Those rules satisfy global rooted path-outcome coherence on the infinite semantic core.

Thus an infinite transition-closed section can sometimes be certified by finite symbolic arithmetic.

Corollary 63.3 — Zero-weight loop coherence.

Every semantic zero-weight loop returns the synthesized physical representation to the same section point.

This is an immediate path-outcome consequence, not a separate confluence theorem.

63.3 Partial canonical sections

Total transition closure may be impossible.

Let

$$G \subseteq M$$

be a semantic region over which a chosen section is transition-closed.

Theorem 63.4 — Canonical interior.

As long as the semantic run remains inside G, the physical run remains exactly on the chosen canonical section.

Representation debt is born only on an edge leaving the canonical interior.

This turns debt creation into a boundary phenomenon.

63.4 Canonical-hit capacity

Suppose many semantic source states share one target under the same input.

Only finitely many source representatives can land directly on one canonical target if the local observation space is finite.

Theorem 63.5 — Canonical-hit capacity.

For a fixed bounded-local policy, the number of source edges that can hit one chosen canonical target directly is bounded by the available local observation capacity.

This theorem no longer assumes total transition closure.

It applies specifically at the boundary of a partial section.

63.5 Quantitative forced debt

Let $\mathcal{S}$ be a finite set of relevant same-target source states and let Ω_a be the input-a observation set.

Corollary 63.6 — Quantitative forced-debt frontier.

At least

$$\boxed{|\mathcal{S}| - |\Omega_a|}$$

source edges are forced to leave the chosen section.

If $\mathcal{S}$ is infinite and Ω_a is finite, infinitely many debt-producing edges are forced.

This is the quantitative form of the earlier fiber-branching theorem.

63.6 Multi-type Dyck frontier

For one bracket type,

$$\boxed{\mathrm{Dyck}_1: \text{ total canonical synthesis.}}$$

For

$$k \geq 2,$$

the multi-type Dyck semantic system has an infinite collapse boundary that exceeds bounded local observation capacity.

Theorem 63.7 — Dyck forced-debt frontier.

For

$$k \geq 2,$$

there is:

$$\boxed{\text{zero-debt live interior} + \text{infinite forced-debt collapse boundary.}}$$

Yet that boundary lies in an absorbing rejecting semantic fiber.

Therefore the forced debt is behaviorally harmless.

This example is crucial because it separates:

$$\boxed{\text{forced physical noncanonicality}}$$

from

$$\boxed{\text{behavioral failure.}}$$

The theory can therefore quantify unavoidable debt without equating debt with error.

Chapter 64

Effective Verification Hierarchy

The General Theory ends with algorithms.

The objective is not to claim new classical automata algorithms.

The contribution is a hierarchy of reductions from normalization correctness to established decision problems.

There are three principal verification levels.

64.1 Level I — Structural safe-reset saturation

Suppose a regular set Bad_{str} contains every structurally unsafe eager target outside a certified safe-reset domain.

Theorem 64.1 — Regular structural safe-reset certification.

If pushdown reachability saturation proves that no configuration in Bad_{str} is reachable under the schedule, then the safe-reset certificate proves policy correctness.

This test is:

$$\boxed{\text{sufficient in general.}}$$

Under activation tightness, it becomes exact.

This is the cheapest level when a good structural reset certificate is available.

64.2 Level II — Exact immediate-disagreement saturation

Suppose the exact immediate disagreement set

$$\mathcal{D}_H$$

is regular and effectively represented.

Theorem 64.2 — Regular immediate-disagreement saturation.

Then

$$\boxed{\mathcal{B}_H = C \setminus \mathrm{Pre}^*_H(\mathcal{D}_H)}$$

is regular and effectively constructible by pushdown predecessor saturation.

Thus the exact behavioral hull can be computed.

Corollary 64.3 — Exact root classification.

The fixed-policy root is correct exactly when it lies outside the saturated predecessor set of $\mathcal{D}_H$.

Corollary 64.4 — Exact finite schedule classification.

For a finite optional-site family, if the regular disagreement construction is effective for every schedule, the full schedule family is exactly decidable by enumerating schedules and applying the saturation test.

This turns the Fatal Frontier theory into a complete finite classifier in the regular subclass.

64.3 Reachable-containment shortcut

Sometimes one only needs to prove correctness of one root, not construct the entire exact hull.

Corollary 64.5 — Reachable-containment root shortcut.

A root can be certified through the reachable semantic containment core without explicitly constructing the whole physical behavioral hull.

This can be substantially cheaper when semantic containment is small.

64.4 Finite semantic containment computation

If the semantic system M is finite and the initial semantically good set G_0 is known:

Theorem 64.6 — Finite containment-core computation.

The maximal forward-invariant semantic containment core is computed by iterative removal of states with outgoing edges leaving the current set.

Once G_0 is known, the edge-processing complexity is

$$\boxed{O(|M||\Sigma|).}$$

Corollary 64.7 — Finite containment-failure certificate.

If a semantic root is removed, there is a semantic witness path of length at most

$$\boxed{|M| - 1}$$

leading from the root to a containment failure.

Thus semantic containment failure has a short certificate.

64.5 Level III — DPDA-equivalence fallback

Regular immediate disagreement is powerful but not universal.

When no useful regular representation of $\mathcal{D}_H$ is available, compare continuation languages directly.

For physical configuration c, construct:

- the physical continuation language from c;
- the semantic continuation language from $\Theta(c)$.

If both have effective deterministic pushdown presentations, classical deterministic pushdown equivalence can decide equality [50, 34].

Theorem 64.8 — DPDA-equivalence behavioral fallback.

Behavioral safety of a fixed physical root reduces exactly to equivalence between its physical and semantic deterministic pushdown continuation languages.

Corollary 64.9 — Exact fixed-policy root verification.

Whenever those two continuation languages are effectively represented as DPDAs, fixed-policy root correctness is decidable.

The imported equivalence algorithm is classical.

The contribution here is the reduction from normalization safety to that equivalence problem.

64.6 Verification hierarchy

The three levels are therefore:

$$\boxed{\text{structural safe-reset certification}}$$

$$\Downarrow$$

$$\boxed{\text{exact regular disagreement saturation}}$$

$$\Downarrow$$

$$\boxed{\text{general DPDA-equivalence fallback.}}$$

They trade:

- structural specificity;
- computational convenience;
- generality.

64.7 General Theory synthesis

The final theoretical chain is now complete:

$$\boxed{\text{semantic fibers}}$$

$$\Downarrow$$

$$\boxed{\text{bounded-local realizability}}$$

$$\Downarrow$$

$$\boxed{\text{representation debt}}$$

$$\Downarrow$$

$$\boxed{\text{behavioral hull / containment}}$$

$$\Downarrow$$

$$\boxed{\text{safe reset / fatal causality}}$$

$$\Downarrow$$

$$\boxed{\text{conflict complexes}}$$

$$\Downarrow$$

$$\boxed{\text{section synthesis / forced debt}}$$

$$\Downarrow$$

$$\boxed{\text{effective verification.}}$$

But the monograph should not end on an abstract verification theorem. The post-theory branches below first record the enumerative, arithmetic, probabilistic, topological, and unrestricted-complexity consequences that emerged from the completed framework.

The final Part then returns one last time to the two 2023 DPDA figures.

The point of the General Theory is not to replace the original machine.

It is to explain it.

Part XIV

Operational Enumeration of Rational Pushdown Normalization

Research provenance. The General Theory separates semantic correctness from physical normalization history. This Part returns to the centered rational normalizer and enumerates that history itself: CANCEL and WRITE events, primitive Schubert statistics, repeated-kernel carry structure, and the exact one-counter compression of arbitrary scale. Classical Narayana, lattice-path matroid, Lorentzian, Bizley–Duchon, Raney, and kernel-method ingredients enter only after the normalization-specific identifications.

Chapter 65

Primitive Operational Statistics and Schubert Cancellation Geometry

The original 2:1 construction has already supplied the provenance for CANCEL, SEARCH, WRITE, and shortest-completion demand. We therefore begin directly with the centered p:q event process rather than repeating the automaton construction.

65.1 The centered p:q event process

Fix coprime positive integers p, q, put

$$m = p + q, \qquad h = \left\lfloor \frac{m-1}{2} \right\rfloor,$$

and write D_d for the centered debt state with $-h \le d \le h$. For an integer residual r, let

$$\mu(r) = (x, y) = \arg\min\{x + y : x, y \in \mathbb{Z}_{\ge 0},\ px - qy = r\},$$

which is unique after restricting to the standard nonnegative kernel strip. The canonical binary realization of a demand pair is

$$\eta(x, y) = \begin{cases} (10)^y 0^{x-y}, & x \ge y, \\ (10)^x 1^{y-x}, & x < y. \end{cases}$$

The centered real-time process has four event classes. If the stack top matches the next input bit, the top cell is popped (MATCH/CANCEL). A mismatch is absorbed by changing d by one while $|d| < h$ (SEARCH$\pm$). At the outer boundary a further same-direction mismatch performs OUTER-WRITE and returns to debt 0. If the stack is empty, the next symbol performs BOTTOM-WRITE using the shortest-demand map μ.

The exact transition rule is the following. With input bit a and stack word S whose leftmost symbol is the top,

$$\begin{array}{ll} \text{top } z = a : & (d, zT) \mapsto (d, T), \\ (a, z) = (1, 0),\ d < h : & (d, 0T) \mapsto (d + 1, T), \\ (a, z) = (0, 1),\ d > -h : & (d, 1T) \mapsto (d - 1, T), \\ (a, z) = (1, 0),\ d = h : & (h, 0T) \mapsto (0, \eta(\mu((h + 1)m))T), \\ (a, z) = (0, 1),\ d = -h : & (-h, 1T) \mapsto (0, \eta(\mu(-(h + 1)m))T), \\ S = \varepsilon : & (d, \varepsilon) \mapsto (0, \eta(\mu(dm + w(a)))), \end{array}$$

where $w(1) = q$ and $w(0) = -p$. The language-recognition correctness of this centered normalization is background for the present event-enumeration problem; all statistics below refer to these displayed transitions.

A word is *rational-Dyck* when every prefix has

$$\rho = qN_1 - pN_0 \geq 0.$$

A scale-k balanced word has kp ones and kq zeros. Let $C(P)$ be its number of MATCH/CANCEL events and $O(P)$ its number of OUTER-WRITE events.

65.2 Primitive balanced words: fixed-stack Hamming geometry

A primitive balanced word has p ones and q zeros.

Lemma 65.1 (No later WRITE in the primitive sector)**.** *After the first input symbol of a primitive balanced word, the shortest-demand stack has length $m-1$. Exactly $m-1$ input symbols remain. Hence no later WRITE occurs: every remaining transition is SEARCH or CANCEL.*

Proof. The first BOTTOM-WRITE realizes the unique shortest completion of the first-symbol residual. Because the completed word has total composition (p, q), the demand length is $m - 1$. Every subsequent non-WRITE transition consumes one input symbol and one stack cell. A new WRITE before the end would create more stack cells than remaining input can remove, contradicting balanced acceptance. □

Thus $C(P)$ is exactly Hamming agreement between the suffix of P and its fixed canonical stack reference.

Theorem 65.2 (Primitive balanced CANCEL distribution)**.** *For every coprime $p, q > 0$ with $p \neq q$,*

$$\boxed{c_{\min}(p, q) = |p - q| - 1.}$$

The number of primitive balanced words attaining this minimum is

$$\boxed{\binom{\max(p, q) - 1}{|p - q| - 1} = \binom{\max(p, q) - 1}{\min(p, q)}.}$$

In the unique coprime diagonal case $(p, q) = (1, 1)$, both primitive balanced words 10 *and* 01 *have exactly one CANCEL, so*

$$\boxed{c_{\min}(1, 1) = 1}$$

with multiplicity 2*. More generally, for $j \geq 0$,*

$$\boxed{\#\{P : C(P) = m - 1 - 2j\} = \binom{p-1}{j}\binom{q}{j} + \binom{p}{j}\binom{q-1}{j},}$$

with all other values zero.

Proof. Condition on the first bit. Lemma 65.1 reduces the suffix to comparison of two binary strings of equal length and fixed composition. If the first bit is 1, the input suffix contains $p - 1$ ones and q zeros; if it is 0, it contains p ones and $q - 1$ zeros. A Hamming disagreement swaps one selected 1-position with one selected 0-position, so disagreements occur in pairs. Choosing j swapped positions in the two composition classes gives the two binomial products. For $p \neq q$, maximizing the number of paired disagreements gives $|p-q|-1$ agreements and the stated multiplicity. In the remaining coprime case $(1,1)$ the distribution formula itself gives two words at CANCEL value 1, yielding the separate diagonal statement. □

Corollary 65.3 (Cancellation-free primitive orbit)**.** *A cancellation-free primitive balanced word exists if and only if $|p-q|=1$. It is unique: for $p=q+1$ it is $(10)^q1$, while for $q=p+1$ it is $0(01)^p$.*

This is the first operational appearance of the Christoffel/mechanical-word boundary: the extremal word is a cyclic representative of the corresponding balanced mechanical pattern, but no Christoffel theorem is needed in the proof.

65.3 Primitive rational-Dyck paths and the Schubert statistic

Assume first $p=q+d$ with $d \geq 1$; the opposite inequality follows by the exact bit/physical duality. Define

$$R_{q,d} = 1(10)^q1^{d-1}, \qquad A_q = \{3,5,\ldots,2q+1\}.$$

For a primitive rational-Dyck path P, let $Z(P)$ be the set of zero positions.

Theorem 65.4 (Canonical-demand agreement theorem)**.** *For every primitive rational-Dyck P of type $(q+d):q$,*

$$\boxed{C(P) = d-1+2|Z(P)\cap A_q|.}$$

Moreover, writing $Z(P)=\{z_1<\cdots<z_q\}$, rational-Dyck nonnegativity is equivalent to

$$\boxed{z_i \geq \left\lceil \frac{(2q+d)i}{q} \right\rceil, \qquad 1 \leq i \leq q.}$$

Hence the zero sets form the bases of a Schubert, equivalently lattice-path, matroid.

Proof. The first BOTTOM-WRITE fixes the entire remaining demand by Lemma 65.1. Direct comparison with that demand yields the reference word $R_{q,d}$ and therefore the displayed agreement formula. At the position z_i of the ith zero, the prefix has z_i-i ones and i zeros; $q(z_i-i)-pi \geq 0$ is exactly the stated lower flag. Such coordinatewise lower flags define the standard Schubert/lattice-path basis family [94]. □

Define the normalized polynomial

$$\mathcal{R}_{q,d}(x) = \sum_B x^{|B\cap A_q|},$$

where B runs over these bases.

Corollary 65.5 (Ultra-log-concavity)**.** *If $\mathcal{R}_{q,d}(x)=\sum_j r_jx^j$, then*

$$\boxed{\left(\frac{r_j}{\binom{q}{j}}\right)^2 \geq \frac{r_{j-1}}{\binom{q}{j-1}}\,\frac{r_{j+1}}{\binom{q}{j+1}}.}$$

Thus the primitive normalized CANCEL coefficients are ultra-log-concave, and hence log-concave and unimodal with no internal zero.

Proof. The multivariate basis polynomial of a matroid is Lorentzian; nonnegative specialization to a marked subset preserves the required Lorentzian consequences. Apply the strongest Mason inequality of Brändén–Huh [95]. $\square$

The support is exact:

$$j_{\max} = \left\lfloor \frac{q(2q+1)}{2q+d} \right\rfloor, \qquad j_{\min} = \begin{cases} 1, & d = 1, \\ 0, & d \geq 2. \end{cases}$$

After reversal to an upper flag, put

$$s_i = 1 + \left\lfloor \frac{(2q+d)(i-1)}{q} \right\rfloor, \qquad \bar{A}_q = \{d, d+2, \ldots, d+2q-2\}.$$

If $u_i = |\bar{A}_q \cap [1, s_i]|$, $v_i = s_i - u_i$, and

$$\sum_{r \geq 0} H_{i,r}(x) z^r = (1 - xz)^{-u_i} (1 - z)^{-v_i},$$

then the standard flagged Jacobi–Trudi/Lindström–Gessel–Viennot specialization [99, 56] gives

$$\boxed{\mathcal{R}_{q,d}(x) = \det[H_{i,1-i+j}(x)]_{1 \leq i,j \leq q}.}$$

Because the determinant is upper Hessenberg it also has the matrix-free recurrence

$$D_{q+1} = 1, \qquad D_i = \sum_{r=1}^{q-i+1} (-1)^{r-1} H_{i,r} D_{i+r}, \qquad \mathcal{R}_{q,d} = D_1.$$

65.3.1 Narayana windows

The near-diagonal cases recover classical distributions after the operational identification. If $d = 1$, deleting the initial 1 gives an ordinary Dyck path of semilength q, and

$$\#\{P : C(P) = 2j\} = \frac{1}{q}\binom{q}{j}\binom{q}{j-1}.$$

If $d = 2$ (so q is odd by coprimality), a last-level decomposition $P = UAUB$ gives

$$\boxed{\#\{P : C(P) = 2j-1\} = \frac{1}{j}\binom{q+1}{j-1}\binom{q-1}{j-1}.}$$

These are ordinary/rational Narayana distributions; the second equality is distributional, not pathwise equality with the classical peak statistic [90].

Chapter 66

Repeated Kernels, Carry Histories, and Algebraic Closure

Primitive normalization has a fixed demand stack. Repeated kernels introduce physical resets, so the event law becomes history-sensitive. The correct decomposition is a Bizley factorization refined by operational events, followed by a Raney carry skeleton and a stationary shifted-Schubert phase kernel.

66.1 Repeated kernels and operational Bizley factorization

For scale k, let $\mathcal{D}_k$ be the rational-Dyck paths with kp ones and kq zeros, and let $\mathcal{I}_k$ be those with no proper zero-residual contact. Put

$$D_k(x,y) = \sum_{P\in\mathcal{D}_k} x^{C(P)}y^{O(P)}, \qquad I_k(x,y) = \sum_{P\in\mathcal{I}_k} x^{C(P)}y^{O(P)}.$$

A zero-residual contact returns the centered machine to the canonical accepting configuration. Both statistics are additive under concatenation, hence

$$\boxed{D(z;x,y) = 1 + \sum_{k\geq 1} D_k(x,y)z^k = \frac{1}{1 - I(z;x,y)}.}$$

Define operational Bizley cumulants by

$$\boxed{\log D(z;x,y) = \sum_{k\geq 1} \frac{A_k(x,y)}{k} z^k.}$$

At $x = y = 1$ this reduces to the classical rational-Dyck/Bizley decomposition; in particular

$$\boxed{A_k(1,1) = \frac{1}{p+q}\binom{k(p+q)}{kp}.}$$

The unweighted factor-free structure is classical [93, 96, 92]; the refinement here is by actual normalization events.

66.2 Physical carry hierarchy and WRITE-weight conservation

Set

$$a = 1 + \left\lfloor \frac{h}{q} \right\rfloor, \qquad c = 1 + \left\lfloor \frac{h}{p} \right\rfloor.$$

A positive OUTER-WRITE is called a *carry.*

Theorem 66.1 (Carry-onset hierarchy)**.** *The first scale at which $r \geq 1$ carries are possible is*

$$\boxed{\kappa_r(p,q) = a + rc.}$$

Equivalently,

$$\boxed{O_{\max}(k) = \max\left\{0, \left\lfloor \frac{k-a}{c} \right\rfloor\right\}.}$$

At $k = \kappa_r$, every path realizing r carries is factor-free, so

$$\boxed{[y^r]A_{\kappa_r}(x,y) = \kappa_r[y^r]D_{\kappa_r}(x,y).}$$

Proof. Before the first positive outer carry, SEARCH+ must accumulate enough positive centered debt to cross the outer boundary. Among rational-Dyck prefixes with a fixed number of completed scale blocks, replacing a zero by a one cannot delay that positive boundary: a zero lowers the rational height and never supplies positive mismatch budget. Hence the all-ones trajectory is extremal for earliest carry onset. Writing $h = q(a-1)+s$ with $0 \leq s < q$ shows that $a = 1 + \lfloor h/q \rfloor$ scale units are necessary and sufficient before the first OUTER+ trigger.

After an OUTER+ reset the written frame has length $cm - 1$, where $c = 1 + \lfloor h/p \rfloor$. The WRITE-weight invariant of Theorem 66.2 assigns this reset weight c. Between two consecutive carries no event can create scale budget; SEARCH and CANCEL only consume existing cells. Thus each further carry costs at least an additional c units of total scale, giving the lower bound $k \geq a + rc$ for r carries. The same extremal positive-boundary trajectory, followed at each stage by its shortest balancing continuation, realizes equality, so $\kappa_r = a + rc$ and the displayed formula for $O_{\max}(k)$ follows.

At $k = \kappa_r$, suppose a path with r carries had a proper zero-residual factorization into positive scales $k_1 + \cdots + k_s = k$, $s \geq 2$, with carry counts $r_1 + \cdots + r_s = r$. Every factor with $r_i > 0$ satisfies $k_i \geq a + r_i c$, while a factor with $r_i = 0$ still has $k_i \geq 1$. If carries occur in two or more factors, summing the onset bounds contributes at least one extra copy of a; if all carries occur in one factor, every remaining factor contributes at least one extra unit of scale. In either case $k > a + rc = \kappa_r$, a contradiction. Hence every onset path with r carries is factor-free. The standard logarithmic relation between all paths and factor-free factors then gives $[y^r]A_{\kappa_r} = \kappa_r[y^r]D_{\kappa_r}$. □

The hierarchy is a consequence of a stronger pathwise invariant.

Theorem 66.2 (WRITE-weight conservation)**.** *Assign every WRITE event a positive integer scale weight ω: a written frame of length $\omega m - 1$ at an empty stack has weight ω, while OUTER+ has fixed weight c and OUTER− fixed weight a. For a balanced accepted scale-k computation,*

$$\boxed{\sum_{\text{WRITE } j} \omega_j = k.}$$

Proof. Every WRITE of weight ω changes physical stack height by $\omega m - 1$, while every SEARCH or CANCEL decreases height by one. If W is the number of WRITEs and N the number of non-WRITEs, empty-to-empty height balance gives

$$N = m \sum_j \omega_j - W.$$

Since every input symbol produces exactly one event, $W + N = km$, and the identity follows. □

For BOTTOM-WRITE at debt d the exact weight is

$$\omega_B(d,1) = \max\left\{1, \left\lceil -\frac{d}{q} \right\rceil, \left\lceil \frac{d+1}{p} \right\rceil \right\},$$

$$\omega_B(d,0) = \max\left\{1, \left\lceil \frac{1-d}{q} \right\rceil, \left\lceil \frac{d}{p} \right\rceil \right\}.$$

Thus scale is literally the total physical shortest-demand WRITE budget.

66.3 Minimal multi-carry paths: Raney signatures and Schubert edge weights

Define the first-carry reserve

$$\boxed{R_0 = aq - h}, \qquad Q = qc.$$

At minimal r-carry onset let z_j be the number of zeros read before the jth carry. The feasible signatures are exactly

$$0 \le z_1 \le \cdots \le z_r, \qquad z_j \le R_0 - 1 + (j-1)Q.$$

Introduce the zero-tail slack

$$\lambda_j = R_0 - 1 + (j-1)Q - z_j.$$

Then

$$\boxed{0 \le \lambda_1 \le R_0 - 1, \qquad 0 \le \lambda_j \le \lambda_{j-1} + Q.}$$

Theorem 66.3 (Raney signature law)**.** *The number of feasible r-carry signatures is*

$$\boxed{|\mathcal{Z}_r| = \frac{R_0}{(Q+1)r + R_0}\binom{(Q+1)r + R_0}{r}.}$$

Hence the signature skeleton is the Raney number $R_r^{(Q+1,R_0)}$.

Proof. Interpret z_j as the number of horizontal steps before the jth vertical step. The displayed bounds are exactly the strict ballot boundary $x < R_0 + Qy$. The classical Raney/cycle lemma gives the formula [98]. □

Before a carry with slack λ, the state is $(D_h, 0^{\lambda+1})$. After the carry, define

$$X = h + 1 + Q, \qquad Y = pc - h - 1.$$

The written frame is

$$W_+ = (10)^Y 0^{X-Y}.$$

For a transition $\lambda \to \mu$ put

$$\delta = Q + \lambda - \mu, \qquad n = pc - 1 + \delta.$$

The no-WRITE phase contains $pc - 1$ ones and δ zeros and is compared with

$$Q_{\lambda,\mu} = (10)^Y 0^{n-2Y}.$$

If $B = \{b_1 < \cdots < b_\delta\}$ is the input-zero set and

$$H_\lambda = (h+1)m + p\lambda,$$

then rational nonnegativity is the lower flag

$$\boxed{b_i \geq \ell_i(\lambda) = \max\left\{i, \left\lceil \frac{mi - H_\lambda}{q} \right\rceil\right\}.}$$

Let $A_{\lambda,\mu}$ be the zero-position set of $Q_{\lambda,\mu}$. The inter-carry polynomial is

$$\boxed{M_{\lambda,\mu}(x) = \sum_B x^{n-|A_{\lambda,\mu}|-\delta+2|A_{\lambda,\mu}\cap B|}.}$$

Thus each edge of the Raney signature skeleton is a shifted-Schubert Hamming enumerator.

After reversal, $M_{\lambda,\mu}$ has the standard flagged determinant and upper-Hessenberg recurrence. This gives an exact finite product/sum over signatures for every minimal-onset polynomial $G^{(r)}_{p,q}(x)$.

66.3.1 Stationary bulk kernel and finite wall

When the rational lower flag is inactive, $M_{\lambda,\mu}$ depends only on $\delta = Q + \lambda - \mu$:

$$\boxed{\widehat{M}_\delta(x) = \sum_{j=\max(0,\delta-Y)}^{\delta} \binom{h+\delta}{j}\binom{Y}{\delta-j} x^{Y-\delta+2j}.}$$

Its generating function is the rational symbol

$$\boxed{\widehat{K}(v;x) = \sum_{\delta\geq 0} \widehat{M}_\delta(x) v^\delta = \frac{(x+(1-x^2)v)^Y}{(1-xv)^{pc}}.}$$

For every $p > q$ the wall is absent. For steep $q > p$, let

$$\Theta = pQ - m(h+1), \qquad b = \left\lceil \frac{\Theta}{p} \right\rceil.$$

If $\Theta > 0$, then

$$\boxed{b = Q - \left\lfloor \frac{m(h+1)}{p} \right\rfloor, \qquad 1 \leq b \leq Q-1,}$$

and only target slacks $0, \ldots, b-1$ differ from the Toeplitz bulk. The exact operator is therefore Toeplitz-Hessenberg bulk plus a finite-rank rational-wall correction.

66.4 Algebraic closure of the minimal-onset family

The bulk kernel equation is

$$\boxed{v^Q = t\widehat{K}(v;x),}$$

or

$$\boxed{v^Q(1-xv)^{pc} = t(x+(1-x^2)v)^Y.}$$

Let $v_1, \ldots, v_Q$ be the Q small Puiseux branches, taken in an algebraic Puiseux extension of $\mathbb{Q}(x)((t^{1/Q}))$. The symmetric Green-function expressions below descend to algebraic formal series over $\mathbb{Q}(x,t)$; see Section 66.6. The bulk excursion Green function is

$$\boxed{\mathcal{E}_Q(t;x) = \frac{(-1)^{Q-1}}{tx^Y} \prod_{i=1}^{Q} v_i,}$$

and from slack 0 to slack μ,

$$\boxed{\mathcal{R}_\mu(t;x) = \mathcal{E}_Q(t;x) h_\mu(v_1, \ldots, v_Q),}$$

where h_μ is the complete homogeneous symmetric polynomial. These are the standard small-root half-line formulas specialized to the DPDA-derived symbol [91].

For a general start $s < Q$, the Green row has the finite form

$$\mathcal{R}_\mu^{(s)} = \sum_{i=1}^{Q} c_i^{(s)} v_i^\mu.$$

The coefficients are determined by a $Q \times Q$ boundary matrix. Writing

$$K_{<n}(v;x) = \sum_{d=0}^{n-1} \widehat{M}_d(x) v^d, \qquad \beta_n(v) = tv^{-n} K_{<n}(v;x),$$

define

$$\mathsf{C}_{\mu i} = \beta_{Q-\mu}(v_i), \qquad 0 \le \mu < Q.$$

Then

$$\boxed{\mathsf{C} c^{(s)} = e_s.}$$

The first-carry prefix vector has finite support $0 \le s < R_0 \le Q$, while the terminal slack generating series is rational. For $p > q$, this boundary system closes the entire minimal-onset family. For $q > p$, the finite wall is added by the matrix inversion lemma; hence:

Theorem 66.4 (Complete minimal-onset algebraicity)**.** *For every fixed coprime p, q, the generating series*

$$\mathcal{G}_{p,q}(t;x) = \sum_{r \ge 1} G_{p,q}^{(r)}(x) t^{r-1}$$

is algebraic over $\mathbb{Q}(x,t)$ and is given by a finite small-root/boundary system. No infinite-dimensional boundary problem remains.

For $q = 1$, $Q = 1$ and ordinary Lagrange inversion gives a scalar coefficient theorem. At 2:1 this simplifies further: if $W = 1 + xtW^3$, then

$$\boxed{\mathcal{G}_{2,1}(t;x) = x^4 W^3,}$$

so at the minimal r-carry scale $\kappa_r = r + 2$,

$$\boxed{G_{2,1}^{(r)}(x) = \frac{1}{r} \binom{3r}{r-1} x^{r+3}.}$$

66.5 Carry-onset lower bound and wall-rank lemma

Let

$$a = 1 + \left\lfloor \frac{h}{q} \right\rfloor, \qquad c = 1 + \left\lfloor \frac{h}{p} \right\rfloor.$$

Before the first positive outer carry, the maximal all-1 trajectory is the cheapest way to accumulate the required positive mismatch debt. It consumes exactly a units of scale before the first carry. After each OUTER+ reset, the written frame has length $cm - 1$, hence, by WRITE-weight conservation, each additional carry consumes at least one further block of scale weight c. Zeros cannot reduce either requirement because they decrease the rational height available to sustain the positive boundary trajectory. The all-1 extremal prefix followed by the unique shortest balancing continuation realizes equality at every level. Consequently the first scale supporting r carries is $\kappa_r = a + rc$.

At $k = \kappa_r$, a proper zero-residual factor would split the total scale into two positive balanced blocks. The carry-onset bound applied separately to the factors would then leave strictly less than the scale required for r carries in the factor containing them, contradicting minimality. Thus every r-carry path at first onset is factor-free.

For steep $q > p$, recall

$$\Theta = pQ - m(h+1) > 0, \qquad b = \left\lceil \frac{\Theta}{p} \right\rceil.$$

Then

$$b = Q - \left\lfloor \frac{m(h+1)}{p} \right\rfloor,$$

and because $m(h+1)/p > 1$ in every genuine wall case,

$$1 \le b \le Q - 1.$$

Hence every row needed by the finite Woodbury wall correction lies inside the already solved $Q \times Q$ boundary Green block. No additional start-state formula is required.

66.6 Formal domain of the small-root formulas

The Q small roots of

$$v^Q = tK_c(v; x)$$

are naturally Puiseux series in an algebraic extension of $\mathbb{Q}(x)((t^{1/Q}))$. The symmetric combinations used for the Green functions descend to algebraic formal series over $\mathbb{Q}(x, t)$; the underlying coefficient recurrence has polynomial coefficients in x. Formulas containing x^{-Y} are therefore interpreted first over $\mathbb{Q}(x)$ and specialized at singular values such as $x = 0$ only through the polynomial recurrence or by a removable-limit argument. Likewise, Vandermonde-denominator expressions are interpreted for distinct generic roots and extended to collisions by symmetric continuation.

Chapter 67

Phase Normality and Exact One-Counter Compression

Despite the history dependence created by WRITE resets, the centered rational chamber has a much smaller exact state description than the literal binary stack suggests. A phase-normal reconstruction theorem reduces the full reachable demand to one signed residual together with a finite phase bit, and a fixed shift makes the residual a nonnegative counter.

67.1 Centered phase normal form and arbitrary-scale compression

Minimal onset still uses a distinguished reset skeleton. At arbitrary scale, BOTTOM and OUTER frames can be nested. The decisive fact is that the apparent binary-stack complexity collapses on the rational chamber.

For a stack word S, write

$$\rho(S) = qN_1(S) - pN_0(S), \qquad R = -\rho(S) = pN_0(S) - qN_1(S).$$

The variable R is signed; on the rational chamber it is bounded below by $-hm$. The shifted counter $J = R + hm$ is therefore nonnegative. Let $A_{10}(S)$ and $A_{01}(S)$ denote the numbers of adjacent factors 10 and 01. A word is called *phase normal* if it consists of an alternating prefix followed by a monochromatic tail. The following theorem is proved by simultaneous induction on the physical transitions; the induction constructs two canonical reconstruction maps ν_0, ν_1 on the reachable shortest-demand pairs.

Theorem 67.1 (Centered phase-normal form)**.** *For every configuration reachable from a rational-Dyck prefix:*

1. *the stack Parikh vector is the shortest-demand pair $\mu(R)$;*
2. *there is a phase bit $b \in \{0,1\}$ such that*

   $$\boxed{S = \nu_b(\mu(R));}$$

   thus the complete bit stack is reconstructible from (R, b);
3. *the centered debt satisfies*

   $$\boxed{A_{10}(S) - h \le d \le h - A_{01}(S).}$$

The three assertions are transition-closed only when proved simultaneously.

Proof. Define the two reconstruction maps explicitly by

$$\nu_0(x,y) = \begin{cases} (10)^y 0^{x-y}, & x \ge y, \\ (10)^x 1^{y-x}, & x < y, \end{cases} \qquad \nu_1(x,y) = \begin{cases} (01)^y 0^{x-y}, & x \ge y, \\ (01)^x 1^{y-x}, & x < y. \end{cases}$$

They have the same Parikh vector (x, y) and are exactly the two phase-normal orderings compatible with a shortest-demand pair. We prove simultaneously by induction over the displayed centered transitions that (i) the current Parikh vector is $\mu(R)$, (ii) the stack equals one of these two words, and (iii) the debt inequality holds.

For a BOTTOM-WRITE the exact shortest-demand formulas are

$$(x,y) = (d + q\omega,\ p\omega - d - 1) \quad \text{on input } 1, \qquad (x,y) = (d + q\omega - 1,\ p\omega - d) \quad \text{on input } 0,$$

with the minimal positive weight ω of Theorem 66.2. Minimality of ω makes the minority coordinate at most h, so the canonical frame lies in the centered alternating interval and satisfies the debt bounds.

For CANCEL, deleting the visible matching cell decreases exactly one Parikh coordinate. Because the pre-transition pair is kernel-reduced and the deleted letter is visible in a phase-normal representative, the remaining pair is again the unique shortest pair for the updated signed demand; deleting the first letter of ν_b leaves a phase-normal word and changes A_{10}, A_{01} by at most one in the direction compensated by the unchanged debt.

For an interior SEARCH, the visible mismatching cell is consumed while d moves by one. If a SEARCH+ sees a stack beginning 00, phase normality forces the stack to be monochromatic from that point onward; hence $A_{01} = 0$ and the upper debt inequality remains strict enough after $d \mapsto d + 1$. The SEARCH$-$ case beginning 11 is the physical dual. All alternating-prefix SEARCH cases simply shorten the alternating prefix and preserve both inequalities.

At OUTER+, the new frame has

$$X_+ = h + 1 + qc, \qquad Y_+ = pc - h - 1,$$

with $Y_+ \le h$ and a terminal zero-run, so it is one of the displayed ν_b words and resets d to 0 inside the required interval. OUTER$-$ is the dual transition at the level of the induction; Theorem 67.2 below then excludes it on the rational-Dyck chamber. These cases exhaust the transition table, proving the three assertions simultaneously. The explicit reconstruction and compressed successor table are collected in Section 67.5. □

Theorem 67.2 (No negative outer carry)**.** *OUTER$-$ never occurs on a rational-Dyck prefix.*

Proof. At a hypothetical OUTER$-$ trigger, $d = -h$ and the top symbol is 1. The lower centered inequality in Theorem 67.1 forces $A_{10}(S) = 0$. Phase normality therefore makes the entire nonempty stack 1^r. Hence the global rational height is

$$\rho_{\text{global}} = dm - \rho(S) = -hm - rq < 0,$$

contradicting rational-Dyck nonnegativity. □

67.1.1 Exact one-counter recurrence

Theorem 67.1 makes the physical stack a derived object. The exact compressed state is

$$\boxed{\sigma = (d, J, b), \qquad J = R + hm,}$$

where $d \in [-h, h]$, the signed demand residual $R = pN_0(S) - qN_1(S)$ satisfies $R \geq -hm$ on the rational chamber, hence $J \in \mathbb{Z}_{\geq 0}$ is a genuine one-counter coordinate, and b is the phase bit. When evaluating transitions we recover $R = J - hm$ and reconstruct $S = \nu_b(\mu(R))$; Section 67.5 gives the exact case table. If $\mu(R) = (x, y)$, popping a top 1 sends

$$\boxed{R \mapsto R + q,}$$

while popping a top 0 sends

$$\boxed{R \mapsto R - p.}$$

The next phase bit is uniquely determined by the reconstructed remaining shortest-demand word. OUTER+ sends

$$\boxed{R' = (R - p) + (h + 1)m, \qquad d' = 0,}$$

and an empty-stack BOTTOM-WRITE on bit a sends

$$\boxed{R' = dm + w(a), \qquad d' = 0.}$$

By Theorem 67.2, no OUTER− transition is present in the rational chamber.

For each positive-height compressed state $\sigma = (d, J, b)$, let $F_\sigma(z, x, y)$ be the generating series of factor-free accepting continuations, with x marking CANCEL, y marking OUTER+, and z marking WRITE scale weight. Section 67.5 gives, for each next bit a, an explicit deterministic successor $T_a(\sigma)$ and monomial $M(\sigma, a)$. Therefore

$$\boxed{F_\sigma = \sum_{a \in \{0,1\}} M(\sigma, a)\, \Xi(T_a(\sigma)),}$$

where Ξ returns 1 at the first legal empty-stack terminal state, 0 if the rational height becomes negative or a proper zero-residual contact occurs, and otherwise returns the corresponding successor series. The initial factor-free series $I(z; x, y)$ is obtained by the two possible first BOTTOM transitions, and

$$\boxed{I_{k,r}(x) = [z^k y^r] I(z; x, y).}$$

The only unbounded coordinate is the nonnegative integer J; (d, b) is finite control. This is thus an exact weighted one-counter recurrence, to which standard one-counter/PDA generating-function methods apply [97, 89].

Corollary 67.3 (Arbitrary-scale operational enumeration)**.** *For every coprime p, q and every scale k, the factor-free bivariate polynomial $I_k(x, y)$ is determined exactly by the compressed one-counter recurrence above. All rational-Dyck paths and operational Bizley cumulants follow from*

$$\boxed{D(z; x, y) = \frac{1}{1 - I(z; x, y)}, \qquad \log D(z; x, y) = \sum_{k \geq 1} \frac{A_k(x, y)}{k} z^k.}$$

The primitive and minimal-onset formulas are specializations of the same compressed dynamics.

67.2 The original 2:1 family at arbitrary scale

The route returns to the motivating ratio with a particularly rigid all-scale law. In a factor-free scale-k 2:1 path, each WRITE has scale weight 1, every nonterminal WRITE episode contains exactly one SEARCH+, and the terminal episode contains none. If W, S, C are the numbers of WRITEs, SEARCHes, and CANCELs, stack-height and input-length balance give

$$C + S = 2W, \qquad W + C + S = 3k.$$

Hence $W = k$, $S = k - 1$, and

$$\boxed{C = k + 1}$$

for every factor-free path.

If an arbitrary rational-Dyck path of scale k has $s(P)$ zero-residual factors, additivity therefore gives the pathwise identity

$$\boxed{C(P) = k + s(P).}$$

This is the operational contribution; the factor counts themselves belong to the classical Fuss–Catalan/factor-free theory.

The number of paths with s factors is

$$\boxed{\#\{P : C(P) = k + s\} = \frac{s}{k}\binom{3k - s - 1}{k - s}, \qquad 1 \le s \le k,}$$

so

$$\boxed{D_k(x, 1) = \sum_{s=1}^{k} \frac{s}{k}\binom{3k - s - 1}{k - s} x^{k+s}.}$$

The operational Bizley cumulants simplify to

$$\boxed{A_k(x, 1) = x^k \sum_{s=1}^{k} \binom{3k - s - 1}{k - s} x^s,}$$

and $A_k(1, 1) = \frac{1}{3}\binom{3k}{k}$ as required.

67.3 Computational validation and structural checks

The proofs above are symbolic. Exact computation was used independently as regression testing while the formulas were developed. The carry-onset hierarchy was checked on coprime $p, q \le 7$ through scales up to 10 (350 scale tests and 138 onset tests for the first four carry layers) with no discrepancy. The stationary inter-carry determinant/Hessenberg formula was compared with direct local-word enumeration on 481 small parameter/slack pairs, and the rational Toeplitz bulk formula on 660 tested transitions, again with zero discrepancies. The scalar p:1 formulas were compared polynomial-by-polynomial with direct centered-machine enumeration in representative 2:1, 3:1, and 4:1 cases. Before Theorem 67.2 was proved, exact state-DP searches over 34 ordered coprime ratios with $p, q \le 7$ and 176 ratio/scale instances found no accepted rational-Dyck example containing OUTER−; the theorem now explains that observation.

These computations are not used as proof. They serve to detect sign, parity, boundary, and reset errors in formulas whose symbolic proofs are independent.

67.4 Operational synthesis: the loss of primitive log-concavity

The event statistics exhibit a sharp structural transition. Primitive normalization has one fixed reference word; CANCEL is therefore a single Hamming/Schubert statistic and inherits the corresponding matroid convexity. A physical WRITE reset replaces that static geometry by sums and products of Schubert pieces. For example, the first-carry polynomial at 5:3 is

$$36x^{10} + 67x^{11} + 153x^{12} + 180x^{13} + 135x^{14} + 39x^{15} + 18x^{16} + 2x^{17},$$

whose coefficients are not log-concave. Thus the loss of primitive Lorentzian behavior is not an artifact of proof technique; it records genuine history sensitivity introduced by a reset.

The repeated-kernel theory can be summarized as

$$\boxed{\begin{array}{c}\text{physical WRITE budget} + \text{Raney carry skeleton}\\ +\text{Schubert cancellation phases} + \text{Bizley zero-contact factorization.}\end{array}}$$

The phase-normal theorem shows that this history-sensitive system is nevertheless much smaller than the literal binary stack suggests: on the rational chamber it is exactly a one-counter process with finite phase control.

67.5 Phase reconstruction and exact compressed transition table

For a shortest-demand pair (x, y) define the two phase-normal words

$$\nu_0(x,y) = \begin{cases} (10)^y 0^{x-y}, & x \geq y, \\ (10)^x 1^{y-x}, & x < y, \end{cases}$$

$$\nu_1(x,y) = \begin{cases} (01)^y 0^{x-y}, & x \geq y, \\ (01)^x 1^{y-x}, & x < y. \end{cases}$$

When one of the formulas coincides with the other, the phase bit is immaterial. The simultaneous induction in Theorem 67.1 proves that every stack reachable from a rational-Dyck prefix is one of these two words for its shortest-demand pair.

It is important that the signed stack-demand residual need not itself be nonnegative. Put

$$R = pN_0(S) - qN_1(S) = -\rho(S).$$

The rational-chamber condition is

$$dm + R \geq 0.$$

Since $-h \leq d \leq h$, it follows that $R \geq -hm$. Thus the genuine one-counter coordinate is the shifted variable

$$J := R + hm \in \mathbb{Z}_{\geq 0}.$$

This shift corrects the misleading shorthand $R \geq 0$: R is signed, whereas J is the nonnegative counter.

Given a compressed state (d, J, b), recover the signed value $R = J - hm$ and reconstruct $S = \nu_b(\mu(R))$ (with the empty-stack case treated separately by the BOTTOM rule). The exact transition and its weight monomial are:

condition	(d', R')	weight
$S = \zeta T,\ \zeta = a$	$(d, R + w_S(a))$	x
$S = 0T,\ a = 1,\ d < h$	$(d + 1, R - p)$	1
$S = 1T,\ a = 0,\ d > -h$	$(d - 1, R + q)$	1
$S = 0T,\ a = 1,\ d = h$	$(0, (R - p) + (h + 1)m)$	yz^c
$S = 1T,\ a = 0,\ d = -h$	excluded on the rational chamber	$--$
$S = \varepsilon$	$(0, dm + w(a))$	$z^{\omega_B(d,a)}$

where $w(1) = q$, $w(0) = -p$, $w_S(1) = q$, $w_S(0) = -p$ records the change in R under a pop, $c = 1 + \lfloor h/p \rfloor$, and ω_B is the exact bottom-WRITE weight from Theorem 66.2. In the first row, equivalently, popping a 1 sends $R \mapsto R + q$ and popping a 0 sends $R \mapsto R - p$. After computing the successor stack by the centered rule, b' is the unique phase for which $S' = \nu_{b'}(\mu(R'))$ (or is ignored when the stack is empty). This gives an executable deterministic recurrence on the finite control (d, b) and the single nonnegative counter $J = R + hm$; after each row one stores $J' = R' + hm$.

For a factor-free positive-height state, the corresponding generating series therefore satisfies

$$F_{d,J,b} = \sum_{a \in \{0,1\}} M(d, J, b; a)\, \Xi(T_a(d, J, b)),$$

where M is the monomial in the table and Ξ is 1 on the first legal empty-stack terminal state, 0 after a negative rational height or a proper zero-residual contact, and otherwise the successor F-series. This is the precise one-counter recurrence used in Corollary 67.3.

67.6 Operational-to-intrinsic bridge

This Part has counted statistics of a chosen physical normalization process. The next enumerative Part changes the viewpoint: the residual becomes the intrinsic rank of a rational path, returns become barrier contacts, and the counting problem is expressed without reference to a physical execution history. The two branches share the same scalar residual but answer different questions—operational event history here, intrinsic path statistics there.

Part XV

From Residual Contacts to Refined Rational-Path Enumeration

Research provenance. This Part first develops the explicit run–return solution and its determinant reduction, then extracts the contact-preserving barrier-compression framework, its four fixed-composition rational-Dyck enumerations, and its arbitrary-boundary and parking-function applications. The determinant technology itself is classical [57, 56, 58]; the contribution is scoped to the contact-sensitive reduction and the resulting refined enumerations.

Chapter 68

From Residual Rank to Explicit Run–Return Enumeration

The preceding Parts developed the residual

$$\rho_{p,q} = qN_1 - pN_0$$

from the completion semantics of the original 2:1 machine and then from the centered real-time p:q family. A later enumerative branch begins by applying the same scalar invariant to a rational Dyck rectangle. Write

$$m = gq, \qquad n = gp.$$

At a lattice point (x, y) the usual rational-path rank is

$$R(x, y) = my - nx = g(qy - px) = g\rho_{p,q}.$$

Thus the rank is exactly the pushdown residual up to the dilation factor g. At a non-terminal return position the rank condition $0 \le R < n$ is therefore equivalent to the low-residual band

$$0 \le \rho_{p,q} < p.$$

This identity is the discovery bridge; none of the determinant proofs below depends logically on an automaton construction.

Dai, Fu, and Qiu introduced the rational run and ratio-run statistics and proved composition-preserving symmetry results with the return statistic [55]. Their concluding remarks posed the explicit enumeration of rational Dyck paths with prescribed run values, of either type, and returns. The next two chapters give that explicit solution.

The key step was not the determinant technology itself. Binomial barrier determinants and the Lindström–Gessel–Viennot method are classical [57, 56, 58]. The new step in the solution was to turn run constraints into avoidance of selected coarea values while *retaining return positions as contacts of the compressed upper barrier*. The resulting discovery chain is

$$\boxed{\begin{array}{c}\text{residual} \to \text{rational rank} \to \text{return as contact} \\ \to \text{forbidden-value compression} \to \text{weighted determinant}\end{array}}$$

This chronology is retained intentionally: the specialized reduction below was discovered while solving that enumeration problem; the subsequent chapter extracts its more general barrier-compression framework.

68.1 Rational Dyck paths and sampled runs

Throughout, m, n are positive integers with $m \geq n$. A path consists of north steps $N = (0, 1)$ and east steps $E = (1, 0)$.

Definition 68.1 — Rational Dyck path.

An $m \times n$ *rational Dyck path* is a path from $(0, 0)$ to (m, n) that remains weakly above $y = nx/m$. Its set is denoted by $\mathcal{D}_{m,n}$.

For $P \in \mathcal{D}_{m,n}$, let

$$\mathcal{U}(P) = (u_1, \ldots, u_n)$$

be its coarea sequence, where u_i is the horizontal coordinate of the ith north step, equivalently the number of cells to its left. Define

$$b_i := \left\lfloor \frac{m(i-1)}{n} \right\rfloor, \qquad 1 \leq i \leq n. \tag{68.1}$$

Proposition 68.2 — Coarea characterization.

A sequence $u = (u_1, \ldots, u_n)$ is the coarea sequence of a path in $\mathcal{D}_{m,n}$ if and only if

$$0 = u_1 \leq u_2 \leq \cdots \leq u_n, \qquad u_i \leq b_i \quad (1 \leq i \leq n). \tag{68.2}$$

Proof. The weak inequalities express the fact that successive north steps occur in nondecreasing columns. Immediately before the ith north step the path is at $(u_i, i-1)$, which lies weakly above the diagonal precisely when

$$i - 1 \geq \frac{n}{m} u_i,$$

or equivalently $u_i \leq m(i-1)/n$. Since u_i is integral, this is $u_i \leq b_i$. Finally $b_1 = 0$, so $u_1 = 0$. □

Dai–Fu–Qiu define a return position as a nonterminal lattice point on the path having rank smaller than n, where the rank of (x, y) is $my - nx$ [55]. Their return statistic counts the corresponding north steps.

Definition 68.3 — Sampled run.

For $d \geq 1$ and $P \in \mathcal{D}_{m,n}$, define

$$\mathrm{run}_d(P) := \min\{a \in [n] : da \notin \{u_1, \ldots, u_n\}\}. \tag{68.3}$$

The minimum exists because the sequence contains only n entries, one of which is 0.

For $d = 1$, Definition 68.3 is the ordinary rational run of [55]. If

$$\kappa = \left\lfloor \frac{m}{n} \right\rfloor,$$

then $d = \kappa$ gives their ratio-run $\widetilde{\mathrm{run}}$.

68.2 Returns are coarea contacts

The following observation is the conceptual bridge used throughout the paper.

Theorem 68.4 — Return–contact correspondence.

For every $P \in \mathcal{D}_{m,n}$ with coarea sequence $(u_1, \ldots, u_n)$,

$$\boxed{\mathrm{ret}(P) = \#\{i \in [n] : u_i = b_i\}.} \tag{68.4}$$

Proof. Immediately before the ith north step the path is at $(u_i, i-1)$ and hence has rank

$$R_i = m(i-1) - nu_i.$$

Write

$$m(i-1) = nb_i + r_i, \qquad 0 \le r_i < n.$$

Then

$$R_i = r_i + n(b_i - u_i). \tag{68.5}$$

By Proposition 68.2, $b_i - u_i \ge 0$. Therefore

$$R_i < n \iff b_i - u_i = 0 \iff u_i = b_i.$$

Every nonterminal path point of rank $< n$ must be followed by a north step: an east step would decrease the rank by n and make it negative, contradicting the rational-Dyck condition. Hence the north steps whose initial points have rank $< n$ are precisely the returns of [55]; the terminal endpoint is not an initial point of a north step. Summing over i gives (68.4). □

Remark 68.R1 — Pushdown residual interpretation.

Let $g = \gcd(m, n)$ and write $m = gq$, $n = gp$ with $\gcd(p, q) = 1$. Under the step encoding $0 = E$, $1 = N$, a prefix containing y north steps and x east steps has the weighted-balance residual

$$\rho = qy - px,$$

which is the scalar residual obtained by generalizing the arithmetic carried by the original 2:1 pushdown machine. The rational-path rank is $R = g\rho$, so a return position satisfies

$$0 \le \rho < p.$$

Thus Theorem 68.4 may equivalently be read as the statement that the finite low-residual band singled out by the pushdown semantics becomes upper-barrier contact in coarea coordinates. This remark records the origin of the viewpoint; the theorem itself depends only on the rank calculation above.

68.3 Deleting forbidden coarea values

Let $F \subset \mathbb{Z}_{>0}$ be finite. We wish to enumerate paths whose coarea sequence avoids F, while retaining the return weight from Theorem 68.4.

Define

$$\phi_F(t) := t - \#(F \cap [1, t]), \qquad t \in \mathbb{Z}_{\ge 0}, \tag{68.6}$$

and

$$c_i(F) := \phi_F(b_i) = b_i - \#(F \cap [1, b_i]). \tag{68.7}$$

Also put

$$\epsilon_i(F) := \mathbf{1}_{\{b_i \notin F\}}. \tag{68.8}$$

Lemma 68.5 — Forbidden-value compression.

The map

$$(u_1, \ldots, u_n) \longmapsto (v_1, \ldots, v_n), \qquad v_i = \phi_F(u_i), \tag{68.9}$$

is a bijection between

$$\left\{ \begin{array}{c} 0 = u_1 \le \cdots \le u_n, \quad u_i \le b_i, \\ \{u_1, \ldots, u_n\} \cap F = \varnothing \end{array} \right\} \tag{68.10}$$

and

$$\{0 = v_1 \leq \cdots \leq v_n : \; v_i \leq c_i(F)\}. \tag{68.11}$$

Moreover, whenever $b_i \notin F$,

$$u_i = b_i \iff v_i = c_i(F). \tag{68.12}$$

If $b_i \in F$, the equality $u_i = b_i$ is impossible on the domain.

Proof. For each $B \geq 0$, deletion of the elements of $F \cap [1, B]$ makes ϕ_F an order-preserving bijection

$$\{0, 1, \ldots, B\} \setminus F \longrightarrow \{0, 1, \ldots, \phi_F(B)\}.$$

Applying this coordinatewise with $B = b_i$ proves that (68.9) is well-defined, preserves weak order, and has a unique inverse obtained by reinserting the deleted values. This proves the bijection. If $b_i \notin F$, the maximum allowed value b_i maps to the maximum compressed value $c_i(F)$, proving (68.12). The final assertion is immediate. □

Chapter 69

Weighted Determinants and Explicit Run–Return Enumeration

Chapter 68 converted returns into upper-barrier contacts and isolated the specialized forbidden-value compression that arose in the explicit-enumeration solution. We now retain the complete determinant and inclusion–exclusion argument, including the coefficient-free form and the independent exact checks. The sampled statistic run_d is used only as a unifying notation: $d = 1$ is ordinary run and $d = \lfloor m/n \rfloor$ is the Dai–Fu–Qiu ratio-run. No independent novelty claim is made for intermediate sampling widths.

69.1 A weighted binomial determinant

We use the convention

$$\binom{a}{b} = 0 \quad \text{if } b < 0 \text{ or } b > a,$$

for integers $a \geq 0$.

Definition 69.1 — Weighted forbidden-value determinant polynomial.

For a finite $F \subset \mathbb{Z}_{>0}$, define

$$\boxed{D_F(z) := \det_{1 \leq i,j \leq n} \left[\binom{c_i(F)+1}{j-i+1} + (z-1)\epsilon_i(F) \binom{c_i(F)}{j-i} \right].} \tag{69.1}$$

Theorem 69.2 — Weighted forbidden-value determinant.

For every finite $F \subset \mathbb{Z}_{>0}$,

$$\boxed{D_F(z) = \sum_{\substack{P \in \mathcal{D}_{m,n} \\ \mathcal{U}(P) \cap F = \varnothing}} z^{\mathrm{ret}(P)}.} \tag{69.2}$$

Proof. Write $c_i = c_i(F)$ and $\epsilon_i = \epsilon_i(F)$. Consider the directed acyclic lattice network with east steps $(1, 0)$ and northeast steps $(1, 1)$. For $1 \leq i, j \leq n$, set

$$S_i = (-c_i - 1, i), \qquad T_j = (0, j + 1).$$

A path from S_i to T_j has horizontal displacement c_i+1 and vertical displacement $j-i+1$, hence there are

$$\binom{c_i+1}{j-i+1}$$

such paths.

If $\epsilon_i = 1$, give weight z to the northeast edge leaving S_i and weight 1 to all other edges. If $\epsilon_i = 0$, all edges leaving S_i have weight 1. The number of $S_i \to T_j$ paths using that first northeast edge is

$$\binom{c_i}{j-i},$$

so the weighted single-path generating function is exactly the (i, j) entry of (69.1).

By the Lindström–Gessel–Viennot lemma [56], the determinant is the signed generating function of vertex-disjoint path families connecting the sources to the targets. The source–target configuration is nonpermutable. Indeed, since c_i is nondecreasing, for $i < k$ we have $x(S_i) \geq x(S_k)$. On the common horizontal range from $x(S_i)$ to 0, an S_i-path starts strictly below an S_k-path. If their assigned targets were inverted, say T_j and T_ℓ with $j > \ell$, their vertical order at $x = 0$ would be reversed. Because every east or northeast step advances the horizontal coordinate by one, each path has one vertex at every integer abscissa in this common range; the change of vertical order therefore forces a common vertex. Thus a vertex-disjoint family has no inversion, and only the identity matching $S_i \to T_i$ contributes.

An identity path $\pi_i : S_i \to T_i$ contains exactly one northeast step and c_i east steps. Let $e_i \in \{0, \dots, c_i\}$ be the number of east steps occurring before its unique northeast step, and set

$$v_i := c_i - e_i. \tag{69.3}$$

For adjacent paths, vertex-disjointness is equivalent to

$$-c_{i+1} - 1 + e_{i+1} < -c_i + e_i,$$

which, since all quantities are integral, is equivalent to

$$c_i - e_i \leq c_{i+1} - e_{i+1},$$

that is, $v_i \leq v_{i+1}$. Thus identity nonintersecting families are in bijection with

$$0 \leq v_1 \leq \cdots \leq v_n, \qquad v_i \leq c_i.$$

This is precisely the compressed sequence class in Lemma 68.5.

Finally, π_i uses its first northeast edge if and only if $e_i = 0$, equivalently $v_i = c_i$. Such an edge is weighted by z exactly when $\epsilon_i = 1$. By Lemma 68.5, these weighted events correspond exactly to the original contacts $u_i = b_i$, and Theorem 68.4 identifies their number with $\operatorname{ret}(P)$. Therefore each family has weight $z^{\operatorname{ret}(P)}$, proving (69.2). □

Corollary 69.3 — Return polynomial.

For $F = \varnothing$,

$$\sum_{P \in \mathcal{D}_{m,n}} z^{\operatorname{ret}(P)} = \det_{1 \leq i,j \leq n} \left[\binom{b_i}{j-i+1} + z \binom{b_i}{j-i} \right]. \tag{69.4}$$

Proof. When $F = \varnothing$, $c_i = b_i$ and $\epsilon_i = 1$. Pascal's identity gives

$$\binom{b_i+1}{j-i+1} + (z-1)\binom{b_i}{j-i} = \binom{b_i}{j-i+1} + z\binom{b_i}{j-i}.$$

Apply Theorem 69.2. □

69.2 Explicit run–return enumeration

For $d \geq 1$ and $a \geq 1$, define the required sampled values

$$S_{d,a-1} := \{d, 2d, \ldots, (a-1)d\}, \tag{69.5}$$

with $S_{d,0} = \varnothing$.

Theorem 69.4 — Main enumeration theorem.

Let $m \geq n \geq 1$, $d \geq 1$, and $a, b \in [n]$. Then

$$\boxed{\begin{aligned} N^{(d)}_{m,n}(a,b) &:= \#\{P \in \mathcal{D}_{m,n} : \mathrm{run}_d(P) = a,\ \mathrm{ret}(P) = b\} \\ &= [z^b] \sum_{T \subseteq S_{d,a-1}} (-1)^{|T|} D_{T \cup \{ad\}}(z) \end{aligned}} \tag{69.6}$$

Proof. By Definition 68.3, $\mathrm{run}_d(P) = a$ if and only if

$$S_{d,a-1} \subseteq \mathcal{U}(P) \qquad \text{and} \qquad ad \notin \mathcal{U}(P). \tag{69.7}$$

First restrict to paths avoiding ad. To force every value of $S_{d,a-1}$ to occur, apply inclusion–exclusion to the events that a specified value of $S_{d,a-1}$ is missing. For $T \subseteq S_{d,a-1}$, the corresponding class avoids the forbidden set $T \cup \{ad\}$ and therefore has return polynomial $D_{T\cup\{ad\}}(z)$ by Theorem 69.2. Thus the return polynomial of paths satisfying (69.7) is

$$\sum_{T \subseteq S_{d,a-1}} (-1)^{|T|} D_{T \cup \{ad\}}(z).$$

Extracting $[z^b]$ gives (69.6). $\square$

Corollary 69.5 — Ordinary run–return.

For $a, b \in [n]$,

$$\#\{P \in \mathcal{D}_{m,n} : \mathrm{run}(P) = a,\ \mathrm{ret}(P) = b\} = [z^b] \sum_{T \subseteq \{1,\ldots,a-1\}} (-1)^{|T|} D_{T \cup \{a\}}(z). \tag{69.8}$$

Corollary 69.6 — Ratio-run–return.

Let $\kappa = \lfloor m/n \rfloor$. For $a, b \in [n]$,

$$\boxed{\begin{aligned} &\#\{P \in \mathcal{D}_{m,n} : \widetilde{\mathrm{run}}(P) = a,\ \mathrm{ret}(P) = b\} \\ &\qquad = [z^b] \sum_{T \subseteq \{\kappa, 2\kappa, \ldots, (a-1)\kappa\}} (-1)^{|T|} D_{T \cup \{a\kappa\}}(z) \end{aligned}} \tag{69.9}$$

Corollaries 69.5 and 69.6 give explicit finite formulas for the two joint enumeration problems posed in [55].

69.2.1 A coefficient-free form

Formula (69.6) is already finite, but the coefficient extraction can be removed. For a finite forbidden set F, define $n \times n$ matrices A^F and B^F by

$$A^F_{ij} := \binom{c_i(F)+1}{j-i+1} - \epsilon_i(F) \binom{c_i(F)}{j-i}, \tag{69.10}$$

$$B^F_{ij} := \epsilon_i(F) \binom{c_i(F)}{j-i}. \tag{69.11}$$

Then

$$D_F(z) = \det(A^F + zB^F). \tag{69.12}$$

For $I \subseteq [n]$, let M_I^F be the matrix whose ith row is the ith row of B^F if $i \in I$, and the ith row of A^F otherwise.

Theorem 69.7 — Coefficient-free explicit formula.

For $a, b \in [n]$,

$$\boxed{N_{m,n}^{(d)}(a,b) = \sum_{T \subseteq S_{d,a-1}} (-1)^{|T|} \sum_{\substack{I \subseteq [n] \\ |I|=b}} \det M_I^{T \cup \{ad\}}.} \tag{69.13}$$

In particular, the joint count is expressed using only finite subset sums, binomial coefficients, and integer determinants.

Proof. The determinant in (69.12) is multilinear in its rows. Choosing the zB^F contribution in exactly the rows indexed by I and the A^F contribution in all remaining rows gives

$$[z^b]D_F(z) = \sum_{\substack{I \subseteq [n] \\ |I|=b}} \det M_I^F.$$

Substitute this identity into Theorem 69.4. □

Remark 69.R1 — Coefficient-free scope.

The formula is effective without symbolic polynomial expansion. For fixed m, n, d, a, b, every determinant in (69.13) is an $n \times n$ integer determinant. The two subset sums have respectively 2^{a-1} and $\binom{n}{b}$ terms before cancellations or structural simplifications.

69.3 Examples

We display joint matrices with rows indexed by the run value and columns by the return value.

Example 69.E1 — Ordinary run, $(m, n) = (12, 4)$.

With $d = 1$, Theorem 69.4 gives

$$\left(N_{12,4}^{(1)}(a,b)\right)_{1 \le a,b \le 4} = \begin{pmatrix} 52 & 30 & 8 & 1 \\ 30 & 8 & 1 & 0 \\ 8 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}. \tag{69.14}$$

This is the ordinary run–return matrix displayed in [55].

Example 69.E2 — Ordinary run, $(m, n) = (13, 5)$.

The formula gives

$$\begin{pmatrix} 106 & 106 & 39 & 9 & 1 \\ 106 & 39 & 9 & 1 & 0 \\ 39 & 9 & 1 & 0 & 0 \\ 9 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}, \tag{69.15}$$

again reproducing the published example [55].

Example 69.E3 — Ratio-run, $(m, n) = (11, 5)$.

Here $\kappa = \lfloor 11/5 \rfloor = 2$. Corollary 69.6 gives

$$\boxed{\begin{pmatrix} 85 & 43 & 13 & 2 & 0 \\ 43 & 32 & 11 & 2 & 0 \\ 13 & 11 & 7 & 2 & 0 \\ 2 & 2 & 2 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.} \tag{69.16}$$

For instance, there are exactly 11 paths with $\widetilde{\mathrm{run}} = 2$ and $\mathrm{ret} = 3$. The sum of all entries is 273.

Example 69.E4 — Ratio-run, $(m, n) = (10, 4)$.

Here $\kappa = 2$ and

$$\begin{pmatrix} 30 & 11 & 2 & 0 \\ 11 & 9 & 4 & 1 \\ 2 & 4 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}. \tag{69.17}$$

The total number of paths is 76.

The symmetry of the displayed matrices is guaranteed independently by the involutions of Dai–Fu–Qiu. In the present approach, that symmetry becomes a family of identities between the determinant sums in Theorem 69.4.

69.4 Computational verification

The proofs above are purely combinatorial and do not depend on computation. We nevertheless used exact computation as an audit of indexing, endpoint conventions, and the inclusion–exclusion signs. Two independent exact checks were performed:

1. direct enumeration of all weakly increasing coarea sequences satisfying (68.2), followed by direct computation of run_d and ret;
2. independent evaluation of Theorem 69.4 from the forbidden-value determinants.

The displayed matrices in Section 69.3 agree entry by entry under these calculations. A small exhaustive determinant audit covered 33 sampled-run cases with no mismatches. An independent audit of the coefficient-free formula covered 86 cases with no mismatches. These computations are consistency checks only and are not used in place of the symbolic proofs.

69.5 Interpretation and scope

The main formula separates the solution into three transparent operations:

$$\boxed{\text{return} = \text{barrier contact}} \longrightarrow \boxed{\text{delete forbidden values}}$$
$$\longrightarrow \boxed{\text{weighted determinant} + \text{inclusion–exclusion}}.$$

The determinant technology itself is classical: the binomial determinant belongs to a well-established family associated with nondecreasing sequences below barriers [57, 58, 56]. The new step in the present solution is the reduction of the run constraints to forbidden-value avoidance while retaining return positions as weighted contacts of the compressed upper barrier; inclusion–exclusion then converts that refinement into the requested joint enumeration.

The sampled statistic run_d is used here as a unifying notation: it packages the ordinary case $d = 1$ and the ratio case $d = \lfloor m/n \rfloor$ into one theorem. No additional structural claim about intermediate sampling widths is needed for the explicit enumeration established here.

The automata-theoretic origin and the combinatorial proof play different roles in the paper. The completion-aware DPDA suggested that a weighted residual, rather than a particular stack representation, is the stable semantic quantity. Under the rational-path encoding this residual becomes the usual rank up to scale, and the low-residual band becomes the contact statistic of Theorem 68.4. No automata construction is needed after this bridge: forbidden-value compression, the weighted LGV determinant, and inclusion–exclusion provide the complete enumeration.

Chapter 70

Contact-Preserving Barrier Compression

The previous two chapters solved the unrestricted run–return enumeration problem. The next step is to separate what was problem-specific from what is a reusable reduction principle. The answer is *contact-preserving barrier compression.*

The novelty boundary is deliberately narrow. Lindström–Gessel–Viennot determinants, Kreweras-type barrier determinants, order-preserving relabeling, and inclusion–exclusion are classical. What is used here as one reduction is the coupling

$$\boxed{\begin{array}{c}\text{value deletion} + \text{induced barrier compression}\\ +\text{exact contact marking} + \text{presence/absence constraints}\end{array}}$$

This coupling permits arbitrary required and forbidden values, exact contact sets, and simultaneous first-missing statistics without losing the original boundary-contact information. The theory from this point is self-contained combinatorics; the residual-to-rank bridge explains its discovery but is not a proof hypothesis.

Throughout this section let

$$\mathbf{b} = (b_1, \dots, b_n), \qquad 0 \le b_1 \le \cdots \le b_n,$$

and define

$$\mathcal{A}_{\mathbf{b}} = \{u \in \mathbb{Z}^n_{\ge 0} : 0 \le u_1 \le \cdots \le u_n,\ u_i \le b_i\}.$$

For $u \in \mathcal{A}_{\mathbf{b}}$, set

$$\mathrm{Val}(u) = \{u_1, \dots, u_n\}, \qquad C_{\mathbf{b}}(u) = \{i \in [n] : u_i = b_i\}.$$

Presence and absence concern values of the sequence; contacts concern row positions. Keeping these two kinds of information separate is the central point of the method. We use the convention $\binom{a}{b} = 0$ whenever $b < 0$ or $b > a$.

70.0.1 Deleting forbidden values

Let $F \subset \mathbb{Z}_{>0}$ be finite and define

$$\phi_F(t) = t - \#(F \cap [1, t]), \qquad c_i(F) = \phi_F(b_i), \qquad \epsilon_i(F) = \mathbf{1}_{\{b_i \notin F\}}. \tag{70.1}$$

Lemma 70.1 — Contact-preserving value compression.

The coordinatewise map

$$\Phi_F(u_1, \dots, u_n) = (\phi_F(u_1), \dots, \phi_F(u_n))$$

is a bijection

$$\{u \in \mathcal{A}_{\mathbf{b}} : \mathrm{Val}(u) \cap F = \varnothing\} \longrightarrow \mathcal{A}_{\mathbf{c}(F)}.$$

Moreover, if $b_i \notin F$, then

$$u_i = b_i \iff \phi_F(u_i) = c_i(F),$$

whereas if $b_i \in F$ then no F-avoiding sequence can contact the original barrier in row i.

Proof. The restriction of ϕ_F to $\mathbb{Z}_{\geq 0} \setminus F$ is the unique order-preserving bijection from the allowed nonnegative integers onto $\mathbb{Z}_{\geq 0}$. Hence an F-avoiding weakly increasing sequence remains weakly increasing after applying ϕ_F, and $u_i \leq b_i$ implies $\phi_F(u_i) \leq \phi_F(b_i) = c_i(F)$. Conversely, applying the inverse order isomorphism coordinatewise to a sequence below $\mathbf{c}(F)$ gives an F-avoiding weakly increasing sequence below $\mathbf{b}$. If $b_i \notin F$, injectivity on the allowed alphabet gives $\phi_F(u_i) = \phi_F(b_i)$ if and only if $u_i = b_i$. If $b_i \in F$, that original contact is forbidden. □

70.0.2 Multivariate contacts

Define

$$D_{\mathbf{b},F}(z_1, \ldots, z_n) = \det_{1 \leq i,j \leq n} \left[\binom{c_i(F)+1}{j-i+1} + (z_i - 1)\epsilon_i(F) \binom{c_i(F)}{j-i} \right]. \tag{70.2}$$

Theorem 70.2 — Multivariate contact determinant.

For every finite $F \subset \mathbb{Z}_{>0}$,

$$\boxed{D_{\mathbf{b},F}(\mathbf{z}) = \sum_{\substack{u \in \mathcal{A}_{\mathbf{b}} \\ \mathrm{Val}(u) \cap F = \varnothing}} \prod_{i \in C_{\mathbf{b}}(u)} z_i.} \tag{70.3}$$

Proof. Write $c_i = c_i(F)$ and $\epsilon_i = \epsilon_i(F)$. Consider the directed acyclic lattice network with east steps $(1, 0)$ and northeast steps $(1, 1)$, sources

$$S_i = (-c_i - 1, i),$$

and targets

$$T_j = (0, j + 1).$$

A path from S_i to T_j has $c_i + 1$ horizontal steps and $j - i + 1$ northeast steps, so there are $\binom{c_i+1}{j-i+1}$ such paths. If $\epsilon_i = 1$, give weight z_i to the northeast edge leaving S_i and weight 1 to all other edges; if $\epsilon_i = 0$, give all initial edges weight 1. The number of S_i–T_j paths using the initial northeast edge is $\binom{c_i}{j-i}$, giving exactly the matrix entry in (70.2).

By the Lindström–Gessel–Viennot lemma [56], the determinant is the signed generating function of vertex-disjoint path families. Since c_i is nondecreasing, the source–target configuration is nonpermutable: an inversion of targets would reverse vertical order over a common horizontal interval and force an intersection. Thus only the identity matching contributes.

An identity path $\pi_i : S_i \to T_i$ has one northeast step and c_i east steps. Let e_i be the number of east steps before that northeast step and set $v_i = c_i - e_i$. Adjacent paths are vertex-disjoint exactly when $v_i \leq v_{i+1}$. Hence nonintersecting identity families are in bijection with

$$0 \leq v_1 \leq \cdots \leq v_n, \qquad v_i \leq c_i.$$

Moreover, π_i uses its initial northeast edge exactly when $v_i = c_i$. Lemma 70.1 identifies the weighted instances of these compressed contacts with the original contacts $u_i = b_i$. Therefore the family corresponding to u has weight $\prod_{i \in C_{\mathbf{b}}(u)} z_i$. □

For exact contacts define

$$B^F_{ij} = \epsilon_i(F)\binom{c_i(F)}{j-i}, \qquad A^F_{ij} = \binom{c_i(F)+1}{j-i+1} - B^F_{ij}. \tag{70.4}$$

For $J \subseteq [n]$, let $\mathsf{M}^{\mathbf{b};F}_J$ be the matrix whose ith row is B^F_i for $i \in J$ and A^F_i for $i \notin J$.

Corollary 70.3 — Exact contact-set determinant.

For every $J \subseteq [n]$,

$$\boxed{\#\{u \in \mathcal{A}_{\mathbf{b}} : \mathrm{Val}(u) \cap F = \varnothing,\ C_{\mathbf{b}}(u) = J\} = \det \mathsf{M}^{\mathbf{b},F}_J.}$$

Proof. The ith row in (70.2) is $A^F_i + z_i B^F_i$. Expand by row multilinearity and compare the coefficient of $\prod_{i\in J} z_i$ with Theorem 70.2. ☐

70.0.3 Presence, absence, and simultaneous first-missing statistics

Let $R, F \subset \mathbb{Z}_{>0}$ be finite and disjoint. Set

$$G_{\mathbf{b};R,F}(\mathbf{z}) = \sum_{T\subseteq R} (-1)^{|T|} D_{\mathbf{b},F\cup T}(\mathbf{z}). \tag{70.5}$$

Theorem 70.4 — Presence–absence contact calculus.

For finite disjoint $R, F \subset \mathbb{Z}_{>0}$,

$$\boxed{G_{\mathbf{b};R,F}(\mathbf{z}) = \sum_{\substack{u\in\mathcal{A}_{\mathbf{b}} \\ R\subseteq \mathrm{Val}(u),\ \mathrm{Val}(u)\cap F=\varnothing}} \prod_{i\in C_{\mathbf{b}}(u)} z_i.}$$

Consequently, for every $J \subseteq [n]$,

$$\boxed{\begin{aligned}\#\{u \in \mathcal{A}_{\mathbf{b}} : R \subseteq \mathrm{Val}(u),\ \mathrm{Val}(u) \cap F = \varnothing,& \\ C_{\mathbf{b}}(u) = J\} = \sum_{T\subseteq R} (-1)^{|T|} \det \mathsf{M}^{\mathbf{b},F\cup T}_J&\end{aligned}} \tag{70.6}$$

Proof. Among the objects avoiding F, require every $r \in R$ to occur. Inclusion–exclusion over the events $r \notin \mathrm{Val}(u)$ gives (70.5); Theorem 70.2 supplies the weighted determinant, and Corollary 70.3 gives (70.6). ☐

Let $s^{(1)}, \dots, s^{(q)}$ be injective sequences of positive integers and define

$$\mathrm{fm}_{s^{(h)}}(u) = \min\{a \geq 1 : s^{(h)}_a \notin \mathrm{Val}(u)\}.$$

For prescribed $a_1, \dots, a_q$, set

$$R(\mathbf{a}) = \bigcup_{h=1}^{q} \{s^{(h)}_1, \dots, s^{(h)}_{a_h-1}\}, \qquad F(\mathbf{a}) = \{s^{(1)}_{a_1}, \dots, s^{(q)}_{a_q}\}.$$

Corollary 70.5 — Simultaneous first-missing statistics.

If $R(\mathbf{a}) \cap F(\mathbf{a}) \neq \varnothing$, no sequence satisfies all prescribed first-missing conditions. Otherwise, for every $J \subseteq [n]$,

$$\boxed{\#\{u \in \mathcal{A}_{\mathbf{b}} : \mathrm{fm}_{s^{(h)}}(u) = a_h\ \forall h,\ C_{\mathbf{b}}(u) = J\} = \sum_{T\subseteq R(\mathbf{a})} (-1)^{|T|} \det \mathsf{M}^{\mathbf{b},F(\mathbf{a})\cup T}_J.}$$

Proof. Each condition $\mathrm{fm}_{s^{(h)}}(u) = a_h$ requires $s_1^{(h)}, \ldots, s_{a_h-1}^{(h)}$ to be present and $s_{a_h}^{(h)}$ to be absent. Combine these conditions and apply Theorem 70.4. □

Remark 70.R1 — Exact-contact specialization.

The ordinary probe $s_a = a$ and the sampled probe $s_a = \delta a$ are immediate special cases. No rational-slope, Dyck-path, or automata assumption occurs anywhere in this section.

70.1 Fixed-composition rational Dyck paths

Assume henceforth that $m > n \geq 1$ and let

$$\alpha = (\alpha_1, \ldots, \alpha_d) \vDash n, \qquad A_0 = 0, \quad A_i = \alpha_1 + \cdots + \alpha_i.$$

Let $\mathcal{D}_{m,\alpha}$ be the set of $m \times n$ rational Dyck paths with ordered vertical-run composition α.

70.1.1 Component-column coordinates and returns

For $P \in \mathcal{D}_{m,\alpha}$, let x_i be the horizontal coordinate of the initial vertex of the ith vertical component and set

$$\beta_i = \left\lfloor \frac{mA_{i-1}}{n} \right\rfloor. \tag{70.7}$$

Lemma 70.6 — Fixed-composition component encoding.

The map $P \mapsto (x_1, \ldots, x_d)$ is a bijection from $\mathcal{D}_{m,\alpha}$ onto

$$\mathcal{X}_{m,\alpha} = \{(x_1, \ldots, x_d) : 0 = x_1 < x_2 < \cdots < x_d,\ x_i \leq \beta_i\}.$$

Proof. The ith component starts at (x_i, A_{i-1}). The Dyck condition gives $A_{i-1} \geq nx_i/m$, hence $x_i \leq \beta_i$. Distinct vertical components are separated by east steps, so their columns are strictly increasing. Conversely, from such a column sequence reconstruct

$$N^{\alpha_1} E^{x_2 - x_1} N^{\alpha_2} \cdots E^{x_d - x_{d-1}} N^{\alpha_d} E^{m - x_d}.$$

Each horizontal segment stays above the diagonal because its right endpoint satisfies the corresponding bound $x_{i+1} \leq \lfloor mA_i/n \rfloor$. □

Let $\mathrm{RetComp}(P)$ be the set of component indices whose first north step is a return.

Lemma 70.7 — Return components are barrier contacts.

For $P \leftrightarrow (x_1, \ldots, x_d)$,

$$\boxed{i \in \mathrm{RetComp}(P) \iff x_i = \beta_i.}$$

Consequently $\mathrm{ret}(P) = |\,\mathrm{RetComp}(P)|$.

Proof. At the start of component i, the rank is

$$R_i = mA_{i-1} - nx_i.$$

Write $mA_{i-1} = n\beta_i + r_i$ with $0 \leq r_i < n$. Then

$$R_i = r_i + n(\beta_i - x_i).$$

Since $x_i \leq \beta_i$, one has $R_i < n$ exactly when $x_i = \beta_i$. This is precisely the return condition of Dai–Fu–Qiu [55]. A later north step in the same component has rank at least $R_i + m \geq n$, so a component contributes at most one return. □

Corollary 70.8 — Exact north-step return positions.

If $J = \mathrm{RetComp}(P)$, then

$$\boxed{\mathrm{ReturnSet}(P) = \{A_{i-1} + 1 : i \in J\}.}$$

70.1.2 Forbidden columns and the effective barrier

For finite $F \subset \mathbb{Z}_{>0}$, set

$$c_i(F) = \beta_i - \#(F \cap [1, \beta_i]), \qquad g_i(F) = c_i(F) - (i-1),$$

and define the suffix-minimum effective barrier

$$\boxed{h_i(F) = \min_{j \geq i} g_j(F).} \tag{70.8}$$

Then $h_1(F) \leq \cdots \leq h_d(F)$.

Lemma 70.9 — Effective-barrier regularization.

Let

$$\mathcal{X}_{m,\alpha}(F) = \{x \in \mathcal{X}_{m,\alpha} : \{x_1, \ldots, x_d\} \cap F = \varnothing\}.$$

If $h_1(F) < 0$, the class is empty. If $h_1(F) \geq 0$, the maps

$$y_i = \phi_F(x_i), \qquad v_i = y_i - (i-1)$$

give a bijection

$$\mathcal{X}_{m,\alpha}(F) \longleftrightarrow \{0 \leq v_1 \leq \cdots \leq v_d : v_i \leq h_i(F)\}.$$

Proof. Because ϕ_F is strictly increasing on the allowed alphabet, strict x-inequalities are equivalent to strict y-inequalities. After the shift $v_i = y_i - (i-1)$, strictness becomes weak monotonicity and the raw upper bound is $v_i \leq g_i(F)$. For a weakly increasing sequence, the full system $v_j \leq g_j(F)$ is equivalent to $v_i \leq \min_{j \geq i} g_j(F) = h_i(F)$ for every i. Since $\beta_1 = 0$, feasibility forces $h_1(F) = 0$; if $h_1(F) < 0$ no nonnegative sequence exists. □

Define the surviving-contact indicator

$$\eta_i(F) = \mathbf{1}_{\{\beta_i \notin F,\ h_i(F) = g_i(F)\}}. \tag{70.9}$$

Lemma 70.10 — Surviving return contacts.

Under Lemma 70.9,

$$\boxed{x_i = \beta_i \iff \eta_i(F) = 1 \text{ and } v_i = h_i(F).}$$

Proof. If $x_i = \beta_i$, then $\beta_i \notin F$ and $v_i = c_i(F) - (i-1) = g_i(F)$. Since $v_i \leq h_i(F) \leq g_i(F)$, equality is forced. Conversely, if $\eta_i(F) = 1$ and $v_i = h_i(F) = g_i(F)$, then $y_i = c_i(F) = \phi_F(\beta_i)$; injectivity on the allowed alphabet gives $x_i = \beta_i$. □

Define

$$D_{\alpha,F}(z_1, \ldots, z_d) = \det_{1 \leq i,j \leq d} \left[\binom{h_i(F) + 1}{j - i + 1} + (z_i - 1)\eta_i(F) \binom{h_i(F)}{j - i} \right] \tag{70.10}$$

when $h_1(F) \geq 0$, and set it equal to zero otherwise. Also define

$$B_{ij}^{\alpha,F} = \eta_i(F) \binom{h_i(F)}{j-i}, \qquad A_{ij}^{\alpha,F} = \binom{h_i(F)+1}{j-i+1} - B_{ij}^{\alpha,F},$$

and let $M_J^{\alpha,F}$ use row $B_i^{\alpha,F}$ when $i \in J$ and $A_i^{\alpha,F}$ otherwise.

Theorem 70.11 — Fixed-composition forbidden-column determinant.

For finite $F \subset \mathbb{Z}_{>0}$,

$$\boxed{D_{\alpha,F}(\mathbf{z}) = \sum_{\substack{P\in\mathcal{D}_{m,\alpha}\\ \{x_i(P)\}\cap F=\varnothing}} \prod_{i\in\mathrm{RetComp}(P)} z_i.}$$

Consequently,

$$\boxed{\#\{P \in \mathcal{D}_{m,\alpha} : \{x_i(P)\} \cap F = \varnothing,\ \mathrm{RetComp}(P) = J\} = \det M_J^{\alpha,F}.} \tag{70.11}$$

Proof. Lemma 70.9 gives a weakly increasing sequence below $\mathbf{h}(F)$. Apply Theorem 70.2 to that effective barrier, but mark row i only when $\eta_i(F) = 1$. Lemma 70.10 identifies exactly those marked effective contacts with the original return components. The exact-set formula follows by row multilinearity. □

Remark 70.R2 — Effective-barrier interpretation.

For $F = \varnothing$, $g_i = \beta_i - (i-1)$ is already nondecreasing because $m > n$ and every part of α is positive. Thus $h_i = g_i$ and $\eta_i = 1$. The suffix-minimum regularization is needed only after forbidden values are deleted.

Chapter 71

Four Composition-Refined Enumeration Problems

With the general contact calculus and the fixed-composition effective barrier in place, the framework yields four distinct rational-Dyck enumeration results at fixed ordered vertical-run composition α:

1. ordinary run together with returns;
2. ratio-run together with the exact return components;
3. simultaneous ordinary run, ratio-run, and exact returns;
4. fixed ratio-signature, where every class on the associated skew diagonal has one common determinant cardinality.

The fourth result also yields a positive complete-homogeneous expansion of the fixed-signature polynomial. The fixed-signature discussion retains an important scope correction: a ratio-signature is *not* generally an initial component prefix; that prefix interpretation becomes valid only after passage to the minimal-return representative used in the proof.

71.1 Explicit fixed-composition run–return enumeration

Dai–Fu–Qiu define a statistic $\mathrm{rr}(\alpha, m)$ depending only on the composition and m; if $I_1(\alpha)$ is the length of the maximal initial string of parts equal to 1, then [55]

$$r_\alpha := \mathrm{rr}(\alpha, m) = \min\left\{ I_1(\alpha) + 1, \left\lceil \frac{n}{m-n} \right\rceil \right\}. \tag{71.1}$$

Every path of composition α has $\mathrm{run}(P), \mathrm{ret}(P) \geq r_\alpha$. Their fixed-composition bijections imply that for $K = k + \ell - r_\alpha$,

$$|\mathcal{D}_{m,\alpha}^{(k,\ell)}| = |\mathcal{D}_{m,\alpha}^{(K,r_\alpha)}|, \tag{71.2}$$

whenever the class is nonempty [55, Theorem 3.2 and Corollary 3.5].

For $K < d$, put $p = d - K$ and define

$$B_i^{(\alpha,K)} = \beta_{K+i} - (K+i) - 1 = \left\lfloor \frac{mA_{K+i-1}}{n} \right\rfloor - (K+i) - 1, \qquad 1 \leq i \leq p. \tag{71.3}$$

Lemma 71.1 — Minimal-return suffix barrier.

For a path in $\mathcal{D}_{m,\alpha}^{(K,r_\alpha)}$, the first K component columns are forced:

$$x_i = i - 1 \qquad (1 \le i \le K).$$

If $K < d$, then after the shift

$$v_i = x_{K+i} - (K+i),$$

the free suffix is in bijection with

$$0 \le v_1 \le \cdots \le v_p, \qquad v_i \le B_i^{(\alpha,K)}.$$

Moreover $B_1^{(\alpha,K)} \le \cdots \le B_p^{(\alpha,K)}$.

Proof. In the minimal-return representative, Dai–Fu–Qiu use the decomposition $M_1 \cdots M_K ET$ with $M_i = N^{\alpha_i}E$ [55, Corollary 3.5]. Hence the first K component columns are $0, 1, \ldots, K-1$ and the forced separator implies $x_{K+1} \ge K+1$. Since the suffix contributes no further returns, Lemma 70.7 gives $x_i \le \beta_i - 1$ for $i > K$. The stated shift yields the weak sequence and its barrier. Finally

$$B_{i+1}^{(\alpha,K)} - B_i^{(\alpha,K)} = \beta_{K+i+1} - \beta_{K+i} - 1 \ge 0,$$

because $m > n$ and $\alpha_{K+i} \ge 1$. □

Theorem 71.2 — Explicit composition-refined run–return enumeration.
Let $k, \ell \ge 1$ and set $K = k + \ell - r_\alpha$. If any of

$$k < r_\alpha, \qquad \ell < r_\alpha, \qquad K > d$$

holds, the class is empty. If $K < d$ and $B_1^{(\alpha,K)} < 0$, it is likewise empty. Otherwise,

$$\boxed{|\mathcal{D}_{m,\alpha}^{(k,\ell)}| = \det_{1 \le i,j \le d-K} \left[\binom{B_i^{(\alpha,K)} + 1}{j - i + 1} \right].} \tag{71.4}$$

For $K = d$, the empty determinant is interpreted as 1.

Proof. Use the Dai–Fu–Qiu reduction (71.2), then apply the classical bounded weak-sequence determinant to the suffix barrier in Lemma 71.1. □

Set

$$C_{\alpha,K} = \det_{1 \le i,j \le d-K} \left[\binom{B_i^{(\alpha,K)} + 1}{j - i + 1} \right].$$

Corollary 71.3 — Positive complete-homogeneous expansion.
With

$$F_{m,\alpha}(u,v) = \sum_{P \in \mathcal{D}_{m,\alpha}} u^{\mathrm{run}(P)} v^{\mathrm{ret}(P)},$$

one has

$$\boxed{F_{m,\alpha}(u,v) = (uv)^{r_\alpha} \sum_{K=r_\alpha}^{d} C_{\alpha,K} h_{K-r_\alpha}(u,v),} \tag{71.5}$$

where $h_j(u,v) = u^j + u^{j-1}v + \cdots + v^j$.

Proof. For fixed K, the feasible statistic pairs are $(r_\alpha + a, K - a)$ for $0 \le a \le K - r_\alpha$, and Theorem 71.2 gives the common cardinality $C_{\alpha,K}$. Summing their monomials gives $(uv)^{r_\alpha} h_{K-r_\alpha}(u,v)$. □

Example 71.E1 — Example.

Let $(m,n) = (11,5)$ and $\alpha = (2,1,1,1)$. Then $\beta = (0,4,6,8)$ and $r_\alpha = 1$. For $(\mathrm{run},\mathrm{ret}) = (1,2)$, $K = 2$ and the suffix barrier is $(2,3)$. Hence

$$|\mathcal{D}^{(1,2)}_{11,(2,1,1,1)}| = \det \begin{pmatrix} 3 & 3 \\ 1 & 4 \end{pmatrix} = 9.$$

Remark 71.R1 — Ordinary-run scope.

The determinant in Theorem 71.2 is classical. The new enumerative step is the reduction from a fixed ordered composition and prescribed $(\mathrm{run},\mathrm{ret})$ to a single bounded monotone suffix. In particular, it makes the cardinality in the partial-path characterization of Dai–Fu–Qiu Corollary 3.5 explicit.

71.2 Composition-refined ratio-run and returns

Set

$$\kappa = \left\lfloor \frac{m}{n} \right\rfloor .$$

For $P \in \mathcal{D}_{m,\alpha}$, let $X(P) = \{x_1(P), \ldots, x_d(P)\}$.

Lemma 71.4 — Coarea values and component columns.

The set of distinct north-step coarea values of P is $X(P)$. Consequently

$$\boxed{\widetilde{\mathrm{run}}(P) = \min\{a \ge 1 : a\kappa \notin X(P)\}.} \tag{71.6}$$

Proof. Every north step in one vertical component has the same horizontal coordinate, and every component contains at least one north step. Thus the distinct coarea values are exactly the component columns. The second statement is the definition of ratio-run [55]. □

For $a \ge 1$, define

$$R_{\kappa,a} = \{\kappa, 2\kappa, \ldots, (a-1)\kappa\}, \qquad F_{\kappa,a} = \{a\kappa\}.$$

Then $\widetilde{\mathrm{run}}(P) = a$ exactly when $R_{\kappa,a} \subseteq X(P)$ and $X(P) \cap F_{\kappa,a} = \varnothing$.

Theorem 71.5 — Composition-refined ratio-run with exact returns.

For every $a \ge 1$ and $J \subseteq [d]$,

$$\boxed{\begin{aligned} &\#\{P \in \mathcal{D}_{m,\alpha} : \widetilde{\mathrm{run}}(P) = a,\ \mathrm{RetComp}(P) = J\} \\ &\quad = \sum_{T \subseteq R_{\kappa,a}} (-1)^{|T|} \det M_J^{\alpha, F_{\kappa,a} \cup T} \end{aligned}} \tag{71.7}$$

Proof. The ratio-run condition is exactly the presence–absence condition above. Inclusion–exclusion over the required values, followed by Theorem 70.11, gives the formula. □

Corollary 71.6 — Composition-refined ratio-run–return enumeration.

For $a, b \geq 1$,

$$\boxed{\#\{P \in \mathcal{D}_{m,\alpha} : \widetilde{\mathrm{run}}(P) = a,\ \mathrm{ret}(P) = b\} = \sum_{\substack{J \subseteq [d] \\ |J| = b}} \sum_{T \subseteq R_{\kappa,a}} (-1)^{|T|} \det M_J^{\alpha, F_{\kappa,a} \cup T}.}$$

Example 71.E2 — Example.

For $(m, n) = (11, 5)$ and $\alpha = (2, 1, 1, 1)$, one has $\beta = (0, 4, 6, 8)$ and $\kappa = 2$. Prescribe $\widetilde{\mathrm{run}} = 2$ and $J = \{1, 4\}$. Then $R_{2,2} = \{2\}$ and $F_{2,2} = \{4\}$. The two inclusion–exclusion determinants are

$$\det M_{\{1,4\}}^{\alpha,\{4\}} = 6, \qquad \det M_{\{1,4\}}^{\alpha,\{2,4\}} = 3,$$

so the required number is $6 - 3 = 3$. The three component-column sequences are

$$(0, 1, 2, 8), \qquad (0, 2, 3, 8), \qquad (0, 2, 5, 8).$$

71.3 Simultaneous run, ratio-run, and exact returns

Lemma 71.7 — Ordinary run as a first-missing component column.

For every $P \in \mathcal{D}_{m,\alpha}$,

$$\boxed{\mathrm{run}(P) = \min\{a \geq 1 : a \notin X(P)\}.}$$

Proof. By Lemma 71.4, $X(P)$ is the set of distinct coarea values. The ordinary run statistic is the first missing positive coarea value [55]. ☐

For positive integers a, c, define

$$R_{a,c} = \{1, \ldots, a-1\} \cup \{\kappa, 2\kappa, \ldots, (c-1)\kappa\}, \qquad F_{a,c} = \{a, c\kappa\}. \tag{71.8}$$

Theorem 71.8 — Simultaneous run–ratio-run–return-set enumeration.

For $J \subseteq [d]$, if $R_{a,c} \cap F_{a,c} \neq \varnothing$, then

$$\#\{P \in \mathcal{D}_{m,\alpha} : \mathrm{run}(P) = a,\ \widetilde{\mathrm{run}}(P) = c,\ \mathrm{RetComp}(P) = J\} = 0.$$

Otherwise,

$$\boxed{\begin{aligned} &\#\{P \in \mathcal{D}_{m,\alpha} : \mathrm{run}(P) = a,\ \widetilde{\mathrm{run}}(P) = c,\ \mathrm{RetComp}(P) = J\} \\ &\quad = \sum_{T \subseteq R_{a,c}} (-1)^{|T|} \det M_J^{\alpha, F_{a,c} \cup T} \end{aligned}} \tag{71.9}$$

Proof. The two first-missing conditions together are precisely $R_{a,c} \subseteq X(P)$ and $X(P) \cap F_{a,c} = \varnothing$. If the sets intersect, the requirements are inconsistent. Otherwise apply inclusion–exclusion and the exact-return determinant (70.11). ☐

Corollary 71.9 — Exact north-step return set.

If $S \subseteq [n]$ is not of the form $S = \{A_{i-1} + 1 : i \in J\}$ for some $J \subseteq [d]$, then no path of composition α has $\mathrm{ReturnSet}(P) = S$. For such a set S and disjoint $R_{a,c}, F_{a,c}$, the number of paths with

$$\mathrm{run}(P) = a, \qquad \widetilde{\mathrm{run}}(P) = c, \qquad \mathrm{ReturnSet}(P) = S$$

is the right-hand side of (71.9).

Example 71.E3 — Simultaneous run and ratio-run.

Again let $(m,n) = (11,5)$ and $\alpha = (2,1,1,1)$, so $\beta = (0,4,6,8)$ and $\kappa = 2$. Prescribe run $= 3$ and $\widetilde{\text{run}} = 2$. Then

$$R_{3,2} = \{1,2\}, \qquad F_{3,2} = \{3,4\}.$$

The admissible component-column sequences are

$$(0,1,2,5),\ (0,1,2,6),\ (0,1,2,7),\ (0,1,2,8).$$

The first three have RetComp $= \{1\}$, while the last has RetComp $= \{1,4\}$. Hence the corresponding exact counts are 3 and 1.

Remark 71.R2 — Automatic compatibility detection.

The condition $R_{a,c} \cap F_{a,c} = \varnothing$ automatically detects incompatibilities. For example, if $\kappa = 2$, then run $= 2$ requires column 2 to be absent while $\widetilde{\text{run}} = 2$ requires it to be present, so the simultaneous class is empty.

71.4 Explicit enumeration at fixed ratio-signature

This section uses the first ratio-signature $\text{sign}_1(P)$ of Dai–Fu–Qiu [55]. Their auxiliary path P_1 is formed by scanning P and collecting each κ-east-step block contributing to ratio-run or return. Thus P_1 is generally a shuffle-selected subpath rather than an initial segment. Let

$$\mathcal{D}_{m,\alpha,s} = \{P \in \mathcal{D}_{m,\alpha} : \text{sign}_1(P) = s\},$$

and assume this class is nonempty. Write

$$r_{\alpha,s} = \widetilde{\text{rr}}(\alpha, m, s),$$

for the minimum value attained by either $\widetilde{\text{run}}$ or ret on this fixed-signature class in the notation of Dai–Fu–Qiu.

Remark 71.R3 — The signature is not generally a prefix.

No argument below identifies a general P_1 with a prefix. The prefix interpretation arises only after passing, via the fixed-signature bijections of Dai–Fu–Qiu Theorem 3.7, to the representative with minimal return.

Lemma 71.10 — Minimal-return prefix collapse.

Suppose $P \in \mathcal{D}_{m,\alpha,s}$ has $\widetilde{\text{run}}(P) = a$ and $\text{ret}(P) = b$. Set

$$h = a + b - r_{\alpha,s}.$$

The fixed-signature bijections of Dai–Fu–Qiu reduce the class to a representative Q with

$$\widetilde{\text{run}}(Q) = h, \qquad \text{ret}(Q) = r_{\alpha,s}, \qquad \text{sign}_1(Q) = s.$$

For this minimal-return representative, Q_1 is exactly the initial segment occupying the first $h\kappa$ east-step columns. Consequently

$$\boxed{|s| = h\kappa.} \tag{71.10}$$

Proof. Dai–Fu–Qiu Theorem 3.7 constructs fixed-signature bijections along the skew diagonal and states that the corresponding minimal value is $r_{\alpha,s}$ [55]. In the minimal-return representative, every return block already belongs to the initial family of blocks contributing simultaneously to ratio-run and return; there are no additional return-only blocks. The ratio-run value h contributes the first h sampled κ-blocks, so the union defining Q_1 is precisely the initial $h\kappa$ east-step segment. Since s is the ordinary east-step signature of Q_1, its length is $h\kappa$. □

Corollary 71.11 — Fixed-signature skew diagonal.

If $\mathcal{D}_{m,\alpha,s} \neq \varnothing$, then $\kappa \mid |s|$. With

$$h_s = \frac{|s|}{\kappa},$$

every nonempty fixed-signature statistic class satisfies

$$\boxed{\widetilde{\mathrm{run}}(P) + \mathrm{ret}(P) = h_s + r_{\alpha,s}.}$$

Let the positions of the 1's in s be

$$1 \le p_1 < \cdots < p_t \le |s|, \qquad t = |s|_1.$$

Lemma 71.12 — Prefix components determined by the signature.

For a minimal-return representative Q as above, the first t component columns are forced:

$$\boxed{x_i = p_i - 1 \qquad (1 \le i \le t).}$$

Proof. By Lemma 71.10, s is the ordinary east-step signature of the first $|s|$ east steps of Q. An east step has signature 1 exactly when it immediately follows a north step. The east step in signature position p_i therefore closes the ith vertical component, whose horizontal coordinate is $p_i - 1$. □

Corollary 71.13 — Reading the minimal return from the signature.

For an admissible signature,

$$\boxed{r_{\alpha,s} = \#\{i \in [t] : p_i - 1 = \beta_i\}.}$$

Proof. In the minimal-return representative, all returns occur among the forced prefix components. Apply Lemma 70.7 and Lemma 71.12. □

Lemma 71.14 — Forced separator and free suffix.

For a minimal-return representative Q,

$$x_{t+1} \ge h_s\kappa + 1 = |s| + 1$$

when $t < d$, and

$$x_{t+j} \le \beta_{t+j} - 1 \qquad (1 \le j \le d - t).$$

Proof. Because $\widetilde{\mathrm{run}}(Q) = h_s$, the sampled column $h_s\kappa$ is missing, so the next component after the prefix starts at least one column later. Since $\mathrm{ret}(Q) = r_{\alpha,s}$ and the prefix already accounts for all returns, no suffix component is a return; Lemma 70.7 gives the strict upper bound. □

Put $p = d - t$ and define

$$c_j^{(\alpha,s)} = \beta_{t+j} - |s| - j - 1, \qquad 1 \le j \le p. \tag{71.11}$$

Then $c_1^{(\alpha,s)} \le \cdots \le c_p^{(\alpha,s)}$ because $\beta_{i+1} - \beta_i \ge 1$.

Theorem 71.15 — Explicit fixed-ratio-signature cardinality.

Let s be admissible and use the notation above. Then

$$\boxed{\begin{aligned} C_{m,\alpha,s} &:= |\{P \in \mathcal{D}_{m,\alpha,s} : \widetilde{\operatorname{run}}(P) = h_s,\ \operatorname{ret}(P) = r_{\alpha,s}\}| \\ &= \det_{1\le i,j\le d-t}\left[\binom{c_i^{(\alpha,s)}+1}{j-i+1}\right] \end{aligned}} \tag{71.12}$$

The empty determinant is 1.

Proof. The signature fixes the prefix columns by Lemma 71.12. For the free suffix set

$$v_j = x_{t+j} - (|s| + j).$$

Lemma 71.14 gives

$$0 \le v_1 \le \cdots \le v_p, \qquad v_j \le c_j^{(\alpha,s)}.$$

The barrier is nondecreasing, so the classical bounded-sequence determinant gives (71.12). □

Corollary 71.16 — Common cardinality on the fixed-signature skew diagonal.

For every $0 \le j \le h_s - r_{\alpha,s}$,

$$\boxed{|\{P \in \mathcal{D}_{m,\alpha,s} : \widetilde{\operatorname{run}}(P) = r_{\alpha,s} + j,\ \operatorname{ret}(P) = h_s - j\}| = C_{m,\alpha,s}.} \tag{71.13}$$

All other fixed-signature statistic classes are empty.

Proof. Apply the fixed-signature skew-diagonal bijection of Dai–Fu–Qiu Theorem 3.7 to the minimal-return class, then use Corollary 71.11 to exhaust the possible pairs. □

Corollary 71.17 — Explicit fixed-signature polynomial.

$$\boxed{\sum_{P\in\mathcal{D}_{m,\alpha,s}} u^{\widetilde{\operatorname{run}}(P)} v^{\operatorname{ret}(P)} = C_{m,\alpha,s}(uv)^{r_{\alpha,s}} h_{h_s - r_{\alpha,s}}(u,v).} \tag{71.14}$$

Example 71.E4 — The fixed signature 11111110**.**

Let

$$(m,n) = (23,11), \qquad \alpha = (1,1,2,1,1,1,1,1,1,1), \qquad s = 11111110.$$

Then $\kappa = 2$, $|s| = 8$, $h_s = 4$, and $t = 7$. The component barrier is

$$\beta = (0,2,4,8,10,12,14,16,18,20),$$

and the signature forces $(x_1,\ldots,x_7) = (0,1,2,3,4,5,6)$. Hence $r_{\alpha,s} = 1$. The free suffix barrier is $(6,7,8)$, so

$$C_{23,\alpha,s} = \det\begin{pmatrix} 7 & 21 & 35 \\ 1 & 8 & 28 \\ 0 & 1 & 9 \end{pmatrix} = 154.$$

Therefore each of the four classes

$$(\widetilde{\operatorname{run}}, \operatorname{ret}) = (1,4),(2,3),(3,2),(4,1)$$

has cardinality 154, and the fixed-signature polynomial is $154\, uv\, h_3(u,v)$.

Chapter 72

Arbitrary Boundaries and Labeled Applications

The framework is not tied to one rational slope. Two application directions from the second paper are retained here in full. First, the contact determinant works below an arbitrary northeast boundary and therefore specializes to partial Dyck paths. Second, fixed-composition path counts transfer to column-increasing rational parking-function labelings by the standard multinomial labeling factor. These applications are mathematically separate from the four core rational-Dyck specializations and are kept visible as such.

72.1 Paths below an arbitrary fixed boundary

Let B be a northeast lattice path whose north-step starting columns are

$$\mathbf{b} = (b_1, \dots, b_n), \qquad b_1 \le \cdots \le b_n.$$

A path P lying weakly northwest of B has a north-step column sequence $u(P) \in \mathcal{A}_{\mathbf{b}}$. Thus Section 70 applies verbatim.

Corollary 72.1 — Arbitrary-boundary presence and exact contacts.
Let $R, F \subset \mathbb{Z}_{>0}$ be finite and disjoint. For $J \subseteq [n]$,

$$\boxed{\begin{aligned}&\#\{P \le B : R \subseteq \mathrm{Val}(P),\ \mathrm{Val}(P) \cap F = \varnothing,\ C_B(P) = J\}\\ &\qquad = \sum_{T \subseteq R} (-1)^{|T|} \det \mathsf{M}_J^{\mathbf{b}, F \cup T}\end{aligned}}$$

72.1.1 Partial Dyck paths

Let $0 \le m \le n$ and let $\mathcal{B}_{m,n}$ be the paths from $(0,0)$ to (m,n) remaining weakly above $y = x$. Their north-step barrier is

$$\boxed{b_i = \min(i-1, m).}$$

For $i \le m+1$, contact $u_i = b_i = i-1$ is a diagonal contact; for $i > m+1$, contact $u_i = b_i = m$ is contact with the terminal vertical wall. The multivariate variables therefore separate two geometrically different kinds of boundary contact without changing the compression argument.

Define

$$\mathrm{Ret}_\Delta(P) = \{i \in [m+1] : u_i = i-1\}, \qquad \mathrm{ret}_\Delta(P) = |\,\mathrm{Ret}_\Delta(P)|.$$

Specializing $z_i = z$ for $i \le m+1$ and $z_i = 1$ afterward gives a polynomial $D^{\mathrm{par}}_{m,n;F}(z)$ satisfying

$$D^{\mathrm{par}}_{m,n;F}(z) = \sum_{\substack{P\in\mathcal{B}_{m,n}\\ \mathrm{Val}(P)\cap F=\varnothing}} z^{\mathrm{ret}_\Delta(P)}.$$

If

$$\mathrm{fm}(P) = \min\{a \ge 1 : a \notin \mathrm{Val}(P)\},$$

then $\mathrm{fm}(P) = a$ is the required/forbidden condition $R_a = \{1,\dots,a-1\}$ and $F_a = \{a\}$. Hence the framework gives both the return-number polynomial and exact diagonal-return-set determinant sums for partial Dyck paths.

72.2 Column-increasing rational parking-function labelings

Classical rational parking functions are commonly represented as rational Dyck paths whose north steps carry labels increasing within each column; see Armstrong, Loehr, and Warrington [54]. For the present application, we use the same column-increasing labeling convention for the $m \times n$ paths considered here, without requiring coprimality. The path statistics run, $\widetilde{\mathrm{run}}$, and ret are inherited from the underlying path.

Lemma 72.2 — Labelings of a fixed composition.

If $P \in \mathcal{D}_{m,\alpha}$, then the number of column-increasing labelings of its n north steps by $1,\dots,n$ is

$$\boxed{L(\alpha) = \frac{n!}{\alpha_1!\cdots\alpha_d!}.}$$

Proof. Choose which α_i labels occupy each vertical component. Once those sets are chosen, the increasing condition determines their within-column order uniquely. ☐

Define the fixed-composition labeled run–return polynomial

$$\mathcal{P}_{m,\alpha}(u,v) = \sum_{\pi:\,\mathrm{comp}(\underline{\pi})=\alpha} u^{\mathrm{run}(\pi)} v^{\mathrm{ret}(\pi)},$$

where $\underline{\pi}$ is the underlying path.

Theorem 72.3 — Composition-refined labeled run–return expansion.

$$\boxed{\mathcal{P}_{m,\alpha}(u,v) = \frac{n!}{\alpha_1!\cdots\alpha_d!}(uv)^{r_\alpha}\sum_{K=r_\alpha}^{d} C_{\alpha,K}h_{K-r_\alpha}(u,v).} \tag{72.1}$$

Proof. Every underlying path of composition α has the same multiplicity $L(\alpha)$ by Lemma 72.2. Multiply Corollary 71.3 by this factor. ☐

Summing over all compositions gives an explicit labeled polynomial

$$\boxed{\mathcal{P}_{m,n}(u,v) = \sum_{\alpha\vDash n}\frac{n!}{\prod_i \alpha_i!}(uv)^{r_\alpha}\sum_{K=r_\alpha}^{\ell(\alpha)} C_{\alpha,K}h_{K-r_\alpha}(u,v).} \tag{72.2}$$

For ratio-run, let $\Sigma_{m,\alpha}$ be the admissible first ratio-signatures. Corollary 71.17 similarly gives

$$\boxed{\widetilde{\mathcal{P}}_{m,\alpha}(u,v) = \frac{n!}{\prod_i \alpha_i!}\sum_{s\in\Sigma_{m,\alpha}} C_{m,\alpha,s}(uv)^{r_{\alpha,s}}h_{h_s-r_{\alpha,s}}(u,v).} \tag{72.3}$$

Thus both labeled distributions have manifest positive complete-homogeneous expansions once the determinant coefficients are fixed.

Remark 72.R1 — Labeling scope.

No symmetric-function coefficient claim is needed here. The application uses only the standard column-increasing labeling rule and the fact that the statistics under consideration depend on the underlying path.

72.3 Computational validation

All results of the paper are established theoretically. As an independent check, we implemented exhaustive enumeration of the underlying bounded sequences and fixed-composition rational Dyck paths and compared the cardinalities with the determinant formulas.

For the general exact-contact theorem, every nondecreasing barrier of length at most 5 with entries in $\{0,1,2,3,4\}$ was tested, together with every forbidden subset of $\{1,2,3,4\}$ and every exact contact set. This produced 88,032 determinant–enumeration comparisons.

For the rational-path specializations, the ordinary fixed-composition formula was tested on all $m > n$ with $n \leq 7$ and $m \leq 12$, every composition of n, and every tested run–return class: 755 parameter instances and 9,983 statistic queries. The ratio-run formula with exact return components was tested on 628 parameter instances and 54,664 exact-return queries. The simultaneous run–ratio-run formula was tested on 501 parameter instances and 246,392 simultaneous queries. No discrepancy was found.

The fixed-ratio-signature construction was tested independently by forming sign_1 directly from the selected κ-column blocks in the Dai–Fu–Qiu definition, rather than by using the determinant reduction. The test covered 2,357 fixed-signature classes over 12,081 underlying paths, with no mismatch.

As a larger stress test, for

$$(m,n) = (23,11), \qquad \alpha = (1,1,2,1,1,1,1,1,1,1),$$

all 63,920 fixed-composition paths were enumerated. Exactly 616 have first ratio-signature 11111110, split into the four classes

$$(1,4),\ (2,3),\ (3,2),\ (4,1)$$

with exactly 154 paths in each, agreeing with Example 71.E4. No mismatch occurred in any exhaustive test.

72.4 Conclusion

We developed a contact-preserving barrier-compression framework for refined lattice-path enumeration. The method separates value constraints, upper-bound geometry, and boundary contacts. Deleting forbidden values produces an order-preserving compression of the alphabet and the barrier, while a multivariate Lindström–Gessel–Viennot determinant retains the surviving contact positions. Inclusion–exclusion then incorporates arbitrary required values and simultaneous first-missing statistics.

The framework arose from a concrete arithmetic route. A completion-aware deterministic pushdown automaton for the 2:1 balance relation led to the scalar residual $\rho_{p,q} = qN_1 - pN_0$. For $m = gq$ and $n = gp$, the identity $R = g\rho_{p,q}$ converted the low-residual condition into the rational-path return condition, which in coarea coordinates became an upper-barrier contact. The present paper isolates the resulting compression mechanism from that discovery route and develops it as a self-contained enumerative method.

For rational Dyck paths with fixed ordered vertical-run composition, the method gives explicit formulas at several levels of refinement. Ordinary run–return classes reduce to one determinant. Ratio-run is handled by sampled presence–absence constraints and exact return-component determinants. Ordinary run and ratio-run can be prescribed simultaneously with exact return positions. At fixed ratio-signature, the minimal-return representative converts the signature data into a forced prefix and a bounded free suffix, giving a single determinant for the common cardinality on the associated skew diagonal. The same framework applies to arbitrary northeast boundaries and partial Dyck paths, while standard column-increasing labeling transfers the fixed-composition formulas to labeled rational parking-function enumerations.

All determinant identities are proved combinatorially, and independent exhaustive computation verified the general framework and the four rational-path specializations over broad finite parameter ranges. The discovery-to-enumeration route can therefore be summarized as

$$\boxed{\begin{array}{c}\text{pushdown residual} \longrightarrow \text{rank} \longrightarrow \text{boundary contact}\\ \longrightarrow \text{contact-preserving compression} \longrightarrow \text{explicit determinants}\end{array}}$$

Part XVI

Farey Arithmetic of Shortest Completion

Research provenance. This Part develops the arithmetic of residual-indexed shortest-completion minimizers, the cross-parameter Ordered Farey Recursion, and the universal completion spectra. Farey/Stern–Brocot, Beatty, Christoffel/mechanical, digital-line, and Dedekind structures are classical neighboring theories [59, 60, 64, 61, 65, 62, 63]; novelty is not claimed for those objects in isolation.

Chapter 73

Canonical Completion Profiles and Farey Unit Lifts

The rational-Dyck branch used the residual as a bridge to enumeration. A different post-theory branch returns instead to the shortest-completion map itself. For coprime positive integers p, q and residual r, define

$$\mu_{p,q}(r) = \arg\min_{x,y \in \mathbb{Z}_{\geq 0}} \{x + y : px - qy = r\}.$$

This is the same minimum-demand semantics developed earlier in the book, now studied as an ordered arithmetic profile in r and across different rational slopes.

The operational provenance remains the original completion-aware DPDA. For example, the prefix 111 has $\rho_{2,1} = 3$, and the machine materializes the completion word 100; its Parikh vector $(2, 1)$ is exactly $\mu_{2,1}(3)$. The automaton is not redrawn here because Chapters 1–8 already preserve it in detail. From this point onward the Farey results are self-contained statements about the minimizer map.

Novelty boundary. The local arithmetic has strong classical shadows. We do not claim novelty for Diophantine-strip reduction, Bézout/Farey unit steps, modular representatives or inverses, mechanical/Christoffel coding, standard Christoffel factorization, digital-line Stern–Brocot updates, Beatty partitions, or Dedekind reciprocity in isolation. Indeed, after a unimodular change of coordinates the local completion walk is a classical rational mechanical/Christoffel path. The principal completion-specific statement appears in Chapter 74: the *cross-parameter Ordered Farey Recursion*, which compares the canonical fixed-residual ℓ^1 minimizer sections of a mediant and both Farey parents and identifies two interleaved child residual subsequences with an unshifted parent profile and a fixed translate of the other. This novelty boundary is a cautious literature assessment, not an absolute priority guarantee.

73.1 Shortest-completion profiles and canonical strips

Fix coprime positive integers p, q. Define

$$W_{p,q}(x, y) = px - qy.$$

The primitive positive kernel vector of $W_{p,q}$ is

$$K_{p,q} = (q, p).$$

Thus any two integer solutions of $px - qy = r$ differ by an integral multiple of (q, p).

Definition 73.1 — Shortest completion.

For $r \in \mathbb{Z}$, define

$$\mu_{p,q}(r) = (x_r, y_r) = \operatorname*{arg\,min}_{x,y \in \mathbb{Z}_{\geq 0}} \{x + y : px - qy = r\},$$

and set

$$L_{p,q}(r) = x_r + y_r.$$

When p, q are fixed, we abbreviate these as $\mu(r)$ and $L(r)$.

Lemma 73.2 — Canonical-strip characterization.

For $r \geq 0$, $\mu(r)$ is the unique nonnegative solution of

$$px - qy = r$$

satisfying

$$0 \leq y < p.$$

For $r \leq 0$, $\mu(r)$ is the unique nonnegative solution satisfying

$$0 \leq x < q.$$

Proof. Suppose $r \geq 0$. Since $\gcd(p, q) = 1$, the congruence

$$qy \equiv -r \pmod{p}$$

has a unique solution $y_0 \in \{0, \ldots, p-1\}$. Put

$$x_0 = \frac{r + qy_0}{p} \geq 0.$$

Every integer solution is

$$(x_0, y_0) + k(q, p), \qquad k \in \mathbb{Z}.$$

Because $0 \leq y_0 < p$, nonnegativity forces $k \geq 0$. Every other nonnegative solution therefore has length

$$x_0 + y_0 + k(p + q),$$

which is strictly larger for $k > 0$. Hence (x_0, y_0) is the unique minimizer. The case $r \leq 0$ is dual, using the unique representative $x_0 \in \{0, \ldots, q-1\}$. □

Corollary 73.3 — Kernel reduction.

If (x, y) is any nonnegative solution of residual r, then

$$(x, y) = \mu(r) + k(q, p)$$

for a unique $k \geq 0$, and

$$x + y = L(r) + k(p + q).$$

Proposition 73.4 — Boundary families.

The cases $p = 1$ or $q = 1$ are explicit. If $p = 1$, then

$$\mu_{1,q}(0) = (0, 0), \qquad \mu_{1,q}(1) = (1, 0),$$

and for $1 \leq s \leq q$,

$$\mu_{1,q}(-s) = (q - s, 1).$$

If $q = 1$, then

$$\mu_{p,1}(0) = (0, 0), \qquad \mu_{p,1}(-1) = (0, 1),$$

and for $1 \leq r \leq p$,

$$\mu_{p,1}(r) = (1, p - r).$$

Proof. Each displayed pair solves the relevant equation and lies in the canonical strip of Lemma 73.2. □

73.2 Canonical Farey unit-lift dynamics

Assume throughout this section that $p, q > 1$. Let

$$(p, q) = (c, u) + (v, d) \tag{73.1}$$

be the oriented Farey-parent decomposition characterized by

$$pu - qc = 1, \qquad qv - pd = 1. \tag{73.2}$$

It follows that

$$uv - cd = 1. \tag{73.3}$$

The existence and uniqueness of this oriented parent pair are classical Farey/Stern–Brocot facts; see, for example, [59].

Theorem 73.5 — Canonical Farey Unit-Lift Theorem.

Define

$$a = (u, c), \qquad b = (-d, -v).$$

Then

$$W_{p,q}(a) = W_{p,q}(b) = 1$$

and

$$a - b = (q, p) = K_{p,q}.$$

If

$$\mu(r) = (x_r, y_r), \qquad 0 \le r < p,$$

then

$$\mu(r+1) = \begin{cases} \mu(r) + (u, c), & 0 \le y_r < v, \\ \mu(r) + (-d, -v), & v \le y_r < p. \end{cases} \tag{73.4}$$

Proof. By equation (73.2),

$$W_{p,q}(u, c) = pu - qc = 1$$

and

$$W_{p,q}(-d, -v) = -pd + qv = 1.$$

Moreover,

$$(u, c) - (-d, -v) = (u + d, c + v) = (q, p).$$

Thus the two vectors are kernel-equivalent lifts of residual increment $+1$.

If $y_r < v$, then

$$0 \le y_r + c < v + c = p,$$

so $\mu(r) + (u, c)$ has residual $r + 1$ and remains in the positive canonical strip. By Lemma 73.2, it is $\mu(r + 1)$.

If $y_r \ge v$, then adding (u, c) leaves the strip because $y_r + c \ge p$. Subtracting one kernel gives

$$(u, c) - (q, p) = (-d, -v),$$

and the new second coordinate is $y_r - v \in [0, p - 1]$. Hence $\mu(r) + (-d, -v)$ is the unique canonical representative of residual $r + 1$. ☐

Remark 73.R1 — Canonical unit lifting.

The mechanism in Theorem 73.5 is intrinsic to shortest completion:

$$\text{unit residual lift} \quad + \quad \text{canonical kernel reduction.}$$

The Farey parent vectors appear as the two kernel-equivalent lifts available to this reduction.

Theorem 73.6 — Negative unit-lift theorem.

Define

$$a^- = (d, v), \qquad b^- = (-u, -c).$$

Then

$$W_{p,q}(a^-) = W_{p,q}(b^-) = -1, \qquad a^- - b^- = (q, p).$$

If

$$\mu(-s) = (x_s, y_s), \qquad 0 \le s < q,$$

then

$$\mu(-(s+1)) = \begin{cases} \mu(-s) + (d, v), & 0 \le x_s < u, \\ \mu(-s) + (-u, -c), & u \le x_s < q. \end{cases} \tag{73.5}$$

Proof. The two increments have residual -1 because

$$pd - qv = -1, \qquad -pu + qc = -1,$$

and their difference is (q, p). If $x_s < u$, adding (d, v) keeps the first coordinate below $u + d = q$. If $x_s \ge u$, this lift leaves the negative canonical strip, and subtracting the kernel replaces (d, v) by $(-u, -c)$. The conclusion follows from Lemma 73.2. □

Corollary 73.7 — Closed vector profiles.

For $0 \le r \le p$,

$$\mu_{p,q}(r) = r(u, c) - \left\lfloor \frac{cr}{p} \right\rfloor (q, p). \tag{73.6}$$

For $0 \le s \le q$,

$$\mu_{p,q}(-s) = s(d, v) - \left\lfloor \frac{ds}{q} \right\rfloor (q, p). \tag{73.7}$$

In particular,

$$\mu(p) = (1, 0), \qquad \mu(-q) = (0, 1).$$

Proof. In the positive walk, the kernel correction occurs exactly when the second coordinate wraps under addition of c modulo p. The number of wraps among the first r steps is $\lfloor cr/p \rfloor$, giving equation (73.6). The negative formula is identical with d modulo q. At $r = p$ and $s = q$, use $pu - qc = 1$ and $qv - pd = 1$. □

Example 73.E1 — The pair 7:4.

For $(p, q) = (7, 4)$ the oriented parents are $(c, u) = (5, 3)$ and $(v, d) = (2, 1)$. Hence the two positive unit lifts are $(3, 5)$ and $(-1, -2)$. The positive fundamental profile is

$$(0, 0), (3, 5), (2, 3), (1, 1), (4, 6), (3, 4), (2, 2), (1, 0),$$

whose successive increments are

$$(3, 5), (-1, -2), (-1, -2), (3, 5), (-1, -2), (-1, -2), (-1, -2).$$

The ordered recursion places the left-parent profile at indices

$$\left\lceil \frac{7r}{5} \right\rceil \in \{0, 2, 3, 5, 6, 7\}$$

and the translated right-parent block at indices

$$\left\lfloor \frac{7s}{2} \right\rfloor + 1 \in \{1, 4\}.$$

Thus the local two-jump law and the global floor/ceiling shuffle are visible in the same small example.

Chapter 74

Ordered Farey Recursion and the Completion Tree

The unit-lift laws identify the two possible local jumps. The central result of the Farey paper is stronger and cross-parametric: it reconstructs the entire mediant profile from the two parent profiles at complementary interleaved residual positions. Parent recovery and the recursively decorated Stern–Brocot family are retained as mechanisms and consequences of that theorem rather than promoted to separate novelty claims.

74.1 Local Farey recovery and ordered parent recursion

74.1.1 Local encoding and parent recovery

Theorem 74.1 — Vector jump law.

The positive fundamental profile has exactly two local increments,

$$\mu(r+1) - \mu(r) \in \{(u, c), (-d, -v)\}, \qquad 0 \le r < p.$$

The jump (u, c) occurs v times and the jump $(-d, -v)$ occurs c times.

The negative fundamental profile has exactly the two increments

$$(d, v), \qquad (-u, -c),$$

with multiplicities u and d, respectively.

Proof. The two-value statements are Theorems 73.5 and 73.6. In the positive profile, the number of kernel corrections is

$$\sum_{r=0}^{p-1} \left(\left\lfloor \frac{c(r+1)}{p} \right\rfloor - \left\lfloor \frac{cr}{p} \right\rfloor \right) = c.$$

Since there are $p = c + v$ jumps in total, the other jump occurs v times. The negative case is identical. □

Corollary 74.2 — Parent recovery.

For $p, q > 1$, the ordered positive completion profile uniquely determines the oriented Farey parents. If its two local increments are

$$(a, b), \qquad (-e, -f),$$

then

$$(c, u) = (b, a), \qquad (v, d) = (f, e),$$

and consequently

$$p = b + f, \qquad q = a + e.$$

Proof. By Theorem 74.1, $(a, b) = (u, c)$ and $(-e, -f) = (-d, -v)$. □

74.1.2 Ordered parent-to-mediant recursion

Write

$$\mathcal{M}^+_{p,q} = (\mu_{p,q}(0), \mu_{p,q}(1), \ldots, \mu_{p,q}(p))$$

and

$$\mathcal{M}^-_{p,q} = (\mu_{p,q}(0), \mu_{p,q}(-1), \ldots, \mu_{p,q}(-q)).$$

Theorem 74.3 — Cross-parameter Ordered Farey Recursion.

Let $(p, q) = (c, u) + (v, d)$ be as in equations (73.1) and (73.2). For $0 \le r \le c$,

$$\mu_{p,q}\Big(\Big\lceil \frac{pr}{c} \Big\rceil\Big) = \mu_{c,u}(r). \tag{74.1}$$

For $0 \le s < v$,

$$\mu_{p,q}\Big(\Big\lfloor \frac{ps}{v} \Big\rfloor + 1\Big) = (u, c) + \mu_{v,d}(s). \tag{74.2}$$

Dually, for $0 \le r \le u$,

$$\mu_{p,q}\Big(-\Big\lfloor \frac{qr}{u} \Big\rfloor\Big) = \mu_{c,u}(-r), \tag{74.3}$$

and for $1 \le s \le d$,

$$\mu_{p,q}\Big(-\Big(\Big\lceil \frac{qs}{d} \Big\rceil - 1\Big)\Big) = (u, c) + \mu_{v,d}(-s). \tag{74.4}$$

Moreover, the positive index families

$$I_L^+ = \Big\{\Big\lceil \frac{pr}{c} \Big\rceil : 0 \le r \le c\Big\}, \qquad I_R^+ = \Big\{\Big\lfloor \frac{ps}{v} \Big\rfloor + 1 : 0 \le s < v\Big\}$$

form a disjoint partition of $\{0, 1, \ldots, p\}$. The corresponding negative index families in equations (74.3) and (74.4) form a disjoint partition of $\{0, 1, \ldots, q\}$.

Proof. For equation (74.1), let $\mu_{c,u}(r) = (x, y)$. Then

$$cx - uy = r, \qquad 0 \le y < c.$$

Set $R = px - qy$. Since $pu = qc + 1$,

$$\frac{pr}{c} = px - \frac{pu}{c}y = px - qy - \frac{y}{c} = R - \frac{y}{c}.$$

As $0 \le y/c < 1$, we get

$$R = \Big\lceil \frac{pr}{c} \Big\rceil.$$

Since $y < c < p$, the same vector lies in the child canonical strip, proving equation (74.1).

For equation (74.2), let $\mu_{v,d}(s) = (x, y)$, so

$$vx - dy = s, \qquad 0 \le y < v.$$

Since $pd = qv - 1$,

$$\frac{ps}{v} = px - \frac{pd}{v}y = px - qy + \frac{y}{v},$$

and therefore

$$px - qy = \left\lfloor \frac{ps}{v} \right\rfloor .$$

Adding (u, c) raises the child residual by one. Moreover $0 \le c + y < c + v = p$, so the translated vector is canonical. This proves equation (74.2).

For equation (74.3), let $\mu_{c,u}(-r) = (x, y)$, so $cx - uy = -r$ and $0 \le x < u$. Then

$$\frac{qr}{u} = qy - \frac{qc}{u}x = qy - px + \frac{x}{u} = -(px - qy) + \frac{x}{u}.$$

Hence $px - qy = -\lfloor qr/u \rfloor$, and the same vector lies in the child negative strip because $x < u < q$.

For equation (74.4), let $\mu_{v,d}(-s) = (x, y)$, so $vx - dy = -s$ and $0 \le x < d$. Using $qv = pd + 1$,

$$\frac{qs}{d} = qy - \frac{qv}{d}x = qy - px - \frac{x}{d} = -(px - qy) - \frac{x}{d}.$$

Thus $\lceil qs/d \rceil = -(px - qy)$. Adding (u, c) raises the residual by one and keeps the first coordinate below $u + d = q$, which gives equation (74.4).

It remains to verify the two index partitions. We give the positive one directly. Both index lists are strictly increasing. For $1 \le n < p$, the number of left indices at most n is

$$\left\lfloor \frac{cn}{p} \right\rfloor + 1,$$

while the number of right indices at most n is

$$\left\lceil \frac{vn}{p} \right\rceil = n - \left\lfloor \frac{cn}{p} \right\rfloor ,$$

because $v = p - c$ and cn/p is nonintegral. Their sum is $n + 1$, exactly the number of positions in $\{0, \dots, n\}$. The endpoint counts at $n = 0, p$ are immediate, so every position $0, \dots, p$ occurs exactly once. The negative partition is identical with $q = u + d$. These index sets are the familiar finite rational Beatty partition; the vector labels supplied by equations (74.1) to (74.4) are the completion-specific content. □

Remark 74.R1 — Vector-valued Beatty shuffle.

Theorem 74.3 says more than set inclusion. The entire residual order is reconstructed: the left parent populates the positions I_L^+ unchanged, while the translated right parent populates the complementary positions I_R^+. The same holds on the negative side. The endpoint $\mu_{v,d}(v) = (1, 0)$ is omitted from equation (74.2) because the child endpoint $\mu_{p,q}(p) = (1, 0)$ is already inherited through the left parent.

Corollary 74.4 — Local–global Farey correspondence.

For a proper pair $p, q > 1$, the oriented Farey parents determine both the local jump law and the global ordered recursion. Conversely, the local positive jump set recovers the oriented parents and hence determines the ordered parent recursion.

Proof. Forward determination is Theorems 73.5 and 74.3; reverse determination is Corollary 74.2. □

74.2 The global Farey completion family

The preceding results concern a single Farey mediant. Iteration over the Stern–Brocot tree produces one recursively coherent completion family.

Definition 74.5 — Farey completion tree.

Decorate every coprime node q/p of the Stern–Brocot tree by the pair of ordered profiles

$$\mathfrak{C}(p,q) = \left(\mathcal{M}^{+}_{p,q}, \mathcal{M}^{-}_{p,q}\right).$$

We call the resulting decorated tree the *Farey completion tree.*

Theorem 74.6 — Recursive coherence.

Whenever

$$(p,q) = (c,u) + (v,d)$$

is a Farey mediant, both ordered profiles in $\mathfrak{C}(p,q)$ are determined uniquely from the two parent profiles by Theorem 74.3. Consequently the family

$$\{\mathfrak{C}(p,q) : p,q \geq 1,\ \gcd(p,q) = 1\}$$

is recursively coherent under Farey mediants.

Proof. Apply Theorem 74.3 at every proper interior node and stop when $p = 1$ or $q = 1$, where Proposition 73.4 supplies the explicit boundary profiles. The two index families partition every proper child profile, so the recursion determines every child entry exactly once. □

Remark 74.R2 — Classical tree, new decoration.

The Stern–Brocot tree and its continued-fraction complexity are classical. The point of Definition 74.5 and Theorem 74.6 is not a new rational tree, but a new vector-valued decoration: the shortest-completion profiles form a single parent-generated system rather than an unrelated collection indexed by $p : q$.

Chapter 75

Universal Completion Spectra and Arithmetic Interfaces

The vector profile remembers the ratio $p{:}q$, but scalar projection reveals a universal phenomenon controlled only by $m = p + q$. This chapter retains the fundamental and shell spectra, their moment consequences, the one-sided reflection law, and the precise interfaces with Beatty, mechanical/Christoffel, digital-line, and Dedekind arithmetic. These classical interfaces are included because they explain what the completion theory projects to; they are not used to overstate the novelty of the local coding.

75.1 Scalar projection and universal spectra

For the Farey-coordinate formulas in the first part of this section assume $p, q > 1$; the boundary families are given by Proposition 73.4. Set

$$m = p + q, \qquad A = c + u, \qquad B = v + d.$$

Then $A + B = m$. From equation (73.2),

$$pA - mc = 1, \qquad qB - md = 1. \tag{75.1}$$

Thus $A \equiv p^{-1} \pmod{m}$ and $B \equiv q^{-1} \pmod{m}$.

Proposition 75.1 — Scalar Farey profiles.

For $0 \le r \le p$,

$$L_{p,q}(r) = Ar - m \left\lfloor \frac{cr}{p} \right\rfloor. \tag{75.2}$$

For $0 \le s \le q$,

$$L_{p,q}(-s) = Bs - m \left\lfloor \frac{ds}{q} \right\rfloor. \tag{75.3}$$

Proof. Take coordinate sums in equations (73.6) and (73.7). □

Corollary 75.2 — Scalar jump law and affine periodicity.

For $0 \le r < p$,

$$L(r+1) - L(r) \in \{A, -B\},$$

with multiplicities v and c. Dually,

$$L(-(s+1)) - L(-s) \in \{B, -A\},$$

with multiplicities u and d. Moreover,

$$L(r+p) = L(r) + 1 \qquad (r \geq 0) \tag{75.4}$$

and

$$L(-(s+q)) = L(-s) + 1 \qquad (s \geq 0). \tag{75.5}$$

Proof. The jump statements are the ℓ^1 projections of Theorem 74.1. Their net positive drift is

$$vA - cB = v(c+u) - c(v+d) = uv - cd = 1,$$

and the negative drift is $uB - dA = 1$. Alternatively, equations (75.4) and (75.5) follow directly from the canonical strips by adding $(1,0)$ or $(0,1)$. $\square$

75.1.1 Fundamental spectra

Define

$$S_+(p,q) = \{L(r) : 0 \leq r < p\}, \qquad S_-(p,q) = \{L(-s) : 0 \leq s < q\}.$$

Proposition 75.3 — Completion–Beatty identification.

As sets,

$$S_+(p,q) = \left\{ \left\lceil \frac{mj}{p} \right\rceil : 0 \leq j < p \right\}, \tag{75.6}$$

and

$$S_-(p,q) = \left\{ \left\lceil \frac{mj}{q} \right\rceil : 0 \leq j < q \right\}. \tag{75.7}$$

Proof. For $0 \leq r < p$, the positive canonical coordinate y_r runs through every residue $j \in \{0, \ldots, p-1\}$ exactly once. For fixed j, the least nonnegative first coordinate satisfying $px - qj \geq 0$ is $\lceil qj/p \rceil$, and therefore the corresponding completion length is

$$j + \left\lceil \frac{qj}{p} \right\rceil = \left\lceil \frac{(p+q)j}{p} \right\rceil.$$

The negative case is dual. $\square$

Theorem 75.4 — Universal Fundamental Spectrum.

For every coprime pair p, q with $m = p + q$,

$$S_+(p,q) \uplus S_-(p,q) = \{0, 0, 2, 3, \ldots, m-1\} \tag{75.8}$$

as multisets.

Proof. By Proposition 75.3, it is enough to show that the two nonzero ceiling sets partition $\{2, \ldots, m-1\}$. For $1 \leq \ell \leq m$, define

$$\chi_p(\ell) = \left\lfloor \frac{p\ell}{m} \right\rfloor - \left\lfloor \frac{p(\ell-1)}{m} \right\rfloor, \qquad \chi_q(\ell) = \left\lfloor \frac{q\ell}{m} \right\rfloor - \left\lfloor \frac{q(\ell-1)}{m} \right\rfloor.$$

Each quantity is 0 or 1. The condition $\chi_p(\ell) = 1$ is equivalent to $\ell = \lceil mj/p \rceil$ for some $1 \leq j \leq p-1$ when $2 \leq \ell \leq m-1$, and similarly for q. Since $q = m-p$ and $\gcd(p,m) = 1$, for every $2 \leq \ell \leq m-1$ neither $p\ell/m$ nor $p(\ell-1)/m$ is integral; using $\lfloor \ell - x \rfloor = \ell - \lceil x \rceil$ gives

$$\chi_p(\ell) + \chi_q(\ell) = 1.$$

Thus every integer $2, \ldots, m-1$ belongs to exactly one of the two nonzero spectra. At $\ell = 1$ neither occurs, while both spectra contain 0. This proves equation (75.8). The resulting partition is the finite rational Beatty complementarity seen in Proposition 75.3. $\square$

75.1.2 Universal shell spectra

For $k \geq 0$, define

$$\Sigma_k^+ = \{L(kp + r) : 0 \leq r < p\}, \qquad \Sigma_k^- = \{L(-kq - s) : 0 \leq s < q\}.$$

Theorem 75.5 — Universal Shell Spectrum.
For every $k \geq 0$,

$$\Sigma_k^+ \uplus \Sigma_k^- = \{k, k, k + 2, k + 3, \dots, k + m - 1\}. \tag{75.9}$$

Proof. By equations (75.4) and (75.5),

$$L(kp + r) = k + L(r), \qquad L(-kq - s) = k + L(-s).$$

Translate Theorem 75.4 by k. □

Corollary 75.6 — Shell generating polynomial.
If

$$C_k(z) = \sum_{r=0}^{p-1} z^{L(kp+r)} + \sum_{s=0}^{q-1} z^{L(-kq-s)},$$

then

$$C_k(z) = z^k\big(2 + z^2 + z^3 + \cdots + z^{m-1}\big),$$

and hence

$$\sum_{k\geq 0} C_k(z)t^k = \frac{2 + z^2 + z^3 + \cdots + z^{m-1}}{1 - tz}.$$

Corollary 75.7 — Universal moment hierarchy.
For every integer $j \geq 1$,

$$M_j(k) := \sum_{\ell\in\Sigma_k^+\uplus\Sigma_k^-} \ell^j = 2k^j + \sum_{\ell=2}^{m-1} (k + \ell)^j.$$

In particular,

$$M_j(0) = \sum_{\ell=2}^{m-1} \ell^j.$$

Thus all combined shell power moments depend only on $m = p + q$.

75.1.3 Reflection symmetry

Write the nonzero positive spectrum in increasing order as

$$\ell_j^+ = \left\lceil \frac{mj}{p} \right\rceil, \qquad 1 \leq j < p,$$

and define ℓ_j^- analogously with q.

Proposition 75.8 — One-sided reflection.
For $1 \leq j < p$,

$$\ell_j^+ + \ell_{p-j}^+ = m + 1,$$

and for $1 \leq j < q$,

$$\ell_j^- + \ell_{q-j}^- = m + 1.$$

Consequently, with $h = (m + 1)/2$, every odd centered one-sided moment vanishes.

Proof. Because $\gcd(m,p)=1$, mj/p is nonintegral for $1 \le j < p$. Hence

$$\left\lceil \frac{m(p-j)}{p} \right\rceil = m - \left\lfloor \frac{mj}{p} \right\rfloor,$$

which sums with $\lceil mj/p \rceil$ to $m+1$. The negative identity is dual, and centered odd powers cancel in symmetric pairs. □

75.2 Classical arithmetic interfaces

The results through section 75.1 were obtained from shortest-completion minimization, canonical kernel reduction, and Farey unimodularity. We now record how classical arithmetic structures appear as projections. The purpose of this section is interpretive rather than foundational.

75.2.1 Mechanical and Christoffel coding

The positive wrap indicator is

$$\varepsilon_r = \left\lfloor \frac{c(r+1)}{p} \right\rfloor - \left\lfloor \frac{cr}{p} \right\rfloor \in \{0,1\}. \tag{75.10}$$

Then

$$\mu(r+1) - \mu(r) = (u,c) - \varepsilon_r(q,p).$$

Thus $\varepsilon_r = 0$ selects (u,c) and $\varepsilon_r = 1$ selects $(-d,-v)$. The finite binary sequence $(\varepsilon_r)_{r=0}^{p-1}$ is a rational lower mechanical word; under the standard finite conventions it is a Christoffel encoding, up to the usual choice of slope orientation and alphabet. Standard factorization, continued fractions, and the relation with the Stern–Brocot tree are classical; see [59].

The completion interpretation is that the mechanical/Christoffel coding records exactly when canonical kernel reduction occurs. This should not be read as a novelty claim for the coding itself.

75.2.2 Unimodular realization and the distinction from standard factorization

Let

$$k_r = \left\lfloor \frac{cr}{p} \right\rfloor \qquad (0 \le r \le p)$$

and define

$$J_+ = \begin{pmatrix} u & -d \\ c & -v \end{pmatrix}.$$

Since $uv - cd = 1$, one has $\det J_+ = -1$. Using $(q,p) = (u,c) - (-d,-v)$ in equation (73.6) gives

$$\mu_{p,q}(r) = (r-k_r)(u,c) + k_r(-d,-v) = J_+ \begin{pmatrix} r-k_r \\ k_r \end{pmatrix}. \tag{75.11}$$

Therefore

$$J_+^{-1}\mu_{p,q}(r) = \begin{pmatrix} r - \lfloor cr/p \rfloor \\ \lfloor cr/p \rfloor \end{pmatrix},$$

which is the vertex sequence of a rational mechanical lattice path with v steps of one type and c of the other. Thus the local completion walk is a unimodular image of classical Christoffel/mechanical geometry. This is consistent with digital-straight-line theory, where arithmetic remainders, minimal $a + b$ parameters, Bézout vectors and Stern–Brocot slope updates already occur together [61], and with classical Christoffel duality and modular-inverse phenomena [60].

It is important, however, not to identify Theorem 74.3 with standard Christoffel factorization. Standard factorization decomposes a proper Christoffel word into a *contiguous concatenation* of two Christoffel factors associated with the Farey parents [59]. By contrast, Theorem 74.3 compares three different canonical minimizer maps,

$$\mu_{p,q}, \qquad \mu_{c,u}, \qquad \mu_{v,d},$$

and partitions the child residual positions into two *interleaved* sets

$$I_L^+ = \left\{ \left\lceil \frac{pr}{c} \right\rceil : 0 \le r \le c \right\}, \qquad I_R^+ = \left\{ \left\lfloor \frac{ps}{v} \right\rfloor + 1 : 0 \le s < v \right\}.$$

The corresponding child-vector subsequences are exactly the left-parent completion profile and a fixed translate of the right-parent completion profile. Hence standard factorization explains why Farey parents and Christoffel structure are nearby, but it is not the same residual-indexed cross-parameter identity.

The two scalar jump amplitudes are

$$A = c + u, \qquad B = v + d,$$

so, under the standard Christoffel factorization conventions, the corresponding factor lengths also appear as the ℓ^1 projections of the two completion unit lifts.

75.2.3 Dedekind structure of the one-sided second moment

The combined spectrum is universal, but its split into positive and negative parts depends on $p : q$. This ratio dependence is already visible in the centered second moment.

Let

$$((x)) = \begin{cases} \{x\} - \frac{1}{2}, & x \notin \mathbb{Z}, \\ 0, & x \in \mathbb{Z}, \end{cases}$$

and define the classical Dedekind sum

$$s(h,k) = \sum_{j=1}^{k-1} \left(\left(\frac{j}{k}\right)\right) \left(\left(\frac{hj}{k}\right)\right).$$

Put $h_0 = (m+1)/2$. For the positive sorted spectrum,

$$\ell_j^+ - h_0 = m \left(\left(\frac{j}{p}\right)\right) - \left(\left(\frac{qj}{p}\right)\right). \tag{75.12}$$

Since multiplication by q permutes the nonzero residue classes modulo p,

$$\sum_{j=1}^{p-1} \left(\left(\frac{qj}{p}\right)\right)^2 = \sum_{j=1}^{p-1} \left(\left(\frac{j}{p}\right)\right)^2 = \frac{(p-1)(p-2)}{12p}.$$

Squaring equation (75.12) therefore gives

$$D_2^+(p,q) := \sum_{r=1}^{p-1} \left(L(r) - \frac{m+1}{2} \right)^2 = \frac{(m^2+1)(p-1)(p-2)}{12p} - 2m\, s(q,p). \tag{75.13}$$

Similarly,

$$D_2^-(p,q) = \frac{(m^2+1)(q-1)(q-2)}{12q} - 2m\,s(p,q). \tag{75.14}$$

By Theorem 75.4,

$$D_2^+(p,q) + D_2^-(p,q) = \frac{(m-1)(m-2)(m-3)}{12}. \tag{75.15}$$

Substituting equations (75.13) and (75.14) into equation (75.15) recovers the classical Dedekind reciprocity law

$$s(q,p) + s(p,q) = -\frac{1}{4} + \frac{1}{12}\left(\frac{q}{p} + \frac{p}{q} + \frac{1}{pq}\right);$$

see [63]. Dedekind reciprocity is not a new result here. Its completion interpretation is that Dedekind sums quantify the ratio-dependent one-sided splitting of an otherwise universal completion spectrum.

Remark 75.R1 — Higher even moments.

By Proposition 75.8, all odd centered moments vanish. Higher even moments expand into products of periodic Bernoulli functions and hence into generalized Dedekind-type sums. Since such sums and their reciprocity laws have an extensive independent literature, including [68], we do not pursue new reciprocity statements here.

75.3 Computational discovery and validation

The results above are proved symbolically and do not depend on computation. Numerical experimentation served first as a discovery device and later as an independent audit of signs, endpoints, and floor/ceiling conventions.

An initial audit covered every coprime ordered pair $2 \le p,q \le 100$. A second reviewer-oriented audit extended the exhaustive range to

$$2 \le p,q \le 200,$$

covering 24,064 coprime ordered pairs, and added 30,000 randomized coprime pairs with $2 \le p,q \le 5000$. Shortest-completion vectors were constructed independently from Lemma 73.2. For every residual in the positive and negative fundamental ranges, these direct solutions were compared with the formulas of sections 73.2 and 74.1. The audits verified, without exception:

- the positive and negative canonical unit-lift laws;
- the closed vector formulas equations (73.6) and (73.7);
- all four ordered parent-to-mediant identities equations (74.1) to (74.4);
- the corresponding positive and negative index partitions.

No counterexample was found in either the exhaustive or randomized vector audits.

A separate scalar audit tested every coprime ordered pair

$$1 \le p,q \le 150,$$

comprising 13,715 ratios. For each pair, the fundamental spectrum and the translated shells $k = 0,1,\ldots,7$ were generated directly from shortest completions. In every case,

$$\Sigma_k^+ \uplus \Sigma_k^- = \{k, k, k+2, k+3, \ldots, k+p+q-1\}.$$

These computations are not proof premises. The direct completion oracle used only the defining equation $px - qy = r$ and the canonical-strip bounds, rather than the closed Farey formulas being tested. The experiments therefore function only as independent consistency checks.

75.4 Conclusion

We have studied the ordered family of shortest nonnegative integer solutions of

$$px - qy = r,$$

with p and q coprime, motivated by shortest completion in rational balance languages.

For a proper Farey decomposition

$$(p, q) = (c, u) + (v, d), \qquad uv - cd = 1,$$

the positive shortest-completion profile advances through two kernel-equivalent unit-residual lifts,

$$(u, c) \qquad \text{and} \qquad (-d, -v).$$

Canonical reduction in the strip $0 \le y < p$ selects between them, and the negative profile has the corresponding dual dynamics. After a unimodular coordinate change this local walk is a classical rational mechanical path, so its Christoffel form is best viewed as the local arithmetic mechanism rather than as the principal novelty.

The principal completion-specific result is the cross-parameter Ordered Farey Recursion Theorem. It reconstructs the entire mediant minimizer section from the two parent minimizer sections through complementary floor/ceiling index maps: the two parent profiles appear as interleaved child subsequences, one after a fixed translation. The local jump vectors recover the oriented parents, and iteration yields a recursively coherent family over the Stern–Brocot tree.

Returning to the motivating automaton, the map $r \mapsto \mu_{2,1}(r)$ is exactly the shortest-demand semantics realized by its distributed state–stack representation; the present results show that this semantic object belongs to a much larger Farey-recursive arithmetic family.

Scalar projection produces a complementary universal phenomenon. Writing $m = p + q$, the combined positive and negative fundamental spectrum is always

$$\{0, 0, 2, 3, \ldots, m - 1\},$$

and every residual shell is an affine translate of this multiset. Hence all combined shell moments depend only on m. Classical Beatty, mechanical/Christoffel, and Dedekind structures arise as arithmetic shadows of this vector theory rather than as proof premises.

Taken together, these results organize the coprime rational balance family as a single Farey-recursive system of shortest-completion profiles, with locally recoverable parent data and universal scalar spectra.

Part XVII

Quantitative LIFO Chronology at Fixed Semantics

Research provenance. The preceding theory separates a unique semantic quotient from nonunique physical lifts. This Part fixes the semantic content completely and randomizes only symbol chronology inside a balanced Parikh fiber. It thereby measures the physical LIFO cost that the residual quotient forgets. Classical bridge, occupation-time, recurrent-chain, and Rayleigh tools are used as probabilistic interfaces; the new content is their coupling to the shortest-demand physical lift.

Chapter 76

Fixed-Parikh Random Ordering and Direct Obstruction

The centered real-time $p{:}q$ normalizer was developed earlier as a deterministic physical lift of shortest-completion semantics. We now keep that semantic content fixed. For coprime p, q, write $m = p + q$ and sample uniformly from

$$\mathcal{F}_{gq,gp} = \{w \in \{0,1\}^{mg} : N_0(w) = gq,\ N_1(w) = gp\}.$$

Every word in this fiber has the same zero endpoint residual and the same endpoint completion demand. Only chronology varies. The resulting experiment therefore isolates costs created by LIFO order rather than by semantic variation. We restate below only the shortest-demand and transition data needed for the quantitative analysis; the original 2:1 provenance figure and the full construction have already appeared in earlier Parts.

76.1 Shortest completion demand

For $r \in \mathbb{Z}$, let

$$\mu(r) = (x_r, y_r)$$

be the unique nonnegative solution of

$$px - qy = r$$

that minimizes $x + y$. All other nonnegative solutions are obtained by adding copies of the primitive zero-weight kernel

$$K = (q, p).$$

We use the canonical stack ordering

$$\eta(x, y) = \begin{cases} (10)^y 0^{x-y}, & x \geq y, \\ (10)^x 1^{y-x}, & y > x. \end{cases} \tag{76.1}$$

The leftmost symbol is the visible stack top.

Let

$$h = \left\lfloor \frac{m-1}{2} \right\rfloor, \qquad D_{-h}, \dots, D_h$$

be the centered debt states. For a logical stack word S define

$$W(S) = pN_0(S) - qN_1(S).$$

The state–stack residual invariant is

$$\rho = W(S) + dm \qquad \text{at configuration } (D_d, S). \tag{76.2}$$

The semantic projection reduces the physical stack plus virtual debt by complete kernel packets and returns exactly $\mu(\rho)$.

76.2 Transition classes

Suppose the current input is $a \in \{0, 1\}$. If $S \neq \varnothing$ and a equals the visible top, the machine performs a *MATCH*: it pops one cell and leaves d unchanged. If the visible top is 0 and $a = 1$, a mismatch increments debt while $d < h$; this is *SEARCH*. At $d = h$ the next such mismatch is an *outer WRITE*: the visible zero is removed, debt is reset to zero, and

$$U_+ = \eta(\mu((h+1)m))$$

is inserted before the surviving suffix. The negative rule is symmetric: input 0 over visible top 1 decrements debt while $d > -h$, and at $d = -h$ an outer WRITE inserts

$$U_- = \eta(\mu(-(h+1)m)).$$

If the logical stack is empty, the next symbol triggers a *bottom WRITE*. For every $|d| \leq h$,

$$(D_d, \varnothing) \xrightarrow{1} \big(D_0, \eta(\mu(dm+q))\big), \tag{76.3}$$

$$(D_d, \varnothing) \xrightarrow{0} \big(D_0, \eta(\mu(dm-p))\big). \tag{76.4}$$

Every WRITE therefore returns the physical stack to the exact canonical shortest-demand section for the new residual.

The construction has the following properties, all consequences of shortest-demand minimality and the centered debt bound.

Lemma 76.1 (WRITE frontier bound)**.** *Every canonical word created by a bottom or outer WRITE has shortest-demand vector (x, y) satisfying*

$$\min(x, y) \leq h.$$

For the outer-WRITE vectors $\mu(\pm(h+1)m)$, the minority coordinate is at most h while the majority coordinate is at least $h+1$; in the positive case the minority one-count is $< p$, and in the negative case the minority zero-count is $< q$.

Proof. For a bottom WRITE the target residual has the form

$$r = jm + q, \qquad -h-1 \leq j \leq h,$$

because $dm - p = (d-1)m + q$. Write $\mu(r) = (x, y)$ and $\ell = x + y$. Modulo m, the relation $px - qy = r \equiv q$ and $p \equiv -q \pmod m$ give

$$-q\ell \equiv q \pmod m.$$

Since $\gcd(q, m) = 1$, $\ell \equiv -1 \pmod m$, so $\ell = km - 1$. If both $x, y \geq h+1$, then $k \geq 2$. The minimum demand is kernel-free, hence either $x < q$ or $y < p$. In the first case $q \geq h+2$ and

$$r = m(x - qk) + q$$

forces $j \le -q-1 \le -h-3$, a contradiction. In the second case $p \ge h+2$ and

$$j = pk - y - 1 \ge p > h,$$

again a contradiction. Thus $\min(x,y) \le h$.

For the positive outer WRITE, put $c = h+1$. The minimum solution is

$$\mu(cm) = (qk + c, pk - c), \qquad k = \left\lceil \frac{c}{p} \right\rceil.$$

Its second coordinate lies in $[0,h]$, is strictly less than p, and its first is at least $c = h+1$. The negative case is symmetric, with the zero-coordinate strictly less than q. ☐

Lemma 76.2 (Boundary guard)**.** *Suppose the current stack is a suffix of a canonical word whose minority count is at most h.*

(a) If the current state is D_h and the stack begins with 0, then an outward $1/0$ mismatch can occur only when the surviving stack is unary zero.

(b) If the current state is D_{-h} and the stack begins with 1, then an outward $0/1$ mismatch can occur only when the surviving stack is unary one.

Proof. Consider the positive case. Since the most recent WRITE, debt started at zero. Reaching D_h requires at least h previous positive mismatches, each of which popped a zero from the pre-existing canonical word. If the current visible zero were still in the mixed alternating part and a one survived later in the stack, then the h previously popped zeros would each have a distinct preceding one in the alternating part. Together with the current zero and a later surviving one, the original canonical word would contain at least $h+1$ zeros and at least $h+1$ ones, contradicting Lemma 76.1. Hence the surviving suffix is unary zero. The negative case is symmetric. ☐

Lemma 76.3 (Canonical-segment property)**.** *Immediately after every WRITE the machine is in state D_0 with stack $\eta(\mu(\rho))$, whose minority count is at most h. Until the next WRITE, every transition removes exactly one cell from that pre-existing canonical word. Hence every reachable logical stack is a suffix of the canonical word produced by the most recent WRITE.*

Proof. A bottom WRITE has the displayed canonical form by definition and satisfies Lemma 76.1. Between WRITEs, MATCH and SEARCH only remove the visible pre-existing cell. At an outer WRITE, Lemma 76.2 shows that the surviving suffix is unary. The newly inserted canonical word and that unary suffix concatenate to the canonical ordering of the new shortest demand; kernel-freeness follows from the minority bound in Lemma 76.1. Induction over WRITE episodes proves the claim. ☐

Lemma 76.4 (Zero-residual correctness)**.** *At the end of every processed binary prefix,*

$$\rho = 0 \quad \Longleftrightarrow \quad (D_d, S) = (D_0, \varnothing).$$

Proof. The reverse implication follows from (76.2). For the forward implication, consider the most recent WRITE. Immediately after it the stack was $\eta(\mu(r_0))$ for some residual r_0. Until the current time no further WRITE has occurred, so every intervening symbol has popped exactly one of those cells. If the current residual is zero, that intervening input is a completion of r_0. By minimality of $\mu(r_0)$, no completion can use fewer symbols than the length of the written stack. Since no more than that many cells can have been popped, equality is forced: the entire stack has been consumed. Equation (76.2) then gives $dm = 0$, hence $d = 0$. ☐

Operational observables. For a word w, let $S(w), O(w), B(w)$ be the numbers of SEARCH, outer-WRITE, and bottom-WRITE transitions. The primary LIFO-obstruction count is

$$R(w) = S(w) + O(w).$$

A bottom WRITE is not counted as an obstruction: it occurs when the physical stack is already empty and represents rematerialization rather than LIFO search.

76.3 The fixed-Parikh model as a critical bridge

For arbitrary $a, b \geq 0$ define the fixed-Parikh fiber

$$\mathcal{F}_{a,b} = \{w \in \{0,1\}^{a+b} : N_0(w) = a,\ N_1(w) = b\}.$$

Our balanced model is

$$W_g \sim \mathrm{Unif}(\mathcal{F}_{gq,gp}), \qquad n = mg.$$

Introduce i.i.d. symbols X_i under the critical law

$$\mathbb{P}(X_i = 1) = \frac{p}{m}, \qquad \mathbb{P}(X_i = 0) = \frac{q}{m},$$

and increments

$$\xi_i = \begin{cases} q, & X_i = 1, \\ -p, & X_i = 0. \end{cases}$$

Then

$$\mathbb{E}\xi_i = 0, \qquad \mathrm{Var}(\xi_i) = pq.$$

Write $\rho_t = \sum_{i \leq t} \xi_i$.

Lemma 76.5 (Canonical critical-bridge representation)**.** *For $n = mg$,*

$$\mathcal{L}(X_1, \dots, X_n \mid \rho_n = 0) = \mathrm{Unif}(\mathcal{F}_{gq,gp}).$$

Proof. The condition $\rho_n = 0$ gives

$$mN_1(n) - pn = 0,$$

so $N_1(n) = pg$ and $N_0(n) = qg$. Every word with these counts has the same i.i.d. probability

$$\left(\frac{p}{m}\right)^{pg} \left(\frac{q}{m}\right)^{qg}.$$

Conditioning therefore makes the fiber uniform. □

At block boundaries set

$$Z_j = \frac{\rho_{mj}}{m}.$$

If K_j is the number of ones in the next m symbols, then under the critical i.i.d. law

$$Z_{j+1} - Z_j = K_j - p, \qquad K_j \sim \mathrm{Bin}\Big(m, \frac{p}{m}\Big). \tag{76.5}$$

Hence

$$\mathbb{E}(Z_{j+1} - Z_j) = 0, \qquad \mathrm{Var}(Z_{j+1} - Z_j) = \frac{pq}{m}. \tag{76.6}$$

The block skeleton is therefore a centered, irreducible, aperiodic, finite-range random walk on $\mathbb{Z}$.

76.4 Direct realization and the first LIFO obstruction

Call a word *direct* if no SEARCH or outer-WRITE transition occurs, equivalently if $R(w) = 0$.

76.4.1 Primitive direct blocks

The shortest demands after the first symbol are

$$\mu(-p) = (q-1, p), \qquad \mu(q) = (q, p-1).$$

Define

$$B_0 = 0\,\eta(q-1, p), \qquad B_1 = 1\,\eta(q, p-1). \tag{76.7}$$

Both words have length m and Parikh vector (q, p).

Theorem 76.6 (Direct-realization factorization)**.** *A balanced word $w \in \mathcal{F}_{gq,gp}$ satisfies $R(w) = 0$ if and only if*

$$w = B_{\varepsilon_1} B_{\varepsilon_2} \cdots B_{\varepsilon_g}$$

for a unique $(\varepsilon_1, \dots, \varepsilon_g) \in \{0,1\}^g$. Consequently

$$\#\{w \in \mathcal{F}_{gq,gp} : R(w) = 0\} = 2^g \tag{76.8}$$

and

$$\mathbb{P}(R(W_g) = 0) = \frac{2^g}{\binom{mg}{qg}}. \tag{76.9}$$

Proof. Start from empty stack and zero debt. The first input symbol triggers a bottom WRITE. If it is 0, the newly written stack is $\eta(\mu(-p))$; if it is 1, the stack is $\eta(\mu(q))$. If no LIFO obstruction occurs thereafter, every subsequent input until the stack empties must match the visible top. Thus the entire first episode is forced to be B_0 or B_1. Each block has length m, consumes exactly (q, p) symbols, and returns the machine to $(D_0, \varnothing)$. Iteration gives the factorization. Conversely, each displayed block is realized by one bottom WRITE followed only by MATCH transitions, so any concatenation is direct. Uniqueness follows from the first symbol of each block. □

By Stirling's formula,

$$\mathbb{P}(R(W_g) = 0) \sim \sqrt{\frac{2\pi pqg}{m}} \left(\frac{2p^p q^q}{m^m}\right)^g. \tag{76.10}$$

The base

$$\lambda_{p,q} = \frac{2p^p q^q}{m^m}$$

is strictly less than one.

76.4.2 Exact first-obstruction law

Let

$$\tau_g = \min\{t \geq 1 : \text{the transition at time } t \text{ is SEARCH or outer WRITE}\},$$

with $\tau_g = mg + 1$ if no obstruction occurs. For $\varepsilon \in \{0, 1\}$ and $0 \leq s < m$, write

$$z_{\varepsilon,s} = N_0(\mathrm{pref}_s(B_\varepsilon)).$$

Theorem 76.7 (Exact first LIFO-obstruction survival law)**.** *For $0 \le k \le g$,*

$$\mathbb{P}(\tau_g > km) = \frac{2^k \binom{m(g-k)}{q(g-k)}}{\binom{mg}{qg}}. \tag{76.11}$$

For $0 \le k < g$ and $1 \le s < m$,

$$\mathbb{P}(\tau_g > km + s) = \frac{2^k \displaystyle\sum_{\varepsilon=0}^{1} \binom{m(g-k) - s}{q(g-k) - z_{\varepsilon,s}}}{\binom{mg}{qg}}. \tag{76.12}$$

Proof. The event $\{\tau_g > km\}$ is exactly the event that the first k primitive blocks are direct. By Theorem 76.6, there are 2^k possible direct prefixes, each consuming (kq, kp) symbols. The remaining suffix is an arbitrary balanced fixed-Parikh word of size $g - k$, giving (76.11). For a partial block, after the first k complete direct blocks the next s symbols must equal $\mathrm{pref}_s(B_0)$ or $\mathrm{pref}_s(B_1)$. These two events are disjoint because B_0 and B_1 begin with different symbols. After fixing such a prefix, the remaining number of zeros is $q(g - k) - z_{\varepsilon,s}$, which gives (76.12). □

Define

$$R_{p,q}(0) = 1,$$

and, for $1 \le s < m$,

$$R_{p,q}(s) = \sum_{\varepsilon=0}^{1} \left(\frac{q}{m}\right)^{z_{\varepsilon,s}} \left(\frac{p}{m}\right)^{s - z_{\varepsilon,s}}.$$

Taking $g \to \infty$ with k, s fixed in Theorem 76.7 gives

$$\mathbb{P}(\tau_g > km + s) \longrightarrow \lambda_{p,q}^k R_{p,q}(s). \tag{76.13}$$

In particular,

$$\boxed{\tau_g = O_{\mathbb{P}}(1).} \tag{76.14}$$

76.4.3 Primitive balanced SEARCH distribution

At $g = 1$ no outer WRITE is required before return, and SEARCH occurs in mismatch pairs.

Proposition 76.8 (Primitive SEARCH law)**.** *On $\mathcal{F}_{q,p}$,*

$$S(W_1) \in \{0, 2, 4, \dots, 2\min(p, q)\},$$

and

$$\#\{w \in \mathcal{F}_{q,p} : S(w) = 2j\} = \binom{p}{j}\binom{q-1}{j} + \binom{q}{j}\binom{p-1}{j}. \tag{76.15}$$

Consequently

$$\mathbb{E}S(W_1) = \frac{2pq(m-2)}{m(m-1)}. \tag{76.16}$$

Proof. Condition on the first symbol. If it is 0, the bottom WRITE creates the deterministic target word $\eta(q - 1, p)$, with $q - 1$ zero-cells and p one-cells. The remaining input suffix has exactly the same Parikh vector. During this primitive episode no outer WRITE can occur: the debt at any prefix is the difference between the number of input ones and target ones seen so far, whose absolute value is at most $\min(p, q-1) \le h$. Thus every step pops one target cell. If exactly j target one-cells are hit by input zero, then conservation of the two symbol counts forces exactly j target zero-cells to be hit by input one, giving $2j$ SEARCH mismatches. The number of such suffixes is

$$\binom{p}{j}\binom{q-1}{j}.$$

If the first symbol is 1, the same argument with target $\eta(q, p-1)$ gives

$$\binom{q}{j}\binom{p-1}{j}.$$

Adding proves (76.15). For the mean, use

$$\sum_j j\binom{a}{j}\binom{b}{j} = \frac{ab}{a+b}\binom{a+b}{a}$$

in the two terms and divide by $\binom{m}{p}$. □

Chapter 77

Physical-Lift Excursions and Rayleigh Rematerialization

The first obstruction is local, but rematerialization of the canonical shortest-demand section is diffusive. The block-boundary lift chain below records precisely that phenomenon.

77.1 The block-boundary physical-lift chain

We now pass from exact finite obstruction counts to the recurrent physical structure that governs rematerialization.

Lemma 77.1 (Empty stack only at block boundaries)**.** *If the logical stack is empty after t processed symbols, then $m \mid t$.*

Proof. If $S_t = \varnothing$, the residual invariant (76.2) gives $\rho_t = dm$ for some integer d. On the other hand,

$$\rho_t = mN_1(t) - pt.$$

Hence $m \mid pt$. Since $\gcd(p, m) = \gcd(p, q) = 1$, we obtain $m \mid t$. ☐

Let X_j be the full physical configuration after jm symbols under the critical i.i.d. input law, and let

$$\pi(X_j) = Z_j = \rho_{jm}/m.$$

Let

$$o = (D_0, \varnothing).$$

By Lemma 76.4, $X_j = o$ exactly when $Z_j = 0$. Since the scalar block walk Z_j is recurrent, so is the reachable class of the physical block chain.

Let

$$T_o^+ = \min\{j \geq 1 : X_j = o\}.$$

Definition 77.2 (Physical-lift invariant measure)**.** For a reachable block-boundary physical configuration x, define

$$\nu(x) = \mathbb{E}_o \left[\sum_{j=0}^{T_o^+ - 1} \mathbf{1}_{\{X_j = x\}} \right]. \tag{77.1}$$

The standard excursion construction for an irreducible recurrent Markov chain makes ν an invariant measure normalized by $\nu(o) = 1$; see, for example, [80, Ch. 1].

Theorem 77.3 (Unit semantic-shell mass)**.** *For every $z \in \mathbb{Z}$,*

$$\boxed{\sum_{x:\pi(x)=z} \nu(x) = 1.} \tag{77.2}$$

Proof. Push ν forward under the factor map π. The result is an invariant measure for the irreducible recurrent random walk Z_j. Such an invariant measure is unique up to a multiplicative constant. Because the block walk is translation-invariant, counting measure on $\mathbb{Z}$ is invariant. At shell zero, Lemma 76.4 shows that the only physical state is o, so the pushed-forward mass equals $\nu(o) = 1$. The normalization therefore fixes the pushed-forward measure to be counting measure, proving (77.2). □

This is the quantitative form of the semantic/physical distinction: the residual quotient assigns one unit of invariant mass to each shell, while physical chronology determines how that unit is split among its lifts.

Definition 77.4 (Physical rematerialization constant)**.** For $-h \le d \le h$, let

$$E_d = (D_d, \varnothing), \qquad \theta_d^{p,q} = \nu(E_d),$$

and define

$$\boxed{\kappa_{p,q} = \sum_{d=-h}^{h} \theta_d^{p,q}.} \tag{77.3}$$

Because E_d belongs to semantic shell d, Theorem 77.3 gives

$$0 \le \theta_d^{p,q} \le 1, \qquad \theta_0^{p,q} = 1, \qquad 1 \le \kappa_{p,q} \le 2h + 1. \tag{77.4}$$

Let $\mathcal{E} = \{E_d : -h \le d \le h\}$ and consider the trace chain obtained by observing X_j only when it enters $\mathcal{E}$. Restriction of an invariant measure to a recurrent trace set is invariant for the trace chain. Therefore its stationary distribution is

$$\pi_d^{\mathcal{E}} = \frac{\theta_d^{p,q}}{\kappa_{p,q}},$$

and in particular

$$\boxed{\pi_0^{\mathcal{E}} = \frac{1}{\kappa_{p,q}}.} \tag{77.5}$$

77.2 Bottom-WRITE rematerialization has a Rayleigh law

The fixed-Parikh bridge can return to physical empty states only at block boundaries by Lemma 77.1. This allows an excursion decomposition at the block level.

A *primitive semantic excursion* is a block path beginning at o, avoiding o at positive intermediate block times, and ending at its first return to o. For such an excursion e, let

$$L(e) = \text{its length in blocks}$$

and let

$$R_B(e) = \#\{\text{empty-stack block states visited in } [0, L(e))\}.$$

Every such visit is followed, unless it is the terminal boundary of the full word, by exactly one bottom WRITE. Concatenating primitive semantic excursions therefore adds the rewards R_B exactly.

Lemma 77.5 (Excursion reward mean and exponential moments)**.** *Under the critical i.i.d. law,*

$$\mathbb{E}R_B(e) = \kappa_{p,q}. \tag{77.6}$$

Moreover there exists $s > 1$ such that

$$\mathbb{E}s^{R_B(e)} < \infty. \tag{77.7}$$

Proof. Equation (77.6) is the excursion formula (77.1) summed over the finite empty-state set $\mathcal{E}$. For (77.7), consider the finite trace chain on reachable empty states. From every such state there is a positive-probability finite input continuation that completes its residual and reaches E_0. Finiteness gives constants $L < \infty$ and $\varepsilon > 0$ such that from every trace state the probability of hitting E_0 within at most L trace transitions is at least ε. Hence the trace hitting time has a geometric tail, which implies an exponential moment for $R_B(e)$. □

Let

$$b_g = \mathbb{P}(Z_g = 0) = \binom{mg}{pg}\left(\frac{p}{m}\right)^{pg}\left(\frac{q}{m}\right)^{qg}.$$

Define the bridge return generating function

$$G(z, 1) = \sum_{g \geq 0} b_g z^g.$$

Let $F(z, u)$ be the probability generating function of primitive semantic excursions, with z marking block length and u marking R_B. Unique decomposition into primitive excursions gives

$$G(z, u) = \frac{1}{1 - F(z, u)}. \tag{77.8}$$

At $u = 1$, Stirling's formula yields

$$b_g \sim \sqrt{\frac{m}{2\pi pq}}\, g^{-1/2}. \tag{77.9}$$

A standard Abelian/Tauberian transfer therefore gives

$$G(z, 1) \sim \sqrt{\frac{m}{2pq}}(1 - z)^{-1/2} \qquad (z \uparrow 1), \tag{77.10}$$

and hence

$$1 - F(z, 1) \sim \sqrt{\frac{2pq}{m}}\sqrt{1 - z}. \tag{77.11}$$

The same square-root singularity is a standard source of Rayleigh local-time laws for zero-drift lattice-path bridges; compare [72]. We use only the elementary regular-variation consequences of this square-root behavior below. Here the reward is not scalar zero-local-time itself but the physical empty-lift reward carried by each primitive semantic excursion.

Theorem 77.6 (Bottom-WRITE Rayleigh law)**.** *Let $B_g = B(W_g)$. Then*

$$\boxed{\frac{B_g}{\sqrt{mg}} \Longrightarrow \mathrm{Rayleigh}\left(\frac{\kappa_{p,q}}{\sqrt{pq}}\right).} \tag{77.12}$$

Equivalently, for every fixed integer $r \geq 1$,

$$\mathbb{E}\left[\left(\frac{B_g}{\sqrt{mg}}\right)^r\right] \longrightarrow \left(\frac{\kappa_{p,q}}{\sqrt{pq}}\right)^r 2^{r/2}\Gamma\left(1 + \frac{r}{2}\right). \tag{77.13}$$

Proof. Write

$$H_j(z) = \partial_u^j F(z,u)\big|_{u=1}.$$

By Lemma 77.5, $H_j(1) < \infty$ for every fixed j, and

$$H_1(1) = \kappa_{p,q}.$$

Differentiating (77.8) r times in u, the term with r factors of H_1 is

$$r!H_1(z)^r G(z,1)^{r+1}. \tag{77.14}$$

Every other Faà di Bruno term contains at most $r-1$ derivative factors of F and hence at most $G(z,1)^r$. More precisely, each such term is a product of some $H_{j_1}(z), \ldots, H_{j_k}(z)$ with $k \le r-1$ and a factor $G(z,1)^{k+1}$. Since every H_j has summable nonnegative coefficients by Lemma 77.5, the convolution theorem for summable sequences against regularly varying coefficients shows that its coefficient is $O(g^{(k-1)/2}) = o(g^{(r-1)/2})$. Thus only (77.14) contributes at leading order.

Using (77.10), Karamata coefficient asymptotics, and the same standard convolution lemma (equivalently, the corresponding singularity-transfer calculation; cf. [74]),

$$[z^g]\partial_u^r G(z,1) \sim \frac{r!\kappa_{p,q}^r}{\Gamma((r+1)/2)} \left(\frac{m}{2pq}\right)^{(r+1)/2} g^{(r-1)/2}.$$

Dividing by b_g in (77.9) gives the conditional factorial moment

$$\mathbb{E}[(B_g)_r] \sim \kappa_{p,q}^r \left(\frac{m}{pq}\right)^{r/2} 2^{r/2}\Gamma\left(1+\frac{r}{2}\right) g^{r/2},$$

where the duplication formula for the Gamma function was used. Ordinary moments have the same leading term. These are exactly the moments of a Rayleigh variable with scale $\kappa_{p,q}\sqrt{m/(pq)}$ for $B_g/\sqrt{g}$, or equivalently scale $\kappa_{p,q}/\sqrt{pq}$ for $B_g/\sqrt{mg}$. The Rayleigh distribution is moment-determinate, so the method of moments proves (77.12). □

Corollary 77.7 (Mean and variance of rematerialization)**.** *With $n = mg$,*

$$\mathbb{E}B_g \sim \kappa_{p,q}\sqrt{\frac{\pi}{2pq}}\sqrt{n}, \tag{77.15}$$

$$\mathrm{Var}(B_g) \sim \kappa_{p,q}^2 \frac{4-\pi}{2pq}\, n. \tag{77.16}$$

Chapter 78

Macroscopic LIFO Cost, Duality, and Finite Kernels

We now pass from the diffusive rematerialization scale to the macroscopic SEARCH/outer-WRITE regime, then return to finite kernels that compute the physical rematerialization constant.

78.1 Two regenerative bulk regimes

We now analyze the n-scale LIFO cost. Put

$$c = h + 1, \qquad k_+ = \left\lceil \frac{c}{p} \right\rceil, \qquad k_- = \left\lceil \frac{c}{q} \right\rceil.$$

The canonical outer-WRITE demands are

$$\mu(cm) = (qk_+ + c,\ pk_+ - c), \tag{78.1}$$
$$\mu(-cm) = (qk_- - c,\ pk_- + c). \tag{78.2}$$

To define the bulk constants, use the translation-invariant positive-tail extension obtained by quotienting complete unary packets (equivalently, adjoin an unbounded unary zero tail and retain the same finite phase dynamics). A finite deep-positive interval in the actual normalizer follows this extension until it hits the central boundary; only its first and last regeneration cycles can be truncated, and those endpoint pieces are handled in Section 78.2. In the homogeneous positive extension, consider the interval from one positive outer WRITE to the next. The mixed head of the newly written word is finite, while the surviving tail is unary zero. The next positive outer WRITE occurs after exactly pk_+ input ones have been read. Under the critical i.i.d. law the cycle length is therefore

$$T_+ \sim \mathrm{NB}\Big(pk_+, \frac{p}{m}\Big), \tag{78.3}$$

where T_+ counts trials through the final success. Similarly,

$$T_- \sim \mathrm{NB}\Big(qk_-, \frac{q}{m}\Big). \tag{78.4}$$

Thus

$$\mathbb{E}T_+ = mk_+, \qquad \mathrm{Var}(T_+) = \frac{k_+ qm}{p}, \tag{78.5}$$
$$\mathbb{E}T_- = mk_-, \qquad \mathrm{Var}(T_-) = \frac{k_- pm}{q}. \tag{78.6}$$

In particular the outer-WRITE rates are

$$o_+ = \frac{1}{mk_+}, \qquad o_- = \frac{1}{mk_-}. \tag{78.7}$$

Theorem 78.1 (Exact bulk LIFO-obstruction rates). *For $R = S + O$, the critical i.i.d. positive- and negative-bulk rates are*

$$\boxed{r_+ = \frac{2pq}{m^2} + \frac{(p-q)c}{m^2k_+},} \tag{78.8}$$

$$\boxed{r_- = \frac{2pq}{m^2} - \frac{(p-q)c}{m^2k_-}.} \tag{78.9}$$

Hence the SEARCH rates are

$$s_+ = r_+ - o_+, \qquad s_- = r_- - o_-. \tag{78.10}$$

Proof. We prove the positive case. Write

$$(x, y) = \mu(cm) = (qk_+ + c, pk_+ - c).$$

A positive cycle ends at the pk_+-th input one. All y one-cells in the canonical head are consumed before the outer boundary can be reached, by Lemma 76.2. Let M be the number of those one-cells that are consumed by input zero; each such event is a negative-direction SEARCH mismatch. Since these cells are reached before stopping and the input is i.i.d.,

$$\mathbb{E}M = y\frac{q}{m}.$$

Among the pk_+ input ones in the cycle, exactly $y - M$ match one-cells. Therefore the number of one-over-zero mismatches is

$$pk_+ - (y - M) = c + M.$$

The total number of LIFO obstructions in the cycle is consequently

$$c + 2M,$$

with mean

$$c + 2(pk_+ - c)\frac{q}{m}.$$

Dividing by the mean cycle length mk_+ gives (78.8). The negative calculation is obtained by interchanging zero and one, p and q, and reversing the debt sign; this gives (78.9). □

The difference

$$r_+ - r_- = \frac{(p-q)c}{m^2}\left(\frac{1}{k_+} + \frac{1}{k_-}\right) \tag{78.11}$$

is nonzero whenever $p \neq q$. Since coprime $p = q$ forces $p = q = 1$, the only symmetric ratio is 1:1.

78.2 Finite-to-macroscopic operational transfer

Let f_t be a bounded operational reward determined by the physical transition at time t; examples include indicators of SEARCH, outer WRITE, MATCH, or bounded functions of the debt state. Put

$$C_g(f) = \sum_{t=1}^{n} f_t, \qquad n = mg.$$

The positive-residual occupation count is

$$A_g = \#\{1 \le t \le n : \rho_t > 0\}.$$

Let $v_+(f)$ and $v_-(f)$ denote the corresponding critical i.i.d. positive- and negative-bulk reward rates. For R, these are r_+ and r_- from Theorem 78.1.

The proof uses a symbol-level finite-phase quotient in the two deep tails. We record the required periodicity explicitly because the rewards in this section are accumulated one input symbol at a time.

Lemma 78.2 (Symbol-level affine tail periodicity)**.** *For every $r \ge 0$,*

$$\mu(r + pm) = \mu(r) + (m, 0), \tag{78.12}$$

and for every $r \le 0$,

$$\mu(r - qm) = \mu(r) + (0, m). \tag{78.13}$$

Consequently, after quotienting complete unary m-packets, the reachable configurations in each sufficiently deep positive or negative residual tail have a finite symbol-level phase space.

Proof. For $r \ge 0$, the minimum solution $\mu(r) = (x, y)$ has $0 \le y < p$: among the p residue classes for y, exactly one makes $r + qy$ divisible by p, and the resulting x is nonnegative. Hence $(x + m, y)$ is a nonnegative solution of weight $r + pm$ with the same $y < p$, so no kernel packet (q, p) can be subtracted. By uniqueness of the minimum demand it is $\mu(r + pm)$. The negative identity is symmetric, using the representative with $0 \le x < q$.

By Lemma 76.3, between WRITEs the stack is a suffix of a canonical shortest-demand word and the mixed alternating part has uniformly bounded length. Equations (78.12)–(78.13) show that translating a deep residual tail changes only the number of complete unary m-packets. Thus a phase may be specified by the debt state, the bounded mixed suffix, the unary symbol and unary length modulo m, together with the residual class modulo pm or qm. Only finitely many phases occur. □

Choose once and for all

$$H \ge \max(p, q)$$

large enough that the finite symbol-level quotients of Lemma 78.2 are homogeneous whenever $\rho > H$ or $\rho < -H$. In the positive tail, feeding sufficiently many 1's first consumes the bounded mixed head and then accumulates positive mismatches on the unary zero tail until the positive outer-WRITE reset; for H large enough there are at least $h+1$ unary zero cells available, and the residual only increases along this forcing word. The negative statement is symmetric, using a run of 0's. Thus every sufficiently deep phase can reach the sign-specific reset phase with positive probability without leaving its tail. Consequently, after transient phases are discarded, the finite phase kernel has a unique recurrent class. Restrict to that class and let P_+ and P_- denote the critical i.i.d. one-symbol phase kernels. For a bounded transition reward f, let

$$\bar{f}_\sigma(x) = \frac{q}{m} f(x, 0) + \frac{p}{m} f(x, 1), \qquad \sigma \in \{+, -\},$$

where $f(x, a)$ denotes the reward of the deterministic transition from phase x on symbol a. If π_σ is the invariant probability of the recurrent phase class, define

$$v_\sigma(f) = \pi_\sigma \bar{f}_\sigma.$$

The finite Poisson equation

$$g_\sigma - P_\sigma g_\sigma = \bar{f}_\sigma - v_\sigma(f) \tag{78.14}$$

has a bounded solution, unique up to an additive constant.

78.2.1 Bridge perturbation

Under the fixed-Parikh bridge, if the current residual is ρ_t, then

$$\mathbb{P}(X_{t+1} = 1 \mid \mathcal{F}_t) = \frac{p}{m} - \frac{\rho_t}{m(n-t)}. \tag{78.15}$$

Thus the deviation from the critical i.i.d. symbol law is

$$\delta_t = -\frac{\rho_t}{m(n-t)}.$$

For a fixed-Parikh bridge,

$$\mathrm{Var}(\rho_t) = pq\frac{t(n-t)}{n-1}. \tag{78.16}$$

Therefore

Lemma 78.3 (Total bridge-kernel perturbation)**.**

$$\mathbb{E}\sum_{t=1}^{n-1} |\delta_t| \le \frac{\pi\sqrt{pq}}{2m}\sqrt{n} + O(1). \tag{78.17}$$

Proof. By Cauchy–Schwarz and (78.16),

$$\mathbb{E}|\delta_t| \le \frac{\sqrt{pq}}{m\sqrt{n-1}}\sqrt{\frac{t}{n-t}}.$$

Summing and comparing with

$$\int_0^1 \sqrt{\frac{x}{1-x}}\, dx = \frac{\pi}{2}$$

gives (78.17). □

78.2.2 The central band

Let

$$L_H = \#\{1 \le t \le n : |\rho_t| \le H\}.$$

Since $N_1(t)$ is hypergeometric with parameters (n, pg, t) and

$$\rho_t = mN_1(t) - pt,$$

a direct Stirling bound on the hypergeometric probability mass function gives, uniformly in admissible r,

$$\mathbb{P}(\rho_t = r) \le C_{p,q}\sqrt{\frac{n}{t(n-t)}}.$$

(The finitely many endpoint values of t are absorbed into the constant.) The band contains only finitely many residual values, hence

$$\boxed{\mathbb{E}L_H = O(\sqrt{n}).} \tag{78.18}$$

Because $H \ge \max(p, q)$, a path cannot jump directly from the deep positive tail $\{\rho > H\}$ to the deep negative tail $\{\rho < -H\}$ in one input step. Therefore the number of maximal deep-tail intervals is at most $L_H + 1$ up to an absolute endpoint constant.

78.2.3 Poisson decomposition under the bridge

At time t in a fixed deep sign, write Q_t for the bridge phase kernel. Since the next-symbol probabilities differ from the critical law only by δ_t,

$$\|(Q_t - P_\sigma)u\|_\infty \le 2\|u\|_\infty |\delta_t| \tag{78.19}$$

for every bounded phase function u, and likewise the bridge conditional reward mean differs from $\bar{f}_\sigma$ by at most $2\|f\|_\infty|\delta_t|$.

For an interior step of a deep-tail interval, combine (78.14) with Q_t to write

$$f_t - v_\sigma(f) = \big(g_\sigma(Y_t) - g_\sigma(Y_{t+1})\big) + \Delta M_t + \varepsilon_t, \tag{78.20}$$

where (ΔM_t) is a martingale-difference sequence with uniformly bounded increments and

$$|\varepsilon_t| \le C_{p,q,f}|\delta_t|. \tag{78.21}$$

Indeed, subtract first the bridge conditional mean of f_t, and then replace both the bridge reward mean and $Q_t g_\sigma$ by their critical counterparts; the two replacement errors are controlled by (78.19). The remaining centered terms form the martingale difference, while the g_σ terms telescope. The at most two transitions adjacent to each deep-tail interval endpoint are charged to the interval endpoint error.

Theorem 78.4 (Finite-to-macroscopic operational transfer)**.** *For every bounded operational reward f there is a constant $K_{p,q,f} < \infty$ such that*

$$\boxed{\mathbb{E}\,|C_g(f) - [nv_-(f) + (v_+(f) - v_-(f))A_g]| \le K_{p,q,f}\sqrt{n}.} \tag{78.22}$$

Proof. Compare f_t with $v_+(f)$ when $\rho_t > 0$ and with $v_-(f)$ when $\rho_t \le 0$. On the central band the absolute discrepancy is bounded by a constant times L_H, hence has expectation $O(\sqrt{n})$ by (78.18).

On every maximal deep positive or negative interval, sum (78.20). The g_σ terms telescope. Since g_σ is bounded, all interval endpoint and omitted boundary-step contributions have total L^1 size $O(\mathbb{E}(L_H + 1)) = O(\sqrt{n})$. The accumulated perturbation has expectation

$$O\left(\mathbb{E}\sum_{t<n}|\delta_t|\right) = O(\sqrt{n})$$

by Lemma 78.3. The sum of all martingale differences has bounded increments and predictable quadratic variation $O(n)$, so Cauchy–Schwarz gives L^1 norm $O(\sqrt{n})$.

Finally,

$$\sum_{t=1}^{n}\left[v_-(f) + (v_+(f) - v_-(f))\mathbf{1}_{\{\rho_t>0\}}\right] = nv_-(f) + (v_+(f) - v_-(f))A_g.$$

Combining the central, endpoint, perturbation, and martingale contributions proves (78.22). ☐

78.3 Macroscopic operational limit laws

The critical bridge invariance principle gives

$$\frac{\rho_{\lfloor nt\rfloor}}{\sqrt{pq\,n}} \Longrightarrow B^{\mathrm{br}}(t)$$

in $C[0,1]$, where B^{br} is a standard Brownian bridge. The zero set of a Brownian bridge has Lebesgue measure zero almost surely, so the positive-occupation functional is almost surely continuous at the limit. The positive occupation time of a Brownian bridge is uniform on $[0,1]$; uniform sojourn laws also admit exchangeable-increment and Lévy-bridge formulations [70, 71]. Hence

$$\frac{A_g}{n} \Longrightarrow U, \qquad U \sim \mathrm{Unif}(0,1). \tag{78.23}$$

Classical fixed-score ballot/election-return results provide a discrete counterpart to this occupation phenomenon [69].

Theorem 78.5 (Fixed-semantics operational limit)**.** *For every bounded operational reward f,*

$$\boxed{\frac{C_g(f)}{n} \Longrightarrow v_-(f) + U\big(v_+(f) - v_-(f)\big), \qquad U \sim \mathrm{Unif}(0,1).} \tag{78.24}$$

Proof. Divide (78.22) by n. The L^1 error tends to zero. Apply (78.23) and Slutsky's theorem. □

Corollary 78.6 (Uniform LIFO-obstruction law)**.** *If $p \neq q$, then*

$$\boxed{\frac{R_g}{n} \Longrightarrow \mathrm{Unif}\big(\min(r_-, r_+), \max(r_-, r_+)\big).} \tag{78.25}$$

If $p = q = 1$, then

$$\frac{R_g}{2g} \longrightarrow \frac{1}{2}$$

in probability.

Corollary 78.7 (Rank-one macroscopic fluctuations)**.** *Let $f_1, \dots, f_k$ be bounded operational rewards and*

$$\boldsymbol{C}_g = (C_g(f_1), \dots, C_g(f_k)).$$

Write

$$V_\pm = (v_\pm(f_1), \dots, v_\pm(f_k)).$$

Then

$$\frac{\boldsymbol{C}_g}{n} \Longrightarrow V_- + U(V_+ - V_-), \tag{78.26}$$

and

$$\boxed{\frac{\mathrm{Cov}(\boldsymbol{C}_g)}{n^2} \longrightarrow \frac{1}{12}(V_+ - V_-)(V_+ - V_-)^T.} \tag{78.27}$$

In particular the limiting covariance matrix has rank at most one.

Proof. Apply Theorem 78.5 componentwise; the same occupation variable U drives every coordinate. Since each normalized count is bounded, second moments converge. The covariance formula follows from $\mathrm{Var}(U) = 1/12$. □

Theorem 78.8 (Three-scale LIFO realization)**.** *For the balanced fixed-Parikh random-ordering model,*

$$\boxed{\begin{aligned} \tau_g &= O_{\mathbb{P}}(1), \\ \frac{B_g}{\sqrt{n}} &\Longrightarrow \mathrm{Rayleigh}\left(\frac{\kappa_{p,q}}{\sqrt{pq}}\right), \\ \frac{R_g}{n} &\Longrightarrow \mathrm{Unif}\big(\min(r_-, r_+), \max(r_-, r_+)\big). \end{aligned}} \tag{78.28}$$

for $p \neq q$, with the last law degenerating to $1/2$ for $p = q = 1$.

78.4 Physical-lift duality

For a word w, let $\bar{w}$ denote the bitwise complement. Interchanging p and q and complementing input reverses the residual:

$$\rho_t^{q,p}(\bar{w}) = -\rho_t^{p,q}(w). \tag{78.29}$$

Moreover shortest demands are coordinate-swapped:

$$\mu_{q,p}(-r) = \mathrm{swap}(\mu_{p,q}(r)). \tag{78.30}$$

The two canonical words $\eta(x,y)$ and $\eta(y,x)$ have the same alternating prefix and opposite unary tails. By the canonical-segment property, outer WRITE cannot occur until the mixed prefix has disappeared. During that common mixed phase, complemented input produces opposite debt. The unary phases are then mirror images.

Theorem 78.9 (Physical-lift duality)**.** *For every finite word w,*

$$B_{p,q}(w) = B_{q,p}(\bar{w}), \qquad O_{p,q}(w) = O_{q,p}(\bar{w}). \tag{78.31}$$

Consequently, for every g,

$$(B_g, O_g)_{p,q} \stackrel{d}{=} (B_g, O_g)_{q,p}$$

under the corresponding balanced fixed-Parikh laws, and

$$\boxed{\kappa_{p,q} = \kappa_{q,p}.} \tag{78.32}$$

Proof. Couple the $p{:}q$ run on w with the $q{:}p$ run on $\bar{w}$. At the start of every WRITE episode the two canonical stacks have the same length, the same alternating-prefix length $2\min(x,y)$, and complementary unary tails. The boundary guard prevents an outer WRITE while that mixed prefix is still present, so both runs necessarily pop the entire mixed prefix. If v ones occur in the original input during those $2\min(x,y)$ steps, then the original debt is $v - \min(x,y)$, whereas the complemented run has debt $2\min(x,y) - v - \min(x,y)$; the two debts are therefore opposite exactly when the unary tails begin. From that point onward the visible unary symbols are complementary and the coupled input symbols are complementary, so MATCH/mismatch status agrees and the debts remain opposite. Hence a positive outer WRITE in one run is paired with a negative outer WRITE at the same time in the other. If no outer WRITE occurs, the equal-length stacks empty simultaneously, so the next input is a bottom WRITE in both runs. Induction over WRITE episodes proves equality of bottom- and outer-WRITE indicators at every time, yielding (78.31). The complement map is a bijection between $\mathcal{F}_{gq,gp}$ and $\mathcal{F}_{gp,gq}$. Finally, (78.32) follows from the bottom-WRITE Rayleigh scale or directly from the mirrored invariant measures. □

78.5 Near-diagonal universality

Let

$$p = q+1, \qquad m = 2q+1, \qquad h = q.$$

The central physical-lift fibers simplify sharply.

Proposition 78.10 (Near-diagonal central fibers)**.** *At block boundaries, for $0 \le z \le q$ the central semantic shell z has only the empty lift*

$$E_z = (D_z, \varnothing).$$

For $z = -r$, $1 \le r \le q$, the only central lifts are

$$E_{-r} = (D_{-r}, \varnothing), \qquad F_{-r} = (D_{q-r}, 1^m). \tag{78.33}$$

Proof. The proof is by most-recent-WRITE analysis. A positive outer WRITE has

$$\mu((q+1)m) = (m, 0),$$

so it creates only a zero tail. Before the next WRITE, a nonempty block-boundary configuration on that tail has stack 0^{km} for some $k \geq 1$ and nonnegative debt, hence semantic shell at least $(q+1)k \geq q+1$. It cannot lie in the central range.

A negative outer WRITE has

$$\mu(-(q+1)m) = (q-1, 3q+3),$$

so its mixed prefix has length $2(q-1) < m$ and its final tail is unary one. Let a later block-boundary suffix have length km and contain a zeros. Debt was reset to zero at the WRITE, and positive debt can subsequently be created only by one-over-zero mismatches that pop zero cells. Since the written word contained only $q-1$ zeros, the current debt satisfies

$$d \leq (q-1) - a.$$

Hence its semantic shell obeys

$$z = a + d - qk \leq q - 1 - qk.$$

If $k \geq 2$, then $z \leq -q-1$, outside the central range. Thus a central nonempty suffix forces $k = 1$; since the mixed prefix is shorter than m, the surviving length-m suffix is exactly 1^m.

It remains to inspect bottom WRITEs. Both possible bottom-WRITE residuals have the form

$$r = jm + q, \qquad -q-1 \leq j \leq q.$$

If $\mu(r) = (x, y)$ and $\ell = x + y$, then

$$mx - q\ell = jm + q,$$

so $\ell \equiv -1 \pmod{m}$. Write $\ell = km - 1$. Solving gives

$$x = j + qk, \qquad y = k(q+1) - 1 - j.$$

For $-q \leq j \leq q$, the minimum is $k = 1$, so the newly written word has length $m-1$ and is completely consumed by the remaining $m-1$ symbols of the block. The single exceptional value $j = -q-1$ requires $k = 2$; it occurs exactly for input zero from E_{-q}, and

$$\mu(r) = (q-1, 3q+2).$$

Its canonical word is $(10)^{q-1}1^{m+2}$, so after the remaining $m-1$ pops the next block boundary contains precisely 1^m. These cases give exactly (78.33). □

Let

$$D_q = \sum_{r=1}^{q} \nu(F_{-r}).$$

Since every central semantic shell has total invariant mass one,

$$\kappa_{q+1,q} = 2q + 1 - D_q. \tag{78.34}$$

Lemma 78.11 (Near-diagonal predecessor bound)**.** *If a block ends in one of the nonempty central lifts F_{-r}, then the preceding semantic shell satisfies*

$$Z_j \leq -q.$$

Proof. If the block contains a WRITE, consider its last WRITE. A positive outer WRITE cannot produce a one-tail. A negative outer WRITE inserts a word of length $2m$ before its surviving suffix, so with at most $m-1$ input symbols left in the block there are too few subsequent pops for the final stack to be exactly 1^m. A bottom WRITE can occur only on the first symbol of a block by Lemma 77.1, and the only such case that can end in a nonempty central lift is the transition from E_{-q}. It remains to consider a block with no WRITE. Then every one of the m input steps pops an old cell, so the starting stack has the form $T1^m$ with $|T| = m$. Let a be the number of zeros in this stack, and let x_0 be the zero-count of the canonical word produced by the most recent WRITE. Because at least m trailing ones survive and the demand is kernel-free, $x_0 \le q-1$. Positive debt can have increased only by popping zero cells through one-over-zero mismatches, so if the current debt is d then

$$d \le x_0 - a.$$

The residual shell of a length-$2m$ stack is

$$Z_j = a + d - 2q \le x_0 - 2q \le -q - 1.$$

This proves the claim. □

Let

$$Y_q = K - (q+1), \qquad K \sim \mathrm{Bin}\left(2q+1, \frac{q+1}{2q+1}\right).$$

Then

$$\mathbb{E}Y_q = 0, \qquad \mathrm{Var}(Y_q) = \frac{q(q+1)}{2q+1}.$$

By Lemma 78.11 and unit shell mass,

$$\begin{aligned} D_q &\le \sum_{z \le -q} \mathbb{P}\{z + Y_q \in [-q,-1]\} \\ &\le \mathbb{E}\big[\min(Y_q+1, q)\mathbf{1}_{\{Y_q \ge 0\}}\big] \\ &\le 1 + \mathbb{E}Y_q^+ \\ &\le 1 + \frac{1}{2}\sqrt{\frac{q(q+1)}{2q+1}}. \end{aligned} \tag{78.35}$$

Theorem 78.12 (Near-diagonal universal Rayleigh scale)**.** *For $p = q+1$,*

$$\boxed{\frac{\kappa_{q+1,q}}{\sqrt{q(q+1)}} \longrightarrow 2.} \tag{78.36}$$

More quantitatively,

$$2\sqrt{\frac{q}{q+1}} - \frac{1}{2\sqrt{2q+1}} \le \frac{\kappa_{q+1,q}}{\sqrt{q(q+1)}} \le \frac{2q+1}{\sqrt{q(q+1)}}. \tag{78.37}$$

Consequently, after first taking $g \to \infty$ and then $q \to \infty$,

$$\frac{B_g}{\sqrt{(2q+1)g}} \Longrightarrow \mathrm{Rayleigh}(2).$$

Proof. Combine (78.34) with (78.35) and divide by $\sqrt{q(q+1)}$. Both sides of (78.37) tend to two. The final statement follows from Theorem 77.6. □

78.6 Finite physical-lift kernels and explicit computation of $\kappa_{p,q}$

The distributional results above require only the invariant-measure definition of $\kappa_{p,q}$. We now explain why the same constant is effectively computable from a finite matrix-kernel system.

78.6.1 Affine shortest-demand tails

At block boundaries the shortest demands have explicit forms. For $z \geq 0$,

$$\mu(mz) = \left(z + q\left\lceil \frac{z}{p} \right\rceil, p\left\lceil \frac{z}{p} \right\rceil - z\right). \tag{78.38}$$

For $n \geq 0$,

$$\mu(-mn) = \left(q\left\lceil \frac{n}{q} \right\rceil - n, n + p\left\lceil \frac{n}{q} \right\rceil\right). \tag{78.39}$$

Hence

$$\mu(m(z+p)) = \mu(mz) + (m, 0), \tag{78.40}$$
$$\mu(-m(n+q)) = \mu(-mn) + (0, m). \tag{78.41}$$

Thus every p positive semantic shells add one unary packet 0^m, while every q negative shells add one unary packet 1^m.

Proof of (78.38)–(78.41). For $mz > 0$, all integer solutions of $px - qy = mz$ are

$$x = z + qk, \qquad y = pk - z.$$

The least k making both coordinates nonnegative is $k = \lceil z/p \rceil$, giving (78.38). The negative formula is symmetric. The affine identities follow by replacing z by $z + p$ and n by $n + q$. □

By Lemma 76.3, a reachable stack is a suffix of a canonical shortest-demand word. The mixed alternating head has bounded length because every freshly written shortest demand has minority count at most h; the only unbounded part is a unary tail. Write the unary-tail length as

$$L = m\ell + r, \qquad 0 \leq r < m.$$

Modulo complete unary packets a^m, the remaining information consists of the debt state, the unary symbol, the bounded mixed head, and r. There are finitely many possibilities.

Theorem 78.13 (Finite physical-lift kernel)**.** *For every coprime p, q, there exist finite positive- and negative-tail phase sets and finite families of rational matrices $\{M_j^+\}$ and $\{M_j^-\}$ such that the invariant shell vectors $v_\ell^\pm$ satisfy, for all sufficiently deep levels,*

$$v_\ell^\pm = \sum_j v_{\ell-j}^\pm M_j^\pm. \tag{78.42}$$

The corresponding matrix Laurent polynomials

$$\mathcal{K}_\pm(\lambda) = I - \sum_j M_j^\pm \lambda^j \tag{78.43}$$

have rational coefficients. Together with finitely many central balance equations and the normalization $\nu(o) = 1$, the bounded nonnegative solutions of (78.42) *uniquely determine $\kappa_{p,q}$.*

Proof. The affine identities (78.40)–(78.41) show that translation in a deep semantic tail adds only complete unary m-packets. The canonical-segment property bounds the mixed head independently of level. Quotienting the unary length modulo m therefore leaves finitely many physical phases and one integer level. During one m-symbol input block, the level can change by only a bounded amount, and once the tail is sufficiently deep the transition probabilities depend only on the phase and the level increment, not on the absolute level. Under the critical law every block word has rational probability $(p/m)^k(q/m)^{m-k}$, so the matrices are rational. The invariant equation in the homogeneous tail is exactly (78.42). Irreducibility and the normalization $\nu(o) = 1$ make the recurrent invariant measure unique, hence the central empty-state coordinates and $\kappa_{p,q}$ are uniquely determined. □

This is a finite-phase, level-homogeneous matrix-analytic structure of the same general type that underlies quasi-birth-and-death and matrix-geometric methods [73]. The matrix-analytic machinery itself is classical; the point here is that the shortest-demand pushdown quotient canonically produces such a kernel whose central boundary values encode physical LIFO rematerialization.

78.6.2 Small explicit kernel examples

The two smallest asymmetric $q = 1$ cases reduce to scalar tail recurrences. We record their eliminations as compact reproducibility checks, and then give the first genuinely multi-phase example at 3:2.

The ratio 2:1. In negative shell $-n$ there are two deep phases,

$$A_n = (D_{-1}, 1^{3(n-1)}), \qquad B_n = (D_0, 1^{3n}).$$

Write $a_n = \nu(A_n)$, $b_n = \nu(B_n)$ and $u_n = a_n - b_n$. The unit-shell identity gives $a_n + b_n = 1$. Exact enumeration of the eight critical three-symbol blocks gives

$$u_{n-2} - 6u_{n-1} + 39u_n - 8u_{n+1} = 0 \tag{78.44}$$

in the homogeneous tail, while the finite central balance equations are

$$31u_1 - 8u_2 = 15, \qquad 4u_1 - 39u_2 + 8u_3 = 3. \tag{78.45}$$

The characteristic polynomial is

$$Q_{2,1}(R) = 8R^3 - 39R^2 + 6R - 1.$$

It has a unique real root $R > 1$. Boundedness of the invariant lift profile cancels that mode, which is equivalent to

$$u_1(31R^2 - 4R + 1) = 15R^2 - 3R.$$

With $\theta = (1 + u_1)/2 = a_1$, elimination of R between this relation and $Q_{2,1}(R) = 0$ gives

$$8\theta^3 - 3\theta - 1 = 0. \tag{78.46}$$

Thus

$$\kappa_{2,1} = 2 + \theta = 2.737843258897860\ldots. \tag{78.47}$$

The ratio 3:1. The same two negative phases occur with unary packet length four. Writing again $u_n = a_n - b_n$, exact sixteen-block enumeration gives

$$81u_{n+1} - 364u_n + 54u_{n-1} - 12u_{n-2} + u_{n-3} = 0, \tag{78.48}$$

and the finite central equations are

$$283u_1 - 81u_2 = 148, \tag{78.49}$$
$$-27u_1 + 364u_2 - 81u_3 = -40, \tag{78.50}$$
$$9u_1 - 54u_2 + 364u_3 - 81u_4 = 4. \tag{78.51}$$

The characteristic polynomial is

$$Q_{3,1}(R) = 81R^4 - 364R^3 + 54R^2 - 12R + 1.$$

Cancelling its unique unstable real mode gives

$$u_1(81R^3 - 81R^2 + 27R - 3) = 148R^2 - 40R + 4.$$

Eliminating R and setting $\theta = (1 + u_1)/2$ yields

$$54\theta^4 - 18\theta^2 - 8\theta - 1 = 0, \tag{78.52}$$

so

$$\kappa_{3,1} = 2 + \theta = 2.750694578264632\ldots. \tag{78.53}$$

The ratio 3:2. Here the deep negative tail has seven phases. Writing the negative semantic shell as $-n$, the four odd-shell phases for $n = 2j + 1$ are

$$\begin{aligned} O_0(j) &= (D_{-1}, 1^{5j}), \\ O_1(j) &= (D_0, 0\,1^{5j+4}), \\ O_2(j) &= (D_0, 10\,1^{5j+3}), \\ O_3(j) &= (D_1, 1^{5(j+1)}), \end{aligned}$$

and the three even-shell phases for $n = 2j$ are

$$\begin{aligned} E_0(j) &= (D_{-2}, 1^{5(j-1)}), \\ E_1(j) &= (D_{-1}, 0\,1^{5j-1}), \\ E_2(j) &= (D_0, 1^{5j}). \end{aligned}$$

Enumerating the 32 five-symbol blocks with their exact critical probabilities $3^k 2^{5-k}/5^5$ gives a rational seven-phase Laurent kernel whose determinant factors as

$$\det \mathcal{K}_{3,2}(\lambda) = \frac{(\lambda - 1)^2 C_3(\lambda) K_5(\lambda)}{5^{20}\lambda^6}, \tag{78.54}$$

where

$$\begin{aligned} C_3(\lambda) &= 59049\lambda^3 - 1531872\lambda^2 + 9472\lambda - 1024, \\ K_5(\lambda) &= 59049\lambda^5 - 1387530\lambda^4 + 16632265\lambda^3 \\ &\quad - 1448160\lambda^2 + 11520\lambda + 1024. \end{aligned}$$

The critical phase kernel has invariant vector

$$\left(\frac{1}{4}, \frac{3}{50}, \frac{1}{10}, \frac{9}{100}, \frac{1}{4}, \frac{1}{25}, \frac{21}{100}\right) \tag{78.55}$$

in the order $(O_0, O_1, O_2, O_3, E_0, E_1, E_2)$. Coupling this exact tail kernel to the finite central balance equations gives numerically

$$\kappa_{3,2} = 4.64089572244120\ldots. \tag{78.56}$$

An independent direct probability propagation, extrapolated from block horizons 100 and 200, gives $4.6408956068\ldots$, independently confirming the first six decimal places. The rematerialization constant need not be rational even though the underlying transition probabilities are rational: already the cubic (78.46) has no rational root, so $\kappa_{2,1}$ is irrational.

Remark 78.14 (Algebraicity and regular kernels). The finite kernels provide an exact rational boundary-value problem for $\kappa_{p,q}$. When the associated matrix Laurent pencil is regular, companion linearization shows that its characteristic modes are algebraic, and the finite central boundary equations then make the normalized bounded solution algebraic. This applies to the explicit kernels above. We do not require, and do not use, an unconditional algebraicity theorem for every coprime p, q in the probabilistic results of this paper.

Part XVIII

Optimal-Fiber Topology and Physical Holonomy

Research provenance. Earlier Parts introduced higher-dimensional minimum-completion fibers and showed how equal-length kernel trades create nontrivial quotient topology. This Part sharpens that picture through the complete active-rank-two classification and then carries the topology into a category-relative physical transport theory. Kernel rank controls when higher topology can appear; the global homotopy type is governed by the translation geometry of the active optimal fiber, while the displacement kernel identifies the pure loops on which physical LIFO holonomy can live.

Chapter 79

Optimal-Fiber Compression and Active Trade Geometry

For an integer counting matrix $A \in \mathbb{Z}^{k\times d}$ and completable residual r, write

$$\ell_A(r) = \min\{\mathbf{1}^\top x : x \in \mathbb{N}^d,\ Ax = -r\}, \qquad \mathcal{O}_A(r) = \{x : Ax = -r,\ \mathbf{1}^\top x = \ell_A(r)\}.$$

The earlier higher-dimensional completion theory supplied the filled shortest-prefix quotient. The purpose here is to compress its global topology to finite optimal-fiber translation data and then determine what active rank two can actually support.

79.1 Minimum-completion geometry

Let $A \in \mathbb{Z}^{k\times d}$. A residual is an element $r \in A\mathbb{Z}^d$ for which the completion equation

$$Ax = -r, \qquad x \in \mathbb{N}^d$$

is feasible. A convenient global hypothesis is the following.

Definition 79.1 (Completion totality)**.** The matrix A is *completion-total* if there exists $h \in \mathbb{N}^d$ with every coordinate positive and $Ah = 0$.

If A is completion-total, every lattice residual is completable: if $r = Az$ with $z \in \mathbb{Z}^d$, then $th - z \in \mathbb{N}^d$ for all sufficiently large t, and $A(th - z) = -r$.

For a completable residual r define

$$\ell_A(r) = \min\{\mathbf{1}^\top x : x \in \mathbb{N}^d,\ Ax = -r\} \tag{79.1}$$

and the optimal Parikh fiber

$$\mathcal{O}_A(r) = \{x \in \mathbb{N}^d : Ax = -r,\ \mathbf{1}^\top x = \ell_A(r)\}. \tag{79.2}$$

The set is finite because all of its points have the same coordinate sum.

The equal-length kernel is

$$K_A = \ker_{\mathbb{Z}} \begin{bmatrix} A \\ \mathbf{1}^\top \end{bmatrix} = \{g \in \mathbb{Z}^d : Ag = 0,\ \mathbf{1}^\top g = 0\}. \tag{79.3}$$

Differences of optimal completions lie in K_A.

79.1.1 The shortest-prefix down-set

For $x \in \mathbb{N}^d$ put

$$B_x = \{u \in \mathbb{N}^d : 0 \le u \le x\},$$

where inequalities are coordinatewise, and define

$$\mathcal{C}_A(r) = \downarrow \mathcal{O}_A(r) = \bigcup_{x \in \mathcal{O}_A(r)} B_x. \tag{79.4}$$

Proposition 79.2 (Exact geodesic-prefix criterion)**.** *For $u \in \mathbb{N}^d$,*

$$u \in \mathcal{C}_A(r) \quad \Longleftrightarrow \quad \ell_A(r + Au) = \ell_A(r) - \mathbf{1}^\top u.$$

Thus $\mathcal{C}_A(r)$ is exactly the set of Parikh vectors that occur as prefixes of shortest completions of r.

Proof. If $u \le x$ for some $x \in \mathcal{O}_A(r)$, then $x - u$ completes $r + Au$ and has length $\ell_A(r) - \mathbf{1}^\top u$. A strictly shorter completion would combine with u to contradict optimality of x.

Conversely, if the displayed equality holds and y is a shortest completion of $r + Au$, then $u + y$ completes r and has length $\ell_A(r)$. Hence $u + y \in \mathcal{O}_A(r)$ and $u \in \mathcal{C}_A(r)$. □

Give every finite down-set $D \subset \mathbb{N}^d$ its standard cubical realization $|D|$: a J-oriented elementary cube based at u is present when $u + \mathbf{1}_J \in D$. The down-set condition supplies all faces automatically.

Two equally oriented cells in $|\mathcal{C}_A(r)|$ are identified when their base vertices differ by an element of K_A. We write

$$I_A(r) = |\mathcal{C}_A(r)|/K_A \tag{79.5}$$

for the resulting filled shortest-prefix quotient. By theorem 79.2, its vertices are precisely shortest-path residual states, and its higher cubes record commuting one-letter shortest moves.

79.2 Active trades and optimal-fiber lattice geometry

The full kernel K_A can contain directions that never occur inside the optimal fiber for a fixed residual. The correct residual-dependent lattice is smaller.

Definition 79.3 (Active trade lattice)**.** Fix a completable residual r. Define

$$L_A(r) = \langle x - y : x, y \in \mathcal{O}_A(r) \rangle_{\mathbb{Z}} \subseteq K_A. \tag{79.6}$$

For $x_0 \in \mathcal{O}_A(r)$ define the normalized optimal fiber

$$P_A(r) = \mathcal{O}_A(r) - x_0 \subseteq L_A(r). \tag{79.7}$$

Its active rank is $\operatorname{rank}_{\mathbb{Z}} L_A(r)$.

Lemma 79.4 (Active collision lemma)**.** *If $u, v \in \mathcal{C}_A(r)$ and $u - v \in K_A$, then*

$$u - v \in \mathcal{O}_A(r) - \mathcal{O}_A(r) \subseteq L_A(r).$$

Proof. The condition $u - v \in K_A$ implies

$$Au = Av, \qquad \mathbf{1}^\top u = \mathbf{1}^\top v.$$

Hence u and v reach the same residual at the same geodesic depth. Choose a shortest completion y of that common residual. By theorem 79.2, both $u + y$ and $v + y$ lie in $\mathcal{O}_A(r)$. Their difference is $u - v$. □

Proposition 79.5 (Active-fiber saturation)**.** *For every $x_0 \in \mathcal{O}_A(r)$,*

$$\mathcal{O}_A(r) = (x_0 + L_A(r)) \cap \mathbb{N}^d. \tag{79.8}$$

Consequently

$$P_A(r) = \operatorname{conv}(P_A(r)) \cap L_A(r). \tag{79.9}$$

Proof. The inclusion $\mathcal{O}_A(r) \subseteq (x_0 + L_A(r)) \cap \mathbb{N}^d$ follows from the definition of $L_A(r)$. Conversely, if $x = x_0 + \lambda \in \mathbb{N}^d$ with $\lambda \in L_A(r) \subseteq K_A$, then

$$Ax = Ax_0 = -r, \qquad \mathbf{1}^\top x = \mathbf{1}^\top x_0 = \ell_A(r),$$

so $x \in \mathcal{O}_A(r)$.

For the second statement, put $P = P_A(r)$. Equation (79.8) says

$$P = L_A(r) \cap \{z \in \mathbb{R}^d : z_i \geq -(x_0)_i \ \forall i\}.$$

The right-hand real region is convex. Since $\operatorname{conv}(P)$ lies inside it, every lattice point of $\operatorname{conv}(P)$ belongs to P. □

Thus every normalized optimal fiber is a finite lattice-convex configuration in its active lattice. This constraint is decisive in active rank two.

79.3 Optimal-fiber nerve compression

Let

$$X = |\mathcal{C}_A(r)|, \qquad L = L_A(r).$$

Form the periodic saturation

$$\widetilde{X} = \bigcup_{\lambda \in L} (X + \lambda). \tag{79.10}$$

By theorem 79.4, the quotient $\widetilde{X}/L$ is canonically the same cell quotient as $I_A(r)$.

Lemma 79.6 (Contractible translate intersections)**.** *Every nonempty finite intersection*

$$\bigcap_{\lambda \in F} (X + \lambda), \qquad F \subset L \text{ finite},$$

is contractible.

Proof. Work first on vertices. Put $C = \mathcal{C}_A(r)$ and

$$D_F = \bigcap_{\lambda \in F} (C + \lambda).$$

Assume $D_F \neq \varnothing$ and set

$$m_i(F) = \max_{\lambda \in F} \lambda_i.$$

Every $u \in D_F$ satisfies $u \geq m(F)$ coordinatewise: for each coordinate choose a translate attaining the maximum. If $0 \leq z \leq u - m(F)$, then for every $\lambda \in F$,

$$0 \leq z + m(F) - \lambda \leq u - \lambda.$$

Since $u - \lambda \in C$ and C is a down-set, $z + m(F) - \lambda \in C$. Thus $D_F - m(F)$ is a finite coordinate down-set in $\mathbb{N}^d$. Its cubical realization is a union of coordinate boxes anchored at the origin and is therefore contractible. Translating back proves the claim. □

Choose $x_0 \in \mathcal{O}_A(r)$ and write $P = P_A(r)$. Define the periodic simplicial complex

$$\mathcal{N}(P) = \{F \subseteq L : \ F \text{ finite and } \exists c \in L \text{ with } c - F \subseteq P\}. \tag{79.11}$$

Equivalently, $\mathcal{N}(P)$ is the nerve of the family of finite convex-lattice sets

$$\{\lambda + P : \lambda \in L\}.$$

The lattice L acts by translation on $\mathcal{N}(P)$.

Theorem 79.7 (Optimal-Fiber Nerve Compression)**.** *For every completable residual r,*

$$I_A(r) \simeq \mathcal{N}(P_A(r))/L_A(r). \tag{79.12}$$

The homotopy type of the global shortest-prefix quotient is therefore determined by the finite optimal fiber and its translation lattice.

Proof. The translates $\{X + \lambda\}_{\lambda \in L}$ cover $\widetilde{X}$ and are permuted by the L-action. By theorem 79.6, every nonempty finite intersection is contractible. The equivariant simplicial nerve theorem [79, Theorem 2.8] gives an L-equivariant homotopy equivalence between $\widetilde{X}$ and the nerve of this cover.

It remains to identify that nerve with (79.11). A finite $F \subset L$ spans a simplex exactly when there exists a common vertex u with

$$u - \lambda \in \mathcal{C}_A(r) \qquad (\lambda \in F).$$

All points $u-\lambda$ have the same residual and the same geodesic depth because their pairwise differences lie in $L \subseteq K_A$. Choose a shortest suffix y from that common residual. Then

$$c = u + y$$

satisfies $c - \lambda \in \mathcal{O}_A(r)$ for every $\lambda \in F$. Conversely, if $c - F \subseteq \mathcal{O}_A(r)$ then c belongs to every translated shortest-prefix set. After translating by x_0, this is exactly the simplex criterion (79.11).

Finally, the L-action is free and cellular, so passage to quotients preserves the equivariant homotopy equivalence. Using theorem 79.4, $\widetilde{X}/L$ is $I_A(r)$. □

Remark 79.8 (Rank-one rigidity). If $\operatorname{rank} L_A(r) = 1$, lattice-convexity forces $P_A(r)$ to be an interval of consecutive points in a rank-one lattice. If the fiber is nontrivial, $\mathcal{N}(P_A(r))$ is contractible and its translation quotient is S^1. Thus the first non-rigid regime is active rank two.

79.4 Finite presentations and first homology

The compressed quotient has one vertex orbit. Its 2-skeleton can be read directly from differences in the optimal fiber.

Proposition 79.9 (Finite fundamental-group presentation)**.** *Assume $0 \in P = P_A(r)$. Then $\pi_1(I_A(r))$ is isomorphic to*

$$G(P) = \left\langle X_p \ (p \in P), \ X_0 = 1 \ \middle| \ X_p^{-1} X_q = X_s^{-1} X_t \ \textit{whenever } q - p = t - s \right\rangle. \tag{79.13}$$

Proof. In $\mathcal{N}(P)/L$, oriented edge orbits are indexed by differences $d = q - p \in P - P$. Write the corresponding edge generator as a_d, with $a_{-d} = a_d^{-1}$. Every triple $p, q, s \in P$ supplies the triangular relation

$$a_{q-p} a_{s-q} = a_{s-p}.$$

Set $X_p = a_p$ for $p \in P$ and $X_0 = 1$. The triangle on $0, p, q$ gives

$$a_{q-p} = X_p^{-1} X_q.$$

Two edges represent the same edge orbit exactly when their differences agree, which gives precisely the relations in (79.13). Higher simplices do not affect the fundamental group. $\square$

There is a canonical displacement homomorphism

$$\delta : G(P) \longrightarrow L, \qquad \delta(X_p) = p. \tag{79.14}$$

It is surjective because $L = \langle P - P \rangle = \langle P \rangle$.

Corollary 79.10 (Parallelogram–commutator principle)**.** *If*

$$p, \quad p+u, \quad p+v, \quad p+u+v \in P,$$

then the corresponding difference generators satisfy

$$[a_u, a_v] = 1.$$

Proof. The two triangles give

$$a_u a_v = a_{u+v} = a_v a_u.$$

$\square$

79.4.1 Freiman dimension and first homology

Let F be a field of characteristic 0. A normalized Freiman 2-homomorphism on P is a function $f : P \to F$ with $f(0) = 0$ such that

$$p + q = s + t \quad \Longrightarrow \quad f(p) + f(q) = f(s) + f(t).$$

Let $\dim_F(P)$ denote the dimension of this vector space. This agrees with the usual normalized Freiman-dimension convention.

Proposition 79.11 (First homology as the Freiman shadow)**.** *For every finite normalized fiber P,*

$$\beta_1(\mathcal{N}(P)/L) = \dim_F(P).$$

Proof. Abelianizing (79.13) replaces X_p by additive variables x_p and the defining relations by

$$x_q - x_p = x_t - x_s \quad \text{whenever} \quad q - p = t - s.$$

Equivalently,

$$x_p + x_t = x_q + x_s \quad \text{whenever} \quad p + t = q + s.$$

After $x_0 = 0$, this is exactly the vector space of normalized Freiman homomorphisms. Tensoring H_1 with F gives the stated equality of ranks. $\square$

Theorem 79.12 (Rank-two first-Betti bound)**.** *If* $\operatorname{rank} L_A(r) = 2$*, then*

$$\beta_1(I_A(r)) \in \{2, 3\}.$$

Proof. Put $P = P_A(r)$, $m = |P|$, and $Q = \mathrm{conv}(P)$ inside $L \otimes \mathbb{R} \cong \mathbb{R}^2$. By theorem 79.5, $P = Q \cap L$. Let B be the number of lattice points on ∂Q. Pick's theorem and the Ehrhart polynomial give

$$|2Q \cap L| = 4m - B - 3 \le 4m - 6. \tag{79.15}$$

Since $P + P \subseteq 2Q \cap L$,

$$|P + P| \le 4m - 6.$$

If $d_F = \dim_F(P)$, Freiman's dimension lemma [81, Chapter 5] gives

$$|P + P| \ge (d_F + 1)m - \binom{d_F + 1}{2}.$$

For $d_F \ge 4$ and $m \ge 5$, the right side is at least $5m - 10 > 4m - 6$. If $m \le 4$, then trivially $d_F \le m - 1 \le 3$. Hence $d_F \le 3$.

On the other hand, the two coordinate linear functionals on the full-rank planar lattice L give two independent normalized Freiman homomorphisms, so $d_F \ge 2$. Apply theorem 79.11 and theorem 79.7. □

79.5 The optimal-fiber Taylor slice

The nerve model has a second description in commutative algebra. This description is classical on the lattice-ideal side; the point here is that the minimum-completion quotient selects a canonical principal slice.

Fix a field $\Bbbk$, put

$$S = \Bbbk[x_1, \ldots, x_d],$$

and let $L = L_A(r) \subseteq \mathbb{Z}^d$. Because $L \subseteq K_A$, every $\lambda \in L$ has coordinate sum 0, hence

$$L \cap \mathbb{N}^d = \{0\}.$$

Define the lattice ideal

$$I_L = \langle x^{u^+} - x^{u^-} : u \in L \rangle \subseteq S. \tag{79.16}$$

Give S the $\mathbb{Z}^d/L$ grading $\deg(x^a) = a + L$. For $\alpha \in \mathbb{N}^d/L$ write

$$\mathrm{fiber}(\alpha) = \{x^a : a \in \mathbb{N}^d,\ a + L = \alpha\}.$$

Choose $x_0 \in \mathcal{O}_A(r)$ and set

$$\alpha = x_0 + L.$$

By theorem 79.5,

$$\mathrm{fiber}(\alpha) \quad \text{has exponent set exactly } \mathcal{O}_A(r). \tag{79.17}$$

For each degree β, let $E_i(\beta)$ be the family of i-element subsets of $\mathrm{fiber}(\beta)$ with monomial gcd equal to 1. Bayer–Sturmfels identify these sets as the homogeneous basis of the lattice-ideal Taylor resolution and give the differential explicitly [78, Proposition 3.10].

Write $\beta \preceq \alpha$ when $\alpha - \beta$ is the degree of a monomial.

Theorem 79.13 (Optimal-Fiber Taylor Slice). *Let*

$$Q(P) = \mathcal{N}(P_A(r))/L_A(r).$$

For every $j \ge 0$ there is a canonical basis correspondence

$$C_j(Q(P); \Bbbk) \cong \bigoplus_{\beta \preceq \alpha} \Bbbk\, E_{j+1}(\beta), \tag{79.18}$$

under which the cellular boundary is the Bayer–Sturmfels Taylor differential specialized by

$$\varepsilon : S \to \Bbbk, \qquad x_i \mapsto 1.$$

Equivalently, $C_\bullet(Q(P); \Bbbk)$ *is the dehomogenized principal degree slice of the lattice-ideal Taylor complex.*

Proof. Represent a j-simplex orbit of $Q(P)$ by a $(j+1)$-element subset

$$F = \{u_0, \dots, u_j\} \subseteq \mathcal{O}_A(r).$$

Let $g = \min(u_0, \dots, u_j)$ coordinatewise. Subtracting g gives a gcd-one set

$$\overline{F} = F - g$$

in the fiber of the degree

$$\beta = \alpha - \deg(x^g) \preceq \alpha.$$

If F is replaced by an L-translate, the gcd changes by the same translate and $\overline{F}$ is unchanged. Hence the construction depends only on the simplex orbit.

Conversely, let $I \in E_{j+1}(\beta)$ with $\beta \preceq \alpha$. Choose $g \in \mathbb{N}^d$ with $\alpha = \beta + \deg(x^g)$. Multiplying every monomial of I by x^g gives a subset of $\mathrm{fiber}(\alpha) = \mathcal{O}_A(r)$. A different choice g' differs from g by an element of L and therefore gives the same simplex orbit. The two constructions are inverse.

If one vertex is removed, the remaining set acquires a new common gcd h. Bayer–Sturmfels' differential is

$$\partial(I) = \sum_{m \in I} \mathrm{sign}(m, I)\ \gcd(I \setminus \{m\})\, [I \setminus \{m\}],$$

where the bracket denotes removal of that common factor. Under $x_i \mapsto 1$, every monomial coefficient becomes 1, leaving exactly the oriented simplicial boundary of the quotient. The shift $E_{j+1} \leftrightarrow C_j$ is the usual one: E_1 is homological degree 0 in the resolution of S/I_L. $\square$

79.5.1 Principal truncation and character twists

Let $\mathbb{T}_\bullet$ be the full lattice Taylor resolution and $\mathbb{F}_\bullet$ a minimal $\mathbb{Z}^d/L$-graded free resolution of S/I_L.

Lemma 79.14 (Principal truncation lemma)**.** *The full graded chain-homotopy equivalence between* $\mathbb{T}_\bullet$ *and* $\mathbb{F}_\bullet$ *restricts to a chain-homotopy equivalence on the principal degree order ideal*

$$\{\beta : \beta \preceq \alpha\}.$$

Proof. Comparison maps and homotopies may be chosen homogeneous of degree 0. A homogeneous map from a basis summand of degree β to one of degree γ has a monomial coefficient of degree $\beta - \gamma$, so it can be nonzero only when $\gamma \preceq \beta$. Hence every map and homotopy preserves every degree down-set, in particular the principal down-set below α. $\square$

Let $\chi : L \to \Bbbk^\times$ be a character. After a faithfully flat extension of the field if necessary, extend χ to a homomorphism

$$\widetilde{\chi} : \mathbb{Z}^d \to \overline{\Bbbk}^\times.$$

Let $\Bbbk_\chi$ be the corresponding rank-one local system on $Q(P)$.

Proposition 79.15 (Character-twisted Taylor slice)**.** *After a degree-dependent rescaling of basis vectors, the complex*

$$C_\bullet(Q(P); \Bbbk_\chi)$$

is isomorphic to the principal Taylor slice evaluated at

$$x_i = \widetilde{\chi}(e_i).$$

Proof. For every $\beta \preceq \alpha$, choose a lift $g_\beta \in \mathbb{N}^d$ with

$$\alpha = \beta + \deg(x^{g_\beta}).$$

Suppose a Taylor boundary term deletes one vertex, produces common gcd exponent h, and passes from degree β to γ. The corresponding face in the periodic nerve differs from the chosen representative in degree γ by the deck translation

$$\tau = h + g_\beta - g_\gamma \in L.$$

After rescaling the degree-β basis by $\widetilde{\chi}(g_\beta)$, the evaluated Taylor coefficient becomes

$$\widetilde{\chi}(h)\frac{\widetilde{\chi}(g_\beta)}{\widetilde{\chi}(g_\gamma)} = \widetilde{\chi}(\tau) = \chi(\tau),$$

which is exactly the local-system coefficient attached to that face. The signs are the common simplicial incidence signs. □

Chapter 80

Lattice-Ideal Slices and Rank-Two Structural Rigidity

The compression theorem admits an algebraic refinement: the cellular chains of the quotient can be read as a principal dehomogenized slice of the Taylor resolution of the active lattice ideal. This bridge controls homology and prepares the planar classification.

80.1 Rank-two homological dimension

Assume from now on that

$$\operatorname{rank}_{\mathbb{Z}} L_A(r) = 2.$$

Peeva–Sturmfels prove the general projective-dimension bound

$$\operatorname{pd}_S(S/I_L) \leq 2^{\operatorname{rank} L} - 1,$$

so in rank two the minimal resolution has no term above homological degree 3 [76].

Lemma 80.1 (Torus-point injectivity)**.** *For every field $\mathbb{k}$ and every character $\chi : L \to \mathbb{k}^\times$, the third differential of the full minimal resolution, after the corresponding torus-point specialization, is injective.*

Proof. Extend scalars to an algebraic closure and represent χ by a point $\xi \in (\overline{\mathbb{k}}^\times)^d$.

If χ is nontrivial, then some lattice binomial evaluates nonzero at ξ, so ξ lies outside the support of S/I_L. The specialized full free resolution is therefore acyclic.

If χ is trivial, localize at the algebraic torus. For a lattice basis g_1, g_2 of L, the Laurent lattice ideal is generated by

$$x^{g_1} - 1, \qquad x^{g_2} - 1.$$

These two independent equations form a regular sequence in the regular Laurent ring. Hence the localized module is a codimension-two complete intersection and its Tor vanishes above degree 2. In both cases the third homology of the specialized full minimal resolution is zero. Since there is no fourth term, the third differential is injective. □

Theorem 80.2 (Rank-Two Homological-Dimension Theorem)**.** *If $\operatorname{rank} L_A(r) = 2$, then*

$$H_j(I_A(r); \mathbb{Z}) = 0 \qquad (j \geq 3), \tag{80.1}$$

and $H_2(I_A(r); \mathbb{Z})$ is free abelian.

Proof. Over any field $\Bbbk$, theorems 79.13 and 79.14 identify the chain complex of $Q(P)$ up to chain homotopy with the principal slice of a minimal free resolution of length at most 3. Thus homology vanishes above degree 3. By theorem 80.1, the third differential remains injective after restriction to the principal slice, so

$$H_j(Q(P); \Bbbk) = 0 \qquad (j \geq 3).$$

The same holds for every prime field and for $\mathbb{Q}$. Since $Q(P)$ is finite, the universal coefficient theorem implies (80.1). Applying the mod-p universal coefficient sequence in degree 3 gives

$$\mathrm{Tor}_1^{\mathbb{Z}}(H_2(Q(P); \mathbb{Z}), \mathbb{F}_p) = 0$$

for every prime p, so $H_2(Q(P); \mathbb{Z})$ has no torsion. Finally use theorem 79.7. □

80.2 Planar lattice geometry and the fundamental group

By theorem 79.5, a rank-two normalized fiber is a finite lattice-convex set in a rank-two lattice. We now show that a single geometric feature—a nondegenerate parallelogram—separates the abelian and free branches of the fundamental group.

80.2.1 The parallelogram branch

Lemma 80.3 (Minimal parallelogram lemma)**.** *Let $P \subset L \cong \mathbb{Z}^2$ be finite and lattice-convex. If P contains a nondegenerate lattice parallelogram, then it contains one of normalized area 1 or 2 whose edge directions are primitive.*

Proof. Choose a nondegenerate parallelogram

$$R = \{a, a+u, a+v, a+u+v\} \subseteq P$$

of minimum positive normalized area $D = |\det_L(u, v)|$. If u were not primitive, an interior lattice point on each of the two u-parallel edges would belong to P by lattice-convexity and would produce a smaller parallelogram. Thus u and v are primitive. Hence the boundary of $\mathrm{conv}(R)$ contains exactly four lattice points. Pick's theorem gives

$$I(R) = D - 1$$

interior lattice points.

Let c be the center of R. Reflection in c preserves the lattice and the parallelogram. If there is an interior lattice point $z \neq c$, its reflection $z' = 2c - z$ is another interior lattice point. One of the two parallelograms formed by the pair z, z' and one of the two opposite vertex pairs of R is nondegenerate; it lies strictly inside R and has smaller normalized area, contradicting minimality. Hence R has at most one interior lattice point, so $D - 1 \leq 1$. □

A finite lattice-convex set $X \subset L$ will be called *parity-complete* if it meets every coset of $2L$.

Lemma 80.4 (Observer chain)**.** *If $X \subsetneq P$ are finite lattice-convex subsets of L, there exists $z \in P \setminus X$ such that*

$$\mathrm{conv}(X \cup \{z\}) \cap L = X \cup \{z\}.$$

Iterating gives a chain from X to P by one-point lattice-convex extensions.

Proof. Choose $z \in P \setminus X$ minimizing the positive Euclidean distance to $\operatorname{conv}(X)$. If a further lattice point $y \notin X \cup \{z\}$ belonged to $\operatorname{conv}(X \cup \{z\})$, then $y \in P$ by lattice-convexity and, along a segment from z toward $\operatorname{conv}(X)$, would have strictly smaller distance to $\operatorname{conv}(X)$, a contradiction. $\square$

Lemma 80.5 (Parity propagation)**.** *Suppose $X \subseteq P$ is lattice-convex and parity-complete, and suppose that in $G(P)$ all X_x with $x \in X$ lie in the subgroup generated by commuting elements U, V whose displacements form a basis of L. If z is an observer of X, then X_z lies in the same subgroup, with exponent vector equal to the lattice coordinates of z.*

Proof. Choose $x \in X$ with $x \equiv z \pmod{2L}$ and put $y = (x+z)/2 \in L$. The observer property forces $y \in X$. Since

$$z - y = y - x,$$

relation (79.13) gives

$$X_y^{-1} X_z = X_x^{-1} X_y, \qquad X_z = X_y X_x^{-1} X_y.$$

If $X_x = U^a V^b$ and $X_y = U^c V^d$, commutativity gives

$$X_z = U^{2c-a} V^{2d-b},$$

whose displacement is $2y - x = z$. $\square$

Lemma 80.6 (Determinant-two parity completion)**.** *Let*

$$C_2 = \{(0,0), (1,0), (0,1), (-1,2), (0,2)\} \subset \mathbb{Z}^2.$$

If a lattice-convex set $P \supseteq C_2$ contains a point in the missing parity class $(1,1) + 2\mathbb{Z}^2$, then it contains at least one point of

$$M = \{(1,-1), (1,1), (-1,1), (-1,3)\}.$$

For every $m \in M$, the set $C_2 \cup \{m\}$ is lattice-convex and parity-complete, and the generator X_m is already a word in the generators from C_2.

Proof. Consider the linear map

$$\Phi(x,y) = (y, 2x + y).$$

It sends $\operatorname{conv}(C_2)$ to the square $[0,2]^2$. If z lies in the missing parity class, write

$$\Phi(z) = (s,t), \qquad A = s - 1, \quad B = t - 1.$$

Then A, B are even and $B - A \equiv 2 \pmod 4$, so $|A| \neq |B|$ and therefore

$$\big| |A| - |B| \big| \geq 2.$$

One of the four inequalities

$$B \geq |A| + 2, \quad -B \geq |A| + 2, \quad A \geq |B| + 2, \quad -A \geq |B| + 2$$

holds. In the first case, the point $(1,3)$ lies in $\operatorname{conv}([0,2]^2 \cup \{\Phi(z)\})$: indeed, with $\theta = 1/(B-1)$, write it as a convex combination of $\Phi(z)$ and a point of the top side using $|A| \leq B - 2$. The other three cases are symmetric. The four resulting points are precisely the images under Φ of the four elements of M. Since P is lattice-convex, the corresponding element of M lies in P.

A direct check of the four small polygons shows

$$\mathrm{conv}(C_2 \cup \{m\}) \cap \mathbb{Z}^2 = C_2 \cup \{m\}$$

for each $m \in M$, and the added point supplies the fourth parity class. Finally, in each case there are $p, a, b \in C_2$ with $m = p + b - a$. The relation

$$X_p^{-1} X_m = X_a^{-1} X_b$$

expresses X_m in the subgroup generated by C_2. □

Theorem 80.7 (Parallelogram–abelian fundamental group)**.** *Let $P = P_A(r)$ have active rank two. If P contains a nondegenerate lattice parallelogram, then*

$$\pi_1(I_A(r)) \cong L_A(r) \cong \mathbb{Z}^2.$$

Proof. By theorem 80.3, choose a minimal parallelogram of normalized area 1 or 2.

If the area is 1, translate one vertex to 0 and let u, v be its primitive edge directions. They form a basis of L, the four vertices represent all four classes in $L/2L$, and theorem 79.10 gives commuting basis generators X_u, X_v. The four-point core is parity-complete. Apply theorems 80.4 and 80.5 inductively to reach all of P.

If the area is 2, let the primitive edge directions be a, b. Modulo $2L$ they are equal, so the center

$$v = \frac{a+b}{2}$$

is a lattice point. With $u = a$, the pair u, v is a lattice basis and the five lattice points of the minimal parallelogram have coordinates

$$C_2 = \{0, u, v, 2v - u, 2v\}.$$

The relations

$$X_{2v} = X_v^2, \qquad X_{2v-u} = X_v X_u^{-1} X_v, \qquad X_u = X_{2v-u}^{-1} X_{2v}$$

force $[X_u, X_v] = 1$. This core represents three parity classes. If P contains only those three classes, theorems 80.4 and 80.5 again propagate the basis expressions to all of P. If the fourth class occurs, theorem 80.6 enlarges the core by one already-dependent generator to a parity-complete lattice-convex core, after which the same induction applies.

Thus $G(P)$ is generated by two commuting elements whose displacements are a basis of L. The displacement map (79.14) is therefore an isomorphism $G(P) \cong L$. Use theorems 79.7 and 79.9. □

80.2.2 The parallelogram-free branch

Call an affine lattice line ℓ *rich* if $|P \cap \ell| \geq 3$. Lattice-convexity implies that $P \cap \ell$ is a consecutive arithmetic progression in the primitive direction of ℓ.

Lemma 80.8 (Line-locality of repeated differences)**.** *If P contains no nondegenerate parallelogram and*

$$q - p = t - s$$

for two distinct ordered pairs of points of P, then the two segments lie on the same affine lattice line.

Proof. If the two segments lie on different parallel lines, the four endpoints form a nondegenerate parallelogram. If the lines are not parallel, equal displacement is impossible. □

A point lying on at least two rich lines will be called *shared*.

Lemma 80.9 (Transverse determinant lemma)**.** *Let P be lattice-convex and parallelogram-free. Suppose*

$$0, u, 2u, -v, v \in P$$

with primitive, nonparallel u, v. Then

$$|\det_L(u, v)| = 1.$$

Proof. Choose coordinates $u = (1, 0)$ and $v = (a, b)$ with $b > 0$ and $\gcd(a, b) = 1$. The triangle

$$T = \mathrm{conv}\{-v, v, 2u\}$$

lies in $\mathrm{conv}(P)$, hence every lattice point of T lies in P.

For $b \geq 3$, the intersection $T \cap (2u - T)$ has, in the u, v coordinate system, the form

$$\{xu + yv : |x - 1| + 2|y| \leq 1\}.$$

Take the ordinary lattice point $z = (m, 1)$ where m is nearest to $1 + a/b$. Then $y = 1/b$ and

$$|x - 1| = \mathrm{dist}(a/b, \mathbb{Z}) \leq \frac{\lfloor b/2 \rfloor}{b} \leq 1 - \frac{2}{b}.$$

Thus z and $2u - z$ both lie in $T \cap L \subseteq P$. They are not on the u-line, and

$$0 + 2u = z + (2u - z)$$

gives a nondegenerate parallelogram, a contradiction.

If $b = 2$, a lattice shear gives $v = (1, 2)$. Then the four points

$$(1, 2), \quad (0, -1), \quad (0, 0), \quad (1, 1)$$

lie in $T \cap L \subseteq P$ and satisfy

$$(1, 2) + (0, -1) = (0, 0) + (1, 1),$$

again a nondegenerate parallelogram. Hence $b = 1$. □

Lemma 80.10 (Endpoint-bridge exclusion)**.** *Let P be lattice-convex and parallelogram-free. A rich segment of lattice length at least 2 cannot have both endpoints interior to transverse rich lines.*

Proof. Write the rich segment as

$$0, u, 2u, \ldots, ku, \qquad k \geq 2,$$

and suppose the two endpoints are interior to transverse rich lines, so

$$-v, 0, v \in P, \qquad ku - w, ku, ku + w \in P.$$

By theorem 80.9,

$$|\det(u, v)| = |\det(u, w)| = 1.$$

Choose coordinates $u = (1, 0)$, $v = (0, 1)$. Then $w = (a, \varepsilon)$ with $\varepsilon = \pm 1$. Hence at the levels $y = 1$ and $y = -1$ the right transverse line contributes points with horizontal coordinates $k + a$ and $k - a$ (in some order). At least one of these is nonzero. On the same level the left transverse line contributes $(0, \pm 1)$. Convexity therefore forces a horizontal unit edge on that level. The main rich segment contains the horizontal unit edge from $(0, 0)$ to $(1, 0)$. The two equal unit edges form a nondegenerate parallelogram, a contradiction. □

Lemma 80.11 (Rich-line incidence)**.** *If P is lattice-convex and parallelogram-free, then every rich line contains at most two shared points; if it contains two, they are consecutive lattice points on that line.*

Proof. A point cannot be interior to two transverse rich lines: the four neighboring points have the same midpoint and form a nondegenerate parallelogram. Thus if a shared point p is interior to a rich line D, it is an endpoint of every transverse rich line through it. Let u be the primitive direction of D and let the transverse line contain $p, p+v, p+2v$. If D extended two steps from p in the same direction, then $p+2u, p+2v \in P$ and lattice-convexity would give

$$p+u+v = \frac{(p+2u)+(p+2v)}{2} \in P,$$

producing the parallelogram $p, p+u, p+v, p+u+v$. Hence a shared interior point can be at most one lattice step from each endpoint of D.

A point cannot be an endpoint of two transverse rich lines either: the two two-step extensions yield the same midpoint argument. Consequently, if a rich line had three shared points, its middle shared point would force the line to consist of exactly three consecutive points, while the two endpoints would be interior to transverse rich lines, contradicting theorem 80.10. If two shared points were nonconsecutive, neither could be an interior point of the rich line, so both would be endpoints and both would be interior to transverse rich lines, again contradicting theorem 80.10. ☐

Theorem 80.12 (Parallelogram-free homotopy theorem)**.** *Let $P = P_A(r)$ have active rank two and contain no nondegenerate lattice parallelogram. Then*

$$I_A(r) \simeq \bigvee^{\beta_1(I_A(r))} S^1. \tag{80.2}$$

In particular,

$$\pi_1(I_A(r)) \cong F_{\beta_1(I_A(r))}.$$

Proof. By theorem 80.8, every nontrivial positive-dimensional face identification in a fundamental simplex on P is supported inside the simplex on $P \cap \ell$ for some rich line ℓ. Let R be the union of these rich-line simplices together with the full 0-skeleton. Then the quotient $Q(P)$ is the pushout obtained from the full simplex $\Delta(P)$ by replacing R by its translation quotient $\overline{R}$. Since $\Delta(P)$ is contractible and $R \hookrightarrow \Delta(P)$ is a cofibration,

$$Q(P) \simeq \mathrm{Cofib}(R \to \overline{R}).$$

Distinct rich-line simplices meet only at shared vertices. By theorem 80.11, each rich line contains at most two such vertices, and if there are two they are consecutive. Hence R deformation retracts to a finite incidence graph. The quotient of a single rich-line simplex is the rank-one completion quotient, hence a circle; no positive-dimensional face is identified across distinct rich lines by theorem 80.8. Thus $\overline{R}$ retracts to a wedge of one circle for each rich line.

Choose a spanning forest in the incidence graph. In the cellular mapping cone of $R \to \overline{R}$, every non-tree incidence edge gives one 2-cell relation. Because the two shared points on the corresponding rich line are consecutive, its attaching word traverses that rich-line circle generator with coefficient ± 1. Choosing fundamental cycles with respect to the forest makes that circle generator unique to the corresponding non-tree relation. Successive elementary 2-cell/1-cell cancellations remove all 2-cells. The mapping cone therefore collapses to a finite graph. Its first Betti number is the first Betti number of $Q(P)$, so the graph is homotopy equivalent to the wedge in (80.2). Use theorem 79.7. ☐

Combining theorems 79.12, 80.7 and 80.12 gives the complete rank-two fundamental-group list

$$F_2, \qquad F_3, \qquad \mathbb{Z}^2.$$

80.3 The parallelogram branch and exact homotopy type

Assume $P = P_A(r)$ contains a nondegenerate parallelogram. By theorem 80.7,

$$\pi_1(Q(P)) \cong L \cong \mathbb{Z}^2.$$

Hence the periodic nerve

$$\widetilde{Q} := \mathcal{N}(P)$$

is the universal cover of $Q(P)$.

Let

$$R_L = \mathbb{Z}[L] \cong \mathbb{Z}[s^{\pm 1}, t^{\pm 1}], \qquad M = H_2(\widetilde{Q}; \mathbb{Z}).$$

The next result is the module-theoretic step that turns the homology classification into a homotopy classification.

Lemma 80.13 (Universal-cover higher homology)**.** *For active rank two,*

$$H_j(\mathcal{N}(P); \mathbb{Z}) = 0 \qquad (j \geq 3).$$

Proof. The complex $\mathcal{N}(P)$ is the nerve of the family of translates $\{\lambda + P\}_{\lambda \in L}$. Since $P = \operatorname{conv}(P) \cap L$, these are convex lattice sets in $L \cong \mathbb{Z}^2$. Alon–Kalai–Matoušek–Meshulam prove that the nerve of every finite family of convex lattice sets in $\mathbb{Z}^d$ is $(2^d - 1)$-collapsible [75]; in dimension two it is therefore 3-Leray. Any finite cellular cycle in $\mathcal{N}(P)$ is supported on a finite induced subcomplex, so homology in degree at least 3 vanishes. □

Proposition 80.14 (Projectivity of the π_2 module)**.** *In the parallelogram branch, M is a finitely generated projective R_L-module.*

Proof. The universal-cover cellular chains are finitely generated free R_L-modules in each degree, and R_L is Noetherian, so M is finitely generated.

Let $\mathfrak{m}$ be a maximal ideal of R_L and let $k(\mathfrak{m})$ be its residue field. Since the group elements of L are units in R_L, their images define a character

$$\chi_{\mathfrak{m}} : L \to k(\mathfrak{m})^{\times}.$$

For the free R_L-chain complex $C_\bullet(\widetilde{Q})$, the hyperhomology spectral sequence gives

$$E^2_{p,q} = \operatorname{Tor}_p^{R_L}(k(\mathfrak{m}), H_q(\widetilde{Q}; \mathbb{Z})) \Longrightarrow H_{p+q}(Q(P); k(\mathfrak{m})_{\chi_{\mathfrak{m}}}).$$

Because $\widetilde{Q}$ is simply connected, $H_1(\widetilde{Q}) = 0$, and by theorem 80.13 only the rows $q = 0, 2$ can be nonzero. The $q = 0$ row is the Koszul homology of the two augmentation elements $s - 1, t - 1$ after evaluation at the residue character; it vanishes in degrees above 2, and for a nontrivial character it is acyclic.

On the other hand, theorems 79.15 and 80.1 give

$$H_3(Q(P); k(\mathfrak{m})_{\chi_{\mathfrak{m}}}) = 0.$$

No differential can kill or create the term $E^2_{1,2}$, so

$$\operatorname{Tor}_1^{R_L}(k(\mathfrak{m}), M) = 0$$

for every maximal ideal $\mathfrak{m}$. Localizing, the finite $R_{L,\mathfrak{m}}$-module $M_\mathfrak{m}$ has vanishing first Tor with its residue field. A minimal local presentation and Nakayama's lemma therefore show that $M_\mathfrak{m}$ is free. Thus M is locally free at every maximal ideal, hence finitely generated projective. □

Theorem 80.15 (Parallelogram-branch homotopy theorem)**.** *Let $P = P_A(r)$ have active rank two and contain a nondegenerate lattice parallelogram. Put*

$$b = \beta_2(I_A(r)).$$

Then $b \geq 1$ and

$$I_A(r) \simeq T^2 \vee \bigvee^{b-1} S^2. \tag{80.3}$$

Proof. By theorem 80.7, $\pi_1(Q(P)) \cong \mathbb{Z}^2$. The Hopf exact sequence therefore gives a surjection

$$H_2(Q(P);\mathbb{Z}) \twoheadrightarrow H_2(\mathbb{Z}^2;\mathbb{Z}) \cong \mathbb{Z},$$

so $b \geq 1$.

By theorem 80.14, M is finitely generated projective over

$$R_L \cong \mathbb{Z}[s^{\pm 1}, t^{\pm 1}].$$

Swan's Laurent polynomial theorem [77] implies that every finitely generated projective module over this ring is free, so

$$M \cong R_L^q$$

for some $q \geq 0$.

To determine q, take any nontrivial character over a field. In the spectral sequence used above, the $q = 0$ row is acyclic and the only nonzero homology of the local-system complex can occur in degree 2. Hence

$$\dim H_2(Q(P); \Bbbk_\chi) = q.$$

The Euler characteristic of a rank-one local-system chain complex is the ordinary Euler characteristic. By theorem 80.2,

$$\chi(Q(P)) = 1 - 2 + b = b - 1.$$

Thus $q = b - 1$, and

$$\pi_2(Q(P)) \cong H_2(\widetilde{Q}) \cong R_L^{b-1}$$

by Hurewicz on the simply connected universal cover.

Let

$$Y = T^2 \vee \bigvee^{b-1} S^2.$$

Choose a map $T^2 \to Q(P)$ inducing the isomorphism on π_1; the torus 2-cell extends because the two basis loops commute. Choose maps of the 2-spheres representing an R_L-basis of $\pi_2(Q(P))$. Together they define

$$f : Y \to Q(P)$$

inducing isomorphisms on π_1 and on the R_L-module π_2. Lift to universal covers. Both universal covers are simply connected; both have zero homology above degree 2 by theorem 80.13, and the lifted map is an isomorphism in degree 2. Hence it is an integral homology equivalence. The simply connected homological Whitehead theorem [29] makes the lifted map a homotopy equivalence, and therefore so is f.

Finally use theorem 79.7. □

Chapter 81

Complete Rank-Two Classification by Lattice Geometry

The remaining distinction is geometric. In active rank two, the presence of a nondegenerate lattice parallelogram separates a toral branch from a free branch. The following results complete the classification and show that fixed rank does not bound second homology.

81.1 Complete active-rank-two classification

We can now combine the geometric and algebraic branches.

Theorem 81.1 (Complete rank-two homotopy classification)**.** *Let r be a completable residual with*

$$\operatorname{rank} L_A(r) = 2.$$

Then exactly one of the following holds.

(a) *The normalized optimal fiber $P_A(r)$ is nondegenerate-parallelogram-free. Then*

$$I_A(r) \simeq \bigvee^{\beta_1} S^1, \qquad \beta_1 \in \{2,3\}.$$

(b) *The normalized optimal fiber contains a nondegenerate lattice parallelogram. Then*

$$I_A(r) \simeq T^2 \vee \bigvee^{\beta_2-1} S^2, \qquad \beta_2 \geq 1.$$

Consequently the possible active-rank-two homotopy types are precisely

$$S^1 \vee S^1, \qquad S^1 \vee S^1 \vee S^1, \qquad T^2 \vee \bigvee^{m} S^2 \quad (m \geq 0). \tag{81.1}$$

Proof. Apply theorems 79.12, 80.12 and 80.15. The two cases are complementary by definition. ∎

Corollary 81.2 (Integral homology profiles)**.** *In active rank two the integral homology is exactly one of*

$$H_0 \cong \mathbb{Z}, \quad H_1 \cong \mathbb{Z}^2, \quad H_j = 0 \ (j \geq 2),$$
$$H_0 \cong \mathbb{Z}, \quad H_1 \cong \mathbb{Z}^3, \quad H_j = 0 \ (j \geq 2),$$

or

$$H_0 \cong \mathbb{Z}, \quad H_1 \cong \mathbb{Z}^2, \quad H_2 \cong \mathbb{Z}^b, \quad H_j = 0 \ (j \geq 3), \qquad b \geq 1.$$

81.2 Explicit scalar realizations

The classification is not merely formal: all allowed types occur in four-letter scalar completion systems.

81.2.1 A fixed-system residual bifurcation

For an integer $d \geq 5$, define

$$\omega_d = (1-2d,\ 1-d,\ 1,\ 1+d). \tag{81.2}$$

This row vector is completion-total; for example

$$h_d = (1,1,2d-3,1) > 0, \qquad \omega_d h_d^\top = 0.$$

The equal-length kernel is

$$K_{\omega_d} = \mathbb{Z}u \oplus \mathbb{Z}v, \qquad u = (1,-2,1,0), \quad v = (2,-3,0,1). \tag{81.3}$$

(We write the first basis vector as u below.)

Theorem 81.3 (Fixed-system rank-two bifurcation)**.** *For the single scalar system* (81.2), *define*

$$r_W = 5d-4, \qquad r_T = 3d-4.$$

Then

$$\ell_{\omega_d}(r_W) = \ell_{\omega_d}(r_T) = 4,$$

and in both cases the active lattice is the full equal-length kernel $K_{\omega_d} \cong \mathbb{Z}^2$. Nevertheless,

$$I_{\omega_d}(r_W) \simeq S^1 \vee S^1, \qquad I_{\omega_d}(r_T) \simeq T^2.$$

Proof. For any $x \in \mathbb{N}^4$ of length n put

$$s = x_2 + 2x_3 + 3x_4.$$

Then

$$\omega_d \cdot x = (1-2d)n + ds.$$

For both residuals the completion equation implies

$$n \equiv 4 \pmod{d}.$$

Since $d \geq 5$ and length-4 solutions exist, both minimum lengths equal 4.

For r_W, setting $n = 4$ reduces the equation to $s = 3$. Its complete set of nonnegative solutions is

$$\mathcal{O}_W = \{(1,3,0,0), (2,1,1,0), (3,0,0,1)\}. \tag{81.4}$$

Relative to the first point, the two nonzero differences are

$$u = (1,-2,1,0), \qquad v = (2,-3,0,1),$$

which form the kernel basis. The normalized fiber has three affinely independent lattice points and no nondegenerate parallelogram, so theorem 81.1 gives $S^1 \vee S^1$.

For r_T, the length-4 equation is $s = 5$ and the complete optimal fiber is

$$\mathcal{O}_T = \{(0,3,1,0), (1,1,2,0), (1,2,0,1), (2,0,1,1)\}. \tag{81.5}$$

Relative to the first point, let

$$g = u, \qquad h = v - u = (1, -1, -1, 1).$$

Then the four normalized points are

$$\{0, g, h, g + h\},$$

a fundamental lattice parallelogram. Its quotient has Euler characteristic 0, so theorems 80.2 and 80.7 give $\beta_2 = 1$, and theorem 80.15 gives T^2. □

Thus even the matrix, the full equal-length kernel, the active lattice, its rank, and the minimum completion length can be held fixed while the global topology changes. The missing datum is the residual-dependent optimal-fiber translation geometry.

81.2.2 Unbounded second homology at fixed rank two

For $m \geq 2$ put

$$\alpha = 2m + 1$$

and define

$$\omega_m = \big(1 - m\alpha,\ 1 - (m-1)\alpha,\ 1,\ 1 + \alpha\big), \tag{81.6}$$

$$r_m = \alpha m(m-1) - 2m. \tag{81.7}$$

The system is completion-total; for example

$$h_m = \big(1, 1, (2m-2)\alpha - 3, 1\big) > 0$$

satisfies $\omega_m h_m^\top = 0$.

Theorem 81.4 (Unbounded rank-two sphere family)**.** *For every $m \geq 2$,*

$$\ell_{\omega_m}(r_m) = 2m.$$

The complete optimal fiber is

$$\mathcal{O}_m = \{A_0, \ldots, A_m, B, C\}, \tag{81.8}$$

where

$$A_t = (t, m-t, m-t, t), \qquad 0 \leq t \leq m,$$

$$B = (0, m+1, 0, m-1), \qquad C = (m-1, 0, m+1, 0).$$

Its active lattice equals the full equal-length kernel and has rank two. Moreover

$$I_{\omega_m}(r_m) \simeq T^2 \vee \bigvee^{m-2} S^2, \tag{81.9}$$

so

$$\beta_2(I_{\omega_m}(r_m)) = m - 1.$$

Proof. All four weights in (81.6) are congruent to 1 modulo α. Hence every completion of r_m with length n satisfies

$$n \equiv -r_m \equiv 2m \pmod{2m+1}.$$

The vectors in (81.8) have length $2m$, so $\ell = 2m$.

At length $2m$, eliminating the common congruence term gives

$$x_2 + mx_3 + (m+1)x_4 = m(m+1).$$

Modulo m,

$$x_2 + x_4 \equiv 0 \pmod{m},$$

and $0 \le x_2 + x_4 \le 2m$. Thus $x_2 + x_4$ is 0, m, or $2m$. The three cases give respectively C, the family A_t, and B, proving (81.8).

Set

$$g = (1, -1, -1, 1), \qquad h = (0, 1, -m, m-1).$$

Every equal-length kernel vector is uniquely $ag + bh$: indeed the equations $\mathbf{1}^\top z = 0$ and $\omega_m \cdot z = 0$ imply

$$z_3 = -(m+1)z_1 - mz_2, \qquad z_4 = mz_1 + (m-1)z_2,$$

which corresponds to $a = z_1$, $b = z_1 + z_2$. Hence g, h form a basis of the full kernel. Relative to A_0,

$$A_t - A_0 = tg, \qquad B - A_0 = h, \qquad C - A_0 = (m-1)g - h.$$

Thus in active coordinates

$$P_m = \{(t, 0) : 0 \le t \le m\} \cup \{(0, 1), (m-1, -1)\}. \tag{81.10}$$

It contains a nondegenerate parallelogram, for example the four points corresponding to $(0, 1)$, $(m-1, -1)$, $(0, 0)$, and $(m-1, 0)$.

It remains to compute β_2. Count simplex orbits in the finite quotient using (81.10). The horizontal-only part is the rank-one quotient and has Euler characteristic 0. The total alternating contribution of faces containing exactly one of the two off-line points, excluding the already-counted singleton vertex, is -1 for each off-line point. Faces containing both off-line points have total alternating contribution 0. The only cross-identifications between the two one-off-line families occur for the m edge pairs

$$\{(0, 1), (t, 0)\} \sim \{(m-1, -1), (m-1-t, 0)\}, \qquad 0 \le t \le m-1.$$

No face of size at least 3 has such a cross-identification because the two off-line levels are unique. Each edge identification raises the Euler characteristic by 1. Therefore

$$\chi(I_{\omega_m}(r_m)) = m - 2.$$

The parallelogram branch has $\beta_1 = 2$, and theorem 80.2 gives no homology above degree 2. Hence

$$m - 2 = 1 - 2 + \beta_2, \qquad \beta_2 = m - 1.$$

Apply theorem 80.15 to obtain (81.9). □

Corollary 81.5 (Fixed-rank unbounded complexity)**.** *There exist completion-total four-letter scalar systems with*

$$\operatorname{rank} K = \operatorname{rank} L = 2, \qquad \pi_1 \cong \mathbb{Z}^2,$$

for which β_2 is arbitrarily large.

81.2.3 A free-rank-three example

Example 81.6 (The F_3 branch)**.** Take

$$\omega = (-11, -8, -5, 4), \qquad r = 25.$$

All weights are 1 modulo 3, so every completion length satisfies $n \equiv 2 \pmod 3$. Length 2 is impossible because the smallest sum of two weights is $-22 > -25$, while length 5 solutions exist. At length 5 the equation becomes

$$x_2 + 2x_3 + 5x_4 = 10,$$

whose complete nonnegative solution set is

$$\mathcal{O} = \{(0,0,5,0), (0,3,1,1), (1,1,2,1), (3,0,0,2)\}.$$

The active lattice has rank two, and the six unordered pair differences are distinct up to sign. Hence there is no nontrivial repeated-difference relation and no nondegenerate parallelogram. Presentation (79.13) is therefore free on the three nonzero normalized fiber points. By theorem 80.12,

$$I_\omega(25) \simeq S^1 \vee S^1 \vee S^1.$$

Together, theorems 81.3, 81.4 and 81.6 realize every homotopy type in (81.1).

81.3 Computational consistency checks

The theorems above are symbolic and do not depend on computation. Small exhaustive enumerations were used only as independent checks of the fiber formulas, cell-orbit conventions, and the rank-two dichotomy.

For the fixed-system family (81.2), direct enumeration for $5 \le d \le 20$ recovers exactly the two fibers in equations (81.4) and (81.5) and minimum length 4 for the stated residuals. For the unbounded family (81.6), direct enumeration for $2 \le m \le 12$ recovers exactly (81.8). The first cases are shown in table 81.1.

m	α	r_m	$\lvert\mathcal{O}_m\rvert$	predicted homotopy type
2	5	6	5	T^2
3	7	36	6	$T^2 \vee S^2$
4	9	100	7	$T^2 \vee 2S^2$
5	11	210	8	$T^2 \vee 3S^2$
6	13	378	9	$T^2 \vee 4S^2$

Table 81.1: Initial cases of the explicit unbounded family. The computations are consistency checks only; theorem 81.4 is proved symbolically.

A separate finite-grid audit of planar lattice-convex fibers was used during discovery to test the parallelogram/free-group dichotomy and the Betti patterns. These experiments motivated the geometric lemmas in section 80.2; they are not proof premises and no computational assumption enters theorem 81.1.

Chapter 82

From Displacement Kernel to Physical Completion Holonomy

The preceding rank-two classification determines the intrinsic homotopy type of the filled optimal-fiber quotient. We now ask what survives when loops are transported by an explicitly specified physical LIFO policy. The displacement map separates lattice motion from pure loops; only the latter can carry holonomy invisible to the semantic endpoint.

82.1 Optimal fibers, displacement, and pure loops

Let $\mathcal{O}(r) \subset \mathbb{N}^s$ be the set of minimum-length completion count vectors at residual r. Fix $x_0 \in \mathcal{O}(r)$ and put

$$P = \mathcal{O}(r) - x_0, \qquad L = \langle \mathcal{O}(r) - \mathcal{O}(r) \rangle_{\mathbb{Z}}.$$

Let $I = I(r)$ denote the filled shortest-prefix quotient associated with this optimal fiber. In active rank two, the completion-fiber construction gives a canonical surjection

$$\delta : \pi_1(I) \twoheadrightarrow L \cong \mathbb{Z}^2,$$

which records the lattice displacement of a loop.

Definition 82.1 (Pure-loop sector)**.** Define

$$K_0 = \ker \delta$$

and

$$K_{\text{lin}} = \ker \big(H_1(I;\mathbb{Z}) \to L\big).$$

The first group is the full zero-displacement sector; the second records the part already visible after abelianization.

Theorem 82.2 (Rank-two displacement-kernel visibility trichotomy)**.** *Assume* $\operatorname{rank} L = 2$. *Then the active-rank-two alternatives have the following pure-loop structure.*

1. *If I is toral, $\pi_1(I) \cong \mathbb{Z}^2$ and δ is an isomorphism. Hence*

$$K_0 = 1, \qquad K_{\text{lin}} = 0.$$

2. *If $I \simeq S^1 \vee S^1$, then $\pi_1(I) \cong F_2$ and*

$$K_0 = [F_2, F_2], \qquad K_{\text{lin}} = 0.$$

Thus pure topology exists but is nonabelian-only.

3. *If $I \simeq S^1 \vee S^1 \vee S^1$, then $\pi_1(I) \cong F_3$ and*

$$K_{\mathrm{lin}} \cong \mathbb{Z}.$$

The full kernel $K_0 = \ker(F_3 \twoheadrightarrow \mathbb{Z}^2)$ is free of countably infinite rank, but one pure direction is already visible in H_1.

Proof. In the toral branch the displacement theorem identifies $\pi_1(I)$ with L. In the F_2 branch, abelianization gives a surjection $\mathbb{Z}^2 \to \mathbb{Z}^2$, hence an isomorphism, so the kernel of $F_2 \to \mathbb{Z}^2$ is exactly the commutator subgroup. In the F_3 branch, abelianization gives a surjection $\mathbb{Z}^3 \to \mathbb{Z}^2$ with primitive rank-one kernel. The assertions about the full kernels follow from Nielsen–Schreier and the standard structure of infinite-index normal subgroups of nonabelian free groups. □

Definition 82.3 (Displacement-compatible physical transport)**.** A flat physical transport is a functor from the fundamental groupoid $\Pi_1(I)$ to a groupoid of invertible physical comparison maps. At a basepoint it induces

$$\mathrm{Hol} : \pi_1(I) \to G_{\mathrm{phys}}.$$

It is displacement-compatible if there is a homomorphism $c : G_{\mathrm{phys}} \to L$ such that

$$c \circ \mathrm{Hol} = \delta.$$

Then $\mathrm{Hol}(K_0)$ lies in the count-invisible subgroup $\ker c$.

Proposition 82.4 (Gauge persistence of pure holonomy)**.** *If $\mathrm{Hol}|_{K_0}$ is nontrivial, no objectwise change of physical trivialization within the same flat invertible transport category can make every pure loop trivial. Such a change only conjugates the holonomy representation.*

This proposition is deliberately category-relative. It does not say that topology forces debt in an arbitrary DPDA.

82.2 A fixed completion system with a free/toral residual bifurcation

Fix $d \geq 5$ and the four weights

$$\omega_d = (1 - 2d,\ 1 - d,\ 1,\ 1 + d).$$

The equal-length kernel contains the basis

$$u = (1, -2, 1, 0), \qquad v = (2, -3, 0, 1).$$

Consider the two residuals

$$r_W = 5d - 4, \qquad r_T = 3d - 4.$$

Both have minimum completion length 4 and the same active rank two.

At r_W the optimal completions are

$$x_0 = (1, 3, 0, 0), \qquad x_u = (2, 1, 1, 0), \qquad x_v = (3, 0, 0, 1),$$

so the normalized fiber is $\{0, u, v\}$ and

$$I(r_W) \simeq S^1 \vee S^1.$$

At r_T the optimal completions are

$$(0,3,1,0),\quad (1,1,2,0),\quad (1,2,0,1),\quad (2,0,1,1),$$

which form a lattice parallelogram; hence

$$I(r_T) \simeq T^2.$$

By Theorem 82.2, the first residual has a nonabelian-only pure sector $[F_2, F_2]$, whereas the second has no pure sector at all.

82.3 Stable-LIFO comparison transport

For an equal-length count vector $x = (x_1, \dots, x_s)$ define the slot word

$$\eta_{\text{slot}}(x) = 1^{x_1} 2^{x_2} \cdots s^{x_s}.$$

Temporarily label its slots from left to right. For two equal-length count vectors x, y, define the Stable-LIFO matching $\tau_{x,y}$ by preserving same-type occurrences stably from left to right and matching all remaining source positions to remaining target positions in reverse order. Then

$$\tau_{y,x} = \tau_{x,y}^{-1}.$$

For the free optimal endpoints, the canonical words are

$$w_0 = 1222, \qquad w_u = 1123, \qquad w_v = 1114.$$

The three comparison permutations are

$$\tau_{0u} = (234), \qquad \tau_{0v} = (24), \qquad \tau_{uv} = (34).$$

On moved slots $\{2,3,4\}$ identify

$$r = (012), \qquad s = (02).$$

Then

$$[r, s] = (021),$$

corresponding on the original slots to (243).

Theorem 82.5 (Pure nonabelian Stable-LIFO holonomy)**.** *The Stable-LIFO comparison transport on the free optimal fiber is flat and sends a zero-displacement commutator in $K_0 = [F_2, F_2]$ to the nontrivial 3-cycle*

$$(243).$$

The image is invisible to every abelian permutation observable.

Proof. The triangle relation is

$$\tau_{0u}\tau_{uv} = \tau_{0v}$$

under chronological right action, so the endpoint comparisons define a flat local system on the three-point free quotient. The two displacement generators map to r and s, and their commutator maps to $(021) \neq 1$. Since every homomorphism from S_4 to an abelian group kills commutators, the certificate is genuinely nonabelian. $\square$

82.4 The full free shortest-completion episode

A prefix count vector y determines

$$n = \mathbf{1}^\top y, \qquad s = y_2 + 2y_3 + 3y_4.$$

For r_W,

$$r_W + \omega_d \cdot y = r_W + (1 - 2d)n + ds.$$

Since $0 \le n \le 4 < d$, residual equality on shortest prefixes is equivalent to equality of (n, s). The full quotient therefore has vertex set

$$V_W = \{(0,0)\} \cup \{(1,s) : 0 \le s \le 3\} \cup \{(2,s) : 0 \le s \le 3\} \cup \{(3,s) : 0 \le s \le 3\} \cup \{(4,3)\},$$

so $|V_W| = 14$. Its cell vector is

$$(f_0, f_1, f_2, f_3) = (14, 28, 15, 2).$$

Theorem 82.6 (Full free-episode realization)**.** *The entire* 14*-vertex shortest-prefix quotient admits a flat S_3 connection whose endpoint products are*

$$w_0 \mapsto e, \qquad w_u \mapsto r, \qquad w_v \mapsto s.$$

All square relations and both 3*-cube boundaries are flat. The pure commutator of Theorem 82.5 therefore survives on the actual one-letter shortest-completion episode, not merely on a compressed comparison complex.*

Construction. Assign edge labels by a spanning-tree gauge and extend them over the remaining edges so that each of the 15 squares has trivial boundary product. The two 3-cubes then have flat boundary automatically. The explicit connection uses only the four local values e, r, r^{-1}, s and the transposition $a = (01)$ at the mixed vertices; direct multiplication gives the endpoint products above for all 20 shortest words. $\square$

Chapter 83

Sharp LIFO Carrier Costs and Toral Correction

The free branch supports genuine pure nonabelian holonomy. The toral branch fills the commutator cell, so global descent instead imposes a correction problem. The resulting comparison gives exact carrier, ancestry, and correction costs inside the stated Stable-LIFO/literal-stack categories.

83.1 Sharp literal-stack carrier cost

Before measuring a carrier, we fix the implementation category. A *buffered ancestry refinement* of capacity b allows at most b temporary finite-control handles that remember distinguished stack-cell ancestries during a one-top macro; the handles are cleared at the endpoint. A macro is ancestry-bijective if every initial distinguished ancestry has exactly one final descendant. Section 83.7 proves the general bottom-move lower bound $b \geq n-1$ when the bottom of an n-cell distinguished prefix changes ancestry. We now impose a specific physical persistence category: no epsilon moves, no permanent finite-control sheet register, common endpoint control and zero potential, literal stack alphabet $\{1, 2, 3, 4\}$ only, and identical endpoint Parikh counts across sheets. Persistent monodromy must therefore live in stack order.

For a zero-weight literal count vector $z \in \mathbb{N}^4$, define its order capacity by the multinomial number

$$M(z) = \frac{|z|_1!}{\prod_i z_i!}.$$

For $k \geq 2$, set

$$g_k^{\mathrm{ord}}(\omega_d) = \min\{|z|_1 : z \neq 0,\ \omega_d z = 0,\ M(z) \geq k\}.$$

Lemma 83.1 (Arithmetic carrier girth)**.** *Every nonempty zero-weight literal word has length at least d. Moreover*

$$g_3^{\mathrm{ord}}(\omega_d) = g_6^{\mathrm{ord}}(\omega_d) = d.$$

Proof. Each weight is congruent to 1 modulo d. Hence $\omega_d z = 0$ implies

$$|z|_1 \equiv 0 \pmod{d},$$

so every nonempty zero-weight word has length at least d. Equality is attained by

$$z^{(3)} = (0, 1, d-1, 0)$$

for three orderings and by

$$z^{(6)} = (0, 2, d-3, 1)$$

for six orderings. □

For the six-state carrier, let

$$T_d = 2\,3^{d-4}$$

and index the six permutations P_g of the three letters 234 by $g \in S_3$. Set

$$Z_g = P_g T_d.$$

All Z_g have the same Parikh vector $z^{(6)}$ and zero total weight.

Theorem 83.2 (Sharp free literal carrier)**.** *There is a real-time one-top realization of the full free connection such that*

$$Z_g \xrightarrow{w_0} Z_g, \qquad Z_g \xrightarrow{w_u} Z_{gr}, \qquad Z_g \xrightarrow{w_v} Z_{gs},$$

with no epsilon moves and no permanent sidecar. The minimum persistent literal order-carrier height in the stated category is exactly

$$\boxed{d}.$$

The chosen realization needs exactly two transient ancestry handles. In the deferred-carrier subclass, exactly three transient path states are necessary and sufficient.

Proof. The lower bound on persistent height is Lemma 83.1: every nonempty zero-weight literal carrier has length divisible by d, hence length at least d. For the upper bound use the six same-Parikh words $Z_g = P_g T_d$, $g \in S_3$. Only the distinguished three-cell prefix P_g is changed; the common suffix T_d remains untouched throughout.

For a shortest word w, let $\sigma(w)$ be the endpoint product supplied by the flat connection of Theorem 82.6. During the four input-synchronous steps, finite control records the current quotient vertex and two temporary ancestry handles. The first two distinguished carrier cells are exposed into those handles, the third remains visible, and on the terminal step the visible third cell together with the two handles determines the incoming prefix permutation g. The transition rewrites that visible cell by the three-cell prefix $P_{g\sigma(w)}$. Thus

$$Z_g \xrightarrow{w_0} Z_g, \qquad Z_g \xrightarrow{w_u} Z_{gr}, \qquad Z_g \xrightarrow{w_v} Z_{gs},$$

with one input symbol consumed per transition and no ε-move. The semantic shortest-episode coordinate in control supplies the unique potential needed to verify the residual identity at each step, so this is an exact lift. Section 83.6 gives the full edge certificate for all twenty shortest words.

Necessity of two ancestry handles follows from Lemma 83.6: the generator $s = (02)$ changes the ancestry occupying the bottom of the distinguished three-cell prefix, so any ancestry-bijective top-access macro needs at least $3 - 1 = 2$ handles. The construction above uses two.

Finally restrict to the deferred-carrier subclass, in which the distinguished carrier order is not changed before the terminal rewrite. The three histories 222, 231, and 411 reach the same preterminal semantic vertex with the same next input but with accumulated products s, a, e; their common last input must therefore produce three distinct endpoint multipliers e, r, s. Determinism forces three transient path states. The explicit connection table has at most three accumulated-product classes at every vertex, so three states suffice. Hence both locality bounds are sharp in the stated subclasses. $\square$

83.2 Toral curvature and the difference between local flattening and torus descent

For the toral residual write the four optimal endpoints as $0, g, h, g+h$. Raw Stable-LIFO endpoint comparisons are

$$A = (1243), \qquad B = (24), \qquad C = (1234), \qquad D = (23).$$

The two chronological route products are

$$AB = (143), \qquad CD = (134) = (143)^{-1}.$$

Thus the raw parallelogram has nontrivial 3-cycle curvature.

On one isolated square, one edge change is enough to make the two route products equal. But a torus local system must descend globally. If normalized path products to the three nonzero lattice points are U, V, W, torus descent requires

$$[U, V] = e, \qquad W = UV = VU.$$

Hence the corrected endpoint tuple must have the form

$$(\widetilde{A}, \widetilde{B}, \widetilde{C}, \widetilde{D}) = (U, V, V, U), \qquad UV = VU.$$

Theorem 83.3 (Minimum number of globally altered endpoint comparisons)**.** *Any torus-descending correction of the raw tuple (A, B, C, D) changes at least three of the four endpoint comparisons. Three changes suffice.*

Proof. Because $A \neq D$, at least one of the g-pair must change; because $B \neq C$, at least one of the h-pair must change. If exactly two comparisons changed, the retained commuting pair would have to be one of $(A, B), (A, C), (D, B), (D, C)$. Each pair has nontrivial commutator (243), so two changes are impossible. Taking $U = A$ and $V = e$ gives (A, e, e, A) and changes exactly three entries. □

Let d_T be Cayley distance in S_4 with respect to transpositions and define

$$F(U, V) = d_T(A, U) + d_T(D, U) + d_T(B, V) + d_T(C, V)$$

subject to $[U, V] = e$.

Theorem 83.4 (Exact global toral correction Pareto law)**.** *The minimum total correction distance is*

$$\boxed{F_{\min} = 4},$$

attained by

$$U = (243), \qquad V = (234) = U^{-1},$$

with four single-transposition corrections. Among corrections changing exactly three endpoint comparisons, the minimum total transposition length is 6 *and the minimum possible maximum support is* 4*. Hence the exact nondominated profiles*

$$(\#\ \textit{altered},\ \textit{total transposition length}, \max\ \textit{support})$$

are

$$\boxed{(3, 6, 4), \qquad (4, 4, 2).}$$

Proof. For any commuting U, V,

$$F(U,V) \geq d_T(A,D) + d_T(B,C) = 2 + 2 = 4$$

by the triangle inequality. Equality is realized by the displayed U, V, using

$$A(12) = U, \quad B(34) = V, \quad C(12) = V, \quad D(34) = U.$$

For the three-change branch, retain one raw edge and minimize over its centralizer. The four symmetric cases give minimum total length 6 and force one correction to move all four slots. These two profiles are incomparable and every other correction is dominated by one of them. □

83.3 The full toral episode and repair/debt tradeoff

For r_T the residual quotient again uses (n, s), now with vertex set

$$V_T = \{(0,0)\} \cup \{(1,s) : 0 \leq s \leq 3\} \cup \{(2,s) : 0 \leq s \leq 5\} \cup \{(3,s) : 2 \leq s \leq 5\} \cup \{(4,5)\}.$$

Thus

$$|V_T| = 16, \qquad (f_0, f_1, f_2, f_3) = (16, 38, 28, 6).$$

There are 40 shortest completion words.

For the minimum-distance correction of Theorem 83.4, $V = U^{-1}$ and the image is the cyclic group

$$\langle U \rangle \cong C_3.$$

Define on the active lattice

$$\rho_T(ag + bh) = U^{a-b}.$$

Because the toral displacement map is an isomorphism, this defines a flat local system on the entire quotient. The 40 shortest words split into endpoint products

$$16 \text{ words} \mapsto e, \qquad 12 \mapsto U, \qquad 12 \mapsto V.$$

Theorem 83.5 (Full toral realization and exact repair/debt tradeoff)**.** *The minimum-distance toral repair admits a full* 16*-vertex real-time realization with nontrivial displacement-visible* C_3 *transport. If this transport is stored persistently in literal stack order, the minimum carrier height is d; the three-slot carrier uses two ancestry handles and, in the deferred subclass, three transient path states.*

Complete erasure of transport is possible by setting $U = V = e$*. Its total correction distance from the raw tuple is*

$$\boxed{8}.$$

Thus toral topology forces no pure debt: one may retain the nearest nontrivial flat transport at distance 4*, or pay distance* 8 *and remove persistent order memory entirely.*

Proof. The corrected image C_3 acts only on three slots, so the free height-d carrier construction applies. The lower bound is again the zero-weight congruence. For complete erasure,

$$\ell_T(A) + \ell_T(B) + \ell_T(C) + \ell_T(D) = 3 + 1 + 3 + 1 = 8.$$

Since $K_0 = 1$ in the toral branch, none of this retained transport is pure zero-displacement holonomy; it factors through the displacement lattice. □

83.4 The free–toral topology-to-LIFO dichotomy

The fixed family gives a clean contrast with every algebraic parameter except the residual held fixed.

free: local flatness permits global pure nonabelian holonomy,
toral: the filled commutator cell forbids pure holonomy
and converts raw noncommutativity into correction cost.

The distinction is not simply "topology requires memory." In the free branch, the missing parallelogram leaves a nonabelian zero-displacement sector in which order memory can live. In the toral branch, the parallelogram fills the commutator relation, so a flat physical representation must commute on the lattice generators. Noncommutativity survives only as the cost of repairing a chosen raw comparison policy.

83.5 Scope and representation dependence

The following distinctions are essential.

- The group K_0, the existence of nontrivial pure holonomy, the free/toral split, and the literal zero-weight height lower bound are intrinsic within the stated completion-fiber and carrier categories.
- The concrete permutations r, s, A, B, C, D and the correction profiles of Theorem 83.4 are relative to the fixed Stable-LIFO endpoint-comparison policy, although they are invariant under simultaneous conjugation of slot labels.
- The two-handle and three-state bounds are exact for the chosen three-slot and deferred-carrier subclasses; they are not claimed as universal lower bounds over all possible pushdown policies.
- No theorem here says that nontrivial topology alone forces physical debt under unrestricted finite control.

83.6 Full free-episode connection certificate

The full shortest-prefix quotient at r_W has vertices

$$V_W = \{(0,0)\} \cup \{(1,s),(2,s),(3,s) : 0 \le s \le 3\} \cup \{(4,3)\}$$

and f-vector $(14, 28, 15, 2)$. A directed edge is written $(n,s) \xrightarrow{i} (n+1, s+i-1)$. With chronological right multiplication, put

$$r = (012), \qquad r^{-1} = (021), \qquad s = (02), \qquad a = (01).$$

The following table is an explicit flat S_3 connection; omitted edges do not exist.

source	i	σ
$(0,0)$	$1,2,3,4$	e
$(1,0)$	$1,2,3,4$	e
$(1,1)$	1	e
$(1,1)$	2	r^{-1}
$(1,1)$	3	a
$(1,2)$	1	e
$(1,2)$	2	a
$(1,3)$	1	e
$(2,0)$	1	e
$(2,0)$	2	e
$(2,0)$	3	r
$(2,0)$	4	e
$(2,1)$	1	e
$(2,1)$	2	e
$(2,1)$	3	a
$(2,2)$	1	r
$(2,2)$	2	a
$(2,3)$	1	e
$(3,0)$	4	s
$(3,1)$	3	r
$(3,2)$	2	e
$(3,3)$	1	s

Substitution into the fifteen square relations

$$\sigma_{v,i}\sigma_{v+e_i,j} = \sigma_{v,j}\sigma_{v+e_j,i}$$

uses only $r^{-1}r = e$ and $as = r$ besides identities with the same factor on both sides. Flatness of all square faces implies compatibility on the two 3-cube boundaries. Endpoint products are e, r, s on the three optimal boxes, so the commutator image is $(021) \neq e$.

83.7 Ancestry model and sharp local-memory bounds

A buffered ancestry refinement of capacity b augments finite control with at most b temporary handles. A one-top microstep may move the visible cell's ancestry into a handle, attach a stored handle to the rewritten visible cell, or push a new decorated cell carrying a stored handle. A closed macro is ancestry-bijective when every initial distinguished ancestry has exactly one final descendant.

Lemma 83.6 (Bottom-move bound)**.** *If an ancestry-bijective top-access macro on n distinguished cells changes the ancestry occupying the bottom distinguished position, then $b \geq n-1$. The bound is sharp.*

Proof. Before the original bottom cell can be inspected, all $n-1$ cells above it have been removed from the live stack. Their ancestry classes must remain available for the final bijection, hence all $n-1$ must be retained externally at that instant. Conversely, pop the upper $n-1$ ancestries into handles, rewrite the exposed bottom, and rebuild the prefix in the target order. ☐

For the free three-slot carrier, the generator $s = (02)$ moves the third distinguished cell, so two handles are necessary and sufficient. The strict-streaming no-go on the full interval follows because, before the third ancestry is visible, every second-layer carrier update lies in a common order-two stabilizer. Flatness then forces the two endpoint generators into a common conjugate of that stabilizer, hence forces them to commute. The noncommuting pair r, s therefore requires transient path memory in addition to the two ancestry handles.

In the deferred-carrier subclass, the three histories 222, 231, and 411 reach the common preterminal vertex with accumulated products s, a, e and share the same final input. They must be distinguished before the final rewrite, giving a lower bound of three transient path states. The edge table above has local accumulated-product sets of size at most three, so three states suffice.

83.8 Global torus-descent audit

For the toral residual, a normalized flat endpoint quadruple has the form

$$(U, V, V, U), \qquad UV = VU.$$

Relative to the raw tuple

$$A = (1243), \quad B = (24), \quad C = (1234), \quad D = (23),$$

at least three entries must change: $A \neq D$, $B \neq C$, and every possible retained pair from $\{A, D\} \times \{B, C\}$ has nontrivial commutator. Three changes are attained by (A, e, e, A).

For Cayley distance d_T with all transpositions as generators,

$$F(U, V) = d_T(A, U) + d_T(D, U) + d_T(B, V) + d_T(C, V) \geq 4.$$

Equality is attained by

$$U = (243), \qquad V = (234) = U^{-1},$$

through

$$A(12) = U, \qquad B(34) = V, \qquad C(12) = V, \qquad D(34) = U.$$

If exactly three raw comparisons are altered, one raw edge is retained. The four centralizer cases give:

retained	fixed generator	$\min F$	min max supp
A	$U = A$	6	4
D	$U = D$	6	4
B	$V = B$	6	4
C	$V = C$	6	4

so the nondominated profiles are $(3, 6, 4)$ and $(4, 4, 2)$. The minimum-distance pair generates C_3 and acts on only three distinguished slots; therefore it reuses the exact height-d three-slot carrier with two ancestry handles. The full toral quotient has vertex set

$$\{(0,0)\} \cup \{(1, s) : 0 \leq s \leq 3\} \cup \{(2, s) : 0 \leq s \leq 5\} \cup \{(3, s) : 2 \leq s \leq 5\} \cup \{(4, 5)\}$$

and f-vector $(16, 38, 28, 6)$. The representation $\rho_T(ag + bh) = U^{a-b}$ therefore gives a flat connection on all 40 shortest words.

83.9 Topology-to-LIFO synthesis

At active rank two the intrinsic topology and the physical transport layer are now cleanly separated. The group π_1 and displacement kernel determine which pure loop sectors exist; the chosen transport category determines whether those sectors appear as nonabelian holonomy, literal carrier depth, transient ancestry cost, or toral correction. No claim is made that topology alone forces a unique physical debt law.

Part XIX

Higher-Rank Optimal-Fiber Realization and Rank Raising

Research provenance. The rank-two classification established a rigid planar picture but did not characterize which finite lattice configurations can occur as optimal fibers or how topology behaves after active rank is raised. This Part supplies an exact realization theorem and two rank-raising operations, then uses them to produce higher tori and fixed-rank families with unbounded higher homology.

Chapter 84

Lattice-Convex Realization and Higher-Rank Topology

For a finite spanning lattice-convex configuration $P \subset L$, let $Q(P)$ denote the periodic translation quotient of the optimal-fiber nerve introduced in the preceding rank-two Part. The completion baseline already proved that every normalized optimal fiber is lattice-convex. The first theorem below proves the converse and therefore turns the intrinsic lattice model into an exact realization theorem.

84.1 Intrinsic periodic-fiber model

A finite set $P \subset L$ in a rank-s lattice is *lattice-convex* when $P = \operatorname{conv}(P) \cap L$, and it spans L when $L = \langle P - P \rangle_{\mathbb{Z}}$. After translation assume $0 \in P$. Let $\mathcal{N}(P)$ be the simplicial complex on vertex set L in which a finite $F \subset L$ is a simplex exactly when $c - F \subseteq P$ for some $c \in L$, and set $Q(P) = |\mathcal{N}(P)|/L$.

For $0 \in P$, the quotient has the difference presentation

$$\pi_1(Q(P)) \cong \left\langle X_p \ (p \in P),\ X_0 = 1 \ \middle|\ X_p^{-1} X_q = X_s^{-1} X_t \text{ whenever } q - p = t - s \right\rangle, \quad (84.1)$$

and a canonical displacement homomorphism

$$\delta : \pi_1(Q(P)) \longrightarrow L, \qquad \delta(X_p) = p. \quad (84.2)$$

This is the presentation established in the preceding optimal-fiber topology Part and will be used below without rederiving the rank-two classification.

84.2 Every lattice-convex configuration is an optimal fiber

The necessity in the completion baseline proved in the preceding Part has a converse.

Definition 84.1 (Completion-total system)**.** An integer counting matrix A is *completion-total on its integer image* if every $b \in A\mathbb{Z}^d$ has a nonnegative preimage. A sufficient condition is the existence of a strictly positive vector $h \in \ker_{\mathbb{Z}} A$: any integer solution can then be translated by a large nonnegative multiple of h.

Theorem 84.2 (Lattice-convex realization)**.** *Let P be a finite lattice-convex configuration spanning a rank-s lattice L. Then there exist an integer matrix A, a completable residual r, and a strictly positive $h \in \ker_{\mathbb{Z}} A$ such that the normalized optimal fiber $P_A(r)$ is affinely lattice-isomorphic to P and its active lattice has rank exactly s.*

Proof. Identify L with $\mathbb{Z}^s$ and translate so that $0 \in P$. Since $\operatorname{conv}(P)$ is a lattice polytope, choose integral affine facet functions

$$\ell_j(z) = b_j - a_j^\top z \qquad (1 \le j \le m)$$

such that

$$\operatorname{conv}(P) = \{z \in \mathbb{R}^s : \ell_j(z) \ge 0 \ \forall j\}.$$

Choose $B \in \mathbb{Z}^s$ so large coordinatewise that $z + B \in \mathbb{N}^s$ for every $z \in P$. Define the integral affine embedding

$$E_0(z) = \bigl(z + B, \ell_1(z), \ldots, \ell_m(z)\bigr) \in \mathbb{Z}^{s+m}.$$

For lattice points $z \in \mathbb{Z}^s$ we have

$$E_0(z) \ge 0 \quad \Longleftrightarrow \quad z \in P.$$

Indeed, points of P satisfy every facet inequality and the chosen coordinate shifts; a lattice point outside $P = \operatorname{conv}(P) \cap \mathbb{Z}^s$ violates at least one facet inequality.

Choose an integer

$$C \ge \max_{z \in P} \mathbf{1}^\top E_0(z)$$

and append one balancing coordinate:

$$E(z) = \bigl(E_0(z),\ C - \mathbf{1}^\top E_0(z)\bigr) \in \mathbb{Z}^n, \qquad n = s + m + 1.$$

Then

$$E(z) \ge 0 \Longleftrightarrow z \in P, \qquad \mathbf{1}^\top E(z) = C \quad (z \in \mathbb{Z}^s). \tag{84.3}$$

Let

$$D = \{E(z) - E(0) : z \in \mathbb{Z}^s\} \subset \mathbb{Z}^n.$$

The first s coordinates of $E(z) - E(0)$ are exactly z, so projection onto those coordinates is a left inverse. Hence $D \cong \mathbb{Z}^s$ is a saturated direct summand of $\mathbb{Z}^n$. By (84.3), every $d \in D$ has coordinate sum zero.

Append one final coordinate and set

$$h = (1, \ldots, 1) \in \mathbb{Z}^{n+1}, \qquad K = (D \times \{0\}) + \mathbb{Z}h \subset \mathbb{Z}^{n+1}.$$

The last coordinate of h is one, so quotienting first by $\mathbb{Z}h$ identifies $\mathbb{Z}^{n+1}/\mathbb{Z}h$ with $\mathbb{Z}^n$; the image of K is the saturated lattice D. Thus K is saturated. Consequently there is an integer matrix A with

$$\ker_{\mathbb{Z}} A = K.$$

The vector h is strictly positive, so A is completion-total on its integer image.

Put

$$x_0 = (E(0), 0) \in \mathbb{N}^{n+1}, \qquad r = -Ax_0.$$

Any integer point in the fiber of Ax_0 has a unique expression

$$x = x_0 + (d, 0) + th, \qquad d \in D,\ t \in \mathbb{Z}.$$

Its last coordinate is t, so nonnegativity forces $t \ge 0$. Writing $d = E(z) - E(0)$ gives

$$\mathbf{1}^\top x = C + t(n+1).$$

Hence every minimum-length point has $t = 0$. At $t = 0$, nonnegativity is equivalent by (84.3) to $z \in P$. Therefore

$$\mathcal{O}_A(r) = \{(E(z), 0) : z \in P\}.$$

The active difference lattice is $D \times \{0\}$, which has rank s, and the map $z \mapsto (E(z), 0)$ is an affine lattice isomorphism from P onto the normalized optimal fiber. $\square$

Corollary 84.3 (Exact fiber characterization)**.** *Up to affine lattice equivalence, the normalized optimal fibers of completion-total integer counting systems are exactly the finite lattice-convex configurations.*

Proof. Necessity is the completion baseline proved in the preceding Part; sufficiency is theorem 84.2. ☐

84.3 Minimal fibers and free groups in every rank

A spanning rank-s configuration has at least $s+1$ points. At equality the quotient is completely rigid.

Theorem 84.4 (Minimal active fiber)**.** *Let $P \subset L$ be a normalized, spanning, lattice-convex configuration with* $\operatorname{rank} L = s$ *and* $|P| = s+1$*. Then*

$$Q(P) \simeq \bigvee^{s} S^1, \qquad \pi_1(Q(P)) \cong F_s, \qquad H_j(Q(P);\mathbb{Z}) = 0 \quad (j \geq 2).$$

Moreover the abelianized displacement map

$$H_1(Q(P);\mathbb{Z}) \longrightarrow L$$

is an isomorphism, so $K_{\mathrm{lin}}(P) = 0$.

Proof. The $s+1$ points are affinely independent. No positive-dimensional subset can be a nontrivial translate of another subset of P: such a translation would give either a repeated nonzero difference among four vertices or three collinear vertices, in either case an affine dependence. Thus the only identifications in a fundamental simplex are the singleton vertices, which form one translation orbit. Consequently

$$Q(P) \cong \Delta^s / P^{(0)}.$$

Collapsing the $s+1$ vertices of a contractible simplex to one point gives the homotopy cofiber of the inclusion of a discrete $(s+1)$-point set into Δ^s, hence a wedge of s circles.

After translating one point to 0, the remaining s points generate L by definition. Their affine independence makes them linearly independent, so they form a $\mathbb{Z}$-basis of L. The s circle generators map to this basis under displacement, proving the final assertion. ☐

Proposition 84.5 (Explicit scalar realization)**.** *For every $s \geq 1$, the scalar weight vector*

$$\omega_s = (\underbrace{1, \ldots, 1}_{s+1 \text{ entries}}, -(s+1))$$

with residual $r = -1$ has minimum completion length one and exactly the first $s+1$ unit vectors as optimal completions. Its active rank is s, and the all-ones vector is a positive kernel vector.

Proof. The equation $\omega_s x = 1$ has the $s+1$ positive unit vectors as length-one solutions; the final coordinate cannot occur in a length-one solution because its weight is negative. Thus the minimum is one and the displayed unit vectors are the complete optimum. Their difference lattice has rank s, while the sum of all $s+2$ columns is zero, so the all-ones vector lies in the kernel. ☐

At $s = 3$ this gives a genuine active-rank-three quotient with $\pi_1 \cong F_3$ and $K_{\mathrm{lin}} = 0$.

84.4 Products and higher tori

The periodic nerve is compatible with Cartesian products in a form that is especially useful for raising active rank.

Theorem 84.6 (Optimal-fiber product theorem)**.** *Let $P \subset L$ and $R \subset M$ be finite lattice-convex configurations spanning their lattices. Then $P \times R$ is lattice-convex in $L \oplus M$, has active rank* $\operatorname{rank} L + \operatorname{rank} M$*, and*

$$Q(P \times R) \simeq Q(P) \times Q(R).$$

The equivalence is $(L \oplus M)$-equivariant before passage to the quotient.

Proof. Write $K = \mathcal{N}(P)$ and $H = \mathcal{N}(R)$. A finite set $F \subset L \times M$ is a simplex of $\mathcal{N}(P \times R)$ exactly when some $(c, d) \in L \times M$ satisfies

$$(c, d) - F \subset P \times R.$$

This is equivalent to

$$c - \pi_L(F) \subset P, \qquad d - \pi_M(F) \subset R.$$

Hence $\mathcal{N}(P \times R)$ is the simplicial categorical product whose simplices are precisely the finite sets whose two coordinate projections are simplices of K and H.

On the posets of nonempty faces define

$$\phi(F) = (\pi_L F, \pi_M F), \qquad \psi(A, B) = A \times B.$$

Then $\phi\psi = \mathrm{id}$ and

$$F \subseteq \psi\phi(F)$$

for every face F. Pointwise comparable poset maps induce homotopic maps on order complexes [102]. Therefore ϕ and ψ induce mutually inverse homotopy equivalences after barycentric subdivision. The order complex of the product face poset is the standard barycentric triangulation of the product of the two barycentric subdivisions, so its realization is naturally homeomorphic to $|K| \times |H|$. The construction and the comparison $F \subseteq \psi\phi(F)$ commute with translations, so the equivalence is $(L \oplus M)$-equivariant. Quotienting gives the asserted product law.

Finally,

$$\operatorname{conv}(P \times R) = \operatorname{conv}(P) \times \operatorname{conv}(R),$$

so

$$\big(\operatorname{conv}(P \times R)\big) \cap (L \times M) = (\operatorname{conv}(P) \cap L) \times (\operatorname{conv}(R) \cap M) = P \times R.$$

The difference lattice is $\langle P - P \rangle \oplus \langle R - R \rangle$, proving the rank statement. ☐

Corollary 84.7 (Iterated products)**.** *For finitely many spanning lattice-convex configurations $P_i \subset L_i$,*

$$Q\left(\prod_i P_i\right) \simeq \prod_i Q(P_i).$$

Every such product is realized as a genuine completion-total optimal fiber by theorem 84.2.

The most symmetric instance is the Boolean cube.

Theorem 84.8 (Higher-torus realization)**.** *For every $s \geq 1$ there is a completion-total integer counting system with active rank s and normalized optimal fiber affinely lattice-isomorphic to $\{0,1\}^s$. Its translation quotient is*

$$Q(\{0,1\}^s) \simeq T^s,$$

so

$$H_j(Q;\mathbb{Z}) \cong \mathbb{Z}^{\binom{s}{j}} \qquad (0 \leq j \leq s).$$

Proof. The topological statement follows immediately from theorem 84.6: $\{0,1\}^s$ is the product of s rank-one intervals, and $Q(\{0,1\}) \simeq S^1$ by theorem 84.4. Thus its quotient is T^s.

For a direct completion-total realization, use $2s+1$ letters grouped into pairs (a_i, b_i), $1 \leq i \leq s$, together with a padding letter c. Let A_s have s rows. For $1 \leq i < s$, row i has coefficient 1 on a_i, b_i, coefficient -1 on a_s, b_s, and 0 elsewhere. The final row has coefficient 1 on the first $2s$ letters and coefficient $-2s$ on c.

Take residual

$$r = (0, \ldots, 0, -s),$$

so the completion equation $A_s x = -r$ has last coordinate s. If $t_i = x_{a_i} + x_{b_i}$ and $z = x_c$, the first $s-1$ equations force

$$t_1 = \cdots = t_s = t,$$

while the last equation gives

$$st - 2sz = s, \qquad t = 1 + 2z.$$

The total length is

$$st + z = s + (2s+1)z,$$

so it is minimized exactly at $z = 0$, $t = 1$. Hence an optimal completion chooses exactly one letter from each pair, giving 2^s optima and normalized cube fiber. The within-pair swaps generate an active lattice of rank s. The all-ones vector is a strictly positive kernel vector because every difference row has total coefficient zero and the last row has $2s - 2s = 0$. □

Corollary 84.9 (First third homology)**.** *Active rank three admits*

$$Q \simeq T^3, \qquad (\beta_0, \beta_1, \beta_2, \beta_3) = (1, 3, 3, 1).$$

This is the first possible active rank for nonzero third homology, because theorem 80.2 gives $H_j = 0$ for $j \geq 3$ in active rank two.

84.5 Two one-rank extensions: apex versus prism

Product with the interval is one natural one-rank extension. A second extension has a different homotopy effect.

Theorem 84.10 (Translation-isolated apex)**.** *Let $P \subset L$ be finite, spanning, and lattice-convex. In $L \oplus \mathbb{Z}e$ put*

$$P^+ = (P \times \{0\}) \cup \{e\},$$

where e has last coordinate one. Then P^+ is lattice-convex, has active rank $\operatorname{rank} L + 1$, *and*

$$Q(P^+) \simeq Q(P) \vee S^1.$$

Proof. A lattice point of the pyramid $\operatorname{conv}(P^+)$ has last coordinate 0 or 1. At height zero the slice is $\operatorname{conv}(P)$, whose lattice points are exactly P; at height one the slice is only the apex. Thus P^+ is lattice-convex, and its difference lattice is $L \oplus \mathbb{Z}e$.

No positive-dimensional face containing the apex has a nontrivial translate inside P^+. Indeed, a translation with last coordinate zero must fix the unique height-one point and is therefore zero; a translation with last coordinate ± 1 sends one of the two height levels outside $\{0, 1\}$. Hence every nontrivial face identification is confined to the base simplex, although all vertices, including the apex, belong to one translation orbit.

Choose the simplex of the periodic nerve corresponding to the translate P^+ itself as a fundamental simplex; it is the cone on the base simplex $\Delta(P)$. Every simplex orbit has a representative in this cone. The preceding height argument has two consequences for its face identifications. First, all identifications between positive-dimensional base faces are exactly those already defining $Q(P)$. Second, a positive-dimensional face containing the apex cannot be identified with a distinct face containing the apex: a translating vector with last coordinate zero would have to fix the unique height-one vertex and hence be zero, while a vector with nonzero last coordinate cannot preserve the two-level pattern of a mixed face. Only the apex vertex itself is identified with the common vertex orbit, because the translation action is transitive on vertices.

Consequently the quotient is obtained from the cone $C\Delta(P)$ by applying the old base quotient

$$q : \Delta(P) \longrightarrow Q(P)$$

and identifying the cone point with the vertex orbit of $Q(P)$. Equivalently it is the homotopy coequalizer of q and the constant map $\Delta(P) \to Q(P)$ at that vertex. Since $\Delta(P)$ is nonempty and contractible, q is homotopic to the constant map. The homotopy coequalizer of two homotopic maps from a nonempty contractible space is homotopy equivalent to $Q(P) \vee S^1$ (for example by the standard mapping-cylinder model). This proves the theorem. ☐

Corollary 84.11 (Prism law). *For every finite spanning lattice-convex P,*

$$Q(P \times \{0, 1\}) \simeq Q(P) \times S^1.$$

Proof. Apply theorem 84.6 and $Q(\{0, 1\}) \simeq S^1$. ☐

Thus apex and prism both add one active lattice direction but act differently:

$$\pi_1(Q(P^+)) \cong \pi_1(Q(P)) * \mathbb{Z}, \qquad \pi_1(Q(P \times \{0, 1\})) \cong \pi_1(Q(P)) \times \mathbb{Z}.$$

For torsion-free homology, the Künneth formula also gives

$$H_j(Q(P) \times S^1) \cong H_j(Q(P)) \oplus H_{j-1}(Q(P)),$$

while apex extension preserves H_j for $j \geq 2$.

Corollary 84.12 (Rank-three branches). *Both operations produce genuine completion-total active-rank-three families.*

(a) *Apexing a rank-two F_2 fiber gives F_3; apexing a rank-two F_3 fiber gives F_4.*

(b) *Apexing a rank-two toral fiber gives fundamental group $\mathbb{Z}^2 * \mathbb{Z}$ and, for the basic torus, homotopy type $T^2 \vee S^1$.*

(c) *Prisming the rank-two minimal F_2 branch gives an aspherical $F_2 \times \mathbb{Z}$ quotient with Betti vector $(1, 3, 2)$. Prisming a rank-two F_3 wedge branch gives an aspherical $F_3 \times \mathbb{Z}$ quotient with Betti vector $(1, 4, 3)$.*

84.6 Fixed-rank higher homology and displacement

We now combine the rank-raising calculus with an explicit rank-two seed.

Proposition 84.13 (Rank-two seed family)**.** *For every $m \geq 2$ there is a completion-total scalar system with active rank two and a normalized fiber affinely equivalent to*

$$P_m = \{(t, 0) : 0 \leq t \leq m\} \cup \{(0, 1), (m - 1, -1)\} \subset \mathbb{Z}^2. \tag{84.4}$$

Its quotient satisfies

$$\pi_1(Q(P_m)) \cong \mathbb{Z}^2, \qquad H_1(Q(P_m); \mathbb{Z}) \cong \mathbb{Z}^2, \qquad H_2(Q(P_m); \mathbb{Z}) \cong \mathbb{Z}^{m-1},$$

and has no homology above degree two. Equivalently, its Betti polynomial is

$$1 + 2t + (m - 1)t^2.$$

Proof. Take $\alpha = 2m + 1$ and the four scalar weights

$$\omega_m = \big(1 - m\alpha,\ 1 - (m - 1)\alpha,\ 1,\ 1 + \alpha\big),$$

with residual

$$r_m = \alpha m(m - 1) - 2m.$$

The strictly positive vector

$$h_m = \big(1, 1, (2m - 2)\alpha - 3, 1\big)$$

lies in the kernel. All four weights are 1 modulo α, so every completion length satisfies

$$n \equiv -r_m \equiv 2m \pmod{2m + 1}.$$

At length $2m$, the complete optimum is

$$A_t = (t, m - t, m - t, t) \quad (0 \leq t \leq m),$$

$$B = (0, m + 1, 0, m - 1), \qquad C = (m - 1, 0, m + 1, 0).$$

Indeed, eliminating the common congruence term gives

$$x_2 + mx_3 + (m + 1)x_4 = m(m + 1),$$

and reduction modulo m forces $x_2 + x_4 \in \{0, m, 2m\}$, yielding exactly C, the A_t, and B.

The equal-length kernel has basis

$$g = (1, -1, -1, 1), \qquad h = (0, 1, -m, m - 1),$$

and relative to A_0 the optimal points have coordinates

$$A_t - A_0 = tg, \qquad B - A_0 = h, \qquad C - A_0 = (m - 1)g - h,$$

which is (84.4). Its convex hull is the lattice quadrilateral

$$(0, 0),\ (m - 1, -1),\ (m, 0),\ (0, 1).$$

All four boundary edges are primitive. The Euclidean area is m, so Pick's theorem gives $m - 1$ interior lattice points. These are exactly $(1, 0), \dots, (m - 1, 0)$; together with the four boundary vertices this is precisely P_m. Hence P_m is lattice-convex.

We next compute the fundamental group directly from the difference presentation (84.1), without invoking a general planar classification. Write

$$x = X_{(1,0)}, \qquad y = X_{(0,1)}, \qquad z = X_{(m-1,-1)}.$$

Repeated horizontal differences give $X_{(t,0)} = x^t$. The repeated difference $(0,1)$ between the pairs $((0,0),(0,1))$ and $((m-1,-1),(m-1,0))$ gives

$$y = z^{-1}x^{m-1}, \qquad z = x^{m-1}y^{-1}.$$

For $1 \le t \le m-1$, the repeated difference $(-t,1)$ between

$$((t,0),(0,1)) \quad \text{and} \quad ((m-1,-1),(m-1-t,0))$$

gives

$$x^{-t}y = yx^{-t}.$$

Taking $t = 1$ shows $xy = yx$. Thus the presentation is generated by the commuting pair x, y. The displacement map (84.2) sends them to the standard basis of $\mathbb{Z}^2$, so it rules out any further relation and yields

$$\pi_1(Q(P_m)) \cong \mathbb{Z}^2.$$

In particular $H_1(Q(P_m);\mathbb{Z}) \cong \mathbb{Z}^2$. The rank-two homological baseline theorem 80.2 gives no homology above degree two and makes H_2 free abelian.

It remains to compute the Euler characteristic by simplex orbits. Put $H = \{(t,0) : 0 \le t \le m\}$. A horizontal-only orbit can be normalized to have minimum horizontal coordinate zero, so its alternating contribution is

$$\sum_{k\ge1}(-1)^{k-1}\binom{m}{k-1} = 0.$$

For either fixed off-line point, faces containing that point and a nonempty horizontal subset contribute

$$\sum_{k=1}^{m+1}(-1)^k\binom{m+1}{k} = -1,$$

while faces containing both off-line points have total alternating contribution

$$\sum_{k=0}^{m+1}(-1)^{k+1}\binom{m+1}{k} = 0.$$

The only cross-identifications between the two one-off-line families are the m edge pairs

$$\{(0,1),(t,0)\} \sim \{(m-1,-1),(m-1-t,0)\}, \qquad 0 \le t \le m-1.$$

A nonzero vertical translation can match at most one horizontal point because each off-line level contains only one point, so there is no higher-dimensional cross-identification. Identifying each duplicate edge orbit raises the alternating count by one. Therefore

$$\chi(Q(P_m)) = 0 - 1 - 1 + m = m - 2.$$

Since $\beta_0 = 1$, $\beta_1 = 2$, and higher homology vanishes,

$$m - 2 = 1 - 2 + \beta_2,$$

so $\beta_2 = m - 1$. Torsion-freeness has already been supplied by theorem 80.2. $\square$

Corollary 84.14 (Apex branch with unbounded second homology)**.** *Let P_m^+ be the apex extension of the seed in theorem 84.13. Then P_m^+ has active rank three and*

$$Q(P_m^+) \simeq Q(P_m) \vee S^1, \qquad \pi_1(Q(P_m^+)) \cong \mathbb{Z}^2 * \mathbb{Z}.$$

Moreover

$$H_2(Q(P_m^+);\mathbb{Z}) \cong \mathbb{Z}^{m-1}, \qquad H_j(Q(P_m^+);\mathbb{Z}) = 0 \quad (j \geq 3).$$

Hence β_2 is unbounded at fixed active rank three and fixed fundamental group $\mathbb{Z}^2 * \mathbb{Z}$.

Proof. Apply theorem 84.10 to theorem 84.13. Wedging on a circle changes only H_1 and takes the fundamental group to the free product with $\mathbb{Z}$. □

The prism operation behaves differently: it moves the seed's second homology into degree three. This leads to the main all-rank phenomenon.

Theorem 84.15 (Fixed rank and fixed fundamental group with unbounded higher homology)**.** *For every $s \geq 2$ there is a one-parameter family of completion-total integer counting systems whose normalized optimal fibers have active rank exactly s and whose quotients $Q_{m,s}$ satisfy*

$$\pi_1(Q_{m,s}) \cong \mathbb{Z}^s,$$

while every Betti number $\beta_j(Q_{m,s})$, $2 \leq j \leq s$, is unbounded as $m \to \infty$.

More precisely, take

$$P_{m,s} = P_m \times \{0,1\}^{s-2}.$$

Then

$$Q(P_{m,s}) \simeq Q(P_m) \times T^{s-2}$$

and its Betti polynomial is

$$\boxed{\big(1 + 2t + (m-1)t^2\big)(1+t)^{s-2}}. \tag{84.5}$$

Equivalently,

$$\boxed{\beta_j = \binom{s-2}{j} + 2\binom{s-2}{j-1} + (m-1)\binom{s-2}{j-2}}, \tag{84.6}$$

with out-of-range binomial coefficients interpreted as zero. In particular

$$\beta_1 = s, \qquad \beta_s = m-1.$$

Proof. The active ranks add under products, so $P_{m,s}$ has rank $2 + (s-2) = s$. By theorems 84.6 and 84.8,

$$Q(P_{m,s}) \simeq Q(P_m) \times T^{s-2}.$$

Its fundamental group is

$$\mathbb{Z}^2 \times \mathbb{Z}^{s-2} = \mathbb{Z}^s.$$

The two factors have torsion-free homology, so the integral Künneth formula has no Tor terms and multiplies Betti polynomials. This gives (84.5) and (84.6). For $2 \leq j \leq s$, the coefficient

$$\binom{s-2}{j-2}$$

of $m-1$ is positive, so every such β_j is unbounded. Finally, theorem 84.2 realizes each product configuration as a completion-total optimal fiber. □

Corollary 84.16 (Unbounded third homology in rank three)**.** *At active rank three there are completion-total systems with*

$$\pi_1 \cong \mathbb{Z}^3, \qquad (H_0, H_1, H_2, H_3) \cong (\mathbb{Z}, \mathbb{Z}^3, \mathbb{Z}^{m+1}, \mathbb{Z}^{m-1}).$$

Thus $\beta_3 = m-1$ is unbounded at fixed active rank and fixed fundamental group. The case $m = 2$ is T^3.

84.6.1 The displacement map contains information beyond π_1

The minimal rank-three fiber from theorem 84.4 has

$$\pi_1 \cong F_3, \qquad H_1 \cong \mathbb{Z}^3 \xrightarrow{\cong} L \cong \mathbb{Z}^3,$$

so $K_{\mathrm{lin}} = 0$. For a rank-two comparison, take

$$P_F = \{(0,0), (0,1), (1,1), (3,2)\} \subset \mathbb{Z}^2.$$

Its convex hull is the triangle with vertices $(0,0), (0,1), (3,2)$ and normalized area 3; Pick's theorem shows that $(1,1)$ is its unique interior lattice point, so P_F is lattice-convex. The six unordered nonzero pair differences

$$(0,1), (1,1), (3,2), (1,0), (3,1), (2,1)$$

are distinct up to sign. Hence the presentation (84.1) has no nontrivial repeated-difference relation and is free on the three nonzero vertices:

$$\pi_1(Q(P_F)) \cong F_3.$$

But the displacement map on abelianization is a surjection

$$\mathbb{Z}^3 \longrightarrow \mathbb{Z}^2,$$

so its kernel is primitive of rank one. By theorem 84.2, P_F also occurs as a genuine completion-total optimal fiber. We therefore obtain the following intrinsic conclusion.

Corollary 84.17 (Same fundamental group, different displacement visibility)**.** *The abstract group $\pi_1(Q(P))$ does not determine the active rank or the linear zero-displacement subgroup $K_{\mathrm{lin}}(P)$. In particular F_3 occurs with both $K_{\mathrm{lin}} = 0$ and $K_{\mathrm{lin}} \cong \mathbb{Z}$.*

This statement is purely about the intrinsic completion quotient. It does not assert a physical stack-carrier lower bound; such claims require a separately specified transport model.

Part XX

Unrestricted State Complexity of $L_{p,q}$

Research provenance. The resource frontiers developed earlier were initially tied either to a residual-faithful representation or to specially designed hierarchy witnesses. The final automata-theoretic branch returns to the natural rational balance family itself and removes the representation assumption. The only semantic coordinate used is the right-quotient residual; the physical proof uses generic LIFO locality, protected pumping, and a regularity contradiction.

Chapter 85

Protected Phase Pumping and Deferred Exposure

Fix coprime positive integers p, q and

$$L_{p,q} = \{w \in \{0,1\}^* : qN_1(w) = pN_0(w)\}.$$

The question is now unrestricted: an arbitrary deterministic pushdown recognizer may distribute the residual among control, visible stack, deep stack, long rewrites, and deterministic ε-normalization. The proof below reconstructs no canonical representation. Instead it uses quotient rigidity to label reachable physical configurations semantically and then forces synchronized phases above protected stack cuts.

85.1 Model and semantic quotient

85.1.1 Pushdown model

A deterministic pushdown automaton has finite control Q, finite stack alphabet Γ, and transitions that inspect the current state, the visible top stack symbol, and either one input symbol or ε. The stack top is written on the right. Stack words, including regular languages of stacks used later, are therefore written and read from bottom to top. Thus a transition

$$\delta(q, X, a) = (q', \beta)$$

acts as

$$(q, \eta X, av) \vdash (q', \eta\beta, v).$$

A single transition may replace X by an arbitrary fixed finite word $\beta \in \Gamma^*$. Determinism means that for each state/top pair there is at most one move on each input letter, and if an ε-move is enabled then no input-consuming move is enabled at that state/top pair.

For the lower bound we use a *regular-target* acceptance convention. Fix a regular set $T \subseteq Q \times \Gamma^*$ of accepting control–stack configurations. After the finite binary input has been consumed, the word is accepted exactly when the deterministic post-input computation reaches T without consuming any further binary input. Reaching T in zero post-input steps is allowed. Ordinary final-state, empty-stack, and final-state-plus-empty-stack acceptance are obtained by the corresponding regular choices of T. A mandatory right-endmarker convention is handled by treating the endmarker as the final symbol of the continuation. The fixed-continuation regularity consequence needed later is proved separately in Theorem 85.12; it is not part of the model definition.

The matching upper bound in Section 86.2 is stated explicitly for a mandatory right endmarker.

Remark 85.1 (Meaning of "unrestricted"). The lower-bound opponent may use an arbitrary finite stack alphabet, arbitrary finite replacement words, deterministic ε-moves, noncanonical intermediate configurations, and any physical encoding compatible with correct recognition. No raw-conjugacy, homogeneous-tail, or canonical normal-form assumption is imposed.

85.1.2 Residuals

For $w \in \{0,1\}^*$ define

$$\rho(w) = qN_1(w) - pN_0(w).$$

Then

$$w \in L_{p,q} \iff \rho(w) = 0,$$

and

$$\rho(wa) = \begin{cases} \rho(w) - p, & a = 0, \\ \rho(w) + q, & a = 1. \end{cases}$$

Lemma 85.2 (Integer residual realizability). *If* $\gcd(p,q) = 1$*, then for every* $t \in \mathbb{Z}$ *there exists* $z \in \{0,1\}^*$ *with*

$$\rho(z) = t.$$

Proof. Choose integers x_0, y_0 with $qx_0 - py_0 = t$. All integer solutions are

$$x = x_0 + pk, \qquad y = y_0 + qk.$$

For sufficiently large k, both x, y are nonnegative. Any binary word with $N_1 = x$ and $N_0 = y$ then has residual t. ☐

Lemma 85.3 (Residual classification of right quotients). *For all* $u, v \in \{0,1\}^*$,

$$u^{-1}L_{p,q} = v^{-1}L_{p,q} \quad \iff \quad \rho(u) = \rho(v).$$

Proof. If $\rho(u) = \rho(v)$, then for every continuation z,

$$uz \in L_{p,q} \iff \rho(z) = -\rho(u) \iff \rho(z) = -\rho(v) \iff vz \in L_{p,q}.$$

Conversely, if $\rho(u) \neq \rho(v)$, choose by Theorem 85.2 a word z with $\rho(z) = -\rho(u)$. Then $uz \in L_{p,q}$ while

$$\rho(vz) = \rho(v) - \rho(u) \neq 0,$$

so $vz \notin L_{p,q}$. ☐

Lemma 85.4 (Physical configurations determine residuals). *If two consumed prefixes* u, v *lead to the same reachable physical configuration of a deterministic recognizer of* $L_{p,q}$*, then*

$$\rho(u) = \rho(v).$$

Consequently every reachable physical configuration C *has a well-defined semantic label* $\Theta(C) \in \mathbb{Z}$*.*

Proof. Starting from one and the same deterministic physical configuration, the set of accepting continuations is identical. Thus

$$u^{-1}L_{p,q} = v^{-1}L_{p,q},$$

and Theorem 85.3 gives $\rho(u) = \rho(v)$. ☐

For every reachable input-consuming transition,

$$\Theta \xrightarrow{1} \Theta + q, \qquad \Theta \xrightarrow{0} \Theta - p,$$

while ε-moves preserve Θ.

85.2 Unary nonrecurrence and height escape

Consider the computation induced by the infinite unary input 1^ω. For every $n \geq 0$, both 1^n and 1^{n+1} have accepting finite extensions: their residuals nq and $(n+1)q$ admit cancelling continuations by Theorem 85.2. Consequently, after consuming 1^n the deterministic machine cannot halt, reject permanently, or become trapped in an infinite ε-computation before consuming the next 1; otherwise no extension beginning with that next 1 could ever be accepted. Hence the unary computation consumes arbitrarily many input symbols, and every finite consumed unary prefix is live.

Lemma 85.5 (Raw nonrecurrence on the positive unary ray)**.** *No full raw configuration occurs twice on the live computation over* 1^ω*.*

Proof. Suppose the same configuration C occurs twice. If the two occurrences have consumed different numbers $m < n$ of input symbols, then Theorem 85.4 assigns C both residuals mq and nq, impossible.

If the two occurrences have consumed the same number of input symbols, the intervening segment consists entirely of ε-moves. Determinism then repeats the same ε-cycle forever, so the next input symbol can never be read. This contradicts liveness of the current unary prefix. □

Corollary 85.6 (Unary height escape)**.** *Along the computation over* 1^ω*, stack height eventually leaves every fixed finite bound.*

Proof. For each fixed height bound H, there are only finitely many state/stack configurations with height at most H. If infinitely many raw positions remained in that bounded region, some full configuration would repeat, contradicting Theorem 85.5. □

Definition 85.7 (Suffix minimum)**.** A raw position C_i on an infinite computation is a suffix minimum if

$$h(C_i) \leq h(C_t) \qquad \text{for every } t \geq i.$$

The following elementary consequence will be used repeatedly.

Lemma 85.8 (Unbounded suffix minima)**.** *If an integer-valued height sequence tends to infinity, then it has suffix-minimum positions at arbitrarily large heights.*

Proof. For every raw position u, the tail $\{h(t) : t \geq u\}$ has a minimum because $h(t) \to \infty$. Choose a position at which this tail minimum is attained. As u tends to infinity, the selected positions cannot remain in a finite set, and their heights must become arbitrarily large. □

85.3 Protected stack pumping and synchronized phases

A nonempty stack configuration $(q, \eta X)$ has *mode* (q, X).

Lemma 85.9 (Protected unary pump)**.** *There exist a state* $r \in Q$*, a symbol* $X \in \Gamma$*, an integer* $c \geq 1$*, a nonempty word* $U \in \Gamma^+$*, and a lower context* η *such that*

$$(r, \eta X) \overset{1^c}{\Longrightarrow} (r, \eta U X),$$

where the displayed macro includes all forced ε*-moves and no raw configuration of the macro has stack height below* $|\eta X|$*.*

Moreover the macro is context-independent: replacing the untouched lower context η *by any other lower context leaves the raw transition sequence unchanged as long as that context is not exposed.*

Proof. By Theorem 85.8, suffix-minimum positions occur at arbitrarily large heights. Since only $|Q||\Gamma|$ state/top modes exist, some mode (r, X) occurs at suffix minima of arbitrarily large height. Choose two such positions $t_1 < t_2$ with the same mode and with the later height strictly larger. Write the earlier stack as ηX.

Because t_1 is a suffix minimum, the computation from t_1 to t_2 never drops below height $|\eta X|$. Exposing any symbol of the lower prefix η would first require popping the visible X and reaching height $|\eta| < |\eta X|$. Hence η is never exposed or modified. At t_2 the state and visible top are again r, X, so the larger stack has the form

$$\eta U X$$

for some $U \in \Gamma^+$.

Let c be the number of input symbols 1 consumed between t_1 and t_2. Every transition of the segment sees only the active suffix above the untouched lower context, so replacing η by any other lower context reproduces the same deterministic raw transition sequence. This proves context independence.

To prove $c \geq 1$, suppose instead that $c = 0$. Then the segment is a context-independent, height-increasing ε-macro from mode (r, X) back to itself:

$$(r, \eta X) \xRightarrow{\varepsilon} (r, \eta U X).$$

Context independence would then repeat the same forced macro from $(r, \eta U^m X)$ for every $m \geq 0$, producing an infinite deterministic ε-computation and preventing the next symbol of the live unary input from ever being consumed. This is impossible. Therefore $c \geq 1$. □

85.3.1 Iteration and protected cuts

By context independence, for every $k \geq 0$,

$$(r, \eta U^k X) \xRightarrow{1^c} (r, \eta U^{k+1} X).$$

Each copied raw segment is identical to the original segment above its lower context, and hence never goes below the height of its own starting configuration. Therefore the cuts

$$H_k = \eta U^k$$

are protected throughout the entire future computation on 1^ω starting at

$$C_k = (r, H_k X).$$

After discarding finitely many initial k and rebasing the index, we may assume every H_k is nonempty.

This explicit repetition argument is important: we do not infer protection of the synthetic endpoints merely from the fact that the original two selected positions were suffix minima.

Lemma 85.10 (Synchronized p-phase family)**.** *For each $j \in \{0, \ldots, p-1\}$ there exist a state q_j and a nonempty upper stack word W_j, independent of k, with the following property. Starting from C_k, consume exactly j additional symbols* 1 *and then take every forced ε-move until the computation is again ready to inspect the next unread input symbol. The resulting raw configuration is*

$$D_{k,j} = (q_j, H_k W_j).$$

There is a constant $A \in \mathbb{Z}$ such that

$$\Theta(D_{k,j}) = A + kcq + jq.$$

Proof. The future 1^ω-computation from every C_k remains above H_k. Hence the relative computation above the cut depends only on the common state/top data and not on k. Liveness excludes an infinite ε-divergence after the jth consumed symbol, so the stated next-input boundary is well defined. For fixed j its state and upper stack are therefore independent of k; call them q_j, W_j. The upper word is nonempty because exposing H_k would contradict protection of the cut along the 1^ω future. The residual formula follows because each pump block contributes cq and each of the additional j symbols contributes q. □

85.4 Exposure, deferral, and the diagonal obstruction

Fix a phase j and follow the unique computation on 0^ω from each $D_{k,j}$. Until the common buried cut H_k is exposed, the machine sees only the state and the stack portion above H_k. Thus the relative raw computation is independent of k.

We say that H_k is *exposed* when the full stack is exactly H_k at a raw position. This event cannot be skipped downward. If the current stack is $H_k Z$ with a nonempty upper suffix, one PDA transition rewrites only the visible top symbol and therefore lowers total stack height by at most one. Consequently any computation that ever reaches below the height of H_k must pass through a raw configuration whose stack is exactly H_k first.

Lemma 85.11 (Uniform exposure dichotomy)**.** *For each phase j, exactly one of the following holds.*

(i) There exist $z_j \geq 0$ and $s_j \in Q$ such that for every k the first exposure of H_k occurs after exactly z_j consumed zeros, at configuration

$$(s_j, H_k).$$

(ii) For every k, the computation on 0^ω from $D_{k,j}$ never exposes H_k.

Proof. Before exposure, every transition acts strictly above H_k. The initial relative configuration (q_j, W_j) is independent of k, so the entire relative raw computation is the same for every k. Either its upper stack becomes empty after one fixed raw history and a fixed number z_j of consumed zeros, or it never becomes empty. In the former case the state at that moment is also fixed. □

85.4.1 Fixed-continuation regularity

Lemma 85.12 (Fixed-continuation stack regularity)**.** *Fix a control state s and a finite continuation v (including the mandatory right endmarker, when that convention is used). Let*

$$R_{s,v} = \{S \in \Gamma^* : \textit{starting from } (s, S), \ \textit{the fixed continuation } v \textit{ leads to acceptance}\}.$$

Then $R_{s,v}$ is regular.

Proof. Write $v = v_1 \cdots v_m$. Augment the finite control by an input-position coordinate $i \in \{0, \dots, m\}$. Original ε-moves leave i fixed; at position $i < m$, only a transition consuming the required letter v_{i+1} advances the coordinate to $i + 1$. Acceptance of the fixed word v is therefore reduced to predecessor reachability of a regular target set in an ordinary pushdown system. If one uses the standard bounded-push presentation of pushdown systems, each transition that writes a longer fixed word is compiled into finitely many auxiliary control states; this normalization does not change the predecessor set on the original configurations. Regular sets of pushdown configurations are effectively closed under predecessor reachability by the standard saturation theorem [8]. Restricting the resulting regular configuration set to the fixed control state s yields a regular language of stacks. ☐

Lemma 85.13 (Deferred exposure is impossible)**.** *No phase $j \in \{0, \ldots, p-1\}$ can satisfy alternative (ii) of Theorem 85.11.*

Proof. Assume phase j is deferred. Fix one k. The 0^ω-computation remains strictly above H_k. Every finite consumed zero-prefix is live, because its current residual has a finite cancelling continuation by Theorem 85.2. Hence the computation consumes arbitrarily many zeros and cannot become trapped in an ε-cycle.

No full raw configuration can repeat: a repetition after different numbers of consumed zeros would assign one physical configuration two residuals differing by a nonzero multiple of p, while a repetition at the same consumed input position would create a deterministic ε-cycle. Thus the relative upper-stack height above H_k escapes every finite bound.

Apply Theorem 85.9, with 0 in place of 1, to this relative zero computation. There exist a state s, a visible symbol Y, an integer $d \geq 1$, a nonempty word $V \in \Gamma^+$, and a fixed upper word β such that

$$(s, H_k\beta Y) \xRightarrow{0^d} (s, H_k\beta V Y).$$

Because the complete relative zero computation before exposure is identical for every k, the same second pump appears above every H_k. Iterating it gives reachable configurations

$$E_{k,\ell} = (s, \eta U^k \beta V^\ell Y), \qquad k, \ell \geq 0.$$

There is a constant $B \in \mathbb{Z}$ such that

$$\Theta(E_{k,\ell}) = B + cqk - pd\ell.$$

Choose, using Theorem 85.2, a fixed binary word v_0 with $\rho(v_0) = -B$. If the model uses a mandatory right endmarker, append it and write $v = v_0 \dashv$; otherwise put $v = v_0$. Since every $E_{k,\ell}$ is reachable and the automaton recognizes $L_{p,q}$,

$$E_{k,\ell} \xRightarrow{v} \text{accept} \quad \iff \quad cqk = pd\ell.$$

By Theorem 85.12, the stack language $R_{s,v}$ is regular. Hence so is

$$K = \{a^k \# b^\ell : \eta U^k \beta V^\ell Y \in R_{s,v}\}.$$

Indeed, if a DFA recognizes $R_{s,v}$, reading each fixed block η, U, β, V, Y induces a fixed transformation of the DFA state set. A finite automaton on the parameter alphabet $\{a, \#, b\}$ can therefore simulate the transformations corresponding to $\eta U^k \beta V^\ell Y$, proving that K is regular.

On the other hand correctness gives

$$K = \{a^k \# b^\ell : cqk = pd\ell\}.$$

With $g = \gcd(cq, pd)$, set

$$\alpha = \frac{pd}{g}, \qquad \beta = \frac{cq}{g}.$$

Then

$$K = \left\{a^{\alpha n} \# b^{\beta n} : n \geq 0\right\}.$$

This language is nonregular. Indeed, if N were a pumping length, the word $a^{\alpha N} \# b^{\beta N}$ would have every admissible pumping segment lying wholly in the initial a-block; pumping that segment changes the number of a's while leaving the number of b's fixed, destroying the equation $\beta k = \alpha \ell$. This contradiction eliminates the deferred alternative. ☐

Chapter 86

Exact Unrestricted State Complexity of $L_{p,q}$

Permanent deferral has now been ruled out, so every synchronized phase must expose the common protected cut. This immediately converts the arithmetic phases into distinct finite-control states. A symmetric argument gives the opposite side, and a three-symbol right-endmarker construction supplies the matching upper bound.

86.1 The unrestricted lower bound

Theorem 86.1 (Positive-side state lower bound). *Every deterministic ε-pushdown recognizer of $L_{p,q}$ in the stated model satisfies*

$$|Q| \geq p.$$

Proof. By Theorem 85.13, every phase $j \in \{0, \ldots, p-1\}$ exposes the common cut. Fix one k. For each phase,

$$D_{k,j} \overset{0^{z_j}}{\Longrightarrow} (s_j, H_k).$$

If $s_j = s_t$ for distinct j, t, the two exposure configurations are physically identical, because their stacks are both exactly H_k. By Theorem 85.4, their residuals are equal:

$$A + kcq + jq - pz_j = A + kcq + tq - pz_t.$$

Thus

$$q(j - t) = p(z_j - z_t).$$

Since $\gcd(p, q) = 1$, this implies $p \mid (j - t)$, impossible for $0 < |j - t| < p$. Therefore $s_0, \ldots, s_{p-1}$ are pairwise distinct and $|Q| \geq p$. □

Theorem 86.2 (Negative-side state lower bound). *Every deterministic ε-pushdown recognizer of $L_{p,q}$ in the stated model satisfies*

$$|Q| \geq q.$$

Proof. Apply the proof of Theorem 86.1 after interchanging the two input letters, interchanging p and q, and replacing the residual by its negative. □

Theorem 86.3 (Unrestricted state lower bound). *For coprime $p, q \geq 1$, every deterministic ε-pushdown recognizer of $L_{p,q}$ in the stated regular-target model satisfies*

$$\boxed{|Q| \geq \max(p, q)}.$$

Proof. Combine Theorems 86.1 and 86.2. □

Remark 86.4. The theorem is stronger than the product lower bound

$$|Q||\Gamma| \geq \max(p, q)$$

that one might first seek. The argument charges the ratio parameter directly to finite control, without any restriction on $|\Gamma|$.

86.2 A matching three-symbol construction

We now give a self-contained upper bound. This section is independent of the lower-bound representation and is included so that the exact state-complexity statement is proved entirely within the present paper.

Assume first that $p \geq q$. Let

$$Q = \{R_0, \ldots, R_{p-1}\}, \qquad \Gamma = \{A, B, Z_0\}.$$

We encode each residual $r \in \mathbb{Z}$ as follows. For $r \geq 0$, write uniquely

$$r = pk + s, \qquad k \geq 0, \quad 0 \leq s < p,$$

and define

$$\zeta(r) = (R_s, Z_0 A^k).$$

For $r < 0$, write uniquely

$$r = u - qh, \qquad h \geq 1, \quad 0 \leq u < q,$$

and define

$$\zeta(r) = (R_u, Z_0 B^h).$$

For $0 \leq s < p$, put

$$s^+ = (s + q) \bmod p, \qquad c_s = \frac{s + q - s^+}{p} \in \{0, 1\},$$

and

$$u_s = (s - p) \bmod q, \qquad h_s = \frac{u_s + p - s}{q} \geq 1.$$

For $0 \leq u < q$, put

$$u^- = (u - p) \bmod q, \qquad d_u = \frac{u^- - u + p}{q} \geq 1.$$

With the stack top on the right, define the following binary rules:

$$\delta(R_s, A, 0) = (R_s, \varepsilon), \qquad \delta(R_s, A, 1) = (R_{s^+}, A^{c_s+1}),$$

$$\delta(R_s, Z_0, 1) = (R_{s^+}, Z_0 A^{c_s}), \qquad \delta(R_s, Z_0, 0) = (R_{u_s}, Z_0 B^{h_s}),$$

and for $0 \leq u < q$,

$$\delta(R_u, B, 1) = (R_u, \varepsilon), \qquad \delta(R_u, B, 0) = (R_{u^-}, B^{d_u+1}).$$

No ε-transition is used.

A mandatory right endmarker $\dashv$ is accepted only from

$$\zeta(0) = (R_0, Z_0).$$

For example, declare R_0 final and define only

$$\delta(R_0, Z_0, \dashv) = (R_0, Z_0)$$

among endmarker moves.

Theorem 86.5 (Three-symbol residual realization)**.** *For every coprime $p, q \geq 1$, there is an ε-free right-endmarker DPDA recognizing $L_{p,q}$ with*

$$|Q| = \max(p, q), \qquad |\Gamma| = 3.$$

Proof. Assume $p \geq q$. We verify

$$\delta(\zeta(r), 0) = \zeta(r - p), \qquad \delta(\zeta(r), 1) = \zeta(r + q)$$

for every residual r.

Let $r = pk + s \geq 0$. If $k \geq 1$, input 0 pops one A, giving

$$pk + s - p = p(k - 1) + s.$$

For input 1,

$$pk + s + q = p(k + c_s) + s^+,$$

so the rule replacing A by A^{c_s+1} gives exactly $\zeta(r + q)$. When $k = 0$, the 1-rule uses the same carry identity, while

$$s - p = u_s - qh_s,$$

so the boundary 0-rule enters exactly the negative encoding.

Now let $r = u - qh < 0$. Input 1 pops one B. If $h > 1$ this gives $u - q(h - 1)$; if $h = 1$ it exposes Z_0 and leaves $(R_u, Z_0) = \zeta(u)$. For input 0,

$$u - p = u^- - qd_u,$$

and hence

$$r - p = u^- - q(h + d_u),$$

which is implemented by replacing one visible B by B^{d_u+1}.

Therefore every binary input symbol realizes the correct residual translation in one ordinary transition. Starting from $\zeta(0)$, after reading w the machine is in $\zeta(\rho(w))$. The right endmarker is accepted exactly at $\zeta(0)$, so the recognized language is $L_{p,q}$.

If $q > p$, exchange 0 and 1, exchange p and q, exchange A and B, and negate the residual. The resulting machine has $q = \max(p, q)$ states. □

86.3 Exact state complexity

Theorem 86.6 (Exact unrestricted right-endmarker state complexity)**.** *For coprime $p, q \geq 1$, among all deterministic pushdown automata in the lower-bound model whose inputs are terminated by the mandatory symbol $\dashv$,*

$$\boxed{Q_{\min}(L_{p,q}) = \max(p, q)}$$

under deterministic pushdown recognition with a mandatory right endmarker.

Proof. Theorem 86.3 gives the lower bound and Theorem 86.5 gives the matching upper bound. □

Corollary 86.7 (Arbitrary positive parameters)**.** *For arbitrary $p, q \geq 1$, let $d = \gcd(p, q)$. Since*

$$L_{p,q} = L_{p/d,q/d},$$

the right-endmarker state complexity is

$$\boxed{Q_{\min}(L_{p,q}) = \max\left(\frac{p}{d}, \frac{q}{d}\right).}$$

Chapter 87

Unrestricted State–Stack Frontiers and Bounded-Push Tradeoffs

The preceding unrestricted-state theorem gives the sharp state coordinate when the stack alphabet is free. We now force the non-bottom work alphabet to be unary and then impose a replacement-length bound. The first result is representation-independent; the exact bounded staircase is deliberately typed to raw residual conjugacy, while the three-symbol bounded construction uses uniquely forced normalization macros.

87.1 Unary protected-pump interface

Fix coprime $p, q \geq 1$, put $M = \max(p, q)$ and $m = \min(p, q)$, and consider a deterministic ε-pushdown recognizer of $L_{p,q}$ with permanent bottom marker. The preceding unrestricted-state theorem gives

$$|Q| \geq M. \tag{87.1}$$

Its protected-pumping proof supplies synchronized positive and negative phase families above hidden cuts.

Lemma 87.1 (Unary protected-pump interface)**.** *On a unary work stack $\Gamma = \{A, Z_0\}$, after rebasing the positive protected cuts have the form*

$$H_k = A^{a+uk} Z_0, \qquad u \geq 1,$$

and the p positive phases expose a common protected cut in pairwise distinct states. Symmetrically, the negative family has affine cut heights $b + v\ell$ with $v \geq 1$ and q pairwise distinct exposure phases.

The lemma is exactly the unary specialization of the protected-pump and phase-exposure construction proved in Chapter 85; the new argument begins by showing that a fixed finite depth above such a cut cannot remain permanently hidden.

87.2 Finite-depth descent in a unary stack

Assume now $\Gamma = \{A, Z_0\}$. The protected pump word in theorem 87.1 is necessarily $U = A^u$ for some $u \geq 1$, so after rebasing the positive cuts have the form

$$H_k = A^{a+uk} Z_0.$$

Lemma 87.2 (Finite-depth unary descent)**.** *Fix an integer $d \geq 0$. For each positive phase j, after discarding finitely many initial pump indices, there exist an integer $e_{j,d} \geq 0$ and a state $s_{j,d}$, independent of k, such that, writing*

$$H_k = A^d L_k,$$

one has

$$(s_j, A^d L_k) \xRightarrow{0^{e_{j,d}}} (s_{j,d}, L_k)$$

for every sufficiently large k.

Proof. Before L_k is exposed, the relative configuration above it is the fixed pair (s_j, A^d). Therefore the entire deterministic raw history above the cut is independent of k. Either the relative stack empties after one fixed history for all sufficiently large k, or it never empties for any such k.

Assume the second alternative. Every finite zero-prefix remains live, raw recurrence is impossible because a repeated physical configuration would carry two residuals, and the relative height therefore escapes. Applying the protected-pump construction to this relative 0^ω computation yields a second context-independent pump. Because the stack alphabet has only the work symbol A above Z_0, write its height increment as $v \geq 1$ and let it consume $e \geq 1$ zeros. After the fixed relative prefix that reaches the second pump mode, iteration of the original 1-pump and of this new 0-pump therefore produces, for all sufficiently large k, ℓ, configurations with one fixed control state, residual

$$B + kcq - \ell ep,$$

and unary stack

$$A^{H+uk+v\ell} Z_0$$

for fixed integers B, H, where the original pump has height increment $u \geq 1$ and consumes $c \geq 1$ ones.

Now compare the reachable parameter pairs $(k + v, \ell)$ and $(k, \ell + u)$. Their stack heights are identical:

$$H + u(k + v) + v\ell = H + uk + v(\ell + u),$$

so, in a unary stack, the complete physical stacks are identical; the control state is identical as well. Their residuals, however, differ by

$$vcq + uep > 0.$$

This contradicts residual rigidity: one reachable physical configuration cannot have two distinct accepting-continuation quotients. Hence the non-exposure alternative is impossible, and the fixed depth is eventually removed. $\square$

Corollary 87.3 (Phase separation survives descent)**.** *For fixed d, all descended phase states are pairwise distinct. In particular,*

$$s_{j,d} \neq s_{t,d} \qquad (0 \leq j < t \leq p - 1).$$

Proof. All descended configurations have the same stack L_k. Equality of the states for phases $j \neq t$ would therefore identify two reachable physical configurations. Their residual difference has the form

$$q(j - t) - p(e_{j,d} - e_{t,d}),$$

which cannot vanish because $\gcd(p, q) = 1$ and $0 < |j - t| < p$. $\square$

The symmetric statement holds for the q negative phases. For fixed descent depth, the positive residuals tend to $+\infty$ with the pump index and the negative residuals tend to $-\infty$.

87.3 Exact unrestricted one-work-symbol complexity

Let the positive and negative protected cut heights be

$$h_k^+ = a + uk, \qquad h_\ell^- = b + v\ell,$$

with $u, v \geq 1$. Put $g = \gcd(u, v)$. Choose large k_0, ℓ_0 and set

$$k_n = k_0 + \frac{v}{g}n, \qquad \ell_n = \ell_0 + \frac{u}{g}n.$$

Then the height difference

$$h_{k_n}^+ - h_{\ell_n}^- = D$$

is constant.

Theorem 87.4 (Unrestricted one-work-symbol lower bound)**.** *Every deterministic ε-pushdown recognizer of $L_{p,q}$ in the permanent-bottom model with exactly one work symbol satisfies*

$$\boxed{|Q| \geq p + q.}$$

No raw residual conjugacy, homogeneous-tail, realtime, or bounded-replacement hypothesis is assumed.

Proof. Suppose $D \geq 0$. Apply theorem 87.2 with depth D to all positive phases and leave the negative family at depth zero. At index n, all $p + q$ phase configurations now have the identical physical stack

$$A^{h_{\ell_n}^-} Z_0.$$

By theorem 87.3, the p positive states are pairwise distinct; the symmetric argument makes the q negative states pairwise distinct. For sufficiently large n, every descended positive residual is positive and every negative residual is negative. Hence no positive state can equal a negative state, because equality together with the common stack would identify two reachable physical configurations carrying different residuals. Thus at least $p + q$ states are required. The case $D < 0$ is symmetric. $\square$

The raw construction of theorem 87.7 at any $B \geq 1 + \lceil M/m \rceil$ uses exactly $p+q$ states and is ε-free, so the bound is sharp.

Theorem 87.5 (Exact unrestricted state–stack frontier)**.** *For every coprime $p, q \geq 1$,*

$$\boxed{\mathbb{F}_{p,q} = \{(p + q, 2), (M, 3)\}, \qquad M = \max(p, q).}$$

Proof. With no work symbol, only finitely many physical configurations are available, so the recognized language is regular. Thus $|\Gamma| = 1$ is impossible. At $|\Gamma| = 2$, theorem 87.4 gives the sharp lower bound $p + q$. For every alphabet size at least three, the baseline theorem (87.1) gives $|Q| \geq M$, while a three-symbol construction attains $(M, 3)$. Every point with at least three stack symbols is dominated by $(M, 3)$, and the two displayed points are incomparable because $M < p + q$. $\square$

For the motivating 2:1 language,

$$\mathbb{F}_{2,1} = \{(3, 2), (2, 3)\}.$$

87.4 Bounded replacement in the raw residual-conjugacy category

We now type the representation more strongly.

Definition 87.6 (Raw B-bounded unary residual conjugacy)**.** Let $B \geq 2$. A raw B-bounded unary residual conjugacy is an injective map

$$\eta : \mathbb{Z} \to Q \times A^* Z_0$$

such that

$$\delta(\eta(r), 0) = \eta(r - p), \qquad \delta(\eta(r), 1) = \eta(r + q)$$

by one ordinary binary transition, with no normalization step on encoded residual configurations, and every replacement word has length at most B.

This category is narrower than the unrestricted opponent of theorem 87.4; the gain is that bounded-push effects become exactly measurable.

87.4.1 A matching bounded construction

Assume $p \geq q$ and define

$$b_B = \max\left(q, \left\lceil \frac{p}{B-1} \right\rceil\right).$$

For $r \geq 0$, write $r = pk + s$, $0 \leq s < p$, and encode

$$\eta(r) = (P_s, A^k Z_0).$$

For $r < 0$, write uniquely

$$r = u - b_B h, \qquad 0 \leq u < b_B, \qquad h \geq 1,$$

and encode

$$\eta(r) = (N_u, A^{h-1} Z_0).$$

The positive transitions use the usual quotient/remainder updates modulo p. On the negative side, input 1 changes the quotient height by at most one because $b_B \geq q$, while input 0 increases it by at most $B - 1$ because

$$b_B \geq \left\lceil \frac{p}{B-1} \right\rceil.$$

The boundary identities are the corresponding Euclidean decompositions. Hence every replacement word has length at most B.

Proposition 87.7 (Bounded raw upper construction)**.** *For $p \geq q$ and $B \geq 2$, there is an ε-free raw B-bounded unary residual conjugacy with*

$$|Q| = p + b_B.$$

By symmetry, for arbitrary coprime p, q,

$$|Q| \leq M + \max\left(m, \left\lceil \frac{M}{B-1} \right\rceil\right), \quad m = \min(p, q).$$

87.4.2 Weighted unit cycles on the two deep tails

Choose nonnegative integers a, b satisfying

$$qa - pb = 1$$

and fix the word

$$w = 1^a 0^b.$$

For all sufficiently large positive residuals, every prefix residual while reading w remains positive. For sufficiently negative residuals, the same is true with "negative" in place of "positive". Raw tail escape makes all corresponding encoded stacks nonempty. Thus on both deep tails the same word w induces one deterministic weighted map on top-A modes,

$$G : (s, h) \mapsto (g(s), h + \omega(s)),$$

and G sends $\eta(r)$ to $\eta(r + 1)$.

Lemma 87.8 (Signed tail cycles)**.** *The positive deep tail is eventually periodic on a directed state cycle C_+ of length L_+ and total height weight $+c_+$ for some $c_+ \geq 1$. The negative left-infinite tail is eventually periodic on a directed cycle C_- of length L_- and total height weight $-c_-$ for some $c_- \geq 1$. The two cycles are state-disjoint.*

Proof. Finite-state determinism gives eventual cycles. A zero total weight would repeat the same state and stack height at residuals differing by the cycle length, contradicting injectivity. On the positive tail, a negative total weight would eventually force work height below zero under repeated forward cycles, so the weight is positive.

For the negative left-infinite tail, a finite functional graph cannot support an infinite backward path outside directed cycles; sufficiently negative indices therefore lie on a cycle. A positive forward weight would force height below zero when followed backward through arbitrarily negative residuals, so its weight is negative.

If the two cycles shared a state, determinism of g would make them the same directed cycle with the same total weight, contradicting the opposite signs. $\square$

Lemma 87.9 (Bounded-push cycle capacity)**.** *Assume $p \geq q$. Then*

$$L_+ \geq p, \qquad L_- \geq \max\left(q, \left\lceil \frac{p}{B-1} \right\rceil\right).$$

Proof. On the positive cycle, periodicity gives

$$h(r + L_+) = h(r) + c_+.$$

Hence, for r sufficiently positive,

$$h(r - pL_+) = h(r) - pc_+.$$

But 0^{L_+} realizes exactly that residual shift, remains on the deep positive tail, and one raw top rewrite can lower work height by at most one. Thus

$$pc_+ \leq L_+,$$

so $L_+ \geq p$.

On the negative cycle,

$$h(r + L_-) = h(r) - c_-.$$

Applying 0^{L_-} shifts the residual by $-pL_-$ and therefore raises height by pc_-. Each raw transition raises work height by at most $B-1$, so

$$pc_- \le L_-(B-1).$$

Applying 1^{L_-} shifts the residual by $+qL_-$ and lowers height by qc_-. One raw top rewrite can lower height by at most one, so

$$qc_- \le L_-.$$

Since $c_- \ge 1$, the result follows. □

Theorem 87.10 (Exact raw bounded unary complexity)**.** *For every coprime $p, q \ge 1$ and every $B \ge 2$,*

$$\boxed{Q^{(B)}_{\mathrm{raw,1work}} = M + \max\left(m, \left\lceil \frac{M}{B-1} \right\rceil\right).}$$

Proof. By symmetry assume $p \ge q$. The disjoint cycles of theorem 87.8 and the capacity bounds of theorem 87.9 require at least

$$p + \max\left(q, \left\lceil \frac{p}{B-1} \right\rceil\right)$$

states. Theorem 87.7 attains exactly that number. □

87.4.3 A separate three-symbol bounded-push optimum

The optimal three-symbol point can also be made replacement-at-most-two, but not within the one-transition raw category of theorem 87.6. The correct interface is a normalized macro-step: one input-consuming move is followed, when necessary, by a uniquely forced terminating ε-burst, after which the stable residual representative has been restored.

Assume $p \ge q$ and use the $M = p$ state, three-symbol stable encoding from the preceding unrestricted-state construction. For $0 \le u < q$, let

$$u^- = (u-p) \bmod q, \qquad d_u = \frac{u^- - u + p}{q},$$

and for a target remainder $v \in \mathbb{Z}_q$ put

$$\mathrm{pre}(v) = (v+p) \bmod q, \qquad L_v = d_{\mathrm{pre}(v)} - 1.$$

Since addition by p permutes $\mathbb{Z}_q$,

$$\sum_{v=0}^{q-1} L_v = \sum_{u=0}^{q-1} (d_u - 1) = p - q.$$

Introduce abstract suffix roles $H_{v,j}$, $1 \le j \le L_v$, meaning "push exactly j further copies of the negative work symbol B, then enter R_v." On a visible B their forced rules are

$$\delta(H_{v,1}, \varepsilon, B) = (R_v, BB), \qquad \delta(H_{v,j}, \varepsilon, B) = (H_{v,j-1}, BB) \quad (j > 1).$$

These are roles rather than new state names. Stable negative configurations use top B only in $R_0, \dots, R_{q-1}$, so the $p-q$ state/top cells

$$(R_i, B), \qquad q \le i < p,$$

are unused by stable semantics. Fix a bijection from the suffix roles onto those cells. A long negative input-0 write is replaced by one consuming $B \mapsto BB$ move into the appropriate suffix role; positive-to-negative boundary writes enter a suffix of the same chain. If $u_s = (s-p) \bmod q$ and $h_s = (u_s + p - s)/q$, then

$$d_{\mathrm{pre}(u_s)} = h_s + \left\lfloor \frac{s}{q} \right\rfloor,$$

so $h_s - 1 \le L_{u_s}$ and every required boundary suffix exists.

Theorem 87.11 (State-preserving push-at-most-two macro compiler)**.** *For every coprime p, q, there is a deterministic right-endmarker DPDA whose stable configurations realize the residual graph and satisfy*

$$\boxed{|Q| = M, \qquad |\Gamma| = 3, \qquad \max|\alpha| \le 2.}$$

Every nontrivial normalization burst is a uniquely forced finite ε-suffix chain. In particular, for every replacement bound $B \ge 2$, the minimum state count among deterministic recognizers with at most three total stack symbols is exactly M.

Proof. The role overlay above proves the case $p \ge q$. Every long negative write is split into one consuming push and a strictly descending suffix chain. The number of roles is exactly $p - q$, all fit into state/top cells unused by stable negative semantics, and those cells carry no competing input transition, so determinism is preserved. Boundary crossings use suffixes of the same chains, every chain terminates, and its macro-result equals the original stable residual transition. The case $q > p$ is symmetric. The baseline lower bound $|Q| \ge M$ from theorem 86.3 proves state optimality. For $B = 1$ stack height cannot increase, leaving only finitely many physical configurations, so no nonregular $L_{p,q}$ can be recognized. ☐

Corollary 87.12 (Critical raw unary push threshold)**.** *The unbounded one-work-symbol optimum $M + m = p + q$ is attained in the raw category if and only if*

$$\boxed{B \ge B_{\mathrm{crit}} := 1 + \left\lceil \frac{M}{m} \right\rceil.}$$

At $B = 2$, the exact raw unary state cost is $2M$.

Corollary 87.13 (Inverse raw unary state-to-push law)**.** *Fix a raw one-work-symbol state budget $S \ge M + m$. The least replacement bound is*

$$\boxed{B_{\min}(S) = \begin{cases} 2, & S \ge 2M, \\ 1 + \left\lceil \dfrac{M}{S - M} \right\rceil, & M + m \le S < 2M. \end{cases}}$$

Theorem 87.14 (Exact raw unary state–push staircase)**.** *Let*

$$Q_B = M + \max\!\left(m, \left\lceil \frac{M}{B-1} \right\rceil\right).$$

At the fixed total stack alphabet size $|\Gamma| = 2$, the Pareto-minimal raw pairs $(|Q|, B)$ are exactly

$$\boxed{\{(Q_B, B) : 2 \le B \le B_{\mathrm{crit}},\ B = 2 \text{ or } Q_B < Q_{B-1}\}.}$$

Proof. A plateau value is dominated by the same state count with the smaller replacement bound. Every point after B_{crit} is dominated by the critical-threshold point. At every strict drop, reducing B forces a larger state count by theorem 87.10. Thus precisely the displayed breakpoints survive. □

Proposition 87.15 (The bounded three-symbol point is globally Pareto-minimal)**.** *In the broad deterministic permanent-bottom model, the resource triple*

$$\boxed{(M, 3, 2)}$$

is Pareto-minimal for $(|Q|, |\Gamma|, B)$.

Proof. It is attained by theorem 87.11. A dominating triple cannot use fewer than M states by theorem 86.3; with M states it cannot use only two total stack symbols because theorem 87.4 requires at least $p + q > M$ states there; and $B = 1$ cannot recognize the nonregular language. Hence no strict coordinate improvement can dominate $(M, 3, 2)$. □

The last proposition and theorem 87.14 deliberately live in different categories. We do not combine them into an "exact raw three-resource frontier": the three-symbol bounded compiler uses forced ε-normalization, whereas theorem 87.6 requires one physical input transition per residual edge.

87.5 Scope of the bounded result

The unrestricted theorem and the bounded raw theorem answer different questions. Theorem 87.4 quantifies over arbitrary recognizers but does not use a replacement bound. Theorem 87.10 measures the exact deformation caused by B after fixing a raw residual section.

For an arbitrary history-dependent one-work-symbol recognizer with bounded replacement, the present paper asserts only the proved envelope

$$p + q \le Q^{(B)}_{\mathrm{unrestricted,1work}} \le M + \max\left(m, \left\lceil \frac{M}{B-1} \right\rceil\right),$$

where the upper bound is supplied by a raw construction. No equality claim is made outside the typed raw category. This prevents a representation-relative cycle argument from being silently promoted to all deterministic pushdown realizations.

Part XXI

Protected Contracts and Pareto Synthesis

Research provenance. Earlier Parts established exact semantic lifts, protected quotient/recoding theory, representation-independent state complexity, and several endpoint resource theorems. This Part reads those results together. Protected contracts order the information a realization must preserve; contract refinement then induces exact monotonicity and strictness principles for the attainable state–stack region.

Chapter 88

Contract Refinement and Protected-Information Capacity

The minimal-core theory becomes quantitatively useful once the protected interface is promoted to part of a representation contract. Refining the contract can only shrink the admissible realization class, and finite protected-role capacity gives explicit certificates that the Pareto frontier must move.

88.1 Protected representation contracts and refinement

Observable refinement alone cannot compare genuinely different physical re-coordinations. The representation class must also be part of the data.

Definition 88.1 (Observable refinement)**.** On one representation class, write

$$\lambda' \succeq_{\mathrm{obs}} \lambda$$

if there is a map h with

$$\lambda = h \circ \lambda'.$$

Then λ' remembers at least as much protected information as λ.

Proposition 88.2 (Quotient antitonicity)**.** *If $\lambda' \succeq_{\mathrm{obs}} \lambda$, every λ'-safe surjective quotient is λ-safe.*

Proof. If $\lambda' = \bar{\lambda}' F$, then $\lambda = h\bar{\lambda}' F$. □

Definition 88.3 (Protected representation contract)**.** A protected contract is

$$\mathfrak{K} = \big(\mathfrak{C}, \{\lambda_{\mathcal{L}}\}_{\mathcal{L}\in\mathfrak{C}}\big),$$

where $\mathfrak{C}$ is a BBR-closed class of finite-coordinate exact sectioned pushdown lifts of one semantic system and each $\lambda_{\mathcal{L}} : C_{\mathcal{L}}^{\mathrm{reach}} \to \Lambda$ is compatible with BBR conjugacy. We suppress the object subscript when harmless. Write $\mathfrak{K}' \succeq \mathfrak{K}$ if $\mathfrak{C}' \subseteq \mathfrak{C}$ and the protected interfaces refine objectwise on every lift of $\mathfrak{C}'$. Section 54.2 records the typing explicitly.

Define

$$\mathcal{R}(\mathfrak{K}) = \{(|Q_{\mathrm{use}}(\mathcal{L})|, |\Gamma_{\mathrm{use}}(\mathcal{L})|) : \mathcal{L} \in \mathfrak{C}\}$$

and

$$\mathrm{Pareto}(\mathfrak{K}) = \mathrm{Min}\,\mathcal{R}(\mathfrak{K}).$$

For finite antichains $A, B \subset \mathbb{N}^2$, write $A \preceq_P B$ if every $b \in B$ is componentwise dominated by some $a \in A$.

Theorem 88.4 (Contract-refinement Pareto monotonicity)**.** *If $\mathfrak{K}' \succeq \mathfrak{K}$, then*

$$\mathcal{R}(\mathfrak{K}') \subseteq \mathcal{R}(\mathfrak{K})$$

and

$$\boxed{\mathrm{Pareto}(\mathfrak{K}) \preceq_P \mathrm{Pareto}(\mathfrak{K}').}$$

Thus a stronger physical contract cannot improve the Pareto frontier.

Proof. Class inclusion gives resource-set inclusion. For any finer-contract Pareto point b, the same point belongs to the coarser resource set. Descending componentwise inside $\mathbb{N}^2$ yields a coarser minimal point $a \le b$. □

88.2 Strictness from protected-information capacity

Weak monotonicity is automatic from class inclusion. Strictness requires an actual obstruction.

Definition 88.5 (Resource down-set)**.** For $r \in \mathbb{N}^2$, let

$$\downarrow r = \{s \in \mathbb{N}^2 : s \le r \text{ componentwise}\}.$$

Theorem 88.6 (Strictness by down-set exclusion)**.** *Let $\mathfrak{K}' \succeq \mathfrak{K}$. If there exists an attainable coarse point $r_0 \in \mathcal{R}(\mathfrak{K})$ such that*

$$\mathcal{R}(\mathfrak{K}') \cap \downarrow r_0 = \varnothing,$$

then

$$\boxed{\mathrm{Pareto}(\mathfrak{K}) \prec_P \mathrm{Pareto}(\mathfrak{K}').}$$

Proof. Theorem 88.4 gives weak dominance. Since r_0 is coarse-attainable, some coarse Pareto point a satisfies $a \le r_0$. Reverse dominance would give a fine Pareto point $b \le a \le r_0$, contradicting the down-set exclusion. □

Corollary 88.7 (Protected-role capacity certificate)**.** *Suppose a fine contract forces N protected roles to occupy distinct local-observation cells, while every realization with resource vector at most r has at most* $\mathrm{cap}(r)$ *such cells, with* cap *monotone. If a coarse attainable point r_0 satisfies*

$$N > \mathrm{cap}(r_0),$$

then r_0 is a down-set exclusion witness and Pareto refinement is strict.

This is the main conceptual bridge:

$$\boxed{\begin{array}{c}\text{protected-information separation}\\ \Longrightarrow\ extresourcedown - setexclusion\\ \Longrightarrow\ extstrictParetopenalty.\end{array}}$$

Chapter 89

Exact Pareto Frontiers for Rational Balance Lifts

We now apply the contract order to the rational-balance endpoint theorems already established in the monograph. The point is synthesis rather than re-proving those endpoint results: broad, deferred, raw residual-faithful, and phase-local architectures become comparable only after their protected information and recoding conventions have been typed.

89.1 Rational balance application I: an exact contract state premium

The abstract SCQ/BBR/protected-contract results above are proved entirely within this development. The rational-balance applications now combine that new comparison framework with four exact endpoint resource theorems from the preceding normalization theory. Section 89.3 states those inputs and records their proof interfaces and scope; in particular, the unrestricted state lower bound and the raw/phase-local optima are not being re-labelled as new results of the present chapter sequence.

Fix coprime $p, q \geq 1$ and a mandatory right-endmarker convention. Let $\mathfrak{K}_{\mathrm{broad}}$ be the broad exact-pushdown contract for the residual semantics of $L_{p,q}$. The previously established broad-class state theorem gives

$$Q_{\min}(\mathfrak{K}_{\mathrm{broad}}) = \max(p, q).$$

A three-symbol exact realization attains this value.

Let $\mathfrak{K}_{\mathrm{def}}$ be the canonical deferred-SEARCH literal-demand contract. Its previously established exact state minimum is

$$D_{p,q} = 2\left\lfloor \frac{p+q-1}{2} \right\rfloor + 1,$$

also attained with three total stack symbols.

Theorem 89.1 (Exact broad/deferred state premium)**.** *The exact state premium of the deferred literal-demand contract over the broad exact-pushdown contract is*

$$\boxed{\Delta_Q(p, q) = D_{p,q} - \max(p, q) = \min(p, q) - \mathbf{1}_{\{p+q \ \mathit{even}\}}.}$$

It is strict except when $\min(p, q) = 1$ *and* $p + q$ *is even.*

Proof. If $p+q$ is odd, $D_{p,q} = p+q$, giving $\Delta_Q = \min(p, q)$. If $p+q$ is even, $D_{p,q} = p+q-1$, giving $\Delta_Q = \min(p, q) - 1$. The equality classification follows immediately. □

The two attaining constructions use the same total stack alphabet size 3, so the displayed witness points isolate a control-state premium rather than a stack-alphabet artifact.

89.2 Rational balance application II: exact two-resource frontiers

Let
$$M = \max(p, q).$$

In raw residual-faithful Model R, let C be the number of work symbols; the total stack alphabet has size $C + 1$ including Z_0. The exact per-C state minimum is

$$S_{\text{raw}}(C) = \begin{cases} p + q, & C = 1, \\ M, & C \geq 2. \end{cases}$$

Theorem 89.2 (Exact raw two-resource frontier)**.** *Across all finite work-alphabet sizes,*

$$\boxed{\text{Pareto}(\mathfrak{K}_{\text{raw}}) = \{(p + q, 2), (M, 3)\}.}$$

Proof. The point $C = 1$ gives $(p+q, 2)$. The point $C = 2$ gives $(M, 3)$ and dominates every $C > 2$ point $(M, C+1)$. Since $M < p+q$, the two displayed points are incomparable. $\square$

Now fix $B \geq M$ and write every residual as $r = s + Bz$, $0 \leq s < B$. Define

$$h_+(B) = \frac{p}{\gcd(B, p)}, \qquad h_-(B) = \frac{q}{\gcd(B, q)}, \qquad H(B) = h_+(B) + h_-(B).$$

The normalized phase-local contract requires the current signed quotient phase to be visible through control state plus one work symbol. Its exact per-C state minimum is

$$Q^{\min}_{\text{phase}}(B, C) = B \left\lceil \frac{H(B)}{C} \right\rceil.$$

Theorem 89.3 (Exact phase-local Pareto staircase)**.** *Let $H = H(B)$ and*

$$\mathcal{B}_H = \{1\} \cup \left\{ 2 \leq C \leq H : \left\lceil \frac{H}{C} \right\rceil < \left\lceil \frac{H}{C - 1} \right\rceil \right\}.$$

Then

$$\boxed{\text{Pareto}(\mathfrak{K}_{\text{phase}}(B)) = \left\{ \left(B \left\lceil \frac{H}{C} \right\rceil, C + 1 \right) : C \in \mathcal{B}_H \right\}.}$$

Proof. The state coordinate is nonincreasing in C, while the alphabet coordinate is strictly increasing. A point inside a plateau of $\lceil H/C \rceil$ is dominated by the smallest C in that plateau. At every strict drop, no earlier point has as small a state coordinate and no later point has as small an alphabet. For $C > H$ the state coordinate is already B, so $C = H$ dominates all later points. Exact attainability holds for every C by the normalized phase-local constructions. $\square$

At fixed C, the phase state premium is

$$\Delta^{\text{phase}}_Q(B, C) = B \left\lceil \frac{H(B)}{C} \right\rceil - S_{\text{raw}}(C) \geq 0.$$

For $C \geq 2$, equality holds exactly when

$$B = M, \qquad C \geq H(B).$$

Since p, q are coprime, $B = M$ gives

$$H(B) = \min(p, q) + 1.$$

Thus additional stack symbols can absorb the stronger protected phase information while retaining raw-optimal state count.

89.2.1 The audited raw-safe nesting

There is one important scope distinction. Model R forbids epsilon-normalization on encoded residual configurations. The phase-local optimum is raw and epsilon-free for $C \geq 2$, but the $C = 1$ optimum uses one forced bottom-cell normalization. Hence full $C \geq 1$ phase-local and raw classes are not literally nested.

Restrict to at least two work symbols. Let $\mathfrak{K}_{\text{raw}}^{\geq 2}$ be Model R with $C \geq 2$, and let $\mathfrak{K}_{\text{phase}}^{\text{raw}}(B)$ be the phase-local contract restricted to $C \geq 2$.

Theorem 89.4 (Exact nested raw-safe comparison)**.** *The restricted phase-local contract refines the restricted raw contract, and*

$$\text{Pareto}(\mathfrak{K}_{\text{raw}}^{\geq 2}) = \{(M, 3)\}.$$

The two restricted Pareto frontiers are equal exactly when

$$\boxed{B = M \quad \textit{and} \quad \min(p, q) = 1.}$$

In every other case the refinement is strict.

Proof. The $C \geq 2$ phase constructions are raw, so the contract inclusion is genuine. Every phase point has state at least $B \geq M$ and alphabet at least 3, hence $(M, 3)$ dominates. Reverse dominance requires a phase point at most $(M, 3)$, so $C = 2$, $B = M$, and $\lceil H/2 \rceil = 1$, equivalently $H = 2$. At $B = M$, coprimality gives $H = \min(p, q) + 1$, hence $\min(p, q) = 1$. $\square$

Theorem 89.5 (Direct full-frontier comparison)**.** *Over all $C \geq 1$,*

$$\text{Pareto}(\mathfrak{K}_{\text{raw}}) \preceq_P \text{Pareto}(\mathfrak{K}_{\text{phase}}(B)),$$

and the dominance is strict except exactly at

$$(p, q, B) = (1, 1, 1),$$

where both frontiers equal

$$\{(2, 2), (1, 3)\}.$$

Proof. For the $C = 1$ phase point,

$$BH(B) \geq 2B \geq 2M \geq p + q,$$

so the raw point $(p + q, 2)$ dominates it. For every phase point with $C \geq 2$, the raw point $(M, 3)$ dominates because its state coordinate is at least $B \geq M$. Reverse dominance forces a phase point below $(p + q, 2)$, necessarily the $C = 1$ point with $BH(B) \leq p + q$. Equality throughout the displayed chain occurs only at $(1, 1, 1)$. $\square$

89.3 Endpoint resource theorems used in the rational-balance application

The rational-balance application combines previously established endpoint optima with the new protected-contract formalism. The logical inputs are:

1. in the broad deterministic pushdown/right-endmarker exact-lift class, the exact state minimum is $\max(p, q)$ and is attained with two work symbols plus the bottom marker;

2. in the canonical deferred-SEARCH literal-demand class, the exact state minimum is
$$D_{p,q} = 2\left\lfloor \frac{p+q-1}{2} \right\rfloor + 1,$$
again attainable with three total stack symbols;

3. in raw residual-faithful Model R, with C work symbols,
$$S_{\text{raw}}(C) = \begin{cases} p+q, & C = 1, \\ \max(p,q), & C \geq 2; \end{cases}$$

4. in the normalized phase-local class,
$$Q^{\min}_{\text{phase}}(B, C) = B\left\lceil \frac{H(B)}{C} \right\rceil, \qquad H(B) = \frac{p}{\gcd(B,p)} + \frac{q}{\gcd(B,q)}.$$

For reviewability, the proof interfaces are as follows. The deferred lower bound exposes the $2h{+}1$ residual debts $-hm, \ldots, 0, \ldots, hm$ at the same exhausted work stack; a common distinguishing continuation then forces distinct control states, while the centered rail attains the bound. The raw frontier uses positive- and negative-side synchronization to give $|Q| \geq \max(p, q)$ for every nonempty work alphabet, strengthens this to $p + q$ with one work symbol by cross-sign unary separation, and is attained by the one- and two-work-symbol residual encodings. The normalized phase-local formula counts, in each residue class modulo B, the $H(B)$ signed phase roles that must be exposed through C work symbols and then sums the sharp per-class bound; the matching construction is raw for $C \geq 2$. The broad unrestricted state theorem combines the positive- and negative-side phase-exposure bounds $|Q| \geq p$ and $|Q| \geq q$ with the explicit three-symbol residual realization having $\max(p, q)$ states.

These endpoint results are inputs, not re-labelled contributions of the present chapter sequence. They are included here at the level needed to audit every numerical comparison, while the new results are the SCQ/BBR reduction theory, protected-contract order, Pareto monotonicity/strictness criteria, and the synthesis of the endpoint optima into exact contract frontiers. The $C = 1$ phase optimum uses one forced bottom-cell normalization and is therefore compared directly with the full raw frontier rather than being treated as a literal subclass of raw Model R. For $C \geq 2$ the phase construction is raw, which is precisely the nested scope of Theorem 89.4.

89.4 Synthesis and scope

The resulting frontiers are exact inside their declared contracts. In particular, the raw Model-R frontier and the normalized phase-local staircase are not promoted to an unrestricted two-resource theorem for all recognizers. The $C = 1$ phase optimum uses one forced bottom-cell normalization, so the full phase-local class is compared directly with raw Model R rather than asserted to be nested inside it; genuine raw/phase nesting is used only for the audited $C \geq 2$ subfamily. This closes the representation-Pareto synthesis needed before the final return to the original 2:1 machine.

Part XXII

Closing the Loop

Chapter 90

The 2023 DPDA Revisited

The monograph began with two diagrams.

They represented the same $C/L/R$ recognition core but different empty-word wrappers:

$$\boxed{L_{2,1} \setminus \{\varepsilon\}}$$

and

$$\boxed{L_{2,1}.}$$

At the beginning, the figures were concrete machines to be understood.

We now return to them after developing:

- shortest-completion geometry and exact residual-faithful resource frontiers;
- representation-independent hierarchy methods;
- higher-dimensional completion geometry;
- finite geodesic compilation and higher-order optimal-compatibility structure theory;
- cancellation/defect geometry and chronological arithmetic refinement;
- the general real-time p:q machine and WSC Structure Theory;
- exact semantic lifts and physical lift freedom;
- arbitrary-coprime causal-schedule compilation and exact fatal-support classification;
- strict coordinate quotients, protected interfaces, bounded bi-recoding, and minimal cores;
- General Theory;
- operational CANCEL/WRITE enumeration and exact one-counter compression;
- explicit rational-Dyck run–return enumeration and contact-preserving barrier compression;
- Farey-recursive shortest-completion arithmetic and universal spectra;
- quantitative LIFO chronology;
- rank-two topology, physical holonomy, and sharp carrier costs;
- exact unrestricted state complexity;
- protected-contract Pareto synthesis.

The purpose of this chapter is not to redraw the machine as an abstract object.

It is to show what every part of the original design means after the full theory has been developed.

90.1 The two origin figures remain distinct

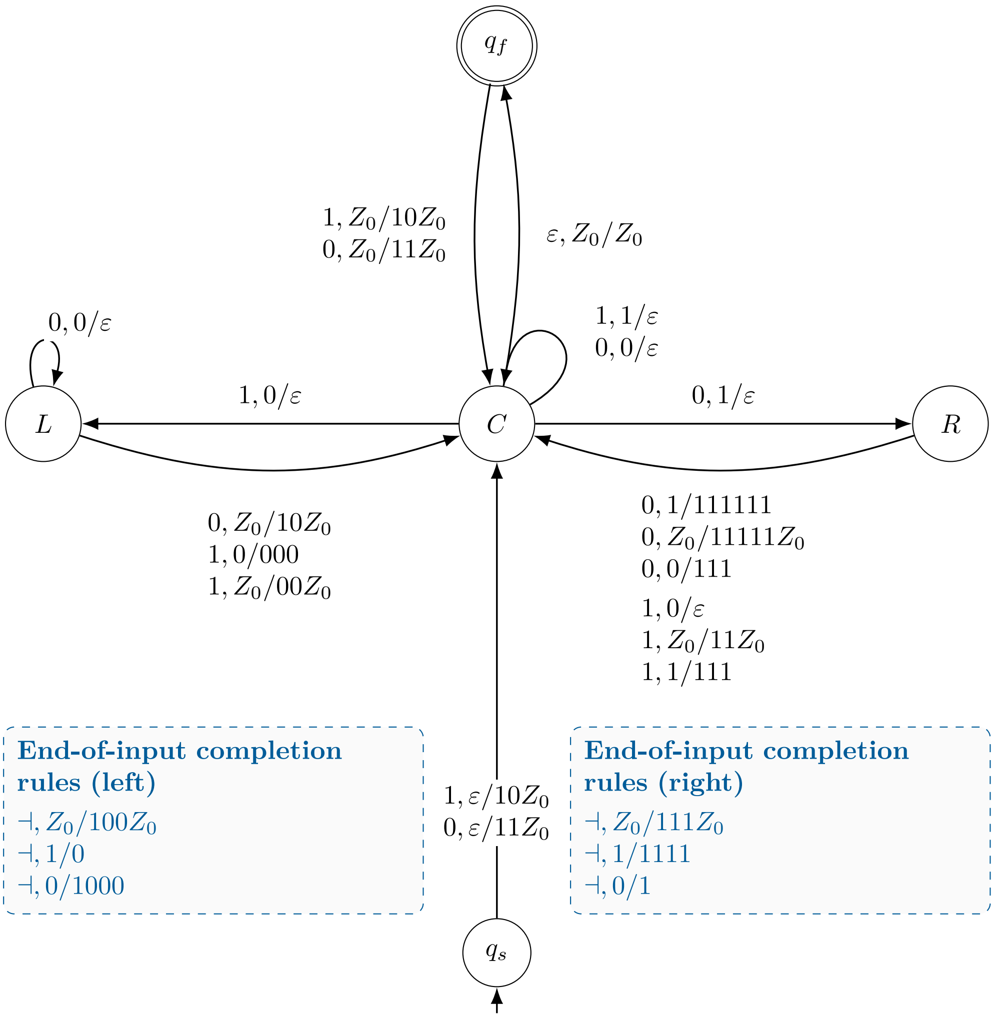


Figure 90.1: The empty-input-excluded original DPDA revisited after the general theory. This is the same formalized machine as Figure 1.1, deliberately repeated so that the final interpretation can be read against the original object.

This version recognizes

$$L_{2,1} \setminus \{\varepsilon\}.$$

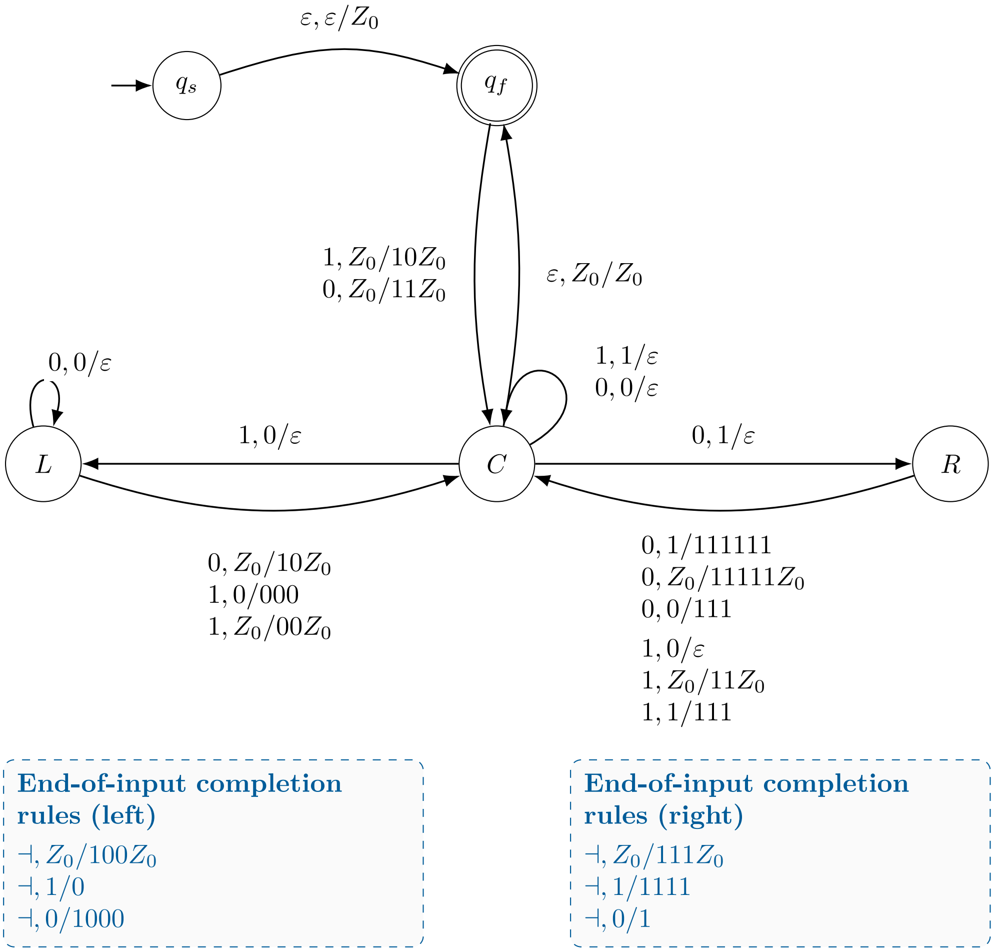


Figure 90.2: The empty-input-included original DPDA revisited. This is the same recognition/completion core as Figure 1.2; the wrapper accepts ε.

This version recognizes

$$L_{2,1}.$$

The common $C/L/R$ operational core is the same.
The difference is the acceptance/initialization wrapper.
This is the first lesson of the completed theory:

$$\boxed{\text{acceptance convention} \neq \text{semantic core.}}$$

90.2 What the stack was storing

The original stack looked like an operational memory structure.
Chapter 1 proved that its reachable form is highly restricted.
The state-potential invariant then gave

$$\boxed{\rho(w) = D(S) + \Phi(q).}$$

For 2:1,

$$D(S) = 2N_0(S) - N_1(S)$$

under the original machine's stack-weight convention, while the state potentials are

$$\Phi(C) = 0, \qquad \Phi(L) = 3, \qquad \Phi(R) = -3.$$

Thus the machine stores the arithmetic imbalance **distributed between control and stack**.
That observation became the prototype for every later semantic/physical decomposition.

90.3 What the completion rules were doing

The right-endmarker completion rules do not create new semantic information.
They materialize information already present in the final state/stack configuration.
After normalization, the physical word decodes a shortest completion demand.
The machine therefore contains an implicit map

$$\boxed{\begin{array}{c}\text{physical configuration} \to \text{residual} \to \text{minimum demand}\\ \to \text{canonical shortest continuation.}\end{array}}$$

This was the discovery that opened the entire completion-geometry branch.

90.4 Why depth two appeared

The original normalized stores lie in only a few structural families.
Therefore the qualitative question

Which demand symbols occur anywhere in the remaining stack?

is determined by the first two cells.

The general centered $p{:}q$ machine later recovered this two-cell qualitative visibility through a completely different canonical ordering.

Thus the original depth-two property was not merely a graphical accident.

It was an early manifestation of a broader principle:

$$\boxed{\text{unbounded quantitative demand can have bounded qualitative visibility.}}$$

90.5 The cancellation transition

The transition

$$\boxed{(L, 0S) \xrightarrow{0} (L, S)}$$

acquired three meanings.

Machine meaning

Delete one matching stack symbol.

Geodesic meaning

Consume one required zero along a shortest completion.

Combinatorial meaning

In the nonnegative 2-Dyck sector, mark a down-step landing at an exact odd-height family.

That third interpretation generated:

$$\begin{aligned} \text{correction signatures} \to \text{defect} \to \text{Catalan boundary} \\ \to \text{binary skeleton} \to \text{enumeration.} \end{aligned}$$

Thus one local transition produced an entire independent research branch.

90.6 Why the improved raw DPDA looked so different

The three-symbol raw residual-conjugate DPDA of Chapters 7–8 is not a cleaned-up drawing of the original machine.

It solves a different physical representation problem.

It asks:

> If the residual itself is the semantic state, what is the most efficient stable raw encoding within the stated model?

That representation moves finite phase information into control and leaves unary magnitude on the stack.

It achieves the exact raw residual-faithful frontier.

The original machine instead stores a richer literal completion-demand process with normalization chronology.

Therefore the two machines are not rivals for one undefined notion of optimality.

They are optimal or explanatory in different representation categories.

90.7 Why the general $p{:}q$ machine returned to the original architecture

The centered real-time $p{:}q$ construction deliberately preserved the operational idea of the original machine:

- literal demand on the stack;
- one-cell LIFO blocker displacement;
- signed debt in finite control;
- later WRITE materialization.

When

$$(p, q) = (2, 1),$$

the general debt rail becomes

$$D_{-1}, D_0, D_1$$

with

$$\boxed{D_{-1} = R, \qquad D_0 = C, \qquad D_1 = L.}$$

The general potentials become

$$-3, 0, +3.$$

The historical WRITE arithmetic is recovered from the kernel-carry compiler.
Thus the original architecture was not merely generalized by analogy.
It was reconstructed from the general arithmetic.

90.8 Why the historical table was not literally the centered specialization

Two reachable right-state matching families remained different.
The centered machine keeps debt lazily in R.
The historical machine immediately rematerializes the same semantic demand in C.
Therefore:

$$\boxed{\text{historical} = \text{eager normalization,}}$$

$$\boxed{\text{centered specialization} = \text{lazy normalization.}}$$

This small transition-table discrepancy turned out to be one of the most important facts in the research program.
It proved concretely that:

$$\boxed{\text{same semantics} \not\Rightarrow \text{unique physical chronology.}}$$

That observation led directly to Structure Theory and Uniqueness Theory.

90.9 C/L/R after Structure Theory

For 2:1,

$$r = p + q = 3.$$

The WSC exchange-sheet coordinate gives

$$\boxed{s(C) = 0, \qquad s(L) = 1, \qquad s(R) = -1.}$$

Thus the original states become SEARCH sheets.

The characteristic stack-length changes

$$-1, +2, +5$$

become materialization levels

$$M = 0, 1, 2.$$

And the historical identity

$$b + 2c = n$$

becomes a specialization of the global materialization budget.

The original table has therefore been reinterpreted from local syntax into semantic/physical accounting.

90.10 C/L/R after Uniqueness Theory

The original eager machine is one reduced lift.

The centered lazy machine is another.

They are:

- semantically equivalent;
- section-preserving incomparable;
- coordinate-reduced;
- non-isomorphic.

Within the selected three-site family, there are six admissible reduced normalization schedules.

The historical machine is therefore not “the” unique physical realization of the semantics.

It is a distinguished point in a nontrivial lift space.

90.11 The original machine after General Theory

The General Theory gives a final vocabulary.

The original configuration space is a physical lift.

The residual is semantic state.

The canonical completion word is a section choice.

The side-state configurations occupy noncanonical semantic fibers.

The one-top LIFO rule creates locality constraints.

Leaving the canonical section creates representation debt.

Some schedule choices preserve the behavioral hull.

Some expose fatal frontiers.

Alternative coordinate systems can remove obstructions that are essential in the literal policy.

Thus the full research program can be read backward into the original figure.

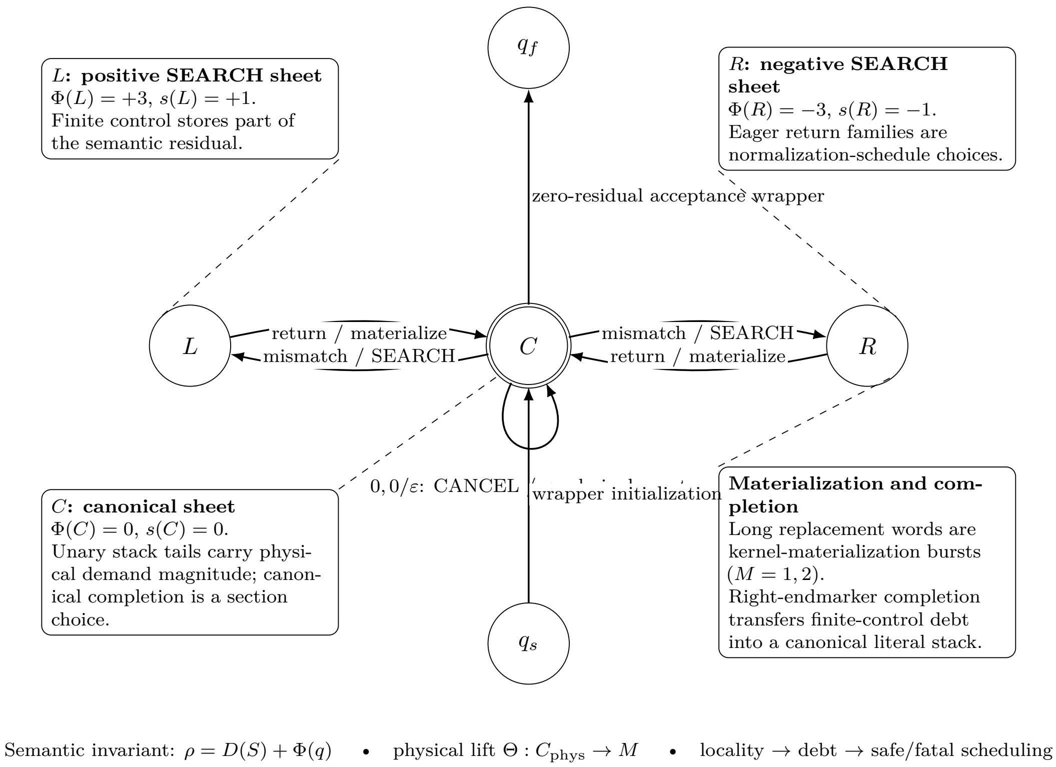


Figure 90.3: The original $C/L/R$ architecture annotated by the complete theory. The figure is interpretive rather than a replacement transition table: it overlays the residual/state-potential invariant, WSC SEARCH-sheet coordinates, literal-demand magnitude, kernel materialization, end-of-input completion, schedule freedom, and the semantic-lift viewpoint on the original machine architecture.

Suggested annotations:

- C: canonical sheet / zero SEARCH charge;
- L: positive SEARCH sheet;
- R: negative SEARCH sheet;
- unary stack tails: physical demand magnitude;
- $0, 0/\varepsilon$: CANCEL / geodesic descent / correction event;
- long replacement words: kernel-materialization bursts;
- right-endmarker completion: final canonical materialization;
- eager right-state families: schedule choice;
- wrapper: acceptance convention, not semantic core.

The final visual object of the monograph should be this reinterpreted original machine.

Chapter 91

What Was Forced, What Was Chosen, What Became General

The entire monograph can now be organized by one question:

$$\boxed{\text{Which features were mathematically forced, and at what level?}}$$

The answer requires several categories.

91.1 Language-semantic necessities

These are determined by $L_{p,q}$ itself.

Residual quotient

$$\boxed{u^{-1}L_{p,q} = v^{-1}L_{p,q} \iff \rho(u) = \rho(v).}$$

The right-residual system is canonical.

Minimum completion demand

Each residual determines one unique minimum nonnegative completion Parikh vector.

Primitive kernel

$$\boxed{K = (q, p).}$$

Every equivalent completion vector differs by a kernel packet.

Semantic carry chronology

For a fixed input word,

$$(\rho_t), \quad (\kappa_t), \quad (\mu_t), \quad (\chi_t)$$

are unique.

These facts do not depend on a pushdown representation.

91.2 Necessities of a chosen locality category

These are not language-intrinsic, but become forced once the physical category is fixed.

Examples include:

One-top suffix obstruction

A hidden suffix cannot be arbitrarily rearranged by one top rewrite.

Literal-policy fiber branching

For nontrivial $p{:}q$, the fixed canonical literal section cannot remain transition-closed.

Deferred-SEARCH control width

$$2\left\lfloor \frac{p+q-1}{2} \right\rfloor + 1$$

states are exact in the canonical deferred-SEARCH model.

Bounded-local collapse capacity

Too many semantic predecessors cannot be distinguished by too few local observations.

These are **representation-category necessities**.

They must always be stated with their category.

91.3 Coordinate artifacts

Some difficulties disappear when the physical coordinates change.

The clearest example is the four-symbol normalized lift.

Its

$$p+q-1$$

raw boundary defects were exact in those coordinates.

But a boundary-compatible coordinate shift removed them and produced a three-symbol raw DPDA.

Therefore:

$$\boxed{\text{locality defect} \not\Rightarrow \text{semantic necessity.}}$$

A defect may belong to the coordinate system.

91.4 Normalization-schedule choices

The semantic chronology can be unique while physical materialization chronology varies.

Examples:

- historical eager 2:1;
- centered lazy 2:1;
- the six admissible reduced 2:1 schedules.

The total kernel budget remains rigid even when its timing changes.

Thus:

$$\boxed{\text{semantic WRITE arrivals are forced;}}$$

$$\boxed{\text{physical WRITE timing may be chosen.}}$$

91.5 Special 2:1 features

Several concrete facts are specializations rather than universal principles.

For

$$p = 2, \qquad q = 1:$$

$$K = (1, 2),$$

$$m = 3,$$

$$h = 1,$$

so the centered debt rail has exactly three states.

These become:

$$R, C, L.$$

The state potentials are

$$-3, 0, +3.$$

The historical materialization changes are

$$-1, +2, +5.$$

The correction transition in the nonnegative path sector lands at

$$3, 5, 7, \ldots$$

These numerical values are 2:1-specific.

The principles they instantiate are broader.

91.6 What generalized to all coprime p:q

The following extend uniformly:

- residual semantics;
- shortest completion demand;
- primitive kernel;
- geodesic rectangles;
- bounded canonical action visibility;
- raw residual-conjugate DPDA;
- exact raw residual-faithful frontier;
- centered real-time literal-demand DPDA;
- SEARCH debt;
- minimal kernel-carry WRITE;
- canonical-segment invariant;
- WSC semantic carry factorization;
- materialization/backlog accounting.

The original 2:1 machine was therefore a genuine first member of a general structural family.

91.7 What generalized beyond rational binary counting

Several branches moved beyond $L_{p,q}$.

Matrix completion geometry

The unique minimum pair became an optimal Parikh fiber.

The rectangle became a cubical quotient by the equal-length kernel.

Topological hierarchy

Kernel rank controlled local/global shortest-schedule topology.

Finite geodesic compilation

The shortest-completion viewpoint generalized further to positive-cost finite-phase periodic systems. Finite arithmetic data compile exact distance, optimal branching, arbitrary shortest-prefix membership, and the complete local optimal-compatibility complex. Across the class, that compatibility hierarchy is strictly non-collapsing at every prescribed order and admits Boolean amplification under products. The universal k-fiber theorem sharpens this: even after exact distance and the complete hierarchy through order $k-1$ are fixed, all $2^{\binom{s}{k}}$ next-order patterns on a distinguished structural s-simplex can occur, including arbitrarily overlapping k-faces.

General exact-lift theory

Semantic fibers, canonical sections, bounded-local obstruction, behavioral hulls, containment, causality, synthesis, and verification no longer depend specifically on weighted binary balance.

Thus the project eventually generalized not just the machine, but the distinction that the machine exposed:

> what must be remembered semantically
>
> *versus*
>
> how a LIFO machine can physically remember it.

91.8 The research story in one chain

The completed narrative has one main discovery spine followed by consequence branches that reuse the semantic objects without pretending that they motivated the earlier construction:

2023 completion-aware 2:1 DPDA

↓

state+stack residual invariant → shortest completion demand

↓

geodesic geometry → better coordinates → resource frontiers

↓

general p:q machine → WRITE–SEARCH–CANCEL

↓

exact lifts → schedule geometry → all-coprime causal classification

↓

SCQ → protected interfaces → BBR → minimal protected cores

↓

typed confluence: schedules → fibers → sheets → protected cores

↓

General Theory: locality, debt, safety, synthesis, verification

↓

shortest completion → geodesic compiler → prescribed-order separation
→ Boolean amplification → universal k-fiber freedom

normalization events → Schubert/Bizley/Raney → one-counter compression

residual → contacts → determinant enumeration

shortest completion → Farey recursion → universal spectra

lift freedom → fixed-semantics chronology → three LIFO scales

optimal fiber → lattice-convex realization → rank raising → higher homology

optimal fiber → π_1 → K_0 → holonomy → carrier cost

CCR-3 → connected quadruples → four-band transport → Stirling shadow

$\{(p+q,2),(M,3)\}$ → bounded-push staircase
→ protected-contract Pareto synthesis

↓

Closing the Loop at the original 2:1 DPDA.

The displayed branches are consequences and syntheses rather than a single linear causation claim. The geodesic-compiler branch is placed earlier in the book, immediately after the higher-dimensional completion geometry from which it is abstracted; the remaining post-theory branches occur after the General Theory. Their order is chosen to preserve mathematical prerequisites and narrative continuity: operational statistics precede intrinsic path statistics, topology precedes physical holonomy, and endpoint resource theorems precede Pareto synthesis.

91.9 Final perspective

The research program began with a concrete deterministic pushdown automaton.

Its most productive feature was not simply that it recognized

$$L_{2,1}.$$

It distributed one arithmetic quantity across:

- finite control;
- a LIFO store;
- completion operations.

Trying to understand that distribution revealed:

$$\boxed{\text{semantic information} \neq \text{physical representation.}}$$

Everything that followed can be traced to that separation. The same semantic extraction later reached beyond the first pushdown representation itself: shortest-completion geometry admitted a finite cubical geodesic compiler whose local optimal-compatibility hierarchy is strict at every prescribed order; normalization history became an operational enumerative object; the residual became rational-path rank and boundary contact; the shortest-completion map became a Farey-recursive arithmetic object with universal scalar spectra; optimal-fiber topology acquired both a rank-raising realization calculus and a distinct physical holonomy/LIFO-cost layer; chronological carry acquired an exact fourth-order deformation parameter; and the state lower-bound branch closed to an unrestricted state–stack frontier with typed bounded-push refinements. The final protected-contract synthesis then makes explicit which resource reductions are genuine and which are only bounded changes of coordinates.

The final lesson is therefore not one specific DPDA construction or one specific theorem.

It is a method:

$$\boxed{\text{extract the semantic object from the machine,}}$$

$$\boxed{\text{identify which physical constraints are truly forced,}}$$

$$\boxed{\text{change coordinates when the obstruction is representational,}}$$

$$\boxed{\text{retain debt when locality forces it,}}$$

$$\boxed{\text{and distinguish semantic correctness from normalization history.}}$$

One additional lesson of the v6 synthesis is that

$$\boxed{\text{shortest-path geometry can be finite in description}}$$

while

$$\boxed{\begin{array}{c}\text{its optimal compatibility is unbounded in}\\ \text{informational order across the class.}\end{array}}$$

The original 2023 DPDA remains the first concrete object in that story—and, after the full theory, the final object through which the story can be read.

Appendix A

Complete Transition Audit of the Original 2023 DPDA

A.1 Purpose and scope

This appendix closes the finite verification step that is deliberately compressed in Chapter 1.

The object audited here is the **original 2023 completion-aware** 2:1 **DPDA**, in the formalized presentation used throughout this monograph. The language convention is

$$L_{2,1} = \{\, w \in \{0,1\}^* : N_1(w) = 2N_0(w) \,\},$$

with signed prefix residual

$$\boxed{\rho(w) = N_1(w) - 2N_0(w).}$$

For a logical stack word $S \in \{0,1\}^*$, define

$$\boxed{D(S) = 2N_0(S) - N_1(S)}$$

and assign the core-state potentials

$$\boxed{\Phi(C) = 0, \qquad \Phi(L) = 3, \qquad \Phi(R) = -3.}$$

The Chapter 1 invariant is

$$\boxed{\rho(w) = D(S) + \Phi(q)} \tag{A.1}$$

for every processed prefix w, where (q, S) is the corresponding core-projected configuration.

Since reading a binary symbol changes the residual by

$$\Delta\rho = \begin{cases} -2, & a = 0, \\ +1, & a = 1, \end{cases} \tag{A.2}$$

it is enough to verify, for every reachable recognition transition,

$$\boxed{\Delta D + \Delta\Phi = \Delta\rho.} \tag{A.3}$$

The audit also checks that every transition preserves the state-dependent reachable-store families

$$\boxed{\mathcal{S}_C = 0^* \cup 1^* \cup \{10\}, \qquad \mathcal{S}_L = 0^*, \qquad \mathcal{S}_R = 1^* \cup \{0\}.} \tag{A.4}$$

The bottom marker Z_0 is omitted from logical-stack words below. The leftmost logical symbol is the top.

A.2 Wrapper projection

The two formal variants differ only in how initialization and empty-input acceptance are wrapped around the semantic core.

The projection used throughout the monograph is

$$\pi(C, SZ_0) = (C, S), \qquad \pi(L, SZ_0) = (L, S), \qquad \pi(R, SZ_0) = (R, S),$$

together with

$$\boxed{\pi(q_s, \varepsilon) = (C, \varepsilon), \qquad \pi(q_f, Z_0) = (C, \varepsilon).} \tag{A.5}$$

Thus the wrappers carry no additional residual semantics.

Empty-input-excluded wrapper

The first binary symbol is consumed directly at the source wrapper.

input	projected source	projected target	ΔD	$\Delta\Phi$	$\Delta(D+\Phi)$	$\Delta\rho$
1	(C, ε)	$(C, 10)$	$+1$	0	$+1$	$+1$
0	(C, ε)	$(C, 11)$	-2	0	-2	-2

Indeed,

$$D(10) = 2 - 1 = 1, \qquad D(11) = -2.$$

Empty-input-included wrapper

The initialization move

$$\varepsilon, \varepsilon/Z_0$$

takes the physical source state to q_f. Under (A.5), both endpoints represent (C, ε), so this move has semantic change 0.

When the first binary symbol is later consumed from q_f,

$$1, Z_0/10Z_0 \quad \text{or} \quad 0, Z_0/11Z_0,$$

the same two projected checks above apply.

Finally, the core-to-final move

$$\varepsilon, Z_0/Z_0$$

is available only at the balanced empty logical stack and projects

$$(C, \varepsilon) \longrightarrow (C, \varepsilon).$$

It therefore also preserves (A.1).

A.3 Center-state recognition transitions

A.3.1 Matching 1-cancellation

$$(C, 1T) \xrightarrow{1} (C, T).$$

One 1 is popped, hence

$$\Delta D = +1, \qquad \Delta\Phi = 0.$$

Therefore

$$\Delta(D + \Phi) = +1 = \Delta\rho.$$

The target remains in $0^* \cup 1^* \cup \{10\}$: from a homogeneous 1-stack it remains homogeneous, while the exceptional source 10 becomes 0.

A.3.2 Matching 0-cancellation

$$(C, 0T) \xrightarrow{0} (C, T).$$

One 0 is popped, so

$$\Delta D = -2, \qquad \Delta\Phi = 0,$$

and therefore

$$\Delta(D + \Phi) = -2 = \Delta\rho.$$

Reachability forces the source to lie in 0^+, so the target lies in 0^*.

A.3.3 Center-to-left mismatch

$$(C, 0T) \xrightarrow{1} (L, T).$$

Popping the top 0 gives

$$\Delta D = -2.$$

The state potential changes by

$$\Delta\Phi = \Phi(L) - \Phi(C) = 3.$$

Thus

$$\boxed{\Delta(D + \Phi) = -2 + 3 = +1 = \Delta\rho.}$$

Because the source is a reachable center configuration with top 0, it belongs to 0^+; hence $T \in 0^*$, exactly the required left-state family.

A.3.4 Center-to-right mismatch

$$(C, 1T) \xrightarrow{0} (R, T).$$

Popping one 1 yields

$$\Delta D = +1,$$

while

$$\Delta\Phi = \Phi(R) - \Phi(C) = -3.$$

Therefore

$$\boxed{\Delta(D + \Phi) = 1 - 3 = -2 = \Delta\rho.}$$

There are two reachable source shapes:

$$1^{m+1} \longmapsto 1^m,$$

which enters the 1^* part of $\mathcal{S}_R$, and

$$10 \longmapsto 0,$$

which produces the exceptional right-state store 0.
These are exactly the two right-state families in (A.4).

A.4 Left-state recognition transitions

Every reachable left-state logical store lies in 0^*.

A.4.1 Left-state 0-cancellation

$$(L, 0T) \xrightarrow{0} (L, T).$$

The stack change is one popped 0:

$$\Delta D = -2, \qquad \Delta\Phi = 0.$$

Hence

$$\Delta(D + \Phi) = -2 = \Delta\rho.$$

The target remains in 0^*.

A.4.2 Empty-left state on input 0

The bottom-marker rule is

$$0, Z_0/10Z_0,$$

so logically

$$(L, \varepsilon) \xrightarrow{0} (C, 10).$$

The source value is

$$D(\varepsilon) + \Phi(L) = 3.$$

The target value is

$$D(10) + \Phi(C) = 1.$$

Thus

$$\Delta(D + \Phi) = 1 - 3 = -2 = \Delta\rho.$$

The target 10 is the exceptional center word permitted by $\mathcal{S}_C$.

A.4.3 Nonempty-left state on input 1

For $T \in 0^*$,

$$(L, 0T) \xrightarrow{1} (C, 000T).$$

Replacing one 0 by three 0's adds two zeros, so

$$\Delta D = +4.$$

The state potential drops by 3:

$$\Delta\Phi = 0 - 3 = -3.$$

Hence

$$\boxed{\Delta(D + \Phi) = 4 - 3 = +1 = \Delta\rho.}$$

The target is again homogeneous 0^*.

A.4.4 Empty-left state on input 1

The bottom-marker rule

$$1, Z_0/00Z_0$$

gives

$$(L, \varepsilon) \xrightarrow{1} (C, 00).$$

Now

$$\Delta D = +4, \qquad \Delta\Phi = -3,$$

so again

$$\Delta(D + \Phi) = +1 = \Delta\rho.$$

The target lies in 0^*.

A.5 Right-state recognition transitions

Every reachable right-state logical store lies in

$$1^* \cup \{0\}.$$

A.5.1 Nonempty homogeneous-right state on input 0

For $T \in 1^*$,

$$(R, 1T) \xrightarrow{0} (C, 111111T).$$

Replacing one 1 by six 1's adds five ones, therefore

$$\Delta D = -5.$$

The state potential rises from -3 to 0:

$$\Delta\Phi = +3.$$

Thus

$$\boxed{\Delta(D + \Phi) = -5 + 3 = -2 = \Delta\rho.}$$

The target is in 1^*.

A.5.2 Empty-right state on input 0

The bottom-marker rule

$$0, Z_0/11111Z_0$$

gives

$$(R, \varepsilon) \xrightarrow{0} (C, 11111).$$

The stack contribution changes by

$$\Delta D = -5,$$

and the potential changes by $+3$. Hence

$$\Delta(D + \Phi) = -2 = \Delta\rho.$$

A.5.3 Exceptional right store 0 on input 0

The rule

$$0, 0/111$$

gives

$$(R, 0) \xrightarrow{0} (C, 111).$$

The stack weight changes from

$$D(0) = 2$$

to

$$D(111) = -3,$$

so

$$\Delta D = -5.$$

Again $\Delta\Phi = +3$, and

$$\Delta(D + \Phi) = -2 = \Delta\rho.$$

A.5.4 Exceptional right store 0 on input 1

The rule

$$1, 0/\varepsilon$$

gives

$$(R, 0) \xrightarrow{1} (C, \varepsilon).$$

The stack contribution changes by

$$\Delta D = -2,$$

while the state-potential change is +3. Therefore

$$\boxed{\Delta(D + \Phi) = -2 + 3 = +1 = \Delta\rho.}$$

A.5.5 Empty-right state on input 1

The bottom-marker rule

$$1, Z_0/11Z_0$$

gives

$$(R, \varepsilon) \xrightarrow{1} (C, 11).$$

Here

$$\Delta D = -2, \qquad \Delta\Phi = +3,$$

and therefore

$$\Delta(D + \Phi) = +1 = \Delta\rho.$$

A.5.6 Nonempty homogeneous-right state on input 1

For $T \in 1^*$,

$$(R, 1T) \xrightarrow{1} (C, 111T).$$

Replacing one 1 by three 1's adds two ones:

$$\Delta D = -2.$$

Together with $\Delta\Phi = +3$,

$$\boxed{\Delta(D + \Phi) = +1 = \Delta\rho.}$$

The target remains homogeneous 1^*.

A.6 Complete recognition-phase audit table

The preceding cases may be compressed into one table.

source	input / top	logical action	target	ΔD	$\Delta\Phi$	total	required
C	1/1	pop 1	C	+1	0	+1	+1
C	0/0	pop 0	C	−2	0	−2	−2
C	1/0	pop 0	L	−2	+3	+1	+1
C	0/1	pop 1	R	+1	−3	−2	−2
L	0/0	pop 0	L	−2	0	−2	−2
L	$0/Z_0$	write 10	C	+1	−3	−2	−2
L	1/0	$0 \mapsto 000$	C	+4	−3	+1	+1
L	$1/Z_0$	write 00	C	+4	−3	+1	+1
R	0/1	$1 \mapsto 111111$	C	−5	+3	−2	−2
R	$0/Z_0$	write 11111	C	−5	+3	−2	−2
R	0/0	$0 \mapsto 111$	C	−5	+3	−2	−2
R	1/0	pop 0	C	−2	+3	+1	+1
R	$1/Z_0$	write 11	C	−2	+3	+1	+1
R	1/1	$1 \mapsto 111$	C	−2	+3	+1	+1

Thus every reachable recognition transition satisfies (A.3).

This proves the finite transition step omitted from the short proof in Chapter 1.

A.7 Reachable-store closure

The same table proves the state-dependent store normal form.

Center

The center can contain only

$$0^*, \qquad 1^*, \qquad 10.$$

Its two matching loops shorten an existing block. A mismatch can leave the center only by popping the visible blocking symbol. Every side-to-center transition returns to a homogeneous block except the left-empty/input-0 case, which creates exactly 10.

Left

The only transition entering L is

$$(C, 0T) \xrightarrow{1} (L, T).$$

Reachability forces the source center stack to belong to 0^+, so

$$T \in 0^*.$$

Therefore no reachable left configuration has top 1.
This proves, in particular, that the listed completion rule

$$\dashv, 1/0$$

has an unreachable source condition.

Right

The only transition entering R is

$$(C, 1T) \xrightarrow{0} (R, T).$$

If the source is 1^{m+1}, then $T = 1^m$.
If the source is the exceptional center word 10, then $T = 0$.
Hence

$$\mathcal{S}_R = 1^* \cup \{0\}.$$

No other right-state store can be produced.

A.8 Audit of the end-of-input completion rules

The endmarker $\dashv$ is not a binary input symbol, so completion normalization must preserve the residual already accumulated by the prefix.

The side-state potential is transferred into literal stack content.

A.8.1 Left-state completion

Every reachable left store is 0^m.

Empty left store.

$$(L, \varepsilon) \xrightarrow{\dashv} 100.$$

Before completion,

$$D(\varepsilon) + \Phi(L) = 0 + 3 = 3.$$

After completion,

$$D(100) = 4 - 1 = 3.$$

Nonempty left store. For $m \geq 1$,

$$(L, 0^m) \xrightarrow{\dashv} 1\,0^{m+2}.$$

Indeed the physical rule replaces the top 0 by 1000, leaving 0^{m-1} untouched.
Before completion,

$$D(0^m) + \Phi(L) = 2m + 3.$$

After completion,

$$D(1\,0^{m+2}) = 2(m+2) - 1 = 2m + 3.$$

Thus left completion preserves the represented residual exactly.
The listed left rule with top 1 is unreachable and plays no role in any run.

A.8.2 Right-state completion

There are three reachable right forms.

Empty right store.

$$(R, \varepsilon) \xrightarrow{\dashv} 111.$$

Before completion,

$$D(\varepsilon) + \Phi(R) = -3,$$

and after completion,

$$D(111) = -3.$$

Nonempty homogeneous right store. For $m \geq 1$,

$$(R, 1^m) \xrightarrow{\dashv} 1^{m+3}.$$

Before completion,

$$D(1^m) + \Phi(R) = -m - 3.$$

After completion,

$$D(1^{m+3}) = -m - 3.$$

Exceptional right store.

$$(R, 0) \xrightarrow{\dashv} 1.$$

Before completion,

$$D(0) + \Phi(R) = 2 - 3 = -1.$$

After completion,

$$D(1) = -1.$$

Therefore every reachable endmarker normalization rule preserves the prefix residual.

A.9 From residual preservation to the canonical shortest completion

After the side-state correction, every normalized logical stack belongs to

$$0^* \cup 1^* \cup 10^+.$$

The three residual cases are therefore:

$$\widehat{S}(w) = \begin{cases} 0^m, & \rho(w) = 2m, \quad m \geq 0, \\ 1\,0^{m+1}, & \rho(w) = 2m+1, \quad m \geq 0, \\ 1^m, & \rho(w) = -m, \quad m \geq 1. \end{cases} \tag{A.6}$$

For any continuation containing x zeros and y ones,

$$wz \in L_{2,1} \iff 2x - y = \rho(w). \tag{A.7}$$

The words in (A.6) realize the unique minimum nonnegative Parikh solution of (A.7):

- $2m$: $(x, y) = (m, 0)$;
- $2m + 1$: $(x, y) = (m + 1, 1)$;
- $-m$: $(x, y) = (0, m)$.

Thus the transition audit not only proves recognition-phase arithmetic consistency; together with the endmarker audit it recovers the shortest-completion semantics of the original design.

A.10 Two representative runs

A.10.1 Accepted word 110

Using the core projection and omitting Z_0,

$$(C, \varepsilon) \xrightarrow{1} (C, 10) \xrightarrow{1} (C, 0) \xrightarrow{0} (C, \varepsilon).$$

The residual sequence is

$$0,\ 1,\ 2,\ 0,$$

while $D(S) + \Phi(q)$ gives the same sequence:

$$0,\ 1,\ 2,\ 0.$$

The final empty center stack therefore represents zero residual and the word is accepted.

A.10.2 Prefix 111

The run is

$$(C, \varepsilon) \xrightarrow{1} (C, 10) \xrightarrow{1} (C, 0) \xrightarrow{1} (L, \varepsilon).$$

The residual is

$$\rho(111) = 3.$$

The final logical stack is empty, but

$$D(\varepsilon) + \Phi(L) = 3.$$

The endmarker rule transfers this finite-control contribution into the literal completion word:

$$(L, \varepsilon) \xrightarrow{\dashv} 100.$$

Since

$$2N_0(100) - N_1(100) = 4 - 1 = 3,$$

the normalized stack represents exactly the outstanding residual. Appending 100 gives

$$111100 \in L_{2,1}.$$

This is the canonical shortest completion selected by the machine.

A.11 Determinism of the completion extension

The completion callouts should not be interpreted as ordinary ε-moves.

For a fixed state/top pair:

- binary recognition transitions are selected by next input 0 or 1;
- completion is selected only by the distinct right endmarker $\dashv$.

Direct all completion rules to a dedicated completion sink with no outgoing transitions.

Then no configuration has two enabled successors.

Hence the completion-extended formalization is deterministic in the standard DPDA sense.

A.12 Exhaustive computational cross-check

The symbolic audit above is complete: it covers every reachable transition family.

The accompanying P1 validation suite provides an independent bounded regression layer. In the final research corpus it checks, among other items,

- 524,287 original-machine binary inputs;
- 8,912,898 prefix transition checks;
- the residual invariant;
- the reachable-store normal form;
- completion normalization;
- independent shortest-completion length;
- depth-two support visibility.

All checks pass.

The computational audit is supporting evidence only; the proof of (A.1) is the finite symbolic transition audit in Sections A.2–A.8.

A.13 Conclusion

The original 2023 design is not merely a correct recognizer.

Its state and stack form an exact split representation of the prefix residual:

$$\boxed{\rho = D + \Phi.}$$

The center state stores the residual entirely in the stack.

The side states store part of the same residual in finite control:

$$\boxed{\Phi(L) = +3, \qquad \Phi(R) = -3.}$$

Every binary transition preserves that semantic quantity with the correct residual increment, and every reachable endmarker rule transfers the finite-control offset back into literal stack form.

Thus the three operational ideas visible in the original machine—

$$\boxed{\text{cancellation}, \qquad \text{side-state mismatch}, \qquad \text{completion},}$$

—are already an exact semantic normalization mechanism.

This is the local machine-level fact from which the later completion geometry, residual quotients, WRITE–SEARCH–CANCEL theory, lift theory, and General Theory are developed.

Appendix B

General Construction Transition Schemas

B.1 Purpose and model separation

The main text deliberately develops the general constructions conceptually before presenting every transition rule. This appendix collects the exact machine syntax in one place so that the book remains reproducible without forcing long transition tables into the narrative chapters.

Fix coprime integers $p, q \geq 1$ and write

$$\rho(w) = qN_1(w) - pN_0(w), \qquad T_0(r) = r - p, \qquad T_1(r) = r + q.$$

The primitive zero-weight kernel is

$$K = (q, p).$$

Three constructions must be kept separate.

1. The *paired-sign four-symbol lift* realizes the residual graph exactly after a deterministic normalization macro-step. It is a normalized residual–configuration conjugacy, not a raw one-step realization.

2. The *boundary-compatible three-symbol lift* changes coordinates so that every residual edge is one ordinary binary-input transition. It is ε-free and gives raw residual–configuration conjugacy with $\max(p, q)$ states.

3. The *centered literal-demand construction* stores the unique shortest completion demand itself, uses a finite-control debt rail to repair the LIFO ordering obstruction, consumes exactly one binary symbol per transition, and is real-time during binary input.

The exact resource-frontier theorems later compare these constructions only inside their stated representation models. In particular, no table below should be read as an unrestricted minimum-state theorem for every DPDA recognizing $L_{p,q}$.

B.2 Four-symbol normalized paired-sign lift

This is the first compact general physical lift used for the residual-geometry and configuration-rectangle results.

Put

$$m_{\mathrm{st}} = \max(p,q), \qquad Q_4 = \{R_0, \dots, R_{m_{\mathrm{st}}-1}\}, \qquad \Gamma_4 = \{0,1,X,Z_0\}.$$

For $0 \le s < p$ and $0 \le t < q$, define the finite residual bases

$$a_s = \begin{cases} 0, & s = 0, \\ s-p, & 1 \le s < p, \end{cases} \qquad b_t = \begin{cases} 0, & t = 0, \\ q-t, & 1 \le t < q. \end{cases}$$

The stable encoding $\eta_4 : \mathbb{Z} \to Q_4 \times \Gamma_4^*$ is

$$\eta_4(0) = (R_0, Z_0).$$

If $r > 0$, let $s = r \bmod p$ and write uniquely

$$r = pk + a_s, \qquad k \ge 1;$$

then

$$\eta_4(r) = (R_s, 0^k Z_0).$$

If $r < 0$, let $t = (-r) \bmod q$ and write uniquely

$$r = -qk + b_t, \qquad k \ge 1;$$

then

$$\eta_4(r) = (R_t, 1^k X).$$

The two sign roles may share a control state; their work symbols and sentinels keep the stable encodings disjoint.

Positive stable rules

For $0 \le s < p$, put

$$s' = (s+q) \bmod p, \qquad c_s = \frac{a_s + q - a_{s'}}{p}.$$

Then

$$\boxed{\delta(R_s, 0, 0) = (R_s, \varepsilon)}$$

and

$$\boxed{\delta(R_s, 1, 0) = (R_{s'}, 0^{c_s+1})}.$$

When the last positive work symbol is removed, a unique forced boundary normalization is used:

$$\boxed{\delta(R_s, \varepsilon, Z_0) = \eta_4(a_s)} \qquad (1 \le s < p).$$

For $s = 0$, popping the final positive 0 already exposes the zero encoding (R_0, Z_0), so no boundary normalization is required.

Negative stable rules

For $0 \le t < q$, put

$$t' = (t+p) \bmod q, \qquad d_t = \frac{p - b_t + b_{t'}}{q}.$$

Then

$$\boxed{\delta(R_t, 1, 1) = (R_t, \varepsilon)}$$

and

$$\boxed{\delta(R_t, 0, 1) = (R_{t'}, 1^{d_t+1})}.$$

When the negative sentinel is exposed,

$$\boxed{\delta(R_t, \varepsilon, X) = \eta_4(b_t)}.$$

Zero rules and acceptance

At residual zero,

$$\boxed{\delta(R_0, 1, Z_0) = \eta_4(q)}, \qquad \boxed{\delta(R_0, 0, Z_0) = \eta_4(-p)}.$$

A mandatory right endmarker is accepted only at the stable zero configuration $\eta_4(0)$.

An expression of the form $\delta(\cdot) = \eta_4(r')$ is semantic shorthand: the exposed boundary symbol is replaced by the stack word of the stable encoding while control enters the corresponding state. Long fixed writes may be compiled into deterministic forced ε-chains if a bounded-write PDA syntax is required. Such a compilation preserves the stable target but is no longer a raw one-binary-transition realization.

Define a normalized input macro-step $c \overset{a}{\Rightarrow} c'$ to mean one consumed binary symbol followed by the unique forced normalization chain. The transition schema satisfies

$$\boxed{\eta_4(r) \overset{0}{\Rightarrow} \eta_4(r-p), \qquad \eta_4(r) \overset{1}{\Rightarrow} \eta_4(r+q).}$$

This is the exact normalized residual–configuration conjugacy used in the configuration-rectangle chapters.

B.3 One-work-symbol raw conjugacy

The one-work-symbol construction is included because it is the matching upper bound for the exact $p+q$-state result in the permanent-bottom raw-conjugacy model.

Let

$$\Gamma_2 = \{A, Z_0\},$$

with disjoint state families

$$P_0, \ldots, P_{p-1}, \qquad N_0, \ldots, N_{q-1}.$$

For $r \geq 0$, write

$$r = pk + s, \qquad k \geq 0, \quad 0 \leq s < p,$$

and define

$$\xi(r) = (P_s, A^k Z_0).$$

For $r < 0$, write uniquely

$$r = u - qh, \qquad 0 \leq u < q, \quad h \geq 1,$$

and define

$$\xi(r) = (N_u, A^{h-1} Z_0).$$

For $0 \leq s < p$, set

$$s^+ = (s+q) \bmod p, \qquad c_s = \frac{s+q-s^+}{p} \geq 0,$$

and write

$$s - p = u_s - qh_s, \qquad 0 \leq u_s < q, \quad h_s \geq 1.$$

The positive and boundary rules are

$$\boxed{\delta(P_s, 0, A) = (P_s, \varepsilon)}, \qquad \boxed{\delta(P_s, 1, A) = (P_{s^+}, A^{c_s+1})},$$

$$\boxed{\delta(P_s, 1, Z_0) = (P_{s^+}, A^{c_s} Z_0)}, \qquad \boxed{\delta(P_s, 0, Z_0) = (N_{u_s}, A^{h_s - 1} Z_0)}.$$

For $0 \le u < q$, put

$$u^- = (u - p) \bmod q, \qquad d_u = \frac{u^- - u + p}{q} \ge 0.$$

The negative rules are

$$\boxed{\delta(N_u, 1, A) = (N_u, \varepsilon)}, \qquad \boxed{\delta(N_u, 0, A) = (N_{u^-}, A^{d_u + 1})},$$

$$\boxed{\delta(N_u, 1, Z_0) = \xi(u)}, \qquad \boxed{\delta(N_u, 0, Z_0) = (N_{u^-}, A^{d_u} Z_0)}.$$

The mandatory right endmarker is accepted only at

$$\xi(0) = (P_0, Z_0).$$

There are no ε-transitions. Direct substitution gives

$$\boxed{\delta(\xi(r), 0) = \xi(r - p), \qquad \delta(\xi(r), 1) = \xi(r + q).}$$

The construction uses exactly $p + q$ states and two total stack symbols, one of which is the permanent bottom marker.

B.4 Boundary-compatible three-symbol raw DPDA

The four-symbol normalized lift shows that the residual graph has a compact physical realization, but it has a finite set of boundary edges that require normalization. The following coordinate change removes those defects completely.

Assume first that $p \ge q$. Let

$$Q_3 = \{R_0, \dots, R_{p-1}\}, \qquad \Gamma_3 = \{0, 1, Z_0\}.$$

For $r \ge 0$, write uniquely

$$r = pk + s, \qquad k \ge 0, \quad 0 \le s < p,$$

and set

$$\boxed{\zeta(r) = (R_s, 0^k Z_0)}.$$

For $r < 0$, write uniquely

$$r = u - qh, \qquad h \ge 1, \quad 0 \le u < q,$$

and set

$$\boxed{\zeta(r) = (R_u, 1^h Z_0)}.$$

The state sets overlap deliberately. The sign is visible from the top of stack: negative encodings have top 1, whereas nonnegative encodings have top 0 or Z_0.

For $0 \le s < p$, define

$$s^+ = (s + q) \bmod p, \qquad c_s = \frac{s + q - s^+}{p} \in \{0, 1\},$$

$$u_s = (s - p) \bmod q, \qquad h_s = \frac{u_s + p - s}{q} \ge 1.$$

For $0 \le u < q$, define

$$u^- = (u - p) \bmod q, \qquad d_u = \frac{u^- - u + p}{q} \ge 1.$$

Complete transition table for $p \geq q$

region	input/top	target	replacement
nonnegative interior	0/0	R_s	ε
nonnegative interior	1/0	R_{s^+}	0^{c_s+1}
nonnegative boundary	$1/Z_0$	R_{s^+}	$0^{c_s} Z_0$
nonnegative boundary	$0/Z_0$	R_{u_s}	$1^{h_s} Z_0$
negative	1/1	R_u	ε
negative	0/1	R_{u^-}	1^{d_u+1}

Equivalently,

$$\boxed{\begin{aligned} \delta(R_s, 0, 0) &= (R_s, \varepsilon), \\ \delta(R_s, 1, 0) &= (R_{s^+}, 0^{c_s+1}), \\ \delta(R_s, 1, Z_0) &= (R_{s^+}, 0^{c_s} Z_0), \\ \delta(R_s, 0, Z_0) &= (R_{u_s}, 1^{h_s} Z_0), \\ \delta(R_u, 1, 1) &= (R_u, \varepsilon), \\ \delta(R_u, 0, 1) &= (R_{u^-}, 1^{d_u+1}). \end{aligned}}$$

The right endmarker is accepted only at

$$\zeta(0) = (R_0, Z_0).$$

No ε-transition is used. Every binary transition consumes exactly one input symbol, and

$$\boxed{\delta(\zeta(r), 0) = \zeta(r - p), \qquad \delta(\zeta(r), 1) = \zeta(r + q).}$$

Hence the raw binary-input configuration graph is label-isomorphic to the residual graph.

If $q > p$, use the symmetric construction obtained by simultaneously exchanging

$$0 \leftrightarrow 1, \qquad p \leftrightarrow q, \qquad r \leftrightarrow -r.$$

The resulting state count is always

$$\boxed{|Q_3| = \max(p, q)},$$

with three total stack symbols including the permanent bottom marker.

B.5 Centered real-time literal-demand construction

The raw residual encodings above store a signed residual coordinate. The centered construction solves a different problem: it keeps a physical representation whose semantic projection is the unique shortest completion demand and repairs one-cell LIFO ordering conflicts by moving a blocker into finite control.

Let

$$m = p + q, \qquad h_{\text{ctr}} = \left\lfloor \frac{m - 1}{2} \right\rfloor.$$

For a residual r, let

$$\mu(r) = (x_r, y_r)$$

be the unique minimum nonnegative solution of

$$px - qy = r.$$

For a demand vector (x, y) use the canonical two-visible ordering

$$\eta(x,y) = \begin{cases} (10)^y 0^{x-y}, & x \geq y, \\ (10)^x 1^{y-x}, & y > x. \end{cases}$$

For debt $d \in \mathbb{Z}$, define the virtual demand

$$V_d = \mu(dm), \qquad \Phi(D_d) = dm.$$

The centered state set is

$$\boxed{Q_{\mathrm{ctr}} = \{D_{-h_{\mathrm{ctr}}}, \ldots, D_{-1}, D_0, D_1, \ldots, D_{h_{\mathrm{ctr}}}\},}$$

with $D_0 = C$, and

$$\Gamma_{\mathrm{ctr}} = \{0, 1, Z_0\}.$$

For a logical stack S, the semantic projection is

$$\Theta(D_d, S) = \mathrm{red}_{p,q}\big(P(S) + V_d\big).$$

The residual invariant is

$$\boxed{ho = W(P(S)) + dm.}$$

MATCH

For $z \in \{0, 1\}$ and every $|d| \leq h_{\mathrm{ctr}}$,

$$\boxed{(D_d, zT) \xrightarrow{z} (D_d, T).}$$

Interior SEARCH

For $d < h_{\mathrm{ctr}}$,

$$\boxed{(D_d, 0T) \xrightarrow{1} (D_{d+1}, T),}$$

and for $d > -h_{\mathrm{ctr}}$,

$$\boxed{(D_d, 1T) \xrightarrow{0} (D_{d-1}, T).}$$

The displaced blocker is not restored immediately. Its signed effect is represented by the debt coordinate d.

Outer-boundary WRITE

Put

$$U_+ = \eta(\mu((h_{\mathrm{ctr}} + 1)m))\,, \qquad U_- = \eta(\mu(-(h_{\mathrm{ctr}} + 1)m))\,.$$

Then

$$\boxed{(D_{h_{\mathrm{ctr}}}, 0T) \xrightarrow{1} (C, U_+T),}$$

and

$$\boxed{(D_{-h_{\mathrm{ctr}}}, 1T) \xrightarrow{0} (C, U_-T).}$$

Bottom WRITE

For every $|d| \leq h_{\mathrm{ctr}}$,

$$\boxed{(D_d, Z_0) \xrightarrow{1} (C, \eta(\mu(dm + q))Z_0)\,,}$$

$$\boxed{(D_d, Z_0) \xrightarrow{0} (C, \eta(\mu(dm - p))Z_0)\,.}$$

Acceptance

A mandatory right endmarker is permitted only at the balanced empty configuration:

$$\boxed{(C, Z_0) \xrightarrow{\dashv} (C, \varepsilon).}$$

There is no right-endmarker rule from any other reachable configuration. Every binary transition consumes exactly one input symbol; the centered construction has no ε-move during binary processing.

B.6 Minimal kernel-carry compiler identity

The WRITE strings in the centered construction are not ad hoc. Suppose a transition consumes input symbol a, removes visible top symbol z, changes state from D_s to D_t, and writes a replacement word u. Let

$$e_0 = (1, 0), \qquad e_1 = (0, 1).$$

Ignoring the untouched suffix, invariant preservation forces

$$P(u) = e_z - e_a + V_s - V_t + \kappa K.$$

At the bottom marker, omit the term e_z. If

$$B = e_z - e_a + V_s - V_t,$$

then the smallest kernel carry making the replacement nonnegative is

$$\boxed{\kappa_* = \max\left(\left\lceil -\frac{B_0}{q} \right\rceil, \left\lceil -\frac{B_1}{p} \right\rceil\right).}$$

The unique shortest compatible replacement vector is

$$\boxed{R_* = B + \kappa_* K.}$$

Every other nonnegative invariant-preserving replacement is $R_* + \ell K$ for some $\ell \geq 1$ and is longer by exactly $\ell(p + q)$ symbols.

This identity is the common arithmetic compiler behind the large literal WRITE words of the original 2:1 machine and the general centered family.

B.7 The 2:1 specialization and the eager–lazy distinction

For $(p, q) = (2, 1)$,

$$m = 3, \qquad h_{\text{ctr}} = 1,$$

so

$$D_{-1} = R, \qquad D_0 = C, \qquad D_1 = L,$$

with potentials

$$-3, \quad 0, \quad +3.$$

The centered schema reproduces the historical center/left/right control skeleton, the center mismatch pops, the left cancellation family, and the arithmetic of the characteristic WRITE words.

Two reachable right-state matching families differ in schedule, not semantics. The centered machine keeps the resulting debt in finite control (lazy materialization), whereas the historical machine immediately materializes the same canonical demand on the stack (eager materialization). If $\mathcal{N}$ denotes canonical materialization, then on exactly those families

$$\delta_{\text{hist}}(c, a) = \mathcal{N}(\delta_{\text{cent}}(c, a)) .$$

Accordingly, direct specialization is not claimed to be literal transition-table identity. The remaining difference is precisely the eager–lazy normalization-schedule choice.

B.8 Determinism and implementation conventions

The constructions in this appendix use three different transition disciplines and should not be conflated.

- In the four-symbol paired-sign lift, forced ε-normalization is enabled only on transient boundary pairs where binary input is disabled. The normalization macro-step is therefore deterministic.
- In the one-work-symbol and boundary-compatible three-symbol raw lifts, there are no ε-transitions at all; the right endmarker is the only nonbinary input symbol.
- In the centered literal-demand construction, every binary step is real-time and the right endmarker is enabled only at the balanced empty configuration.

All replacement words are fixed finite constants once (p, q) is fixed. If a separate PDA convention imposes a bound on replacement length, long fixed writes can be compiled using auxiliary deterministic microstates. Such compilation changes the physical transition granularity and must not be silently substituted into a theorem whose statement concerns raw one-input transitions or exact state counts in a fixed model.

B.9 Summary table

construction	states	stack alphabet	exact role
paired-sign normalized lift	$\max(p, q)$	$\{0, 1, X, Z_0\}$	exact residual conjugacy after a unique normalization macro-step
one-work-symbol raw lift	$p + q$	$\{A, Z_0\}$	exact raw residual conjugacy; matching upper bound for the one-work-symbol state theorem
boundary-compatible raw lift	$\max(p, q)$	$\{0, 1, Z_0\}$	ε-free raw residual conjugacy by ordinary binary transitions
centered literal-demand lift	$2\lfloor (p+q-1)/2 \rfloor + 1$	$\{0, 1, Z_0\}$	real-time shortest-demand normalization with finite-control SEARCH debt; exact width only in the canonical deferred-SEARCH model

The purpose of the table is organizational. The constructions optimize different interfaces, and their state counts are not interchangeable unrestricted complexity claims.

Appendix C

Scoped Lower Bounds and Finite Certificates

C.1 Why this appendix is necessary

Several chapters use the word *optimal*, but they do so in different representation categories. The purpose of this appendix is to make the scope of every principal lower bound explicit and to place the one genuinely computer-assisted finite elimination in an explicitly auditable finite form.

The central distinction is:

$$\boxed{\text{representation-relative lower bound} \neq \text{unrestricted recognition lower bound.}}$$

A machine may recognize the same language while abandoning the residual coordinates, stack-tail form, literal-demand interface, or normalization schedule used by another construction. A lower bound obtained by counting coordinates in one architecture therefore cannot automatically be promoted to all DPDAs.

C.2 Lower-bound scope table

result	opponent class	exact bound	excluded overclaim
one-work-symbol raw residual conjugacy	permanent bottom, one work symbol, ε-free raw residual–configuration conjugacy	$\|Q\| \geq p+q$	not an unrestricted recognition bound for $L_{p,q}$
homogeneous-tail raw frontier	permanent bottom, sign-separated homogeneous tails, at least two work symbols	$\|Q\| \geq \max(p,q)$	not a theorem about arbitrary stack encodings
arbitrary three-symbol raw residual encoding	permanent bottom, three total stack symbols, arbitrary injective raw residual encoding	$\|Q\| \geq \max(p,q)$ for every coprime (p,q)	Theorem 8.7 is representation-shape-free within Model R; it is not an unrestricted recognition bound
canonical deferred-SEARCH width	literal shortest-demand representation with deferred SEARCH interface	$2\lfloor (p+q-1)/2\rfloor+1$ states	not an unrestricted state minimum for every DPDA recognizing $L_{p,q}$
phase-local capacity	specified local observation depth and remainder architecture	$\|Q\|g^{\lambda} \geq BH(B)$	not a universal Pareto frontier

result	opponent class	exact bound	excluded overclaim
binary stack-alphabet hierarchy witness	arbitrary general DPDA in the stated single-initial-symbol model, deterministic ε-moves allowed	$\lvert Q\rvert\lvert\Gamma\rvert \geq mk+1$ for $B_{m,k}$	applies to the engineered witness language, not automatically to $L_{p,q}$

C.3 Raw residual-conjugacy frontier

For the permanent-bottom raw residual-conjugacy models developed in the resource chapters, the exact structured frontier is:

$$\boxed{S_{\mathrm{raw}}(g) = \begin{cases} \infty, & g = 0, \\ p+q, & g = 1, \\ \max(p,q), & g \geq 2, \end{cases}}$$

where g denotes the number of work symbols above the permanent bottom marker in the homogeneous-tail model.

The upper bounds are realized by the constructions collected in Appendix B. The lower bounds rely on the raw-conjugacy interface: every residual must have one injective physical configuration, and each binary residual edge must be realized by one ordinary input transition.

The $g = 1$ lower bound is exact because a one-work-symbol store can encode unbounded magnitude only through height, while the p positive-end and q negative-end arithmetic phases require disjoint recurrent state resources. The resulting state requirement is

$$\boxed{|Q| \geq p+q.}$$

With two distinct work symbols, sign can be placed in the visible top symbol. The boundary-compatible construction then attains

$$\boxed{|Q| = \max(p,q), \qquad |\Gamma| = 3}$$

including the permanent bottom marker.

The homogeneous-tail hypothesis is removed completely in Chapter 8. The representation-shape-free Model-R argument proves the same three-symbol state optimum $|Q| = \max(p,q)$ for every coprime pair. The earlier small-ratio eliminations remain useful finite certificates, but they are no longer the endpoint of the theory.

C.4 The 3:2 finite reduction

The first genuinely two-sided arbitrary-encoding case is $(p,q) = (3,2)$. The theorem concerns the full permanent-bottom three-symbol model

$$\Gamma_3 = \{A, B, Z_0\}$$

with no ε-moves and an injective raw residual–configuration conjugacy. No homogeneous-tail or sign-separated assumption is imposed.

Assume, for contradiction, that two control states suffice. Choose an encoded configuration of minimum work-stack height and write the two parity chains relative to its residual r_* as

$$A_k = \eta(r_* + 2k), \qquad B_k = \eta(r_* + 1 + 2k).$$

Let G denote the input-1 action and F the input-0 action. Since input 1 adds 2 to the residual and input 0 subtracts 3,

$$G(A_k) = A_{k+1}, \qquad G(B_k) = B_{k+1},$$

$$F(A_{k+2}) = B_k, \qquad F(B_{k+1}) = A_k.$$

Therefore, on the encoded residual graph,

$$\boxed{F^2G^3 = G^3F^2 = \mathrm{id}\,.} \tag{C.1}$$

Equation (C.1) is the key finiteness mechanism. One consuming PDA transition deletes at most one work symbol. If a work-top rule writes m work symbols in place of one, its first step increases work height by $m-1$. The remaining four moves of either five-step inverse identity can delete at most four symbols before the original configuration must be recovered. Hence

$$\boxed{m \le 5}$$

for every work-top replacement encountered in the search. From the bare bottom, the first step creates rather than replaces its work word, so every bottom write has length at most four.

Thus an a priori infinite transition search reduces to a finite deterministic rule-synthesis problem.

C.5 Normalization of the minimum configuration

There are exactly two minimum-height cases.

Positive minimum height. The suffix strictly below the visible top of a minimum encoded configuration can never be exposed by another encoded configuration; it is inert and may be stripped. Rename states and work symbols so that

$$\boxed{A_0 = (0, A\#),}$$

where # denotes the fixed inert suffix.

Zero minimum height. Normalize instead to

$$\boxed{A_0 = (0, \#),}$$

where # is the bare bottom marker.

These two cases exhaust the possible minimum work heights.

C.6 Legal branching and pruning rules

The finite certificate assigns a transition rule only when a state/top pair is encountered for the first time. At that first occurrence it branches over every legal target state and every replacement word satisfying the proven length bounds. A repeated state/top pair must reuse the rule already fixed by determinism.

Branches are rejected only by consequences already forced by the raw conjugacy and the inverse identities:

1. a prescribed F-relation or G-relation has the wrong target;

2. suffix preservation makes the required replacement impossible;
3. two distinct residuals obtain the same encoded configuration, violating injectivity;
4. a configuration is too tall to return to a prescribed target in the remaining number of one-pop transitions.

Representative deletion-capacity bounds are

$$F^2(A_3) = A_0, \qquad F(B_1) = A_0,$$

which imply

$$|A_3| \leq |A_0| + 2, \qquad |B_1| \leq |A_0| + 1.$$

More generally,

$$F(B_j) = A_{j-1}, \qquad F(A_{j+2}) = B_j,$$

and the corresponding two-deletion relations

$$F^2(B_j) = B_{j-3}, \qquad F^2(A_{j+2}) = A_{j-1}$$

provide the remaining legal height-pruning bounds.

No heuristic pruning is permitted: every discarded branch violates one of these proved necessary conditions.

C.7 Exact survivor vectors

For positive minimum height, the exact numbers of surviving partial rule systems at successive synthesis stages are

$$\boxed{1, 123, 11643, 1691, 2120, 1591, 348, 43, 13, 0.} \tag{C.2}$$

The search examines exactly

$$\boxed{\text{1,130,430}}$$

branch attempts in this case.

For minimum height zero, the survivor vector is

$$\boxed{1, 61, 7556, 1134, 2146, 2966, 6846, 7032, 2376, 650, 190, 26, 0.} \tag{C.3}$$

This case examines exactly

$$\boxed{\text{1,696,120}}$$

branch attempts.

Both exhaustive searches terminate with zero survivors. Since the symbolic reduction proves that every possible two-state raw encoding must occur in one of the two finite searches, two states are impossible. The explicit three-state boundary-compatible construction supplies the matching upper bound. Hence

$$\boxed{|Q|_{\min} = 3}$$

for the arbitrary permanent-bottom three-symbol raw-conjugacy model at $(3, 2)$, and symmetrically at $(2, 3)$.

C.8 Computational certificate validation

The P2 computational validation was independently rerun from the final research corpus. The complete audit terminates with

$$\boxed{\text{80,717,789 checks – PASS.}}$$

Among the independently checked components are

component	count
brute minimum-demand cases	11,011
broad arithmetic cases	1,548,547
compact normalized macro cases	667,110
normalized rectangle vertices	1,859,269
normalized rectangle diamonds	1,286,746
raw four-symbol immediate cases	1,111,110
one-work-symbol raw cases	1,111,110
three-symbol raw cases	1,111,110
raw rectangle classifications	61,455
three-symbol raw rectangle vertices	731,091
three-symbol raw rectangle diamonds	474,684

The same validation also confirms the exact branch counts and survivor vectors in (C.2)–(C.3).

The later resource-frontier computational validation was likewise rerun and terminates with

$$\boxed{\text{79,606,679 checks – PASS.}}$$

These computational audits support the symbolic results and certify the bounded search. Except for the explicitly computer-assisted $(3, 2)$ elimination, they are regression evidence rather than substitutes for proof.

C.9 Architecture-relative deferred-SEARCH width

For the centered literal-demand machine, put

$$h_{\text{ctr}} = \left\lfloor \frac{p+q-1}{2} \right\rfloor .$$

In the canonical deferred-SEARCH model, after a witness demand block has been consumed, the remaining work stack is the same bare bottom marker for every debt value

$$-h_{\text{ctr}}, \ldots, h_{\text{ctr}}.$$

Distinct residuals are continuation-distinguishable. Determinism therefore forces distinct control states for all these debt values. The exact width is

$$\boxed{|Q|_{\min}^{\text{deferred}} = 2h_{\text{ctr}} + 1 = 2\left\lfloor \frac{p+q-1}{2} \right\rfloor + 1.}$$

The centered construction attains the bound.

The deferred condition is essential: the final consumed witness cell is a pure pop, and materialization is postponed to a later transition. Eager encodings, alternate stack coordinates, or machines that do not store literal minimum demand are outside this lower-bound category.

C.10 Representation-independent hierarchy product bound

A different chapter deliberately leaves the counting-language representation class. For every $m \geq 2$ and $k \geq 1$, the binary witness language $B_{m,k}$ is recognized by a realtime DPDA with exactly m states and $k+1$ pushdown symbols. Every competing DPDA in the stated *general* deterministic model—including opponents with arbitrary deterministic ε-moves and unrelated internal stack encodings—satisfies

$$\boxed{|Q|\,|\Gamma| \geq mk+1.} \tag{C.4}$$

Consequently the fixed-state stack-alphabet classes obey the strict hierarchy

$$\boxed{\mathcal{D}_{m,k} \subsetneq \mathcal{D}_{m,k+1} \qquad (m \geq 2,\ k \geq 1).}$$

The proof does not count residual phases in a prescribed code. It isolates $mk+1$ externally distinguishable affine completion slopes. Repetition, height-escape, and step-pump arguments show that one recurrent state/top-symbol mode cannot realize two distinct slopes. Since a DPDA has at most $|Q||\Gamma|$ nonempty modes, (C.4) follows.

This is representation-independent for the witness language $B_{m,k}$, but it must not be retroactively read as an unrestricted product lower bound for the rational counting languages $L_{p,q}$.

C.11 Final scope rule

The lower-bound hierarchy in this monograph is intentionally layered:

$$\boxed{\begin{array}{c}\text{coordinate-relative exact frontiers}\\ \Downarrow\\ \text{selected arbitrary-encoding raw bounds}\\ \Downarrow\\ \text{architecture-relative literal-demand width}\\ \Downarrow\\ \text{separate representation-independent witness hierarchy.}\end{array}}$$

The arrows describe the historical research route, not logical implication between the theorem classes. Each lower bound remains attached to the opponent model stated in its theorem.

Appendix D

Computational Validation

D.1 Role of computation

The symbolic proofs are the mathematical basis of the monograph. Computational audits serve three narrower purposes:

1. detect transcription errors in long transition tables and closed forms;
2. test boundary cases across large finite parameter ranges;
3. certify a finite bounded elimination when a theorem explicitly has a computer-assisted final step.

The discussion below records the scope and numerical outcome of each audit. Implementation-specific identifiers and storage locations are deliberately omitted: the relevant information for the mathematical record is what was tested, whether the run was independently repeated, and what result was obtained.

D.2 Validation-status terminology

Three statuses are used throughout this appendix.

Independently rerun.
The reported validation was repeated during the monograph audit and the stated outcome was reproduced.

Reconstructed and rerun.
The exact historical executable was unavailable, but an equivalent regression audit was reconstructed from the authoritative mathematical specification and archived checkpoints, then run independently. Such a reconstruction is not claimed to be byte-for-byte identical to the historical implementation.

Source-reported only.
A manuscript or checkpoint reports a numerical audit, but the monograph does not claim an independent rerun of that exact historical computation.

This distinction concerns the provenance of the computational evidence only; it does not alter the symbolic theorem statements.

D.3 Independently rerun audits

P1 — completion/frontier/original-machine audit

The independent audit passes and includes:

check family	count
original-machine binary inputs	524,287
prefix transition checks	8,912,898
independent quotient-distance checks	25,281,628
compact-DPDA runs	1,282,610
parameter formula audits	3,931

The audit checks the original-machine residual invariant, reachable-store structure, completion normalization, shortest-completion formulas, compact realization, and related finite-frontier claims.

P2 — structure/resource audit

The independent audit terminates with

$$\boxed{\text{80,717,789 checks – PASS.}}$$

Principal reported subcounts are:

check family	count
brute minimum-demand cases	11,011
broad arithmetic cases	1,548,547
compact macro cases	667,110
configuration-rectangle vertices	1,859,269
configuration-rectangle diamonds	1,286,746
phase samples	44,047
raw four-symbol cases	1,111,110
raw four-symbol defects encountered	16,247
one-work-symbol cases	1,111,110
raw three-symbol cases	1,111,110
raw rectangle cases	61,455
raw three-symbol rectangle vertices	731,091
raw three-symbol rectangle diamonds	474,684
ladder paths	11,778

The same audit contains the bounded $(3, 2)$ rule-synthesis certificate recorded in Appendix C.

P5 — exact resource-frontier audit

The independent audit terminates with

$$\boxed{\text{79,606,679 checks – PASS.}}$$

It independently rechecks the minimum-demand arithmetic, normalized and raw configuration rectangles, raw-conjugacy boundaries, and the state–alphabet resource formulas used in the exact frontier analysis.

P9 — centered real-time family audit

The independent audit covers every ordered coprime pair

$$1 \leq p, q \leq 20$$

and reports

$$\boxed{\text{255 ordered coprime ratios}},$$

$$\boxed{\text{1,909,191 reached configurations}},$$

and

$$\boxed{\text{3,738,864 binary edges}}.$$

The audit checks transition totality on reached binary configurations, the residual invariant, residual-collision consistency, two-cell visibility, and zero-residual correctness. The rerun passes.

D.4 Reconstructed validation audits

P10 — WRITE–SEARCH–CANCEL audit

The P10 study reports

$$\boxed{\text{477,033 assertion-level checks.}}$$

The exact historical executable was unavailable in the final research corpus. An equivalent regression audit was therefore reconstructed on 19 August 2026 from the authoritative manuscript's explicit audit scope and archived research checkpoints. The reconstructed audit checks:

- all coprime $1 \leq p, q \leq 8$ and all binary prefixes of length at most nine for the closed-form minimum-demand formula, semantic carry criterion, and demand clock;
- the bounded canonical one-top direct-realizability classification;
- WSC defect-lattice coordinate formulas;
- count-shadow fiber inequalities;
- physical-factorization reconstruction formulas;
- bounded carry-order debt instances;
- supplementary kernel-fiber identities.

It reproduces the manuscript-reported total exactly:

$$\boxed{\text{477,033 checks – PASS.}}$$

This is a functional reconstruction of the validation, not a claim of recovering the historical implementation.

P11 — semantic-uniqueness/lift audit

The P11 study reports

$$\boxed{\text{458,776 assertion-level checks.}}$$

The exact historical executable was unavailable in the final research corpus. An equivalent validation audit was reconstructed while preserving the exhaustive earlier core and adding the later schedule/causality checks stated in the authoritative manuscript. In particular it checks:

- the six admissible and two inadmissible 2:1 schedules;
- pairwise distinct reduced schedule fingerprints;
- the two-maxima/non-lattice schedule-poset structure;
- the directed $A \rightsquigarrow B$ causal certificate;
- the one-kernel excess at the eager B successor;
- the first forbidden recognition witness.

The reconstructed audit reproduces the reported total exactly:

$$\boxed{\text{458,776 checks – PASS.}}$$

Again, the claim concerns equivalence of the tested mathematical conditions and reproduced count, not identity with the historical implementation.

D.5 P12 and computational support

The General Theory is formulated and proved symbolically. Its verification chapter imports classical decision mechanisms only after reducing correctness to an exact structural target. No external computation is required to establish the main P12 proofs.

Finite catalogues used as stress tests, including the 3:2 and odd-p:2 schedule families, are recorded separately in the General-Theory catalogue appendix rather than being treated as substitutes for the abstract theorems.

D.6 Post-lock P13–P15 computational audits

The three post-lock manuscripts retain symbolic proofs as primary evidence. Their numerical audits are recorded below as source-reported unless an independent rerun is explicitly stated.

P13 — explicit run–return determinants

The source paper reports direct coarea enumeration against the weighted determinant formula and an independent coefficient-free check. The final manuscript records 33 sampled-run determinant cases and 86 coefficient-free cases with no mismatches; an earlier full audit record additionally reports an extended 220-case determinant/direct-enumeration pass. These computations audit indexing and signs and are not proof substitutes.

P14 — contact-preserving barrier compression

The source paper reports 88,032 determinant/enumeration comparisons for the general exact-contact theorem. Its rational-path specializations report 755 ordinary fixed-composition instances with 9,983 statistic queries, 628 ratio-run instances with 54,664 exact-return queries, and 501 simultaneous instances with 246,392 queries, all without discrepancy. The fixed-signature audit covers 2,357 classes over 12,081 paths. A larger $(23, 11)$ stress test enumerates all 63,920 fixed-composition paths; 616 paths of signature 11111110 split into four classes of 154 each, matching the determinant.

P15 — Farey recursion and spectra

The reviewer-oriented source audit exhaustively checks all 24,064 ordered coprime pairs with $2 \leq p, q \leq 200$ and adds 30,000 randomized coprime pairs with $p, q \leq 5000$ for the unit-lift laws, closed profiles, all four ordered-recursion identities, and the complementary index partitions, with no counterexample. The scalar audit covers all 13,715 coprime ratios with $1 \leq p, q \leq 150$ through residual shells $k = 0, \ldots, 7$, again with no counterexample. The novelty stress test separately confirms that the local walk is a unimodular image of classical mechanical/Christoffel geometry and therefore keeps the novelty claim on the cross-parameter minimizer recursion.

D.7 Interpretation rule

Throughout the book, the phrase "validated computationally" must be read according to the hierarchy

$$\boxed{\text{symbolic theorem first,} \qquad \text{computational evidence second.}}$$

The sole exception is an explicitly labeled computer-assisted theorem whose symbolic argument reduces the remaining obligation to a finite exhaustive search. In that case, the finite certificate is part of the proof and its pruning assumptions, survivor vectors, and terminal zero-survivor result must be preserved together.

D.8 Post-theory validation added in v3

D.8.1 Fixed-Parikh quantitative LIFO chronology

The probabilistic results of Part XV are symbolic. Computation was used only as an independent transcription and boundary-case audit of the finite physical-lift kernels and small-parameter constants. Direct probability propagation and kernel calculations were compared on overlapping horizons and parameter pairs; the numerical checks do not enter any proof premise.

D.8.2 Optimal-fiber rank-two topology

The rank-two classification of Part XVI is proved symbolically. Finite-grid computations were used as consistency checks for active-fiber saturation, nerve compression, cellular homology, and the explicit scalar realization families. These calculations serve only as audits of examples and boundary cases.

D.8.3 Unrestricted state complexity

The lower bound of Part XVII is entirely symbolic. During preparation, the matching three-symbol construction was independently sanity-checked on finite ranges of coprime parameters and residuals by verifying the two translation identities for every tested configuration. No computational assertion is used in the proof of the lower bound or exactness theorem.

D.9 Geodesic-compiler validation added in v6

P27 — prescribed-order separation and Boolean amplification

The geodesic-compiler and non-factorization theorems are proved symbolically. For the v6 integration, an independent bounded regression was reconstructed directly from the explicit translation actions in the theorem statements; the corresponding regression materials are archived with the source bundle. The audit computes shortest distances by reverse breadth-first search rather than by using the closed distance formulas being checked.

For every $1 \le k \le 10$, the audit verifies the Binary-Deficit Distance Lemma on every pair

$$0 \le D < 2^k, \qquad k \le \ell \le 2k,$$

for a total of

$$\boxed{\text{20,480 binary-deficit instances.}}$$

For the same ten blocks it reconstructs the complete structural/optimal face data and confirms

$$\mathcal{H}(r) = 2^{A_-}, \qquad \mathcal{H}(r') = 2^{A_-} \setminus \{J_\star, A_-\},$$

including exact agreement through order $k-1$ and the unique order-k discrepancy.

A separate product-graph reverse-BFS audit checks

$$(k, m) = (1, 3), (2, 2), (3, 2), (4, 2).$$

In every case all 2^m selected product states have the same exact distance and the same order-$(k-1)$ signature, while the order-k signatures are pairwise distinct and each coordinate toggles exactly its distinguished face $J_\star^{(t)}$. The archived run terminates with

$$\boxed{\text{OVERALL PASS.}}$$

These computations are independent regression evidence only and are not premises of the compiler, non-factorization, or amplification theorems.

D.10 Universal k-fiber validation added in v7

The Universal k-Fiber Theorem and exact capacity theorem in Chapter 19 are proved symbolically. As an independent regression check, the explicit two-phase construction was exhaustively evaluated for several small parameter pairs. For every witness state in each tested family, the audit checked the common closed-form distance, reconstructed the complete structural/optimal truncation through order $k-1$, and tested every distinguished k-face against the Bellman equality.

s	k	$\binom{s}{k}$	witnesses	common distance	patterns
2	2	1	2	6	2
3	2	3	8	11	8
4	2	6	64	18	64
5	2	10	1024	27	1024
3	3	1	2	8	2
4	3	4	16	18	16
5	3	10	1024	37	1024
4	4	1	2	10	2
5	4	5	32	27	32

In every tested case, all witness states had one common exact distance and one common complete lower-order compatibility truncation, while the distinguished order-k layer ranged over all $2^{\binom{s}{k}}$ prescribed patterns. The $s = 3, k = 2$ row independently reproduces all eight simple graphs in the worked universal fiber of Chapter 19. These computations are regression evidence only and are not premises of any theorem.

Appendix E

Lagrange–Bürmann Derivation of the Exact Correction Triangle

E.1 The marked first-return grammar

This appendix supplies the coefficient derivation deferred from Chapter 20. Fix an integer $d \geq 1$. Let $F(z,t)$ denote the correction-marked excursion generating function at baseline zero, and let $A(z,t)$ and $B(z,t)$ be the two periodic tail classes introduced by the first-return decomposition. They satisfy

$$\boxed{F = 1 + zA^d F,} \tag{E.1}$$

$$\boxed{A = 1 + zA^d B,} \tag{E.2}$$

$$\boxed{B = 1 + ztA^{d-1}B^2.} \tag{E.3}$$

The variable z marks down-steps and t marks the distinguished correction events.

Define

$$\boxed{Q_d(n,k) = [z^n t^k]F(z,t).}$$

The goal is to derive the closed form for $Q_d(n,k)$ directly from (E.1)–(E.3).

E.2 The auxiliary kernel variable

Set

$$Y = zA^{d-1}B.$$

Equation (E.2) gives

$$A = 1 + AY,$$

so

$$\boxed{A = \frac{1}{1-Y}.} \tag{E.4}$$

Similarly, divide (E.3) by B and use $Y = zA^{d-1}B$:

$$1 = B^{-1} + tY,$$

whence

$$\boxed{B = \frac{1}{1-tY}.} \tag{E.5}$$

Substituting (E.4)–(E.5) into the definition of Y yields

$$Y = \frac{z}{(1-Y)^{d-1}(1-tY)}.$$

Equivalently,

$$\boxed{Y(1-Y)^{d-1}(1-tY) = z.} \tag{E.6}$$

Thus Y is the unique formal power series with zero constant term satisfying

$$\boxed{Y = z\,\Phi(Y), \qquad \Phi(y) = (1-y)^{-(d-1)}(1-ty)^{-1}.} \tag{E.7}$$

This is exactly the form required for Lagrange inversion.

E.3 Coefficient formula for the auxiliary kernel

Define

$$K_d(z,t) = \frac{Y}{z}.$$

Since

$$[z^n t^k]K_d = [z^{n+1}t^k]Y,$$

the basic Lagrange formula applied to (E.7) gives

$$[z^{n+1}t^k]Y = \frac{1}{n+1}[y^n t^k]\Phi(y)^{n+1}.$$

Now

$$\Phi(y)^{n+1} = (1-y)^{-(d-1)(n+1)}(1-ty)^{-(n+1)}.$$

Extracting the t^k coefficient gives

$$[t^k](1-ty)^{-(n+1)} = \binom{n+k}{k}y^k.$$

Therefore

$$[z^n t^k]K_d = \frac{1}{n+1}\binom{n+k}{k}[y^{n-k}](1-y)^{-(d-1)(n+1)}.$$

Using

$$[y^r](1-y)^{-a} = \binom{a+r-1}{r},$$

we obtain

$$\boxed{[z^n t^k]K_d = \frac{1}{n+1}\binom{n+k}{k}\binom{dn+d-k-2}{n-k}.} \tag{E.8}$$

For $d=2$ this reduces to

$$\boxed{[z^n t^k]K_2 = \frac{1}{n+1}\binom{n+k}{k}\binom{2n-k}{n-k}.} \tag{E.9}$$

This is the auxiliary kernel triangle identified in the main text. It is not the actual correction distribution.

E.4 Recovering the correction generating function

From (E.1),

$$F = \frac{1}{1 - zA^d}.$$

Using (E.4),

$$\boxed{F = \sum_{m \geq 0} z^m A^{dm} = \sum_{m \geq 0} z^m (1 - Y)^{-dm}.} \tag{E.10}$$

This expansion is the convenient starting point for the exact correction triangle.

For fixed $n \geq 1$,

$$Q_d(n,k) = \sum_{m=0}^{n} [z^{n-m} t^k](1 - Y)^{-dm}. \tag{E.11}$$

The term $m = n$ contains no positive power of Y, so it contributes

$$\boxed{\mathbf{1}_{\{k=0\}}.} \tag{E.12}$$

The term $m = 0$ contributes nothing when $n \geq 1$. It remains to treat $1 \leq m \leq n - 1$.

E.5 Lagrange–Bürmann extraction for a fixed summand

Fix $1 \leq m \leq n - 1$ and set

$$N = n - m > 0.$$

Apply Lagrange–Bürmann to

$$f(y) = (1 - y)^{-dm}.$$

Since

$$f'(y) = dm(1 - y)^{-dm-1},$$

we obtain

$$[z^N](1 - Y)^{-dm} = \frac{dm}{N}[y^{N-1}](1 - y)^{-dm-1}\Phi(y)^N.$$

Substitute (E.7):

$$[z^N](1 - Y)^{-dm} = \frac{dm}{N}[y^{N-1}](1 - y)^{-dm-1-(d-1)N}(1 - ty)^{-N}. \tag{E.13}$$

Because $N = n - m$,

$$dm + 1 + (d - 1)N = (d - 1)n + m + 1.$$

Thus

$$[z^N t^k](1 - Y)^{-dm} = \frac{dm}{N}[y^{N-1} t^k](1 - y)^{-((d-1)n+m+1)}(1 - ty)^{-N}.$$

Now

$$[t^k](1 - ty)^{-N} = \binom{N + k - 1}{k} y^k,$$

so

$$[z^N t^k](1 - Y)^{-dm} = \frac{dm}{N}\binom{N + k - 1}{k}[y^{N-k-1}](1 - y)^{-((d-1)n+m+1)}.$$

A final binomial extraction gives

$$[y^{N-k-1}](1 - y)^{-((d-1)n+m+1)} = \binom{dn - k - 1}{N - k - 1}.$$

Hence

$$\boxed{[z^{n-m}t^k](1-Y)^{-dm} = \frac{dm}{n-m}\binom{n-m+k-1}{k}\binom{dn-k-1}{n-m-k-1}.} \tag{E.14}$$

The coefficient is automatically zero unless

$$n-m \geq k+1.$$

E.6 Exact correction triangle

Insert (E.12) and (E.14) into (E.11). Make the change of variable

$$j = n-m.$$

Then

$$m = n-j,$$

and the nonzero range is

$$k+1 \leq j \leq n-1.$$

Therefore

$$\boxed{Q_d(n,k) = \mathbf{1}_{\{k=0\}} + \sum_{j=k+1}^{n-1} \frac{d(n-j)}{j}\binom{j+k-1}{k}\binom{dn-k-1}{j-k-1}.} \tag{E.15}$$

This is the formula stated in Chapter 20.

The upper support bound follows immediately. If $k > n-2$, then there is no integer j with

$$k+1 \leq j \leq n-1,$$

and the indicator term vanishes because $k > 0$. Thus

$$\boxed{Q_d(n,k) = 0 \qquad (k > n-2).} \tag{E.16}$$

E.7 Equivalent elimination through $U = A - 1$

For completeness, the scalar equation quoted in the main chapter follows directly from the same grammar. Put

$$U = A - 1.$$

From (E.2),

$$U = zA^dB = AY.$$

Since $A = 1 + U$, this gives

$$Y = \frac{U}{1+U}.$$

Equation (E.3) can be written as

$$B = 1 + tYB,$$

so

$$B = \frac{1}{1-tY} = \frac{1+U}{1+(1-t)U}.$$

Substituting into $U = zA^dB$ with $A = 1 + U$ gives the convenient form

$$\boxed{U\big(1 + (1-t)U\big) = z(1+U)^{d+1}.} \tag{E.17}$$

Equation (E.17) and the kernel equation (E.6) are related by

$$\boxed{Y = \frac{U}{1+U}.}$$

Thus the two inversion routes are exactly the same formal system in different coordinates.

E.8 Row sum

Setting $t = 1$ forgets the correction marking. Equation (E.17) becomes

$$U = z(1+U)^{d+1}.$$

At $t = 1$, the correction generating function simplifies to

$$F(z, 1) = 1 + U.$$

Lagrange inversion gives, for $n \geq 1$,

$$[z^n]U = \frac{1}{n}[u^{n-1}](1+u)^{(d+1)n} = \frac{1}{n}\binom{(d+1)n}{n-1}.$$

Using

$$\frac{1}{n}\binom{(d+1)n}{n-1} = \frac{1}{dn+1}\binom{(d+1)n}{n},$$

we recover the Fuss–Catalan row sum

$$\boxed{\sum_{k\geq 0} Q_d(n,k) = \frac{1}{dn+1}\binom{(d+1)n}{n}.} \tag{E.18}$$

E.9 Why the kernel and correction arrays differ

The kernel coefficients (E.8) arise from the single implicit series Y and record the algebraic kernel introduced to solve the periodic grammar. The correction coefficients (E.15) arise only after the full zero-baseline excursion series

$$F = \sum_{m\geq 0} z^m (1-Y)^{-dm}$$

is reconstructed.

Therefore

$$\boxed{K_d \text{ is an algebraic waypoint, whereas } Q_d \text{ is the path statistic.}}$$

The distinction is not terminological: the two coefficient arrays have different combinatorial meanings and diverge as soon as the correction marking becomes nontrivial.

Appendix F

General-Theory Stress-Test Catalogues

F.1 Purpose and scope

The abstract General Theory developed in Part XI is not defined by the weighted-balance family. Its semantic system may be non-additive and need not have a residual space isomorphic to $\mathbb{Z}$. The centered $p{:}q$ constructions are instead a deliberately rigid stress-test family in which safe-reset domains, source-fiber expansion, directed exposure, and robust fatal supports can all be calculated exactly.

This appendix records the full finite 3:2 certificate and the infinite odd-p:2 arithmetic/source catalogues used in the stress-test theorems. The purpose is reproducibility: the main chapters can state the structural consequences without interrupting the narrative with long case catalogues.

Throughout this appendix, for a canonical literal-demand pair (x, y) we use

$$\eta(x, y) = \begin{cases} (10)^y 0^{x-y}, & x \geq y, \\ (10)^x 1^{y-x}, & y > x. \end{cases}$$

For the weighted-balance residual, the primitive nonnegative kernel vector is $K = (q, p)$.

F.2 The exact centered 3:2 certificate

Set

$$p = 3, \qquad q = 2, \qquad m = p + q = 5,$$

and use the centered states

$$D_{-2}, D_{-1}, \mathsf{C}, D_1, D_2.$$

The six natural eager sites are indexed by the current centered state, the next input symbol, and the visible top symbol. Their eager replacement words are

site	(d, a, X)	γ
A	$(-2, 0, 0)$	11111
B	$(-2, 1, 0)$	10111
C	$(-2, 1, 1)$	11111
D	$(1, 0, 0)$	10100
E	$(1, 1, 0)$	10000
F	$(2, 0, 0)$	10000

(F.1)

F.2.1 Finite reset repertoires

Define

$$\boxed{\begin{aligned}\mathcal{R}_0^{(3,2)} &= \{(2,2)\} \cup \{(0,y) : y \geq 4\} \cup \{(1,y) : y \geq 3\} \\ &\cup \{(x,1) : x \geq 3\} \cup \{(x,0) : x \geq 4\}\end{aligned}} \tag{F.2}$$

and

$$\boxed{\mathcal{R}_D^{(3,2)} = \mathcal{R}_0^{(3,2)} \cup \{(x,2) : x \geq 3\}.} \tag{F.3}$$

Before the first reset event of an episode begun from one of these canonical pairs, the relevant eager-source shapes are

local context	$\mathcal{R}_0^{(3,2)}$	$\mathcal{R}_D^{(3,2)}$
D_{-2}, top 0	0	0^k
D_{-2}, top 1	1^k	1^k
D_1, top 0	0^k	0^k
D_2, top 0	0^k	0^k

$(k \geq 1)$. (F.4)

The bottom-WRITE targets are

d	0	1
-2	$(1,8)$	$(0,4)$
-1	$(0,4)$	$(1,3)$
0	$(1,3)$	$(2,2)$
1	$(2,2)$	$(3,1)$
2	$(3,1)$	$(4,0)$

(F.5)

and every one lies in $\mathcal{R}_0^{(3,2)}$.

Writing the unary source tail as $t \geq 0$, the remaining reset targets are

reset action	canonical target pair
positive boundary WRITE	$(t+5,0)$
negative boundary WRITE	$(1,t+9)$
A	$(0,5)$
B	$(1,4)$
C	$(0,t+5)$
D	$(t+3,2)$
E	$(t+4,1)$
F	$(t+4,1)$

(F.6)

These tables give two closed reset regimes:

(a) with D disabled, every baseline reset and every enabled eager action among A, B, C, E, F returns to $\mathcal{R}_0^{(3,2)}$;

(b) with D enabled and A, B disabled, every baseline reset and every enabled eager action among C, D, E, F returns to $\mathcal{R}_D^{(3,2)}$.

The only new reset ray in the second regime is

$$\{(x,2) : x \geq 3\},$$

and this is exactly the ray that enlarges the D_{-2}-top-0 source from the singleton stack 0 to the unary family 0^k.

F.2.2 Direct verification of reset closure

The canonical words in $\mathcal{R}_0^{(3,2)}$ are respectively

$$1010, \qquad 1^y, \qquad 10\,1^{y-1}, \qquad 1\,0^x, \qquad 0^x,$$

while the added D-ray has

$$\eta(x,2) = 101\,0^{x-1}.$$

Before the first reset, MATCH and interior SEARCH only remove the visible top cell, so the current stack is always a suffix of the starting canonical word. The source catalogue in (F.4) follows immediately. In particular, without the D-ray only the word 1010 can reach D_{-2} while a zero remains visible, and exactly its final zero remains.

For the non-bottom reset events, the centered boundary blocks and the six eager replacement words yield the canonical words

$$0^{t+5}, \quad 10\,1^{t+8}, \quad 1^5, \quad 10\,1^3, \quad 1^{t+5}, \quad 101\,0^{t+2}, \quad 1\,0^{t+4}.$$

These are precisely the encodings listed in (F.6). Every row except D lies in $\mathcal{R}_0^{(3,2)}$; the D row is precisely the additional ray of $\mathcal{R}_D^{(3,2)}$.

F.2.3 Exact schedule classification and fatal witnesses

The admissible schedules are exactly

$$\boxed{\mathrm{Adm}_{3,2} = \{H : \{A,D\} \not\subseteq H,\ \{B,D\} \not\subseteq H\}\,.} \tag{F.7}$$

Therefore

$$\boxed{\text{40 schedules are admissible and 24 are inadmissible.}} \tag{F.8}$$

The minimal directed fatal fork is

$$\boxed{D \to A, \qquad D \to B.} \tag{F.9}$$

The maximal admissible faces are

$$\{A,B,C,E,F\}, \qquad \{C,D,E,F\},$$

so the admissible complex is non-pure and the schedule poset is not a lattice. Its independence polynomial is

$$\boxed{(1+x)^3(1+3x+x^2).} \tag{F.10}$$

Sufficiency follows from the two reset-closed regimes above: the $D = 0$ regime contributes $2^5 = 32$ schedules, while $D = 1$ with $A = B = 0$ contributes $2^3 = 8$ more.

Necessity is certified by two complete balanced words,

$$\boxed{w_B = 1111000011}, \qquad \boxed{w_A = 011111110000011}. \tag{F.11}$$

Their critical portions are

branch	reaching prefix	victim step	diagnostic
$D \to B$	$11110000 : (D_{-2}, 00)$	$1 : (\mathsf{C}, 101110)$	$1 : (\mathsf{C}, 01110)$
$D \to A$	$011111110000 : (D_{-2}, 000)$	$0 : (\mathsf{C}, 1111100)$	$11 : (\mathsf{C}, 11100)$

(F.12)

In the first row, the earlier eager D step occurs on prefix 11110; in the second it occurs on prefix 011111110. Before the victim step the only eager optional site encountered is therefore D. After the victim step the listed diagnostic uses only baseline center transitions. Both terminal configurations have semantic residual zero but a nonempty center stack, so they are nonaccepting under the fixed centered acceptance convention. The witnesses are therefore extension-stable under arbitrary choices at C, E, F and at the other victim site.

F.3 The infinite odd-p:2 family

Fix

$$\boxed{p = 2h - 1, \qquad q = 2, \qquad h \geq 2,} \tag{F.13}$$

so that

$$m = p + q = 2h + 1.$$

For $t \geq 1$ set

$$\epsilon_t = t \bmod 2, \qquad k_t = \left\lceil \frac{t}{2} \right\rceil, \qquad \beta_t = t + p\, k_t,$$

and define

$$\boxed{\Gamma_t = \eta(\epsilon_t, \beta_t).} \tag{F.14}$$

The six eager sites are

$$\begin{aligned}
A &: (D_{-h}, 0T) \xrightarrow{0} (\mathsf{C}, \Gamma_h T), \\
B &: (D_{-h}, 0T) \xrightarrow{1} (\mathsf{C}, \Gamma_{h-1} T), \\
C &: (D_{-h}, 1T) \xrightarrow{1} (\mathsf{C}, \Gamma_h T), \\
D &: (D_{h-1}, 0T) \xrightarrow{0} (\mathsf{C}, (10)^h 0T), \\
E &: (D_{h-1}, 0T) \xrightarrow{1} (\mathsf{C}, (10)^{h-1} 000T), \\
F &: (D_h, 0T) \xrightarrow{0} (\mathsf{C}, (10)^{h-1} 000T).
\end{aligned} \tag{F.15}$$

F.3.1 Arithmetic catalogue

For the minimum-demand map μ the exact identities are:

(i) for every $t \geq 1$,

$$\boxed{\mu(-tm) = (\epsilon_t, \beta_t);} \tag{F.16}$$

(ii) for $0 \leq j \leq h$,

$$\boxed{\mu(jm + 2) = (j + 2, p - j - 1);} \tag{F.17}$$

(iii) for $1 \leq t \leq h + 1$,

$$\boxed{\mu(-tm + 2) = (\epsilon_t, \beta_t - 1);} \tag{F.18}$$

(iv) at the positive outer boundary,

$$\boxed{\mu((h + 1)m) = (h + 3, h - 2).} \tag{F.19}$$

For completeness, if $t = 2r$ then $(0, \beta_t)$ has weight

$$-2(2r + pr) = -2r(p + 2) = -tm,$$

while if $t = 2r + 1$ then $(1, \beta_t)$ has weight

$$p - 2(2r + 1 + p(r + 1)) = -(2r + 1)(p + 2) = -tm.$$

In both cases the first coordinate is below the kernel step $q = 2$, so the pair is kernel-reduced. Equation (F.17) follows from

$$p(j + 2) - 2(p - j - 1) = j(p + 2) + 2 = jm + 2,$$

with second coordinate below p. Equation (F.18) is obtained by lowering the second coordinate in (F.16) by one, and (F.19) follows from

$$p(h+3) - 2(h-2) = (h+1)(2h+1).$$

The complete family of canonical baseline reset pairs is therefore

$$\boxed{\begin{aligned} \mathcal{R}_h^{\text{base}} &= \{\mu(jm+2) : -h-1 \le j \le h\} \\ &\quad \cup \{(x, h-2) : x \ge h+3\} \\ &\quad \cup \{(\epsilon_{h+1}, y) : y \ge \beta_{h+1}\}. \end{aligned}} \tag{F.20}$$

The two schedule-regime repertoires are

$$\boxed{\begin{aligned} \mathcal{R}_{\neg D,h} = \mathcal{R}_h^{\text{base}} &\cup \{(\epsilon_h, y) : y \ge \beta_h\} \\ &\cup \{(\epsilon_{h-1}, \beta_{h-1})\} \cup \{(x, h-1) : x \ge h+2\}, \end{aligned}} \tag{F.21}$$

and

$$\boxed{\begin{aligned} \mathcal{R}_{D,h} = \mathcal{R}_h^{\text{base}} &\cup \{(\epsilon_h, y) : y \ge \beta_h\} \\ &\cup \{(x, h-1) : x \ge h+2\} \cup \{(x, h) : x \ge h+1\}. \end{aligned}} \tag{F.22}$$

Every pair in either repertoire has minority size at most h.

F.3.2 Source-geometry catalogue

Consider an episode beginning at a canonical center pair in one of the two repertoires and stopping at its first reset event. The exact source geometry is:

(i) in $\mathcal{R}_{\neg D,h}$,

$$D_{-h} \text{ with top } 0 \Longrightarrow \boxed{\text{complete remaining stack } 0}; \tag{F.23}$$

(ii) in $\mathcal{R}_{D,h}$,

$$D_{-h} \text{ with top } 0 \Longrightarrow \boxed{\text{complete remaining stack } 0^n,\ n \ge 1}, \tag{F.24}$$

and lengths $n \ge 2$ arise only from the additional D-ray $\{(x, h) : x \ge h+1\}$;

(iii) in either repertoire,

$$\begin{aligned} D_{-h} \text{ with top } 1 &\Longrightarrow 1^n, \\ D_{h-1} \text{ with top } 0 &\Longrightarrow 0^n, \\ D_h \text{ with top } 0 &\Longrightarrow 0^n \end{aligned} \tag{F.25}$$

for some $n \ge 1$.

The key observation is that, before the first reset event, MATCH and interior SEARCH only delete the visible top cell, so the current stack is a suffix of the canonical reset word from which the episode began. Reaching D_{-h} requires at least h more negative than positive SEARCHes. In $\mathcal{R}_{\neg D,h}$ the only canonical pair with enough of both symbols to leave a zero visible at that boundary is (h, h), whose word is $(10)^h$; all h one-cells must have been negative SEARCHes, leaving exactly its final zero. The only new family in $\mathcal{R}_{D,h}$ that can leave a longer zero suffix is (x, h), whose word is $(10)^h 0^{x-h}$.

The remaining unary-source conclusions follow from the same minority-size bound. A suffix containing the opposite symbol beyond the visible boundary cell would force minority size at least $h+1$, contradicting the defining families in (F.21)–(F.22).

F.3.3 Reset-target catalogue and closure

Writing the unary source stack length as $n \geq 1$, all eager reset targets are

$$\begin{array}{c|c|c} \text{site} & \text{source stack} & \text{canonical target pair} \\ \hline A & 0 & (\epsilon_h, \beta_h) \\ B & 0 & (\epsilon_{h-1}, \beta_{h-1}) \\ C & 1^n & (\epsilon_h, \beta_h + n - 1) \\ D & 0^n & (n+h, h) \\ E & 0^n & (n+h+1, h-1) \\ F & 0^n & (n+h+1, h-1) \end{array} \tag{F.26}$$

The identities are literal: for instance

$$(10)^h 0\, 0^{n-1} = \eta(n+h, h)$$

and

$$(10)^{h-1} 000\, 0^{n-1} = \eta(n+h+1, h-1).$$

Consequently:

(a) if D is disabled, baseline resets and enabled actions among A, B, C, E, F preserve $\mathcal{R}_{\neg D,h}$;

(b) if D is enabled and A, B are disabled, baseline resets and enabled actions among C, D, E, F preserve $\mathcal{R}_{D,h}$.

F.3.4 Uniform D-induced source-fiber expansion

For every $x \geq h+1$,

$$\boxed{(\mathsf{C}, \eta(x,h)) \xrightarrow{0^{2h-1}} (D_{-h}, 0^{x-h+1}).} \tag{F.27}$$

Indeed,

$$\eta(x,h) = (10)^h 0^{x-h}.$$

The first $2h-2$ zeros remove $h-1$ complete 10 pairs, and the final zero SEARCHes through the last 1, leaving the last alternating zero together with the zero tail. Thus enabling D expands the common A/B source fiber from

$$\boxed{\{0\}}$$

to

$$\boxed{\{0, 00, 000, \ldots\}}. \tag{F.28}$$

F.3.5 Uniform reachability of the length-three frontier

There is, for every $h \geq 2$, a word u_h that reaches

$$\boxed{(D_{-h}, 000)} \tag{F.29}$$

from the initial centered configuration and encounters no eager optional site other than D.

Define

$$v_h = \begin{cases} 1^{(h-1)m/2}, & h \text{ odd}, \\ 0\,1^{2h+(h-2)m/2}, & h \text{ even}. \end{cases} \tag{F.30}$$

Then

$$\boxed{u_h = v_h\, 1^{\,2h-1}\, 0^{\,2h}.} \tag{F.31}$$

The useful increment block is

$$(D_d, \varepsilon) \xrightarrow{1^m} (D_{d+2}, \varepsilon), \qquad 0 \le d \le h-2. \tag{F.32}$$

The first symbol bottom-WRITEs

$$\mu(dm+2) = (d+2, p-d-1),$$

whose canonical word has length $p+1 = 2h$; the remaining $2h$ ones erase the word, and its $d+2$ zeros contribute the required positive SEARCHes. This builds v_h so that the machine reaches (D_{h-1}, ε). The next 1^{2h-1} segment leaves $(D_{h-1}, 00)$, the following zero invokes D and produces $(\mathsf{C}, (10)^h 00)$, and the remaining 0^{2h-1} reaches $(D_{-h}, 000)$. No site among A, B, C, E, F is encountered before that endpoint.

F.3.6 Uniform kernel-suffix fatality

For every $t \ge 1$, Γ_t has suffix 1^p. Hence there is a unique word z_t such that

$$\boxed{\Gamma_t = z_t 1^p.} \tag{F.33}$$

If $t = 2r$ then

$$\Gamma_t = 1^{\beta_t}, \qquad \beta_t = rm \ge m > p,$$

while if $t = 2r+1$ then

$$\Gamma_t = 10\, 1^{\beta_t - 1}, \qquad \beta_t - 1 = p + rm \ge p.$$

Thus the suffix claim is uniform in t.

At the common source $(D_{-h}, 000)$, eager A and B yield

$$A : (D_{-h}, 000) \xrightarrow{0} (\mathsf{C}, \Gamma_h 00) = (\mathsf{C}, z_h 1^p 00),$$

$$B : (D_{-h}, 000) \xrightarrow{1} (\mathsf{C}, \Gamma_{h-1} 00) = (\mathsf{C}, z_{h-1} 1^p 00).$$

Reading z_h or z_{h-1} from the center uses only MATCHes and reaches

$$\boxed{(\mathsf{C}, 1^p 00).} \tag{F.34}$$

The stack has two zeros and p ones, hence weight

$$p \cdot 2 - 2 \cdot p = 0,$$

but is nonempty. Under centered acceptance it is therefore a site-free fatal diagnostic.

F.3.7 Exact odd-p:2 classification

For every odd $p \ge 3$,

$$\boxed{\mathrm{Adm}_{p,2} = \mathrm{Ind}\big\{\{A, D\}, \{B, D\}\big\}.} \tag{F.35}$$

Equivalently,

$$\boxed{H \in \mathrm{Adm}_{p,2} \iff \neg(A \wedge D) \wedge \neg(B \wedge D).} \tag{F.36}$$

Every member of the family therefore has exactly 40 admissible and 24 inadmissible schedules, with facets

$$\{A, B, C, E, F\}, \qquad \{C, D, E, F\},$$

and the same fatal fork

$$\boxed{D \to A, \qquad D \to B.} \tag{F.37}$$

Sufficiency is the reset-closure induction: $D = 0$ gives $2^5 = 32$ schedules, and $D = 1$ with $A = B = 0$ gives another $2^3 = 8$. Necessity uses the common reaching prefix u_h from (F.31) together with the complete counterexample words

$$\boxed{w_A^{(h)} = u_h\, 0\, z_h,} \qquad \boxed{w_B^{(h)} = u_h\, 1\, z_{h-1}.} \tag{F.38}$$

The prefix u_h encounters no optional eager site except D, and the post-victim suffixes are site-free. The two forbidden supports are therefore robust under arbitrary additions of C, E, F and of the other victim site.

F.4 Interpretive boundary

The finite 3:2 calculation and the infinite odd-p:2 family have a single role in the monograph: they certify that the abstract mechanisms of Part XI are not artifacts of one numerical instance. They exhibit, in exact families,

$$\boxed{\begin{array}{c}\text{safe reset} \longrightarrow \text{fiber expansion}\\ \longrightarrow \text{directed exposure} \longrightarrow \text{robust fatal support.}\end{array}}$$

They do *not* define the General Theory. The General Theory applies to an exact physical lift over a deterministic semantic system through

$$\Theta\delta_a = U_a\Theta,$$

with no requirement that the semantic system be a weighted-balance residual space or even an additive group.

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